Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3120 SCREEN BUILDING

CIVIL DESIGN CALCULATION

AS BUILT

MD-216-3120-EG-CV-CAL-1005

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	16-10-2023	AS BUILT	RPK / APL	SBY	AG	
C01	24-07-2020	ISSUED FOR CONSTRUCTION	RPK / APL	SBY	AG	
A02	19-06-2020	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	RPK / APL	SBY	AG	
A01	07-03-2020	FIRST ISSUE	RPK / APL	SBY	AG	

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Ma'aden Document Number:

MD-216-3120-EG-CV-CAL-1005 D01

Outotec Document Number:

900707 GAU01 ZC01 CED01 31209 OU100672909 3

L&T Document Number:

O19045 100 C K 24070 001 3

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK / APL	RPK / APL	RPK / APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1 INTENT 4

2 EXECUTIVE SUMMARY 4

3 SCOPE 4

4 CONCRETE STRUCTURE 4

4.1 Design 4
4.2 Materials 4
4.3 Reference 4

5 LOAD CALCULATIONS 4-9

5.1 Dead Load 4
5.2 Live Load 4
5.3 Superimposed Dead Load 4
5.4 Wind Load 5
5.5 Temperature Load 8
5.6 Earthquake Load 8
5.7 Load combinations 9

6 GEOMETRY 10-12
6.1 Geometry of Screen Building & Grids 10
6.2 Geometry of Secondary Crusher House Floors 12

7 ANALYSIS & RESULTS OF SECONDARY CRUSHER HOUSE 13

7.1 Deflection of Column 13
7.2 Deflection of Beam 13
7.3 Bending monent diagram of Frames 13

8 DESIGN OF CONCRETE COMPONENTS 14-115

8.1 Design Summary of Column 14
8.2 Design of Slab 14
8.3 Design of Foundation 15
8.4 Design of Beams 34

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWING 124

ATTACHMENT 2 STAAD INPUT FILE OF SCREEN BUILDING 134

ATTACHMENT 3 STAAD OUTPUT FILE OF SCREEN BUILDING 190

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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
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Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning ar

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1. INTENT

The document is prepared for Structural design Calculation of Screen building. This building have the one no Secondary Screen & two nos Tertiary Screen, a Secondary feed bin of 153.6 T capacity, a tertiary feed bin of 144 T capacity, 3 nos vibrating feeder, feed conveyors & discharge conveyors .

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for design of Concrete structure of Screen Building.
This building is partly of concrete structure and partly steel structure.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
3D- Model and analysis have been performed in Staad-Pro Software. All columns of structure have been designed in Staad-Pro by ACI 318M.
Beams, slabs and foundations have been designed based on results from staad analysis in adherance to the above mentioned codes.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1.5m depth below FGL (ie. RL 912.50m) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 LOAD CALCULATONS

5.1 DEAD LOAD

Selfweight of structure is applied through Staad Pro.

Dead load on floor @ 926.500m

Selfweight of 6 o/p chequered plate	=	0.55	kN/m^2
Selfweight of stiffener	=	0.25	kN/m^2
Total dead load from chequered plate	=	0.80	kN/m^2

Dead Load Of RCC Slab

@ 916.3 m Level =	0.15 X 24 = 3.6 kN/m^2
@ 922.75 m Level =	0.175 X 24 = 4.2 kN/m^2
@ 926.5 m Level =	0.175 X 24 = 4.2 kN/m^2

Dead load on floor @ 938.20m

Selfweight of chequered plate	=	0.55	kN/m^2
Selfweight of stiffener	=	0.25	kN/m^2
Total dead load from chequered plate	=	0.80	kN/m^2

Dead load of stairs

Selfweight of Grating	=	0.35	kN/m^2
Selfweight of stringer	=	0.33	kN/m^2
Stair width	=	1.2	m
Stringer Span	=	6	m
Stringer Length	=	7	m

Concentrated load on beam supporting stringer = 1.2 x 6 x 0.35 / 4 + 0.33 x 7 / 2 = 1.8 kN

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5.2 LIVE LOAD

Live load on all floors = 5.00 kN/m^2
Live load on Staircase = 4.00 kN/m^2 Concentrated load on beam supporting stringer
=1.2 x 6 x 4 / 4 7.2 kN

5.3 SUPERIMPOSED DEAD LOAD

All Euipment Loads are considered as superimposed dead load i.e. Screen Load, Feed Bin Load, Vibro Feeder Load, Conveyor Load as per Clause 4.2.1 of Civil, Structural Design Criteria.
For All equipment loads, Refer Attachment 1 - Mechanical General Arrangement drawings and for Bunker Loads Refer Attachement 5 & 6.

SDL1 : Equipment load -Case I = Chute Empty & Feed Bin Empty Condition
SDL2 : Equipment load -Case II = Chute bolcked & Feed Bin filled Condition + Monorail Load

Monorail Load (Cap. of 8.5 MT) = Capacity + Wt.of Hoist = 85+10 = 95 kN
Impact = 20%
So, Design Load = = 1.2 x 95 = 114 kN

5.4 WIND LOAD

WIND LOAD CALCULATION OF SECONDARY CRUSHER HOUSE STRUCTURE :

Wind Load Parameters (As per design Criteria) :

Basic wind speed, V_b = 43 m/s = 155 Km/h
Exposure category = C (For Process Plant Area)
Building Category = II (Ref. Table 1.6-1 : SBC 301-2007)
Wind directionablity factor, K_d = 0.85 (Ref. Table 6.4-1 : SBC 301-2007)

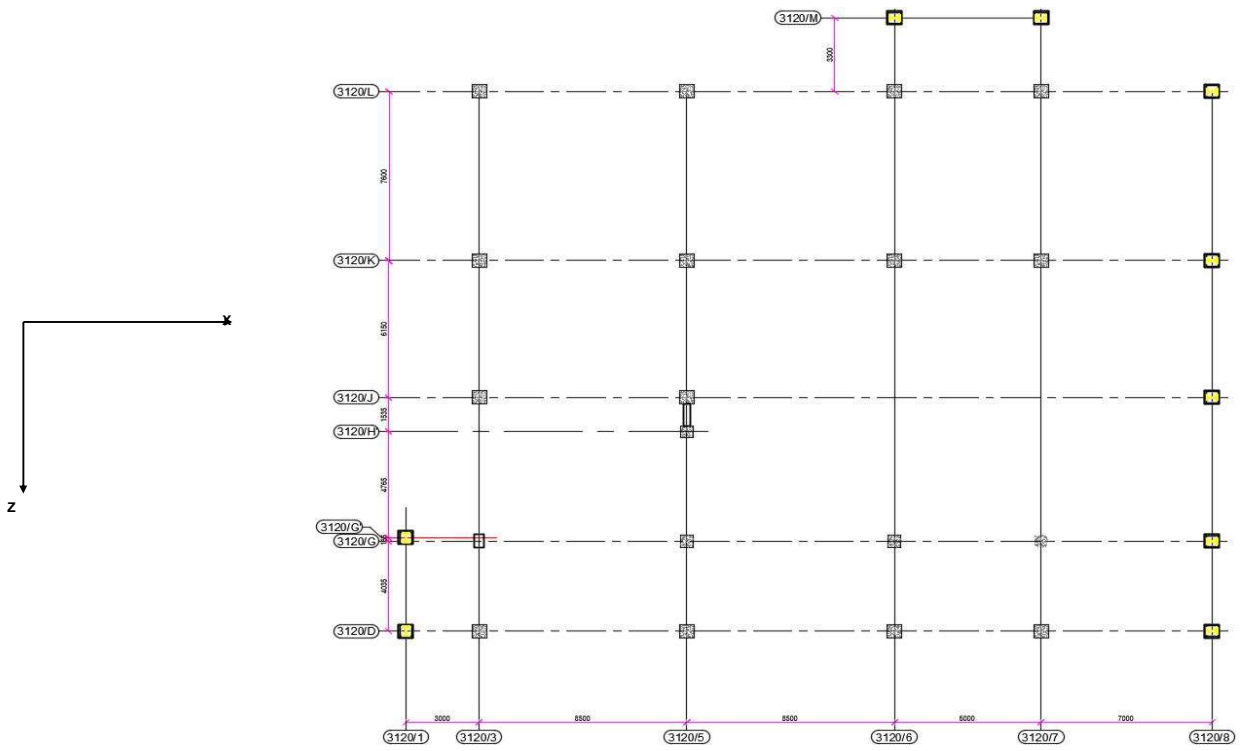
Topography Factor, K_{zt} = 1 (As The project is not located in isolated hills or escarpments)
Importance Factor, I = 1 (Ref. Table 6.5-1 : SBC 301-2007)
Velocity Pressure Co-efficient = Kz (Ref. Table 7.2-2 : SBC 301-2007)

Gust Effect Factor, G = 0.85 (For Rigid Structure)
(Ref. Section 7.2.7.1 : SBC 301-2007)

Wind Velocity Pressure, qz = $0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ (kN/m^2)
(Ref. Eq. 7.2-12 : SBC 301-2007)

Level	Height	kz	Pressure	Unit
EL +922.75	9.75m	1.0	0.966	kN/m^2
EL +926.5	13.5m	1.07	1.034	kN/m^2
EL +938.2	25.2m	1.21	1.169	kN/m^2

Average Pressure on wall, qz = **1.056** kN/m^2



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WIND LOAD IN +X DIRECTION (From Axis 3120/1 to 3120/8)

1) Wind Load due to Structural Building Structure

Area for Structural Frame work at Axis 3120 / 6

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Concrete Column	0.65	13.80	8.97	4	35.88
Concrete Column	0.5	13.80	6.90	1	6.90
Steel Column	0.16	58.20	9.31	1	9.31
Steel Column	0.55	11.40	6.27	2	12.54
Concrete Beam	0.65	24.25	15.76	3	47.29
Concrete Beam	0.5	20.20	10.10	1	10.10
Steel Beam	0.3	24.25	7.28	1	7.28
Inclined Bracing	0.18	28.60	5.15	2	10.30
Horz. Bracing	0.15	12.00	1.80	1	1.80
Stair Supporting Beam	0.2	3.00	0.60	5	3.00
Stair Stringer	0.075	12.6	0.95	2	1.89
Stair Grating	0.03	1.2	0.04	60	2.16
Conveyor Structure & Hood	1	24.25	24.25	1	24.25
Total Effective Projected Area for Framework=				A_f =	**172.69**

Height of Building	=	25.2	m	Width(m)=	27.55
Frontal Area	=	694.3	m^2		
Solidity ratio, Φ	=	0.25			
Force Co-efficient, C_f	=	1.80	(Ref. Table 7.2-17 : SBC 301-2007)		
Design Wind Load	F =	$q_z.G.C_f.A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)		
Total Wind load for Frame	=	279.0	KN		

Wind load in column	Influence Width	Force	Unit
At Row 3120/M	1.65	0.66	kN/m
At Row 3120/L	5.45	2.19	kN/m
At Row 3120/K	6.88	2.76	kN/m
At Row 3120/J	6.31	2.54	kN/m
At Row 3120/G	5.25	2.11	kN/m
At Row 3120/D	2.02	0.81	kN/m

2) Wind Load due to Equipment Structure

a) Secondary screen Feed Bin

Total Height of Bin	=	5.8	m			Height of Hopper	=	3.8 m
Height of Straight Portion	=	2	m	Width =	13.75	Top Width of Hopper	=	5.5 m
Area of Straight Portion	=	27.5	m^2			Bottom Width of Hopper	=	2.8 m
h/d	=	1.40				Area	=	15.8 m^2
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)			Wind Load for Hopper Portion	=	24.1 kN
Wind pressue at Bin	=	1.169	kN/m^2			Load in Bunker Girder	=	**4.39** kN/m
Total Wind Load for Stright Portion	=	42.1	kN					
Load in Beam at axis 3120/6	=	**3.06**	kN/m					

b) Tertiary screen Feed Bin

Total Height of Bin	=	6.7	m			Height of Hopper	=	4.7 m
Height of Straight Portion	=	2	m	Width =	10.5	Top Width of Hopper	=	7.25 m
Area of Straight Portion	=	21	m^2			Bottom Width of Hopper	=	4.5 m
h/d	=	1.14				Area	=	27.6 m^2
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)			Wind Load for Hopper Portion	=	42.3 kN
Wind pressue at Bin	=	1.169	kN/m^2			Load in Bunker Girder	=	**5.83** kN/m
Total Wind Load for Stright Portion	=	32.2	kN					
Load in Beam at axis 3120/6	=	**3.06**	kN/m					

c) Secondary Crusher Vibrating Feeder

Height	=	4.5	m
Width	=	3	m
Area of Straight Portion	=	13.5	m^2
h/d	=	1.50	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	1.034	kN/m^2
Total Wind Load	=	18.3	kN
Total Wind Load / Wheel	H =	**3.05**	kN
	V =	**6.85**	kN (up/dwn)

d) Tertiary Crusher Vibrating Feeder

Height	=	2.6	m
Width	=	2.2	m
Area of Straight Portion	=	5.72	m^2
h/d	=	1.18	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	1.034	kN/m^2
Total Wind Load	=	7.7	kN
Total Wind Load / Wheel	H =	**1.94**	kN
	V =	**1.68**	kN (up/dwn)

e) Secondary Feed Conveyor Head End & Motor

Height of Chute	=	2.3 m
Width of Chute	=	1.9 m
Height of motor	=	3.5 m
Width of motor	=	1.3 m
Area	=	8.92 m^2
h/d	=	1.2
Cf	=	1.3
Wind pressue	=	1.169 kN/m^2
Total Wind Load	=	**13.6** kN
Total Wind Load / support	H =	**2.26** kN
	V =	**3.71** kN (up/dwn)

f) Tertiary Feed Conveyor Head End & Motor

Height of Chute	=	2.3 m
Width of Chute	=	1.5 m
Height of motor	=	2.6 m
Width of motor	=	1.5 m
Area	=	7.35 m^2
h/d	=	1.5
Cf	=	1.3
Wind pressue	=	1.169 kN/m^2
Total Wind Load	=	**11.2** kN
Total Wind Load / support	H =	**1.86** kN
	V =	**2.07** kN (up/dwn)

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g) Secondary Screen

Height	=	5.8	m
Width	=	5.15	m
Area of Straight Portion	=	29.87	m^2
h/d	=	1.13	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	1.034	kN/m^2
Total Wind Load	=	40.4	kN
Total Wind Load / support	H =	**10.11**	kN
	V =	**10.86**	kN (up/dwn)

h) Tertiary Screen

Height	=	2.8	m
Width	=	3.6	m
Area of Straight Portion	=	10.08	m^2
h/d	=	1.29	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	1.034	kN/m^2
Total Wind Load	=	13.6	kN
Total Wind Load / Support	H =	**3.41**	kN
	V =	**5.33**	kN (up/dwn)

i) Secondary undersized Chute

Height	=	5.8	m
Average Width	=	2.925	m
Area of Straight Portion	=	16.965	m^2
h/d	=	1.98	
Force Co-efficient, Cf	=	1.33	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	0.966	kN/m^2
Total Wind Load	=	21.8	kN
Total Wind Load / support	H =	**5.45**	kN

j) Tertiary Undersized Chute

Height	=	7.2	m
Average Width	=	2.2	m
Area of Straight Portion	=	15.84	m^2
h/d	=	3.27	
Force Co-efficient, Cf	=	1.35	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	0.966	kN/m^2
Total Wind Load	=	20.7	kN
Total Wind Load / Support	H =	**5.16**	kN

k) Maintance / access Plaform , Column Width 0.2 m Cf = 2.00
For Bracing & Beam = 0.1 m

Wind Load = 0.85 x (0.2+0.1) x 2 x 0.966 **0.5** kN/m^2

WIND LOAD IN +Z DIRECTION (From Axis 3120/L to 3120/D)

1) Wind Load due to Structural Building Structure

Area for Structural Frame work at Axis 3120/ E

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Concrete Column	0.65	13.95	9.07	2	18.14
Concrete Column	0.6	9.90	5.94	2	11.88
Steel Column	0.45	11.40	5.13	2	10.26
Steel Column	0.36	23.09	8.31	1	8.31
Concrete Beam	0.9	23.00	20.70	1	20.70
Concrete Beam	0.6	30.00	18.00	1	18.00
Steel Beam	0.3	27.00	8.10	1	8.10
Bracing	0.2	21.90	4.38	2	8.76
Bracket	0.2	4.00	0.80	2	1.60
Stair Stringer	0.24	6	1.44	10	14.40
Conveyor Structure & Hood	1.6	1.2	1.92	2	3.84
Total Effective Projected Area =				A_f =	**123.99**

Height of Building	=	25.2	m		
h1	=	9.75	m	w1 (m)=	17
h2	=	15.45	m	w2 (m)=	13
Frontal Area	=	366.6	m^2		
Solidity ratio, Φ	=	0.34			
Force Co-efficient, C_f	=	**1.60**	(Ref. Table 7.2-17 : SBC 301-2007)		
Design Wind Load	F =	$q_z.G.C_f.A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)		
Total Wind load	=	178.1	KN		

Wind load in column	Influence Width	Force	Unit
At Row 3120/3	4.25	2.06	kN/m
At Row 3120/5	8.50	4.13	kN/m
At Row 3120/6 (Below 926.5m)	7.25	3.52	kN/m
At Row 3120/6 (Above 926.5m)	3.00	1.46	kN/m
At Row 3120/7	6.50	3.16	kN/m
At Row 3120/8	3.50	1.70	kN/m

2) Wind Load due to Equipment Structure

a) Secondary screen Feed Bin

Total Height of Bin	=	5.8	m		
Height of Straight Portion	=	2	m	Width =	6
Area of Straight Portion	=	12	m^2		
h/d	=	1.81			
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)		
Wind pressue at Bin	=	1.169	kN/m^2		
Total Wind Load for Stright Portion	=	18.4	kN		
Load in Bunker Gider	=	**3.06**	kN/m		

Height of Hopper	=	3.8	m
Top Width of Hopper	=	4.8	m
Bottom Width of Hopper	=	1.7	m
Area	=	12.2	m^2
Wind Load for Hopper Portion	=	18.7	kN
Load in Bunker Gider	=	**3.93**	kN/m

b) Tertiary screen Feed Bin

Total Height of Bin	=	6.7	m		
Height of Straight Portion	=	2	m	Width =	6
Area of Straight Portion	=	12	m^2		
h/d	=	2.59			
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)		
Wind pressue at Bin	=	1.169	kN/m^2		
Total Wind Load for Stright Portion	=	18.4	kN		
Load in Bunker Gider	=	**3.06**	kN/m		

Height of Hopper	=	4.7	m
Top Width of Hopper	=	3.70	m
Bottom Width of Hopper	=	1.5	m
Area	=	12.1	m^2
Wind Load for Hopper Portion	=	18.6	kN
Load in Bunker Gider	=	**5.02**	kN/m

c) Secondary Crusher Vibrating Feeder

Height	=	2.8	m
Width	=	5.5	m
Area of Straight Portion	=	15.4	m^2

d) Tertiary Crusher Vibrating Feeder

Height	=	2.6	m
Width	=	2.3	m
Area of Straight Portion	=	5.98	m^2

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
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h/d	=	1.96	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	1.034	kN/m^2
Total Wind Load	=	20.9	kN
Total Wind Load / Wheel	H =	**3.48**	kN
	V =	**2.70**	kN (up/dwn)

h/d	=	1.13	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	1.034	kN/m^2
Total Wind Load	=	8.1	kN
Total Wind Load / Wheel	H =	**2.02**	kN
	V =	**1.75**	kN (up/dwn)

e) Secondary Feed Conveyor Head End & Motor

Height of Chute	=	2.7	m
Width of Chute	=	3.3	m
Area	=	8.91	m^2
h/d	=	1.2	
Cf	=	1.3	
Wind pressue	=	1.169	kN/m^2
Total Wind Load	=	**13.5**	kN
Total Wind Load / support	H =	**2.26**	kN
	V =	**3.21**	kN (up/dwn)

f) Tertiary Feed Conveyor Head End & Motor

Height of Chute	=	2.3	m
Width of Chute	=	2	m
Area	=	4.6	m^2
h/d	=	1.2	
Cf	=	1.3	
Wind pressue	=	1.169	kN/m^2
Total Wind Load	=	**7.0**	kN
Total Wind Load / support	H =	**1.16**	kN
	V =	**1.49**	kN (up/dwn)

g) Secondary Screen

Height	=	5.8	m
Width	=	5.15	m
Area of Straight Portion	=	29.87	m^2
h/d	=	1.13	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	0.966	kN/m^2
Total Wind Load	=	37.8	kN
Total Wind Load / support	H =	**9.45**	kN
	V =	**10.15**	kN (up/dwn)

h) Tertiary Screen

Height	=	2.8	m
Width	=	6.8	m
Area of Straight Portion	=	19.04	m^2
h/d	=	2.43	
Force Co-efficient, Cf	=	1.31	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	0.966	kN/m^2
Total Wind Load	=	24.1	kN
Total Wind Load / Support	H =	**6.02**	kN
	V =	**9.40**	kN (up/dwn)

i) Secondary undersized Chute

Height	=	5.8	m
Average Width	=	4.3	m
Area of Straight Portion	=	24.94	m^2
h/d	=	1.35	
Force Co-efficient, Cf	=	1.33	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	0.966	kN/m^2
Total Wind Load	=	32.0	kN
Total Wind Load / support	H =	**8.01**	kN

j) Tertiary Undersized Chute

Height	=	7.2	m
Average Width	=	3.7	m
Area of Straight Portion	=	26.64	m^2
h/d	=	1.95	
Force Co-efficient, Cf	=	1.35	(Ref. Table 7.2-15 : SBC 301-2007)
Wind pressue	=	0.966	kN/m^2
Total Wind Load	=	34.7	kN
Total Wind Load / Support	H =	**8.68**	kN

k) Maintance / access Plaform , Column Width = 0.2 m Cf = 2.00

For Bracing & Beam = 0.1 m

Wind Load = 0.85 x (0.2+0.1) x 2 x 0.966 = **0.5** kN/m^2

5.5 TEMPERATURE LOAD

There are two temperature Load case wehich are considered as TL1 & TL2 in structural design

TL1 - For Temperature Difference + 25^0 C

TL2 - For Temperature Difference -25^0 C

5.6 EARTHQUAKE LOAD

Seismic Design Category		=	A	
Site class		=	B	It is corresponds to Site Class 2 in Staad
Spectral response acceleration (1-sec)	S1	=	0.02	
Spectral response acceleration (.02-sec)	Ss	=	0.1	
Importance factor	I	=	1	(From Table 9.5, SBC 301:2007)
Response Reduction factor along X	Rx	=	2.5	(From Table 10.2, SBC 301:2007)
Response Reduction factor along Z	Rz	=	2.5	(From Table 10.2, SBC 301:2007)
Height of building	h	=	25.2	m
Time period along X	Tx	=	0.619	(eqn. 10.9.3.2-1, SBC 301:2007)
Time period along Z	Tz	=	0.619	(eqn. 10.9.3.2-1, SBC 301:2007)
Factor Fa	Fa	=	1	
Factor Fv	Fv	=	1	
SM1	Fa x S1	=	0.02	
SMS	Fa x Ss	=	0.1	
Design spectral acceleration (1-sec)	SD1 = 2/3 x SM1	=	0.0133	
Design spectral acceleration (0.2-sec)	SDS = 2/3 x SMS	=	0.0667	

The above parameters are used in the staad input to find the final earthquake load of the structure as per IBC-2003

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5.7 LOAD COMBINATIONS

5.7.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD 301 (DL + SDL1)	(Eq. 2.4.1-1 of SBC 301)
LOAD 302 (DL + SDL2)	
LOAD 303 (DL + SDL1 + TL1) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 304 (DL + SDL2 + TL1) + LL	
LOAD 305 (DL + SDL1 + TL2) + LL	
LOAD 306 (DL + SDL2 + TL2) + LL	
LOAD 307 (DL+ SDL1) + 0.75 LL + 0.75 WL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 308 (DL+ SDL1) + 0.75 LL + 0.75 WL(-X)	
LOAD 309 (DL+ SDL1) + 0.75 LL + 0.75 WL(+Z)	
LOAD 310 (DL+ SDL1) + 0.75 LL + 0.75 WL(-Z)	
LOAD 311 (DL+ SDL2) + 0.75 LL + 0.75 WL(+X)	
LOAD 312 (DL+ SDL2) + 0.75 LL + 0.75 WL(-X)	
LOAD 313 (DL+ SDL2) + 0.75 LL + 0.75 WL(+Z)	
LOAD 314 (DL+ SDL2) + 0.75 LL + 0.75 WL(-Z)	
LOAD 315 (DL+ SDL1) + 0.75 LL + 0.525 EL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 316 (DL+ SDL1) + 0.75 LL + 0.525 EL(-X)	
LOAD 317 (DL+ SDL1) + 0.75 LL + 0.525 EL(+Z)	
LOAD 318 (DL+ SDL1) + 0.75 LL + 0.525 EL(-Z)	
LOAD 319 (DL+ SDL2) + 0.75 LL + 0.525 EL(+X)	
LOAD 320 (DL+ SDL2) + 0.75 LL + 0.525 EL(-X)	
LOAD 321 (DL+ SDL2) + 0.75 LL + 0.525 EL(+Z)	
LOAD 322 (DL+ SDL2) + 0.75 LL + 0.525 EL(-Z)	
LOAD 323 0.6 (DL+SDL1) + 1.0 WL(+X)	(Eq. 2.4.1-7 of SBC 301)
LOAD 324 0.6 (DL+SDL1) + 1.0 WL(-X)	
LOAD 325 0.6 (DL+SDL1) + 1.0 WL(+Z)	
LOAD 326 0.6 (DL+SDL1) + 1.0 WL(-Z)	
LOAD 327 0.6 (DL+SDL2) + 1.0 WL(+X)	
LOAD 328 0.6 (DL+SDL2) + 1.0 WL(-X)	
LOAD 329 0.6 (DL+SDL2) + 1.0 WL(+Z)	
LOAD 330 0.6 (DL+SDL2) + 1.0 WL(-Z)	
LOAD 331 0.6 (DL+SDL1) + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD 332 0.6 (DL+SDL1) + 0.7 EL(-X)	
LOAD 333 0.6 (DL+SDL1) + 0.7 EL(+Z)	
LOAD 334 0.6 (DL+SDL1) + 0.7 EL(-Z)	
LOAD 335 0.6 (DL+SDL2) + 0.7 EL(+X)	
LOAD 336 0.6 (DL+SDL2) + 0.7 EL(-X)	
LOAD 337 0.6 (DL+SDL2) + 0.7 EL(+Z)	
LOAD 338 0.6 (DL+SDL2) + 0.7 EL(-Z)	

5.7.2 Load Combination for concrete design

Load Case	Reference
LOAD 401 1.4 (DL + SDL1)	(Eq. 2.3.2-1 of SBC 301)
LOAD 402 1.4 (DL + SDL2)	
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL	
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL	
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL	
LOAD 407 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+X)	(Eq. 2.3.2-4 of SBC 301)
LOAD 408 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(-X)	
LOAD 409 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+Z)	
LOAD 410 1.2 (DL+ SDL1)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 411 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+X)	
LOAD 412 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(-X)	
LOAD 413 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+Z)	
LOAD 414 1.2 (DL+ SDL2)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 415 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+X)	(Eq. 2.3.2-5 of SBC 301)
LOAD 416 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(-X)	
LOAD 417 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+Z)	
LOAD 418 1.2 (DL+ SDL1)+ 0.5 LL + 1.0 EL(-Z)	
LOAD 419 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+X)	
LOAD 420 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-X)	
LOAD 421 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+Z)	
LOAD 422 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-Z)	
LOAD 423 0.9 (DL+ SDL1) + 1.6 WL(+X)	(Eq. 2.3.2-6 of SBC 301)
LOAD 424 0.9 (DL+ SDL1) + 1.6 WL(-X)	
LOAD 425 0.9 (DL+ SDL1) + 1.6 WL(+Z)	
LOAD 426 0.9 (DL+ SDL1) + 1.6 WL(-Z)	
LOAD 427 0.9 (DL+ SDL2) + 1.6 WL(+X)	
LOAD 428 0.9 (DL+ SDL2) + 1.6 WL(-X)	
LOAD 429 0.9 (DL+ SDL2) + 1.6 WL(+Z)	
LOAD 430 0.9 (DL+ SDL2) + 1.6 WL(-Z)	
LOAD 431 0.9 (DL+ SDL1) + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD 432 0.9 (DL+ SDL1) + 1.0 EL(-X)	
LOAD 433 0.9 (DL+ SDL1) + 1.0 EL(+Z)	
LOAD 434 0.9 (DL+ SDL1) + 1.0 EL(-Z)	
LOAD 435 0.9 (DL+ SDL2) + 1.0 EL(+X)	
LOAD 436 0.9 (DL+ SDL2) + 1.0 EL(-X)	
LOAD 437 0.9 (DL+ SDL2) + 1.0 EL(+Z)	
LOAD 438 0.9 (DL+ SDL2) + 1.0 EL(-Z)	

Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

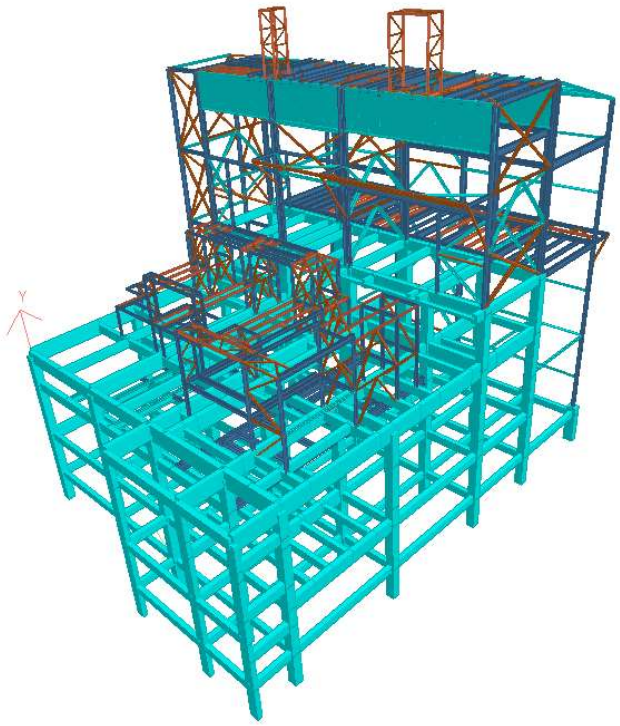
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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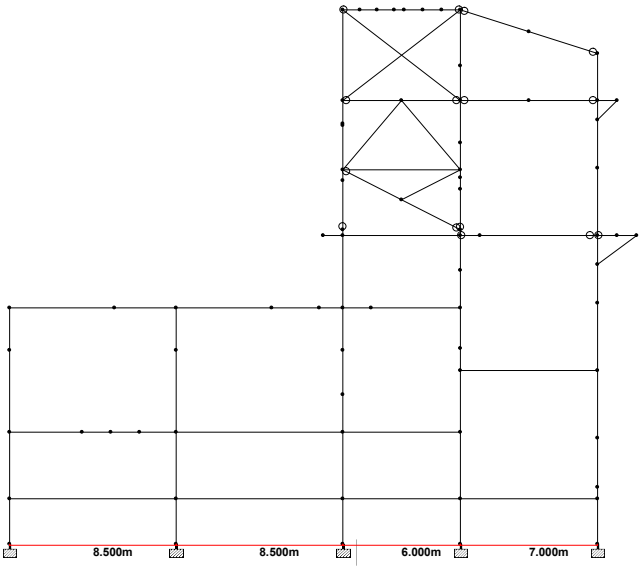
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6 GEOMETRY

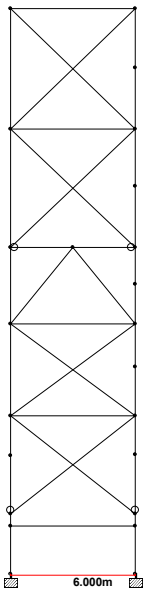
6.1 Geometry of Screen Building & Grids



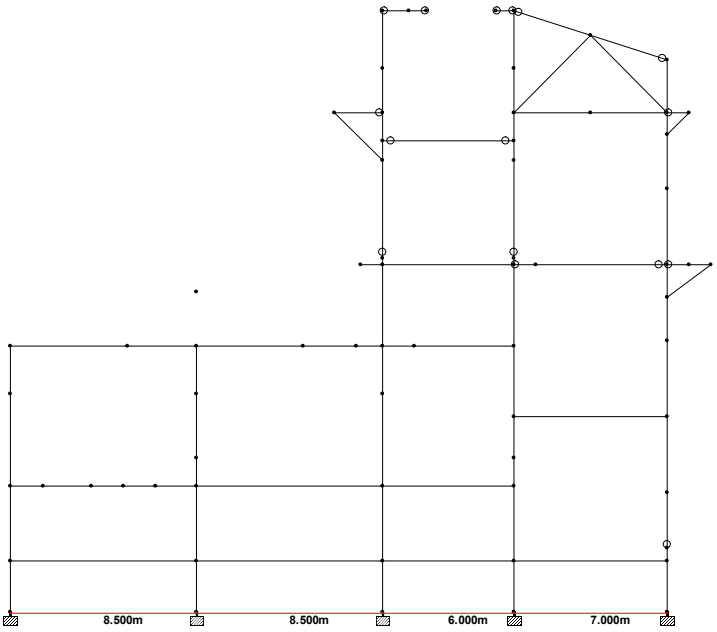
3D-Model of Screen Building



Elevation Along Axis - 3120 / L



Elevation Along Row - 3120 / M

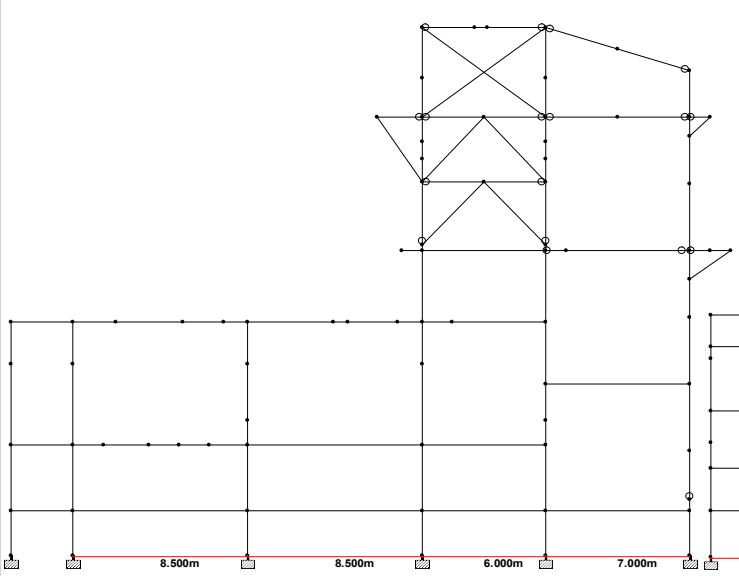


Elevation Along Axis - 3120 / K

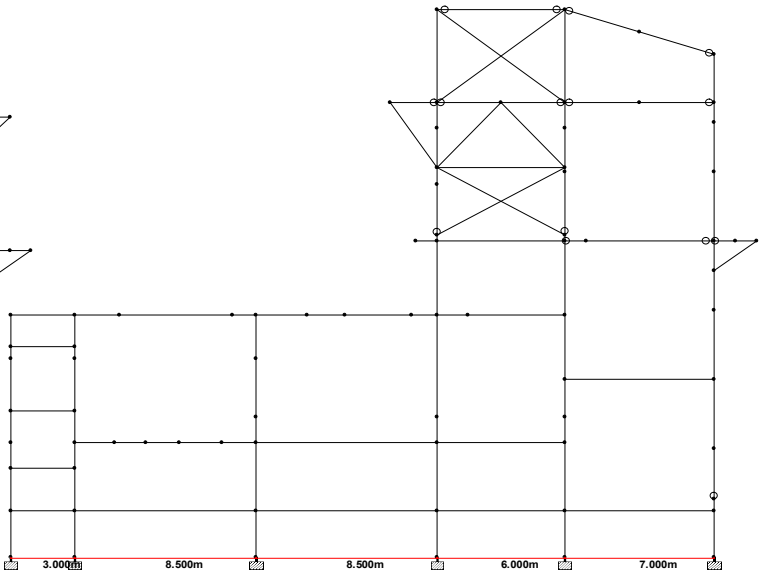
Global Project ID	900707
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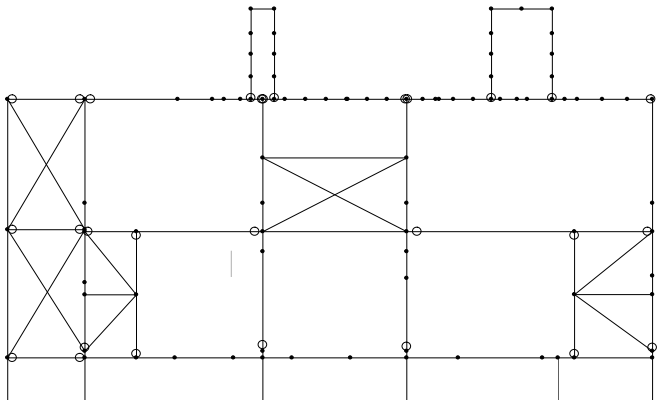
Outotec / L&T Revision No.:	0	1	2	3
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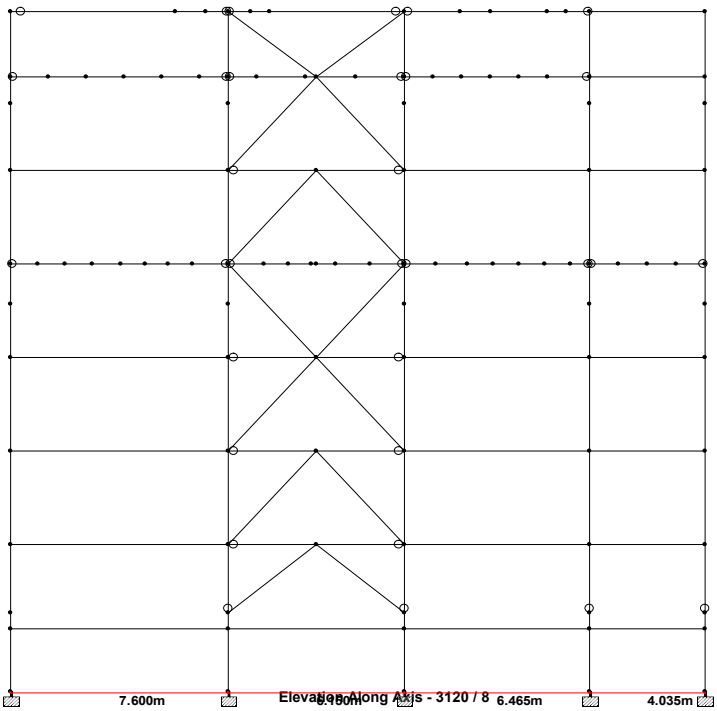
Elevation Along Axis - 3120 / J



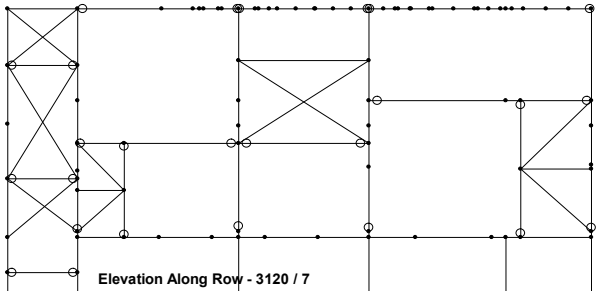
Elevation Along Axis - 3120 / D



Elevation Along Row - 3120 /6



Elevation Along Axis - 3120 / 8



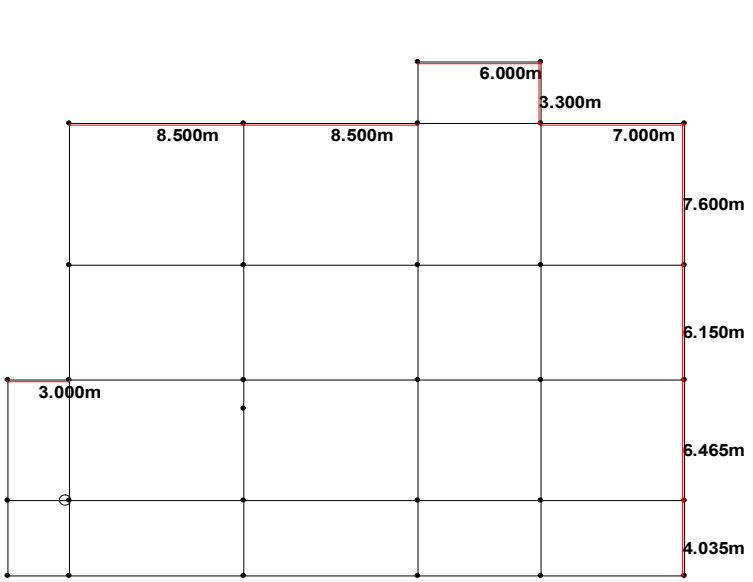
Elevation Along Row - 3120 / 7

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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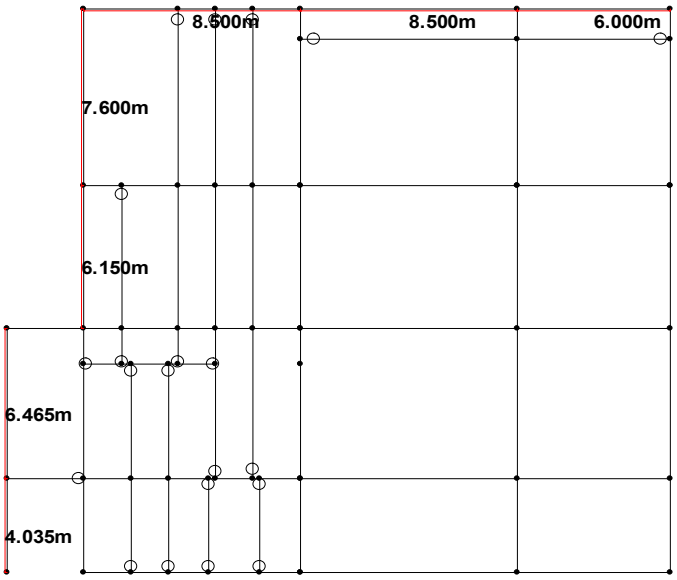
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Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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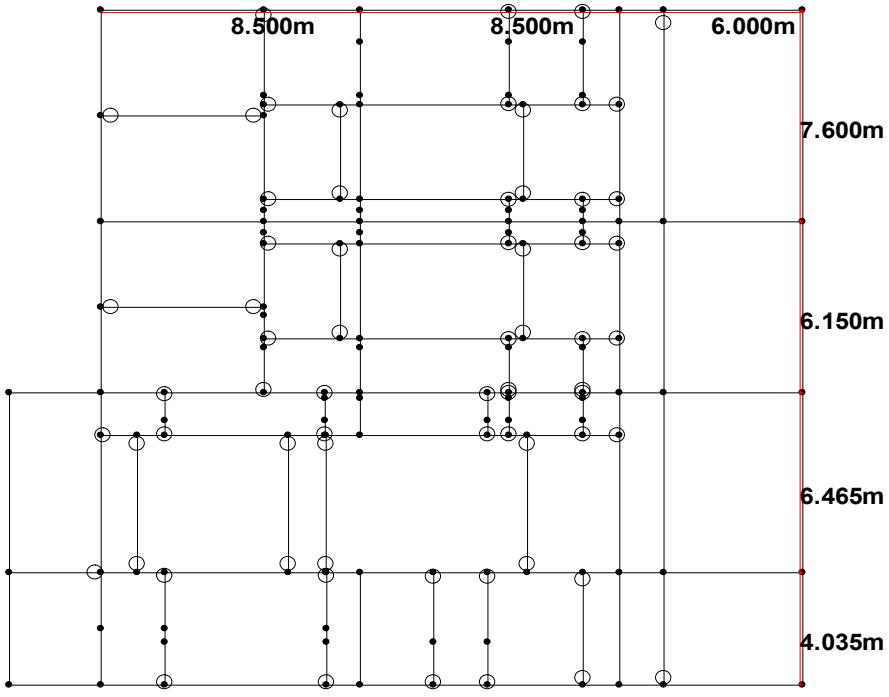
6.2 Geometry of Screen Building Concrete Floors



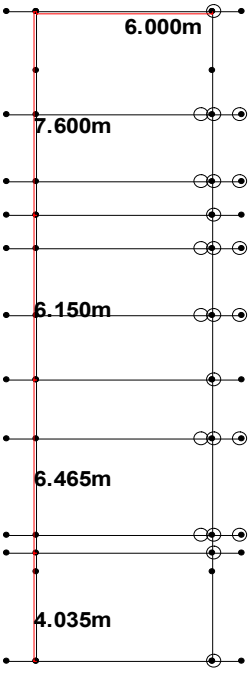
PLAN AT RL 912.850M (TOC)



PLAN AT RL 916.300M & 917.700M (TOC)



PLAN AT RL 922.750 M(TOC)



PLAN AT RL 926.500 M(TOC)

Global Project ID	900707
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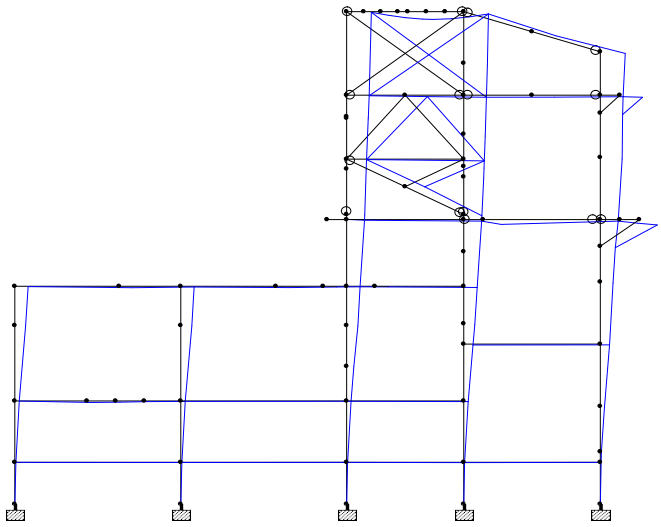
7 ANALYSIS & RESULTS OF SCREEN BUILDING

For 3D analysis and results of Screen Building, refer Staad file - **Screen Building R0.STD**

7.1 Deflection of column

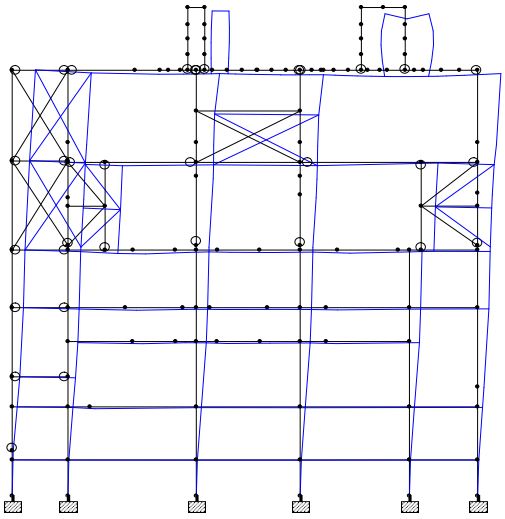
Nodal Deflection at Column Top is obtained from Staad-pro and checked as per clause 7.8.1 of Maaden Design criteria Doc. No. MD-101-SMEM-EG-ST-CRT-0001

Column at Row / Axis	Height (m)	Max Deflection in X (mm)	Load Case	Max Deflection in Z (mm)	Load Case	Allowable Deflection H/400 (mm)	Remarks
3120/L & 3120/7	16.3	23.0	323			40.75	OK
3120/L & 3120/7	16.3			21.1	329	40.75	OK
3120/L & 3120/5	12.25	17.3	323			30.6	OK
3120/G & 3120/5	12.25			10.4	329	30.6	OK



Load 107 :

Deflection diagram of Row - 3120/L, 3120/K and 3120/ D



Load 113 :

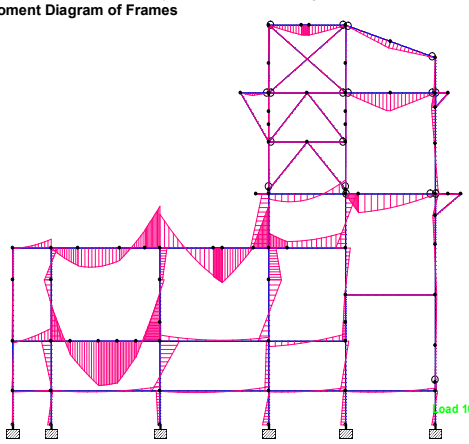
Deflection diagram of Axis - 3120/6 and 3120/ 7

7.2 Deflection of Beam

Deflection of all floor has been checked in parameter 2 of Staad-Pro design file considering below deflection limit

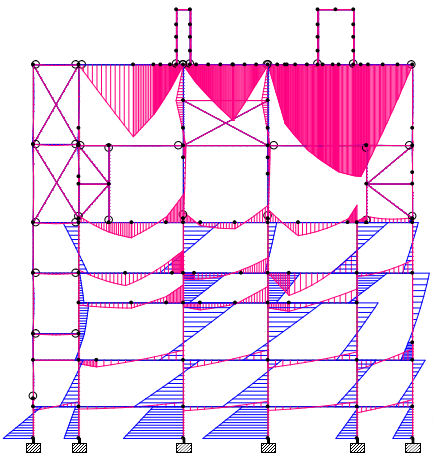
1) General Floor Beam - Span/ 360
2) Equipment Supporting Beam - Span/ 500

7.3 Bending Moment Diagram of Frames



Load 104 : Bending Z : Bending Y

Bending Moment diagram of Row - 3120/L, 3120/K and 3120/ D



Load 113 : Bending Z : Bending Y

Bending Moment diagram of Axis - 3120/6 and 3120/ 7

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8. DESIGN OF CONCRETE COMPONENTS

All Major components of building i.e. beams, slabs, foundations have been designed using worksheet in accordance to SBC-304.

All columns of structure have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 3 - Staad Output file.

8.1 Design Summary of Column

8.1.1 Column Reinforcement Summary

Column mark	Width	Depth	Top reinforcement	Stirrup	Grids	Numbers
C1	350	600	12 No.- D20	3 Nos D10 @ 250c/c	D-1, G'-1 & J-1	3
C2	550	550	4 No.- D32 + 8 No.- D25	3 Nos D10 @ 250c/c	D-3, J-3, L-3, D-5, J-5 & D-6	6
C2A	550	550	4 No.- D25 + 8 No.- D20	3 Nos D10 @ 250c/c	G-5 & H'-5	2
C2B	550	550	12 No.- D32	3 Nos D10 @ 250c/c	G-3	1
C2C	550	550	12 No.- D25	3 Nos D10 @ 250c/c	G-7 & L-7	3
C3	600	850	8 No.- D32 + 8 No.- D25	4 Nos D10 @ 250c/c	K-3, K-5, L-5, G-6, J-6, K-6 & L-6	6
C3A	600	850	12 No.- D25	3 Nos D10 @ 250c/c	J-7 & K-7	2
C3B	600	850	8 No.- D32 + 8 No.- D32	4 Nos D10 @ 250c/c	K-3, K-5, L-5, G-6, J-6, K-6 & L-6	1
P1, P2 & P3	600	850	12 No.- D25	3 Nos D10 @ 250c/c	M-6 & M-7; D-8, G-8 & L-8; J-8 & K-8	2; 3 & 2

8.2 Design of Slab

8.2.1 Design of Slab 1- 175mm thick

Slab is designed as simply supported slab along shorter span

Span of slab	=	4	m	max span considered	Clear cover	=	40	mm
Thickness of slab	=	0.175	m		Bar Dia	=	12	mm
Grade of steel	=	420	N/mm2		Unit weight of concrete	=	24	KN/m3
Grade of concrete	=	35	N/mm2		Live load on slab	=	5	KN/m2
Mimimum thickness of one way slab	=	span/24		*Table 7.3.1.1 SBC 304*	β_1	=	0.8	
	=	0.167	m	Hence slab thickness okay				

Load calculation on slab

Dead load from slab	=	0.175 x 24	=	4.2	KN/m2		
Live load from slab	=			5	KN/m2		
Load combination	=	1.4DL + 1.7LL	=	1.4 x 4.2 + 1.7 x 5	=	14.4	KN/m2

Calculation of bending moment and shear force

BM at midspan	=	$wl^2/8$	=	28.8	KNm		
SF at support	=	wl/2	=	28.8	KN		

Calculation of flexural reinforcement

					Effective Depth Along X	=	129	mm
	M	=		28.76	KNm			
	M_{lim}	=	$\emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$			For tension controlled section $\left(\frac{c}{d}\right)_{max}$	= $\frac{0.003}{0.003 + 0.005}$ =	0.375
		=		113.62	KNm			
M_{lim} > M	Tension controlled section					Min reinforcement	A_{stmin} = $\frac{0.0018 \times 420}{f_y} bd$	*Table 7.6.1.1 SBC 304*
Therefore	Φ	=		0.9	*Table 21.2.2 SBC 304*	greater of or	A_{stmin} =	0.0014bd
	A_{st}	=	$\frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$				A_{stmin} =	232.2 mm^-
		=		609.4	mm^2		A_{req} =	609.4 mm^-
						Required Spacing	=	185.6 mm
	Provide	***12mm dia @***		***150***	***c/c***	***Top & bottom both directions***	A_{prov} =	754.0 mm^-

Shear Check

							Lesser of the following		
						i)	V_c =	$\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$	*Table 22.5.5.1 SBC 304*
Reinforcement ratio	ρ_w	=	0.0058						
Effective Depth	d	=	129	mm			λ	= 1	Normal concrete
Shear Force	V_u	=	28.76	KN			Mu	= 28.76	KNm
Allowable shear strength	φVc	=	92.82	KN	O.K		φ	= 0.75	for shear
							φVc	= 92.82	KN
						ii)	V_c =	$\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$	
							φVc	= 101.19 KN	*Table 22.5.5.1 SBC 304*
						iii)	V_c =	$0.29\lambda\sqrt{f_c'} b_w d$	
							φVc	= 165.99 KN	

8.2.2 Design of Slab 2- 150mm thick

Slab is designed as simply supported slab along shorter span

Span of slab	=	3.5	m	max span considered	Clear cover	=	40	mm
Thickness of slab	=	0.2	m		Bar Dia	=	10	mm
Grade of steel	=	420	N/mm2		Unit weight of concrete	=	24	KN/m3
Grade of concrete	=	35	N/mm2		Live load on slab	=	5	KN/m2
Mimimum thickness of one way slab	=	span/24		*Table 7.3.1.1 SBC 304*	β_1	=	0.8	
	=	0.146	m	Hence slab thickness okay				

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Load calculation on slab

Dead load from slab = 0.2 x 24 = 4.8 KN/m2

Live load from slab = 5 KN/m2

Load combination = 1.4DL + 1.7LL = 1.4 x 4.8 + 1.7 x 5 = 15.2 KN/m2

Calculation of bending moment and shear force

BM at midspan = wl2/8 = 23.3 KNm

SF at support = wl/2 = 26.6 KN

Calculation of flexural reinforcement

Effective Depth Along X = 155 mm

M = 23.31 KNm

$M_{lim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 164.03 KNm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

M_{lim} > M Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 404.7 mm^2

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 7.6.1.1 SBC 304*

greater of or

A_{stmin} = 0.0014bd

A_{stmin} = 279.0 mm^2

A_{req} = 404.7 mm^2

Required Spacing = 194.1 mm

Provide 10mm dia @ 150 c/c Top & bottom both directions A_{prov} = 523.6 mm^2

Shear Check

Reinforcement ratio ρ_w = 0.0034

Effective Depth d = 155 mm

Shear Force V_u = 26.64 KN

Allowable shear strength φVc = 111.22 KN O.K

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 23.31 KNm

φ = 0.75 for shear

φVc = 111.22 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 116.71 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

φVc = 199.45 KN

8.3 Design of Foundation

8.3.1 Foundation marking and Max/min Base pressure

Grid	Foundation mark	length	Width	Thickness	Max pressure	Min Pressure	Top reinforcement	Bottom reinforcement	F.O.S.
3120/M & 3120/6	F2	3000	3000	600	107	-5	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.20
3120/M & 3120/7	F2	3000	3000	600	148	-7	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.28
3120/L & 3120/3	F1A	2500	2500	450	154	8	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.29
3120/L & 3120/5	F2B	3000	3000	600	227	3	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.42
3120/L & 3120/6	**F2A**	**3000**	**3000**	**600**	**280**	**11**	**D12 @ 150c/c B/W**	**D16 @ 150c/c B/W**	**0.52**
3120/L & 3120/7	F2A	3000	3000	600	246	39	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.46
3120/L & 3120/8	F2	3000	3000	600	126	9	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.24
3120/K & 3120/3	F2B	3000	3000	600	224	9	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.42
C01 **3120/K & 3120/5**	**F4A**	**4000**	**4000**	**800**	**341**	**83**	**D12 @ 150c/c B/W**	**D20 @ 150c/c B/W**	**0.64**
3120/K & 3120/6	**F3**	**3200**	**3200**	**700**	**371**	**24**	**D12 @ 150c/c B/W**	**D20 @ 150c/c B/W**	**0.69**
3120/K & 3120/7	F3	3200	3200	700	321	41	D12 @ 150c/c B/W	D20 @ 150c/c B/W	0.60
3120/K & 3120/8	F4	4000	4000	600	133	-5	D12 @ 200c/c B/W	D20 @ 200c/c B/W	0.25
3120/J & 3120/1	F1A	2500	2500	450	102	3	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.19
3120/J & 3120/3	F2B	3000	3000	600	249	75	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.46
C01 **3120/J and 3120/H' & 3120/5**	**CF1**	**3000**	**3535**	**650**	**482**	**24**	**D12 @ 200c/c B/W**	**D25 @ 200c/c B/W**	**0.90**
3120/J & 3120/6	F3	3200	3200	700	381	33	D12 @ 150c/c B/W	D20 @ 150c/c B/W	0.71
3120/J & 3120/7	F3	3200	3200	700	400	69	D12 @ 150c/c B/W	D20 @ 150c/c B/W	0.75
3120/J & 3120/8	**F4**	**4000**	**4000**	**600**	**170**	**20**	**D12 @ 200c/c B/W**	**D20 @ 200c/c B/W**	**0.32**
3120/G' & 3120/1	F1A	2500	2500	450	113	34	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.21
3120/G & 3120/3	F2B	3000	3000	600	224	69	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.42
3120/G & 3120/5	F3A	3200	3200	700	260	88	D12 @ 150c/c B/W	D20 @ 150c/c B/W	0.49
3120/G & 3120/6	F3	3200	3200	700	285	15	D12 @ 150c/c B/W	D20 @ 150c/c B/W	0.53
3120/G & 3120/7	F2A	3000	3000	600	205	55	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.38
3120/G & 3120/8	**F1**	**2500**	**2500**	**450**	**242**	**8**	**D12 @ 150c/c B/W**	**D16 @ 150c/c B/W**	**0.45**
3120/D & 3120/1	F2B	3000	3000	600	84	8	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.16
3120/D & 3120/3	F1A	2500	2500	450	220	31	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.41
3120/D & 3120/5	F1A	2500	2500	450	200	21	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.37
3120/D & 3120/6	F3	3200	3200	700	176	11	D12 @ 150c/c B/W	D20 @ 150c/c B/W	0.33
3120/D & 3120/7	F2A	3000	3000	600	227	46	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.42
3120/D & 3120/8	F1	2500	2500	450	143	-10	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.27

8.3.2 Foundation groupwise Reinforcement summary

Foundation mark	Width	Depth	Top reinforcement	Bottom reinforcement	Grids	Numbers
F1 & F1A -450 mm Thickness	2500	2500	D12 @ 150c/c B/W	D16 @ 150c/c B/W	L-3, J-1, G'-1, G-8, D-3, D-5 & D-8	7
F2, F2A & F2B -600 mm Thickness	3000	3000	D12 @ 150c/c B/W	D16 @ 150c/c B/W	M-6, M-7, L-5, L-6, L-7, L-8, K-3, J-3, G-3, G-7, D-1, D-7 & D-8	12
F3 & F3A -700 mm Thickness	3200	3200	D12 @ 150c/c B/W	D20 @ 150c/c B/W	K-6, K-7, J-6, J-7, G-5, G-6 & D-6	7
F4 -600 mm Thickness	4000	4000	D12 @ 200c/c B/W	D20 @ 200c/c B/W	K-8 & J-8	2
C01 F4A -800 mm Thickness	4000	4000	D16 @ 200c/c B/W	D25 @ 200c/c B/W	K-5	1
CF1 -650 mm Thickness	3000	3535	D12 @ 200c/c B/W	D25 @ 200c/c B/W	H'-5 & J-5	1

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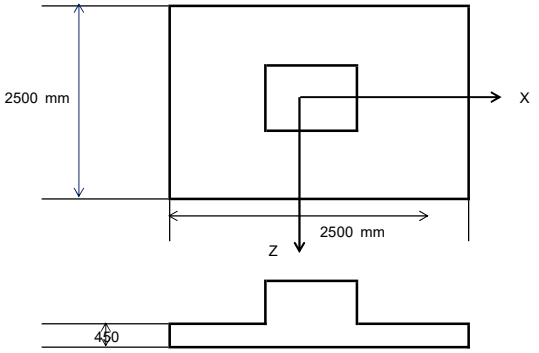
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8.3.3 Design of foundation F1 at Grid 3120/G-3120/8

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	-2.050	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	9.000	KN/m^2
Length of footing	Lz	=	2.50	m
Width of footing	Lx	=	2.50	m
Thickness of footing	D	=	0.45	m
Depth of pedestal	Bx	=	0.70	m
Width of pedestal	Bz	=	0.55	m
Height of Load point from bottom of footing	H	=	0.450	m
Section Modulus of Pedestal	Zxx	=	2.60	m^3
Section Modulus of Pedestal	Zzz	=	2.60	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	536	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	333	
Weight of soil		=	163.63	KN
Weight of foundation		=	67.50	KN
Uplift		=	0.00	KN
Surcharge		=	52.785	KN
Pressure on top for tension		=	45.4	KN/m^2

*inputs



$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	32.8	353.957	12.7	18.67	-0.1	-57.0	637.9	131.10	73.02	1	0.24
302	30.1	612.634	14.5	21.31	-0.158	-51.7	896.6	171.49	115.41	1	0.32
303	33.7	572.822	17.4	26.32	-0.208	-59.5	856.7	170.02	104.14	1	0.32
304	31.0	831.499	19.2	28.96	-0.305	-54.2	1115.4	210.40	146.53	1	0.39
305	30.1	571.822	11.4	16.05	0.12	-52.1	855.7	163.10	110.74	1	0.30
306	27.4	830.499	13.2	18.69	0.0	-46.9	1114.4	203.49	153.13	1	0.38
307	80.9	526.219	17.1	25.77	-0.679	-160.4	810.1	201.10	58.14	1	0.38
308	-16.0	509.348	10.8	15.13	0.584	47.0	793.3	150.77	103.08	1	0.28
309	29.2	525.954	77.8	129.73	0.054	-50.2	809.9	198.65	60.50	1	0.37
310	35.0	509.508	-49.9	-88.62	-0.15	-62.0	793.4	184.79	69.11	1	0.34
311	78.2	784.895	18.8	28.41	-0.8	-155.1	1068.8	241.49	100.53	1	0.45
312	-18.7	768.025	12.6	17.77	0.487	52.2	1051.9	195.18	141.44	1	0.36
313	26.5	784.631	79.6	132.37	-0.043	-44.9	1068.5	239.04	102.89	1	0.45
314	32.3	768.185	-48.1	-85.98	-0.2	-56.8	1052.1	223.15	113.52	1	0.42
315	32.1	517.731	14.0	20.56	-0.048	-56.1	801.6	157.69	98.84	1	0.29
316	32.1	517.731	14.0	20.56	-0.048	-56.1	801.6	157.69	98.84	1	0.29
317	32.1	517.731	14.0	20.56	-0.048	-56.1	801.6	157.69	98.84	1	0.29
318	32.1	517.731	14.0	20.56	-0.048	-56.1	801.6	157.69	98.84	1	0.29
319	29.4	776.408	15.8	23.20	-0.1	-50.8	1060.3	198.08	141.22	1	0.37
320	29.4	776.408	15.8	23.20	-0.145	-50.8	1060.3	198.08	141.22	1	0.37
321	29.4	776.408	15.8	23.20	-0.145	-50.8	1060.3	198.08	141.22	1	0.37
322	29.4	776.408	15.8	23.20	-0.145	-50.8	1060.3	198.08	141.22	1	0.37
323	84.7	223.691	11.7	18.15	-0.878	-173.2	507.6	154.71	7.73	1	0.29
324	-44.5	201.198	3.4	3.97	0.8	103.2	485.1	118.78	36.46	1	0.22
325	15.8	223.339	92.8	156.77	0.1	-26.3	507.3	151.45	10.87	1	0.28
326	23.6	201.41	-77.5	-134.36	-0.172	-42.1	485.3	145.40	9.90	1	0.27
327	83.1	378.898	12.8	19.73	-0.936	-170.1	662.8	178.94	33.16	1	0.33
328	-46.1	356.404	4.4	5.55	0.7	106.4	640.3	145.43	59.48	1	0.27
329	14.2	378.545	93.8	158.35	0.041	-23.1	662.5	175.68	36.31	1	0.33
330	22.0	356.616	-76.4	-132.78	-0.231	-38.9	640.5	168.42	36.55	1	0.31
331	19.7	212.374	7.6	11.20	-0.036	-34.2	496.3	96.83	61.98	1	0.18
332	19.7	212.374	7.6	11.20	-0.036	-34.2	496.3	96.83	61.98	1	0.18
333	19.7	212.374	7.6	11.20	0.0	-34.2	496.3	96.83	61.98	1	0.18
334	19.7	212.374	7.6	11.20	-0.036	-34.2	496.3	96.83	61.98	1	0.18
335	18.1	367.581	8.7	12.78	-0.095	-31.0	651.5	121.06	87.42	1	0.23
336	18.1	367.581	8.7	12.78	-0.095	-31.0	651.5	121.06	87.42	1	0.23
337	18.1	367.581	8.7	12.78	-0.095	-31.0	651.5	121.06	87.42	1	0.23
338	18.1	367.581	8.7	12.78	-0.1	-31.0	651.5	121.06	87.42	1	0.23

Max P_{max}	=	241.49	KN/m^2
Max P_{min}	=	7.73	KN/m^2
Resultant F.O.S	=	0.45	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK / APL	RPK/APL	RPK / APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T	0	1	2	3
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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	45.9	495.5	17.8	30.14	-0.1	-90.07	125.45	33.13
402	42.2	857.7	20.3	34.40	-0.221	-81.88	181.88	92.58
403	46.9	867.5	24.9	43.19	-0.286	-93.43	191.26	86.33
404	41.9	866.1	16.5	26.94	0.173	-82.05	180.42	96.72
405	43.1	1229.6	27.4	47.45	-0.423	-85.24	247.69	145.78
406	38.1	1228.2	19.0	31.20	0.037	-73.86	236.86	156.17
407	142.9	552.0	22.7	39.87	-1.41	-322.43	227.45	-50.80
408	-63.8	516.0	9.3	14.18	1.285	166.42	151.92	13.22
409	32.7	551.5	152.3	290.84	0.153	-62.48	223.91	-47.44
410	45.1	516.4	-120.1	-236.27	-0.282	-90.56	208.13	-42.88
411	139.7	862.5	24.8	43.52	-1.527	-315.41	275.82	0.16
412	-67.1	826.5	11.4	17.83	1.168	173.44	205.68	58.79
413	29.5	861.9	154.5	294.49	0.037	-55.46	272.28	3.52
414	41.9	826.8	-118.0	-232.62	-0.399	-83.54	253.69	10.88
415	38.9	533.9	16.1	27.28	-0.064	-76.52	125.29	45.57
416	38.9	533.9	16.1	27.28	-0.064	-76.52	125.29	45.57
417	38.9	533.9	16.1	27.28	-0.064	-76.52	125.29	45.57
418	38.9	533.9	16.1	27.28	-0.064	-76.52	125.29	45.57
419	35.7	844.3	18.2	30.93	-0.181	-69.50	173.66	96.53
420	35.7	844.3	18.2	30.93	-0.181	-69.50	173.66	96.53
421	35.7	844.3	18.2	30.93	-0.181	-69.50	173.66	96.53
422	35.7	844.3	18.2	30.93	-0.181	-69.50	173.66	96.53
423	133.6	336.7	18.0	31.97	-1.4	-303.81	182.81	-75.07
424	-73.2	300.7	4.6	6.27	1.295	185.04	121.57	-25.35
425	23.3	336.1	147.7	282.93	0.163	-43.86	179.27	-71.71
426	35.7	301.0	-124.8	-244.18	-0.272	-71.94	169.55	-73.23
427	131.1	569.5	19.6	34.70	-1.488	-298.55	219.08	-36.85
428	-75.6	533.5	6.2	9.01	1.208	190.30	161.89	8.82
429	20.9	568.9	149.3	285.67	0.076	-38.60	215.55	-33.49
430	33.3	533.8	-123.2	-241.44	-0.36	-66.68	203.73	-32.91
431	29.5	318.6	11.4	19.38	-0.055	-57.90	80.65	21.29
432	29.5	318.6	11.4	19.38	-0.055	-57.90	80.65	21.29
433	29.5	318.6	11.4	19.38	-0.055	-57.90	80.65	21.29
434	29.5	318.6	11.4	19.38	-0.055	-57.90	80.65	21.29
435	27.1	551.4	13.0	22.11	-0.142	-52.64	116.92	59.51
436	27.1	551.4	13.0	22.11	-0.142	-52.64	116.92	59.51
437	27.1	551.4	13.0	22.11	-0.142	-52.64	116.92	59.51
438	27.1	551.4	13.0	22.11	-0.142	-52.64	116.92	59.51

Max Net factored B.P = 275.82 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z: Muz = 111.71 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 817.1 mm^2

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 817.1 mm^2

Required Spacing = **246.1** mm

Length of cantilever along Z = 0.975 m

Moment about X: Mux = 131.10 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 1007.3 mm^2

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 1007.3 mm^2

Required Spacing = **199.6** mm

Provide Bottom Reinforcement = ***16 mm dia @ 150 c/c***

Ast prov = 1340.4 mm^2

Global Project ID	**900707**
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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T	0	1	2	3
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Top Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or

greater of or A_{stmin} = 50% of 0.0014bd more prov in bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 332.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **340.6** mm

Length of cantilever along Z = 0.975 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or

greater of or A_{stmin} = 50% of 0.0014bd more prov in bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 321.3 mm^2

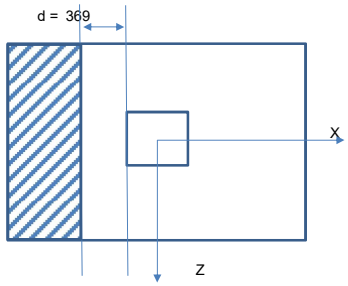
A_{req} = 0.0 mm^2

Required Spacing = **352.0** mm

Provide top Reinforcement = ***12 mm dia @ 150 c/c***

Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever L = 0.531 m

Reinforcement ratio ρ_w = 0.0036

Effective Depth Along X d = 369 mm

Shear Force at d distance V_u = 146.46 KN

Allowable shear strength φVc = 279.05 KN O.K

Utility ratio = 0.52

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 38.89 KNm

φ = 0.75 for shear

φVc = 285.7 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 279.1 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

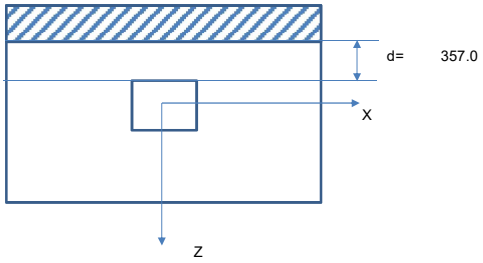
= 474.8 KN

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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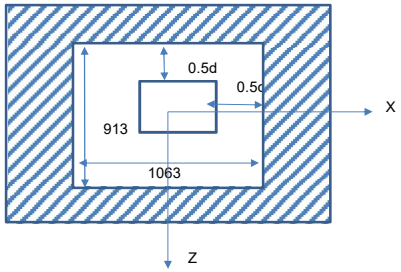
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.618	m		Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0038						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	357	mm				λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	170.46	KN				Mu	= 52.67 KNm
Allowable shear strength		φVc	=	270.54	KN	O.K			φ	= 0.75 for shear
Utility ratio			=	0.63					φVc	= 273.2 KN
								ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	= 270.5 KN
								iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									φVc	= 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	6.25	m2
Punching area		=	0.97	m2
	λ	=	1	
	β	=	1.272727273	
	αs	=	40	
	b0	=	3952.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	2.59 N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.79 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.95	m		Vc	= $v_c b_0 d$
Depth at section	d	=	363	mm			= 2800.73 KN
Shear Force at section	Vu	=	1456.2	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	2100.55	KN	O.K	φV_n	= φVc
Utility ratio			0.69			φVc	= 2100.5 KN

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

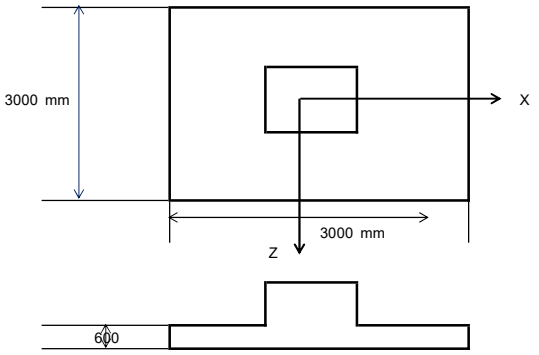
Outotec / L&T	0	1	2	3
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8.3.4 Design of foundation F2 at Grid 3120/L-3120/6

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	-1.900	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	9.000	KN/m^2
Length of footing	Lz	=	3.00	m
Width of footing	Lx	=	3.00	m
Thickness of footing	D	=	0.60	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.85	m
Height of Load point from bottom of footing	H	=	0.600	m
Section Modulus of Pedestal	Zxx	=	4.50	m^3
Section Modulus of Pedestal	Zzz	=	4.50	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	536	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	11	
Weight of soil		=	213.95	KN
Weight of foundation		=	129.60	KN
Uplift		=	0.00	KN
Surcharge		=	76.41	KN
Pressure on top for tension		=	46.7	KN/m^2

*inputs



$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	19.2	811.29	-1.2	11.88	0.9	-57.9	1231.2	152.31	121.30	1	0.28
302	18.6	1219.32	-2.4	7.02	0.51	-52.7	1639.3	195.41	168.88	1	0.36
303	21.0	1037.64	-2.7	5.48	4.17	-56.3	1457.6	175.68	148.23	1	0.33
304	20.4	1445.67	-3.9	0.63	3.79	-51.1	1865.6	218.78	195.80	1	0.41
305	20.5	1307.87	-1.2	13.71	-2.99	-62.1	1727.8	208.83	175.13	1	0.39
306	19.8	1715.9	-2.4	8.85	-3.4	-56.9	2135.9	251.93	222.71	1	0.47
307	63.9	861.93	-1.6	9.16	3.46	-207.6	1281.9	190.59	94.27	1	0.36
308	-23.1	1308.09	-1.8	11.50	-1.9	89.5	1728.0	214.45	169.56	1	0.40
309	25.8	1098.47	56.2	298.75	-1.25	-78.1	1518.4	252.45	84.98	1	0.47
310	15.0	1065.92	-59.7	-278.43	2.58	-39.7	1485.9	235.80	94.40	1	0.44
311	63.2	1269.96	-2.9	4.30	3.1	-202.3	1689.9	233.68	141.86	1	0.44
312	-23.7	1716.12	-3.0	6.64	-2.28	94.7	2136.1	259.87	214.81	1	0.48
313	25.1	1506.49	55.0	293.90	-1.63	-72.8	1926.4	295.54	132.56	1	0.55
314	14.3	1473.95	-60.9	-283.29	2.2	-34.5	1893.9	281.05	139.82	1	0.52
315	20.4	1082.39	-1.7	10.17	0.66	-58.9	1502.3	182.27	151.58	1	0.34
316	20.4	1082.39	-1.7	10.17	0.66	-58.9	1502.3	182.27	151.58	1	0.34
317	20.4	1082.39	-1.7	10.17	0.66	-58.9	1502.3	182.27	151.58	1	0.34
318	20.4	1082.39	-1.7	10.17	0.66	-58.9	1502.3	182.27	151.58	1	0.34
319	19.7	1490.42	-3.0	5.31	0.3	-53.7	1910.4	225.37	199.16	1	0.42
320	19.7	1490.42	-3.0	5.31	0.29	-53.7	1910.4	225.37	199.16	1	0.42
321	19.7	1490.42	-3.0	5.31	0.29	-53.7	1910.4	225.37	199.16	1	0.42
322	19.7	1490.42	-3.0	5.31	0.29	-53.7	1910.4	225.37	199.16	1	0.42
323	69.5	192.83	-0.6	5.78	4.26	-233.0	612.8	121.14	15.03	1	0.23
324	-46.4	787.7	-0.8	8.90	-2.9	163.1	1207.7	172.41	95.96	1	0.32
325	18.7	508.21	76.6	391.92	-2.03	-60.3	928.2	203.63	2.63	1	0.38
326	4.3	464.82	-78.0	-377.67	3.09	-9.2	884.8	184.28	12.34	1	0.34
327	69.2	437.65	-1.3	2.87	4.03	-229.8	857.6	147.00	43.58	1	0.27
328	-46.8	1032.52	-1.5	5.99	-3.1	166.3	1452.5	199.66	123.11	1	0.37
329	18.4	753.03	75.8	389.01	-2.25	-57.2	1173.0	229.48	31.18	1	0.43
330	4.0	709.64	-78.7	-380.59	2.86	-6.0	1129.6	211.43	39.59	1	0.39
331	11.5	486.77	-0.7	7.13	0.53	-34.7	906.7	110.05	91.44	1	0.21
332	11.5	486.77	-0.7	7.13	0.53	-34.7	906.7	110.05	91.44	1	0.21
333	11.5	486.77	-0.7	7.13	0.5	-34.7	906.7	110.05	91.44	1	0.21
334	11.5	486.77	-0.7	7.13	0.53	-34.7	906.7	110.05	91.44	1	0.21
335	11.2	731.59	-1.5	4.22	0.31	-31.6	1151.5	135.91	119.99	1	0.25
336	11.2	731.59	-1.5	4.22	0.31	-31.6	1151.5	135.91	119.99	1	0.25
337	11.2	731.59	-1.5	4.22	0.31	-31.6	1151.5	135.91	119.99	1	0.25
338	11.2	731.59	-1.5	4.22	0.3	-31.6	1151.5	135.91	119.99	1	0.25

Max P_{max}	=	295.54	KN/m^2
Max P_{min}	=	2.63	KN/m^2
Resultant F.O.S	=	0.55	

Global Project ID	900707
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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	26.9	1135.8	-1.6	16.14	1.2	-89.15	149.60	102.80
402	26.0	1707.0	-3.4	8.82	0.71	-81.56	209.75	169.59
403	29.9	1561.1	-4.0	5.80	5.74	-88.17	194.34	152.58
404	29.1	1939.5	-1.9	17.93	-4.27	-96.10	240.83	190.16
405	29.0	2132.4	-5.7	-1.52	5.22	-80.57	255.17	218.69
406	28.2	2510.7	-3.6	10.62	-4.8	-88.50	300.99	256.94
407	116.6	684.0	-1.6	10.49	6.87	-422.28	172.17	-20.18
408	-68.9	1635.8	-1.9	15.39	-4.56	267.09	244.52	118.98
409	35.3	1188.6	121.9	665.33	-3.18	-121.64	306.95	-42.82
410	12.3	1119.2	-125.4	-640.20	5.01	-32.94	273.94	-25.23
411	115.9	1173.6	-3.0	4.23	6.42	-415.78	223.74	37.06
412	-69.6	2125.4	-3.4	9.12	-5.01	273.60	298.98	173.33
413	34.6	1678.2	120.4	659.07	-3.63	-115.14	358.52	14.42
414	11.6	1608.8	-126.9	-646.46	4.55	-26.43	328.28	29.23
415	23.8	1154.3	-1.8	12.57	0.91	-77.29	148.22	108.29
416	23.8	1154.3	-1.8	12.57	0.91	-77.29	148.22	108.29
417	23.8	1154.3	-1.8	12.57	0.91	-77.29	148.22	108.29
418	23.8	1154.3	-1.8	12.57	0.91	-77.29	148.22	108.29
419	23.1	1643.9	-3.3	6.31	0.46	-70.78	199.79	165.53
420	23.1	1643.9	-3.3	6.31	0.46	-70.78	199.79	165.53
421	23.1	1643.9	-3.3	6.31	0.46	-70.78	199.79	165.53
422	23.1	1643.9	-3.3	6.31	0.46	-70.78	199.79	165.53
423	110.1	259.9	-0.8	8.29	6.76	-402.31	120.12	-62.37
424	-75.4	1211.7	-1.1	13.19	-4.67	287.07	201.35	67.90
425	28.8	764.5	122.6	663.13	-3.29	-101.67	254.90	-85.02
426	5.8	695.0	-124.7	-642.40	4.89	-12.95	222.86	-68.41
427	109.6	627.1	-1.9	3.59	6.42	-397.43	158.79	-19.44
428	-76.0	1578.9	-2.2	8.49	-5.01	291.95	242.19	108.67
429	28.3	1131.7	121.5	658.43	-3.63	-96.79	293.57	-42.08
430	5.2	1062.3	-125.8	-647.10	4.55	-8.07	263.62	-27.56
431	17.3	730.2	-1.1	10.36	0.8	-57.31	96.17	66.09
432	17.3	730.2	-1.1	10.36	0.8	-57.31	96.17	66.09
433	17.3	730.2	-1.1	10.36	0.8	-57.31	96.17	66.09
434	17.3	730.2	-1.1	10.36	0.8	-57.31	96.17	66.09
435	16.7	1097.4	-2.2	5.67	0.46	-52.43	134.84	109.02
436	16.7	1097.4	-2.2	5.67	0.46	-52.43	134.84	109.02
437	16.7	1097.4	-2.2	5.67	0.46	-52.43	134.84	109.02
438	16.7	1097.4	-2.2	5.67	0.46	-52.43	134.84	109.02

Max Net factored B.P = 358.52 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 258.13 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1824.95 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1343.9 mm^2

Effective Depth Along X = 517 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 930.6 mm^2

A_{req} = 1343.9 mm^2

Required Spacing = **149.6** mm

Length of cantilever along Z = 1.075 m

Moment about X: Mux = 207.15 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1713.74 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1109.9 mm^2

Effective Depth Along X = 501 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 901.8 mm^2

A_{req} = 1109.9 mm^2

Required Spacing = **181.2** mm

Provide Bottom Reinforcement = ***16 mm dia @ 150 c/c***

Ast prov = 1340.4 mm^2

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Top Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1839.10 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 519 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 467.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **242.1** mm

Length of cantilever along Z = 1.075 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1755.03 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 507 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

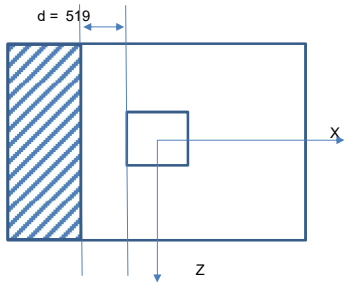
A_{stmin} = 456.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **247.9** mm

Provide top Reinforcement = ***12 mm dia @ 150 c/c*** Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.681 m

Reinforcement ratio: ρ_w = 0.0026

Effective Depth Along X: d = 519 mm

Shear Force at d distance: V_u = 244.15 KN

Allowable shear strength: φVc = 385.54 KN O.K

Utility ratio = 0.63

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 83.13 KNm

φ = 0.75 for shear

φVc = 394.5 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 385.5 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

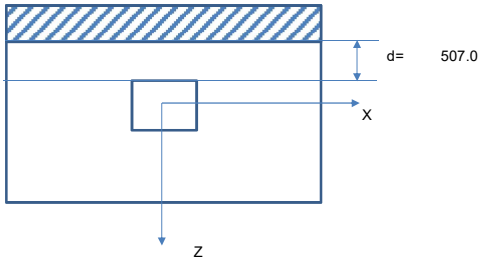
= 667.8 KN

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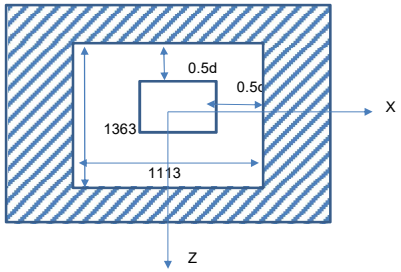
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.568	m		Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0026						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	507	mm				λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	203.64	KN				Mu	= 57.83 KNm
Allowable shear strength		φVc	=	377.02	KN	O.K			φ	= 0.75 for shear
Utility ratio			=	0.54					φVc	= 390.4 KN
								ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	= 377.0 KN
								iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									φVc	= 652.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	9	m2
Punching area		=	1.52	m2
	λ	=	1	
	β	=	1.416666667	
	αs	=	40	
	b0	=	4952.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95	N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	2.43	N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	3.02	N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.95	m		Vc	=	$v_c b_0 d$
Depth at section	d	=	513	mm			=	4959.59 KN
Shear Force at section	Vu	=	2682.8	KN		φ	=	0.75 for shear
Allowable shear strength	φVc	=	3719.69	KN	O.K	φV_n	=	φVc
Utility ratio		=	0.72			φVc	=	3719.7 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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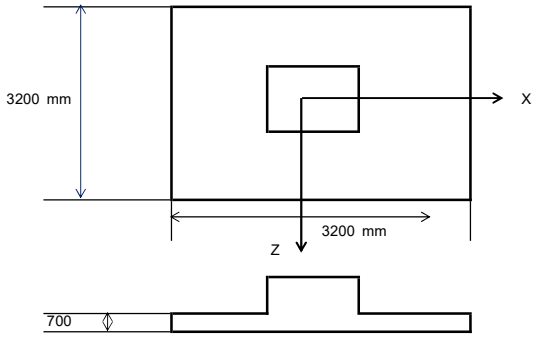
Outotec / L&T	0	1	2	3
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8.3.5 Design of foundation F3 at Grid 3120/K-3120/6

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	-1.800	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	3.20	m
Width of footing	Lx	=	3.20	m
Thickness of footing	D	=	0.70	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.85	m
Height of Load point from bottom of footing	H	=	0.700	m
Section Modulus of Pedestal	Zxx	=	5.46	m^3
Section Modulus of Pedestal	Zzz	=	5.46	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	536	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	20	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no	=	12	
Weight of soil	=	227.68	KN
Weight of foundation	=	172.03	KN
Uplift	=	0.00	KN
Surcharge	=	0	KN
Pressure on top for tension	=	39.0	KN/m^2

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	53.4	1479.74	4.6	20.76	0.0	-99.5	1879.5	205.56	161.52	1	0.38
302	55.9	2272.47	3.6	15.30	0.07	-105.7	2672.2	283.11	238.80	1	0.53
303	73.9	1854.54	2.5	15.07	0.2	-117.2	2254.3	244.36	195.93	1	0.46
304	76.4	2647.27	1.4	9.62	0.22	-123.4	3047.0	321.91	273.21	1	0.60
305	62.2	1899.19	5.1	24.18	-0.32	-109.2	2298.9	248.92	200.08	1	0.46
306	64.7	2691.92	4.1	18.72	-0.3	-115.4	3091.6	326.48	277.36	1	0.61
307	115.1	1684.33	3.8	18.38	-0.81	-274.3	2084.0	257.11	149.93	1	0.48
308	13.8	1867.68	4.3	21.80	0.8	53.9	2267.4	235.28	207.57	1	0.44
309	67.1	1714.16	72.8	336.65	-3.97	-118.9	2113.9	289.85	123.01	1	0.54
310	61.6	1841.78	-64.8	-296.83	3.9	-100.6	2241.5	291.66	146.13	1	0.54
311	117.7	2477.06	2.7	12.93	-0.8	-280.5	2876.8	334.67	227.20	1	0.62
312	16.4	2660.41	3.3	16.34	0.83	47.7	3060.1	310.56	287.12	1	0.58
313	69.6	2506.89	71.7	331.20	-3.94	-125.2	2906.6	367.41	200.29	1	0.69
314	64.2	2634.51	-65.8	-302.29	3.9	-106.8	3034.2	371.21	221.41	1	0.69
315	64.4	1777.58	4.0	19.91	-0.04	-109.8	2177.3	236.37	188.88	1	0.44
316	64.4	1777.58	4.0	19.91	-0.04	-109.8	2177.3	236.37	188.88	1	0.44
317	64.4	1777.58	4.0	19.91	-0.04	-109.8	2177.3	236.37	188.88	1	0.44
318	64.4	1777.58	4.0	19.91	-0.04	-109.8	2177.3	236.37	188.88	1	0.44
319	66.9	2570.31	3.0	14.45	0.0	-116.0	2970.0	313.92	266.16	1	0.59
320	66.9	2570.31	3.0	14.45	-0.01	-116.0	2970.0	313.92	266.16	1	0.59
321	66.9	2570.31	3.0	14.45	-0.01	-116.0	2970.0	313.92	266.16	1	0.59
322	66.9	2570.31	3.0	14.45	-0.01	-116.0	2970.0	313.92	266.16	1	0.59
323	99.7	763.5	2.4	10.41	-1.01	-279.1	1163.2	166.60	60.59	1	0.31
324	-35.4	1007.97	3.2	14.97	1.1	158.5	1407.7	169.23	105.70	1	0.32
325	35.6	803.28	94.5	434.78	-5.22	-71.9	1203.0	210.26	24.70	1	0.39
326	28.4	973.44	-88.9	-409.87	5.27	-47.4	1373.2	217.83	50.36	1	0.41
327	101.3	1239.14	1.8	7.14	-0.99	-282.8	1638.9	213.14	106.95	1	0.40
328	-33.8	1483.61	2.6	11.69	1.2	154.8	1883.3	214.40	153.43	1	0.40
329	37.2	1278.92	93.8	431.51	-5.21	-75.7	1678.6	256.79	71.06	1	0.48
330	29.9	1449.07	-89.6	-413.14	5.29	-51.2	1848.8	265.56	95.53	1	0.50
331	32.0	887.85	2.8	12.46	0.02	-59.7	1287.6	138.95	112.53	1	0.26
332	32.0	887.85	2.8	12.46	0.02	-59.7	1287.6	138.95	112.53	1	0.26
333	32.0	887.85	2.8	12.46	0.0	-59.7	1287.6	138.95	112.53	1	0.26
334	32.0	887.85	2.8	12.46	0.02	-59.7	1287.6	138.95	112.53	1	0.26
335	33.5	1363.48	2.1	9.19	0.04	-63.4	1763.2	185.48	158.89	1	0.35
336	33.5	1363.48	2.1	9.19	0.04	-63.4	1763.2	185.48	158.89	1	0.35
337	33.5	1363.48	2.1	9.19	0.04	-63.4	1763.2	185.48	158.89	1	0.35
338	33.5	1363.48	2.1	9.19	0.0	-63.4	1763.2	185.48	158.89	1	0.35

Max P_{max}	=	371.21	KN/m^2
Max P_{min}	=	24.70	KN/m^2
Resultant F.O.S	=	0.69	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	74.7	2071.6	6.4	31.33	0.1	-165.42	238.33	166.28
402	78.3	3181.5	5.0	23.17	0.1	-175.35	347.04	274.34
403	107.8	2715.5	3.2	21.89	0.24	-205.87	306.89	223.48
404	91.4	2778.0	7.0	35.95	-0.48	-188.99	312.48	230.10
405	111.4	3825.3	1.8	13.75	0.28	-215.81	415.60	331.53

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406	95.0	3887.8	5.5	27.79	-0.44	-198.93	421.18	338.16
407	179.7	1775.3	4.6	22.68	-1.65	-540.17	276.43	70.31
408	-36.4	2166.5	5.8	30.40	1.78	235.64	260.28	162.85
409	77.2	1839.0	151.8	753.20	-8.4	-172.83	349.15	10.02
410	65.6	2111.2	-141.6	-700.93	8.39	-129.58	358.24	54.10
411	182.8	2726.6	3.3	15.69	-1.62	-548.68	369.61	162.93
412	-33.4	3117.7	4.6	23.42	1.82	227.13	350.34	258.59
413	80.2	2790.2	150.6	746.22	-8.36	-181.34	442.32	102.64
414	68.6	3062.5	-142.9	-707.92	8.42	-138.09	453.98	144.16
415	71.4	1974.3	5.1	26.14	0	-151.20	225.27	160.32
416	71.4	1974.3	5.1	26.14	0	-151.20	225.27	160.32
417	71.4	1974.3	5.1	26.14	0	-151.20	225.27	160.32
418	71.4	1974.3	5.1	26.14	0	-151.20	225.27	160.32
419	74.4	2925.5	3.9	19.16	0.03	-159.72	318.45	252.94
420	74.4	2925.5	3.9	19.16	0.03	-159.72	318.45	252.94
421	74.4	2925.5	3.9	19.16	0.03	-159.72	318.45	252.94
422	74.4	2925.5	3.9	19.16	0.03	-159.72	318.45	252.94
423	156.4	1132.8	3.6	16.67	-1.61	-495.31	204.37	16.88
424	-59.8	1524.0	4.8	24.40	1.82	280.51	204.66	93.00
425	53.8	1196.5	150.9	747.20	-8.36	-127.96	277.09	-43.40
426	42.2	1468.7	-142.6	-706.94	8.43	-84.71	288.38	-1.53
427	158.7	1846.3	2.7	11.43	-1.59	-501.69	274.26	86.34
428	-57.5	2237.4	3.9	19.16	1.85	274.12	272.20	164.80
429	56.1	1909.9	149.9	741.96	-8.33	-134.36	346.97	26.06
430	44.5	2182.2	-143.5	-712.17	8.46	-91.09	360.18	66.02
431	48.0	1331.8	4.1	20.14	0.04	-106.33	153.21	106.90
432	48.0	1331.8	4.1	20.14	0.04	-106.33	153.21	106.90
433	48.0	1331.8	4.1	20.14	0.04	-106.33	153.21	106.90
434	48.0	1331.8	4.1	20.14	0.04	-106.33	153.21	106.90
435	50.3	2045.2	3.2	14.90	0.06	-112.72	223.10	176.36
436	50.3	2045.2	3.2	14.90	0.06	-112.72	223.10	176.36
437	50.3	2045.2	3.2	14.90	0.06	-112.72	223.10	176.36
438	50.3	2045.2	3.2	14.90	0.06	-112.72	223.10	176.36

Max Net factored B.P = 453.98 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.3 m

Moment about Z: Muz = 383.61 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 2582.38 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1680.5 mm^2

Effective Depth Along X = 615 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or greater of

A_{stmin} = 0.0014bd

A_{stmin} = 1107.0 mm^2

A_{req} = 1680.5 mm^2

Required Spacing = **186.9** mm

Length of cantilever along Z = 1.175 m

Moment about X: Mux = 313.39 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 2417.15 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1415.4 mm^2

Effective Depth Along X = 595 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or greater of

A_{stmin} = 0.0014bd

A_{stmin} = 1071.0 mm^2

A_{req} = 1415.4 mm^2

Required Spacing = **222.0** mm

Provide Bottom Reinforcement = ***20 mm dia @ 150 c/c***

Ast prov = 2094.4 mm^2

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Top Reinforcement

Length of cantilever along X = 1.3 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 2616.08 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 619 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of:

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 557.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **203.0** mm

Length of cantilever along Z = 1.175 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 2515.63 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 607 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of:

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 546.3 mm^2

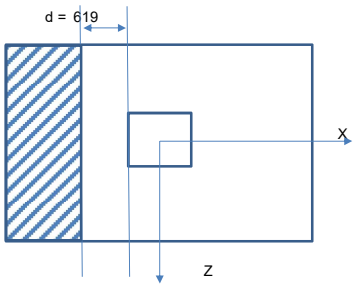
A_{req} = 0.0 mm^2

Required Spacing = **207.0** mm

Provide top Reinforcement = ***12 mm dia @ 150 c/c***

Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.681 m

Reinforcement ratio: ρ_w = 0.0034

Effective Depth Along X: d = 619 mm

Shear Force at d distance: V_u = 309.16 KN

Allowable shear strength: φVc = 466.15 KN O.K

Utility ratio = 0.66

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 105.27 KNm

φ = 0.75 for shear

φVc = 488.0 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 466.1 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

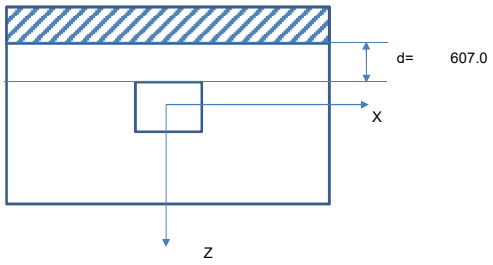
= 796.5 KN

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Outotec / L&T	0	1	2	3
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Revision Tex	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK /APL	RPK/APL	RPK / APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

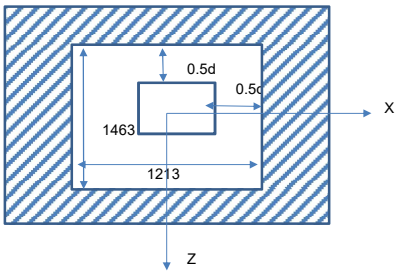
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.568	m		Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0035						Table 22.5.5.1 SBC 304
Effective Depth Along Z		d	=	607	mm				λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	257.86	KN				Mu	= 73.23 KNm
Allowable shear strength		φVc	=	457.63	KN	O.K			φ	= 0.75 for shear
Utility ratio			=	0.56					φVc	= 488.0 KN
								ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	= 457.6 KN
								iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									φVc	= 781.1 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	10.24	m2
Punching area		=	1.77	m2
	λ	=	1	
	β	=	1.416666667	
	αs	=	40	
	b0	=	5352.0	mm

Allowable shear stress — least of — Table 22.6.5.2 SBC 304

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	2.43 N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	3.23 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	5.35	m		Vc	= $v_c b_0 d$
Depth at section	d	=	613	mm			= 6405.08 KN
Shear Force at section	Vu	=	3843.1	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	4803.81	KN	O.K	φV_n	= φVc
Utility ratio		=	0.80			φVc	= 4803.8 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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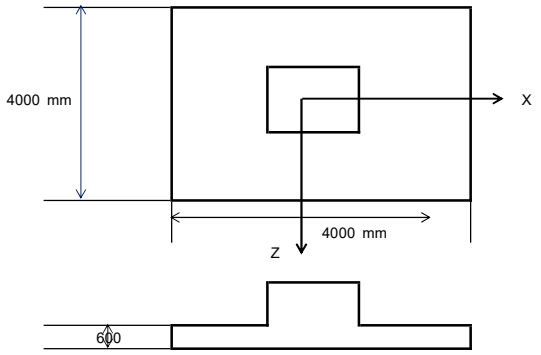
Outotec / L&T	0	2	3
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Approved	AG	AG	AG

8.3.6 Design of foundation F4 at Grid 3120/J-3120/8

Finished Ground level	F.G.L	=	0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	-1.900	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	9.000	KN/m^2
Length of footing	Lz	=	4.00	m
Width of footing	Lx	=	4.00	m
Thickness of footing	D	=	0.60	m
Depth of pedestal	Bx	=	0.70	m
Width of pedestal	Bz	=	0.50	m
Height of Load point from bottom of footing	H	=	0.600	m
Section Modulus of Pedestal	Zxx	=	10.67	m^3
Section Modulus of Pedestal	Zzz	=	10.67	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	554	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	20	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	332	
Weight of soil		=	676.08	KN
Weight of foundation		=	230.40	KN
Uplift		=	0.00	KN
Surcharge		=	140.85	KN
Pressure on top for tension		=	65.5	KN/m^2

*inputs



$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	33.2	470.67	4.4	13.61	-0.3	-59.3	1518.0	101.71	88.04	1	0.18
302	33.2	664.71	4.9	15.34	-0.4	-58.9	1712.0	113.96	100.04	1	0.21
303	34.4	755.86	6.7	19.96	-0.44	-63.0	1803.2	120.48	104.92	1	0.22
304	34.4	949.9	7.1	21.67	-0.54	-62.6	1997.2	132.73	116.93	1	0.24
305	29.6	663.9	3.8	11.46	-0.21	-51.6	1711.2	112.87	101.04	1	0.20
306	29.6	857.94	4.2	13.18	-0.3	-51.2	1905.3	125.11	113.05	1	0.23
307	92.4	703.16	6.8	19.89	-0.6	-190.1	1750.5	129.09	89.72	1	0.23
308	-27.3	595.5	3.2	10.29	-0.04	73.4	1642.8	110.52	94.84	1	0.20
309	30.4	1219.22	43.2	114.79	-0.43	-54.9	2266.6	157.57	125.75	1	0.28
310	34.2	80.93	-33.2	-84.43	-0.21	-60.7	1128.3	84.13	56.91	1	0.15
311	92.4	897.21	7.3	21.60	-0.7	-189.6	1944.5	141.34	101.73	1	0.26
312	-27.4	789.54	3.6	12.01	-0.15	73.8	1836.9	122.85	106.76	1	0.22
313	30.4	1413.26	43.7	116.51	-0.54	-54.5	2460.6	169.82	137.76	1	0.31
314	34.2	274.98	-32.7	-82.71	-0.3	-60.3	1322.3	96.05	69.24	1	0.17
315	32.3	650.08	5.0	15.18	-0.32	-57.8	1697.4	112.93	99.25	1	0.20
316	32.3	650.08	5.0	15.18	-0.32	-57.8	1697.4	112.93	99.25	1	0.20
317	32.3	650.08	5.0	15.18	-0.32	-57.8	1697.4	112.93	99.25	1	0.20
318	32.3	650.08	5.0	15.18	-0.32	-57.8	1697.4	112.93	99.25	1	0.20
319	32.3	844.12	5.5	16.90	-0.4	-57.4	1891.5	125.18	111.25	1	0.23
320	32.3	844.12	5.5	16.90	-0.42	-57.4	1891.5	125.18	111.25	1	0.23
321	32.3	844.12	5.5	16.90	-0.42	-57.4	1891.5	125.18	111.25	1	0.23
322	32.3	844.12	5.5	16.90	-0.42	-57.4	1891.5	125.18	111.25	1	0.23
323	100.1	353.18	5.1	14.44	-0.55	-211.9	1400.5	108.75	66.31	1	0.20
324	-59.6	209.63	0.2	1.64	0.2	139.3	1257.0	91.77	65.35	1	0.17
325	17.4	1041.26	53.6	140.98	-0.33	-31.7	2088.6	146.72	114.35	1	0.26
326	22.4	-476.46	-48.3	-124.64	-0.03	-39.5	570.9	51.07	20.29	1	0.09
327	100.1	469.61	5.3	15.47	-0.61	-211.6	1516.9	116.10	73.52	1	0.21
328	-59.6	326.05	0.4	2.68	0.1	139.6	1373.4	99.17	72.50	1	0.18
329	17.4	1157.68	53.9	142.02	-0.39	-31.4	2205.0	154.07	121.55	1	0.28
330	22.4	-360.03	-48.0	-123.61	-0.09	-39.2	687.3	58.22	27.69	1	0.11
331	19.9	282.4	2.7	8.17	-0.18	-35.6	1329.7	87.21	79.01	1	0.16
332	19.9	282.4	2.7	8.17	-0.18	-35.6	1329.7	87.21	79.01	1	0.16
333	19.9	282.4	2.7	8.17	-0.2	-35.6	1329.7	87.21	79.01	1	0.16
334	19.9	282.4	2.7	8.17	-0.18	-35.6	1329.7	87.21	79.01	1	0.16
335	19.9	398.82	2.9	9.20	-0.24	-35.3	1446.2	94.56	86.21	1	0.17
336	19.9	398.82	2.9	9.20	-0.24	-35.3	1446.2	94.56	86.21	1	0.17
337	19.9	398.82	2.9	9.20	-0.24	-35.3	1446.2	94.56	86.21	1	0.17
338	19.9	398.82	2.9	9.20	-0.2	-35.3	1446.2	94.56	86.21	1	0.17

Max P_{max}	=	169.82	KN/m^2
Max P_{min}	=	20.29	KN/m^2
Resultant F.O.S	=	0.31	

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	46.2	657.9	6.2	20.80	-0.4	-96.25	52.09	30.14
402	46.5	930.9	6.8	23.54	-0.588	-96.45	69.43	46.93
403	47.5	1128.8	9.6	31.30	-0.657	-101.38	82.99	58.11
404	40.7	999.6	5.5	18.13	-0.342	-83.15	71.97	52.98
405	47.8	1401.8	10.2	34.04	-0.807	-101.58	100.33	74.90
406	41.0	1272.7	6.1	20.87	-0.492	-83.35	89.31	69.77
407	167.2	796.5	9.5	30.13	-1.001	-401.69	90.26	9.30
408	-88.2	567.1	1.7	7.37	0.212	236.45	58.30	12.59
409	35.0	1898.8	87.2	256.08	-0.621	-73.95	149.62	87.74
410	43.0	-532.0	-75.9	-218.13	-0.166	-88.71	-4.48	-62.02
411	167.5	1030.5	10.1	32.48	-1.129	-401.87	105.13	23.69
412	-87.9	801.2	2.3	9.72	0.083	236.28	73.14	27.01
413	35.3	2132.9	87.8	258.43	-0.75	-74.12	164.48	102.13
414	43.3	-297.9	-75.3	-215.78	-0.294	-88.88	9.94	-47.18
415	39.0	683.4	5.7	18.98	-0.394	-81.33	52.12	33.31
416	39.0	683.4	5.7	18.98	-0.394	-81.33	52.12	33.31
417	39.0	683.4	5.7	18.98	-0.394	-81.33	52.12	33.31
418	39.0	683.4	5.7	18.98	-0.394	-81.33	52.12	33.31
419	39.3	917.5	6.3	21.33	-0.522	-81.50	66.98	47.70
420	39.3	917.5	6.3	21.33	-0.522	-81.50	66.98	47.70
421	39.3	917.5	6.3	21.33	-0.522	-81.50	66.98	47.70
422	39.3	917.5	6.3	21.33	-0.522	-81.50	66.98	47.70
423	157.9	536.0	7.8	24.52	-0.889	-382.24	71.63	-4.63
424	-97.4	306.7	0.0	1.76	0.323	255.91	43.32	-4.99
425	25.7	1638.3	85.5	250.48	-0.51	-54.49	130.99	73.81
426	33.7	-792.5	-77.6	-223.74	-0.054	-69.25	-22.06	-77.00
427	158.1	711.5	8.2	26.28	-0.985	-382.37	82.78	6.16
428	-97.2	482.2	0.4	3.52	0.227	255.78	54.45	5.83
429	25.9	1813.9	85.9	252.24	-0.606	-54.62	142.14	84.60
430	33.9	-616.9	-77.1	-221.97	-0.15	-69.38	-11.24	-65.87
431	29.7	422.9	4.0	13.37	-0.282	-61.88	33.49	19.38
432	29.7	422.9	4.0	13.37	-0.282	-61.88	33.49	19.38
433	29.7	422.9	4.0	13.37	-0.282	-61.88	33.49	19.38
434	29.7	422.9	4.0	13.37	-0.282	-61.88	33.49	19.38
435	29.9	598.5	4.4	15.13	-0.378	-62.00	44.64	30.17
436	29.9	598.5	4.4	15.13	-0.378	-62.00	44.64	30.17
437	29.9	598.5	4.4	15.13	-0.378	-62.00	44.64	30.17
438	29.9	598.5	4.4	15.13	-0.378	-62.00	44.64	30.17

Max Net factored B.P = 164.48 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.65 m

Moment about Z: Muz = 223.90 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1810.86 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1167.4 mm^2

Effective Depth Along X = 515 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 927.0 mm^2

A_{req} = 1167.4 mm^2

Required Spacing = **269.1** mm

Length of cantilever along Z = 1.75 m

Moment about X: Mux = 251.86 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1672.94 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1371.2 mm^2

Effective Depth Along X = 495 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 891.0 mm^2

A_{req} = 1371.2 mm^2

Required Spacing = **229.1** mm

Provide Bottom Reinforcement = ***20 mm dia @ 200 c/c***

Ast prov = 1570.8 mm^2

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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01	Outotec / L&T	0	2	3
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Checked	SBY	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	AG	AG	AG	AG	Approved	AG	AG	AG

Top Reinforcement

Length of cantilever along X = 1.65 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1839.10 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 519 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or

greater of or more prov in

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 467.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **242.1** mm

Length of cantilever along Z = 1.75 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1755.03 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 507 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or

greater of or more prov in

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 456.3 mm^2

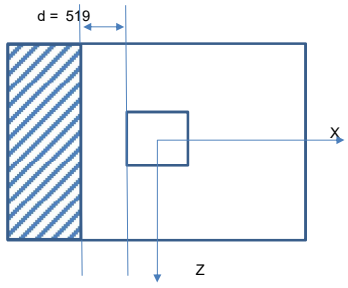
A_{req} = 0.0 mm^2

Required Spacing = **247.9** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 1.131 m

Reinforcement ratio: ρ_w = 0.0030

Effective Depth Along X: d = 519 mm

Shear Force at d distance: V_u = 186.03 KN

Allowable shear strength: φVc = 386.83 KN O.K

Utility ratio = 0.48

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 105.20 KNm

φ = 0.75 for shear

φVc = 386.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 388.5 KN *Table 22.5.5.1 SBC 304*

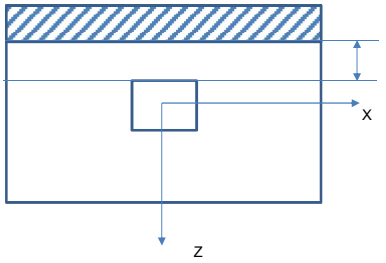
iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

= 667.8 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01	Outotec / L&T	0	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023	Date	07-03-2020	24-07-2020	16-10-2023
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Prepared By	RPK / APL	RPK/APL	RPK / APL	RPK / APL	Prepared By	RPK /APL	RPK / APL	RPK / APL
Checked	SBY	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	AG	AG	AG	AG	Approved	AG	AG	AG

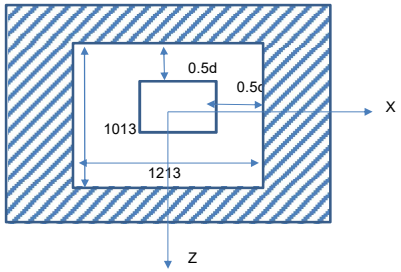
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	1.243	m		Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0031						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	507	mm				λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	204.45	KN				Mu	= 127.07 KNm
Allowable shear strength		φVc	=	376.27	KN	O.K			φ	= 0.75 for shear
Utility ratio			=	0.54					φVc	= 376.3 KN
								ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	= 380.0 KN
								iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									φVc	= 652.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	16	m2
Punching area		=	1.23	m2
	λ	=	1	
	β	=	1.4	
	αs	=	40	
	b0	=	4452.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	2.44 N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	3.25 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.45	m		Vc	= $v_c b_0 d$
Depth at section	d	=	513	mm			= 4458.83 KN
Shear Force at section	Vu	=	2429.6	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	3344.12	KN	O.K	φV_n	= φVc
Utility ratio		=	0.73			φVc	= 3344.1 KN

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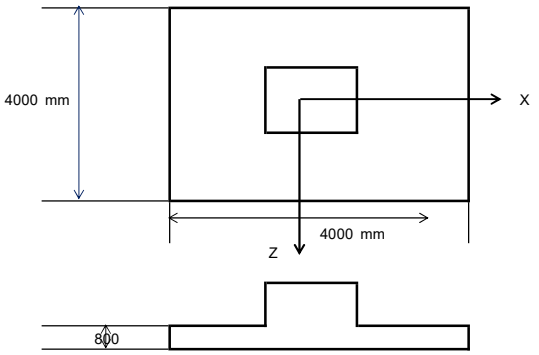
Outotec / L&T	0	1	2	3
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Approved	AG	AG	AG	AG

8.3.7 Design of foundation F4A at Grid 3120/K-3120/5

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	-1.700	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	9.000	KN/m^2
Length of footing	Lz	=	4.00	m
Width of footing	Lx	=	4.00	m
Thickness of footing	D	=	0.80	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.85	m
Height of Load point from bottom of footing	H	=	0.800	m
Section Modulus of Pedestal	Zxx	=	10.67	m^3
Section Modulus of Pedestal	Zzz	=	10.67	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	536	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	16	mm
Bottom Reinforcement		=	25	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	7	
Weight of soil		=	334.58	KN
Weight of foundation		=	307.20	KN
Uplift		=	0.00	KN
Surcharge		=	139.41	KN
Pressure on top for tension		=	48.8	KN/m^2

*inputs



$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-22.8	2003.54	-4.7	-13.95	0.9	-33.6	2784.7	178.51	169.58	1	0.33
302	-22.8	3833.85	-6.4	-21.94	1.25	-37.2	4615.0	293.98	282.90	1	0.55
303	-45.9	2458.27	-6.3	-18.52	1.14	-11.5	3239.5	205.28	199.65	1	0.38
304	-46.0	4288.58	-8.0	-26.51	1.51	-15.0	5069.8	320.75	312.97	1	0.60
305	-41.7	2454.83	-5.0	-12.33	0.99	-15.0	3236.0	204.81	199.69	1	0.38
306	-41.7	4285.13	-6.7	-20.32	1.4	-18.5	5066.3	320.29	313.00	1	0.60
307	14.1	2369.95	-5.8	-17.27	0.55	-203.8	3151.1	217.67	176.22	1	0.41
308	-90.9	2316.7	-5.0	-12.54	1.53	166.1	3097.9	210.36	176.87	1	0.39
309	-35.5	2319.98	71.1	274.27	-5.98	-29.2	3101.2	222.27	165.37	1	0.41
310	-41.6	2366.62	-81.9	-304.39	8.03	-7.5	3147.8	225.98	167.50	1	0.42
311	14.1	4200.26	-7.4	-25.27	0.9	-207.3	4981.5	333.15	289.54	1	0.62
312	-91.0	4147.01	-6.7	-20.53	1.89	162.6	4928.2	325.18	290.85	1	0.61
313	-35.5	4150.29	69.4	266.28	-5.62	-32.7	4931.5	336.25	280.19	1	0.63
314	-41.7	4196.93	-83.6	-312.38	8.4	-11.0	4978.1	341.45	280.81	1	0.64
315	-38.5	2343.3	-5.4	-15.06	1.02	-18.4	3124.5	198.41	192.15	1	0.37
316	-38.5	2343.3	-5.4	-15.06	1.02	-18.4	3124.5	198.41	192.15	1	0.37
317	-38.5	2343.3	-5.4	-15.06	1.02	-18.4	3124.5	198.41	192.15	1	0.37
318	-38.5	2343.3	-5.4	-15.06	1.02	-18.4	3124.5	198.41	192.15	1	0.37
319	-38.6	4173.6	-7.1	-23.06	1.4	-21.9	4954.8	313.89	305.46	1	0.59
320	-38.6	4173.6	-7.1	-23.06	1.38	-21.9	4954.8	313.89	305.46	1	0.59
321	-38.6	4173.6	-7.1	-23.06	1.38	-21.9	4954.8	313.89	305.46	1	0.59
322	-38.6	4173.6	-7.1	-23.06	1.38	-21.9	4954.8	313.89	305.46	1	0.59
323	56.6	1237.67	-3.3	-11.32	-0.1	-267.5	2018.9	152.32	100.04	1	0.28
324	-83.5	1166.67	-2.3	-5.00	1.2	225.7	1947.9	143.37	100.11	1	0.27
325	-9.6	1171.04	99.2	377.40	-8.8	-34.7	1952.2	160.65	83.38	1	0.30
326	-17.8	1233.23	-104.8	-394.15	9.87	-5.7	2014.4	163.39	88.41	1	0.30
327	56.6	2335.85	-4.3	-16.12	0.12	-269.6	3117.0	221.60	168.03	1	0.41
328	-83.5	2264.85	-3.3	-9.80	1.4	223.6	3046.0	212.26	168.49	1	0.40
329	-9.6	2269.23	98.2	372.61	-8.59	-36.8	3050.4	229.03	152.27	1	0.43
330	-17.8	2331.41	-105.8	-398.94	10.08	-7.8	3112.6	232.67	156.40	1	0.43
331	-13.7	1202.12	-2.8	-8.37	0.53	-20.2	1983.3	126.63	121.28	1	0.24
332	-13.7	1202.12	-2.8	-8.37	0.53	-20.2	1983.3	126.63	121.28	1	0.24
333	-13.7	1202.12	-2.8	-8.37	0.5	-20.2	1983.3	126.63	121.28	1	0.24
334	-13.7	1202.12	-2.8	-8.37	0.53	-20.2	1983.3	126.63	121.28	1	0.24
335	-13.7	2300.31	-3.8	-13.17	0.75	-22.3	3081.5	195.92	189.27	1	0.37
336	-13.7	2300.31	-3.8	-13.17	0.75	-22.3	3081.5	195.92	189.27	1	0.37
337	-13.7	2300.31	-3.8	-13.17	0.75	-22.3	3081.5	195.92	189.27	1	0.37
338	-13.7	2300.31	-3.8	-13.17	0.8	-22.3	3081.5	195.92	189.27	1	0.37

Max P_{max}	=	341.45	KN/m^2
Max P_{min}	=	83.38	KN/m^2
Resultant F.O.S	=	0.64	

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-31.9	2805.0	-6.6	-22.15	1.2	-34.36	180.61	170.01
402	-31.9	5367.4	-8.9	-34.29	1.75	-39.26	342.36	328.57
403	-70.6	3577.5	-9.1	-30.01	1.66	18.26	228.12	219.07
404	-64.6	3572.7	-7.4	-20.65	1.45	10.96	226.25	220.33
405	-70.7	6139.9	-11.4	-42.14	2.16	13.36	388.95	378.54
406	-64.7	6135.1	-9.7	-32.78	1.95	6.07	387.09	379.80
407	74.5	2687.6	-6.8	-24.93	0.15	-455.64	213.03	122.92
408	-149.6	2574.0	-5.3	-14.20	2.23	423.14	201.88	119.88
409	-31.3	2581.0	157.1	662.59	-13.78	-40.83	227.26	95.37
410	-44.4	2680.5	-169.3	-702.43	16.09	10.74	234.39	100.67
411	74.5	4884.0	-8.8	-35.33	0.58	-459.84	351.67	258.83
412	-149.7	4770.4	-7.3	-24.59	2.67	418.92	339.73	256.57
413	-31.3	4777.4	155.0	652.18	-13.35	-45.04	363.95	233.22
414	-44.5	4876.9	-171.3	-712.83	16.53	6.54	372.25	237.36
415	-37.8	2630.8	-6.1	-19.93	1.16	-15.05	167.70	161.14
416	-37.8	2630.8	-6.1	-19.93	1.16	-15.05	167.70	161.14
417	-37.8	2630.8	-6.1	-19.93	1.16	-15.05	167.70	161.14
418	-37.8	2630.8	-6.1	-19.93	1.16	-15.05	167.70	161.14
419	-37.9	4827.1	-8.1	-30.32	1.59	-19.25	306.34	297.05
420	-37.9	4827.1	-8.1	-30.32	1.59	-19.25	306.34	297.05
421	-37.9	4827.1	-8.1	-30.32	1.59	-19.25	306.34	297.05
422	-37.9	4827.1	-8.1	-30.32	1.59	-19.25	306.34	297.05
423	91.9	1860.1	-4.9	-19.25	-0.21	-462.67	161.43	71.07
424	-132.3	1746.5	-3.4	-8.51	1.88	416.09	148.96	69.35
425	-13.9	1753.5	158.9	668.26	-14.14	-47.87	176.73	42.45
426	-27.1	1853.0	-167.4	-696.75	15.74	3.70	181.48	50.14
427	91.9	3507.3	-6.4	-27.04	0.11	-465.84	265.42	173.00
428	-132.3	3393.7	-4.9	-16.31	2.2	412.93	252.35	171.87
429	-13.9	3400.7	157.4	660.47	-13.81	-51.03	279.25	145.84
430	-27.1	3500.2	-168.9	-704.54	16.06	0.55	284.87	152.66
431	-20.5	1803.2	-4.2	-14.24	0.8	-22.09	116.10	109.29
432	-20.5	1803.2	-4.2	-14.24	0.8	-22.09	116.10	109.29
433	-20.5	1803.2	-4.2	-14.24	0.8	-22.09	116.10	109.29
434	-20.5	1803.2	-4.2	-14.24	0.8	-22.09	116.10	109.29
435	-20.5	3450.5	-5.7	-22.04	1.12	-25.23	220.09	211.22
436	-20.5	3450.5	-5.7	-22.04	1.12	-25.23	220.09	211.22
437	-20.5	3450.5	-5.7	-22.04	1.12	-25.23	220.09	211.22
438	-20.5	3450.5	-5.7	-22.04	1.12	-25.23	220.09	211.22

Max Net factored B.P = 388.95 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.7 m

Moment about Z: Muz = 562.03 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 3466.09 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 2129.1 mm^2

Effective Depth Along X = 712.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 1282.5 mm^2

A_{req} = 2129.1 mm^2

Required Spacing = **230.6** mm

Length of cantilever along Z = 1.575 m

Moment about X: Mux = 482.42 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 3227.12 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 1890.7 mm^2

Effective Depth Along X = 687.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 1237.5 mm^2

A_{req} = 1890.7 mm^2

Required Spacing = **259.6** mm

Provide Bottom Reinforcement = ***25 mm dia @ 200 c/c***

Ast prov = 2454.4 mm^2

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Top Reinforcement

Length of cantilever along X = 1.7 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 3510.01 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 717 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 645.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **311.6** mm

Length of cantilever along Z = 1.575 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 3355.10 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 701 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 630.9 mm^2

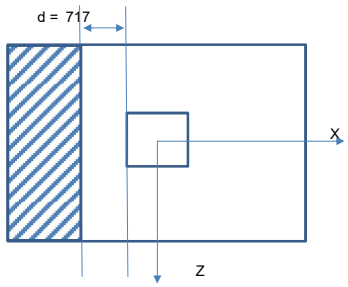
A_{req} = 0.0 mm^2

Required Spacing = **318.7** mm

Provide top Reinforcement = ***16 mm dia @ 200 c/c***

Ast prov = 1005.3 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.983 m

Reinforcement ratio: ρ_w = 0.0034

Effective Depth Along X: d = 717 mm

Shear Force at d distance: V_u = 382.34 KN

Allowable shear strength: φVc = 540.31 KN O.K

Utility ratio = 0.71

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 187.92 KNm

φ = 0.75 for shear

φVc = 554.7 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 540.3 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

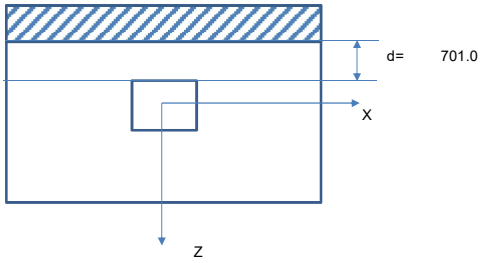
= 922.6 KN

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK / APL	RPK/APL	RPK / APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T	0	1	2	3
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ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.874	m				
Reinforcement ratio		ρ_w	=	0.0035					
Effective Depth Along Z		d	=	701	mm				
Shear Force at d distance		V_u	=	339.94	KN				
Allowable shear strength		φVc	=	528.95	KN	O.K			
Utility ratio			=	0.64					

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete	
Mu	=	148.55	KNm	
φ	=	0.75	for shear	
φVc	=	547.9	KN	

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

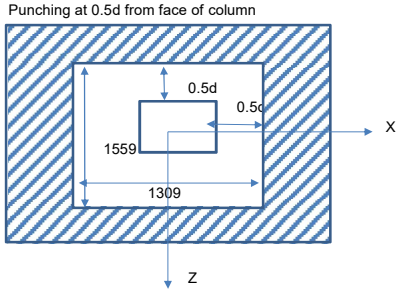
φVc	=	529.0	KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}\,b_w d$

φVc	=	902.0	KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	16	m2
Punching area	=	2.04	m2
λ	=	1	
β	=	1.416666667	
αs	=	40	
b0	=	5736.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 2.43 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 3.41 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face		u	=	5.74	m		Vc	=	$v_c b_0 d$
Depth at section		d	=	709	mm			=	7939.69 KN
Shear Force at section		Vu	=	5429.4	KN		φ	=	0.75 for shear
Allowable shear strength		φVc	=	5954.76	KN	O.K	φV_n	=	φVc
Utility ratio			=	0.91			φVc	=	5954.8 KN

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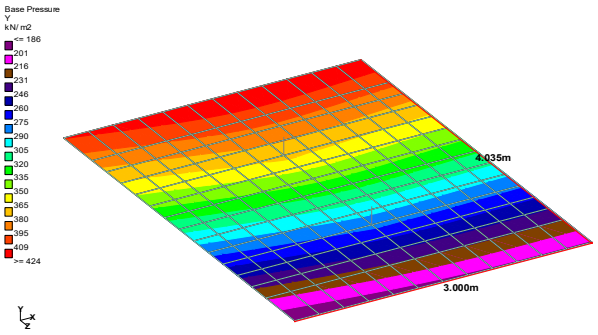
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

8.3.8 Design of combined footing for column at 3120J_3120/5 & 3120/H'_3120/5 and foundation mkd. as CF1

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF-1 GR-J-5.STD**

Staad Supoort Node for column at **3120J_3120/5** **8**
3120/H'_3120/5 **120**

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.000	m	
Width of footing	Lx	=	4.035	m	
Thickness of footing	D	=	0.65	m	
Depth of pedestal	Bx	=	0.55	m	
Width of pedestal	Bz	=	0.55	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	24.3	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressure	=	482.3	KN/m^2	Load Case 314
Minimum Base Pressure	=	22.4	KN/m^2	Load Case 326

2) Design of foundation Raft :

Slab Thickness =	650 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

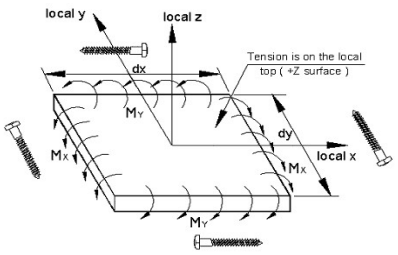
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	-0.700	kN-m	Corresponding Mxy	-1.220	kN-m	*(Plate No. - 259 , L/C No. - 423)*
Min. Mx	-362.900	kN-m	Corresponding Mxy	-35.990	kN-m	*(Plate No. - 292 , L/C No. - 405)*
Max. My	26.340	kN-m	Corresponding Mxy	29.210	kN-m	*(Plate No. - 222 , L/C No. - 410)*
Min. My	-314.110	kN-m	Corresponding Mxy	63.070	kN-m	*(Plate No. - 281 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	-0.7 + 1.22 =	0.520 kN-m
My1 =	My + abs(Mxy) =	26.34 + 29.21 =	55.550 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	-0.7 + 0.057 =	-0.643 kN-m
My2 =	My+abs(Mxy^2/Mx) =	26.34 + 7.121 =	33.461 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 0.52 kN-m
Myd = 55.55 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 0.52 kN-m
Φ*Kn = Mu/bd^2 deff. = 569 mm
Mu/bd^2 = 0.52X10^6/(1000X569^2) = 0.002 N/mm^2 < 6.9 N/mm^2
Pt = 0.0000 Reqd. Ast = 0X1000X569/100 = 0 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 **O.K.**
Total Provide Pt = 0.001

ALONG LOCAL Y - DIRECTION :

Mu = 55.55 kN-m
Φ*Kn = Mu/bd^2 deff. = 557 mm
Mu/bd^2 = 55.55X10^6/(1000X557^2) = 0.172 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X569/100 = 2.79 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 **O.K.**
Total Provide Pt = 0.001

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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -362.9 - 35.99 = -398.89 kN-m
My1 = My - abs(Mxy) = -314.11 - 63.07 = -377.18 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -362.9 - 4.124 = -367.024 kN-m
My2 = My-abs(Mxy^2/Mx) = -314.11 - 10.961 = -303.149 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -398.89 kN-m
Myd = -377.18 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 398.89 kN-m
Φ*Kn = Mu/bd^2 deff. = 562.5 mm
Mu/bd^2 = 398.89X10^6/(1000X562.5^2) = 1.261 N/mm^2 < 6.9 N/mm^2
Pt = 0.0034 Reqd. Ast = 0034X1000X562.5/100 = 19.13 cm^2
Provided 25 mm @ 200 C/C (Provide Ast = 24.54)cm^2

Total Provide Ast = 24.54 cm^2 O.K.
Total Provide Pt = 0.0044

ALONG LOCAL Y - DIRECTION :

Mu = 377.18 kN-m
Φ*Kn = Mu/bd^2 deff. = 537.5 mm
Mu/bd^2 = 377.18X10^6/(1000X537.5^2) = 1.306 N/mm^2 < 6.9 N/mm^2
Pt = 0.0035 Reqd. Ast = 0035X1000X537.5/100 = 18.81 cm^2
Provided 25 mm @ 200 C/C (Provide Ast = 24.54)cm^2

Total Provide Ast = 24.54 cm^2 O.K.
Total Provide Pt = 0.0046

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.66 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 271 , L/C No. - 405)*
Vu = 371.25 kN

Provided Pt = 0.0044
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0044X371.25X10^3X562.5/398.89/10^6)X1000X562.5/10^3
= 554.474 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0044)X1000X562.5/10^3
= 574.522 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X562.5/10^3
= 965.061 kN
HENCE, Vc = 554.474 kN
Φt*Vc = 0.75X554.474 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 415.856 kN **> 371.25T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.34 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 163 , L/C No. - 405)*
Vu = 182.75 kN

Pt provided = 0.0046
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0046X182.75X10^3X537.5/377.18/10^3)X1000X537.5/10^3
= 519.729 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0046)X1000X537.5/10^3
= 550.815 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X537.5/10^3
= 922.169 kN
HENCE, Vc = 519.729 kN
Φt*Vc = 0.75X519.729 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 389.797 kN **> 182.75T , Hence O.K.**

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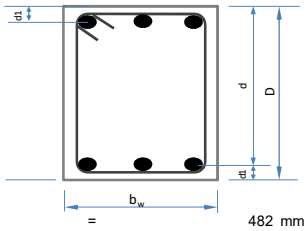
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

8.4 Design of Beam

8.4.1 Beam marked TB1

Staad Beam Group TB1

Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	482	mm	
Dia of longitudinal bar	=	**16**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	482	mm
d'	=	68	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	412	Staad beam number -	1509	=	**39**	KNm
B.M for bottom reinforcement	Mz	L/C -	407	Staad beam number -	1509	=	**15**	KNm
Shear Force	Fy	L/C -	407	Staad beam number -	1509	=	**23**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 482 x 482 (1 - 0.8/2 x 0.375)

= 475.9 KNm

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 39.39 KNm

Mu_2 = Mu - Mu_{lim} = 39.39 - 475.9

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 482 x (1-√(1-(2.35 x 39.4 x 10^6)/(0.9 x 35 x 300 x 482^2)))/420

= 218.3 mm^2

= 0 x 10^6/(0.9 x 420 x (482 - 68))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 509.2 mm^2 or 482 mm^2 = 509.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 509.3 + 0.00 + 0

= 509.3 mm^2

Top layer1	**2**	*nos*	**16**	*mm*	*dia*	+	**1**	*nos*	**12**	*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 515.221195 mm^2 Safe

ρ_w = 0.003563

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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 482 x 482 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 475.9 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 15.22 KNm

Mu_2 = Mu - Mu_{lim} = 15.22 - 475.9

= 0 KNm

A_{st} = $\frac{0.85 f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset f_s'(d - d')}$

= 0.85 x 35x 300 x 482 x (1-√(1-(2.35 x 15.3 x 10^6)/(0.9 x 35 x 300 x 482^2)))/420

= 0 x 10^6/(0.9 x 420 x (482 - 68))

= 83.8 mm^2

= 0 mm^2

e_{sc} = 0.0019

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 509.2 mm^2 or 482 mm^2 = 509.205438 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 509.3 + 0 + 0

= 509.3 mm^2

bottom layer1	2	nos	16	mm	dia	+	1	nos	12	mm	dia
bottom layer2		nos		mm	dia	+		nos		mm	dia

Ast provided = 515.2211952 mm^2 Safe

ρ_w = 0.003563079

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 139.3 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 145.6 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 248.1 KN

fV_c = 0.75 x 139 = 104.47 KN > 22.55 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((22.55-104.5) x 10^3)/(0.75 x 420 x 482) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	0 mm	*if Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	241 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

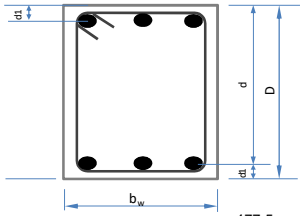
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.2 Beam marked TB2 & TB2A

Staad Beam Group: TB2 & TB2A

Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	477.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	477.5 mm
d'	=	72.5 mm
h	=	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	411	Staad beam number -	1506	=	**241**	KNm
B.M at span	Mz	L/C -	406	Staad beam number -	1506	=	**155**	KNm
Shear Force	Fy	L/C -	403	Staad beam number -	1506	=	**215**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 241.2 KNm

Mu_2 = Mu - Mu_{lim} = 241.2 - 467.1

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 241.2 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 1436.3 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0018

Asc = 0.0000 mm^2

= 504.5 mm^2 or 477.5 mm^2 = 504.5 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1436.4 + 0.00 + 0

= 1436.4 mm^2

Top layer1	3	*nos*	25	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1472.62156 mm^2 Safe

ρ_w = 0.01028

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 155.43 KNm

Mu_2 = Mu - Mu_{lim} = 155.43 - 467.1

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 155.5 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 900.0 mm^2

= 0 mm^2

e_{sc} = 0.0018

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 504.5 mm^2 or 477.5 mm^2 = 504.451446 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 900 + 0 + 0

= 900 mm^2

bottom layer1	3	nos	20	mm	dia	+		nos		mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.006579252

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 146.3 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 160.6 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 245.8 KN

fV_c = 0.75 x 146 = 109.69 KN < 215 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((215-109.7) x 10^3)/(0.75 x 420 x 477.5) = 0.700147589 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	224.3521724	mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737	mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	238.75	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0	mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

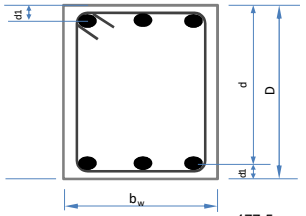
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.3 Beam marked TB3 & TB3A

Staad Beam Group TB3 & TB3A

Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	477.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	477.5 mm
d'	=	72.5 mm
h	=	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	411	Staad beam number -	1512	=	**236**	KNm
B.M for bottom reinforcement	Mz	L/C -	407	Staad beam number -	1510	=	**166**	KNm
Shear Force	Fy	L/C -	403	Staad beam number -	1517	=	**212**	KN
						=	**0**	KNm

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 236.42 KNm

Mu_2 = Mu - Mu_{lim} = 236.42 - 467.1

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 236.5 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 1405.6 mm^2

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0018

Asc = 0.0000 mm^2

= 504.5 mm^2 or 477.5 mm^2

= 504.5 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1405.6 + 0.00 + 0

= 1405.6 mm^2

Top layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1472.62156 mm^2 Safe

ρ_w = 0.01028

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 166.14 KNm

Mu_2 = Mu - Mu_{lim} = 166.14 - 467.1

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 166.2 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 965.2 mm^2

= 0 mm^2

e_{sc} = 0.0018

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 504.5 mm^2 or 477.5 mm^2 = 504.451446 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 965.3 + 0 + 0

= 965.3 mm^2

bottom layer1	2	nos	25	mm	dia	+	1	nos	12	mm	dia
bottom layer2		nos		mm	dia	+		nos		mm	dia

Ast provided = 1094.84504 mm^2 Safe

ρ_w = 0.007642897

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 146.3 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 160.6 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 245.8 KN

fV_c = 0.75 x 146 = 109.75 KN < 212.32 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((212.32-109.8) x 10^3)/(0.75 x 420 x 477.5) = 0.681931376 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	230.3452198	mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s =	599.5455737	mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	238.75	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0	mm	

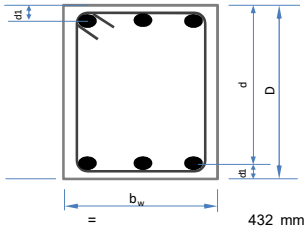
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023			Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL			Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

8.4.4 Beam marked TB4

Staad Beam Group TB4

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	432	mm	
Dia of longitudinal bar	=	**16**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	432	mm
d'	=	68	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	409	Staad beam number -	1532	=	**64**	KNm
B.M for bottom reinforcement	Mz	L/C -	409	Staad beam number -	1532	=	**51**	KNm
Shear Force	Fy	L/C -	409	Staad beam number -	1532	=	**37**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 63.77 KNm

Mu_2 = Mu - Mu_{lim} = 63.77 - 382.3

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 63.8 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 398.7 mm^2

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 456.4 mm^2 or 432 mm^2 = 456.4 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 456.4 + 0.00 + 0

= 456.4 mm^2

Top layer1	**2**	*nos*	**16**	*mm*	*dia*	+	**1**	*nos*	**12**	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 515.221195 mm^2 Safe

ρ_w = 0.003975

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c' \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 50.51 KNm

Mu_2 = Mu - Mu_{lim} = 50.51 - 382.3

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 50.6 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 314.3 mm^2

= 0 mm^2

e_{sc} = 0.0017

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 456.4 mm^2 or 432 mm^2 = 456.383298 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 456.4 + 0 + 0

= 456.4 mm^2

bottom layer1	2	nos	16	mm	dia	+	1	nos	12	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 515.2211952 mm^2 Safe

ρ_w = 0.003975472

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 124.9 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 131.4 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 222.3 KN

fV_c = 0.75 x 124 = 93.65 KN > 36.88 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((36.88-93.7) x 10^3)/(0.75 x 420 x 432) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	0 mm	*if Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	216 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

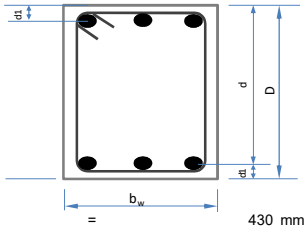
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023			Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL			Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

8.4.5 Beam marked TB5

Staad Beam Group TB5

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	430	mm	
Dia of longitudinal bar	=	**20**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	430	mm
d'	=	70	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	413	Staad beam number -	1553	=	**104**	KNm
B.M for bottom reinforcement	Mz	L/C -	413	Staad beam number -	1536	=	**79**	KNm
Shear Force	Fy	L/C -	413	Staad beam number -	1536	=	**51**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

Singly reinforced beam

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 104.45 KNm

Mu_2 = Mu - Mu_{lim} = 104.45 - 378.8 = 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 104.5 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 666.1 mm^2

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 454.3 mm^2 or 430 mm^2 = 454.3 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 666.1 + 0.00 + 0

= 666.1 mm^2

Top layer1	**2**	*nos*	**20**	*mm*	*dia*	+	**1**	*nos*	**12**	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 741.415866 mm^2 Safe

ρ_w = 0.005747

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01		
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 78.98 KNm

Mu_2 = Mu - Mu_{lim} = 78.98 - 378.8

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 79 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 498.9 mm²

= 0 mm²

e_{sc} = 0.0017

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm²

= 454.3 mm² or 430 mm² = 454.270412 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 499 + 0 + 0

= 499 mm²

bottom layer1	2	nos	16	mm	dia	+	1	nos	12	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 515.2211952 mm² Safe

ρ_w = 0.003993963

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 124.8 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 134.7 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 221.3 KN

fV_c = 0.75 x 124 = 93.57 KN > 51.02 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((51.02-93.6) x 10^3)/(0.75 x 420 x 430) = 0 mm²/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	0 mm	*if Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	215 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

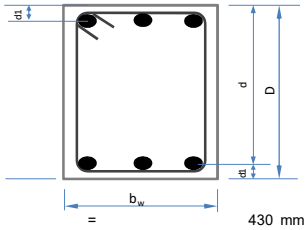
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023			Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL			Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

8.4.6 Beam marked TB5A

Staad Beam Group TB5A

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	430	mm	
Dia of longitudinal bar	=	**20**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	430	mm
d'	=	70	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	409	Staad beam number -	1539	=	**172**	KNm
B.M for bottom reinforcement	Mz	L/C -	409	Staad beam number -	1539	=	**173**	KNm
Shear Force	Fy	L/C -	409	Staad beam number -	1539	=	**228**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

Singly reinforced beam

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 172.25 KNm

Mu_2 = Mu - Mu_{lim} = 172.25 - 378.8

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 172.3 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 1128.0 mm^2

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 454.3 mm^2 or 430 mm^2 = 454.3 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1128.1 + 0.00 + 0

= 1128.1 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+	**2**	*nos*	**12**	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1168.67247 mm^2 Safe

ρ_w = 0.009059

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 173.17 KNm

Mu_2 = Mu - Mu_{lim} = 173.17 - 378.8

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 173.2 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 1134.5 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 0 mm^2

e_{sc} = 0.0017

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 454.3 mm^2 or 430 mm^2 = 454.270412 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1134.5 + 0 + 0

= 1134.5 mm^2

bottom layer1	3	nos	20	mm	dia	+	2	nos	12	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1168.672467 mm^2 Safe

ρ_w = 0.009059476

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 133.4 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 142.0 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 221.3 KN

fV_c = 0.75 x 133 = 100.07 KN < 228.28 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((228.28-100.1) x 10^3)/(0.75 x 420 x 430) = 0.946531337 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	165.9529131	mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c}\frac{b_w S}{f_t}$	s =	599.5455737	mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	215	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0	mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

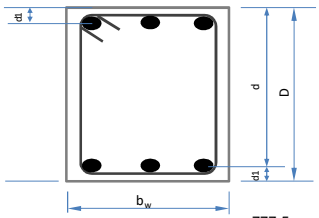
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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8.4.7 Beam marked 1FB1

Staad beam Group 1FB1

Width of beam	=	**350**	mm	
Depth of beam	=	**850**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	777.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	777.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	411	Staad beam number -	1804	=	**682**	KNm
B.M for bottom reinforcement	Mz	L/C -	423	Staad beam number -	1801	=	**338**	KNm
Shear Force	Fy	L/C -	411	Staad beam number -	1804	=	**264**	KN
						=	**0**	KNm

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 777.5 x 777.5 (1 - 0.8/2 x 0.375)

= 1444.6 KNm

$\frac{c}{d}_{max} = \frac{0.003}{0.003+0.005} = 0.375$

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 682.38 KNm

Mu_2 = Mu - Mu_{lim} = 682.38 - 1444.6

= 0.00 KNm

$A_{st1} = \frac{0.85\ f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c' bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 777.5 x (1-√(1-(2.35 x 682.4 x 10^6)/(0.9 x 35 x 350 x 777.5^2)))/420

= 2478.3 mm^2

= 0 x 10^6/(0.9 x 420 x (777.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0023

Asc = 0.0000 mm^2

= 958.3 mm^2 or 907.0833333 mm^2

= 958.3 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2478.3 + 0.00 + 0

= 2478.3 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 2767.743128 mm^2 Safe

ρ_w = 0.010171

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
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Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ — *For tension-controlled section & following the assuptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 777.5 x 777.5 (1 - 0.8/2 x 0.375)

= 1444.6 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 337.58 KNm

Mu_2 = Mu - Mu_{lim} = 337.58 - 1444.6

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 350 x 777.5 x (1-√(1-(2.35 x 337.6 x 10^6)/(0.9 x 35 x 350 x 777.5^2)))/420

= 1183.5 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (777.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0023

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 958.3 mm^2 or 907.0833333 mm^2 = 958.281673 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1183.6 + 0 + 0

= 1183.6 mm^2

bottom layer1	2	*nos*	25	*mm*	*dia*	+	2	*nos*	16	*mm* *dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1383.871564 mm^2 Safe

ρ_w = 0.005085426

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc — *Corresponding to table 22.5.5.1 of SBC 304*

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 271.8 KN — λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 304.6 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 466.9 KN

fV_c = 0.75 x 271. = 203.82 KN < 264.46 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((264.46-203.9) x 10^3)/(0.75 x 420 x 777.5) = 0.247586651 mm^2/mm

No. of legs = 4 Spacing provided = 150 mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement — s = 1268.886121 mm — *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ — s = 1027.792412 mm — *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ — s = 579.986336 mm — *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 — s = 388.75 mm — *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion — s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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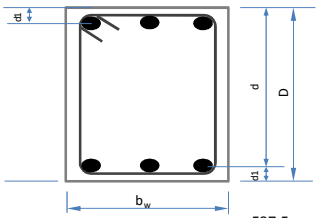
Ma'aden Revision No:	A01	A02	C01	D01
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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8.4.8 Beam marked 1FB2

Staad beam Group 1FB2

Width of beam	=	**350**	mm	
Depth of beam	=	**600**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	527.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	527.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	407	Staad beam number -	1806	=	**302**	KNm
B.M for bottom reinforcement	Mz	L/C -	407	Staad beam number -	1806	=	**239**	KNm
Shear Force	Fy	L/C -	407	Staad beam number -	1806	=	**120**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

Singly reinforced beam

$\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003+0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 302.03 KNm

Mu_2 = Mu - Mu_{lim} = 302.03 - 665

= 0.00 KNm

$A_{st1} = \frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 302.1 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1612.2 mm^2

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.4166667 mm^2

= 650.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1612.3 + 0.00 + 0

= 1612.3 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1963.495408 mm^2 Safe

ρ_w = 0.010635

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 239.17 KNm

Mu_2 = Mu - Mu_{lim} = 239.17 - 665

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 239.2 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1258.5 mm^2

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0019

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_wd$ or $\frac{1.4}{f_y}b_wd$

= 650.2 mm^2 or 615.4166667 mm^2 = 650.152518 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1258.6 + 0 + 0

= 1258.6 mm^2

bottom layer1	**2**	*nos*	**25**	*mm*	*dia*	+	**2**	*nos*	**16**	*mm* *dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1383.871564 mm^2 Safe

ρ_w = 0.007495581

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_wd$ = 181.7 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_wd$ = 208.1 KN

V_c = $0.29\lambda\sqrt{f_c'}b_wd$ = 316.8 KN

fV_c = 0.75 x 181 = 136.31 KN > 119.87 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U-\emptyset V_C}{\emptyset\ f_y d}$ = ((119.87-136.4) x 10^3)/(0.75 x 420 x 527.5) = 0 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	0 mm	*if Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	1027.792412 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	579.986336 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	263.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
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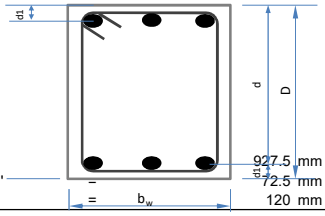
Ma'aden Revision No:	A01	A02	C01	D01
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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8.4.9 Beam marked 1FB3

Staad beam Group 1FB3

Width of beam	=	**350**	mm	
Depth of beam	=	**1000**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	927.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	927.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	411	Staad beam number -	1811	=	**1104**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	1824	=	**665**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	1811	=	**558**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 927.5 x 927.5 (1 - 0.8/2 x 0.375)

= 2055.7 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 1103.97 KNm

Mu_2 = Mu - Mu_{lim} = 1103.97 - 2055.8

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 350 x 927.5 x (1-√(1-(2.35 x 1104 x 10^6)/(0.9 x 35 x 350 x 927.5^2)))/420

= 3395.6 mm^2

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'\,(d - d')}$

= 0 x 10^6/(0.9 x 420 x (927.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1143.2 mm^2 or 1082.083333 mm^2 = 1143.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3395.7 + 0.00 + 0

= 3395.7 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 3926.990817 mm^2 Safe

ρ_w = 0.012097

Global Project ID	900707
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Ma'aden Revision No:	A01	A02	C01	D01
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Outotec / L&T Revision No.:	0	1	2	3
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 927.5 x 927.5 (1 - 0.8/2 x 0.375)

= 2055.7 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 664.78 KNm

Mu_2 = Mu - Mu_{lim} = 664.78 - 2055.8

= 0 KNm

A_{st} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 927.5 x (1-√(1-(2.35 x 664.8 x 10^6)/(0.9 x 35 x 350 x 927.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (927.5 - 72.5))

= 1978.9 mm^2

= 0 mm^2

e_{sc} = 0.0024

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1143.2 mm^2 or 1082.083333 mm^2 = 1143.15917 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1979 + 0 + 0

= 1979 mm^2

bottom layer1	4	*nos*	25	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1963.495408 mm^2 Unsafe

ρ_w = 0.006048503

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 338.6 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 374.0 KN

V_c = $0.29\lambda\sqrt{f_c'}\, b_w d$ = 556.9 KN

fV_c = 0.75 x 338 = 253.94 KN < 558.11 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((558.11-254) x 10^3)/(0.75 x 420 x 927.5) = 1.041105207 mm^2/mm

No. of legs = 4 Spacing provided = 150 mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 301.7555413 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 1027.792412 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 579.986336 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 463.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

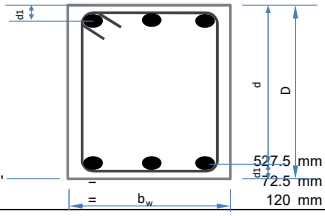
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.10 Beam marked 1FB4

Staad beam Group 1FB4

Width of beam	=	**350**	mm	
Depth of beam	=	**600**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	527.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	527.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	407	Staad beam number -	1813	=	**331**	KNm
B.M for bottom reinforcement	Mz	L/C -	407	Staad beam number -	1813	=	**284**	KNm
Shear Force	Fy	L/C -	407	Staad beam number -	1813	=	**120**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 331.13 KNm

Mu_2 = Mu - Mu_{lim} = 331.13 - 665

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'\,(d - d')}$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 331.2 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1779.7 mm^2

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.4166667 mm^2

= 650.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1779.7 + 0.00 + 0

= 1779.7 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1963.495408 mm^2 Safe

ρ_w = 0.010635

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 283.67 KNm

Mu_2 = Mu - Mu_{lim} = 283.67 - 665

= 0 KNm

A_{st} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 283.7 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 1507.8 mm^2

= 0 mm^2

e_{sc} = 0.0019

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.4166667 mm^2 = 650.152518 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1507.8 + 0 + 0

= 1507.8 mm^2

bottom layer1	**2**	*nos*	**25**	*mm*	*dia*	+	**2**	*nos*	**20**	*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1610.066235 mm^2 Safe

ρ_w = 0.008720738

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 181.2 KN

λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 208.1 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 316.8 KN

fV_c = 0.75 x 181 = 135.87 KN > 120.28 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((120.28-135.9) x 10^3)/(0.75 x 420 x 527.5) = 0 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 0 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 1027.792412 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 579.986336 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 263.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

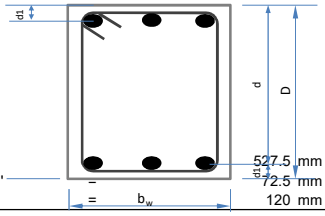
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.11 Beam marked 1FB5

Staad beam Group 1FB5

Width of beam	=	**350**	mm	
Depth of beam	=	**600**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	527.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	527.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	407	Staad beam number -	1821	=	**316**	KNm
B.M for bottom reinforcement	Mz	L/C -	407	Staad beam number -	1821	=	**269**	KNm
Shear Force	Fy	L/C -	407	Staad beam number -	1814	=	**186**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

Singly reinforced beam

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 316.17 KNm

Mu_2 = Mu - Mu_{lim} = 316.17 - 665

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 316.2 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1693.3 mm^2

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'\,(d - d')}$

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.4166667 mm^2 = 650.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1693.3 + 0.00 + 0

= 1693.3 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1963.495408 mm^2 Safe

ρ_w = 0.010635

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 269.23 KNm

Mu_2 = Mu - Mu_{lim} = 269.23 - 665

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 269.3 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1426.3 mm^2

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0019

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_wd$ or $\frac{1.4}{f_y}b_wd$

= 650.2 mm^2 or 615.4166667 mm^2 = 650.152518 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1426.4 + 0 + 0

= 1426.4 mm^2

bottom layer1	2	nos	25	mm	dia	+	2	nos	20	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1610.066235 mm^2 Safe

ρ_w = 0.008720738

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_wd$ = 185.1 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_wd$ = 208.1 KN

V_c = $0.29\lambda\sqrt{f_c'}b_wd$ = 316.8 KN

fV_c = 0.75 x 185 = 138.86 KN < 186.43 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((186.43-138.9) x 10^3)/(0.75 x 420 x 527.5) = 0.286300919 mm^2/mm

No. of legs = 4 Spacing provided = 150 mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 1097.304425 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$ s = 1027.792412 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 579.986336 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 263.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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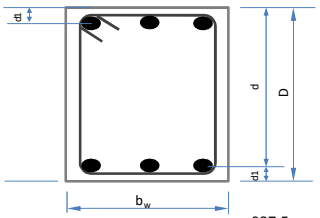
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.12 Beam marked 1FB6

Staad beam Group 1FB6

Width of beam	=	**350**	mm	
Depth of beam	=	**1000**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	927.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	927.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	411	Staad beam number -	1819	=	**932**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	1817	=	**774**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	1819	=	**579**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 927.5 x 927.5 (1 - 0.8/2 x 0.375)

= 2055.7 KNm

Singly reinforced beam

$\left(\frac{c}{d}\right)_{max}$ = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 932.41 KNm

Mu_2 = Mu - Mu_{lim} = 932.41 - 2055.8

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 927.5 x (1-√(1-(2.35 x 932.5 x 10^6)/(0.9 x 35 x 350 x 927.5^2)))/420

= 2830.4 mm^2

= 0 x 10^6/(0.9 x 420 x (927.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1143.2 mm^2 or 1082.083333 mm^2

= 1143.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2830.4 + 0.00 + 0

= 2830.4 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 3220.13247 mm^2 Safe

ρ_w = 0.00992

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 927.5 x 927.5 (1 - 0.8/2 x 0.375)

= 2055.7 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 773.59 KNm

Mu_2 = Mu - Mu_{lim} = 773.59 - 2055.8

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 927.5 x (1-√(1-(2.35 x 773.6 x 10^6)/(0.9 x 35 x 350 x 927.5^2)))/420

= 2320.9 mm^2

= 0 x 10^6/(0.9 x 420 x (927.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0024

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 1143.2 mm^2 or 1082.083333 mm^2 = 1143.15917 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2320.9 + 0 + 0

= 2320.9 mm^2

bottom layer1	**2**	*nos*	**25**	*mm*	*dia*	+	**2**	*nos*	**20**	*mm* *dia*
bottom layer2	**2**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 2591.813939 mm^2 Safe

ρ_w = 0.007984024

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 338.8 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 362.0 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 556.9 KN

fV_c = 0.75 x 338 = 254.09 KN < 578.67 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((578.67-254.1) x 10^3)/(0.75 x 420 x 927.5) = 1.110943215 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement — s = 282.7860696 mm — *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ — s = 1027.792412 mm — *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ — s = 579.986336 mm — *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 — s = 463.75 mm — *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion — s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

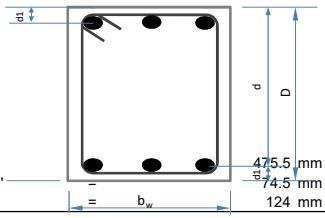
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.13 Beam marked 1FB7

Staad beam Group 1FB7

Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	475.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**12**	mm	



d	=	475.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	414	Staad beam number -	1663	=	**268**	KNm
B.M for bottom reinforcement	Mz	L/C -	409	Staad beam number -	1667	=	**251**	KNm
Shear Force	Fy	L/C -	413	Staad beam number -	1661	=	**301**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 475.5 x 475.5 (1 - 0.8/2 x 0.375)

= 463.1 KNm

$\frac{c}{d}max = \frac{0.003}{0.003 + 0.005} = 0.375$

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 268.15 KNm

Mu_2 = Mu - Mu_{lim} = 268.15 - 463.2

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'\,(d - d')}$

= 0.85 x 35x 300 x 475.5 x (1-√(1-(2.35 x 268.2 x 10^6)/(0.9 x 35 x 300 x 475.5^2)))/420

= 1619.9 mm^2

= 0 x 10^6/(0.9 x 420 x (475.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 502.3 mm^2 or 475.5 mm^2 = 502.3 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1619.9 + 0.00 + 0

= 1619.9 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+	**3**	*nos*	**20**	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1884.955592 mm^2 Safe

ρ_w = 0.013214

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 475.5 x 475.5 (1 - 0.8/2 x 0.375)

= 463.1 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 250.82 KNm

Mu_2 = Mu - Mu_{lim} = 250.82 - 463.2

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 300 x 475.5 x (1-√(1-(2.35 x 250.9 x 10^6)/(0.9 x 35 x 300 x 475.5^2)))/420

= 1505.9 mm^2

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (475.5 - 74.5))

= 0 mm^2

e_{sc} = 0.0017

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

= 502.3 mm^2 or 475.5 mm^2 = 502.33856 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1506 + 0 + 0

= 1506 mm^2

bottom layer1	3	nos	20	mm	dia	+	3	nos	16	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1545.663586 mm^2 Safe

ρ_w = 0.010835356

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 152.1 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 167.1 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 244.7 KN

fV_c = 0.75 x 152 = 114.10 KN < 301.02 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((301.02-114.2) x 10^3)/(0.75 x 420 x 475.5) = 1.247942469 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 181.2540856 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 863.3456261 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 487.1885223 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 237.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

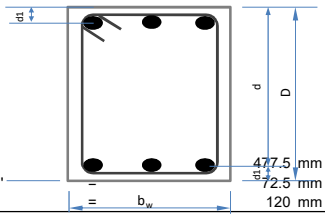
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.14 Beam marked 1FB8

Staad beam Group 1FB8

Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	477.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	477.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	409	Staad beam number -	1651	=	**94**	KNm
B.M for bottom reinforcement	Mz	L/C -	409	Staad beam number -	1652	=	**58**	KNm
Shear Force	Fy	L/C -	403	Staad beam number -	1651	=	**73**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 94.15 KNm

Mu_2 = Mu - Mu_{lim} = 94.15 - 467.1

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 94.2 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 535.1 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0018

Asc = 0.0000 mm^2

= 504.5 mm^2 or 477.5 mm^2 = 504.5 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 535.1 + 0.00 + 0

= 535.1 mm^2

Top layer1	3	*nos*	16	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004211

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 58.28 KNm

Mu_2 = Mu - Mu_{lim} = 58.28 - 467.1

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 58.3 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 327.8 mm^2

= 0 mm^2

e_{sc} = 0.0018

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 504.5 mm^2 or 477.5 mm^2 = 504.451446 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 504.5 + 0 + 0

= 504.5 mm^2

bottom layer1	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004210721

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 139.4 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 145.9 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 245.8 KN

fV_c = 0.75 x 139 = 104.55 KN > 73.19 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((73.19-104.6) x 10^3)/(0.75 x 420 x 477.5) = 0 mm^2/mm

No. of legs = **2** Spacing provided = **150** mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 0 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$ s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 238.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

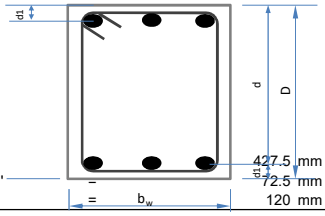
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.15 Beam marked 1B1

Staad beam Group 1B1

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	427.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	401	Staad beam number -	1695	=	**0**	KNm
B.M for bottom reinforcement	Mz	L/C -	404	Staad beam number -	1699	=	**84**	KNm
Shear Force	Fy	L/C -	404	Staad beam number -	1699	=	**62**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 0 KNm

Mu_2 = Mu - Mu_{lim} = 0 - 374.4

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 0 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 0.0 mm^2

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'\,(d - d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0016

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 451.7 + 0.00 + 0

= 451.7 mm^2

Top layer1	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004703

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 84.25 KNm

Mu_2 = Mu - Mu_{lim} = 84.25 - 374.4

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 84.3 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 536.6 mm^2

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0016

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.629305 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 536.6 + 0 + 0

= 536.6 mm^2

bottom layer1	3	*nos*	16	*mm*	*dia*	+		*nos*		*mm* *dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004703203

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = #DIV/0! KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 131.7 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 220.0 KN

fV_c = 0.75 x 131 = 98.74 KN > 62.17 KN *Shear Reinforcement not required*

Vs = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = $\frac{\emptyset\ A_V f_y d}{s}$

= ((62.17-98.8) x 10^3)/(0.75 x 420 x 427.5) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

$\frac{A_V}{S}$

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 0 mm *if Vu > φVc*

ii) Min Asv = s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 213.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion $0.65\frac{b_w s}{f_{yt}}$ s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

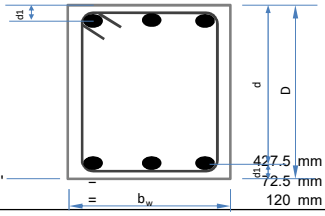
Ma'aden Revision No:	A01	A02	C01	D01
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.16 Beam marked 1B2

Staad beam Group 1B2

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	427.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	401	Staad beam number -	1675	=	**0**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	1677	=	**73**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	1676	=	**78**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 0 KNm

Mu_2 = Mu - Mu_{lim} = 0 - 374.4 = 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 0 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 0.0 mm²

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm²

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0016

Asc = 0.0000 mm²

= 451.6 mm² or 427.5 mm²

= 451.6 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 451.7 + 0.00 + 0

= 451.7 mm²

Top layer1	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 603.1857895 mm² Safe

ρ_w = 0.004703

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 72.68 KNm

Mu_2 = Mu - Mu_{lim} = 72.68 - 374.4

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 72.7 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 460.9 mm^2

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0016

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.629305 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 460.9 + 0 + 0

= 460.9 mm^2

bottom layer1	3	nos	16	mm	dia	+		nos		mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004703203

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = #DIV/0! KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 131.7 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 220.0 KN

fV_c = 0.75 x 131 = 98.74 KN > 78.28 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((78.28-98.8) x 10^3)/(0.75 x 420 x 427.5) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 0 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 213.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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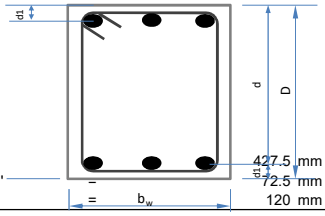
Ma'aden Revision No:	A01	A02	C01	D01
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.17 Beam marked 1B3

Staad beam Group 1B3

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	427.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	406	Staad beam number -	1682	=	**182**	KNm
B.M for bottom reinforcement	Mz	L/C -	403	Staad beam number -	1682	=	**93**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	1682	=	**152**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 182.35 KNm

Mu_2 = Mu - Mu_{lim} = 182.35 - 374.4

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 182.4 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 1207.2 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'\,(d - d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0016

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1207.3 + 0.00 + 0

= 1207.3 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**3**	*nos*	**12**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1281.769803 mm^2 Safe

ρ_w = 0.009994

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 93.32 KNm

Mu_2 = Mu - Mu_{lim} = 93.32 - 374.4

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 93.4 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 596.3 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0016

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

= 451.6 mm^2 or 427.5 mm^2 = 451.629305 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 596.4 + 0 + 0

= 596.4 mm^2

bottom layer1	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004703203

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 129.2 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 143.2 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 220.0 KN

fV_c = 0.75 x 129 = 96.88 KN < 152.23 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((152.23-96.9) x 10^3)/(0.75 x 420 x 427.5) = 0.411020773 mm^2/mm

No. of legs = **2** Spacing provided = **150** mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 382.1695711 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 213.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

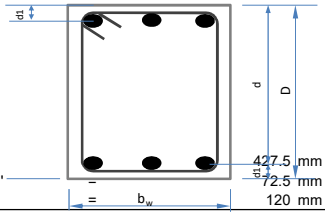
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

8.4.18 Beam marked 1B4

Staad beam Group 1B4

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	427.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	405	Staad beam number -	1696	=	**210**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	1696	=	**157**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	1693	=	**153**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 209.52 KNm

Mu_2 = Mu - Mu_{lim} = 209.52 - 374.4

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 209.6 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 1403.3 mm^2

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0016

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1403.4 + 0.00 + 0

= 1403.4 mm^2

Top layer1	**2**	nos	**25**	mm	dia	+	**1**	nos	**20**	mm dia
Top layer2	**2**	nos	**16**	mm	dia	+		nos		mm dia

Ast provided = 1698.030829 mm^2 Safe

ρ_w = 0.01324

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 156.92 KNm

Mu_2 = Mu - Mu_{lim} = 156.92 - 374.4

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 157 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 1028.0 mm^2

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0016

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

= 451.6 mm^2 or 427.5 mm^2 = 451.629305 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1028.1 + 0 + 0

= 1028.1 mm^2

bottom layer1	3	nos	20	mm	dia	+		nos		mm dia
bottom layer2	2	nos	12	mm	dia	+		nos		mm dia

Ast provided = 1168.672467 mm^2 Safe

ρ_w = 0.009112456

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 130.4 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 150.3 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 220.0 KN

fV_c = 0.75 x 130 = 97.79 KN < 152.71 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((152.71-97.8) x 10^3)/(0.75 x 420 x 427.5) = 0.407802661 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 385.1854039 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 213.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

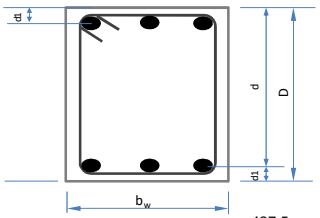
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.19 Beam marked 1B5

Staad beam Group 1B5

Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	427.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	403	Staad beam number -	1680	=	**109**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	1680	=	**75**	KNm
Shear Force	Fy	L/C -	403	Staad beam number -	1680	=	**68**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

$\left(\frac{c}{d}\right)_{max}$ = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 109.25 KNm

Mu_2 = Mu - Mu_{lim} = 109.25 - 374.4

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 109.3 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 702.4 mm^2

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0016

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 702.4 + 0.00 + 0

= 702.4 mm^2

Top layer1	**3**	nos	**16**	mm	dia	+	**2**	nos	**12**	mm dia
Top layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 829.3804605 mm^2 Safe

ρ_w = 0.006467

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Outotec / L&T Revision No.:	0	1	2	3
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

= 374.3 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 74.63 KNm

Mu_2 = Mu - Mu_{lim} = 74.63 - 374.4

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 74.7 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 473.6 mm^2

= 0 mm^2

e_{sc} = 0.0016

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.629305 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 473.7 + 0 + 0

= 473.7 mm^2

bottom layer1	3	nos	16	mm	dia	+		nos		mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004703203

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 125.2 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 135.5 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 220.0 KN

fV_c = 0.75 x 125 = 93.88 KN > 68.43 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((68.43-93.9) x 10^3)/(0.75 x 420 x 427.5) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s =	0 mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s =	213.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s =	0 mm	

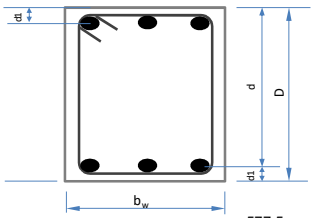
Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.20 Beam marked 2FB1

Staad beam Group 2FB1



Width of beam	=	**350**	mm	
Depth of beam	=	**650**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	577.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	577.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	414	Staad beam number -	2104	=	**277**	KNm
B.M for bottom reinforcement	Mz	L/C -	413	Staad beam number -	2104	=	**162**	KNm
Shear Force	Fy	L/C -	402	Staad beam number -	2104	=	**283**	KN
						=	**0**	KNm

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 577.5 x 577.5 (1 - 0.8/2 x 0.375)

= 797.0 KNm

Singly reinforced beam

$\left(\frac{c}{d}\right)_{max}$ = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 277.33 KNm

Mu_2 = Mu - Mu_{lim} = 277.33 - 797

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 350 x 577.5 x (1-√(1-(2.35 x 277.4 x 10^6)/(0.9 x 35 x 350 x 577.5^2)))/420

= 1330.7 mm^2

= 0 x 10^6/(0.9 x 420 x (577.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0020

Asc = 0.0000 mm^2

= 711.8 mm^2 or 673.75 mm^2 = 711.8 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1330.7 + 0.00 + 0

= 1330.7 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1545.663586 mm^2 Safe

ρ_w = 0.007647

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 577.5 x 577.5 (1 - 0.8/2 x 0.375)

= 797.0 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 161.94 KNm

Mu_2 = Mu - Mu_{lim} = 161.94 - 797

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 350 x 577.5 x (1-√(1-(2.35 x 162 x 10^6)/(0.9 x 35 x 350 x 577.5^2)))/420

= 761.1 mm^2

= 0 x 10^6/(0.9 x 420 x (577.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0020

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 711.8 mm^2 or 673.75 mm^2 = 711.778349 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 761.2 + 0 + 0

= 761.2 mm^2

bottom layer1	3	*nos*	20	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.004662846

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 206.8 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 217.6 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 346.8 KN

fV_c = 0.75 x 20(= 155.11 KN < 283.08 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((283.08-155.2) x 10^3)/(0.75 x 420 x 577.5) = 0.703462408 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s	=	223.2949917 mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s	=	513.896206 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s	=	289.993168 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	288.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

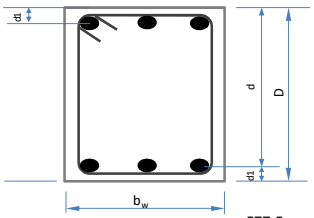
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.21 Beam marked 2FB2

Staad beam Group 2FB2



Width of beam	=	**350**	mm	
Depth of beam	=	**650**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	577.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	577.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	414	Staad beam number -	1725	=	**363**	KNm
B.M for bottom reinforcement	Mz	L/C -	413	Staad beam number -	1726	=	**217**	KNm
Shear Force	Fy	L/C -	414	Staad beam number -	1725	=	**235**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 577.5 x 577.5 (1 - 0.8/2 x 0.375)

= 797.0 KNm

Singly reinforced beam

β_{max} = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 362.64 KNm

Mu_2 = Mu - Mu_{lim} = 362.64 - 797

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 577.5 x (1-√(1-(2.35 x 362.7 x 10^6)/(0.9 x 35 x 350 x 577.5^2)))/420

= 1768.4 mm^2

= 0 x 10^6/(0.9 x 420 x (577.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0020

Asc = 0.0000 mm^2

= 711.8 mm^2 or 673.75 mm^2 = 711.8 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1768.4 + 0.00 + 0

= 1768.4 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1884.955592 mm^2 Safe

ρ_w = 0.009326

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK / APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 577.5 x 577.5 (1 - 0.8/2 x 0.375)

= 797.0 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 217.04 KNm

Mu_2 = Mu - Mu_{lim} = 217.04 - 797

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 350 x 577.5 x (1-√(1-(2.35 x 217.1 x 10^6)/(0.9 x 35 x 350 x 577.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (577.5 - 72.5))

= 1030.1 mm^2

= 0 mm^2

e_{sc} = 0.0020

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 711.8 mm^2 or 673.75 mm^2 = 711.778349 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1030.1 + 0 + 0

= 1030.1 mm^2

bottom layer1	2	nos	25	mm	dia	+	1	nos	16	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1182.809634 mm^2 Safe

ρ_w = 0.005851872

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 203.3 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 223.4 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 346.8 KN

fV_c = 0.75 x 203 = 152.49 KN < 235.11 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((235.11-152.5) x 10^3)/(0.75 x 420 x 577.5) = 0.454159124 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	345.8691554	mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s =	513.896206	mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	289.993168	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	288.75	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0	mm	

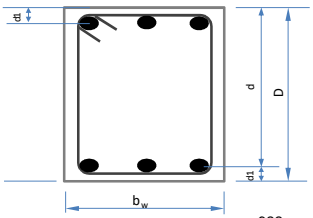
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.22 Beam marked 2FB3

Staad beam Group 2FB3



Width of beam	=	**350**	mm	
Depth of beam	=	**1000**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	922	mm	
Dia of longitudinal bar	=	**32**	mm	
Dia of stirrup bar	=	**12**	mm	

d	=	922	mm
d'	=	78	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	413	Staad beam number -	1712	=	**990**	KNm
B.M for bottom reinforcement	Mz	L/C -	402	Staad beam number -	1710	=	**568**	KNm
Shear Force	Fy	L/C -	402	Staad beam number -	1712	=	**838**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 922 x 922 (1 - 0.8/2 x 0.375)

= 2031.4 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003+0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 989.82 KNm

Mu_2 = Mu - Mu_{lim} = 989.82 - 2031.5 = 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 350 x 922 x (1-√(1-(2.35 x 989.9 x 10^6)/(0.9 x 35 x 350 x 922^2)))/420

= 3038.5 mm^2

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (922 - 78))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 1136.4 mm^2 or 1075.666667 mm^2

e_{sc} = 0.0023

Asc = 0.0000 mm^2

= 1136.4 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3038.6 + 0.00 + 0

= 3038.6 mm^2

Top layer1	**3**	*nos*	**32**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 3355.220954 mm^2 Safe

ρ_w = 0.010397

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 922 x 922 (1 - 0.8/2 x 0.375)

= 2031.4 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 568.07 KNm

Mu_2 = Mu - Mu_{lim} = 568.07 - 2031.5

= 0 KNm

A_{st} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 922 x (1-√(1-(2.35 x 568.1 x 10^6)/(0.9 x 35 x 350 x 922^2)))/420

= 0 x 10^6/(0.9 x 420 x (922 - 78))

= 1690.4 mm^2

= 0 mm^2

e_{sc} = 0.0023

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1136.4 mm^2 or 1075.666667 mm^2 = 1136.38033 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1690.5 + 0 + 0

= 1690.5 mm^2

bottom layer1	3	nos	25	mm	dia	+		nos		mm	dia
bottom layer2	3	nos	16	mm	dia	+		nos		mm	dia

Ast provided = 2075.807346 mm^2 Safe

ρ_w = 0.006432623

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 350.0 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 362.5 KN

$V_c = 0.29\lambda\sqrt{f_c'}\, b_w d$ = 553.6 KN

fV_c = 0.75 x 350 = 262.50 KN < 838.27 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S} = \frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((838.27-262.5) x 10^3)/(0.75 x 420 x 922) = 1.98248247 mm^2/mm

No. of legs = 2 Spacing provided = 125 mm

Provide 2L -12 mm stirrups @ 125 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s	=	114.0966816 mm	*if Vu > φVc*
ii) Min Asv or	= $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s	=	740.0105367 mm	*Corresponding to table 9.6.3.3 of SBC 304*
Min Asv	= $0.65\frac{b_w S}{f_{yt}}$	s	=	417.590162 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	461 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

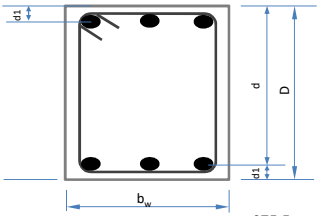
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.23 Beam marked 3FB1 Staad beam Group 3FB1



Width of beam	=	**300**	mm		
Depth of beam	=	**1050**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	975.5	mm		
Dia of longitudinal bar	=	**25**	mm	d	= 975.5 mm
Dia of stirrup bar	=	**12**	mm	d'	= 74.5 mm
				h	= 124 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	411	Staad beam number -	2205	=	**837**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2203	=	**477**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	2205	=	**485**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 975.5 x 975.5 (1 - 0.8/2 x 0.375)
= 1949.2 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 836.68 KNm

Mu_2 = Mu - Mu_{lim} = 836.68 - 1949.2
= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 975.5 x (1-√(1-(2.35 x 836.7 x 10^6)/(0.9 x 35 x 300 x 975.5^2)))/420
= 2405.8 mm^2

= 0 x 10^6/(0.9 x 420 x (975.5 - 74.5))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1030.6 mm^2 or 975.5 mm^2 = 1030.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 2405.8 + 0.00 + 0
= 2405.8 mm^2

Top layer1	3	nos	25	mm	dia	+		nos		mm dia
Top layer2	3	nos	25	mm	dia	+		nos		mm dia

Ast provided = 2945.243113 mm^2 Safe

ρ_w = 0.010064

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 975.5 x 975.5 (1 - 0.8/2 x 0.375)

= 1949.2 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 477.12 KNm

Mu_2 = Mu - Mu_{lim} = 477.12 - 1949.2

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 300 x 975.5 x (1-√(1-(2.35 x 477.2 x 10^6)/(0.9 x 35 x 300 x 975.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (975.5 - 74.5))

= 1335.3 mm^2

= 0 mm^2

e_{sc} = 0.0024

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1030.6 mm^2 or 975.5 mm^2 = 1030.56 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1335.4 + 0 + 0

= 1335.4 mm^2

bottom layer1	3	nos		20	mm	dia	+		nos		mm dia
bottom layer2	3	nos		20	mm	dia	+		nos		mm dia

Ast provided = 1884.955592 mm^2 Safe

ρ_w = 0.00644099

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 305.3 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 327.1 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 502.1 KN

fV_c = 0.75 x 305. = 228.98 KN < 484.68 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((484.68-229) x 10^3)/(0.75 x 420 x 975.5) = 0.83212898 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s	=	271.8264553 mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s	=	863.3456261 mm	*Corresponding to table 9.6.3.3 of SBC 304*
	s	=	487.1885223 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s	=	487.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s	=	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

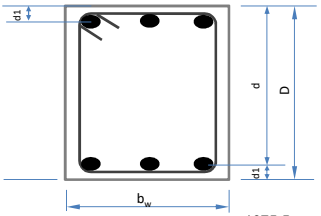
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.24 Beam marked 3FB2 Staad beam Group 3FB2



Width of beam	=	**400**	mm	
Depth of beam	=	**1150**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	1075.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**12**	mm	

d	=	1075.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	406	Staad beam number -	2237	=	**1382**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2236	=	**848**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	2233	=	**904**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 1075.5 x 1075.5 (1 - 0.8/2 x 0.375)

= 3159.0 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 1382.29 KNm

Mu_2 = Mu - Mu_{lim} = 1382.29 - 3159.1

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 400 x 1075.5 x (1-√(1-(2.35 x 1382.3 x 10^6)/(0.9 x 35 x 400 x 1075.5^2)))/420

= 3609.7 mm^2

= 0 x 10^6/(0.9 x 420 x (1075.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1514.9 mm^2 or 1434 mm^2 = 1514.9 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3609.7 + 0.00 + 0

= 3609.7 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3926.990817 mm^2 Safe

ρ_w = 0.009128

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 1075.5 x 1075.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

= 3159.0 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 847.84 KNm

Mu_2 = Mu - Mu_{lim} = 847.84 - 3159.1

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 400 x 1075.5 x (1-√(1-(2.35 x 847.9 x 10^6)/(0.9 x 35 x 400 x 1075.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (1075.5 - 74.5))

= 2159.4 mm^2

= 0 mm^2

e_{sc} = 0.0024

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 1514.9 mm^2 or 1434 mm^2 = 1514.939 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2159.5 + 0 + 0

= 2159.5 mm^2

bottom layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3220.13247 mm^2 Safe

ρ_w = 0.007485199

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 454.2 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 474.0 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 738.1 KN

fV_c = 0.75 x 454. = 340.64 KN < 904.41 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((904.41-340.7) x 10^3)/(0.75 x 420 x 1075.5) = 1.664092852 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s	=	271.8534255 mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s	=	1295.018439 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s	=	730.7827834 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	537.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

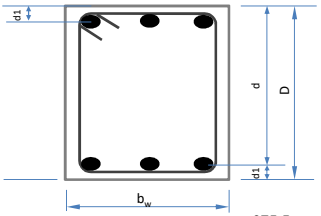
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.25 Beam marked 3FB2A Staad beam Group 3FB2A



Width of beam	=	**400**	mm		
Depth of beam	=	**1050**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	975.5	mm		
Dia of longitudinal bar	=	**25**	mm	D01	
Dia of stirrup bar	=	**12**	mm	16-10-2023	

d	=	975.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS — AS BUILT

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	411	Staad beam number -	RPK / APL	2223	=	**914**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	SBY	2221	=	**628**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	AG	2223	=	**613**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 975.5 x 975.5 (1 - 0.8/2 x 0.375)

= 2598.9 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 913.79 KNm

Mu_2 = Mu - Mu_{lim} = 913.79 - 2598.9 = 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 400 x 975.5 x (1-√(1-(2.35 x 913.8 x 10^6)/(0.9 x 35 x 400 x 975.5^2)))/420

= 2597.1 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (975.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1374.1 mm^2 or 1300.666667 mm^2 = 1374.1 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2597.1 + 0.00 + 0

= 2597.1 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 3220.13247 mm^2 Safe

ρ_w = 0.008253

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 975.5 x 975.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

= 2598.9 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 628.11 KNm

Mu_2 = Mu - Mu_{lim} = 628.11 - 2598.9

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 400 x 975.5 x (1-√(1-(2.35 x 628.2 x 10^6)/(0.9 x 35 x 400 x 975.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (975.5 - 74.5))

= 1757.1 mm^2

= 0 mm^2

e_{sc} = 0.0024

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1374.1 mm^2 or 1300.666667 mm^2 = 1374.08 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1757.2 + 0 + 0

= 1757.2 mm^2

bottom layer1	**4**	*nos*		**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**4**	*nos*		**16**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 2767.743128 mm^2 Safe

ρ_w = 0.00709314

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 405.1 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 424.1 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 669.5 KN

fV_c = 0.75 x 405. = 303.86 KN < 612.52 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((612.52-303.9) x 10^3)/(0.75 x 420 x 975.5) = 1.004480051 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s	=	450.3716541 mm	*if Vu > φVc*
ii) Min Asv or Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or = $0.65\frac{b_w s}{f_{yt}}$	s	=	1295.018439 mm	*Corresponding to table 9.6.3.3 of SBC 304*
		s	=	730.7827834 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	487.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

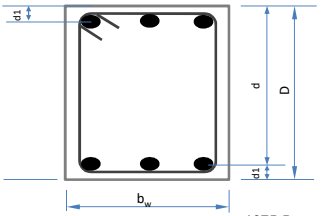
Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.26 Beam marked 3FB2B Staad beam Group 3FB2B

Width of beam	=	**400**	mm	
Depth of beam	=	**1150**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	1075.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**12**	mm	



d	=	1075.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	406	Staad beam number -	1980	=	**1534**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	1982	=	**1390**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	1980	=	**968**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 1075.5 x 1075.5 (1 - 0.8/2 x 0.375)

= 3159.0 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 1534.35 KNm

Mu_2 = Mu - Mu_{lim} = 1534.35 - 3159.1 = 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 400 x 1075.5 x (1-√(1-(2.35 x 1534.4 x 10^6)/(0.9 x 35 x 400 x 1075.5^2)))/420

= 4036.9 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (1075.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1514.9 mm^2 or 1434 mm^2 = 1514.9 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 4036.9 + 0.00 + 0

= 4036.9 mm^2

Top layer1	**4**	*nos*	**32**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 5180.486286 mm^2 Safe

ρ_w = 0.012042

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 1075.5 x 1075.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 3159.0 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 1389.74 KNm

Mu_2 = Mu - Mu_{lim} = 1389.74 - 3159.1

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 400 x 1075.5 x (1-√(1-(2.35 x 1389.8 x 10^6)/(0.9 x 35 x 400 x 1075.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (1075.5 - 74.5))

= 3630.5 mm^2

= 0 mm^2

e_{sc} = 0.0024

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1514.9 mm^2 or 1434 mm^2 = 1514.939 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3630.5 + 0 + 0

= 3630.5 mm^2

bottom layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3926.990817 mm^2 Safe

ρ_w = 0.009128291

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 467.0 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 495.3 KN

$V_c = 0.29\lambda\sqrt{f_c'} b_w d$ = 738.1 KN

fV_c = 0.75 x 467 = 350.22 KN < 967.79 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S} = \frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((967.79-350.3) x 10^3)/(0.75 x 420 x 1075.5) = 1.822913214 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	248.1683378 mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	1295.018439 mm	*Corresponding to table 9.6.3.3 of SBC 304*
	s =	730.7827834 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	537.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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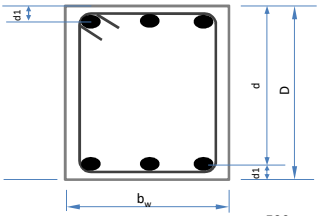
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.27 Beam marked 3FB3 Staad beam Group 3FB3



Width of beam	=	**350**	mm	
Depth of beam	=	**650**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	580	mm	
Dia of longitudinal bar	=	**20**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	580	mm
d'	=	70	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	413	Staad beam number -	2174	=	**149**	KNm
B.M for bottom reinforcement	Mz	L/C -	413	Staad beam number -	2174	=	**128**	KNm
Shear Force	Fy	L/C -	413	Staad beam number -	2174	=	**102**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 580 x 580 (1 - 0.8/2 x 0.375)
= 803.9 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 149.28 KNm

Mu_2 = Mu - Mu_{lim} = 149.28 - 803.9
= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 580 x (1-√(1-(2.35 x 149.3 x 10^6)/(0.9 x 35 x 350 x 580^2)))/420
= 696.9 mm^2

= 0 x 10^6/(0.9 x 420 x (580 - 70))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0020

Asc = 0.0000 mm^2

= 714.9 mm^2 or 676.6666667 mm^2 = 714.9 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 714.9 + 0.00 + 0
= 714.9 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.004643

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Ma'aden Revision No:	A01	A02	C01	D01
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Outotec / L&T Revision No.:	0	1	2	3
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 580 x 580 (1 - 0.8/2 x 0.375)

= 803.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 128.06 KNm

Mu_2 = Mu - Mu_{lim} = 128.06 - 803.9

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 350 x 580 x (1-√(1-(2.35 x 128.1 x 10^6)/(0.9 x 35 x 350 x 580^2)))/420

= 0 x 10^6/(0.9 x 420 x (580 - 70))

= 595.7 mm^2

= 0 mm^2

e_{sc} = 0.0020

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 714.9 mm^2 or 676.6666667 mm^2 = 714.85964 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 714.9 + 0 + 0

= 714.9 mm^2

bottom layer1	3	nos		20	mm	dia	+		nos		mm dia
bottom layer2		nos			mm	dia	+		nos		mm dia

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.004642748

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 198.5 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 208.2 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 348.3 KN

fV_c = 0.75 x 198. = 148.87 KN > 101.79 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((101.79-148.9) x 10^3)/(0.75 x 420 x 580) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	0 mm	*if Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	513.896206 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
	s =	289.993168 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	290 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

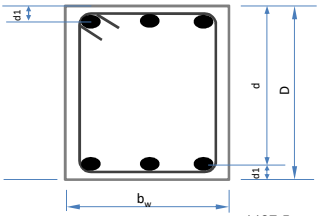
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.28 Beam marked 3FB4 Staad beam Group 3FB4



Width of beam	=	**350**	mm	
Depth of beam	=	**1200**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	1127.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	1127.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	413	Staad beam number -	2182	=	**820**	KNm
B.M for bottom reinforcement	Mz	L/C -	413	Staad beam number -	2182	=	**596**	KNm
Shear Force	Fy	L/C -	414	Staad beam number -	2181	=	**573**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 1127.5 x 1127.5 (1 - 0.8/2 x 0.375)

= 3037.9 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 819.63 KNm

Mu_2 = Mu - Mu_{lim} = 819.63 - 3037.9

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 1127.5 x (1-√(1-(2.35 x 819.7 x 10^6)/(0.9 x 35 x 350 x 1127.5^2)))/420

= 1991.7 mm^2

= 0 x 10^6/(0.9 x 420 x (1127.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0025

Asc = 0.0000 mm^2

= 1389.7 mm^2 or 1315.416667 mm^2

= 1389.7 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1991.7 + 0.00 + 0

= 1991.7 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2	**4**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 2767.743128 mm^2 Safe

ρ_w = 0.007014

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 1127.5 x 1127.5 (1 - 0.8/2 x 0.375)

= 3037.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 595.83 KNm

Mu_2 = Mu - Mu_{lim} = 595.83 - 3037.9

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 1127.5 x (1-√(1-(2.35 x 595.9 x 10^6)/(0.9 x 35 x 350 x 1127.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (1127.5 - 72.5))

= 1433.0 mm^2

= 0 mm^2

e_{sc} = 0.0025

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1389.7 mm^2 or 1315.416667 mm^2 = 1389.6625 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1433.1 + 0 + 0

= 1433.1 mm^2

bottom layer1	**4**	*nos*		**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**4**	*nos*		**16**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 2060.884781 mm^2 Safe

ρ_w = 0.005222388

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 410.6 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 420.6 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 677.0 KN

fV_c = 0.75 x 410. = 307.96 KN < 572.84 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((572.84-308) x 10^3)/(0.75 x 420 x 1127.5) = 0.745788629 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 421.2443758 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 1027.792412 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 579.986336 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 563.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

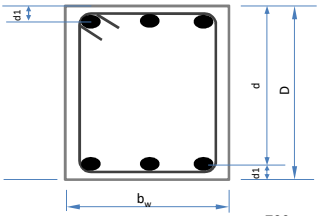
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.29 Beam marked 3FB5 Staad beam Group 3FB5



Width of beam	=	**350**	mm		
Depth of beam	=	**850**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	780	mm		
Dia of longitudinal bar	=	**20**	mm	d =	780 mm
Dia of stirrup bar	=	**10**	mm	d' =	70 mm
				h =	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	413	Staad beam number -	2177	=	**659**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	2176	=	**382**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	2177	=	**590**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 780 x 780 (1 - 0.8/2 x 0.375)

= 1453.9 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 658.91 KNm

Mu_2 = Mu - Mu_{lim} = 658.91 - 1453.9

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 350 x 780 x (1-√(1-(2.35 x 659 x 10^6)/(0.9 x 35 x 350 x 780^2)))/420

= 2378.3 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (780 - 70))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0023

Asc = 0.0000 mm^2

= 961.4 mm^2 or 910 mm^2 = 961.4 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2378.3 + 0.00 + 0

= 2378.3 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 2767.743128 mm^2 Safe

ρ_w = 0.010138

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 780 x 780 (1 - 0.8/2 x 0.375)

= 1453.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 381.7 KNm

Mu_2 = Mu - Mu_{lim} = 381.7 - 1453.9

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 350 x 780 x (1-√(1-(2.35 x 381.7 x 10^6)/(0.9 x 35 x 350 x 780^2)))/420

= 0 x 10^6/(0.9 x 420 x (780 - 70))

= 1339.4 mm^2

= 0 mm^2

e_{sc} = 0.0023

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 961.4 mm^2 or 910 mm^2 = 961.36296 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1339.4 + 0 + 0

= 1339.4 mm^2

bottom layer1	**4**	*nos*		**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**2**	*nos*		**16**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1658.760921 mm^2 Safe

ρ_w = 0.006076047

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 291.3 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 305.5 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 468.4 KN

fV_c = 0.75 x 291. = 218.46 KN < 590.17 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((590.17-218.5) x 10^3)/(0.75 x 420 x 780) = 1.512843122 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 207.6614956 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 1027.792412 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 579.986336 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 390 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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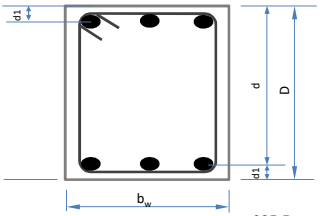
Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.30 Beam marked 3FB5A Staad beam Group 3FB5A

Width of beam	=	**350**	mm	
Depth of beam	=	**1000**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	925.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**12**	mm	



d	=	925.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	406	Staad beam number -	2165	=	**1116**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	2163	=	**1287**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	2165	=	**1573**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 925.5 x 925.5 (1 - 0.8/2 x 0.375)

= 2046.9 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 1115.75 KNm

Mu_2 = Mu - Mu_{lim} = 1115.75 - 2046.9

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 925.5 x (1-√(1-(2.35 x 1115.8 x 10^6)/(0.9 x 35 x 350 x 925.5^2)))/420

= 3443.8 mm^2

= 0 x 10^6/(0.9 x 420 x (925.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1140.7 mm^2 or 1079.75 mm^2

= 1140.7 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3443.8 + 0.00 + 0

= 3443.8 mm^2

Top layer1	**4**	*nos*	**32**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 4473.627939 mm^2 Safe

ρ_w = 0.013811

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 925.5 x 925.5 (1 - 0.8/2 x 0.375)

= 2046.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 1287.09 KNm

Mu_2 = Mu - Mu_{lim} = 1287.09 - 2046.9

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 350 x 925.5 x (1-√(1-(2.35 x 1287.1 x 10^6)/(0.9 x 35 x 350 x 925.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (925.5 - 74.5))

= 4028.1 mm^2

= 0 mm^2

e_{sc} = 0.0024

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1140.7 mm^2 or 1079.75 mm^2 = 1140.6941 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 4028.1 + 0 + 0

= 4028.1 mm^2

bottom layer1	**4**	*nos*		**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**4**	*nos*		**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3220.13247 mm^2 Unsafe

ρ_w = 0.009940982

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 405.8 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 382.7 KN

$V_c = 0.29\lambda\sqrt{f_c'} b_w d$ = 555.7 KN

fV_c = 0.75 x 382. = 287.00 KN < 1572.66 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((1572.66-287.1) x 10^3)/(0.75 x 420 x 925.5) = 4.409996507 mm^2/mm

No. of legs = **4** Spacing provided = **100** mm

Provide 4L -12 mm stirrups @ 100 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 102.5826985 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 1480.021073 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 835.1803239 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 462.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

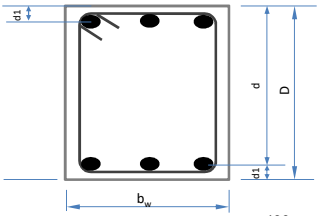
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.31 Beam marked 3FB6 Staad beam Group 3FB6



Width of beam	=	**300**	mm		
Depth of beam	=	**500**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	430	mm		
Dia of longitudinal bar	=	**20**	mm	d =	430 mm
Dia of stirrup bar	=	**10**	mm	d' =	70 mm
				h =	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	414	Staad beam number -	2189	=	**147**	KNm
B.M for bottom reinforcement	Mz	L/C -	414	Staad beam number -	2191	=	**136**	KNm
Shear Force	Fy	L/C -	414	Staad beam number -	2191	=	**76**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 146.656 KNm

Mu_2 = Mu - Mu_{lim} = 146.656 - 378.8

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 146.7 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 950.6 mm^2

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 454.3 mm^2 or 430 mm^2 = 454.3 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 950.6 + 0.00 + 0

= 950.6 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1884.955592 mm^2 Safe

ρ_w = 0.014612

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c' \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 136.149 KNm

Mu_2 = Mu - Mu_{lim} = 136.149 - 378.8

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 136.2 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 878.9 mm^2

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 0 mm^2

e_{sc} = 0.0017

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 454.3 mm^2 or 430 mm^2 = 454.27041 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 878.9 + 0 + 0

= 878.9 mm^2

bottom layer1	3	nos	16	mm	dia	+		nos		mm	dia
bottom layer2	3	nos	16	mm	dia	+		nos		mm	dia

Ast provided = 1206.371579 mm^2 Safe

ρ_w = 0.009351718

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 129.3 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 154.2 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 221.3 KN

fV_c = 0.75 x 129. = 96.94 KN > 76.105 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((76.105-97) x 10^3)/(0.75 x 420 x 430) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s	=	0 mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s	=	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
	s	=	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s	=	215 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s	=	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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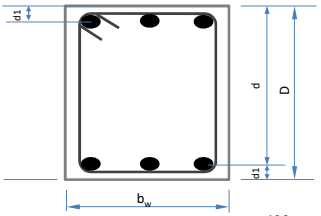
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.32 Beam marked 3B1 Staad beam Group 3B1



Width of beam	=	**300**	mm		
Depth of beam	=	**500**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	430	mm		
Dia of longitudinal bar	=	**20**	mm	d =	430 mm
Dia of stirrup bar	=	**10**	mm	d' =	70 mm
				h =	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	404	Staad beam number -	2184	=	**99**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2184	=	**66**	KNm
Shear Force	Fy	L/C -	404	Staad beam number -	2184	=	**68**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 98.85 KNm

Mu_2 = Mu - Mu_{lim} = 98.85 - 378.8

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 98.9 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 629.1 mm^2

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 454.3 mm^2 or 430 mm^2 = 454.3 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 629.1 + 0.00 + 0

= 629.1 mm^2

Top layer1	**3**	nos	**16**	mm	dia	+		nos		mm dia
Top layer2	**3**	nos	**16**	mm	dia	+		nos		mm dia

Ast provided = 1206.371579 mm^2 Safe

ρ_w = 0.009352

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 66.172 KNm

Mu_2 = Mu - Mu_{lim} = 66.172 - 378.8

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 66.2 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 416.1 mm²

= 0 mm²

e_{sc} = 0.0017

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm²

= 454.3 mm² or 430 mm² = 454.27041 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 454.3 + 0 + 0

= 454.3 mm²

bottom layer1	3	nos	20	mm	dia	+		nos		mm	dia
bottom layer2		nos		mm	dia	+		nos		mm	dia

Ast provided = 942.4777961 mm² Safe

ρ_w = 0.007306029

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 128.1 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 142.6 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 221.3 KN

fV_c = 0.75 x 128. = 96.11 KN > 67.646 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S} = \frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((67.646-96.2) x 10^3)/(0.75 x 420 x 430) = 0 mm²/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement — s = 0 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ — s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ — s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 — s = 215 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion — s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

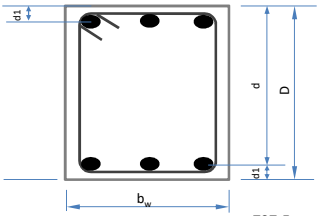
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.33 Beam marked 3B2 Staad beam Group 3B2



Width of beam	=	**300**	mm	
Depth of beam	=	**800**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	727.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	727.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	405	Staad beam number -	2255	=	**316**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2252	=	**304**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	2253	=	**251**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 727.5 x 727.5 (1 - 0.8/2 x 0.375)

= 1084.1 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 316.19 KNm

Mu_2 = Mu - Mu_{lim} = 316.19 - 1084.1

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 300 x 727.5 x (1-√(1-(2.35 x 316.2 x 10^6)/(0.9 x 35 x 300 x 727.5^2)))/420

= 1194.5 mm^2

= 0 x 10^6/(0.9 x 420 x (727.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0022

Asc = 0.0000 mm^2

= 768.6 mm^2 or 727.5 mm^2 = 768.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1194.6 + 0.00 + 0

= 1194.6 mm^2

Top layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 2075.807346 mm^2 Safe

ρ_w = 0.009511

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 727.5 x 727.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 1084.1 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 304.4 KNm

Mu_2 = Mu - Mu_{lim} = 304.4 - 1084.1

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 727.5 x (1-√(1-(2.35 x 304.4 x 10^6)/(0.9 x 35 x 300 x 727.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (727.5 - 72.5))

= 1148.2 mm^2

= 0 mm^2

e_{sc} = 0.0022

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 768.6 mm^2 or 727.5 mm^2 = 768.56215 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1148.2 + 0 + 0

= 1148.2 mm^2

bottom layer1	3	nos		20	mm	dia	+		nos		mm dia
bottom layer2	3	nos		16	mm	dia	+		nos		mm dia

Ast provided = 1545.663586 mm^2 Safe

ρ_w = 0.007082078

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 227.0 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 241.9 KN

$V_c = 0.29\lambda\sqrt{f_c'} b_w d$ = 374.4 KN

fV_c = 0.75 x 227 = 170.21 KN < 250.78 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset A_V f_y d}{s}$

$\frac{A_V}{S} = \frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((250.78-170.3) x 10^3)/(0.75 x 420 x 727.5) = 0.351569764 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	446.7950562	mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737	mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	363.75	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0	mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

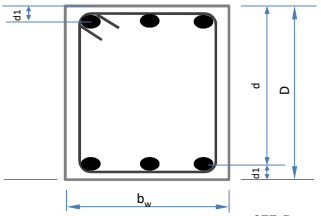
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.34 Beam marked 3B3 Staad beam Group 3B3



Width of beam	=	**440**	mm		
Depth of beam	=	**750**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	677.5	mm		
Dia of longitudinal bar	=	**25**	mm	d	= 677.5 mm
Dia of stirrup bar	=	**10**	mm	d'	= 72.5 mm
				h	= 120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	406	Staad beam number -	1907	=	**902**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	1909	=	**800**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	1907	=	**571**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 440 x 677.5 x 677.5 (1 - 0.8/2 x 0.375)

= 1378.9 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 902.1 KNm

Mu_2 = Mu - Mu_{lim} = 902.1 - 1379

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 440 x 677.5 x (1-√(1-(2.35 x 902.1 x 10^6)/(0.9 x 35 x 440 x 677.5^2)))/420

= 3873.4 mm^2

= 0 x 10^6/(0.9 x 420 x (677.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0021

Asc = 0.0000 mm^2

= 1049.8 mm^2 or 993.6666667 mm^2

= 1049.8 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3873.4 + 0.00 + 0

= 3873.4 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3926.990817 mm^2 Safe

ρ_w = 0.013173

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c' \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 440 x 677.5 x 677.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 1378.9 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 800.09 KNm

Mu_2 = Mu - Mu_{lim} = 800.09 - 1379

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 440 x 677.5 x (1-√(1-(2.35 x 800.1 x 10^6)/(0.9 x 35 x 440 x 677.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (677.5 - 72.5))

= 3392.9 mm^2

= 0 mm^2

e_{sc} = 0.0021

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 1049.8 mm^2 or 993.6666667 mm^2 = 1049.752 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3392.9 + 0 + 0

= 3392.9 mm^2

bottom layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3926.990817 mm^2 Safe

ρ_w = 0.013173401

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 310.8 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 348.9 KN

$V_c = 0.29\lambda\sqrt{f_c'} b_w d$ = 511.4 KN

fV_c = 0.75 x 310. = 233.10 KN < 570.83 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((570.83-233.1) x 10^3)/(0.75 x 420 x 677.5) = 1.582545376 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s	=	198.5151706 mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s	=	817.562146 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s	=	461.3527673 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	338.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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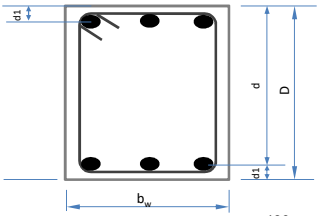
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.35 Beam marked 3B4 Staad beam Group 3B4



Width of beam	=	**300**	mm		
Depth of beam	=	**550**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	480	mm		
Dia of longitudinal bar	=	**20**	mm	d =	480 mm
Dia of stirrup bar	=	**10**	mm	d' =	70 mm
				h =	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	428	Staad beam number -	2295	=	**7**	KNm
B.M for bottom reinforcement	Mz	L/C -	406	Staad beam number -	2309	=	**239**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	2284	=	**259**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 480 x 480 (1 - 0.8/2 x 0.375)

= 471.9 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 7.49 KNm

Mu_2 = Mu - Mu_{lim} = 7.49 - 472

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 480 x (1-√(1-(2.35 x 7.5 x 10^6)/(0.9 x 35 x 300 x 480^2)))/420

= 41.3 mm^2

= 0 x 10^6/(0.9 x 420 x (480 - 70))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0018

Asc = 0.0000 mm^2

= 507.1 mm^2 or 480 mm^2 = 507.1 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 507.1 + 0.00 + 0

= 507.1 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.006545

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
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Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 480 x 480 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 471.9 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 239.32 KNm

Mu_2 = Mu - Mu_{lim} = 239.32 - 472

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 480 x (1-√(1-(2.35 x 239.4 x 10^6)/(0.9 x 35 x 300 x 480^2)))/420

= 0 x 10^6/(0.9 x 420 x (480 - 70))

= 1415.6 mm^2

= 0 mm^2

e_{sc} = 0.0018

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 507.1 mm^2 or 480 mm^2 = 507.09255 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1415.6 + 0 + 0

= 1415.6 mm^2

bottom layer1	**3**	*nos*		**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
bottom layer2	**3**	*nos*		**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1884.955592 mm^2 Safe

ρ_w = 0.013089969

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 402.0 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 152.3 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 247.1 KN

fV_c = 0.75 x 152. = 114.25 KN < 258.8 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((258.8-114.3) x 10^3)/(0.75 x 420 x 480) = 0.956041994 mm^2/mm

No. of legs = **2** Spacing provided = **150** mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s	=	164.302022 mm	*if Vu > φVc*
ii) Min Asv or Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or = $0.65\frac{b_w s}{f_{yt}}$	s	=	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
		s	=	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	240 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

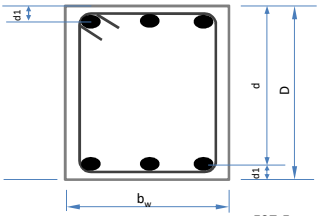
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.36 Beam marked 3B5 Staad beam Group 3B5



Width of beam	=	**300**	mm	
Depth of beam	=	**600**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	527.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	527.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	401	Staad beam number -	1892	=	**0.1**	KNm
B.M for bottom reinforcement	Mz	L/C -	403	Staad beam number -	1892	=	**319**	KNm
Shear Force	Fy	L/C -	403	Staad beam number -	1892	=	**180**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 569.9 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 0.1 KNm

Mu_2 = Mu - Mu_{lim} = 0.1 - 570

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 527.5 x (1-√(1-(2.35 x 0.1 x 10^6)/(0.9 x 35 x 300 x 527.5^2)))/420

= 0.5 mm^2

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 557.3 mm^2 or 527.5 mm^2

= 557.3 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 557.3 + 0.00 + 0

= 557.3 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.005956

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 569.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 318.604 KNm

Mu_2 = Mu - Mu_{lim} = 318.604 - 570

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 527.5 x (1-√(1-(2.35 x 318.7 x 10^6)/(0.9 x 35 x 300 x 527.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 1729.2 mm²

= 0 mm²

e_{sc} = 0.0019

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm²

= 557.3 mm² or 527.5 mm² = 557.27359 mm²

clause 9.6.1.2 of SBC 304

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1729.3 + 0 + 0

= 1729.3 mm²

bottom layer1	3	nos	25	mm	dia	+		nos		mm dia
bottom layer2	3	nos	16	mm	dia	+		nos		mm dia

Ast provided = 2075.807346 mm² Safe

ρ_w = 0.013117266

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 15334.1 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 165.8 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 271.5 KN

fV_c = 0.75 x 165. = 124.36 KN < 179.66 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((179.66-124.4) x 10^3)/(0.75 x 420 x 527.5) = 0.332789005 mm²/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 472.009683 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 263.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

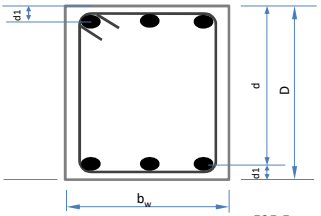
Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.37 Beam marked 3B6 Staad beam Group 3B6

Width of beam	=	**300**	mm	
Depth of beam	=	**600**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	525.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**12**	mm	



d	=	525.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	405	Staad beam number -	2285	=	**148**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2277	=	**464**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	2285	=	**457**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 525.5 x 525.5 (1 - 0.8/2 x 0.375)

= 565.6 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 147.85 KNm

Mu_2 = Mu - Mu_{lim} = 147.85 - 565.7

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 525.5 x (1-√(1-(2.35 x 147.9 x 10^6)/(0.9 x 35 x 300 x 525.5^2)))/420

= 769.9 mm^2

= 0 x 10^6/(0.9 x 420 x (525.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 555.2 mm^2 or 525.5 mm^2

= 555.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 770 + 0.00 + 0

= 770 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.005978

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 525.5 x 525.5 (1 - 0.8/2 x 0.375)

= 565.6 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 464.41 KNm

Mu_2 = Mu - Mu_{lim} = 464.41 - 565.7

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 525.5 x (1-√(1-(2.35 x 464.5 x 10^6)/(0.9 x 35 x 300 x 525.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (525.5 - 74.5))

= 2649.3 mm^2

= 0 mm^2

e_{sc} = 0.0019

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 555.2 mm^2 or 525.5 mm^2 = 555.1607 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2649.4 + 0 + 0

= 2649.4 mm^2

bottom layer1	**3**	*nos*		**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2	**3**	*nos*		**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 2945.243113 mm^2 Safe

ρ_w = 0.018682164

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 175.2 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 165.2 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 270.5 KN

fV_c = 0.75 x 165. = 123.94 KN < 456.9 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((456.9-124) x 10^3)/(0.75 x 420 x 525.5) = 2.011466091 mm^2/mm

No. of legs = **2** Spacing provided = **100** mm

Provide 2L -12 mm stirrups @ 100 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 112.4526394 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 863.3456261 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 487.1885223 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 262.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

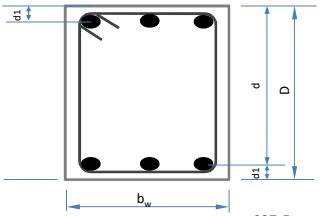
Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.38 Beam marked 3B7 Staad beam Group 3B7



Width of beam	=	**300**	mm	
Depth of beam	=	**700**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	627.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	627.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	401	Staad beam number -	2290	=	**0**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2290	=	**530**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	2290	=	**342**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$ *For tension-controlled section & following the assumptions in clause 22.2 of SBC 304*

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 627.5 x 627.5 (1 - 0.8/2 x 0.375)
= 806.5 KNm
Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 0.1 KNm

Mu_2 = Mu - Mu_{lim} = 0.1 - 806.6 = 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 627.5 x (1-√(1-(2.35 x 0.1 x 10^6)/(0.9 x 35 x 300 x 627.5^2)))/420
= 0.4 mm^2

= 0 x 10^6/(0.9 x 420 x (627.5 - 72.5))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0021

Asc = 0.0000 mm^2

= 662.9 mm^2 or 627.5 mm^2 = 662.9 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 663 + 0.00 + 0
= 663 mm^2

Top layer1	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm^2 Unsafe

ρ_w = 0.003204

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	03-05-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assumptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 627.5 x 627.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 806.5 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 529.56 KNm

Mu_2 = Mu - Mu_{lim} = 529.56 - 806.6

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 627.5 x (1-√(1-(2.35 x 529.6 x 10^6)/(0.9 x 35 x 300 x 627.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (627.5 - 72.5))

= 2456.0 mm^2

= 0 mm^2

e_{sc} = 0.0021

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 662.9 mm^2 or 627.5 mm^2 = 662.91787 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2456.0 + 0 + 0

= 2456 mm^2

bottom layer1	3	nos	25	mm	dia	+		nos		mm	dia
bottom layer2	3	nos	25	mm	dia	+		nos		mm	dia

Ast provided = 2945.243113 mm^2 Safe

ρ_w = 0.015645382

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 22192.5 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 188.4 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 323.0 KN

fV_c = 0.75 x 188. = 141.33 KN < 342.13 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((342.13-141.4) x 10^3)/(0.75 x 420 x 627.5) = 1.015848423 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 154.6290068 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 599.5455737 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338.3253627 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 313.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

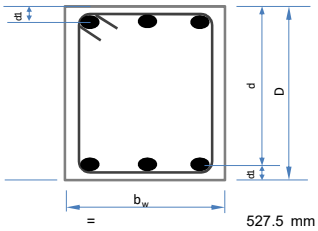
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.37 Beam marked 4FB1

Staad beam Group 4FB1

Width of beam	=	**350**	mm	
Depth of beam	=	**600**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	527.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	527.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	407	Staad beam number -	2036	=	**306**	KNm
B.M for bottom reinforcement	Mz	L/C -	407	Staad beam number -	2036	=	**227**	KNm
Shear Force	Fy	L/C -	407	Staad beam number -	2036	=	**136**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

$\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Singly reinforced beam

Mu = 306.46 KNm

Mu_2 = Mu - Mu_{lim} = 306.46 - 665

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 306.5 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1637.6 mm^2

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.416667 mm^2

= 650.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1637.6 + 0.00 + 0

= 1637.6 mm^2

Top layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**2**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1874.745416 mm^2 Safe

ρ_w = 0.010154

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 226.72 KNm

Mu_2 = Mu - Mu_{lim} = 226.72 - 665

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 226.8 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1189.7 mm^2

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0019

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.416667 mm^2 = 650.152518 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1189.8 + 0 + 0

= 1189.8 mm^2

bottom layer1	2	nos	25	mm	dia	+	1	nos	20	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1295.90697 mm^2 Safe

ρ_w = 0.007019131

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 182.2 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 206.6 KN

$V_c = 0.29\lambda\sqrt{f_c'}b_w d$ = 316.8 KN

fV_c = 0.75 x 182. = 136.68 KN > 136.25 KN *Shear Reinforcement not require*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S} = \frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((136.25-136.7) x 10^3)/(0.75 x 420 x 527.5) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 0 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 513.896206 mm / 289.993168 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 263.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

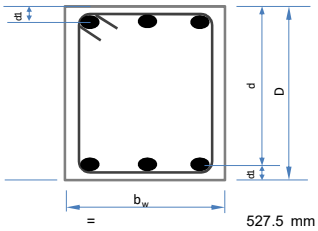
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.38 Beam marked 4FB2

Staad beam Group 4FB2

Width of beam	=	**350**	mm
Depth of beam	=	**600**	mm
Grade of steel	=	420	MPa
Grade of concrete	=	35	MPa
Cover	=	50	mm
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*
Effective Depth	=	527.5	mm
Dia of longitudinal bar	=	**25**	mm
Dia of stirrup bar	=	**10**	mm



d	=	527.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	407	Staad beam number -	2040	=	**334**	KNm
B.M for bottom reinforcement	Mz	L/C -	407	Staad beam number -	2040	=	**234**	KNm
Shear Force	Fy	L/C -	404	Staad beam number -	2040	=	**158**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

$\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 333.81 KNm

Mu_2 = Mu - Mu_{lim} = 333.81 - 665

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 333.9 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 1795.2 mm^2

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0019

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.416667 mm^2

= 650.2 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1795.3 + 0.00 + 0

= 1795.3 mm^2

Top layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**2**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1874.745416 mm^2 Safe

ρ_w = 0.010154

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Outotec / L&T Revision No.:	0	1	2	3
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 527.5 x 527.5 (1 - 0.8/2 x 0.375)

= 664.9 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 233.61 KNm

Mu_2 = Mu - Mu_{lim} = 233.61 - 665

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 527.5 x (1-√(1-(2.35 x 233.7 x 10^6)/(0.9 x 35 x 350 x 527.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (527.5 - 72.5))

= 1227.8 mm^2

= 0 mm^2

e_{sc} = 0.0019

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 650.2 mm^2 or 615.416667 mm^2 = 650.152518 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1227.8 + 0 + 0

= 1227.8 mm^2

bottom layer1	2	nos	25	mm	dia	+	1	nos	20	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1295.90697 mm^2 Safe

ρ_w = 0.007019131

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 182.7 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 206.6 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 316.8 KN

fV_c = 0.75 x 182. = 137.05 KN < 158.42 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((158.42-137.1) x 10^3)/(0.75 x 420 x 527.5) = 0.128580965 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 1221.639861 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 513.896206 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 289.993168 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 263.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

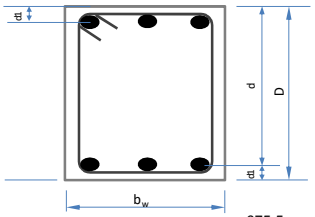
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.39 Beam marked 4FB3

Staad beam Group 4FB3

Width of beam	=	**350**	mm	
Depth of beam	=	**750**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	675.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**12**	mm	



d	=	675.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	404	Staad beam number -	2008	=	**600**	KNm
B.M for bottom reinforcement	Mz	L/C -	403	Staad beam number -	2009	=	**400**	KNm
Shear Force	Fy	L/C -	403	Staad beam number -	2010	=	**696**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 675.5 x 675.5 (1 - 0.8/2 x 0.375)

= 1090.4 KNm

$\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 599.95 KNm

Mu_2 = Mu - Mu_{lim} = 599.95 - 1090.5

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 350 x 675.5 x (1-√(1-(2.35 x 600 x 10^6)/(0.9 x 35 x 350 x 675.5^2)))/420

= 2539.2 mm^2

= 0 x 10^6/(0.9 x 420 x (675.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0021

Asc = 0.0000 mm^2

= 832.6 mm^2 or 788.083333 mm^2 = 832.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2539.2 + 0.00 + 0

= 2539.2 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 3220.13247 mm^2 Safe

ρ_w = 0.01362

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 675.5 x 675.5 (1 - 0.8/2 x 0.375)

= 1090.4 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 399.72 KNm

Mu_2 = Mu - Mu_{lim} = 399.72 - 1090.5

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 350 x 675.5 x (1-√(1-(2.35 x 399.8 x 10^6)/(0.9 x 35 x 350 x 675.5^2)))/420

= 1644.2 mm^2

= 0 x 10^6/(0.9 x 420 x (675.5 - 74.5))

= 0 mm^2

e_{sc} = 0.0021

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 832.6 mm^2 or 788.083333 mm^2 = 832.564978 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1644.3 + 0 + 0

= 1644.3 mm^2

bottom layer1	4	nos	20	mm	dia	+		nos		mm	dia
bottom layer2	2	nos	20	mm	dia	+		nos		mm	dia

Ast provided = 1884.955592 mm^2 Safe

ρ_w = 0.007972742

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 266.7 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 278.5 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 405.6 KN

fV_c = 0.75 x 266. = 200.01 KN < 695.83 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((695.83-200.1) x 10^3)/(0.75 x 420 x 675.5) = 2.330167554 mm^2/mm

No. of legs = 4 Spacing provided = 150 mm

Provide 4L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 194.1445547 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 1480.021073 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 835.1803239 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 337.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

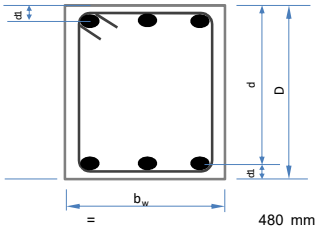
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.40 Beam marked 4B1

Staad beam Group 4B1

Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	480	mm	
Dia of longitudinal bar	=	**20**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	480	mm
d'	=	70	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	406	Staad beam number -	2046	=	**125**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2046	=	**273**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	2046	=	**185**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 480 x 480 (1 - 0.8/2 x 0.375)

= 471.9 KNm

$\left(\frac{c}{d}\right)_{max}$ = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 125.32 KNm

Mu_2 = Mu - Mu_{lim} = 125.32 - 472

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 480 x (1-√(1-(2.35 x 125.4 x 10^6)/(0.9 x 35 x 300 x 480^2)))/420

= 714.9 mm^2

= 0 x 10^6/(0.9 x 420 x (480 - 70))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0018

Asc = 0.0000 mm^2

= 507.1 mm^2 or 480 mm^2 = 507.1 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 714.9 + 0.00 + 0

= 714.9 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 942.4777961 mm^2 Safe

ρ_w = 0.006545

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 480 x 480 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

= 471.9 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 272.8 KNm

Mu_2 = Mu - Mu_{lim} = 272.8 - 472

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 300 x 480 x (1-√(1-(2.35 x 272.8 x 10^6)/(0.9 x 35 x 300 x 480^2)))/420

= 0 x 10^6/(0.9 x 420 x (480 - 70))

= 1632.2 mm^2

= 0 mm^2

e_{sc} = 0.0018

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 507.1 mm^2 or 480 mm^2 = 507.092553 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1632.3 + 0 + 0

= 1632.3 mm^2

bottom layer1	3	nos	20	mm	dia	+		nos		mm dia
bottom layer2	3	nos	20	mm	dia	+		nos		mm dia

Ast provided = 1884.955592 mm^2 Safe

ρ_w = 0.013089969

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 147.7 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 152.3 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 247.1 KN

fV_c = 0.75 x 147. = 110.76 KN < 185.31 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((185.31-110.8) x 10^3)/(0.75 x 420 x 480) = 0.493062727 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s =	318.5794101	mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737	mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s =	240	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s =	0	mm	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

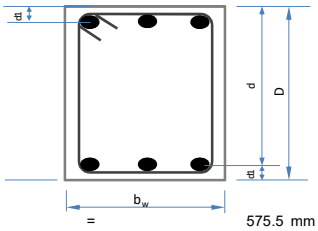
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

8.4.41 Beam marked 4B2

Staad beam Group 4B2

Width of beam	=	**300**	mm	
Depth of beam	=	**650**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	575.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**12**	mm	



d	=	575.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M for top reinforcement	Mz	L/C -	404	Staad beam number -	2056	=	**220**	KNm
B.M for bottom reinforcement	Mz	L/C -	405	Staad beam number -	2054	=	**441**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	2054	=	**385**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 575.5 x 575.5 (1 - 0.8/2 x 0.375)

= 678.4 KNm

$\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 219.92 KNm

Mu_2 = Mu - Mu_{lim} = 219.92 - 678.4

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 575.5 x (1-√(1-(2.35 x 220 x 10^6)/(0.9 x 35 x 300 x 575.5^2)))/420

= 1055.2 mm^2

= 0 x 10^6/(0.9 x 420 x (575.5 - 74.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0020

Asc = 0.0000 mm^2

= 608.0 mm^2 or 575.5 mm^2 = 608.0 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1055.3 + 0.00 + 0

= 1055.3 mm^2

Top layer1	**2**	*nos*	**25**	*mm*	*dia*	+	**1**	*nos*	**16**	*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1182.809634 mm^2 Safe

ρ_w = 0.006851

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Bottom reinforcement

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 575.5 x 575.5 (1 - 0.8/2 x 0.375)

= 678.4 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 440.74 KNm

Mu_2 = Mu - Mu_{lim} = 440.74 - 678.4

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 575.5 x (1-√(1-(2.35 x 440.8 x 10^6)/(0.9 x 35 x 300 x 575.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (575.5 - 74.5))

= 2226.1 mm^2

= 0 mm^2

e_{sc} = 0.0020

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 608.0 mm^2 or 575.5 mm^2 = 607.982842 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2226.2 + 0 + 0

= 2226.2 mm^2

bottom layer1	3	nos		25	mm	dia	+		nos		mm dia
bottom layer2	3	nos		20	mm	dia	+		nos		mm dia

Ast provided = 2415.099352 mm^2 Safe

ρ_w = 0.013988412

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 183.7 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 183.5 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 296.2 KN

fV_c = 0.75 x 183. = 137.65 KN < 385.34 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((385.34-137.7) x 10^3)/(0.75 x 420 x 575.5) = 1.366319618 mm^2/mm

No. of legs = 2 Spacing provided = 150 mm

Provide 2L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s =	165.5503354	mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	863.3456261	mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s =	487.1885223	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s =	287.75	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s =	0	mm	

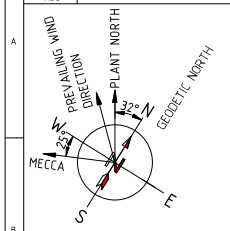
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01		
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL		
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Approved	AG	AG	AG	AG	

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWING

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1004 | REV: A01



BECHTEL

Document Number
MD-216-3120-EG-ME-LPX-1004

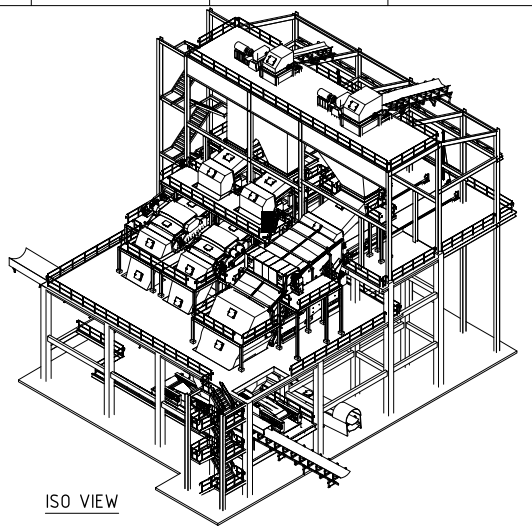
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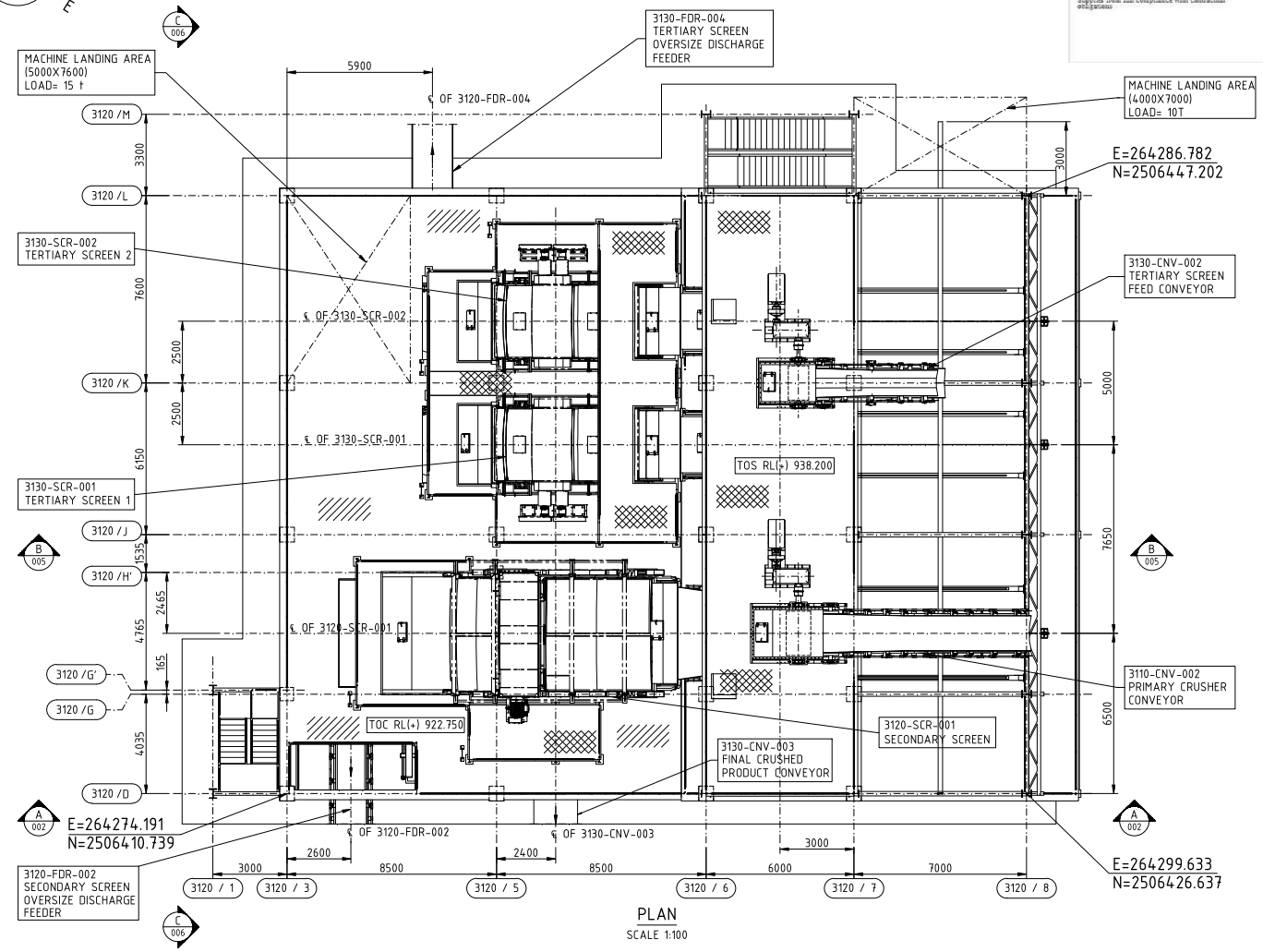
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ISO VIEW



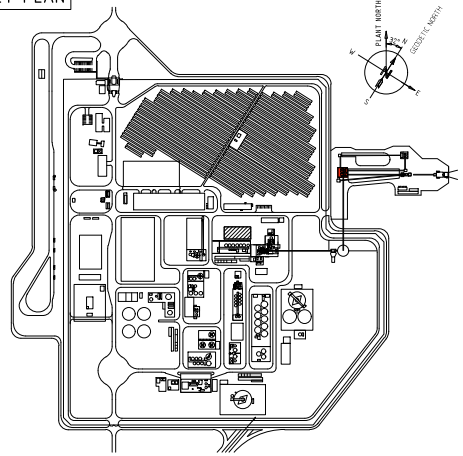
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SCALE 1:100

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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
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5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1009	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1008	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1007	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1010	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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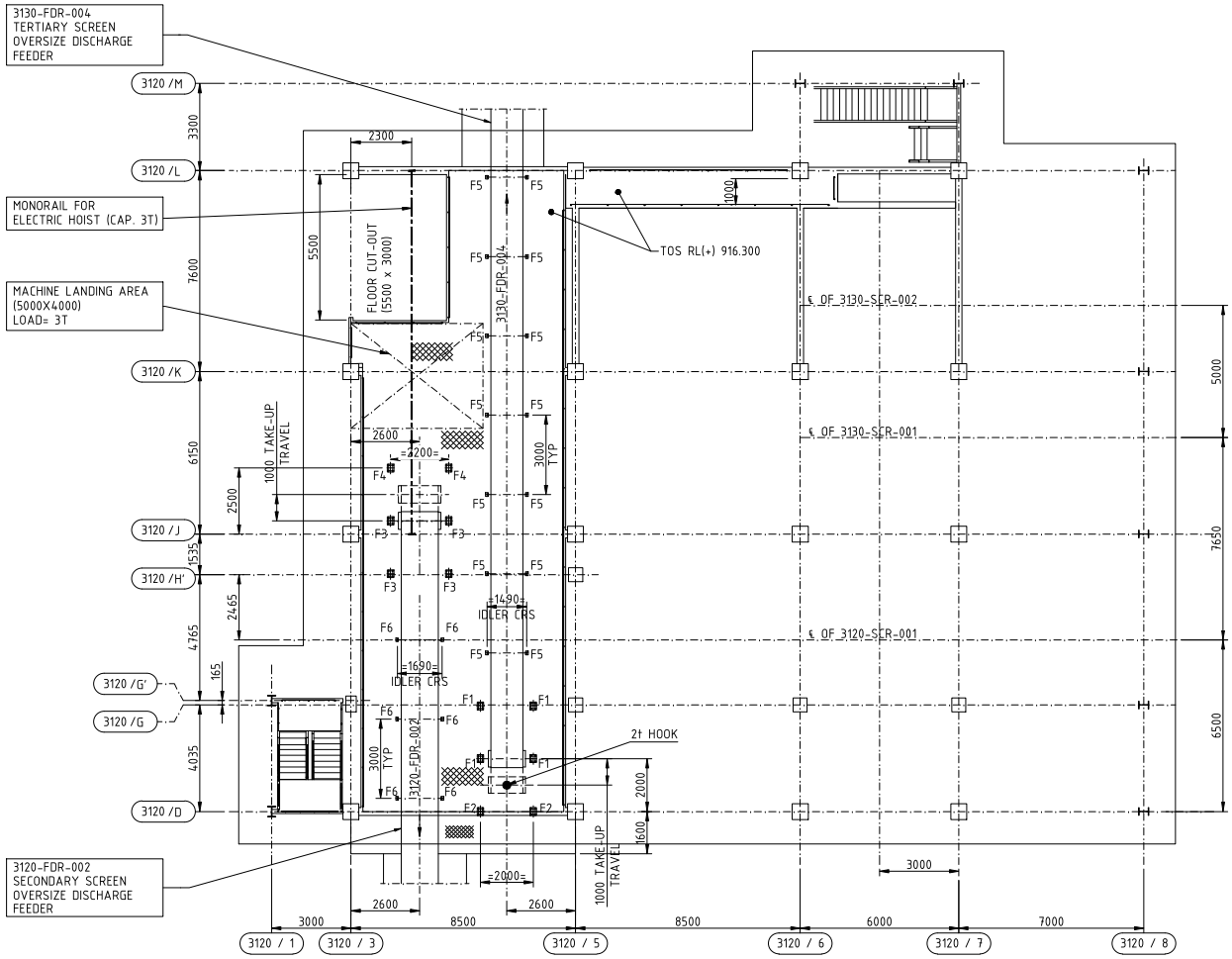
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OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31111	DOCUMENT ID OU100666775	REVISION 0
DRAWN	ANN	06/01/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	08/01/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	08/01/2020	SECONDARY CRUSHING
APPROVED	BGR	10/01/2020	SCREEN HOUSE-PLAN VIEW
EMPLOYER			
SCALE	1:100	A1	DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1004 REV: A01

DRAWING NUMBER MD-216-3120-EG-ME-LPX-1005 REV A01



PLAN AT RL(+) 916.300

SCALE 1:100

Document Number
MD-216-3120-EG-ME-LPX-1005

Revision A01

Approved with Comments

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TABLE-1: LOAD FOR SCREW TAKE-UP FRAME

FOR BELT FEEDER : 3130-FDR-004

Loads are normally specified in kN.

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F1	(+)916.300	(+)2.0	(+)2.0	(±)0.20
F2		(+)2.0	(+)2.0	(±)0.20

FOR BELT FEEDER : 3120-FDR-002

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F3	(+)916.300	(+)2.0	(+)3.0	(±)0.20
F4		(+)2.0	(+)3.0	(±)0.20

NOTE:

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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN

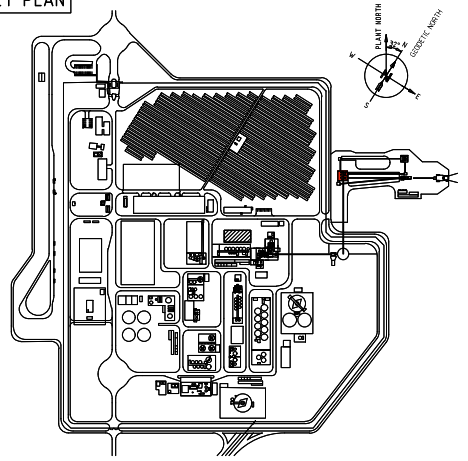


TABLE-2: LOAD FOR CONVEYOR SHORT POST

BELT FEEDER : 3130-FDR-004

	POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F5	(+)916.300	(+) 0.98	(+) 0.55	(±) 0.17	(±) 0.09

BELT FEEDER : 3120-FDR-002

	POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F6	(+)916.300	(+) 1.12	(+) 0.55	(±) 0.17	(±) 0.09

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1009	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1008	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1007	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1010	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1004	SCREEN HOUSE-PLAN VIEW	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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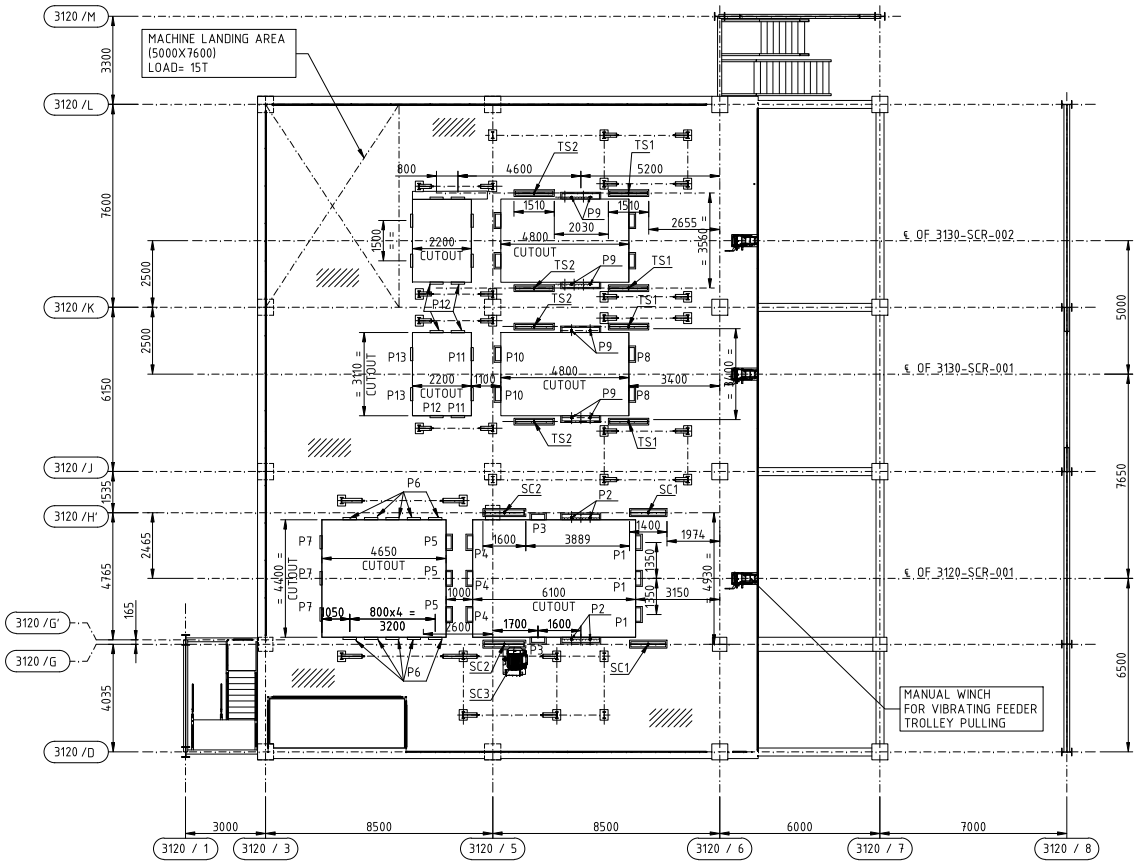
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OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31113	DOCUMENT ID OU100666777	REVISION 0
DRAWN	ANN	06/01/2020	
CHECKED	BRD	08/01/2020	
ENGINEER	BD	08/01/2020	
APPROVED	BGR	10/01/2020	
EMPLOYER			
SCALE	1:100	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SCREEN HOUSE- PLAN VIEW

DRAWING NUMBER MD-216-3120-EG-ME-LPX-1005 REV A01

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1006 REV A01



PLAN AT RL(+) 922.750

SCALE 1:100

TABLE-3: LOAD FOR SECONDARY SCREEN

3120-SCR-001

Loads are normally specified in kN.

LIVE LOAD DUE TO MATERIAL	SECONDARY SCREEN LOADS AT EACH SUPPORT POINT IN "t", WITH ISOLATION FRAME & WITHOUT MATERIAL INCLUDES FACTOR OF SAFETY			
STATIC	STATIC	DYNAMIC		
VERTICAL	VERTICAL	VERTICAL	LONGITUDINAL	TRANSVERSE
FEED END SC1 1.039	FEED END SC1 14.414	FEED END SC1 0.116	FEED END SC1 0.017	FEED END SC1 0.009
DISCHARGE END SC2 2.663	DISCHARGE END SC2 19.630	DISCHARGE END SC2 0.135	DISCHARGE END SC2 0.019	DISCHARGE END SC2 0.010
	DRIVE SC3 0.630	DRIVE SC3 0.023	DRIVE SC3 0.023	DRIVE SC3 0.008
MAX OVERTURNING MOMENT AT SC3: 0.156 t-m				

Please provide equipment operation frequency for structural design purpose.

Note: Wind load to be considered during civil & structural design

TABLE-4: LOAD FOR TERTIARY SCREEN

3130-SCR-001 & 002

LOAD WITHOUT MATERIAL

TOTAL WEIGHT : 14.03 t

MOVING MASS : 13.20 t

TS1 - STATIC : 3.16 t TS2 - STATIC : 3.16 t

DYNAMIC LOAD, RUNNING (+/-)

TS1 - DYNAMIC VERTICAL : 1.224 t TS2 - DYNAMIC VERTICAL : 1.224 t

TS1 - DYNAMIC HOR / LAT : 0.612 t TS2 - DYNAMIC VERTICAL : 0.612 t

DYNAMIC LOAD, STOP AND START (+/-)

TS1 - DYNAMIC VERTICAL : 3.059 t TS2 - DYNAMIC VERTICAL : 3.059 t

TS1 - DYNAMIC HOR / LAT : 1.530 t TS2 - DYNAMIC VERTICAL : 1.530 t

DRIVE UNIT LOADS

DRIVE UNIT MASS : 1.360 t

STATIC LOAD : 1.360 t

DYNAMIC LOAD, START AND STOP (+/-)

DYNAMIC VERTICAL : 0.510 t

DYNAMIC HOR / LAT : 0.510 t

Loads are normally specified in kN.

TABLE-5: LOAD FOR CHUTE SADDLE

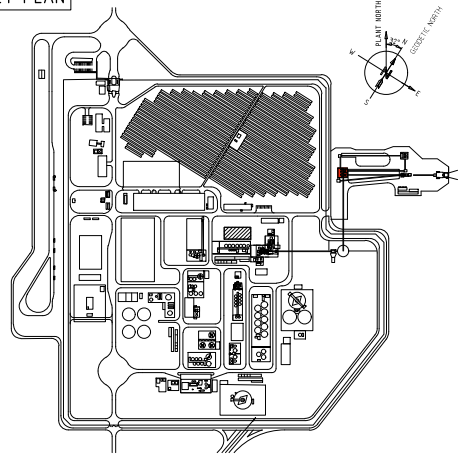
POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
		CHUTE EMPTY	CHUTE BLOCKED
P1	(+)922.750	(+)2.1	(+)8.5
P2		(+)2.1	(+)8.5
P3		(+)2.1	(+)8.5
P4		(+)2.1	(+)8.5
P5		(+)1.9	(+)3.7
P6		(+)1.9	(+)3.7
P7		(+)1.9	(+)3.7
P8		(+)2.2	(+)6.0
P9		(+)2.2	(+)6.0
P10		(+)2.2	(+)6.0
P11		(+)1.5	(+)3.2
P12		(+)1.5	(+)3.2
P13		(+)1.5	(+)3.2

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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



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Document Number MD-216-3120-EG-ME-LPX-1006

Revision A01

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MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1004	SCREEN HOUSE-PLAN VIEW	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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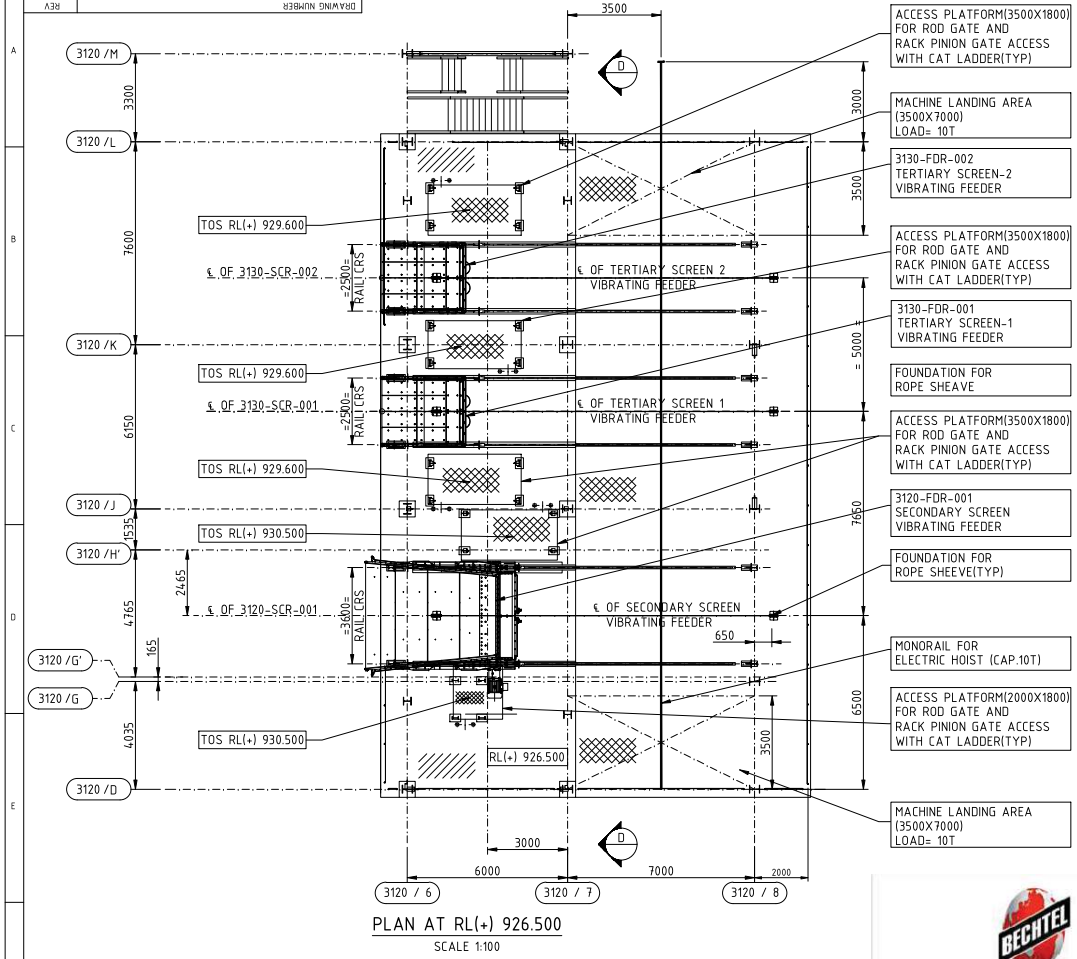
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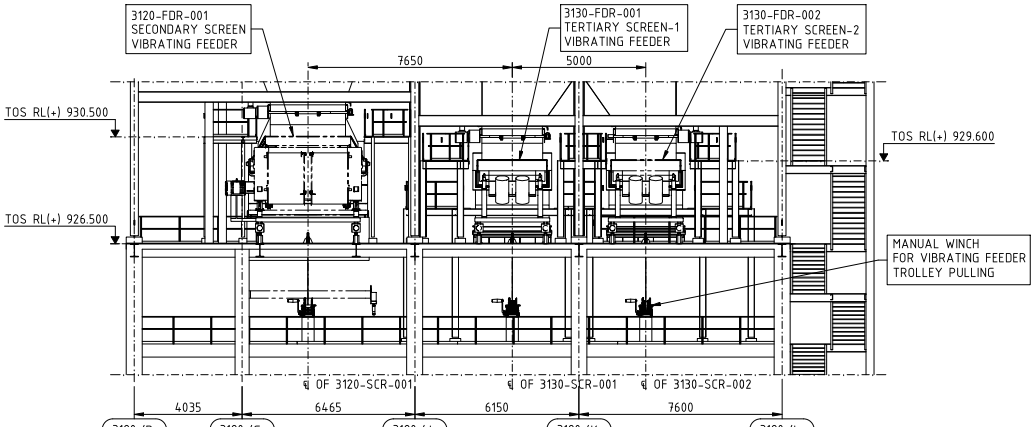
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CHECKED	BRD	08/01/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	08/01/2020	SECONDARY CRUSHING
APPROVED	BGR	10/01/2020	SCREEN HOUSE- PLAN VIEW
EMPLOYER			
SCALE 1:100		A1	DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1006 REV: A01

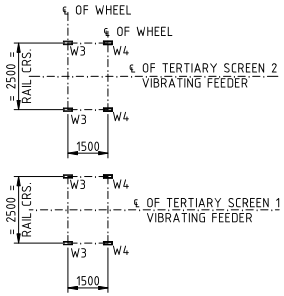
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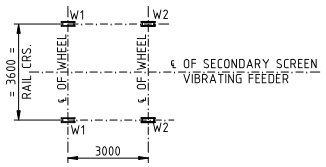


PLAN AT RL(+) 926.500
SCALE 1:100



SECTION D-D
SCALE 1:100





WHEEL LOAD DIAGRAM OF VIBRATING FEEDERS
SCALE 1:100

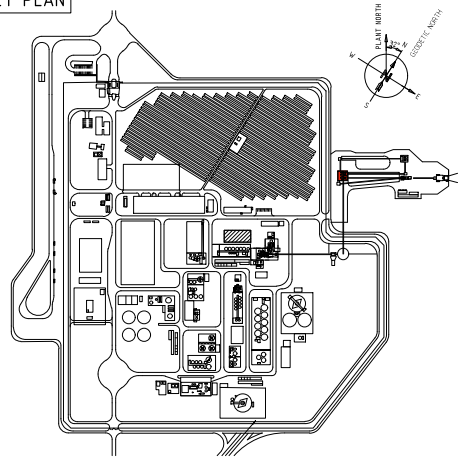
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

TABLE-6: LOAD FOR WHEEL
(DURING TRAVEL OF FEEDER AT MAINTENANCE TIME)

NO.	WHEEL MARKING	LOAD IN TON
1	W1	15
2	W2	15
3	W3	4
4	W4	4

KEY PLAN



Document Number
MD-216-3120-EG-ME-LPX-1007

Revision A01

Approved

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MD-216-3120-EG-ME-LPX-1009	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1008	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1010	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1004	SCREEN HOUSE-PLAN VIEW	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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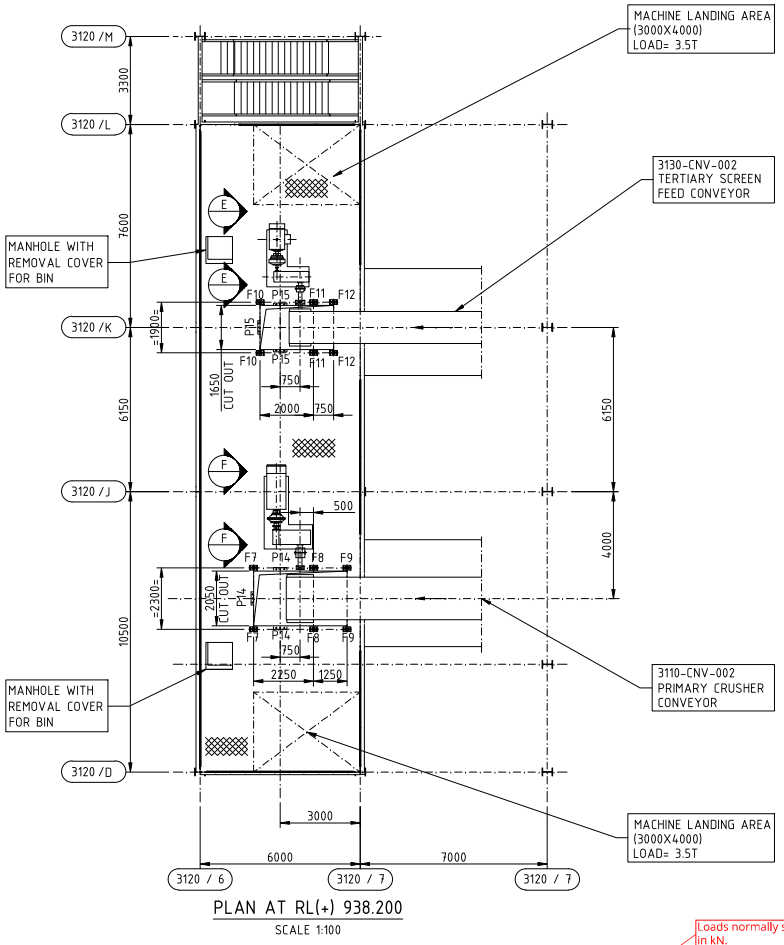
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OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31236 — DOCUMENT ID: OU100669061 — REVISION: 0

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APPROVED	BGR	10/01/2020
EMPLOYER		
SCALE	1:100	A1

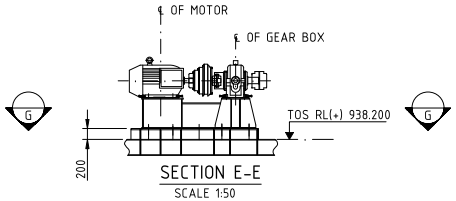
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SECONDARY CRUSHING
SCREEN HOUSE- PLAN VIEW

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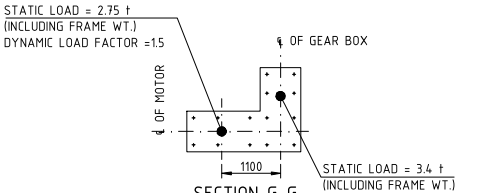
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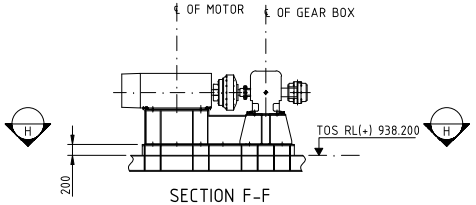
PLAN AT RL(+) 938.200
SCALE 1:100



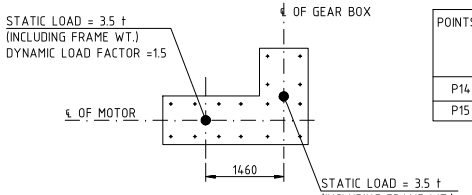
SECTION E-E
SCALE 1:50



SECTION G-G
SCALE 1:50



SECTION F-F
SCALE 1:50



SECTION H-H
SCALE 1:50

TABLE-9: LOAD FOR CHUTE SADDLE

POINTS	ELEVATIONS IN "m"	V (IN "t")
P14	(+)938.200	(+)2.0
P15		(+)1.5

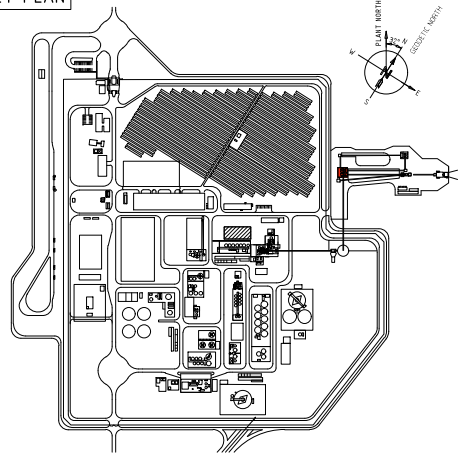
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5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)
(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

Loads normally specified in kN.

TABLE-7 : LOAD FOR DRIVE PULLEY FRAME

FOR CONV. 3110-CNV-002

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F7	(+)938.200	(+) 1.90	(-) 2.30	(±) 0.20	(-) 3.80	(-) 5.60	(±) 0.40
F8		(+) 7.90	(-) 2.30	(±) 0.80	(+) 8.10	(-) 1.90	(±) 0.82
F9		(+) 5.40	(-) 6.10	(±) 0.55	(+) 7.50	(-) 8.40	(±) 0.75

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

TABLE-8 : LOAD FOR DRIVE PULLEY FRAME

FOR CONV. 3130-CNV-002

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F10	(+)938.200	(+) 2.10	(-) 1.20	(±) 0.20	(-) 1.60	(-) 2.60	(±) 0.20
F11		(+) 4.10	(-) 1.60	(±) 0.40	(+) 3.60	(-) 1.50	(±) 0.36
F12		(+) 4.60	(-) 4.10	(±) 0.50	(+) 5.90	(-) 5.20	(±) 0.60

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1009	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1007	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1010	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1004	SCREEN HOUSE-PLAN VIEW	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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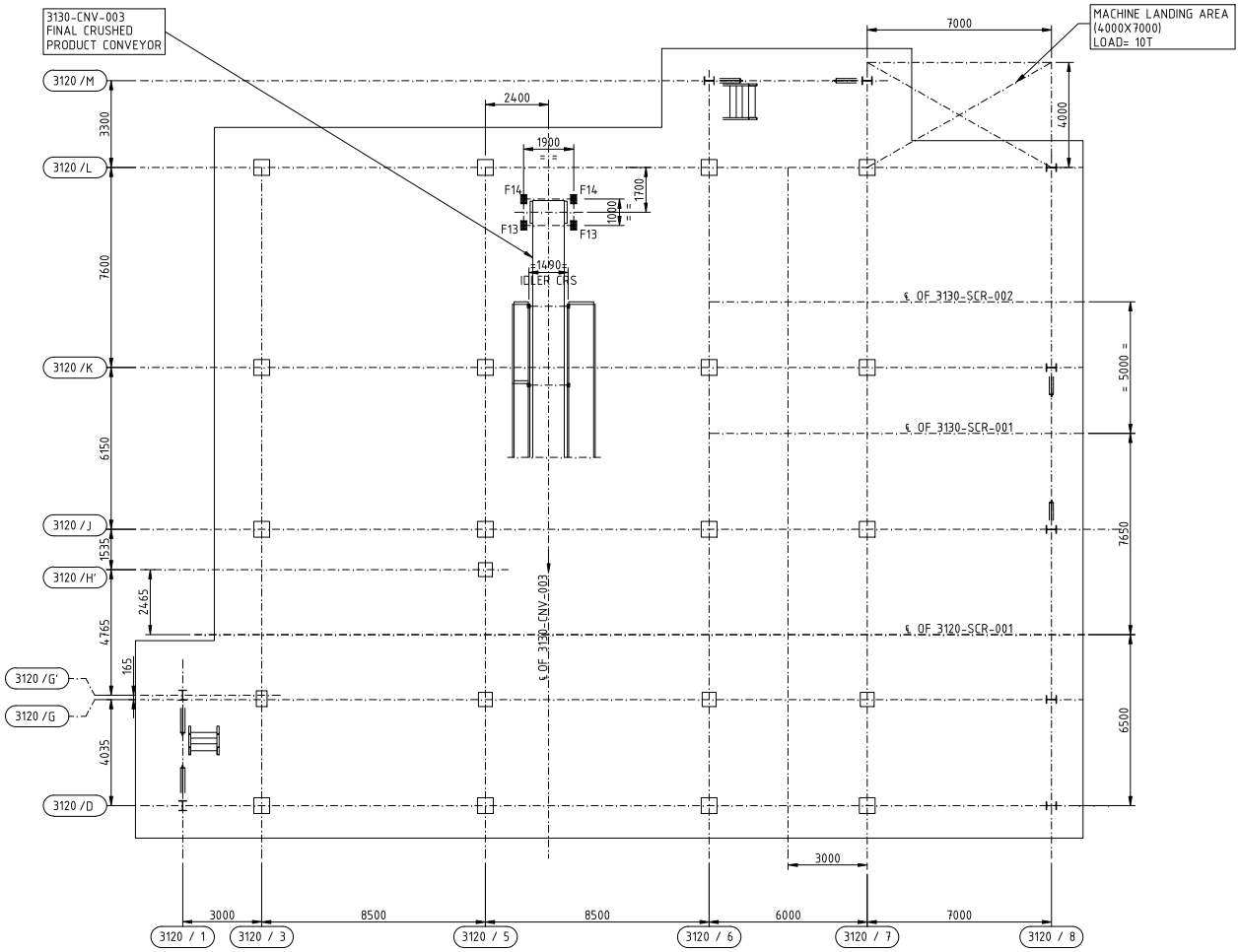
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EMPLOYER		
SCALE	1:100 & 1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SCREEN HOUSE- PLAN VIEW

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1008 REV: A01

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PLAN AT RL(+) 913.000

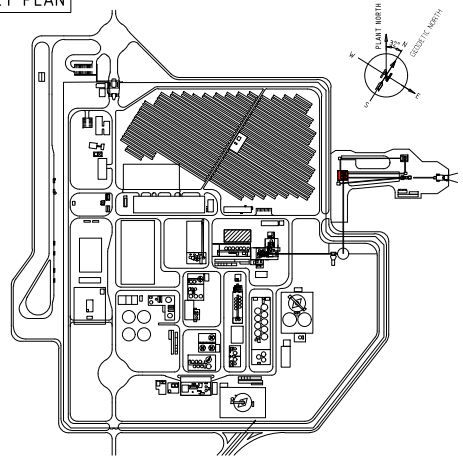
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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



Loads are normally specified in kN.

TABLE-10: LOAD FOR TAIL PULLEY FRAME

FOR CONVEYOR : 3130-CNV-003

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F13	(+)913.000	(+)5.90	(+)4.32	(±)0.20
F14		(-)4.80	(+)3.90	(±)0.20

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

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Document Number
MD-216-3120-EG-ME-LPX-1009

Revision A01

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MD-216-3120-EG-ME-LPX-1008	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1007	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1010	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1004	SCREEN HOUSE-PLAN VIEW	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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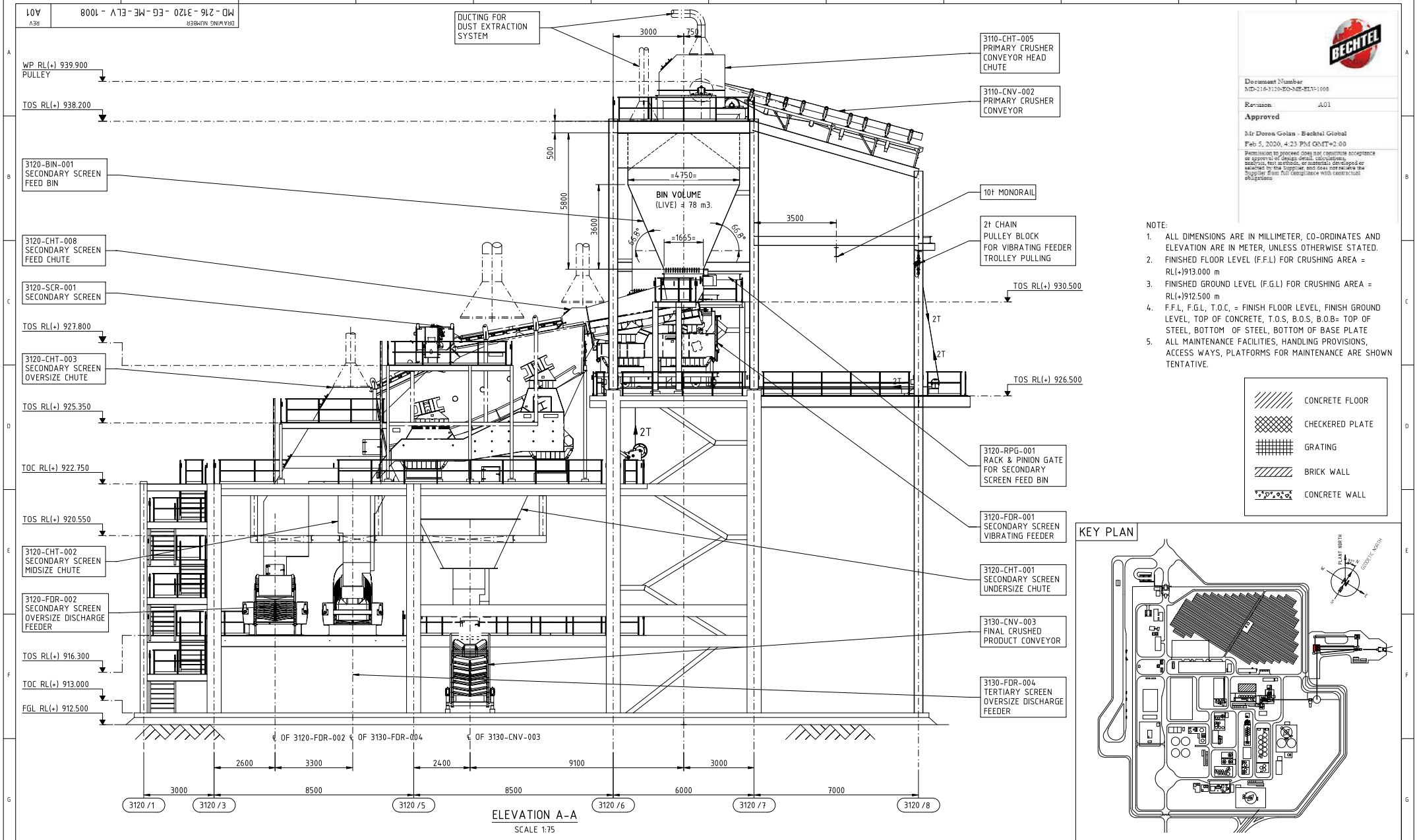
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EMPLOYER		
SCALE	1:100	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SCREEN HOUSE- PLAN VIEW

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1009 — REV: A01

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ELEVATION A-A

SCALE 1:75

NOTE:

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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN

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MD-216-3120-EG-ME-LPX-1009	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1008	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1007	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1010	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
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EMPLOYER			
SCALE	1:75		A1

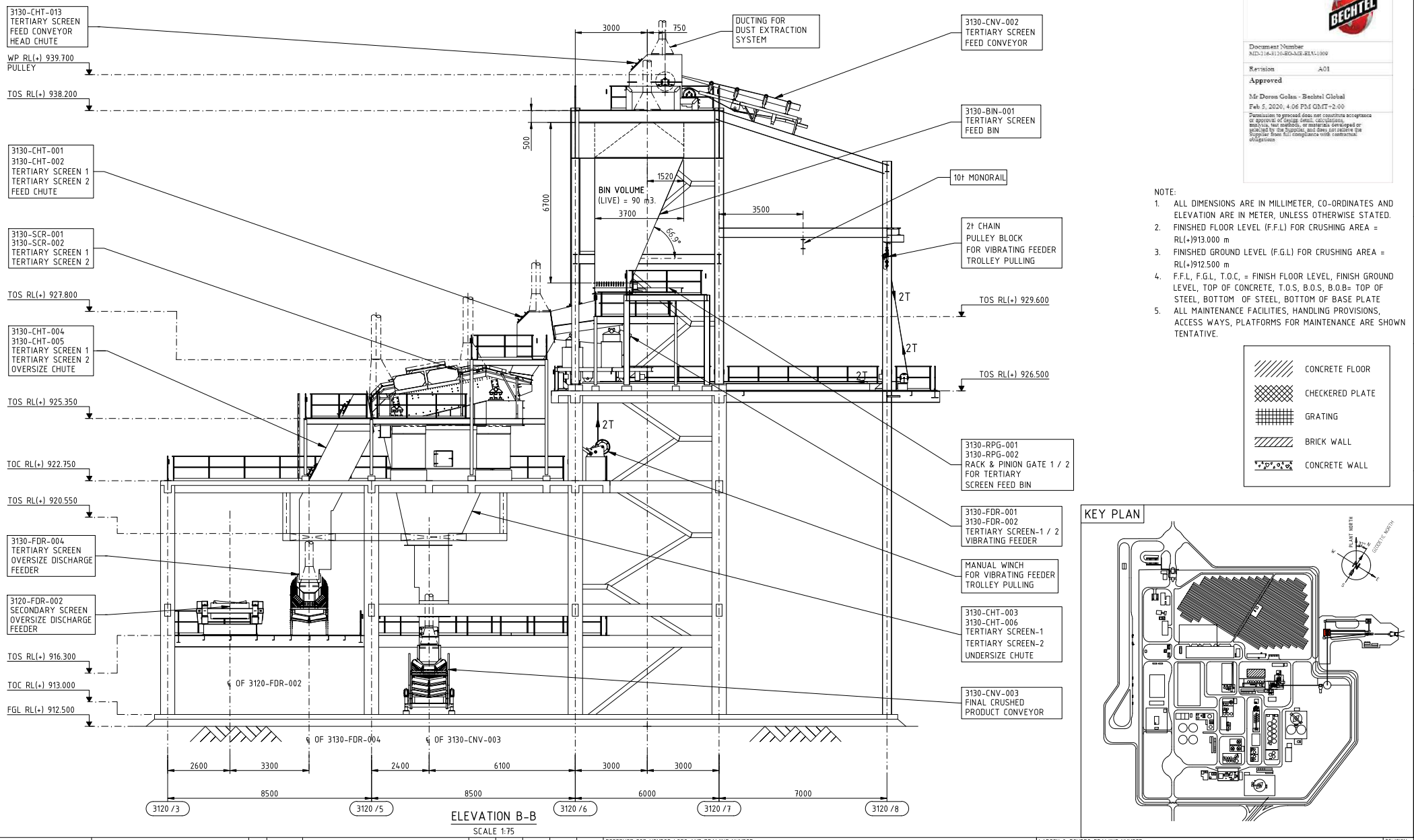
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SCREEN HOUSE-ELEVATION

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1008 — REV: A01

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ELEVATION B-B

SCALE 1:75

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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN

BECHTEL

Document Number MD-216-3120-EG-ME-ELV-1009
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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1009	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1008	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1007	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1010	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1004	SCREEN HOUSE-PLAN VIEW	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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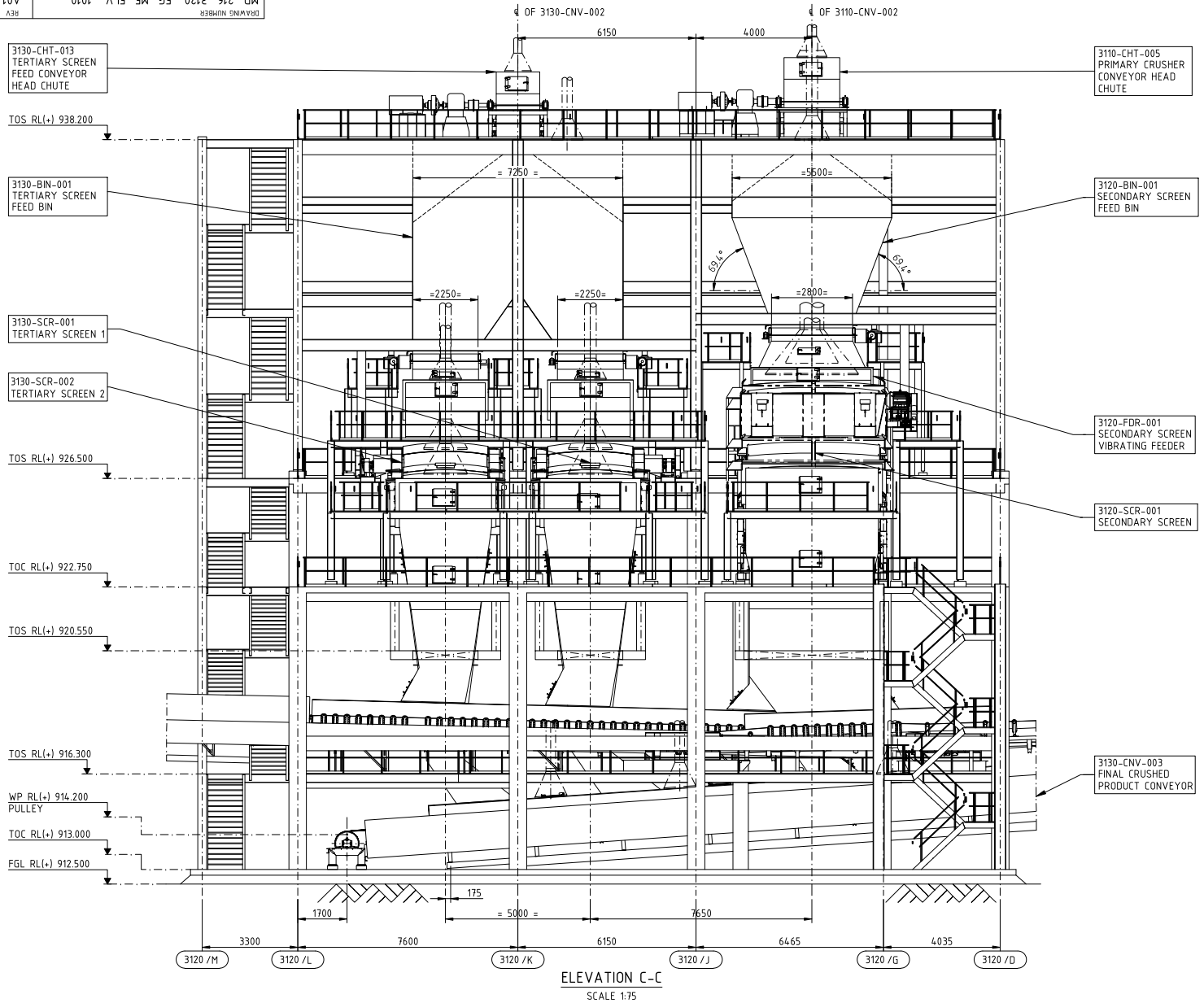
LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16124_005		REVISION 0
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31115	DOCUMENT ID OU100666779	REVISION 0

DRAWN	ANN	06/01/2020
CHECKED	BRD	08/01/2020
ENGINEER	BD	08/01/2020
APPROVED	BGR	10/01/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SCREEN HOUSE- SECTION

DRAWING NUMBER MD-216-3120-EG-ME-ELV-1009 REV A01

DRAWING NUMBER MD-216-3120-EG-ME-ELV-1010 REV A01



ELEVATION C-C

SCALE 1:75

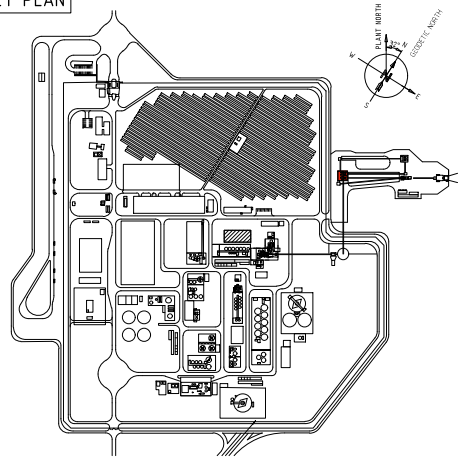
Document Number MD-216-3120-EG-ME-ELV-1010
Revision A01
Approved
Mr Doron Golan - Bechtel Global
Feb 5, 2020, 4:07 PM GMT+2:00
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NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1009	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1008	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1007	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1009	SCREEN HOUSE-SECTION								
MD-216-3120-EG-ME-LPX-1006	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1005	SCREEN HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1008	SCREEN HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1004	SCREEN HOUSE-PLAN VIEW	A01	10/01/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16124_006 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31219 — DOCUMENT ID: OU100669037 — REVISION: 0

DRAWN	ANN	06/01/2019	
CHECKED	BRD	08/01/2020	
ENGINEER	BD	08/01/2020	
APPROVED	BGR	10/01/2020	
EMPLOYER			
SCALE	1:75	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SCREEN HOUSE- ELEVATION

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1010 — REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01		
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL	
Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG	

ATTACHMENT 2 STAAD INPUT FILE OF SCREEN BUILDING

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -2.5 0; 2 0 -2.5 7.6; 3 0 -2.5 13.75; 4 0 -2.5 20.215; 5 0 -2.5 24.25;
6 8.5 -2.5 0; 7 8.5 -2.5 7.6; 8 8.5 -2.5 13.75; 9 8.5 -2.5 20.215;
10 8.5 -2.5 24.25; 11 17 -2.5 0; 12 17 -2.5 7.6; 13 17 -2.5 13.75;
14 17 -2.5 20.215; 15 17 -2.5 24.25; 16 23 -2.5 0; 17 23 -2.5 7.6;
18 23 -2.5 13.75; 19 23 -2.5 20.215; 20 23 -2.5 24.25; 21 30 0.45 0;
22 30 0.45 7.6; 23 30 0.45 13.75; 24 30 0.45 20.215; 25 30 0.45 24.25;
26 17 0.45 -3.3; 27 23 0.45 -3.3; 30 0 -0.15 0; 31 0 -0.15 7.6;
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36 8.5 -0.15 7.6; 37 8.5 -0.15 13.75; 38 8.5 -0.15 20.215; 39 8.5 -0.15 24.25;
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44 17 -0.15 24.25; 45 23 -0.15 0; 46 23 -0.15 7.6; 47 23 -0.15 13.75;
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57 0 9.75 13.75; 58 0 9.75 20.215; 59 0 9.75 24.25; 60 8.5 3.3 0;
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_1FB4 1812 1813 1830 1831
_1FB5 1814 1820 TO 1822
_1FB6 1815 TO 1819
_1FB7 1653 TO 1672 2321
_1FB8 1651 1652
_1B1 1695 1699
_1B2 1675 TO 1679
_1B3 1682 TO 1690
_1B4 1691 TO 1694 1696 TO 1698
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_1B5 1680 1681
_2FB1 1701 TO 1706 2104 2105
_2FB2 1719 TO 1726
_2FB3 1710 TO 1716
_3FB1 2201 TO 2207 2240 TO 2249
_3FB2 2208 TO 2214 2227 TO 2239 2329
_3FB2A 2215 TO 2225
_3BFB2B 1322 1389 1390 1974 TO 1978 1980 TO 1984 2320 2328
_3FB3 2151 2152 2174 2226
_3FB4 2157 TO 2160 2181 TO 2183
_3FB5 2153 TO 2156 2175 TO 2180
_3FB5A 2161 TO 2173
_3FB6 2188 TO 2191
_3B1 2184 TO 2187
_3B2 2250 TO 2262
_3B3 1893 TO 1910 1912 TO 1917
_3B4 2263 TO 2312 2330
_3B5 1892 1911
_4FB1 2035 2036 2043 2044
_4FB2 2037 TO 2042
_4FB3 2001 TO 2024
_4B1 2045 TO 2052
_4B2 2053 TO 2056
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
***Column
1301 TO 1316 PRIS YD 0.35 ZD 0.6
1951 TO 1964 PRIS YD 0.7 ZD 0.5
1331 TO 1350 1357 TO 1387 2313 2316 2319 PRIS YD 0.6 ZD 0.85
1351 TO 1356 1388 TO 1446 2320 PRIS YD 0.55 ZD 0.55
***Tie Beams
1501 TO 1524 PRIS YD 0.55 ZD 0.3
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1531 TO 1555 PRIS YD 0.5 ZD 0.3
****916 m Floor
1801 TO 1804 1832 TO 1836 PRIS YD 0.85 ZD 0.35
1807 TO 1811 1815 TO 1819 1823 TO 1829 PRIS YD 1 ZD 0.35
1805 1806 1812 TO 1814 1820 1821 1830 1831 1837 1838 PRIS YD 0.6 ZD 0.35
1651 TO 1672 2321 PRIS YD 0.55 ZD 0.3
1675 TO 1699 1822 PRIS YD 0.5 ZD 0.3
****920m floor
1248 1249 1319 1320 1701 TO 1706 1719 TO 1727 2100 TO 2106 PRIS YD 0.65 ZD 0.35
1710 TO 1716 PRIS YD 1 ZD 0.35
****922.75
2201 TO 2207 2240 TO 2249 PRIS YD 1.05 ZD 0.3
1322 1974 TO 1978 1980 TO 1984 2208 TO 2225 2227 TO 2239 2328 -
2329 PRIS YD 1.05 ZD 0.4
2157 TO 2160 2181 TO 2183 PRIS YD 1.1 ZD 0.35
2153 TO 2156 2161 TO 2173 2175 TO 2180 PRIS YD 0.85 ZD 0.35
2250 TO 2262 PRIS YD 0.65 ZD 0.3
2151 2152 2174 2226 PRIS YD 0.65 ZD 0.35
1892 1911 PRIS YD 0.6 ZD 0.3
2263 TO 2312 2330 PRIS YD 0.55 ZD 0.3
2188 TO 2191 PRIS YD 0.5 ZD 0.3
2184 TO 2187 PRIS YD 0.5 ZD 0.3
****Floor 926.5
2025 TO 2034 PRIS YD 0.18 ZD 0.3
2001 TO 2024 PRIS YD 0.75 ZD 0.35
2035 TO 2044 PRIS YD 0.6 ZD 0.35
2045 TO 2052 PRIS YD 0.55 ZD 0.3
MEMBER PROPERTY EUROPEAN
1 TO 11 TABLE ST HE500A
12 TO 27 TABLE ST HE500A
28 TO 38 TABLE ST HE400A
63 TO 74 76 78 82 2322 2323 2325 TABLE ST HE300A
75 77 81 83 TO 93 2326 2327 TABLE ST HE200A
107 TO 114 TABLE ST IPE450
115 TO 128 TABLE ST IPE400
129 TO 145 TABLE ST IPE400
146 TO 155 TABLE ST IPE300
156 TO 192 261 TO 265 2107 2108 TABLE ST IPE360
193 TO 226 TABLE ST IPE300
227 TABLE ST IPE300
228 230 TO 236 TABLE ST IPE300
229 237 TO 260 266 TO 277 2118 TABLE ST IPE300
278 TO 300 302 TO 307 TABLE ST IPE200
301 308 TO 340 365 TABLE ST IPE240
341 TO 364 366 TO 384 TABLE ST IPE200
385 TO 424 TABLE ST IPE160
425 TO 494 TABLE ST IPE160
****
495 525 TO 579 1122 2110 TO 2117 TAPERED 2.5 0.01 2.5 0.3 0.02 0.3 0.02
496 TO 511 TAPERED 2 0.012 2 0.3 0.016 0.3 0.016
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512 TO 516 519 TO 521 523 524 594 595 TABLE ST UPN240
580 TO 593 596 TO 605 TABLE ST UPN200
517 518 606 TO 628 TABLE ST UPN160
629 TO 678 680 TO 691 TABLE ST UPN120
692 TO 707 TABLE ST IPE550
712 TO 724 TABLE ST IPE450
729 TO 736 TABLE ST IPE400
708 TO 711 725 TO 728 737 TO 750 TABLE ST IPE500
751 TO 769 TABLE ST IPE300
770 TO 773 TABLE ST HE360A
774 775 TABLE ST HE400A
776 TO 782 TABLE ST IPE600
785 TO 793 TABLE LD L130X130X10 SP 0.01
783 784 794 TO 807 TABLE LD L100X100X10 SP 0.01
808 TO 823 TABLE LD L100X100X8 SP 0.01
824 TO 827 TABLE LD L90X90X8 SP 0.01
828 TO 837 TABLE LD L90X90X8 SP 0.01
838 TO 875 879 TO 897 TABLE LD L65X65X6 SP 0.01
898 TO 916 TABLE LD L50X50X6 SP 0.01
917 TO 924 TABLE ST L100X100X8
877 929 TO 939 944 TO 950 TABLE ST L90X90X8
876 878 951 TO 967 TABLE ST L75X75X8
968 TO 990 TABLE ST L65X65X6
991 TO 1010 2109 TABLE ST L65X65X6
1011 TO 1016 1036 1037 1041 1050 1051 TABLE SA L75X75X8
1021 TO 1029 1052 TO 1072 1097 TABLE SA L90X90X8
1073 TO 1084 TABLE SA L130X130X10
1085 TO 1096 TABLE SA L100X100X8
1099 TO 1112 TABLE ST IPE500
1113 TO 1116 TABLE TB IPE500 WP 0.15 TH 0.012 BW 0.15 BT 0.012
94 TO 106 1117 TO 1121 TABLE ST HE300A
MEMBER PROPERTY EUROPEAN
1123 TO 1133 TAPERED 0.25 0.008 0.25 0.15 0.012 0.15 0.012
1134 TO 1171 1250 TABLE ST UPN100
1172 TO 1238 TABLE ST L50X50X6
MEMBER PROPERTY EUROPEAN
1239 TO 1241 TABLE LD L75X75X8 SP 0.01
1017 TO 1020 1039 1040 1043 TO 1045 1048 1049 1098 1242 TO 1246 -
1247 TABLE SA L65X65X6
MEMBER PROPERTY EUROPEAN
39 TO 44 46 TO 50 52 TO 62 TAPERED 0.5 0.01 0.5 0.3 0.016 0.3 0.016
1030 TO 1035 1046 1047 TABLE ST L130X130X10
925 TO 928 940 TO 943 TABLE ST L100X100X10
MEMBER PROPERTY EUROPEAN
2053 TO 2056 PRIS YD 0.65 ZD 0.3
MEMBER PROPERTY EUROPEAN
1893 TO 1910 1912 TO 1917 PRIS YD 0.6 ZD 0.44
MEMBER PROPERTY
79 80 1042 2324 TAPERED 0.5 0.012 0.5 0.3 0.025 0.3 0.025
CONSTANTS
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BETA 90 MEMB 1 TO 11 28 TO 38 83 TO 88 281 286 293 294 341 345 346 -
386 TO 388 393 TO 395 420 TO 422 425 TO 427 434 442 471 TO 476 478 TO 482 -
791 TO 793 826 827 1134 TO 1141 1144 TO 1159 1164 TO 1171 1242 TO 1245
MATERIAL CONCRETE MEMB 1248 1249 1301 TO 1316 1319 1320 1322 1331 TO 1446 -
1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2201 TO 2313 2316 2319 TO 2321 2328 TO 2330
MATERIAL STEEL MEMB 1 TO 44 46 TO 50 52 TO 521 523 TO 678 680 TO 1037 1039 -
1040 TO 1247 1250 2107 TO 2118 2322 TO 2327
SUPPORTS
1 TO 20 120 293 294 330 TO 334 886 888 890 FIXED
MEMBER RELEASE
1 6 28 33 94 100 115 240 255 258 261 297 299 300 316 417 420 435 474 480 493 -
517 596 598 602 605 606 611 615 661 1117 1134 1138 1144 1148 1152 1156 1164 -
1168 1186 1250 1892 1893 1899 1905 1911 1912 1974 2025 2184 2250 2263 2266 -
2270 2272 2276 2279 2280 2284 2285 2287 2290 2291 2293 2295 2297 2298 2302 -
2303 2308 TO 2310 2326 2327 2330 START MY MZ
89 TO 91 241 257 260 265 296 298 491 493 518 597 599 603 604 607 612 616 1186 -
1250 1515 1822 1892 1898 1904 1910 1911 1917 1984 2034 2046 2048 2050 2052 -
2054 2056 2102 2187 2226 2262 2265 2269 2271 2274 2278 2279 2283 2284 2286 -
2289 TO 2291 2294 2296 TO 2298 2302 2305 2308 2309 2312 2326 2327 -
2330 END MY MZ
12 25 73 74 308 512 TO 515 519 TO 521 600 601 727 750 1675 1680 1682 1684 -
1686 1689 1691 1695 1696 1699 START MY MZ
19 24 73 74 308 512 TO 515 519 TO 521 600 601 728 736 750 807 1679 1681 1683 -
1685 1688 1690 1694 1695 1698 1699 END MY MZ
16 17 20 39 57 63 TO 67 70 75 TO 83 85 87 107 111 120 129 136 142 146 149 -
152 153 156 163 165 166 173 178 184 188 193 199 218 227 TO 229 231 232 234 -
235 254 266 272 276 295 301 317 TO 323 325 326 328 TO 334 336 337 339 365 -
366 368 TO 379 389 391 496 499 TO 501 504 507 TO 509 516 523 525 526 536 -
537 553 554 594 595 628 637 638 641 643 654 655 658 659 668 TO 672 -
676 TO 678 692 700 708 712 713 725 729 731 734 735 737 739 741 748 -
751 TO 754 759 TO 761 764 767 770 TO 776 785 796 800 801 804 808 TO 810 824 -
916 1023 1027 1066 1113 1115 1123 1126 1129 START MY MZ
27 44 50 56 62 TO 66 69 72 75 77 79 81 84 86 88 110 114 135 141 145 148 151 -
152 155 162 164 168 172 177 183 187 192 227 229 TO 233 235 254 271 275 277 -
301 317 TO 320 322 TO 327 329 TO 334 340 365 367 TO 373 379 498 TO 500 503 -
506 TO 508 511 516 524 535 551 552 578 579 594 595 628 643 644 647 648 658 -
659 662 663 668 671 TO 674 680 TO 682 699 707 709 723 724 726 730 732 733 -
738 740 747 749 751 TO 753 758 761 764 767 TO 775 782 786 797 802 803 811 -
812 TO 813 825 915 1026 1027 1068 1125 1128 1133 2111 2322 TO 2324 -
2325 END MY MZ
89 TO 92 237 242 245 248 251 278 TO 282 289 291 293 302 304 306 341 TO 343 -
350 351 355 TO 358 363 385 TO 388 396 TO 402 412 418 419 425 TO 427 -
430 TO 434 436 TO 440 442 TO 445 447 448 457 458 463 465 467 469 477 483 -
580 583 586 589 592 608 TO 610 613 614 617 TO 619 621 622 629 631 632 635 -
636 639 642 645 649 TO 651 656 657 660 666 684 685 687 TO 689 691 898 901 -
903 905 907 909 911 948 2275 2299 2306 START MY MZ
93 239 244 247 250 253 303 305 307 350 353 364 412 431 436 437 440 -
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444 TO 452 457 458 464 466 468 470 477 483 492 581 584 587 590 593 -
608 TO 610 613 614 617 618 620 626 627 630 631 634 TO 636 640 642 646 649 -
650 653 660 661 664 665 667 685 TO 690 899 902 904 906 908 910 913 949 2292 -
2301 2307 END MY MZ
309 311 312 349 675 683 START MP 0.99
311 TO 313 354 675 683 END MP 0.99
MEMBER TRUSS
783 784 787 TO 790 794 795 798 799 805 806 814 TO 823 828 TO 837 872 873 876 -
877 TO 884 887 TO 897 900 912 914 917 TO 919 929 TO 932 934 935 939 946 950 -
959 968 TO 977 1003 1010 TO 1022 1024 1025 1028 1029 1036 1037 1039 TO 1045 -
1048 TO 1065 1067 1069 TO 1098 1172 1181 TO 1184 1189 1199 TO 1214 1223 1225 -
1228 1237 TO 1247
MEMBER TRUSS
838 TO 875 885 886 920 TO 928 936 TO 938 940 TO 945 951 TO 958 960 TO 967 -
980 TO 1009 1030 TO 1035 1046 1047 1173 TO 1180 1187 1188 1191 TO 1198 1215 -
1216 TO 1222 1224 1227 1229 TO 1236 2109
****LOAD APPLICATION
MEMBER OFFSET
496 TO 511 START 0 -0.5 0
496 TO 511 END 0 -0.5 0
DEFINE IBC 2003
SDS 0.066 SD1 0.0133 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.619 PZ 0.619
SELFWEIGHT 1
JOINT WEIGHT
*3130 FDR-004 Load AT +916.3 M LEVEL
226 267 WEIGHT 20
261 264 WEIGHT 20
*Tertiary FEEDERS
124 126 128 130 WEIGHT 40
*Secondary FEEDERS
132 134 WEIGHT 150
****HEAD END OF 3110-CNV-002 AT +938.2
520 521 WEIGHT 19
522 523 WEIGHT 54
****HEAD END OF 3130-CNV-002 AT +938.2
472 473 WEIGHT 21
474 475 WEIGHT 41
****Bunker Loads WITH MATERIAL
456 TO 459 WEIGHT 419.4
452 454 WEIGHT 482.2
453 455 WEIGHT 920.2
MEMBER WEIGHT
***Conv Gallery Loads 3110-CNV-001 & 3130-CNV-002
*Dead Load
89 91 CON 28.8
90 92 93 CON 33.3
*Live Load (25%)
89 91 CON 5
90 92 93 CON 10.4
*Conveyor Dead Load
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89 91 CON 15.5
90 92 93 CON 22.9
*Conveyor Material Load (25%)
89 91 CON 4
90 92 93 CON 5.7
***Short Post Dead Load
596 TO 599 602 TO 605 CON 10
***Short Post Live Load (25%)
596 TO 599 602 TO 605 CON 1.3
*** Walk way Dead Load
517 518 596 TO 599 602 TO 607 611 612 615 616 UNI 0.5
*** Walk way Live Load (25%)
517 518 596 TO 599 602 TO 607 611 612 615 616 UNI 0.5
*****Stair case Dead Load
317 TO 320 329 TO 333 CON 1.8 0.2
317 TO 320 329 TO 333 CON 1.8 1.4
301 317 TO 320 329 TO 333 CON 1.8 1.7
301 317 TO 320 329 TO 333 CON 1.8 2.9
***Front Stair case Dead Load
1657 1822 2100 TO 2103 2226 CON 1.8 0.3
1657 1822 2100 TO 2103 2226 CON 1.8 1.3
1657 1822 2100 TO 2103 2226 CON 1.8 1.6
1657 1822 2100 TO 2103 2226 CON 1.8 2.6
*
*****Stair case Live Load (25%)
317 TO 320 329 TO 333 CON 1.8 0.2
317 TO 320 329 TO 333 CON 1.8 1.4
301 317 TO 320 329 TO 333 CON 1.8 1.7
301 317 TO 320 329 TO 333 CON 1.8 2.9
***Front Stair case Live Load (25%)
1657 1822 2100 TO 2103 2226 CON 1.8 0.3
1657 1822 2100 TO 2103 2226 CON 1.8 1.3
1657 1822 2100 TO 2103 2226 CON 1.8 1.6
1657 1822 2100 TO 2103 2226 CON 1.8 2.6
*3130 FDR-004 Load AT +916.3 M LEVEL
1695 1699 CON 20 2.035
1691 1696 CON 15.3 0.4
1691 1696 CON 15.3 3.4
1691 1696 CON 15.3 6.4
1692 1697 CON 15.3 1.8
1692 1697 CON 15.3 4.8
1694 CON 15.3 0.05
1694 CON 15.3 3.05
1698 CON 15.3 1.55
1698 CON 15.3 4.55
****3120 FDR-002
***TAIL END
1682 1687 CON 20 3.65
1682 1687 CON 20 5.65
1683 1688 CON 20 1.5
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**IMPACT ZONE
1684 1689 CON 16.7 2.5
1685 1690 CON 16.7 0.57
1685 1690 CON 16.7 3.57
****HEAD END OF 3110-CNV-002 AT +938.2
109 113 CON 79 2.25
****HEAD END OF 3130-CNV-002 AT +938.2
140 CON 41 0.3
144 CON 41 2
****Motor n Gearbox load
365 CON 21 4.9
324 CON 21 2.375
134 CON 25.5 0.15
229 CON 25.5 4.5
669 670 CON 26.25 0.1
151 CON 26.25 0.4
152 CON 26.25 3.91
****Chute Load P15& P16
138 144 CON 15 0.9
109 113 CON 20 1
***Secondary Screen Load
1980 2233 CON 154.53 0.7
1983 2236 CON 222.93 0.5
****Empty Chute load
****P15 P16 & P17
2290 CON 20 1.115
2290 CON 20 2.465
2290 CON 20 3.81
2233 2234 CON 20 1.2
2233 CON 20 2.3
1980 CON 20 1.2
1980 CON 20 2.3
1980 CON 20 3.6
*****P20 & P21
2309 CON 15 1.11
2309 CON 15 2.465
2309 CON 15 3.81
1976 2229 2328 2329 CON 15 0.8
1976 2229 2328 2329 CON 15 1.6
1976 2229 2328 2329 CON 15 2.4
1976 2229 2328 2329 CON 15 3.2
1976 2229 2328 2329 CON 15 3.9
**** Tertiary Screen Load
1895 1901 1907 1914 CON 35 1.35
1896 1902 1908 1915 CON 35 0.15
**** EMpty Chute Load
****P22 P23 & P24
2164 2169 2279 2284 CON 15 0.8
2164 2169 2279 2284 CON 15 2.3
1895 1901 1907 1914 CON 15 2.05
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1895 1901 1907 1914 CON 15 2.85
****P25 P26 & P27
1893 1899 1905 1912 2253 2258 2297 2298 CON 10 0.8
1893 1899 1905 1912 2253 2297 2298 CON 10 2.3
*Tertiary FEEDERS
2046 2048 2050 2052 CON 40 1.5
*Secondary FEEDERS
2054 2056 CON 150 3
FLOOR WEIGHT
*Floor DL
_FL_916.3 FLOAD 3.6
_FL_916.3_2 FLOAD 4.2
_FL_922.75_1 FLOAD 4.2
_FL_922.75_2 FLOAD 4.2
_FL_926.5_1 FLOAD 4.2
****steel Floor
_FL_926.5_2 FLOAD 0.8
_FL_938.2_1 FLOAD 0.8
_FL_938.2_2 FLOAD 0.8
YRANGE 12.2 12.3 FLOAD 0.8 XRANGE 2 12.7 ZRANGE 1.1 20.3
YRANGE 15.15 15.25 FLOAD 0.8 XRANGE 7.35 12.7 ZRANGE 20.2 22.8
YRANGE 14.65 14.75 FLOAD 0.8 XRANGE 12.6 15.8 ZRANGE 1.1 13.99
**Floor 25% Of LL
_FL_916.3 FLOAD 1.25
_FL_916.3_2 FLOAD 1.25
_FL_922.75_1 FLOAD 1.25
_FL_922.75_2 FLOAD 1.25
_FL_926.5_1 FLOAD 1.25
****steel Floor
*
_FL_938.2_1 FLOAD 1.25
_FL_938.2_2 FLOAD 1.25
YRANGE 12.2 12.3 FLOAD 0.5 XRANGE 2 12.7 ZRANGE 1.1 20.3
YRANGE 15.15 15.25 FLOAD 0.5 XRANGE 7.35 12.7 ZRANGE 20.2 22.8
YRANGE 14.65 14.75 FLOAD 0.5 XRANGE 12.6 15.8 ZRANGE 1.1 13.99
*
LOAD 11 EARTHQUAKE LOAD (+X)
IBC LOAD X 1
LOAD 12 EARTHQUAKE LOAD (+Z)
IBC LOAD Z 1
LOAD 1 DEAD LOAD (DL)
SELFWEIGHT Y -1
FLOOR LOAD
_FL_916.3 FLOAD -3.6 GY
_FL_916.3_2 FLOAD -4.2 GY
_FL_922.75_1 FLOAD -4.2 GY
_FL_922.75_2 FLOAD -4.2 GY
_FL_926.5_1 FLOAD -4.2 GY
****steel Floor
_FL_926.5_2 FLOAD -0.8 GY
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_FL_938.2_1 FLOAD -0.8 GY
_FL_938.2_2 FLOAD -0.8 GY
_FL_925.35 FLOAD -0.8 GY
YRANGE 12.2 12.3 FLOAD -0.8 XRANGE 2 12.7  ZRANGE 1.1 20.3  GY
YRANGE 15.15 15.25 FLOAD -0.8 XRANGE 7.35 12.7  ZRANGE 20.2 22.8  GY
YRANGE 14.65 14.75 FLOAD -0.8 XRANGE 12.6 15.8  ZRANGE 1.1 13.99  GY
****
***Conv Gallery Loads 3110-CNV-002 & 3130-CNV-002
MEMBER LOAD
89 91 CON GY -28.8
89 91 CON GZ 1.6
90 CON GY -33.3
92 CON GY -33.3 0.49
90 CON GZ 1.6
92 CON GZ 1.6 0.49
***Short Post Load
596 TO 599 602 TO 605 CON GY -10
*** Walk way Load
517 518 596 TO 599 602 TO 607 611 612 615 616 UNI GY -0.5
*****Stair case Load
317 TO 320 329 TO 333 CON GY -1.8 0.2
317 TO 320 329 TO 333 CON GY -1.8 1.4
301 317 TO 320 329 TO 333 CON GY -1.8 1.7
301 317 TO 320 329 TO 333 CON GY -1.8 2.9
***Front Stair case
1657 1822 2100 TO 2103 2226 CON GY -1.8 0.3
1657 1822 2100 TO 2103 2226 CON GY -1.8 1.3
1657 1822 2100 TO 2103 2226 CON GY -1.8 1.6
1657 1822 2100 TO 2103 2226 CON GY -1.8 2.6
***Load from ground portals 3130 CNV-003
1506 1512 1517 CON GY -10 0.4
1506 1512 1517 CON GY -10 4.5
LOAD 2 LIVE LOAD (LL)
FLOOR LOAD
_FL_916.3 FLOAD -5 GY
_FL_916.3_2 FLOAD -5 GY
_FL_922.75_1 FLOAD -5 GY
_FL_922.75_2 FLOAD -5 GY
_FL_926.5_1 FLOAD -5 GY
****steel Floor
_FL_926.5_2 FLOAD -5 GY
_FL_938.2_1 FLOAD -5 GY
_FL_938.2_2 FLOAD -5 GY
_FL_925.35 FLOAD -2 GY
YRANGE 12.2 12.3 FLOAD -2 XRANGE 2 12.7  ZRANGE 1.1 20.3  GY
YRANGE 15.15 15.25 FLOAD -2 XRANGE 7.35 12.7  ZRANGE 20.2 22.8  GY
YRANGE 14.65 14.75 FLOAD -2 XRANGE 12.6 15.8  ZRANGE 1.1 13.99  GY
****
***Conv Gallery Loads 3110-CNV-002 & 3130-CNV-002
MEMBER LOAD
```

```
89 91 CON GY -20
89 91 CON GZ 0.7
90 CON GY -41.3
92 CON GY -41.3 0.49
90 CON GZ 0.7
92 CON GZ 0.7 0.49
MEMBER LOAD
***Short Post Load
596 TO 599 602 TO 605 CON GY -5.5
***Walkway Load
517 518 596 TO 599 602 TO 607 611 612 615 616 UNI GY -2
*****Stair case Load
317 TO 320 329 TO 333 CON GY -7.2 0.2
317 TO 320 329 TO 333 CON GY -7.2 1.4
301 317 TO 320 329 TO 333 CON GY -7.2 1.7
301 317 TO 320 329 TO 333 CON GY -7.2 2.9
***Front Stair case
1657 1822 2100 TO 2103 2226 CON GY -7.2 0.3
1657 1822 2100 TO 2103 2226 CON GY -7.2 1.3
1657 1822 2100 TO 2103 2226 CON GY -7.2 1.6
1657 1822 2100 TO 2103 2226 CON GY -7.2 2.6
***Load from ground portals 3130 CNV-003
1506 1512 1517 CON GY -27 0.4
1506 1512 1517 CON GY -27 4.5
LOAD 3 SUPERIMPOSED DEAD LOAD 1 (SDL1) - (BUNKER EMPTY & CHUTE EMPTY CONDITION)
****3130 FDR-004 Load AT +916.3 M LEVEL
JOINT LOAD
226 267 FX -2 FY -20 FZ -20
MEMBER LOAD
1695 1699 CON GX -2 2.035
1695 1699 CON GY -20 2.035
1695 1699 CON GZ -20 2.035
JOINT LOAD
261 264 FX -2 FY -20 FZ -20
MEMBER LOAD
1691 1696 CON GY -9.8 0.4
1691 1696 CON GY -9.8 3.4
1691 1696 CON GY -9.8 6.4
1692 1697 CON GY -9.8 1.8 0
1692 1697 CON GY -9.8 4.8
1694 CON GY -9.8 0.05
1694 CON GY -9.8 3.05
1698 CON GY -9.8 1.55 0
1698 CON GY -9.8 4.55
****Along
1691 1696 CON GZ -1.7 0.4
1691 1696 CON GZ -1.7 3.4
1691 1696 CON GZ -1.7 6.4
1692 1697 CON GZ -1.7 1.8 0
1692 1697 CON GZ -1.7 4.8
```

```
1694 CON GZ -1.7 0.05
1694 CON GZ -1.7 3.05
1698 CON GZ -1.7 1.55 0
1698 CON GZ -1.7 4.55
****ACROSS
1691 1696 CON GX -1 0.4
1691 1696 CON GX -1 3.4
1691 1696 CON GX -1 6.4
1692 1697 CON GX -1 1.8 0
1692 1697 CON GX -1 4.8
1694 CON GX -1 0.05
1694 CON GX -1 3.05
1698 CON GX -1 1.55 0
1698 CON GX -1 4.55
****3120 FDR-002 Load AT +916.3 M LEVEL
***TAIL END VERTICAL
1682 1687 CON GY -20 3.65
1682 1687 CON GY -20 5.65
1683 1688 CON GY -20 1.5
***TAIL END ACROSS
1682 1687 CON GX -2 3.65
1682 1687 CON GX -2 5.65
1683 1688 CON GX -2 1.5
***TAIL END ALONG
1682 1687 CON GZ 30 3.65
1682 1687 CON GZ 30 5.65
1683 1688 CON GZ 30 1.5
**IMPACT ZONE
1684 1689 CON GY -11.2 2.5
1685 1690 CON GY -11.2 0.57
1685 1690 CON GY -11.2 3.57
****ACROSS
1684 1689 CON GX -1 2.5
1685 1690 CON GX -1 0.57
1685 1690 CON GX -1 3.57
****Along
1684 1689 CON GZ 1.7 2.5
1685 1690 CON GZ 1.7 0.57
1685 1690 CON GZ 1.7 3.57
****
****VIBRO FEEDERS AT +926.5
*AT FRAME
JOINT LOAD
124 TO 131 FY -40
425 427 429 431 FY -40
MEMBER LOAD
2046 2048 2050 2052 CON GY -40 1.5
770 TO 773 CON GY -40 0.5
2053 2055 CON GY -150 0.5
2054 2056 CON GY -150 2.5
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774 775 CON GY -150 2
JOINT LOAD
133 135 FY -150
****HEAD END OF 3110-CNV-002 AT +938.2
****CASE II-STARTING, CHUTE EMPTY
JOINT LOAD
520 521 FX 56 FY 38 FZ 4
MEMBER LOAD
109 113 CON GY -81 2.25
109 113 CON GX 19 2.25
109 113 CON GZ 8.2 2.25
JOINT LOAD
522 523 FX 84 FY -75 FZ 7.5
****HEAD END OF 3130-CNV-002 AT +938.2
****CASE II-STARTING, CHUTE EMPTY
JOINT LOAD
472 473 FX 26 FY 16 FZ 2
474 475 FX 52 FY -59 FZ 6
MEMBER LOAD
140 CON GY -36 0.3
144 CON GY -36 2
140 CON GX 15 0.3
144 CON GX 15 2
140 CON GZ 3.6 0.3
144 CON GZ 3.6 2
****
****Motor n Gearbox load
365 CON GY -21 4.9
324 CON GY -21 2.375
134 CON GY -25.5 0.15
229 CON GY -25.5 4.5 0
669 670 CON GY -26.25 0.1
151 CON GY -26.25 0.4
152 CON GY -26.25 3.91
****Chute Load P15& P16
138 144 CON GY -15 0.9
109 113 CON GY -20 1
****
***Conv Gallery Loads 3110-CNV-001 & 3130-CNV-002
*CDL
MEMBER LOAD
89 91 CON GY -15.5
89 91 CON GZ -0.3
90 CON GY -22.9
92 CON GY -22.9 0.49
90 CON GZ 3.8
92 CON GZ 3.8 0.49
*ML
MEMBER LOAD
89 91 CON GY -15
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89 91 CON GZ -1.7
90 CON GY -22.9
92 CON GY -22.9 0.49
90 CON GZ 1.7
92 CON GZ 1.7 0.49
****
****Loading at 922.75m
***Secondary Screen Load
MEMBER LOAD
1980 2233 CON GY -144.14 0.7
1983 2236 CON GY -196.3 0.5
***Drive Load
MEMBER LOAD
2293 2295 CON GY -3.15 0.5
****Empty Chute load
****P15 P16 & P17
2290 CON GY -20 1.115
2290 CON GY -20 2.465
2290 CON GY -20 3.81
2233 2234 CON GY -20 1.2
2233 CON GY -20 2.3
1980 CON GY -20 1.2
1980 CON GY -20 2.3
1980 CON GY -20 3.6
*****P20 & P21
2309 CON GY -15 1.11
2309 CON GY -15 2.465
2309 CON GY -15 3.81
1976 2229 CON GY -15 0.8
1976 2229 CON GY -15 1.6
2329 CON GY -15 0.25
2328 CON GY -15 0.25
2329 CON GY -15 1.05
2328 CON GY -15 1.05
2329 CON GY -15 1.75
2328 CON GY -15 1.75
**** Tertiary Screen Load
1895 1901 1907 1914 CON GY -35 1.35
1896 1902 1908 1915 CON GY -35 0.15
**** EMpty Chute Load
****P22 P23 & P24
2164 2169 2279 2284 CON GY -15 0.8
2164 2169 2279 2284 CON GY -15 2.3
1895 1901 1907 1914 CON GY -15 2.05
1895 1901 1907 1914 CON GY -15 2.85
****P25 P26 & P27
1893 1899 1905 1912 2253 2258 2297 2298 CON GY -10 0.8
1893 1899 1905 1912 2253 2297 2298 CON GY -10 2.3
JOINT LOAD
153 FY -10
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******
******Load of Empty chute at 920.55m lvl
****P1 P2 P3
MEMBER LOAD
68 CON GY -7.5 0.69
71 CON GY -7.5 0.69
68 CON GY -7.5 2.19
71 CON GY -7.5 2.19
73 74 CON GY -7.5 1.25
73 74 CON GY -7.5 1.75
73 74 CON GY -7.5 2.25
****P4 & P5
63 64 CON GY -15 1.715
63 64 CON GY -15 3.215
15 23 CON GY -15
****P6 P7 P8
65 66 CON GY -15 1.715
65 66 CON GY -15 3.215
13 21 CON GY -15 0.65
13 21 CON GY -15 1.55
****P9 P10 P11
75 TO 78 CON GY -10 0.775
75 TO 78 CON GY -10 1.775
43 49 55 61 CON GY -10 0.8
43 49 55 61 CON GY -10 2
****P12 P13 P14
79 TO 82 CON GY -15 0.775
79 TO 82 CON GY -15 1.775
40 46 52 58 CON GY -15
****P28 P29 P30 & P31
464 466 468 470 CON GY -5
457 458 CON GY -5 1.2
457 458 CON GY -5 2.3
238 311 312 CON GY -5
238 311 312 CON GY -5 1.45
238 311 312 CON GY -5 3.05
*****P32 P33 P34
243 246 249 252 CON GY -21 0.85
243 246 249 252 CON GY -21 2.35
635 636 649 650 CON GY -21
***Machine Loading Area
FLOOR LOAD
YRANGE 9.7 9.8 FLOAD -4 XRANGE -0.1 5.4  ZRANGE -0.1 7.65  GY
YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 0 3.8  GY
YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 20.1 24.3  GY
MEMBER LOAD
321 325 326 CON GY -12 1.5
231 232 CON GY -12 4
230 CON GY -12 0.9
****Bunker WEIGHT IN EMPTY
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JOINT LOAD
456 TO 459 FY -56.7
452 454 FY -69
453 455 FY -108.1
***Duct Loads
MEMBER LOAD
1142 CON GY -11
1162 CON GY -9
1162 CON GY -9 0
535 536 565 566 CON GY -5
535 536 565 566 CMOM GZ 6.5
299 CON GY -27 1.3
300 CON GY -52 1.3
316 CON GY -75 1.3
477 483 CON GY -15
240 CON GY -75
350 351 CON GY -25
***Monorail at 916
2217 CON GY -42.5 0.3
1892 1911 2208 CON GY -42.5 2.4
2242 CON GY -30 3.8
2329 CON GY -30 1.65
***Load from ground portals 3130 CNV-003
1506 1512 1517 CON GY -55 0.4
1506 1512 1517 CON GY -55 4.5
LOAD 4 SUPERIMPOSED DEAD LOAD 2 (SDL2) - (BUNKER & CHUTE FILLED CONDITION)
**** WITH MATERIAL LOAD
****3130 FDR-004
JOINT LOAD
226 267 FX 2 FY -20 FZ -20
MEMBER LOAD
1695 1699 CON GX 2 2.035
1695 1699 CON GY -20 2.035
1695 1699 CON GZ -20 2.035
JOINT LOAD
261 264 FX 2 FY -20 FZ -20
MEMBER LOAD
1691 1696 CON GY -15.3 0.4
1691 1696 CON GY -15.3 3.4
1691 1696 CON GY -15.3 6.4
1692 1697 CON GY -15.3 1.8 0
1692 1697 CON GY -15.3 4.8
1694 CON GY -15.3 0.05
1694 CON GY -15.3 3.05
1698 CON GY -15.3 1.55 0
1698 CON GY -15.3 4.55
****Along
1691 1696 CON GZ -1.7 0.4
1691 1696 CON GZ -1.7 3.4
1691 1696 CON GZ -1.7 6.4
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1692 1697 CON GZ -1.7 1.8 0
1692 1697 CON GZ -1.7 4.8
1694 CON GZ -1.7 0.05
1694 CON GZ -1.7 3.05
1698 CON GZ -1.7 1.55 0
1698 CON GZ -1.7 4.55
****ACROSS
1691 1696 CON GX 1 0.4
1691 1696 CON GX 1 3.4
1691 1696 CON GX 1 6.4
1692 1697 CON GX 1 1.8 0
1692 1697 CON GX 1 4.8
1694 CON GX 1 0.05
1694 CON GX 1 3.05
1698 CON GX 1 1.55 0
1698 CON GX 1 4.55
****3120 FDR-002
***TAIL END VERTICAL
1682 1687 CON GY -20 3.65
1682 1687 CON GY -20 5.65
1683 1688 CON GY -20 1.5
***TAIL END ACROSS
1682 1687 CON GX 2 3.65
1682 1687 CON GX 2 5.65
1683 1688 CON GX 2 1.5
***TAIL END ALONG
1682 1687 CON GZ 30 3.65
1682 1687 CON GZ 30 5.65
1683 1688 CON GZ 30 1.5
**IMPACT ZONE
1684 1689 CON GY -16.7 2.5
1685 1690 CON GY -16.7 0.57
1685 1690 CON GY -16.7 3.57
****ACROSS
1684 1689 CON GX 1 2.5
1685 1690 CON GX 1 0.57
1685 1690 CON GX 1 3.57
****Along
1684 1689 CON GZ 1.7 2.5
1685 1690 CON GZ 1.7 0.57
1685 1690 CON GZ 1.7 3.57
****VIBRO FEEDERS AT +926.5
*AT MID SPAN (Max B.M. Position)
770 TO 773 2046 2048 2050 2052 CON GY -40 2.625
770 TO 773 2046 2048 2050 2052 CON GY -40 4.125
774 775 2054 2056 CON GY -150 2.75
774 775 2054 2056 CON GY -150 5.75
****HEAD END OF 3110-CNV-002 AT +938.2
****CASE I-RUNNING, MATERIAL LOAD
JOINT LOAD
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520 521 FX 23 FY -19 FZ 2
522 523 FX 61 FY -54 FZ 5.5
MEMBER LOAD
109 113 CON GY -79 2.25
109 113 CON GX 23 2.25
109 113 CON GZ 8 2.25
****HEAD END OF 3130-CNV-002 AT +938.2
****CASE I-RUNNING, MATERIAL LOAD
JOINT LOAD
472 473 FX 12 FY -21 FZ 2
474 475 FX 41 FY -46 FZ 5
MEMBER LOAD
140 CON GY -41 0.3
144 CON GY -41 2
140 CON GX 16 0.3
144 CON GX 16 2
140 CON GZ 4 0.3
144 CON GZ 4 2
****
****Motor n Gearbox load
365 CON GY -21 4.9
324 CON GY -21 2.375
134 CON GY -25.5 0.15
229 CON GY -25.5 4.5 0
669 670 CON GY -26.25 0.1
151 CON GY -26.25 0.4
152 CON GY -26.25 3.91
****Chute Load P15& P16
138 144 CON GY -15 0.9
109 113 CON GY -20 1
***Conv Gallery Loads 3110-CNV-002 & 3130-CNV-002
*CDL
MEMBER LOAD
89 91 CON GY -15.5
89 91 CON GZ -0.3
90 CON GY -22.9
92 CON GY -22.9 0.49
90 CON GZ 3.8
92 CON GZ 3.8 0.49
*ML
MEMBER LOAD
89 91 CON GY -15
89 91 CON GZ -1.7
90 CON GY -22.9
1048 CON GY -22.9 0.285
90 CON GZ 1.7
1048 CON GZ 1.7 0.285
****
***Secondary Screen Load
MEMBER LOAD
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1980 2233 CON GY -154.53 0.7
1983 2236 CON GY -222.93 0.5
***Drive Load
MEMBER LOAD
2293 2295 CON GY -3.15 0.5
****Blocked Chute load
****P15 P16 & P17
2290 CON GY -45 1.115
2290 CON GY -45 2.465
2290 CON GY -45 3.81
2233 2234 CON GY -45 1.2
2233 CON GY -45 2.3
1980 CON GY -45 1.2
1980 CON GY -45 2.3
1980 CON GY -45 3.6
*****P20 & P21
2309 CON GY -60 1.11
2309 CON GY -60 2.465
2309 CON GY -60 3.81
1976 2229 CON GY -60 0.8
1976 2229 CON GY -60 1.6
2329 CON GY -60 0.25
2328 CON GY -60 0.25
2329 CON GY -60 1.05
2328 CON GY -60 1.05
2329 CON GY -60 1.75
2328 CON GY -60 1.75
**** Tertiary Screen Load
1895 1901 1907 1914 CON GY -35 1.35
***Dynamic Load
1895 1901 1907 1914 CON GY -12.24 1.35
1895 1901 1907 1914 CON GX 6.12 1.35
***Dynamic Load
1896 1902 1908 1915 CON GY -35 0.15
1896 1902 1908 1915 CON GY -12.24 0.15
1896 1902 1908 1915 CON GX 6.12 0.15
****Blocked Chute Load
****P22 P23 & P24
2164 2169 2279 2284 CON GY -50 0.8
2164 2169 2279 2284 CON GY -50 2.3
1895 1901 1907 1914 CON GY -50 2.05
1895 1901 1907 1914 CON GY -50 2.85
****P25 P26 & P27
1893 1899 1905 1912 2253 2258 2297 2298 CON GY -30 0.8
1893 1899 1905 1912 2253 2297 2298 CON GY -30 2.3
JOINT LOAD
153 FY -30
******Load of Blocked chute at 920.55m lvl
****P1 P2 P3
MEMBER LOAD
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68 CON GY -55 0.69
71 CON GY -55 0.69
68 CON GY -55 2.19
71 CON GY -55 2.19
73 74 CON GY -55 1.25
73 74 CON GY -55 1.75
73 74 CON GY -55 2.25
****P4 & P5
63 64 CON GY -60 1.715
63 64 CON GY -60 3.215
15 23 CON GY -60
****P6 P7 P8
65 66 CON GY -60 1.715
65 66 CON GY -60 3.215
13 21 CON GY -60 0.65
13 21 CON GY -60 1.55
****P9 P10 P11
75 TO 78 CON GY -50 0.775
75 TO 78 CON GY -50 1.775
43 49 55 61 CON GY -50 0.8
43 49 55 61 CON GY -50 2
****P12 P13 P14
79 TO 82 CON GY -60 0.75
79 TO 82 CON GY -60 1.775
40 46 52 58 CON GY -60
****P28 P29 P30 & P31
464 466 468 470 CON GY -5
457 458 CON GY -15 1.2
457 458 CON GY -15 2.3
238 311 312 CON GY -15
238 311 312 CON GY -15 1.45
238 311 312 CON GY -15 3.05
*****P32 P33 P34
243 246 249 252 CON GY -35 0.85
243 246 249 252 CON GY -35 2.35
635 636 649 650 CON GY -35
***Machine Loading Area
FLOOR LOAD
YRANGE 9.7 9.8 FLOAD -4 XRANGE -0.1 5.4  ZRANGE -0.1 7.65  GY
YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 0 3.8  GY
YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 20.1 24.3  GY
MEMBER LOAD
321 325 326 CON GY -12 1.5
231 232 CON GY -12 4
230 CON GY -12 0.9
****Bunker Loads WITH MATERIAL
JOINT LOAD
456 TO 459 FY -419.4
452 454 FY -482.2
453 455 FY -920.2
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*Monorail Load
MEMBER LOAD
708 710 725 727 CON GY -114 3.35
750 CON GY -114
***Duct Loads
MEMBER LOAD
1142 CON GY -11
1162 CON GY -9
1162 CON GY -9 0
535 536 565 566 CON GY -5
535 536 565 566 CMOM GZ 6.5
299 CON GY -27 1.3
300 CON GY -52 1.3
316 CON GY -75 1.3
477 483 CON GY -15
240 CON GY -75
350 351 CON GY -25
***Load from ground portals 3130 CNV-003
1506 1512 1517 CON GY -55 0.4
1506 1512 1517 CON GY -55 4.5
LOAD 5 WIND LOAD IN +X DIRN
MEMBER LOAD
193 TO 208 UNI GX 0.66
94 TO 106 209 TO 217 1336 TO 1338 1340 TO 1344 1392 TO 1394 1434 TO 1439 -
1440 UNI GX 2.19
1 TO 5 28 TO 32 1099 TO 1105 1113 1114 1332 TO 1334 1358 TO 1361 1363 TO 1367 -
1369 TO 1374 2313 UNI GX 2.76
6 TO 11 33 TO 38 1106 TO 1112 1115 1116 1302 TO 1304 1376 TO 1380 -
1382 TO 1387 1396 TO 1398 1417 TO 1420 2316 UNI GX 2.54
120 TO 128 1306 TO 1309 1347 TO 1350 1400 TO 1403 1422 TO 1425 1442 TO 1446 -
2319 UNI GX 2.11
115 TO 119 218 TO 226 1117 TO 1121 1311 TO 1316 1352 TO 1356 1405 TO 1410 -
1412 TO 1415 1427 TO 1432 UNI GX 0.81
*WL on Bunkers
525 527 529 531 533 535 536 538 539 541 543 545 547 549 551 553 555 557 559 -
561 563 565 566 568 570 TO 572 574 576 578 UNI GX 3.06
507 UNI GX 4.39
499 UNI GX 5.83
***WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
520 521 FX 2.6 FY 3.71
522 523 FX 2.6 FY -3.71
MEMBER LOAD
109 113 CON GX 2.6 2.25
****WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
472 473 FX 1.86 FY 2.07
474 475 FX 1.86 FY -2.07
MEMBER LOAD
140 CON GX 1.86 0.3
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144 CON GX 1.86 2
***WL on VIBRO FEEDERS AT +926.5
*Tertiary
770 TO 773 CON GX 1.94 2.375
770 TO 773 CON GX 1.94 3.875
770 TO 773 CON GY 1.68 2.375
770 TO 773 CON GY -1.68 3.875
*Secondary
774 775 CON GX 3.05 1.5
774 775 CON GX 3.05 4.5
774 775 CON GY 6.85 1.5
774 775 CON GY -6.85 4.5
***WL on Secondary Screen
1980 2233 CON GX 10.11 0.45
1983 2236 CON GX 10.11 0.5
1980 2233 CON GY 10.86 0.45
1983 2236 CON GY -10.86 0.5
****WL on Tertiary Screen
1895 1901 1907 1914 CON GX 3.41 1.35
1896 1902 1908 1915 CON GX 3.41 0.15
1895 1901 1907 1914 CON GY 5.33 1.35
1896 1902 1908 1915 CON GY -5.33 0.15
***WL on Secondary Chute
2233 2234 CON GX 5.45 1.2
1980 CON GX 5.45 1.2
1980 CON GX 5.45 3.6
***WL on Tertiary Chute
1895 1901 1907 1914 CON GX 5.16 2.05
1895 1901 1907 1914 CON GX 5.16 2.85
***WL ON PLATFORMS
278 TO 287 289 TO 294 341 TO 346 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 1134 TO 1141 -
1144 TO 1159 1164 TO 1171 UNI GX 0.5
LOAD 6 WIND LOAD IN -X DIRN
MEMBER LOAD
193 TO 208 UNI GX -0.66
94 TO 106 209 TO 217 1336 TO 1338 1340 TO 1344 1392 TO 1394 1434 TO 1439 -
1440 UNI GX -2.19
1 TO 5 28 TO 32 1099 TO 1105 1113 1114 1332 TO 1334 1358 TO 1361 1363 TO 1367 -
1369 TO 1374 2313 UNI GX -2.76
6 TO 11 33 TO 38 1106 TO 1112 1115 1116 1302 TO 1304 1376 TO 1380 -
1382 TO 1387 1396 TO 1398 1417 TO 1420 2316 UNI GX -2.54
120 TO 128 1306 TO 1309 1347 TO 1350 1400 TO 1403 1422 TO 1425 1442 TO 1446 -
2319 UNI GX -2.11
115 TO 119 218 TO 226 1117 TO 1121 1311 TO 1316 1352 TO 1356 1405 TO 1410 -
1412 TO 1415 1427 TO 1432 UNI GX -0.81
*WL on Bunkers
495 526 528 530 532 534 537 540 542 544 546 548 550 552 554 556 558 560 562 -
564 567 569 573 575 577 579 933 1122 2110 TO 2117 UNI GX -3.06
508 UNI GX -4.39
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500 UNI GX -5.83
***WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
520 521 FX -2.6 FY -3.71
522 523 FX -2.6 FY 3.71
MEMBER LOAD
109 113 CON GX -2.6 2.25
****WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
472 473 FX -1.86 FY -2.07
474 475 FX -1.86 FY 2.07
MEMBER LOAD
140 CON GX -1.86 0.3
144 CON GX -1.86 2
***WL on VIBRO FEEDERS AT +926.5
*Tertiary
770 TO 773 CON GX -1.94 2.375
770 TO 773 CON GX -1.94 3.875
770 TO 773 CON GY -1.68 2.375
770 TO 773 CON GY 1.68 3.875
*Secondary
774 775 CON GX -3.05 1.5
774 775 CON GX -3.05 4.5
774 775 CON GY -6.85 1.5
774 775 CON GY 6.85 4.5
***WL on Secondary Screen
1980 2233 CON GX -10.11 0.45
1983 2236 CON GX -10.11 0.5
1980 2233 CON GY -10.86 0.45
1983 2236 CON GY 10.86 0.5
****WL on Tertiary Screen
1895 1901 1907 1914 CON GX -3.41 1.35
1896 1902 1908 1915 CON GX -3.41 0.15
1895 1901 1907 1914 CON GY -5.33 1.35
1896 1902 1908 1915 CON GY 5.33 0.15
***WL on Secondary Chute
2233 2234 CON GX -5.45 1.2
1980 CON GX -5.45 1.2
1980 CON GX -5.45 3.6
***WL on Tertiary Chute
1895 1901 1907 1914 CON GX -5.16 2.05
1895 1901 1907 1914 CON GX -5.16 2.85
***WL ON PLATFORMS
278 TO 287 289 TO 294 341 TO 346 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 UNI GX -0.5
LOAD 7 WIND LOAD IN +Z DIRN
MEMBER LOAD
1332 TO 1334 1392 TO 1394 1396 TO 1398 1400 TO 1403 1405 TO 1410 UNI GZ 2.06
1336 TO 1338 1358 TO 1361 1389 1390 1417 TO 1420 1422 TO 1425 1427 TO 1430 -
2320 UNI GZ 4.13
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193 TO 195 1340 TO 1343 1347 TO 1349 1352 TO 1354 1363 TO 1365 1376 TO 1378 -
2313 2316 2319 UNI GZ 3.52
28 TO 38 94 TO 99 115 TO 119 196 TO 198 1344 1345 1355 1356 1366 1367 1379 -
1380 UNI GZ 1.46
1 TO 11 100 TO 106 199 TO 208 1117 TO 1121 1369 TO 1374 1382 TO 1387 1412 -
1413 TO 1415 1431 1432 1434 TO 1440 1442 TO 1446 UNI GZ 3.16
120 TO 128 209 TO 226 1099 TO 1116 1302 TO 1304 1306 TO 1309 1311 TO 1315 -
1316 UNI GZ 1.7
*WL on Bunkers
496 TO 498 504 TO 506 UNI GZ 3.06
497 UNI GZ 5.02
505 UNI GZ 3.93
***WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
520 522 FY 3.21 FZ 2.25
521 523 FY -3.21 FZ 2.25
MEMBER LOAD
109 113 CON GZ 2.25 2.25
109 CON GY 3.21 2.25
113 CON GY -3.21 2.25
****WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
472 474 FY 1.49 FZ 1.16
473 475 FY -1.49 FZ 1.16
MEMBER LOAD
140 CON GZ 1.16 0.3
140 CON GY 1.49 0.3
144 CON GZ 1.16 2
144 CON GY -1.49 2
***WL on VIBRO FEEDERS AT +926.5
*Tertiary
770 TO 773 CON GZ 2.02 2.375
770 TO 773 CON GZ 2.02 3.875
770 772 CON GY 1.75 2.375
770 772 CON GY 1.75 3.875
771 773 CON GY -1.75 2.375
771 773 CON GY -1.75 3.875
*Secondary
774 775 CON GZ 3.48 1.5
774 775 CON GZ 3.48 4.5
774 CON GY 2.7 1.5
774 CON GY 2.7 4.5
775 CON GY -2.7 1.5
775 CON GY -2.7 4.5
***WL on Secondary Screen
1980 2233 CON GZ 9.45 0.45
1983 2236 CON GZ 9.45 0.5
1980 CON GY 10.15 0.45
1983 CON GY 10.15 0.5
2233 CON GY -10.15 0.45
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2236 CON GY -10.15 0.5
****WL on Tertiary Screen
1895 1901 1907 1914 CON GZ 6.02 1.35
1896 1902 1908 1915 CON GZ 6.02 0.15
1895 1907 CON GY 9.4 1.35
1896 1908 CON GY 9.4 0.15
1901 1914 CON GY -9.4 1.35
1902 1915 CON GY -9.4 0.15
***WL on Secondary Chute
2233 2234 CON GZ 8.01 1.2
1980 CON GZ 8.01 1.2
1980 CON GZ 8.01 3.6
***WL on Tertiary Chute
1895 1901 1907 1914 CON GZ 8.68 2.05
1895 1901 1907 1914 CON GZ 8.68 2.85
***WL ON PLATFORMS
278 TO 287 289 TO 294 341 TO 348 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 UNI GZ 0.5
**WL FROM CONVEYOR GALLERY 3110-002 & 3130-002
*JOINT LOAD
*745 758 FY 59.1 FZ 19.3
*749 760 FY -59.1 FZ 19.3
MEMBER LOAD
89 91 CON GY 59.1
89 91 CON GZ 19.3
90 CON GY -59.1
92 CON GY -59.1 0.49
90 CON GZ 19.3
92 CON GZ 19.3 0.49
LOAD 8 WIND LOAD IN -Z DIRN
MEMBER LOAD
1332 TO 1334 1392 TO 1394 1396 TO 1398 1400 TO 1403 1405 TO 1410 UNI GZ -2.06
1336 TO 1338 1358 TO 1361 1389 1390 1417 TO 1420 1422 TO 1425 1427 TO 1430 -
2320 UNI GZ -4.13
193 TO 195 1340 TO 1343 1347 TO 1349 1352 TO 1354 1363 TO 1365 1376 TO 1378 -
2313 2316 2319 UNI GZ -3.52
28 TO 38 94 TO 99 115 TO 119 196 TO 198 1344 1345 1355 1356 1366 1367 1379 -
1380 UNI GZ -1.46
1 TO 11 100 TO 106 199 TO 208 1117 TO 1121 1369 TO 1374 1382 TO 1387 1412 -
1413 TO 1415 1431 1432 1434 TO 1440 1442 TO 1446 UNI GZ -3.16
120 TO 128 209 TO 226 1099 TO 1116 1302 TO 1304 1306 TO 1309 1311 TO 1315 -
1316 UNI GZ -1.7
*WL on Bunkers
501 TO 503 509 TO 511 UNI GZ -3.06
502 UNI GZ -5.02
510 UNI GZ -3.93
***WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
520 522 FY -3.21 FZ -2.25
521 523 FY 3.21 FZ -2.25
```

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MEMBER LOAD
109 113 CON GZ -2.25 2.25
109 CON GY -3.21 2.25
113 CON GY 3.21 2.25
****WL on HEAD END OF 3110-CNV-002 AT +938.2
JOINT LOAD
472 474 FY -1.49 FZ -1.16
473 475 FY 1.49 FZ -1.16
MEMBER LOAD
140 CON GZ -1.16 0.3
140 CON GY -1.49 0.3
144 CON GZ -1.16 2
144 CON GY 1.49 2
***WL on VIBRO FEEDERS AT +926.5
*Tertiary
770 TO 773 CON GZ -2.02 2.375
770 TO 773 CON GZ -2.02 3.875
770 772 CON GY -1.75 2.375
770 772 CON GY -1.75 3.875
771 773 CON GY 1.75 2.375
771 773 CON GY 1.75 3.875
*Secondary
774 775 CON GZ -3.48 1.5
774 775 CON GZ -3.48 4.5
774 CON GY -2.7 1.5
774 CON GY -2.7 4.5
775 CON GY 2.7 1.5
775 CON GY 2.7 4.5
***WL on Secondary Screen
1980 2233 CON GZ -9.45 0.45
1983 2236 CON GZ -9.45 0.5
1980 CON GY -10.15 0.45
1983 CON GY -10.15 0.5
2233 CON GY 10.15 0.45
2236 CON GY 10.15 0.5
****WL on Tertiary Screen
1895 1901 1907 1914 CON GZ -6.02 1.35
1896 1902 1908 1915 CON GZ -6.02 0.15
1895 1907 CON GY -9.4 1.35
1896 1908 CON GY -9.4 0.15
1901 1914 CON GY 9.4 1.35
1902 1915 CON GY 9.4 0.15
***WL on Secondary Chute
2233 2234 CON GZ -8.01 1.2
1980 CON GZ -8.01 1.2
1980 CON GZ -8.01 3.6
***WL on Tertiary Chute
1895 1901 1907 1914 CON GZ -8.68 2.05
1895 1901 1907 1914 CON GZ -8.68 2.85
***WL ON PLATFORMS
```

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278 TO 287 289 TO 294 341 TO 348 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 UNI GZ -0.5
**WL FROM CONVEYOR GALLERY 3110-002 & 3130-002
*JOINT LOAD
*745 758 FY -59.1 FZ -19.3
*749 760 FY 59.1 FZ -19.3
MEMBER LOAD
89 91 CON GY -59.1
89 91 CON GZ -19.3
90 CON GY 59.1
92 CON GY 59.1 0.49
90 CON GZ -19.3
92 CON GZ -19.3 0.49
LOAD 9 TEMERATURE LOAD 1
TEMPERATURE LOAD
1 TO 30 32 TO 44 46 TO 50 52 TO 67 70 75 TO 92 94 TO 107 109 TO 111 -
113 TO 118 120 TO 124 133 TO 142 144 TO 193 195 TO 235 237 TO 239 -
242 TO 287 289 TO 298 301 308 TO 314 317 TO 323 325 TO 342 349 352 -
354 TO 417 425 TO 471 496 TO 511 516 TO 518 525 TO 528 530 532 TO 534 536 -
537 539 TO 564 566 567 569 TO 591 594 TO 619 628 TO 633 635 TO 642 -
645 TO 647 649 TO 652 654 TO 660 662 TO 667 669 TO 678 681 TO 696 -
699 TO 704 707 TO 715 720 TO 732 735 TO 743 747 TO 749 751 TO 778 780 782 -
783 TO 790 794 TO 825 828 TO 884 898 899 901 TO 945 951 TO 968 973 974 980 -
981 TO 1009 1011 TO 1037 1039 TO 1045 1052 TO 1097 1099 TO 1133 TEMP 25
LOAD 10 TEMERATURE LOAD 2
TEMPERATURE LOAD
1 TO 30 32 TO 44 46 TO 50 52 TO 67 70 75 TO 92 94 TO 107 109 TO 111 -
113 TO 118 120 TO 124 133 TO 142 144 TO 193 195 TO 235 237 TO 239 -
242 TO 287 289 TO 298 301 308 TO 314 317 TO 323 325 TO 342 349 352 -
354 TO 417 425 TO 471 496 TO 511 516 TO 518 525 TO 528 530 532 TO 534 536 -
537 539 TO 564 566 567 569 TO 591 594 TO 619 628 TO 633 635 TO 642 -
645 TO 647 649 TO 652 654 TO 660 662 TO 667 669 TO 678 681 TO 696 -
699 TO 704 707 TO 715 720 TO 732 735 TO 743 747 TO 749 751 TO 778 780 782 -
783 TO 790 794 TO 825 828 TO 884 898 899 901 TO 945 951 TO 968 973 974 980 -
981 TO 1009 1011 TO 1037 1039 TO 1045 1052 TO 1097 1099 TO 1133 TEMP -25
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
```

```
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
```

```
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
****
```

```
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
```

```
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
```

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LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
PERFORM ANALYSIS
DEFINE ENVELOPE
301 TO 338 ENVELOPE 1 TYPE STRENGTH
401 TO 438 ENVELOPE 2 TYPE STRENGTH
END DEFINE ENVELOPE
LOAD LIST 401 TO 438
START CONCRETE DESIGN
CODE ACI
CLB 0.05 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
CLS 0.05 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
CLT 0.05 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
FC 35000 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
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1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
FYMAIN 420000 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 -
317 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
FYSEC 420000 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 -
317 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
DESIGN BEAM 1501 TO 1524 1531 TO 1541 1543 TO 1546 1548 TO 1555
DESIGN BEAM 333 946 1651 TO 1672 1675 TO 1699 1801 TO 1838 2321
DESIGN BEAM 1701 TO 1706 1710 TO 1716 1719 TO 1726 2104 2105
DESIGN BEAM 318 1024 1322 1892 TO 1917 1974 TO 1978 1980 TO 1984 2151 TO 2191 -
2201 TO 2312 2328 TO 2330
DESIGN BEAM 2001 TO 2024 2035 TO 2056
DESIGN COLUMN 1301 TO 1316
DESIGN COLUMN 1331 TO 1387 2313 2316 2319
DESIGN COLUMN 1388 TO 1446 2320
DESIGN COLUMN 1951 TO 1964
END CONCRETE DESIGN
FINISH
*Load Combination for servicibilty
*LOAD 101 DL+SDL1
*REPEAT LOAD
*1 1.0 3 1.0
*LOAD 102 DL+SDL2
*REPEAT LOAD
*1 1.0 4 1.0
*LOAD 103 DL + SDL1 + TL1 + LL
*REPEAT LOAD
*1 1.0 3 1.0 9 1.0 2 1.0
*LOAD 104 DL + SDL1 + TL2 + LL
*REPEAT LOAD
*1 1.0 4 1.0 9 1.0 2 1.0
*LOAD 105 DL + SDL2 + TL1 + LL
*REPEAT LOAD
*1 1.0 3 1.0 10 1.0 2 1.0
*LOAD 106 DL + SDL2 + TL2 + LL
```

```
*REPEAT LOAD
*1 1.0 4 1.0 10 1.0 2 1.0
*LOAD 107 DL + SDL1 + LL + WL(+X)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 5 1.0
*LOAD 108 DL + SDL1 + LL + WL(-X)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 6 1.0
*LOAD 109 DL + SDL1 + LL + WL(+Z)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 7 1.0
*LOAD 110 DL + SDL1 + LL + WL(-Z)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 8 1.0
*LOAD 111 DL + SDL2 + LL + WL(+X)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 5 1.0
*LOAD 112 DL + SDL2 + LL + WL(-X)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 6 1.0
*LOAD 113 DL + SDL2 + LL + WL(+Z)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 7 1.0
*LOAD 114 DL + SDL2 + LL + WL(-Z)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 8 1.0
*LOAD 115 DL + SDL1 + LL + EL(+X)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 11 1.0
*LOAD 116 DL + SDL1 + LL + EL(-X)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 11 -1.0
*LOAD 117 DL + SDL1 + LL + EL(+Z)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 12 1.0
*LOAD 118 DL + SDL1 + LL + EL(-Z)
*REPEAT LOAD
*1 1.0 3 1.0 2 1.0 12 -1.0
*LOAD 119 DL + SDL2 + LL + EL(+X)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 11 1.0
*LOAD 120 DL + SDL2 + LL + EL(-X)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 11 -1.0
*LOAD 121 DL + SDL2 + LL + EL(+Z)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 12 1.0
*LOAD 122 DL + SDL2 + LL + EL(-Z)
*REPEAT LOAD
*1 1.0 4 1.0 2 1.0 12 -1.0
```

```
*LOAD 123 DL + SDL1 + WL(+X)
*REPEAT LOAD
*1 1.0 3 1.0 5 1.0
*LOAD 124 DL + SDL1 + WL(-X)
*REPEAT LOAD
*1 1.0 3 1.0 6 1.0
*LOAD 125 DL + SDL1 + WL(+Z)
*REPEAT LOAD
*1 1.0 3 1.0 7 1.0
*LOAD 126 DL + SDL1 + WL(-Z)
*REPEAT LOAD
*1 1.0 3 1.0 8 1.0
*LOAD 127 DL + SDL2 + WL(+X)
*REPEAT LOAD
*1 1.0 4 1.0 5 1.0
*LOAD 128 DL + SDL2 + WL(-X)
*REPEAT LOAD
*1 1.0 4 1.0 6 1.0
*LOAD 129 DL + SDL2 + WL(+Z)
*REPEAT LOAD
*1 1.0 4 1.0 7 1.0
*LOAD 130 DL + SDL2 + WL(-Z)
*REPEAT LOAD
*1 1.0 4 1.0 8 1.0
*LOAD 131 DL + SDL1 + EL(+X)
*REPEAT LOAD
*1 1.0 3 1.0 11 1.0
*LOAD 132 DL + SDL1 + EL(-X)
*REPEAT LOAD
*1 1.0 3 1.0 11 -1.0
*LOAD 133 DL + SDL1 + EL(+Z)
*REPEAT LOAD
*1 1.0 3 1.0 12 1.0
*LOAD 134 DL + SDL1 + EL(-Z)
*REPEAT LOAD
*1 1.0 3 1.0 12 -1.0
*LOAD 135 DL + SDL2 + EL(+X)
*REPEAT LOAD
*1 1.0 4 1.0 11 1.0
*LOAD 136 DL + SDL2 + EL(-X)
*REPEAT LOAD
*1 1.0 4 1.0 11 -1.0
*LOAD 137 DL + SDL2 + EL(+Z)
*REPEAT LOAD
*1 1.0 4 1.0 12 1.0
*LOAD 138 DL + SDL2 + EL(-Z)
*REPEAT LOAD
*1 1.0 4 1.0 12 -1.0
*****
**Load Combination for Structural design
```

```
*LOAD 201 1.4 (DL+SDL1)
*REPEAT LOAD
*1 1.4 3 1.4
*LOAD 202 1.4 (DL+SDL2)
*REPEAT LOAD
*1 1.4 4 1.4
*LOAD 203 1.2 (DL + SDL1 + TL1) + 1.6 LL
*REPEAT LOAD
*1 1.2 3 1.2 9 1.2 2 1.6
*LOAD 204 1.2 (DL + SDL1 + TL2) + 1.6 LL
*REPEAT LOAD
*1 1.2 3 1.2 10 1.2 2 1.6
*LOAD 205 1.2 (DL + SDL2 + TL1) + 1.6 LL
*REPEAT LOAD
*1 1.2 4 1.2 9 1.2 2 1.6
*LOAD 206 1.2 (DL + SDL2 + TL2) + 1.6 LL
*REPEAT LOAD
*1 1.2 4 1.2 10 1.2 2 1.6
*LOAD 207 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
*REPEAT LOAD
*1 1.2 3 1.2 2 0.5 5 1.6
*LOAD 208 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
*REPEAT LOAD
*1 1.2 3 1.2 2 0.5 6 1.6
*LOAD 209 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
*REPEAT LOAD
*1 1.2 3 1.2 2 0.5 7 1.6
*LOAD 210 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
*REPEAT LOAD
*1 1.2 3 1.2 2 0.5 8 1.6
*LOAD 211 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 5 1.6
*LOAD 212 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 6 1.6
*LOAD 213 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 7 1.6
*LOAD 214 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 8 1.6
*LOAD 215 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
*REPEAT LOAD
*1 1.2 3 1.2 2 0.5 11 1.0
*LOAD 216 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
*REPEAT LOAD
*1 1.2 3 1.2 2 0.5 11 -1.0
*LOAD 217 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
*REPEAT LOAD
```

```
*1 1.2 3 1.2 2 0.5 12 1.0
*LOAD 218 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
*REPEAT LOAD
*1 1.2 3 1.2 2 0.5 12 -1.0
*LOAD 219 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 11 1.0
*LOAD 220 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 11 -1.0
*LOAD 221 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 12 1.0
*LOAD 222 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
*REPEAT LOAD
*1 1.2 4 1.2 2 0.5 12 -1.0
*LOAD 223 0.9 (DL + SDL1) + 1.6 WL(+X)
*REPEAT LOAD
*1 0.9 3 0.9 5 1.6
*LOAD 224 0.9 (DL + SDL1) + 1.6 WL(-X)
*REPEAT LOAD
*1 0.9 3 0.9 6 1.6
*LOAD 225 0.9 (DL + SDL1) + 1.6 WL(+Z)
*REPEAT LOAD
*1 0.9 3 0.9 7 1.6
*LOAD 226 0.9 (DL + SDL1) + 1.6 WL(-Z)
*REPEAT LOAD
*1 0.9 3 0.9 8 1.6
*LOAD 227 0.9 (DL + SDL2) + 1.6 WL(+X)
*REPEAT LOAD
*1 0.9 4 0.9 5 1.6
*LOAD 228 0.9 (DL + SDL2) + 1.6 WL(-X)
*REPEAT LOAD
*1 0.9 4 0.9 6 1.6
*LOAD 229 0.9 (DL + SDL2) + 1.6 WL(+Z)
*REPEAT LOAD
*1 0.9 4 0.9 7 1.6
*LOAD 230 0.9 (DL + SDL2) + 1.6 WL(-Z)
*REPEAT LOAD
*1 0.9 4 0.9 8 1.6
*LOAD 231 0.9 (DL + SDL1) + 1.0 EL(+X)
*REPEAT LOAD
*1 0.9 3 0.9 11 1.0
*LOAD 232 0.9 (DL + SDL1) + 1.0 EL(-X)
*REPEAT LOAD
*1 0.9 3 0.9 11 -1.0
*LOAD 233 0.9 (DL + SDL1) + 1.0 EL(+Z)
*REPEAT LOAD
*1 0.9 3 0.9 12 1.0
*LOAD 234 0.9 (DL + SDL1) + 1.0 EL(-Z)
```

```
*REPEAT LOAD
*1 0.9 3 0.9 12 -1.0
*LOAD 235 0.9 (DL + SDL2) + 1.0 EL(+X)
*REPEAT LOAD
*1 0.9 3 0.9 11 1.0
*LOAD 236 0.9 (DL + SDL2) + 1.0 EL(-X)
*REPEAT LOAD
*1 0.9 3 0.9 11 -1.0
*LOAD 237 0.9 (DL + SDL2) + 1.0 EL(+Z)
*REPEAT LOAD
*1 0.9 3 0.9 12 1.0
*LOAD 238 0.9 (DL + SDL2) + 1.0 EL(-Z)
*REPEAT LOAD
*1 0.9 3 0.9 12 -1.0
*PERFORM ANALYSIS
*STEEL TAKE OFF LIST 1 TO 19 21 TO 24 26 TO 72 75 TO 126 128 133 TO 192 197 -
*198 205 TO 301 308 TO 316 320 TO 329 332 334 TO 495 516 TO 518 523 TO 642 -
*644 TO 667 669 TO 678 680 TO 749 751 TO 888 898 899 901 TO 945 947 TO 969 -
*972 TO 974 980 TO 1009 1011 TO 1022 1024 1025 1027 TO 1037 1048 TO 1184 1187 -
*1188 TO 1189 1191 TO 1247 2107 TO 2118
*LOAD LIST 201 TO 238
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 1 TO 72 75 TO 124 129 TO 521 523 TO 678 680 TO 893 898 899 -
*901 TO 1009 1011 TO 1171 1173 TO 1184 1187 TO 1189 1191 TO 1245 2107 TO 2118
*FU 410000 MEMB 1 TO 72 75 TO 521 523 TO 678 680 TO 893 898 899 901 TO 945 -
*947 TO 1009 1011 TO 1171 1173 TO 1184 1187 TO 1189 1191 TO 1245 2107 TO 2118
*TRACK 0 MEMB 1 TO 72 75 TO 521 523 TO 678 680 TO 893 898 899 901 TO 945 947 -
*948 TO 1009 1011 TO 1171 1173 TO 1184 1187 TO 1189 1191 TO 1225 1227 TO 1245 -
*2107 TO 2118
*BEAM 1 MEMB 1 TO 72 75 TO 521 523 TO 678 680 TO 893 898 899 901 TO 945 947 -
*948 TO 1009 1011 TO 1171 1173 TO 1184 1187 TO 1189 1191 TO 1225 1227 TO 1245 -
*2107 TO 2118
*TMAIN 300 MEMB 1 TO 72 75 TO 301 308 TO 492 494 TO 521 523 TO 678 680 TO 893 -
*898 899 901 TO 945 947 TO 1009 1011 TO 1133 1181 TO 1184 1189 1199 TO 1214 -
*1223 1225 1228 1237 TO 1245 2107 TO 2118
*MAIN 200 MEMB 1 TO 72 75 TO 301 308 TO 492 494 TO 521 523 TO 678 680 TO 893 -
*898 899 901 TO 945 947 TO 1009 1011 TO 1133 1181 TO 1184 1189 1199 TO 1214 -
*1223 1225 1228 1237 TO 1245 2107 TO 2118
*NSF 0.9 MEMB 1 TO 72 75 TO 301 308 TO 492 494 TO 521 523 TO 678 680 TO 749 -
*751 TO 893 898 899 901 TO 945 947 TO 1009 1011 TO 1133 1181 TO 1184 1189 -
*1199 TO 1214 1223 1225 1228 1237 TO 1245 1504 1508 1514 1519 1524 -
*1552 TO 1555 1931 1945 1955 1962 1973 2107 TO 2118
*KZ 0.5 MEMB 783 TO 790 794 795 798 799 805 806 814 TO 823 828 829 832 TO 837 -
*870 TO 875 879 880 890 TO 897
*KY 2 MEMB 592 593 684 686 690 691 785 786 796 797 800 TO 804 807 TO 813 824 -
*825 TO 827 898 899 901 TO 911 913 915 916 1023 1026
*LY 0.1 MEMB 107 TO 114 129 TO 162 165 TO 172 227 TO 232 237 TO 239 -
*242 TO 254 308 TO 313 321 TO 326 334 339 340 349 352 354 363 TO 365 -
*374 TO 384 389 TO 391 412 TO 415 428 429 432 433 435 436 438 439 441 443 -
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*444 TO 470 491 492 495 512 TO 516 519 TO 521 523 TO 591 594 595 600 601 608 -
*609 TO 610 613 614 617 TO 620 628 TO 667 669 TO 678 680 TO 707 712 TO 724 -
*730 736 738 740 TO 747 749 751 TO 782 920 TO 924 936 TO 938 955 TO 958 980 -
*981 TO 990 1122 2110 TO 2117
*LX 0.1 MEMB 107 TO 114 129 TO 162 165 TO 172 227 TO 232 237 TO 239 -
*242 TO 254 308 TO 313 321 TO 326 334 339 340 349 352 354 363 TO 365 -
*374 TO 384 389 TO 391 412 TO 415 428 429 432 433 435 436 438 439 441 443 -
*444 TO 470 491 492 495 512 TO 516 519 TO 521 523 TO 591 594 595 600 601 608 -
*609 TO 610 613 614 617 TO 620 628 TO 667 669 TO 674 676 TO 678 680 TO 682 -
*684 TO 707 712 TO 724 730 736 738 740 TO 747 749 751 TO 782 920 TO 924 936 -
*937 TO 938 955 TO 958 980 TO 990 1122 2110 TO 2117
*UNT 0.1 MEMB 107 TO 114 133 TO 162 165 TO 172 227 TO 232 237 TO 239 -
*242 TO 254 308 TO 313 321 TO 326 334 339 340 349 352 354 363 TO 365 -
*374 TO 384 389 TO 391 412 TO 415 428 429 432 433 435 436 438 439 441 443 -
*444 TO 470 491 492 495 512 TO 516 519 TO 521 523 TO 591 594 595 600 601 608 -
*609 TO 610 613 614 617 TO 620 628 TO 642 644 TO 667 669 TO 678 680 TO 682 -
*684 TO 707 712 TO 724 730 736 738 740 TO 747 749 751 TO 782 920 TO 928 936 -
*937 TO 938 940 TO 945 951 TO 958 964 TO 967 980 TO 990 1030 TO 1035 1122 2110 -
*2111 TO 2117
**COLUMNS
*LX 5.8 MEMB 30 TO 32 94 TO 99 206 207
*LY 5.8 MEMB 30 TO 32 94 TO 97
*LX 4.9 MEMB 4 5 10 11 37 38 98 99 118 119 1120 1121
*LY 4.9 MEMB 4 5 10 11 37 38 98 99 118 119 1120 1121
*LY 3 MEMB 116 117 1118 1119
*LX 3 MEMB 116 117 1118 1119
*LY 4 MEMB 104 105
*LX 4 MEMB 104 105
*LZ 4.9 MEMB 2 TO 4 6 TO 10 29 TO 31 33 TO 37 105 106 366 367
*LY 3.5 MEMB 7 TO 9 34 TO 36 126 127 215 216 224 225 1103 1104 1110 1111
*LZ 3.6 MEMB 94 TO 98 100 TO 104 115 TO 117 1117 TO 1120
*LX 3.5 MEMB 7 TO 9 34 TO 36 126 127 215 216 224 225 1103 1104 1110 1111
*LZ 6.65 MEMB 120 121 209 210 218 219 1113 TO 1116
*LZ 7 MEMB 122 TO 127 211 TO 216 220 TO 225 327 328 1099 TO 1104 1106 TO 1111
*KZ 2 MEMB 201 TO 208
*LZ 5.4 MEMB 193 194 199 200
*LY 5.4 MEMB 1 2 28 29
*LY 10.5 MEMB 791 TO 793
*LX 2.5 MEMB 279 TO 281 284 TO 286 396 416 430 434 471 494 974
*LZ 2.5 MEMB 279 TO 281 284 TO 286 396 416 430 434 471 494 974 1152 TO 1159 -
*1164 TO 1171
*LY 2.5 MEMB 279 TO 281 284 TO 286 396 416 430 434 471 494 974
*LZ 3.3 MEMB 345 TO 348 420 TO 422 472 TO 476 478 TO 482 1160 TO 1163
*LZ 4.3 MEMB 1134 TO 1141 1144 TO 1159 1164 TO 1171
**FLOOR BEAMS
*LZ 6 MEMB 107 TO 114 129 TO 151 156 TO 162 166 TO 168
*LZ 1.9 MEMB 637 647 654 655 662 663 669 670 673 674 676 677 681 682
*LZ 10.5 MEMB 173 TO 177 188 TO 192 495 553 TO 579 1048 1122 2107 2108 2110 -
*2115 TO 2117
*LZ 6.465 MEMB 153 TO 155 173 TO 177 188 TO 191 261 TO 265 741 TO 747 -
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*776 TO 782 1129 TO 1133 2107 2108
*LZ 4.035 MEMB 339 340 759 760 762 763 765 766 768 769
*LZ 6.15 MEMB 184 TO 187 258 TO 260 272 TO 275 314 315 536 TO 552 712 TO 724 -
*754 TO 758 1126 TO 1128 2113 2114 2118
*LZ 7.6 MEMB 165 169 TO 172 178 TO 183 255 TO 257 266 TO 271 525 TO 535 692 -
*693 TO 707 1123 TO 1125 2111 2112
*LZ 7 MEMB 276 277 708 TO 711 725 TO 740 748 749
*LY 7 MEMB 708 TO 711 725 TO 728
*LX 7 MEMB 708 TO 711 725 TO 728
*LZ 7.9 MEMB 12 TO 27
*LZ 8.5 MEMB 39 TO 62
*LZ 4.93 MEMB 67 TO 72 484 TO 490
*LZ 3.15 MEMB 463 TO 470 632 TO 634 639 640 645 646 651 TO 653
*LZ 5.3 MEMB 240 241 309 310 313 349 351 TO 354
*LZ 2.5 MEMB 389 TO 391 413 TO 415 432 433 438 439 449 TO 452 580 TO 591 661
*LZ 4.13 MEMB 242 TO 253
*LZ 5.465 MEMB 237 TO 239 621 TO 627
*CHECK CODE MEMB 1 TO 11 13 TO 16 18 TO 24 26 TO 30 32 TO 298 301 TO 314 317 -
*318 TO 491 496 TO 521 523 524 580 TO 619 621 TO 633 635 TO 652 654 TO 678 -
*680 TO 790 794 TO 825 828 TO 890 898 899 901 TO 1009 1030 TO 1035 1046 1047 -
*1099 TO 1121 1123 TO 1133 1181 TO 1184 1189 1199 TO 1214 1223 1225 1228 1237 -
*1238 2107 TO 2109 2118
*CHECK CODE MEMB 891 TO 897
*CHECK CODE MEMB 315 492 TO 494 620 634 653 1250
*CHECK CODE MEMB 31 299 300 316 791 TO 793 826 827 1134 TO 1171 1173 TO 1180 -
*1187 1188 1191 TO 1198 1215 TO 1222 1224 1226 1227 1229 TO 1236 1239 TO 1241
*LOAD LIST 101 TO 138
*PARAMETER 2
*CODE LRFD
**Floor @ 938.2 M Level
*DJ1 268 MEMB 156 TO 162
*DJ2 319 MEMB 156 TO 162
*DJ1 930 MEMB 129 TO 135
*DJ2 486 MEMB 129 TO 135
*DJ1 468 MEMB 136 TO 141
*DJ2 470 MEMB 136 TO 141
*DJ1 469 MEMB 142 144 145
*DJ2 471 MEMB 142 144 145
*DJ1 493 MEMB 146 TO 148
*DJ2 498 MEMB 146 TO 148
*DJ1 270 MEMB 166 TO 168
*DJ2 321 MEMB 166 TO 168
*DJ1 502 MEMB 149 TO 151
*DJ2 507 MEMB 149 TO 151
*DJ1 504 MEMB 107 109 110
*DJ2 509 MEMB 107 109 110
*DJ1 506 MEMB 111 113 114
*DJ2 511 MEMB 111 113 114
*DJ1 512 MEMB 228 230
*DJ2 515 MEMB 228 230
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*DJ1 472 MEMB 637 647
*DJ2 473 MEMB 637 647
*DJ1 474 MEMB 638 648
*DJ2 475 MEMB 638 648
*DJ1 526 MEMB 654 662
*DJ2 530 MEMB 654 662
*DJ1 528 MEMB 655 663
*DJ2 531 MEMB 655 663
*DJ1 530 MEMB 669 673
*DJ2 527 MEMB 669 673
*DJ1 531 MEMB 670 674
*DJ2 529 MEMB 670 674
*DJ1 520 MEMB 676 682
*DJ2 521 MEMB 676 682
*DJ1 522 MEMB 677 681
*DJ2 523 MEMB 677 681
*DFF 360 MEMB 129 TO 135 146 TO 152 156 TO 162 166 TO 168 227 TO 232 254 321 -
*322 TO 326 365 523 525 TO 534 536 537 539 TO 564 566 567 569 TO 579 594 595 -
*637 638 641 643 647 648 654 655 658 659 662 663 669 TO 674 676 TO 678 681 -
*682 1122
*DFF 500 MEMB 107 109 TO 111 113 114 136 TO 142 144 145
**Floor @ 926.5 M Level
*DJ1 102 MEMB 729 730
*DJ2 335 MEMB 729 730
*DJ1 335 MEMB 374 380
*DJ2 409 MEMB 374 380
*DJ1 103 MEMB 737 738
*DJ2 336 MEMB 737 738
*DJ1 419 MEMB 692 TO 699
*DJ2 420 MEMB 692 TO 699
*DJ1 335 MEMB 700 TO 707
*DJ2 336 MEMB 700 TO 707
*DJ1 409 MEMB 165 169 TO 172
*DJ2 410 MEMB 165 169 TO 172
*DJ1 104 MEMB 739 740
*DJ2 337 MEMB 739 740
*DJ1 337 MEMB 376 382
*DJ2 411 MEMB 376 382
*DJ1 420 MEMB 713 714 716 719 721 723
*DJ2 421 MEMB 713 714 716 719 721 723
*DJ1 336 MEMB 712 715 717 718 720 722 724
*DJ2 337 MEMB 712 715 717 718 720 722 724
*DJ1 410 MEMB 754 TO 758
*DJ2 411 MEMB 754 TO 758
*DJ1 105 MEMB 748 749
*DJ2 338 MEMB 748 749
*DJ1 338 MEMB 377 383
*DJ2 412 MEMB 377 383
*DJ1 421 MEMB 741 TO 747
*DJ2 422 MEMB 741 TO 747
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*DJ1 337 MEMB 776 TO 782
*DJ2 338 MEMB 776 TO 782
*DJ1 411 MEMB 153 TO 155
*DJ2 412 MEMB 153 TO 155
*DJ1 106 MEMB 735 736
*DJ2 339 MEMB 735 736
*DJ1 339 MEMB 378 384
*DJ2 413 MEMB 378 384
*DJ1 422 MEMB 760 762 765 768
*DJ2 423 MEMB 760 762 765 768
*DJ1 338 MEMB 759 763 766 769
*DJ2 339 MEMB 759 763 766 769
*DJ1 412 MEMB 339 340
*DJ2 413 MEMB 339 340
*DFF 360 MEMB 153 TO 155 165 169 TO 172 334 339 340 374 TO 384 512 TO 516 519 -
*520 TO 521 600 601 628 692 TO 707 712 TO 724 729 730 735 TO 749 751 TO 769 -
*776 TO 782 980 TO 982 987 TO 990
*DFF 500 MEMB 368 TO 373 770 TO 775
**Floor @ 926.5 M Level
*DJ1 799 MEMB 39 TO 41
*DJ2 798 MEMB 39 TO 41
*DJ1 798 MEMB 42 TO 44
*DJ2 800 MEMB 42 TO 44
*DJ1 802 MEMB 45 TO 47
*DJ2 801 MEMB 45 TO 47
*DJ1 801 MEMB 48 TO 50
*DJ2 803 MEMB 48 TO 50
*DJ1 793 MEMB 51 52 59
*DJ2 792 MEMB 51 52 59
*DJ1 792 MEMB 54 TO 56
*DJ2 794 MEMB 54 TO 56
*DJ1 796 MEMB 57 TO 59
*DJ2 795 MEMB 57 TO 59
*DJ1 795 MEMB 60 TO 62
*DJ2 797 MEMB 60 TO 62
*DJ1 791 MEMB 12 TO 16
*DJ2 877 MEMB 12 TO 16
*DJ1 878 MEMB 17 TO 19
*DJ2 790 MEMB 17 TO 19
*DJ1 884 MEMB 20 TO 24
*DJ2 882 MEMB 20 TO 24
*DJ1 883 MEMB 25 TO 27
*DJ2 885 MEMB 25 TO 27
*DJ1 809 MEMB 67 TO 69
*DJ2 823 MEMB 67 TO 69
*DJ1 804 MEMB 70 TO 72
*DJ2 822 MEMB 70 TO 72
*DFF 360 MEMB 12 TO 27 39 TO 82 1038 TO 1045 2106
*DJ1 375 MEMB 708 709
*DJ2 345 MEMB 708 709
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*DJ1 389 MEMB 710 711
*DJ2 346 MEMB 710 711
*DJ1 393 MEMB 725 726
*DJ2 347 MEMB 725 726
*DJ1 404 MEMB 727 728
*DJ2 348 MEMB 727 728
*DJ1 345 MEMB 266 TO 271
*DJ2 346 MEMB 266 TO 271
*DJ1 346 MEMB 272 TO 275 2118
*DJ2 347 MEMB 272 TO 275 2118
*DJ1 347 MEMB 173 TO 177 2107
*DJ2 956 MEMB 173 TO 177 2107
*DJ1 732 MEMB 178 TO 183
*DJ2 733 MEMB 178 TO 183
*DJ1 733 MEMB 184 TO 187
*DJ2 734 MEMB 184 TO 187
*DJ1 734 MEMB 188 TO 192 2108
*DJ2 735 MEMB 188 TO 192 2108
*DFF 500 MEMB 173 TO 192 266 TO 275 335 TO 338 708 TO 711 725 TO 728 750 2107 -
*2108 2118
***CONVEYOR WALKWAY
*DJ1 319 MEMB 276 277
*DJ2 340 MEMB 276 277
*DJ1 320 MEMB 163 164
*DJ2 341 MEMB 163 164
*DJ1 321 MEMB 731 732
*DJ2 342 MEMB 731 732
*DJ1 340 MEMB 1123 TO 1125
*DJ2 341 MEMB 1123 TO 1125
*DJ1 768 MEMB 255 TO 257
*DJ2 391 MEMB 255 TO 257
*DJ1 341 MEMB 1126 TO 1128
*DJ2 342 MEMB 1126 TO 1128
*DJ1 342 MEMB 1129 TO 1133
*DJ2 957 MEMB 1129 TO 1133
*DJ1 391 MEMB 258 TO 260
*DJ2 777 MEMB 258 TO 260
*DJ1 777 MEMB 261 TO 265
*DJ2 961 MEMB 261 TO 265
*DJ1 960 MEMB 733 734
*DJ2 957 MEMB 733 734
*DFF 360 MEMB 163 164 173 TO 192 255 TO 277 517 518 606 607 611 612 615 616 -
*731 TO 734 1123 TO 1133 2107 2108 2118
*DFF 500 MEMB 596 TO 599 602 TO 605
*CHECK CODE MEMB 89 TO 93 107 109 TO 111 113 129 TO 142 144 TO 152 156 TO 164 -
*166 TO 168 173 TO 192 227 TO 232 254 TO 277 301 321 TO 326 335 TO 338 365 -
*517 518 523 594 TO 599 602 TO 607 611 612 615 616 637 638 641 643 647 648 -
*654 655 658 659 662 663 669 TO 674 676 TO 678 681 682 708 TO 711 725 TO 728 -
*731 TO 734 750 1123 TO 1133 2107 2108 2118
FINISH
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023			Date	07-03-2020	19-06-2020	24-07-2020	16-10-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 04-04-2020	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL			Prepared By	RPK/APL	RPK/APL	RPK/APL	RPK / APL	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 3 STAAD OUTPUT FILE OF SCREEN BUILDING

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUN 19, 2020                *
*           Time=    13:19:12                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
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     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Screen Building\Civil\Document\Staad\R1\Screen Building
     2. START JOB INFORMATION
     3. ENGINEER DATE 04-JAN-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -2.5 0; 2 0 -2.5 7.6; 3 0 -2.5 13.75; 4 0 -2.5 20.215; 5 0 -2.5 24.25
     9. 6 8.5 -2.5 0; 7 8.5 -2.5 7.6; 8 8.5 -2.5 13.75; 9 8.5 -2.5 20.215
    10. 10 8.5 -2.5 24.25; 11 17 -2.5 0; 12 17 -2.5 7.6; 13 17 -2.5 13.75
    11. 14 17 -2.5 20.215; 15 17 -2.5 24.25; 16 23 -2.5 0; 17 23 -2.5 7.6
    12. 18 23 -2.5 13.75; 19 23 -2.5 20.215; 20 23 -2.5 24.25; 21 30 0.45 0
    13. 22 30 0.45 7.6; 23 30 0.45 13.75; 24 30 0.45 20.215; 25 30 0.45 24.25
    14. 26 17 0.45 -3.3; 27 23 0.45 -3.3; 30 0 -0.15 0; 31 0 -0.15 7.6
    15. 32 0 -0.15 13.75; 33 0 -0.15 20.215; 34 0 -0.15 24.25; 35 8.5 -0.15 0
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466. 1074 372 23; 1075 352 371; 1076 371 355; 1077 353 370; 1078 370 356
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543. 1710 111 798; 1711 798 801; 1712 801 119; 1713 119 792; 1714 792 795
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609. START GROUP DEFINITION
610. FLOOR
611. _FL_938.2_1 107 TO 114 129 TO 152 156 TO 162 166 TO 168 227 TO 232 254 321 -
612. 322 TO 326 365 495 523 TO 533 535 536 538 TO 563 565 566 568 TO 579 594 595 -
613. 637 641 644 647 654 655 658 659 662 663 669 TO 674 676 678 680 682 1122 2110 -
614. 2112 2114 2117
615. _FL_938.2_2 110 114 141 145 534 537 564 567 638 648 677 681 2111 2113 2115 2116
616. _FL_926.5_1 2001 TO 2056
617. _FL_926.5_2 153 TO 155 165 169 TO 172 334 339 340 368 TO 384 512 TO 516 519 -
618. 520 TO 521 600 601 628 692 TO 707 712 TO 724 729 730 735 TO 749 751 TO 782 -
619. 980 TO 982 987 TO 990 2013 TO 2024
620. _FL_922.75_1 1322 1892 TO 1917 1974 TO 1978 1980 TO 1984 2151 2153 TO 2163 -
621. 2165 TO 2168 2170 TO 2175 2177 2178 2180 2181 2183 2201 TO 2205 2208 TO 2212 -
622. 2215 TO 2223 2226 TO 2237 2242 TO 2247 2250 TO 2278 2280 TO 2283 -
623. 2285 TO 2289 2291 TO 2296 2299 TO 2301 2303 TO 2307 2309 TO 2312 2328 2329
624. _FL_922.75_2 1894 1897 1898 1900 1903 1904 1906 1909 1910 1913 1916 1917 1977 -
625. 1978 1983 1984 2164 2169 2175 TO 2187 2206 2213 2224 2230 2231 2236 TO 2238 -
626. 2248 2279 2284 2290 2297 2298 2302 2308 2330
627. _FL_916.3 1654 TO 1663 1668 1675 TO 1699 1802 TO 1811 1815 TO 1819 -
628. 1823 TO 1829 1832 TO 1836 2321
629. _FL_916.3_2 1651 1655 1656 1814 1822
630. _FL_925.35 432 433 435 436 438 439 449 TO 452 491 TO 493 608 TO 610 613 614 -
631. 617 TO 620 634 653 661 668
632. MEMBER
633. _TB1 1509
634. _TB2 1501 TO 1508
635. _TB3 1510 TO 1524
636. _TB4 1531 1532
637. _TB5 1533 TO 1536 1542 1547 1552 TO 1555
638. _TB5A 1537 TO 1541 1543 TO 1546 1548 TO 1551
639. _1FB1 1801 TO 1804 1832 TO 1836
640. _1FB2 1805 1806 1837 1838
641. _1FB3 1807 TO 1811 1823 TO 1829
642. _1FB4 1812 1813 1830 1831
643. _1FB5 1814 1820 TO 1822
644. _1FB6 1815 TO 1819
645. _1FB7 1653 TO 1672 2321
646. _1FB8 1651 1652
647. _1B1 1695 1699
648. _1B2 1675 TO 1679
649. _1B3 1682 TO 1690
650. _1B4 1691 TO 1694 1696 TO 1698
651. _1B5 1680 1681
652. _2FB1 1701 TO 1706 2104 2105
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655. _3FB1 2201 TO 2207 2240 TO 2249
656. _3FB2 2208 TO 2214 2227 TO 2239 2329
657. _3FB2A 2215 TO 2225
658. _3BFB2B 1322 1389 1390 1974 TO 1978 1980 TO 1984 2320 2328
659. _3FB3 2151 2152 2174 2226
660. _3FB4 2157 TO 2160 2181 TO 2183
661. _3FB5 2153 TO 2156 2175 TO 2180
662. _3FB5A 2161 TO 2173
663. _3FB6 2188 TO 2191
664. _3B1 2184 TO 2187
665. _3B2 2250 TO 2262
666. _3B3 1893 TO 1910 1912 TO 1917
667. _3B4 2263 TO 2312 2330
668. _3B5 1892 1911
669. _4FB1 2035 2036 2043 2044
670. _4FB2 2037 TO 2042
671. _4FB3 2001 TO 2024
672. _4B1 2045 TO 2052
673. _4B2 2053 TO 2056
674. END GROUP DEFINITION
675. DEFINE MATERIAL START
676. ISOTROPIC CONCRETE
677. E 2.17185E+007
678. POISSON 0.17
679. DENSITY 23.5616
680. ALPHA 1E-005
681. DAMP 0.05
682. TYPE CONCRETE
683. STRENGTH FCU 27579
684. ISOTROPIC STEEL
685. E 2.05E+008
686. POISSON 0.3
687. DENSITY 76.8195
688. ALPHA 1.2E-005
689. DAMP 0.03
690. TYPE STEEL
691. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
692. END DEFINE MATERIAL
693. MEMBER PROPERTY EUROPEAN
694. ***COLUMN
695. 1301 TO 1316 PRIS YD 0.35 ZD 0.6
696. 1951 TO 1964 PRIS YD 0.7 ZD 0.5
697. 1331 TO 1350 1357 TO 1387 2313 2316 2319 PRIS YD 0.6 ZD 0.85
698. 1351 TO 1356 1388 TO 1446 2320 PRIS YD 0.55 ZD 0.55
699. ***TIE BEAMS
700. 1501 TO 1524 PRIS YD 0.55 ZD 0.3
701. 1531 TO 1555 PRIS YD 0.5 ZD 0.3
702. ****916 M FLOOR
703. 1801 TO 1804 1832 TO 1836 PRIS YD 0.85 ZD 0.35
704. 1807 TO 1811 1815 TO 1819 1823 TO 1829 PRIS YD 1 ZD 0.35
705. 1805 1806 1812 TO 1814 1820 1821 1830 1831 1837 1838 PRIS YD 0.6 ZD 0.35
706. 1651 TO 1672 2321 PRIS YD 0.55 ZD 0.3
707. 1675 TO 1699 1822 PRIS YD 0.5 ZD 0.3
708. ****920M FLOOR
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711. ****922.75
712. 2201 TO 2207 2240 TO 2249 PRIS YD 1.05 ZD 0.3
713. 1322 1974 TO 1978 1980 TO 1984 2208 TO 2225 2227 TO 2239 2328 -
714. 2329 PRIS YD 1.05 ZD 0.4
715. 2157 TO 2160 2181 TO 2183 PRIS YD 1.1 ZD 0.35
716. 2153 TO 2156 2161 TO 2173 2175 TO 2180 PRIS YD 0.85 ZD 0.35
717. 2250 TO 2262 PRIS YD 0.65 ZD 0.3
718. 2151 2152 2174 2226 PRIS YD 0.65 ZD 0.35
719. 1892 1911 PRIS YD 0.6 ZD 0.3
720. 2263 TO 2312 2330 PRIS YD 0.55 ZD 0.3
721. 2188 TO 2191 PRIS YD 0.5 ZD 0.3
722. 2184 TO 2187 PRIS YD 0.5 ZD 0.3
723. ****FLOOR 926.5
724. 2025 TO 2034 PRIS YD 0.18 ZD 0.3
725. 2001 TO 2024 PRIS YD 0.75 ZD 0.35
726. 2035 TO 2044 PRIS YD 0.6 ZD 0.35
727. 2045 TO 2052 PRIS YD 0.55 ZD 0.3
728. MEMBER PROPERTY EUROPEAN
729. 1 TO 11 TABLE ST HE500A
730. 12 TO 27 TABLE ST HE500A
731. 28 TO 38 TABLE ST HE400A
732. 63 TO 74 76 78 82 2322 2323 2325 TABLE ST HE300A
733. 75 77 81 83 TO 93 2326 2327 TABLE ST HE200A
734. 107 TO 114 TABLE ST IPE450
735. 115 TO 128 TABLE ST IPE400
736. 129 TO 145 TABLE ST IPE400
737. 146 TO 155 TABLE ST IPE300
738. 156 TO 192 261 TO 265 2107 2108 TABLE ST IPE360
739. 193 TO 226 TABLE ST IPE300
740. 227 TABLE ST IPE300
741. 228 230 TO 236 TABLE ST IPE300
742. 229 237 TO 260 266 TO 277 2118 TABLE ST IPE300
743. 278 TO 300 302 TO 307 TABLE ST IPE200
744. 301 308 TO 340 365 TABLE ST IPE240
745. 341 TO 364 366 TO 384 TABLE ST IPE200
746. 385 TO 424 TABLE ST IPE160
747. 425 TO 494 TABLE ST IPE160
748. ****
749. 495 525 TO 579 1122 2110 TO 2117 TAPERED 2.5 0.01 2.5 0.3 0.02 0.3 0.02
750. 496 TO 511 TAPERED 2 0.012 2 0.3 0.016 0.3 0.016
751. 512 TO 516 519 TO 521 523 524 594 595 TABLE ST UPN240
752. 580 TO 593 596 TO 605 TABLE ST UPN200
753. 517 518 606 TO 628 TABLE ST UPN160
754. 629 TO 678 680 TO 691 TABLE ST UPN120
755. 692 TO 707 TABLE ST IPE550
756. 712 TO 724 TABLE ST IPE450
757. 729 TO 736 TABLE ST IPE400
758. 708 TO 711 725 TO 728 737 TO 750 TABLE ST IPE500
759. 751 TO 769 TABLE ST IPE300
760. 770 TO 773 TABLE ST HE360A
761. 774 775 TABLE ST HE400A
762. 776 TO 782 TABLE ST IPE600
763. 785 TO 793 TABLE LD L130X130X10 SP 0.01
764. 783 784 794 TO 807 TABLE LD L100X100X10 SP 0.01
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767. 828 TO 837 TABLE LD L90X90X8 SP 0.01
768. 838 TO 875 879 TO 897 TABLE LD L65X65X6 SP 0.01
769. 898 TO 916 TABLE LD L50X50X6 SP 0.01
770. 917 TO 924 TABLE ST L100X100X8
771. 877 929 TO 939 944 TO 950 TABLE ST L90X90X8
772. 876 878 951 TO 967 TABLE ST L75X75X8
773. 968 TO 990 TABLE ST L65X65X6
774. 991 TO 1010 2109 TABLE ST L65X65X6
775. 1011 TO 1016 1036 1037 1041 1050 1051 TABLE SA L75X75X8
776. 1021 TO 1029 1052 TO 1072 1097 TABLE SA L90X90X8
777. 1073 TO 1084 TABLE SA L130X130X10
778. 1085 TO 1096 TABLE SA L100X100X8
779. 1099 TO 1112 TABLE ST IPE500
780. 1113 TO 1116 TABLE TB IPE500 WP 0.15 TH 0.012 BW 0.15 BT 0.012
781. 94 TO 106 1117 TO 1121 TABLE ST HE300A
782. MEMBER PROPERTY EUROPEAN
783. 1123 TO 1133 TAPERED 0.25 0.008 0.25 0.15 0.012 0.15 0.012
784. 1134 TO 1171 1250 TABLE ST UPN100
785. 1172 TO 1238 TABLE ST L50X50X6
786. MEMBER PROPERTY EUROPEAN
787. 1239 TO 1241 TABLE LD L75X75X8 SP 0.01
788. 1017 TO 1020 1039 1040 1043 TO 1045 1048 1049 1098 1242 TO 1246 -
789. 1247 TABLE SA L65X65X6
790. MEMBER PROPERTY EUROPEAN
791. 39 TO 44 46 TO 50 52 TO 62 TAPERED 0.5 0.01 0.5 0.3 0.016 0.3 0.016
792. 1030 TO 1035 1046 1047 TABLE ST L130X130X10
793. 925 TO 928 940 TO 943 TABLE ST L100X100X10
794. MEMBER PROPERTY EUROPEAN
795. 2053 TO 2056 PRIS YD 0.65 ZD 0.3
796. MEMBER PROPERTY EUROPEAN
797. 1893 TO 1910 1912 TO 1917 PRIS YD 0.6 ZD 0.44
798. MEMBER PROPERTY
799. 79 80 1042 2324 TAPERED 0.5 0.012 0.5 0.3 0.025 0.3 0.025
800. CONSTANTS
801. BETA 90 MEMB 1 TO 11 28 TO 38 83 TO 88 281 286 293 294 341 345 346 -
802. 386 TO 388 393 TO 395 420 TO 422 425 TO 427 434 442 471 TO 476 478 TO 482 -
803. 791 TO 793 826 827 1134 TO 1141 1144 TO 1159 1164 TO 1171 1242 TO 1245
804. MATERIAL CONCRETE MEMB 1248 1249 1301 TO 1316 1319 1320 1322 1331 TO 1446 -
805. 1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
806. 1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
807. 1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
808. 2201 TO 2313 2316 2319 TO 2321 2328 TO 2330
809. MATERIAL STEEL MEMB 1 TO 44 46 TO 50 52 TO 521 523 TO 678 680 TO 1037 1039 -
810. 1040 TO 1247 1250 2107 TO 2118 2322 TO 2327
811. SUPPORTS
812. 1 TO 20 120 293 294 330 TO 334 886 888 890 FIXED
813. MEMBER RELEASE
814. 1 6 28 33 94 100 115 240 255 258 261 297 299 300 316 417 420 435 474 480 493 -
815. 517 596 598 602 605 606 611 615 661 1117 1134 1138 1144 1148 1152 1156 1164 -
816. 1168 1186 1250 1892 1893 1899 1905 1911 1912 1974 2025 2184 2250 2263 2266 -
817. 2270 2272 2276 2279 2280 2284 2285 2287 2290 2291 2293 2295 2297 2298 2302 -
818. 2303 2308 TO 2310 2326 2327 2330 START MY MZ
819. 89 TO 91 241 257 260 265 296 298 491 493 518 597 599 603 604 607 612 616 1186 -
820. 1250 1515 1822 1892 1898 1904 1910 1911 1917 1984 2034 2046 2048 2050 2052 -
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823. 2330 END MY MZ
824. 12 25 73 74 308 512 TO 515 519 TO 521 600 601 727 750 1675 1680 1682 1684 -
825. 1686 1689 1691 1695 1696 1699 START MY MZ
826. 19 24 73 74 308 512 TO 515 519 TO 521 600 601 728 736 750 807 1679 1681 1683 -
827. 1685 1688 1690 1694 1695 1698 1699 END MY MZ
828. 16 17 20 39 57 63 TO 67 70 75 TO 83 85 87 107 111 120 129 136 142 146 149 -
829. 152 153 156 163 165 166 173 178 184 188 193 199 218 227 TO 229 231 232 234 -
830. 235 254 266 272 276 295 301 317 TO 323 325 326 328 TO 334 336 337 339 365 -
831. 366 368 TO 379 389 391 496 499 TO 501 504 507 TO 509 516 523 525 526 536 -
832. 537 553 554 594 595 628 637 638 641 643 654 655 658 659 668 TO 672 -
833. 676 TO 678 692 700 708 712 713 725 729 731 734 735 737 739 741 748 -
834. 751 TO 754 759 TO 761 764 767 770 TO 776 785 796 800 801 804 808 TO 810 824 -
835. 916 1023 1027 1066 1113 1115 1123 1126 1129 START MY MZ
836. 27 44 50 56 62 TO 66 69 72 75 77 79 81 84 86 88 110 114 135 141 145 148 151 -
837. 152 155 162 164 168 172 177 183 187 192 227 229 TO 233 235 254 271 275 277 -
838. 301 317 TO 320 322 TO 327 329 TO 334 340 365 367 TO 373 379 498 TO 500 503 -
839. 506 TO 508 511 516 524 535 551 552 578 579 594 595 628 643 644 647 648 658 -
840. 659 662 663 668 671 TO 674 680 TO 682 699 707 709 723 724 726 730 732 733 -
841. 738 740 747 749 751 TO 753 758 761 764 767 TO 775 782 786 797 802 803 811 -
842. 812 TO 813 825 915 1026 1027 1068 1125 1128 1133 2111 2322 TO 2324 -
843. 2325 END MY MZ
844. 89 TO 92 237 242 245 248 251 278 TO 282 289 291 293 302 304 306 341 TO 343 -
845. 350 351 355 TO 358 363 385 TO 388 396 TO 402 412 418 419 425 TO 427 -
846. 430 TO 434 436 TO 440 442 TO 445 447 448 457 458 463 465 467 469 477 483 -
847. 580 583 586 589 592 608 TO 610 613 614 617 TO 619 621 622 629 631 632 635 -
848. 636 639 642 645 649 TO 651 656 657 660 666 684 685 687 TO 689 691 898 901 -
849. 903 905 907 909 911 948 2275 2299 2306 START MY MZ
850. 93 239 244 247 250 253 303 305 307 350 353 364 412 431 436 437 440 -
851. 444 TO 452 457 458 464 466 468 470 477 483 492 581 584 587 590 593 -
852. 608 TO 610 613 614 617 618 620 626 627 630 631 634 TO 636 640 642 646 649 -
853. 650 653 660 661 664 665 667 685 TO 690 899 902 904 906 908 910 913 949 2292 -
854. 2301 2307 END MY MZ
855. 309 311 312 349 675 683 START MP 0.99
856. 311 TO 313 354 675 683 END MP 0.99
857. MEMBER TRUSS
858. 783 784 787 TO 790 794 795 798 799 805 806 814 TO 823 828 TO 837 872 873 876 -
859. 877 TO 884 887 TO 897 900 912 914 917 TO 919 929 TO 932 934 935 939 946 950 -
860. 959 968 TO 977 1003 1010 TO 1022 1024 1025 1028 1029 1036 1037 1039 TO 1045 -
861. 1048 TO 1065 1067 1069 TO 1098 1172 1181 TO 1184 1189 1199 TO 1214 1223 1225 -
862. 1228 1237 TO 1247
863. MEMBER TRUSS
864. 838 TO 875 885 886 920 TO 928 936 TO 938 940 TO 945 951 TO 958 960 TO 967 -
865. 980 TO 1009 1030 TO 1035 1046 1047 1173 TO 1180 1187 1188 1191 TO 1198 1215 -
866. 1216 TO 1222 1224 1227 1229 TO 1236 2109
867. ****LOAD APPLICATION
868. MEMBER OFFSET
869. 496 TO 511 START 0 -0.5 0
870. 496 TO 511 END 0 -0.5 0
871. DEFINE IBC 2003
872. SDS 0.066 SD1 0.0133 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.619 PZ 0.619
873. SELFWEIGHT 1
874. JOINT WEIGHT
875. *3130 FDR-004 LOAD AT +916.3 M LEVEL
876. 226 267 WEIGHT 20
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879. 124 126 128 130 WEIGHT 40
880. *SECONDARY FEEDERS
881. 132 134 WEIGHT 150
882. ****HEAD END OF 3110-CNV-002 AT +938.2
883. 520 521 WEIGHT 19
884. 522 523 WEIGHT 54
885. ****HEAD END OF 3130-CNV-002 AT +938.2
886. 472 473 WEIGHT 21
887. 474 475 WEIGHT 41
888. ****BUNKER LOADS WITH MATERIAL
889. 456 TO 459 WEIGHT 419.4
890. 452 454 WEIGHT 482.2
891. 453 455 WEIGHT 920.2
892. MEMBER WEIGHT
893. ***CONV GALLERY LOADS 3110-CNV-001 _3130-CNV-002
894. *DEAD LOAD
895. 89 91 CON 28.8
896. 90 92 93 CON 33.3
897. *LIVE LOAD (25%)
898. 89 91 CON 5
899. 90 92 93 CON 10.4
900. *CONVEYOR DEAD LOAD
901. 89 91 CON 15.5
902. 90 92 93 CON 22.9
903. *CONVEYOR MATERIAL LOAD (25%)
904. 89 91 CON 4
905. 90 92 93 CON 5.7
906. ***SHORT POST DEAD LOAD
907. 596 TO 599 602 TO 605 CON 10
908. ***SHORT POST LIVE LOAD (25%)
909. 596 TO 599 602 TO 605 CON 1.3
910. *** WALK WAY DEAD LOAD
911. 517 518 596 TO 599 602 TO 607 611 612 615 616 UNI 0.5
912. *** WALK WAY LIVE LOAD (25%)
913. 517 518 596 TO 599 602 TO 607 611 612 615 616 UNI 0.5
914. *****STAIR CASE DEAD LOAD
915. 317 TO 320 329 TO 333 CON 1.8 0.2
916. 317 TO 320 329 TO 333 CON 1.8 1.4
917. 301 317 TO 320 329 TO 333 CON 1.8 1.7
918. 301 317 TO 320 329 TO 333 CON 1.8 2.9
919. ***FRONT STAIR CASE DEAD LOAD
920. 1657 1822 2100 TO 2103 2226 CON 1.8 0.3
921. 1657 1822 2100 TO 2103 2226 CON 1.8 1.3
922. 1657 1822 2100 TO 2103 2226 CON 1.8 1.6
923. 1657 1822 2100 TO 2103 2226 CON 1.8 2.6
924. *
925. *****STAIR CASE LIVE LOAD (25%)
926. 317 TO 320 329 TO 333 CON 1.8 0.2
927. 317 TO 320 329 TO 333 CON 1.8 1.4
928. 301 317 TO 320 329 TO 333 CON 1.8 1.7
929. 301 317 TO 320 329 TO 333 CON 1.8 2.9
930. ***FRONT STAIR CASE LIVE LOAD (25%)
931. 1657 1822 2100 TO 2103 2226 CON 1.8 0.3
932. 1657 1822 2100 TO 2103 2226 CON 1.8 1.3
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935. *3130 FDR-004 LOAD AT +916.3 M LEVEL
936. 1695 1699 CON 20 2.035
937. 1691 1696 CON 15.3 0.4
938. 1691 1696 CON 15.3 3.4
939. 1691 1696 CON 15.3 6.4
940. 1692 1697 CON 15.3 1.8
941. 1692 1697 CON 15.3 4.8
942. 1694 CON 15.3 0.05
943. 1694 CON 15.3 3.05
944. 1698 CON 15.3 1.55
945. 1698 CON 15.3 4.55
946. ****3120 FDR-002
947. ***TAIL END
948. 1682 1687 CON 20 3.65
949. 1682 1687 CON 20 5.65
950. 1683 1688 CON 20 1.5
951. **IMPACT ZONE
952. 1684 1689 CON 16.7 2.5
953. 1685 1690 CON 16.7 0.57
954. 1685 1690 CON 16.7 3.57
955. ****HEAD END OF 3110-CNV-002 AT +938.2
956. 109 113 CON 79 2.25
957. ****HEAD END OF 3130-CNV-002 AT +938.2
958. 140 CON 41 0.3
959. 144 CON 41 2
960. ****MOTOR N GEARBOX LOAD
961. 365 CON 21 4.9
962. 324 CON 21 2.375
963. 134 CON 25.5 0.15
964. 229 CON 25.5 4.5
965. 669 670 CON 26.25 0.1
966. 151 CON 26.25 0.4
967. 152 CON 26.25 3.91
968. ****CHUTE LOAD P15_P16
969. 138 144 CON 15 0.9
970. 109 113 CON 20 1
971. ***SECONDARY SCREEN LOAD
972. 1980 2233 CON 154.53 0.7
973. 1983 2236 CON 222.93 0.5
974. ****EMPTY CHUTE LOAD
975. ****P15 P16 _P17
976. 2290 CON 20 1.115
977. 2290 CON 20 2.465
978. 2290 CON 20 3.81
979. 2233 2234 CON 20 1.2
980. 2233 CON 20 2.3
981. 1980 CON 20 1.2
982. 1980 CON 20 2.3
983. 1980 CON 20 3.6
984. *****P20 _P21
985. 2309 CON 15 1.11
986. 2309 CON 15 2.465
987. 2309 CON 15 3.81
988. 1976 2229 2328 2329 CON 15 0.8
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 991. 1976 2229 2328 2329 CON 15 3.2
 992. 1976 2229 2328 2329 CON 15 3.9
 993. **** TERTIARY SCREEN LOAD
 994. 1895 1901 1907 1914 CON 35 1.35
 995. 1896 1902 1908 1915 CON 35 0.15
 996. **** EMPTY CHUTE LOAD
 997. ****P22 P23 _P24
 998. 2164 2169 2279 2284 CON 15 0.8
 999. 2164 2169 2279 2284 CON 15 2.3
1000. 1895 1901 1907 1914 CON 15 2.05
1001. 1895 1901 1907 1914 CON 15 2.85
1002. ****P25 P26 _P27
1003. 1893 1899 1905 1912 2253 2258 2297 2298 CON 10 0.8
1004. 1893 1899 1905 1912 2253 2297 2298 CON 10 2.3
1005. *TERTIARY FEEDERS
1006. 2046 2048 2050 2052 CON 40 1.5
1007. *SECONDARY FEEDERS
1008. 2054 2056 CON 150 3
1009. FLOOR WEIGHT
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

1010. *FLOOR DL
1011. _FL_916.3 FLOAD 3.6
1012. _FL_916.3_2 FLOAD 4.2
1013. _FL_922.75_1 FLOAD 4.2
1014. _FL_922.75_2 FLOAD 4.2
1015. _FL_926.5_1 FLOAD 4.2
1016. ****STEEL FLOOR
1017. _FL_926.5_2 FLOAD 0.8
1018. _FL_938.2_1 FLOAD 0.8
1019. _FL_938.2_2 FLOAD 0.8
1020. YRANGE 12.2 12.3 FLOAD 0.8 XRANGE 2 12.7 ZRANGE 1.1 20.3
1021. YRANGE 15.15 15.25 FLOAD 0.8 XRANGE 7.35 12.7 ZRANGE 20.2 22.8
1022. YRANGE 14.65 14.75 FLOAD 0.8 XRANGE 12.6 15.8 ZRANGE 1.1 13.99
1023. **FLOOR 25% OF LL
1024. _FL_916.3 FLOAD 1.25
1025. _FL_916.3_2 FLOAD 1.25
1026. _FL_922.75_1 FLOAD 1.25
1027. _FL_922.75_2 FLOAD 1.25
1028. _FL_926.5_1 FLOAD 1.25
1029. ****STEEL FLOOR
1030. *
1031. _FL_938.2_1 FLOAD 1.25
1032. _FL_938.2_2 FLOAD 1.25
1033. YRANGE 12.2 12.3 FLOAD 0.5 XRANGE 2 12.7 ZRANGE 1.1 20.3
1034. YRANGE 15.15 15.25 FLOAD 0.5 XRANGE 7.35 12.7 ZRANGE 20.2 22.8
1035. YRANGE 14.65 14.75 FLOAD 0.5 XRANGE 12.6 15.8 ZRANGE 1.1 13.99
1036. *
1037. LOAD 11 EARTHQUAKE LOAD (+X)
1038. IBC LOAD X 1
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 1039. LOAD 12 EARTHQUAKE LOAD (+Z)
 1040. IBC LOAD Z 1
 1041. LOAD 1 DEAD LOAD (DL)
 1042. SELFWEIGHT Y -1
 1043. FLOOR LOAD
 1044. _FL_916.3 FLOAD -3.6 GY
**NOTE** about FLOOR GROUP command
  All floor panels in a floor group should lie in a single plane.
  If not lying in a single plane break single FLOOR GROUP
  command into multiple FLOOR GROUP commands where each
  floor group lie in a single plane / height.

 1045. _FL_916.3_2 FLOAD -4.2 GY
 1046. _FL_922.75_1 FLOAD -4.2 GY
 1047. _FL_922.75_2 FLOAD -4.2 GY
 1048. _FL_926.5_1 FLOAD -4.2 GY
 1049. ****STEEL FLOOR
 1050. _FL_926.5_2 FLOAD -0.8 GY
 1051. _FL_938.2_1 FLOAD -0.8 GY
 1052. _FL_938.2_2 FLOAD -0.8 GY
 1053. _FL_925.35 FLOAD -0.8 GY
 1054. YRANGE 12.2 12.3 FLOAD -0.8 XRANGE 2 12.7  ZRANGE 1.1 20.3  GY
 1055. YRANGE 15.15 15.25 FLOAD -0.8 XRANGE 7.35 12.7  ZRANGE 20.2 22.8  GY
 1056. YRANGE 14.65 14.75 FLOAD -0.8 XRANGE 12.6 15.8  ZRANGE 1.1 13.99  GY
 1057. ****
 1058. ***CONV GALLERY LOADS 3110-CNV-002 _3130-CNV-002
 1059. MEMBER LOAD
 1060. 89 91 CON GY -28.8
 1061. 89 91 CON GZ 1.6
 1062. 90 CON GY -33.3
 1063. 92 CON GY -33.3 0.49
 1064. 90 CON GZ 1.6
 1065. 92 CON GZ 1.6 0.49
 1066. ***SHORT POST LOAD
 1067. 596 TO 599 602 TO 605 CON GY -10
 1068. *** WALK WAY LOAD
 1069. 517 518 596 TO 599 602 TO 607 611 612 615 616 UNI GY -0.5
 1070. *****STAIR CASE LOAD
 1071. 317 TO 320 329 TO 333 CON GY -1.8 0.2
 1072. 317 TO 320 329 TO 333 CON GY -1.8 1.4
 1073. 301 317 TO 320 329 TO 333 CON GY -1.8 1.7
 1074. 301 317 TO 320 329 TO 333 CON GY -1.8 2.9
 1075. ***FRONT STAIR CASE
 1076. 1657 1822 2100 TO 2103 2226 CON GY -1.8 0.3
 1077. 1657 1822 2100 TO 2103 2226 CON GY -1.8 1.3
 1078. 1657 1822 2100 TO 2103 2226 CON GY -1.8 1.6
 1079. 1657 1822 2100 TO 2103 2226 CON GY -1.8 2.6
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1080. ***LOAD FROM GROUND PORTALS 3130 CNV-003
1081. 1506 1512 1517 CON GY -10 0.4
1082. 1506 1512 1517 CON GY -10 4.5
1083. LOAD 2 LIVE LOAD (LL)
1084. FLOOR LOAD
1085. _FL_916.3 FLOAD -5 GY
1086. _FL_916.3_2 FLOAD -5 GY
1087. _FL_922.75_1 FLOAD -5 GY
1088. _FL_922.75_2 FLOAD -5 GY
1089. _FL_926.5_1 FLOAD -5 GY
1090. ****STEEL FLOOR
1091. _FL_926.5_2 FLOAD -5 GY
1092. _FL_938.2_1 FLOAD -5 GY
1093. _FL_938.2_2 FLOAD -5 GY
1094. _FL_925.35 FLOAD -2 GY
1095. YRANGE 12.2 12.3 FLOAD -2 XRANGE 2 12.7  ZRANGE 1.1 20.3  GY
1096. YRANGE 15.15 15.25 FLOAD -2 XRANGE 7.35 12.7  ZRANGE 20.2 22.8  GY
1097. YRANGE 14.65 14.75 FLOAD -2 XRANGE 12.6 15.8  ZRANGE 1.1 13.99  GY
1098. ****
1099. ***CONV GALLERY LOADS 3110-CNV-002 _3130-CNV-002
1100. MEMBER LOAD
1101. 89 91 CON GY -20
1102. 89 91 CON GZ 0.7
1103. 90 CON GY -41.3
1104. 92 CON GY -41.3 0.49
1105. 90 CON GZ 0.7
1106. 92 CON GZ 0.7 0.49
1107. MEMBER LOAD
1108. ***SHORT POST LOAD
1109. 596 TO 599 602 TO 605 CON GY -5.5
1110. ***WALKWAY LOAD
1111. 517 518 596 TO 599 602 TO 607 611 612 615 616 UNI GY -2
1112. *****STAIR CASE LOAD
1113. 317 TO 320 329 TO 333 CON GY -7.2 0.2
1114. 317 TO 320 329 TO 333 CON GY -7.2 1.4
1115. 301 317 TO 320 329 TO 333 CON GY -7.2 1.7
1116. 301 317 TO 320 329 TO 333 CON GY -7.2 2.9
1117. ***FRONT STAIR CASE
1118. 1657 1822 2100 TO 2103 2226 CON GY -7.2 0.3
1119. 1657 1822 2100 TO 2103 2226 CON GY -7.2 1.3
1120. 1657 1822 2100 TO 2103 2226 CON GY -7.2 1.6
1121. 1657 1822 2100 TO 2103 2226 CON GY -7.2 2.6
1122. ***LOAD FROM GROUND PORTALS 3130 CNV-003
1123. 1506 1512 1517 CON GY -27 0.4
1124. 1506 1512 1517 CON GY -27 4.5
1125. LOAD 3 SUPERIMPOSED DEAD LOAD 1 (SDL1) - (BUNKER EMPTY _CHUTE EMPTY CONDITION)
1126. ****3130 FDR-004 LOAD AT +916.3 M LEVEL
1127. JOINT LOAD
1128. 226 267 FX -2 FY -20 FZ -20
1129. MEMBER LOAD
1130. 1695 1699 CON GX -2 2.035
1131. 1695 1699 CON GY -20 2.035
1132. 1695 1699 CON GZ -20 2.035
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1133. JOINT LOAD
1134. 261 264 FX -2 FY -20 FZ -20
1135. MEMBER LOAD
1136. 1691 1696 CON GY -9.8 0.4
1137. 1691 1696 CON GY -9.8 3.4
1138. 1691 1696 CON GY -9.8 6.4
1139. 1692 1697 CON GY -9.8 1.8 0
1140. 1692 1697 CON GY -9.8 4.8
1141. 1694 CON GY -9.8 0.05
1142. 1694 CON GY -9.8 3.05
1143. 1698 CON GY -9.8 1.55 0
1144. 1698 CON GY -9.8 4.55
1145. ****ALONG
1146. 1691 1696 CON GZ -1.7 0.4
1147. 1691 1696 CON GZ -1.7 3.4
1148. 1691 1696 CON GZ -1.7 6.4
1149. 1692 1697 CON GZ -1.7 1.8 0
1150. 1692 1697 CON GZ -1.7 4.8
1151. 1694 CON GZ -1.7 0.05
1152. 1694 CON GZ -1.7 3.05
1153. 1698 CON GZ -1.7 1.55 0
1154. 1698 CON GZ -1.7 4.55
1155. ****ACROSS
1156. 1691 1696 CON GX -1 0.4
1157. 1691 1696 CON GX -1 3.4
1158. 1691 1696 CON GX -1 6.4
1159. 1692 1697 CON GX -1 1.8 0
1160. 1692 1697 CON GX -1 4.8
1161. 1694 CON GX -1 0.05
1162. 1694 CON GX -1 3.05
1163. 1698 CON GX -1 1.55 0
1164. 1698 CON GX -1 4.55
1165. ****3120 FDR-002 LOAD AT +916.3 M LEVEL
1166. ***TAIL END VERTICAL
1167. 1682 1687 CON GY -20 3.65
1168. 1682 1687 CON GY -20 5.65
1169. 1683 1688 CON GY -20 1.5
1170. ***TAIL END ACROSS
1171. 1682 1687 CON GX -2 3.65
1172. 1682 1687 CON GX -2 5.65
1173. 1683 1688 CON GX -2 1.5
1174. ***TAIL END ALONG
1175. 1682 1687 CON GZ 30 3.65
1176. 1682 1687 CON GZ 30 5.65
1177. 1683 1688 CON GZ 30 1.5
1178. **IMPACT ZONE
1179. 1684 1689 CON GY -11.2 2.5
1180. 1685 1690 CON GY -11.2 0.57
1181. 1685 1690 CON GY -11.2 3.57
1182. ****ACROSS
1183. 1684 1689 CON GX -1 2.5
1184. 1685 1690 CON GX -1 0.57
1185. 1685 1690 CON GX -1 3.57
1186. ****ALONG
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1189. 1685 1690 CON GZ 1.7 3.57
1190. ****
1191. ****VIBRO FEEDERS AT +926.5
1192. *AT FRAME
1193. JOINT LOAD
1194. 124 TO 131 FY -40
1195. 425 427 429 431 FY -40
1196. MEMBER LOAD
1197. 2046 2048 2050 2052 CON GY -40 1.5
1198. 770 TO 773 CON GY -40 0.5
1199. 2053 2055 CON GY -150 0.5
1200. 2054 2056 CON GY -150 2.5
1201. 774 775 CON GY -150 2
1202. JOINT LOAD
1203. 133 135 FY -150
1204. ****HEAD END OF 3110-CNV-002 AT +938.2
1205. ****CASE II-STARTING, CHUTE EMPTY
1206. JOINT LOAD
1207. 520 521 FX 56 FY 38 FZ 4
1208. MEMBER LOAD
1209. 109 113 CON GY -81 2.25
1210. 109 113 CON GX 19 2.25
1211. 109 113 CON GZ 8.2 2.25
1212. JOINT LOAD
1213. 522 523 FX 84 FY -75 FZ 7.5
1214. ****HEAD END OF 3130-CNV-002 AT +938.2
1215. ****CASE II-STARTING, CHUTE EMPTY
1216. JOINT LOAD
1217. 472 473 FX 26 FY 16 FZ 2
1218. 474 475 FX 52 FY -59 FZ 6
1219. MEMBER LOAD
1220. 140 CON GY -36 0.3
1221. 144 CON GY -36 2
1222. 140 CON GX 15 0.3
1223. 144 CON GX 15 2
1224. 140 CON GZ 3.6 0.3
1225. 144 CON GZ 3.6 2
1226. ****
1227. ****MOTOR N GEARBOX LOAD
1228. 365 CON GY -21 4.9
1229. 324 CON GY -21 2.375
1230. 134 CON GY -25.5 0.15
1231. 229 CON GY -25.5 4.5 0
1232. 669 670 CON GY -26.25 0.1
1233. 151 CON GY -26.25 0.4
1234. 152 CON GY -26.25 3.91
1235. ****CHUTE LOAD P15_P16
1236. 138 144 CON GY -15 0.9
1237. 109 113 CON GY -20 1
1238. ****
1239. ***CONV GALLERY LOADS 3110-CNV-001 _3130-CNV-002
1240. *CDL
1241. MEMBER LOAD
1242. 89 91 CON GY -15.5
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1245. 92 CON GY -22.9 0.49
1246. 90 CON GZ 3.8
1247. 92 CON GZ 3.8 0.49
1248. *ML
1249. MEMBER LOAD
1250. 89 91 CON GY -15
1251. 89 91 CON GZ -1.7
1252. 90 CON GY -22.9
1253. 92 CON GY -22.9 0.49
1254. 90 CON GZ 1.7
1255. 92 CON GZ 1.7 0.49
1256. ****
1257. ****LOADING AT 922.75M
1258. ***SECONDARY SCREEN LOAD
1259. MEMBER LOAD
1260. 1980 2233 CON GY -144.14 0.7
1261. 1983 2236 CON GY -196.3 0.5
1262. ***DRIVE LOAD
1263. MEMBER LOAD
1264. 2293 2295 CON GY -3.15 0.5
1265. ****EMPTY CHUTE LOAD
1266. ****P15 P16 _P17
1267. 2290 CON GY -20 1.115
1268. 2290 CON GY -20 2.465
1269. 2290 CON GY -20 3.81
1270. 2233 2234 CON GY -20 1.2
1271. 2233 CON GY -20 2.3
1272. 1980 CON GY -20 1.2
1273. 1980 CON GY -20 2.3
1274. 1980 CON GY -20 3.6
1275. *****P20 _P21
1276. 2309 CON GY -15 1.11
1277. 2309 CON GY -15 2.465
1278. 2309 CON GY -15 3.81
1279. 1976 2229 CON GY -15 0.8
1280. 1976 2229 CON GY -15 1.6
1281. 2329 CON GY -15 0.25
1282. 2328 CON GY -15 0.25
1283. 2329 CON GY -15 1.05
1284. 2328 CON GY -15 1.05
1285. 2329 CON GY -15 1.75
1286. 2328 CON GY -15 1.75
1287. **** TERTIARY SCREEN LOAD
1288. 1895 1901 1907 1914 CON GY -35 1.35
1289. 1896 1902 1908 1915 CON GY -35 0.15
1290. **** EMPTY CHUTE LOAD
1291. ****P22 P23 _P24
1292. 2164 2169 2279 2284 CON GY -15 0.8
1293. 2164 2169 2279 2284 CON GY -15 2.3
1294. 1895 1901 1907 1914 CON GY -15 2.05
1295. 1895 1901 1907 1914 CON GY -15 2.85
1296. ****P25 P26 _P27
1297. 1893 1899 1905 1912 2253 2258 2297 2298 CON GY -10 0.8
1298. 1893 1899 1905 1912 2253 2297 2298 CON GY -10 2.3
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1301. ******
1302. ******LOAD OF EMPTY CHUTE AT 920.55M LVL
1303. ****P1 P2 P3
1304. MEMBER LOAD
1305. 68 CON GY -7.5 0.69
1306. 71 CON GY -7.5 0.69
1307. 68 CON GY -7.5 2.19
1308. 71 CON GY -7.5 2.19
1309. 73 74 CON GY -7.5 1.25
1310. 73 74 CON GY -7.5 1.75
1311. 73 74 CON GY -7.5 2.25
1312. ****P4 _P5
1313. 63 64 CON GY -15 1.715
1314. 63 64 CON GY -15 3.215
1315. 15 23 CON GY -15
1316. ****P6 P7 P8
1317. 65 66 CON GY -15 1.715
1318. 65 66 CON GY -15 3.215
1319. 13 21 CON GY -15 0.65
1320. 13 21 CON GY -15 1.55
1321. ****P9 P10 P11
1322. 75 TO 78 CON GY -10 0.775
1323. 75 TO 78 CON GY -10 1.775
1324. 43 49 55 61 CON GY -10 0.8
1325. 43 49 55 61 CON GY -10 2
1326. ****P12 P13 P14
1327. 79 TO 82 CON GY -15 0.775
1328. 79 TO 82 CON GY -15 1.775
1329. 40 46 52 58 CON GY -15
1330. ****P28 P29 P30 _P31
1331. 464 466 468 470 CON GY -5
1332. 457 458 CON GY -5 1.2
1333. 457 458 CON GY -5 2.3
1334. 238 311 312 CON GY -5
1335. 238 311 312 CON GY -5 1.45
1336. 238 311 312 CON GY -5 3.05
1337. *****P32 P33 P34
1338. 243 246 249 252 CON GY -21 0.85
1339. 243 246 249 252 CON GY -21 2.35
1340. 635 636 649 650 CON GY -21
1341. ***MACHINE LOADING AREA
1342. FLOOR LOAD
1343. YRANGE 9.7 9.8 FLOAD -4 XRANGE -0.1 5.4  ZRANGE -0.1 7.65  GY
1344. YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 0 3.8  GY
1345. YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 20.1 24.3  GY
1346. MEMBER LOAD
1347. 321 325 326 CON GY -12 1.5
1348. 231 232 CON GY -12 4
1349. 230 CON GY -12 0.9
1350. ****BUNKER WEIGHT IN EMPTY
1351. JOINT LOAD
1352. 456 TO 459 FY -56.7
1353. 452 454 FY -69
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1354. 453 455 FY -108.1
1355. ***DUCT LOADS
1356. MEMBER LOAD
1357. 1142 CON GY -11
1358. 1162 CON GY -9
1359. 1162 CON GY -9 0
1360. 535 536 565 566 CON GY -5
1361. 535 536 565 566 CMOM GZ 6.5
1362. 299 CON GY -27 1.3
1363. 300 CON GY -52 1.3
1364. 316 CON GY -75 1.3
1365. 477 483 CON GY -15
1366. 240 CON GY -75
1367. 350 351 CON GY -25
1368. ***MONORAIL AT 916
1369. 2217 CON GY -42.5 0.3
1370. 1892 1911 2208 CON GY -42.5 2.4
1371. 2242 CON GY -30 3.8
1372. 2329 CON GY -30 1.65
1373. ***LOAD FROM GROUND PORTALS 3130 CNV-003
1374. 1506 1512 1517 CON GY -55 0.4
1375. 1506 1512 1517 CON GY -55 4.5
1376. LOAD 4 SUPERIMPOSED DEAD LOAD 2 (SDL2) - (BUNKER _CHUTE FILLED CONDITION)
1377. **** WITH MATERIAL LOAD
1378. ****3130 FDR-004
1379. JOINT LOAD
1380. 226 267 FX 2 FY -20 FZ -20
1381. MEMBER LOAD
1382. 1695 1699 CON GX 2 2.035
1383. 1695 1699 CON GY -20 2.035
1384. 1695 1699 CON GZ -20 2.035
1385. JOINT LOAD
1386. 261 264 FX 2 FY -20 FZ -20
1387. MEMBER LOAD
1388. 1691 1696 CON GY -15.3 0.4
1389. 1691 1696 CON GY -15.3 3.4
1390. 1691 1696 CON GY -15.3 6.4
1391. 1692 1697 CON GY -15.3 1.8 0
1392. 1692 1697 CON GY -15.3 4.8
1393. 1694 CON GY -15.3 0.05
1394. 1694 CON GY -15.3 3.05
1395. 1698 CON GY -15.3 1.55 0
1396. 1698 CON GY -15.3 4.55
1397. ****ALONG
1398. 1691 1696 CON GZ -1.7 0.4
1399. 1691 1696 CON GZ -1.7 3.4
1400. 1691 1696 CON GZ -1.7 6.4
1401. 1692 1697 CON GZ -1.7 1.8 0
1402. 1692 1697 CON GZ -1.7 4.8
1403. 1694 CON GZ -1.7 0.05
1404. 1694 CON GZ -1.7 3.05
1405. 1698 CON GZ -1.7 1.55 0
1406. 1698 CON GZ -1.7 4.55
1407. ****ACROSS
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1410. 1691 1696 CON GX 1 6.4
1411. 1692 1697 CON GX 1 1.8 0
1412. 1692 1697 CON GX 1 4.8
1413. 1694 CON GX 1 0.05
1414. 1694 CON GX 1 3.05
1415. 1698 CON GX 1 1.55 0
1416. 1698 CON GX 1 4.55
1417. ****3120 FDR-002
1418. ***TAIL END VERTICAL
1419. 1682 1687 CON GY -20 3.65
1420. 1682 1687 CON GY -20 5.65
1421. 1683 1688 CON GY -20 1.5
1422. ***TAIL END ACROSS
1423. 1682 1687 CON GX 2 3.65
1424. 1682 1687 CON GX 2 5.65
1425. 1683 1688 CON GX 2 1.5
1426. ***TAIL END ALONG
1427. 1682 1687 CON GZ 30 3.65
1428. 1682 1687 CON GZ 30 5.65
1429. 1683 1688 CON GZ 30 1.5
1430. **IMPACT ZONE
1431. 1684 1689 CON GY -16.7 2.5
1432. 1685 1690 CON GY -16.7 0.57
1433. 1685 1690 CON GY -16.7 3.57
1434. ****ACROSS
1435. 1684 1689 CON GX 1 2.5
1436. 1685 1690 CON GX 1 0.57
1437. 1685 1690 CON GX 1 3.57
1438. ****ALONG
1439. 1684 1689 CON GZ 1.7 2.5
1440. 1685 1690 CON GZ 1.7 0.57
1441. 1685 1690 CON GZ 1.7 3.57
1442. ****VIBRO FEEDERS AT +926.5
1443. *AT MID SPAN (MAX B.M. POSITION)
1444. 770 TO 773 2046 2048 2050 2052 CON GY -40 2.625
1445. 770 TO 773 2046 2048 2050 2052 CON GY -40 4.125
1446. 774 775 2054 2056 CON GY -150 2.75
1447. 774 775 2054 2056 CON GY -150 5.75
1448. ****HEAD END OF 3110-CNV-002 AT +938.2
1449. ****CASE I-RUNNING, MATERIAL LOAD
1450. JOINT LOAD
1451. 520 521 FX 23 FY -19 FZ 2
1452. 522 523 FX 61 FY -54 FZ 5.5
1453. MEMBER LOAD
1454. 109 113 CON GY -79 2.25
1455. 109 113 CON GX 23 2.25
1456. 109 113 CON GZ 8 2.25
1457. ****HEAD END OF 3130-CNV-002 AT +938.2
1458. ****CASE I-RUNNING, MATERIAL LOAD
1459. JOINT LOAD
1460. 472 473 FX 12 FY -21 FZ 2
1461. 474 475 FX 41 FY -46 FZ 5
1462. MEMBER LOAD
1463. 140 CON GY -41 0.3
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1466. 144 CON GX 16 2
1467. 140 CON GZ 4 0.3
1468. 144 CON GZ 4 2
1469. ****
1470. ****MOTOR N GEARBOX LOAD
1471. 365 CON GY -21 4.9
1472. 324 CON GY -21 2.375
1473. 134 CON GY -25.5 0.15
1474. 229 CON GY -25.5 4.5 0
1475. 669 670 CON GY -26.25 0.1
1476. 151 CON GY -26.25 0.4
1477. 152 CON GY -26.25 3.91
1478. ****CHUTE LOAD P15_P16
1479. 138 144 CON GY -15 0.9
1480. 109 113 CON GY -20 1
1481. ***CONV GALLERY LOADS 3110-CNV-002 _3130-CNV-002
1482. *CDL
1483. MEMBER LOAD
1484. 89 91 CON GY -15.5
1485. 89 91 CON GZ -0.3
1486. 90 CON GY -22.9
1487. 92 CON GY -22.9 0.49
1488. 90 CON GZ 3.8
1489. 92 CON GZ 3.8 0.49
1490. *ML
1491. MEMBER LOAD
1492. 89 91 CON GY -15
1493. 89 91 CON GZ -1.7
1494. 90 CON GY -22.9
1495. 1048 CON GY -22.9 0.285
1496. 90 CON GZ 1.7
1497. 1048 CON GZ 1.7 0.285
1498. ****
1499. ***SECONDARY SCREEN LOAD
1500. MEMBER LOAD
1501. 1980 2233 CON GY -154.53 0.7
1502. 1983 2236 CON GY -222.93 0.5
1503. ***DRIVE LOAD
1504. MEMBER LOAD
1505. 2293 2295 CON GY -3.15 0.5
1506. ****BLOCKED CHUTE LOAD
1507. ****P15 P16 _P17
1508. 2290 CON GY -45 1.115
1509. 2290 CON GY -45 2.465
1510. 2290 CON GY -45 3.81
1511. 2233 2234 CON GY -45 1.2
1512. 2233 CON GY -45 2.3
1513. 1980 CON GY -45 1.2
1514. 1980 CON GY -45 2.3
1515. 1980 CON GY -45 3.6
1516. *****P20 _P21
1517. 2309 CON GY -60 1.11
1518. 2309 CON GY -60 2.465
1519. 2309 CON GY -60 3.81
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1522. 2329 CON GY -60 0.25
1523. 2328 CON GY -60 0.25
1524. 2329 CON GY -60 1.05
1525. 2328 CON GY -60 1.05
1526. 2329 CON GY -60 1.75
1527. 2328 CON GY -60 1.75
1528. **** TERTIARY SCREEN LOAD
1529. 1895 1901 1907 1914 CON GY -35 1.35
1530. ***DYNAMIC LOAD
1531. 1895 1901 1907 1914 CON GY -12.24 1.35
1532. 1895 1901 1907 1914 CON GX 6.12 1.35
1533. ***DYNAMIC LOAD
1534. 1896 1902 1908 1915 CON GY -35 0.15
1535. 1896 1902 1908 1915 CON GY -12.24 0.15
1536. 1896 1902 1908 1915 CON GX 6.12 0.15
1537. ****BLOCKED CHUTE LOAD
1538. ****P22 P23 _P24
1539. 2164 2169 2279 2284 CON GY -50 0.8
1540. 2164 2169 2279 2284 CON GY -50 2.3
1541. 1895 1901 1907 1914 CON GY -50 2.05
1542. 1895 1901 1907 1914 CON GY -50 2.85
1543. ****P25 P26 _P27
1544. 1893 1899 1905 1912 2253 2258 2297 2298 CON GY -30 0.8
1545. 1893 1899 1905 1912 2253 2297 2298 CON GY -30 2.3
1546. JOINT LOAD
1547. 153 FY -30
1548. ******LOAD OF BLOCKED CHUTE AT 920.55M LVL
1549. ****P1 P2 P3
1550. MEMBER LOAD
1551. 68 CON GY -55 0.69
1552. 71 CON GY -55 0.69
1553. 68 CON GY -55 2.19
1554. 71 CON GY -55 2.19
1555. 73 74 CON GY -55 1.25
1556. 73 74 CON GY -55 1.75
1557. 73 74 CON GY -55 2.25
1558. ****P4 _P5
1559. 63 64 CON GY -60 1.715
1560. 63 64 CON GY -60 3.215
1561. 15 23 CON GY -60
1562. ****P6 P7 P8
1563. 65 66 CON GY -60 1.715
1564. 65 66 CON GY -60 3.215
1565. 13 21 CON GY -60 0.65
1566. 13 21 CON GY -60 1.55
1567. ****P9 P10 P11
1568. 75 TO 78 CON GY -50 0.775
1569. 75 TO 78 CON GY -50 1.775
1570. 43 49 55 61 CON GY -50 0.8
1571. 43 49 55 61 CON GY -50 2
1572. ****P12 P13 P14
1573. 79 TO 82 CON GY -60 0.75
1574. 79 TO 82 CON GY -60 1.775
1575. 40 46 52 58 CON GY -60
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1578. 457 458 CON GY -15 1.2
1579. 457 458 CON GY -15 2.3
1580. 238 311 312 CON GY -15
1581. 238 311 312 CON GY -15 1.45
1582. 238 311 312 CON GY -15 3.05
1583. *****P32 P33 P34
1584. 243 246 249 252 CON GY -35 0.85
1585. 243 246 249 252 CON GY -35 2.35
1586. 635 636 649 650 CON GY -35
1587. ***MACHINE LOADING AREA
1588. FLOOR LOAD
1589. YRANGE 9.7 9.8 FLOAD -4 XRANGE -0.1 5.4  ZRANGE -0.1 7.65  GY
1590. YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 0 3.8  GY
1591. YRANGE 13.45 13.55 FLOAD -4 XRANGE 22.9 30.1  ZRANGE 20.1 24.3  GY
1592. MEMBER LOAD
1593. 321 325 326 CON GY -12 1.5
1594. 231 232 CON GY -12 4
1595. 230 CON GY -12 0.9
1596. ****BUNKER LOADS WITH MATERIAL
1597. JOINT LOAD
1598. 456 TO 459 FY -419.4
1599. 452 454 FY -482.2
1600. 453 455 FY -920.2
1601. *MONORAIL LOAD
1602. MEMBER LOAD
1603. 708 710 725 727 CON GY -114 3.35
1604. 750 CON GY -114
1605. ***DUCT LOADS
1606. MEMBER LOAD
1607. 1142 CON GY -11
1608. 1162 CON GY -9
1609. 1162 CON GY -9 0
1610. 535 536 565 566 CON GY -5
1611. 535 536 565 566 CMOM GZ 6.5
1612. 299 CON GY -27 1.3
1613. 300 CON GY -52 1.3
1614. 316 CON GY -75 1.3
1615. 477 483 CON GY -15
1616. 240 CON GY -75
1617. 350 351 CON GY -25
1618. ***LOAD FROM GROUND PORTALS 3130 CNV-003
1619. 1506 1512 1517 CON GY -55 0.4
1620. 1506 1512 1517 CON GY -55 4.5
1621. LOAD 5 WIND LOAD IN +X DIRN
1622. MEMBER LOAD
1623. 193 TO 208 UNI GX 0.66
1624. 94 TO 106 209 TO 217 1336 TO 1338 1340 TO 1344 1392 TO 1394 1434 TO 1439 -
1625. 1440 UNI GX 2.19
1626. 1 TO 5 28 TO 32 1099 TO 1105 1113 1114 1332 TO 1334 1358 TO 1361 1363 TO 1367 -
1627. 1369 TO 1374 2313 UNI GX 2.76
1628. 6 TO 11 33 TO 38 1106 TO 1112 1115 1116 1302 TO 1304 1376 TO 1380 -
1629. 1382 TO 1387 1396 TO 1398 1417 TO 1420 2316 UNI GX 2.54
1630. 120 TO 128 1306 TO 1309 1347 TO 1350 1400 TO 1403 1422 TO 1425 1442 TO 1446 -
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1631. 2319 UNI GX 2.11
1632. 115 TO 119 218 TO 226 1117 TO 1121 1311 TO 1316 1352 TO 1356 1405 TO 1410 -
1633. 1412 TO 1415 1427 TO 1432 UNI GX 0.81
1634. *WL ON BUNKERS
1635. 525 527 529 531 533 535 536 538 539 541 543 545 547 549 551 553 555 557 559 -
1636. 561 563 565 566 568 570 TO 572 574 576 578 UNI GX 3.06
1637. 507 UNI GX 4.39
1638. 499 UNI GX 5.83
1639. ***WL ON HEAD END OF 3110-CNV-002 AT +938.2
1640. JOINT LOAD
1641. 520 521 FX 2.6 FY 3.71
1642. 522 523 FX 2.6 FY -3.71
1643. MEMBER LOAD
1644. 109 113 CON GX 2.6 2.25
1645. ****WL ON HEAD END OF 3110-CNV-002 AT +938.2
1646. JOINT LOAD
1647. 472 473 FX 1.86 FY 2.07
1648. 474 475 FX 1.86 FY -2.07
1649. MEMBER LOAD
1650. 140 CON GX 1.86 0.3
1651. 144 CON GX 1.86 2
1652. ***WL ON VIBRO FEEDERS AT +926.5
1653. *TERTIARY
1654. 770 TO 773 CON GX 1.94 2.375
1655. 770 TO 773 CON GX 1.94 3.875
1656. 770 TO 773 CON GY 1.68 2.375
1657. 770 TO 773 CON GY -1.68 3.875
1658. *SECONDARY
1659. 774 775 CON GX 3.05 1.5
1660. 774 775 CON GX 3.05 4.5
1661. 774 775 CON GY 6.85 1.5
1662. 774 775 CON GY -6.85 4.5
1663. ***WL ON SECONDARY SCREEN
1664. 1980 2233 CON GX 10.11 0.45
1665. 1983 2236 CON GX 10.11 0.5
1666. 1980 2233 CON GY 10.86 0.45
1667. 1983 2236 CON GY -10.86 0.5
1668. ****WL ON TERTIARY SCREEN
1669. 1895 1901 1907 1914 CON GX 3.41 1.35
1670. 1896 1902 1908 1915 CON GX 3.41 0.15
1671. 1895 1901 1907 1914 CON GY 5.33 1.35
1672. 1896 1902 1908 1915 CON GY -5.33 0.15
1673. ***WL ON SECONDARY CHUTE
1674. 2233 2234 CON GX 5.45 1.2
1675. 1980 CON GX 5.45 1.2
1676. 1980 CON GX 5.45 3.6
1677. ***WL ON TERTIARY CHUTE
1678. 1895 1901 1907 1914 CON GX 5.16 2.05
1679. 1895 1901 1907 1914 CON GX 5.16 2.85
1680. ***WL ON PLATFORMS
1681. 278 TO 287 289 TO 294 341 TO 346 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
1682. 416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 1134 TO 1141 -
1683. 1144 TO 1159 1164 TO 1171 UNI GX 0.5
1684. LOAD 6 WIND LOAD IN -X DIRN
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1687. 94 TO 106 209 TO 217 1336 TO 1338 1340 TO 1344 1392 TO 1394 1434 TO 1439 -
1688. 1440 UNI GX -2.19
1689. 1 TO 5 28 TO 32 1099 TO 1105 1113 1114 1332 TO 1334 1358 TO 1361 1363 TO 1367 -
1690. 1369 TO 1374 2313 UNI GX -2.76
1691. 6 TO 11 33 TO 38 1106 TO 1112 1115 1116 1302 TO 1304 1376 TO 1380 -
1692. 1382 TO 1387 1396 TO 1398 1417 TO 1420 2316 UNI GX -2.54
1693. 120 TO 128 1306 TO 1309 1347 TO 1350 1400 TO 1403 1422 TO 1425 1442 TO 1446 -
1694. 2319 UNI GX -2.11
1695. 115 TO 119 218 TO 226 1117 TO 1121 1311 TO 1316 1352 TO 1356 1405 TO 1410 -
1696. 1412 TO 1415 1427 TO 1432 UNI GX -0.81
1697. *WL ON BUNKERS
1698. 495 526 528 530 532 534 537 540 542 544 546 548 550 552 554 556 558 560 562 -
1699. 564 567 569 573 575 577 579 933 1122 2110 TO 2117 UNI GX -3.06
1700. 508 UNI GX -4.39
1701. 500 UNI GX -5.83
1702. ***WL ON HEAD END OF 3110-CNV-002 AT +938.2
1703. JOINT LOAD
1704. 520 521 FX -2.6 FY -3.71
1705. 522 523 FX -2.6 FY 3.71
1706. MEMBER LOAD
1707. 109 113 CON GX -2.6 2.25
1708. ****WL ON HEAD END OF 3110-CNV-002 AT +938.2
1709. JOINT LOAD
1710. 472 473 FX -1.86 FY -2.07
1711. 474 475 FX -1.86 FY 2.07
1712. MEMBER LOAD
1713. 140 CON GX -1.86 0.3
1714. 144 CON GX -1.86 2
1715. ***WL ON VIBRO FEEDERS AT +926.5
1716. *TERTIARY
1717. 770 TO 773 CON GX -1.94 2.375
1718. 770 TO 773 CON GX -1.94 3.875
1719. 770 TO 773 CON GY -1.68 2.375
1720. 770 TO 773 CON GY 1.68 3.875
1721. *SECONDARY
1722. 774 775 CON GX -3.05 1.5
1723. 774 775 CON GX -3.05 4.5
1724. 774 775 CON GY -6.85 1.5
1725. 774 775 CON GY 6.85 4.5
1726. ***WL ON SECONDARY SCREEN
1727. 1980 2233 CON GX -10.11 0.45
1728. 1983 2236 CON GX -10.11 0.5
1729. 1980 2233 CON GY -10.86 0.45
1730. 1983 2236 CON GY 10.86 0.5
1731. ****WL ON TERTIARY SCREEN
1732. 1895 1901 1907 1914 CON GX -3.41 1.35
1733. 1896 1902 1908 1915 CON GX -3.41 0.15
1734. 1895 1901 1907 1914 CON GY -5.33 1.35
1735. 1896 1902 1908 1915 CON GY 5.33 0.15
1736. ***WL ON SECONDARY CHUTE
1737. 2233 2234 CON GX -5.45 1.2
1738. 1980 CON GX -5.45 1.2
1739. 1980 CON GX -5.45 3.6
1740. ***WL ON TERTIARY CHUTE
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1743. ***WL ON PLATFORMS
1744. 278 TO 287 289 TO 294 341 TO 346 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
1745. 416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 UNI GX -0.5
1746. LOAD 7 WIND LOAD IN +Z DIRN
1747. MEMBER LOAD
1748. 1332 TO 1334 1392 TO 1394 1396 TO 1398 1400 TO 1403 1405 TO 1410 UNI GZ 2.06
1749. 1336 TO 1338 1358 TO 1361 1389 1390 1417 TO 1420 1422 TO 1425 1427 TO 1430 -
1750. 2320 UNI GZ 4.13
1751. 193 TO 195 1340 TO 1343 1347 TO 1349 1352 TO 1354 1363 TO 1365 1376 TO 1378 -
1752. 2313 2316 2319 UNI GZ 3.52
1753. 28 TO 38 94 TO 99 115 TO 119 196 TO 198 1344 1345 1355 1356 1366 1367 1379 -
1754. 1380 UNI GZ 1.46
1755. 1 TO 11 100 TO 106 199 TO 208 1117 TO 1121 1369 TO 1374 1382 TO 1387 1412 -
1756. 1413 TO 1415 1431 1432 1434 TO 1440 1442 TO 1446 UNI GZ 3.16
1757. 120 TO 128 209 TO 226 1099 TO 1116 1302 TO 1304 1306 TO 1309 1311 TO 1315 -
1758. 1316 UNI GZ 1.7
1759. *WL ON BUNKERS
1760. 496 TO 498 504 TO 506 UNI GZ 3.06
1761. 497 UNI GZ 5.02
1762. 505 UNI GZ 3.93
1763. ***WL ON HEAD END OF 3110-CNV-002 AT +938.2
1764. JOINT LOAD
1765. 520 522 FY 3.21 FZ 2.25
1766. 521 523 FY -3.21 FZ 2.25
1767. MEMBER LOAD
1768. 109 113 CON GZ 2.25 2.25
1769. 109 CON GY 3.21 2.25
1770. 113 CON GY -3.21 2.25
1771. ****WL ON HEAD END OF 3110-CNV-002 AT +938.2
1772. JOINT LOAD
1773. 472 474 FY 1.49 FZ 1.16
1774. 473 475 FY -1.49 FZ 1.16
1775. MEMBER LOAD
1776. 140 CON GZ 1.16 0.3
1777. 140 CON GY 1.49 0.3
1778. 144 CON GZ 1.16 2
1779. 144 CON GY -1.49 2
1780. ***WL ON VIBRO FEEDERS AT +926.5
1781. *TERTIARY
1782. 770 TO 773 CON GZ 2.02 2.375
1783. 770 TO 773 CON GZ 2.02 3.875
1784. 770 772 CON GY 1.75 2.375
1785. 770 772 CON GY 1.75 3.875
1786. 771 773 CON GY -1.75 2.375
1787. 771 773 CON GY -1.75 3.875
1788. *SECONDARY
1789. 774 775 CON GZ 3.48 1.5
1790. 774 775 CON GZ 3.48 4.5
1791. 774 CON GY 2.7 1.5
1792. 774 CON GY 2.7 4.5
1793. 775 CON GY -2.7 1.5
1794. 775 CON GY -2.7 4.5
1795. ***WL ON SECONDARY SCREEN
1796. 1980 2233 CON GZ 9.45 0.45
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1799. 1983 CON GY 10.15 0.5
1800. 2233 CON GY -10.15 0.45
1801. 2236 CON GY -10.15 0.5
1802. ****WL ON TERTIARY SCREEN
1803. 1895 1901 1907 1914 CON GZ 6.02 1.35
1804. 1896 1902 1908 1915 CON GZ 6.02 0.15
1805. 1895 1907 CON GY 9.4 1.35
1806. 1896 1908 CON GY 9.4 0.15
1807. 1901 1914 CON GY -9.4 1.35
1808. 1902 1915 CON GY -9.4 0.15
1809. ***WL ON SECONDARY CHUTE
1810. 2233 2234 CON GZ 8.01 1.2
1811. 1980 CON GZ 8.01 1.2
1812. 1980 CON GZ 8.01 3.6
1813. ***WL ON TERTIARY CHUTE
1814. 1895 1901 1907 1914 CON GZ 8.68 2.05
1815. 1895 1901 1907 1914 CON GZ 8.68 2.85
1816. ***WL ON PLATFORMS
1817. 278 TO 287 289 TO 294 341 TO 348 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
1818. 416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 UNI GZ 0.5
1819. **WL FROM CONVEYOR GALLERY 3110-002 _3130-002
1820. *JOINT LOAD
1821. *745 758 FY 59.1 FZ 19.3
1822. *749 760 FY -59.1 FZ 19.3
1823. MEMBER LOAD
1824. 89 91 CON GY 59.1
1825. 89 91 CON GZ 19.3
1826. 90 CON GY -59.1
1827. 92 CON GY -59.1 0.49
1828. 90 CON GZ 19.3
1829. 92 CON GZ 19.3 0.49
1830. LOAD 8 WIND LOAD IN -Z DIRN
1831. MEMBER LOAD
1832. 1332 TO 1334 1392 TO 1394 1396 TO 1398 1400 TO 1403 1405 TO 1410 UNI GZ -2.06
1833. 1336 TO 1338 1358 TO 1361 1389 1390 1417 TO 1420 1422 TO 1425 1427 TO 1430 -
1834. 2320 UNI GZ -4.13
1835. 193 TO 195 1340 TO 1343 1347 TO 1349 1352 TO 1354 1363 TO 1365 1376 TO 1378 -
1836. 2313 2316 2319 UNI GZ -3.52
1837. 28 TO 38 94 TO 99 115 TO 119 196 TO 198 1344 1345 1355 1356 1366 1367 1379 -
1838. 1380 UNI GZ -1.46
1839. 1 TO 11 100 TO 106 199 TO 208 1117 TO 1121 1369 TO 1374 1382 TO 1387 1412 -
1840. 1413 TO 1415 1431 1432 1434 TO 1440 1442 TO 1446 UNI GZ -3.16
1841. 120 TO 128 209 TO 226 1099 TO 1116 1302 TO 1304 1306 TO 1309 1311 TO 1315 -
1842. 1316 UNI GZ -1.7
1843. *WL ON BUNKERS
1844. 501 TO 503 509 TO 511 UNI GZ -3.06
1845. 502 UNI GZ -5.02
1846. 510 UNI GZ -3.93
1847. ***WL ON HEAD END OF 3110-CNV-002 AT +938.2
1848. JOINT LOAD
1849. 520 522 FY -3.21 FZ -2.25
1850. 521 523 FY 3.21 FZ -2.25
1851. MEMBER LOAD
1852. 109 113 CON GZ -2.25 2.25
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1855. ****WL ON HEAD END OF 3110-CNV-002 AT +938.2
1856. JOINT LOAD
1857. 472 474 FY -1.49 FZ -1.16
1858. 473 475 FY 1.49 FZ -1.16
1859. MEMBER LOAD
1860. 140 CON GZ -1.16 0.3
1861. 140 CON GY -1.49 0.3
1862. 144 CON GZ -1.16 2
1863. 144 CON GY 1.49 2
1864. ***WL ON VIBRO FEEDERS AT +926.5
1865. *TERTIARY
1866. 770 TO 773 CON GZ -2.02 2.375
1867. 770 TO 773 CON GZ -2.02 3.875
1868. 770 772 CON GY -1.75 2.375
1869. 770 772 CON GY -1.75 3.875
1870. 771 773 CON GY 1.75 2.375
1871. 771 773 CON GY 1.75 3.875
1872. *SECONDARY
1873. 774 775 CON GZ -3.48 1.5
1874. 774 775 CON GZ -3.48 4.5
1875. 774 CON GY -2.7 1.5
1876. 774 CON GY -2.7 4.5
1877. 775 CON GY 2.7 1.5
1878. 775 CON GY 2.7 4.5
1879. ***WL ON SECONDARY SCREEN
1880. 1980 2233 CON GZ -9.45 0.45
1881. 1983 2236 CON GZ -9.45 0.5
1882. 1980 CON GY -10.15 0.45
1883. 1983 CON GY -10.15 0.5
1884. 2233 CON GY 10.15 0.45
1885. 2236 CON GY 10.15 0.5
1886. ****WL ON TERTIARY SCREEN
1887. 1895 1901 1907 1914 CON GZ -6.02 1.35
1888. 1896 1902 1908 1915 CON GZ -6.02 0.15
1889. 1895 1907 CON GY -9.4 1.35
1890. 1896 1908 CON GY -9.4 0.15
1891. 1901 1914 CON GY 9.4 1.35
1892. 1902 1915 CON GY 9.4 0.15
1893. ***WL ON SECONDARY CHUTE
1894. 2233 2234 CON GZ -8.01 1.2
1895. 1980 CON GZ -8.01 1.2
1896. 1980 CON GZ -8.01 3.6
1897. ***WL ON TERTIARY CHUTE
1898. 1895 1901 1907 1914 CON GZ -8.68 2.05
1899. 1895 1901 1907 1914 CON GZ -8.68 2.85
1900. ***WL ON PLATFORMS
1901. 278 TO 287 289 TO 294 341 TO 348 355 TO 362 385 TO 388 392 TO 402 406 TO 411 -
1902. 416 TO 422 424 TO 427 430 434 442 471 TO 476 478 TO 482 494 UNI GZ -0.5
1903. **WL FROM CONVEYOR GALLERY 3110-002 _3130-002
1904. *JOINT LOAD
1905. *745 758 FY -59.1 FZ -19.3
1906. *749 760 FY 59.1 FZ -19.3
1907. MEMBER LOAD
1908. 89 91 CON GY -59.1
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1911. 92 CON GY 59.1 0.49
1912. 90 CON GZ -19.3
1913. 92 CON GZ -19.3 0.49
1914. LOAD 9 TEMERATURE LOAD 1
1915. TEMPERATURE LOAD
1916. 1 TO 30 32 TO 44 46 TO 50 52 TO 67 70 75 TO 92 94 TO 107 109 TO 111 -
1917. 113 TO 118 120 TO 124 133 TO 142 144 TO 193 195 TO 235 237 TO 239 -
1918. 242 TO 287 289 TO 298 301 308 TO 314 317 TO 323 325 TO 342 349 352 -
1919. 354 TO 417 425 TO 471 496 TO 511 516 TO 518 525 TO 528 530 532 TO 534 536 -
1920. 537 539 TO 564 566 567 569 TO 591 594 TO 619 628 TO 633 635 TO 642 -
1921. 645 TO 647 649 TO 652 654 TO 660 662 TO 667 669 TO 678 681 TO 696 -
1922. 699 TO 704 707 TO 715 720 TO 732 735 TO 743 747 TO 749 751 TO 778 780 782 -
1923. 783 TO 790 794 TO 825 828 TO 884 898 899 901 TO 945 951 TO 968 973 974 980 -
1924. 981 TO 1009 1011 TO 1037 1039 TO 1045 1052 TO 1097 1099 TO 1133 TEMP 25
1925. LOAD 10 TEMERATURE LOAD 2
1926. TEMPERATURE LOAD
1927. 1 TO 30 32 TO 44 46 TO 50 52 TO 67 70 75 TO 92 94 TO 107 109 TO 111 -
1928. 113 TO 118 120 TO 124 133 TO 142 144 TO 193 195 TO 235 237 TO 239 -
1929. 242 TO 287 289 TO 298 301 308 TO 314 317 TO 323 325 TO 342 349 352 -
1930. 354 TO 417 425 TO 471 496 TO 511 516 TO 518 525 TO 528 530 532 TO 534 536 -
1931. 537 539 TO 564 566 567 569 TO 591 594 TO 619 628 TO 633 635 TO 642 -
1932. 645 TO 647 649 TO 652 654 TO 660 662 TO 667 669 TO 678 681 TO 696 -
1933. 699 TO 704 707 TO 715 720 TO 732 735 TO 743 747 TO 749 751 TO 778 780 782 -
1934. 783 TO 790 794 TO 825 828 TO 884 898 899 901 TO 945 951 TO 968 973 974 980 -
1935. 981 TO 1009 1011 TO 1037 1039 TO 1045 1052 TO 1097 1099 TO 1133 TEMP -25
1936. LOAD 301 DL+SDL1
1937. REPEAT LOAD
1938. 1 1.0 3 1.0
1939. LOAD 302 DL+SDL2
1940. REPEAT LOAD
1941. 1 1.0 4 1.0
1942. LOAD 303 DL + SDL1 + TL1 + LL
1943. REPEAT LOAD
1944. 1 1.0 3 1.0 9 1.0 2 1.0
1945. LOAD 304 DL + SDL1 + TL2 + LL
1946. REPEAT LOAD
1947. 1 1.0 4 1.0 9 1.0 2 1.0
1948. LOAD 305 DL + SDL2 + TL1 + LL
1949. REPEAT LOAD
1950. 1 1.0 3 1.0 10 1.0 2 1.0
1951. LOAD 306 DL + SDL2 + TL2 + LL
1952. REPEAT LOAD
1953. 1 1.0 4 1.0 10 1.0 2 1.0
1954. LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
1955. REPEAT LOAD
1956. 1 1.0 3 1.0 2 0.75 5 0.75
1957. LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
1958. REPEAT LOAD
1959. 1 1.0 3 1.0 2 0.75 6 0.75
1960. LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
1961. REPEAT LOAD
1962. 1 1.0 3 1.0 2 0.75 7 0.75
1963. LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
1964. REPEAT LOAD
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1967. REPEAT LOAD
1968. 1 1.0 4 1.0 2 0.75 5 0.75
1969. LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
1970. REPEAT LOAD
1971. 1 1.0 4 1.0 2 0.75 6 0.75
1972. LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
1973. REPEAT LOAD
1974. 1 1.0 4 1.0 2 0.75 7 0.75
1975. LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
1976. REPEAT LOAD
1977. 1 1.0 4 1.0 2 0.75 8 0.75
1978. LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
1979. REPEAT LOAD
1980. 1 1.0 3 1.0 2 0.75 11 0.525
1981. LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
1982. REPEAT LOAD
1983. 1 1.0 3 1.0 2 0.75 11 -0.525
1984. LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
1985. REPEAT LOAD
1986. 1 1.0 3 1.0 2 0.75 12 0.525
1987. LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
1988. REPEAT LOAD
1989. 1 1.0 3 1.0 2 0.75 12 -0.525
1990. LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
1991. REPEAT LOAD
1992. 1 1.0 4 1.0 2 0.75 11 0.525
1993. LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
1994. REPEAT LOAD
1995. 1 1.0 4 1.0 2 0.75 11 -0.525
1996. LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
1997. REPEAT LOAD
1998. 1 1.0 4 1.0 2 0.75 12 0.525
1999. LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
2000. REPEAT LOAD
2001. 1 1.0 4 1.0 2 0.75 12 -0.525
2002. LOAD 323 0.6 (DL + SDL1) + WL(+X)
2003. REPEAT LOAD
2004. 1 0.6 3 0.6 5 1.0
2005. LOAD 324 0.6 (DL + SDL1) + WL(-X)
2006. REPEAT LOAD
2007. 1 0.6 3 0.6 6 1.0
2008. LOAD 325 0.6 (DL + SDL1) + WL(+Z)
2009. REPEAT LOAD
2010. 1 0.6 3 0.6 7 1.0
2011. LOAD 326 0.6 (DL + SDL1) + WL(-Z)
2012. REPEAT LOAD
2013. 1 0.6 3 0.6 8 1.0
2014. LOAD 327 0.6 (DL + SDL2) + WL(+X)
2015. REPEAT LOAD
2016. 1 0.6 4 0.6 5 1.0
2017. LOAD 328 0.6 (DL + SDL2) + WL(-X)
2018. REPEAT LOAD
2019. 1 0.6 4 0.6 6 1.0
2020. LOAD 329 0.6 (DL + SDL2) + WL(+Z)
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2023. LOAD 330 0.6 (DL + SDL2) + WL(-Z)
2024. REPEAT LOAD
2025. 1 0.6 4 0.6 8 1.0
2026. LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
2027. REPEAT LOAD
2028. 1 0.6 3 0.6 11 0.7
2029. LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
2030. REPEAT LOAD
2031. 1 0.6 3 0.6 11 -0.7
2032. LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
2033. REPEAT LOAD
2034. 1 0.6 3 0.6 12 0.7
2035. LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
2036. REPEAT LOAD
2037. 1 0.6 3 0.6 12 -0.7
2038. LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
2039. REPEAT LOAD
2040. 1 0.6 4 0.6 11 0.7
2041. LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
2042. REPEAT LOAD
2043. 1 0.6 4 0.6 11 -0.7
2044. LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
2045. REPEAT LOAD
2046. 1 0.6 4 0.6 11 -0.7
2047. LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
2048. REPEAT LOAD
2049. 1 0.6 4 0.6 11 -0.7
2050. ****
2051. *LOAD COMBINATION FOR CONCRETE DESIGN
2052. LOAD 401 1.4 (DL+SDL1)
2053. REPEAT LOAD
2054. 1 1.4 3 1.4
2055. LOAD 402 1.4 (DL+SDL2)
2056. REPEAT LOAD
2057. 1 1.4 4 1.4
2058. LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
2059. REPEAT LOAD
2060. 1 1.4 3 1.4 9 1.4 2 1.7
2061. LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL
2062. REPEAT LOAD
2063. 1 1.4 3 1.4 10 1.4 2 1.7
2064. LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL
2065. REPEAT LOAD
2066. 1 1.4 4 1.4 9 1.4 2 1.7
2067. LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
2068. REPEAT LOAD
2069. 1 1.4 4 1.4 10 1.4 2 1.7
2070. LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
2071. REPEAT LOAD
2072. 1 1.2 3 1.2 2 0.5 5 1.6
2073. LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
2074. REPEAT LOAD
2075. 1 1.2 3 1.2 2 0.5 6 1.6
2076. LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
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2079. LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
2080. REPEAT LOAD
2081. 1 1.2 3 1.2 2 0.5 8 1.6
2082. LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
2083. REPEAT LOAD
2084. 1 1.2 4 1.2 2 0.5 5 1.6
2085. LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
2086. REPEAT LOAD
2087. 1 1.2 4 1.2 2 0.5 6 1.6
2088. LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
2089. REPEAT LOAD
2090. 1 1.2 4 1.2 2 0.5 7 1.6
2091. LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
2092. REPEAT LOAD
2093. 1 1.2 4 1.2 2 0.5 8 1.6
2094. LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
2095. REPEAT LOAD
2096. 1 1.2 3 1.2 2 0.5 11 1.0
2097. LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
2098. REPEAT LOAD
2099. 1 1.2 3 1.2 2 0.5 11 -1.0
2100. LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
2101. REPEAT LOAD
2102. 1 1.2 3 1.2 2 0.5 12 1.0
2103. LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
2104. REPEAT LOAD
2105. 1 1.2 3 1.2 2 0.5 12 -1.0
2106. LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
2107. REPEAT LOAD
2108. 1 1.2 4 1.2 2 0.5 11 1.0
2109. LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
2110. REPEAT LOAD
2111. 1 1.2 4 1.2 2 0.5 11 -1.0
2112. LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
2113. REPEAT LOAD
2114. 1 1.2 4 1.2 2 0.5 12 1.0
2115. LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
2116. REPEAT LOAD
2117. 1 1.2 4 1.2 2 0.5 12 -1.0
2118. LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
2119. REPEAT LOAD
2120. 1 0.9 3 0.9 5 1.6
2121. LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
2122. REPEAT LOAD
2123. 1 0.9 3 0.9 6 1.6
2124. LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
2125. REPEAT LOAD
2126. 1 0.9 3 0.9 7 1.6
2127. LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
2128. REPEAT LOAD
2129. 1 0.9 3 0.9 8 1.6
2130. LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
2131. REPEAT LOAD
2132. 1 0.9 4 0.9 5 1.6
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2135. 1 0.9 4 0.9 6 1.6
2136. LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
2137. REPEAT LOAD
2138. 1 0.9 4 0.9 7 1.6
2139. LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
2140. REPEAT LOAD
2141. 1 0.9 4 0.9 8 1.6
2142. LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
2143. REPEAT LOAD
2144. 1 0.9 3 0.9 11 1.0
2145. LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
2146. REPEAT LOAD
2147. 1 0.9 3 0.9 11 -1.0
2148. LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
2149. REPEAT LOAD
2150. 1 0.9 3 0.9 12 1.0
2151. LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
2152. REPEAT LOAD
2153. 1 0.9 3 0.9 12 -1.0
2154. LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
2155. REPEAT LOAD
2156. 1 0.9 4 0.9 11 1.0
2157. LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
2158. REPEAT LOAD
2159. 1 0.9 4 0.9 11 -1.0
2160. LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
2161. REPEAT LOAD
2162. 1 0.9 4 0.9 12 1.0
2163. LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
2164. REPEAT LOAD
2165. 1 0.9 4 0.9 12 -1.0
2166. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**976**	**NUMBER OF MEMBERS**	**1828**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**31**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 88, TOTAL DEGREES OF FREEDOM = 5670
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.

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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.028 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  1.153 T =  0.619 Tuser = 0.619                 *
*  TIME PERIOD USED (T) =  0.619                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.009 X     24713.62   *
*                       =       212.40  KN                 *
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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.028 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  1.153 T =  0.619 Tuser = 0.619                 *
*  TIME PERIOD USED (T) =  0.619                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.009 X     24713.62   *
*                       =       212.40  KN                 *
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2167. DEFINE ENVELOPE
2168. 301 TO 338 ENVELOPE 1 TYPE STRENGTH
2169. 401 TO 438 ENVELOPE 2 TYPE STRENGTH
2170. END DEFINE ENVELOPE
2171. LOAD LIST 401 TO 438
2172. START CONCRETE DESIGN
2173. CODE ACI
2174. CLB 0.05 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
2175. 318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
2176. 1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
2177. 1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
2178. 1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
2179. 1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
2180. 1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2181. 2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
2182. CLS 0.05 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
2183. 318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
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2186. 1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
2187. 1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
2188. 1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2189. 2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
2190. CLT 0.05 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
2191. 318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
2192. 1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
2193. 1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
2194. 1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
2195. 1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
2196. 1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2197. 2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
2198. FC 35000 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 317 -
2199. 318 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
2200. 1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
2201. 1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
2202. 1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
2203. 1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
2204. 1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2205. 2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
2206. FYMAIN 420000 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 -
2207. 317 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
2208. 1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
2209. 1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
2210. 1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
2211. 1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
2212. 1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2213. 2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
2214. FYSEC 420000 MEMB 12 TO 27 39 TO 44 46 TO 50 52 TO 82 193 TO 196 199 TO 204 -
2215. 317 TO 319 330 331 333 818 819 828 TO 831 946 1023 TO 1026 1039 TO 1045 1248 -
2216. 1249 1301 TO 1316 1319 1320 1322 1331 TO 1344 1346 TO 1355 1357 TO 1366 1368 -
2217. 1369 TO 1373 1375 TO 1379 1381 TO 1386 1388 TO 1431 1433 TO 1439 1441 TO 1446 -
2218. 1501 TO 1524 1531 TO 1555 1651 TO 1672 1675 TO 1699 1701 TO 1706 -
2219. 1710 TO 1716 1719 TO 1727 1801 TO 1838 1892 TO 1917 1951 TO 1964 -
2220. 1974 TO 1978 1980 TO 1984 2001 TO 2056 2100 TO 2106 2151 TO 2191 -
2221. 2201 TO 2313 2316 2319 TO 2325 2328 TO 2330
2222. DESIGN BEAM 1501 TO 1524 1531 TO 1541 1543 TO 1546 1548 TO 1555
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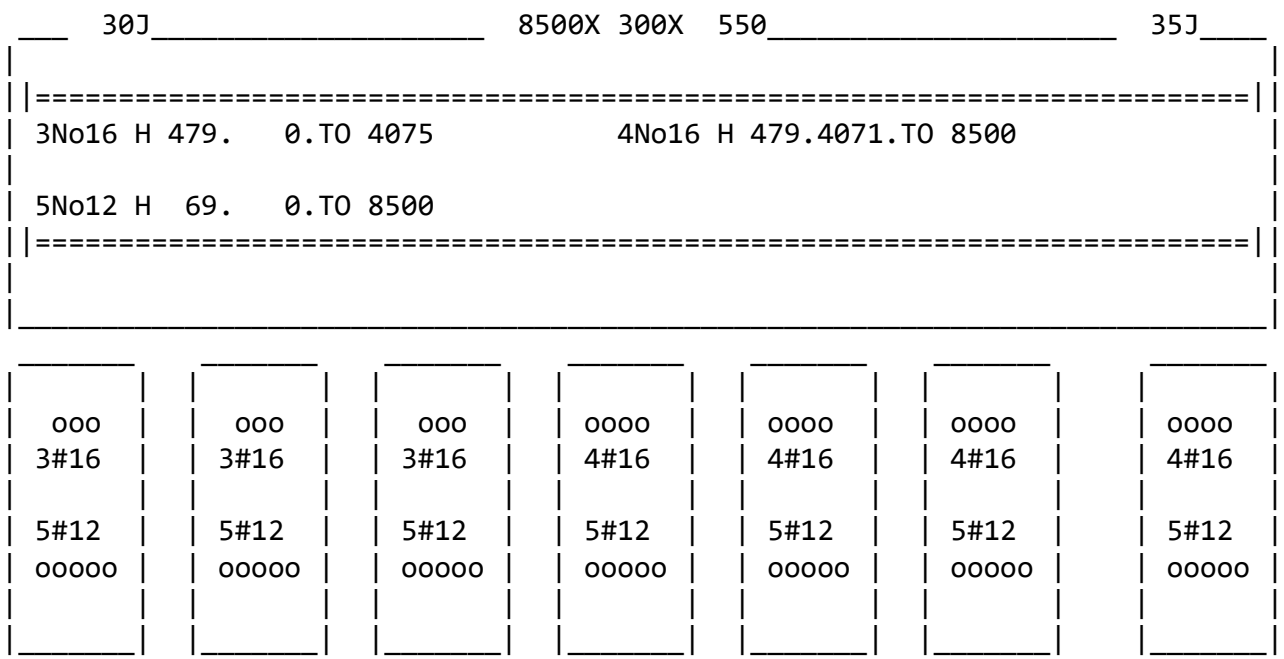
ACI 318-11 BEAM NO. 1501 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	8500.	YES	YES
2	479.	3 - 16MM	0.	4075.	YES	NO
3	479.	4 - 16MM	4071.	8500.	NO	YES

B E A M N O. 1501 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.23 KNS Vc= 138.85 KNS Vs= 0.00 KNS
Tu= 1.04 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 25.87 KNS Vc= 138.85 KNS Vs= 0.00 KNS
Tu= 1.04 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



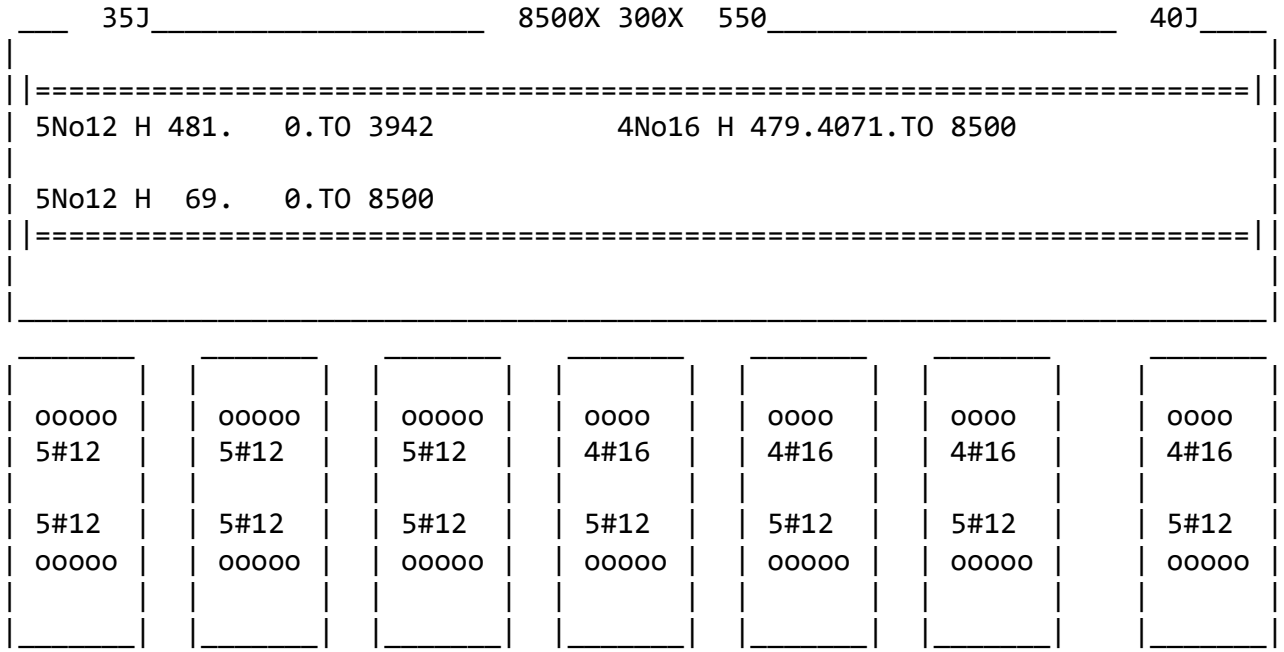
ACI 318-11 BEAM NO. 1502 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	8500.	YES	YES
2	481.	5 - 12MM	0.	3942.	YES	NO
3	479.	4 - 16MM	4071.	8500.	NO	YES

B E A M N O. 1502 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.53 KNS Vc= 254.36 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 25.57 KNS Vc= 138.04 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



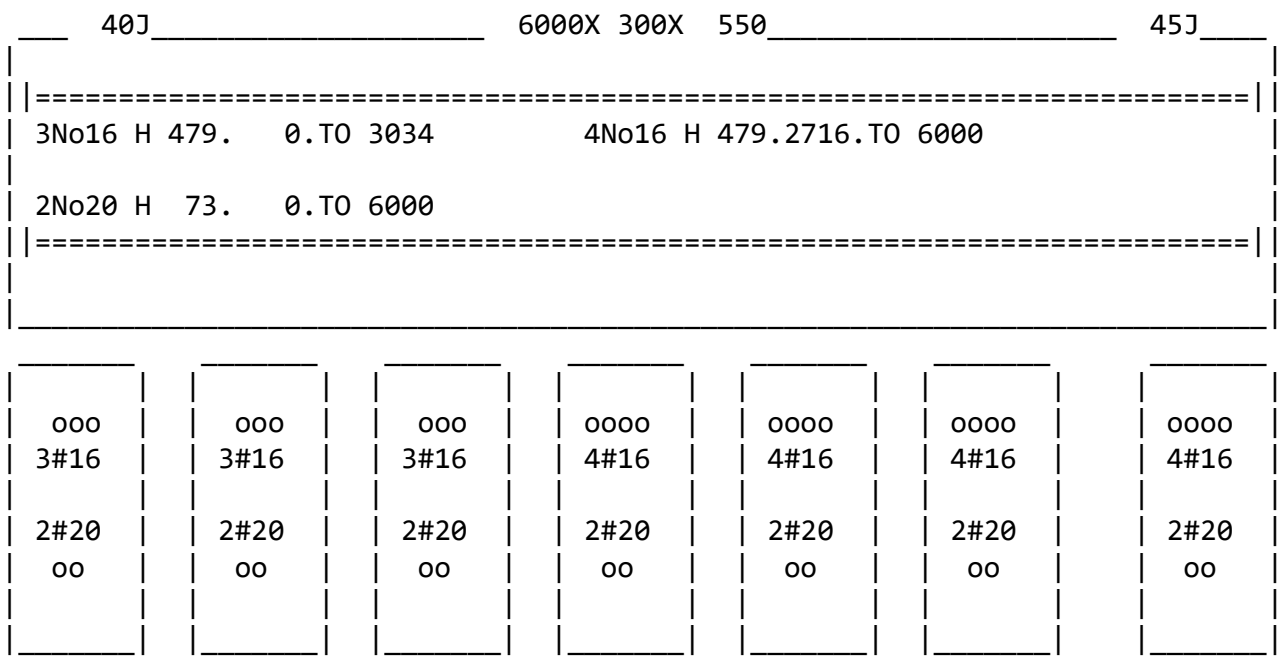
ACI 318-11 BEAM NO. 1503 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	2 - 20MM	0.	6000.	YES	YES
2	479.	3 - 16MM	0.	3034.	YES	NO
3	479.	4 - 16MM	2716.	6000.	NO	YES

B E A M N O. 1503 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.17 KNS Vc= 141.16 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 21.32 KNS Vc= 138.65 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



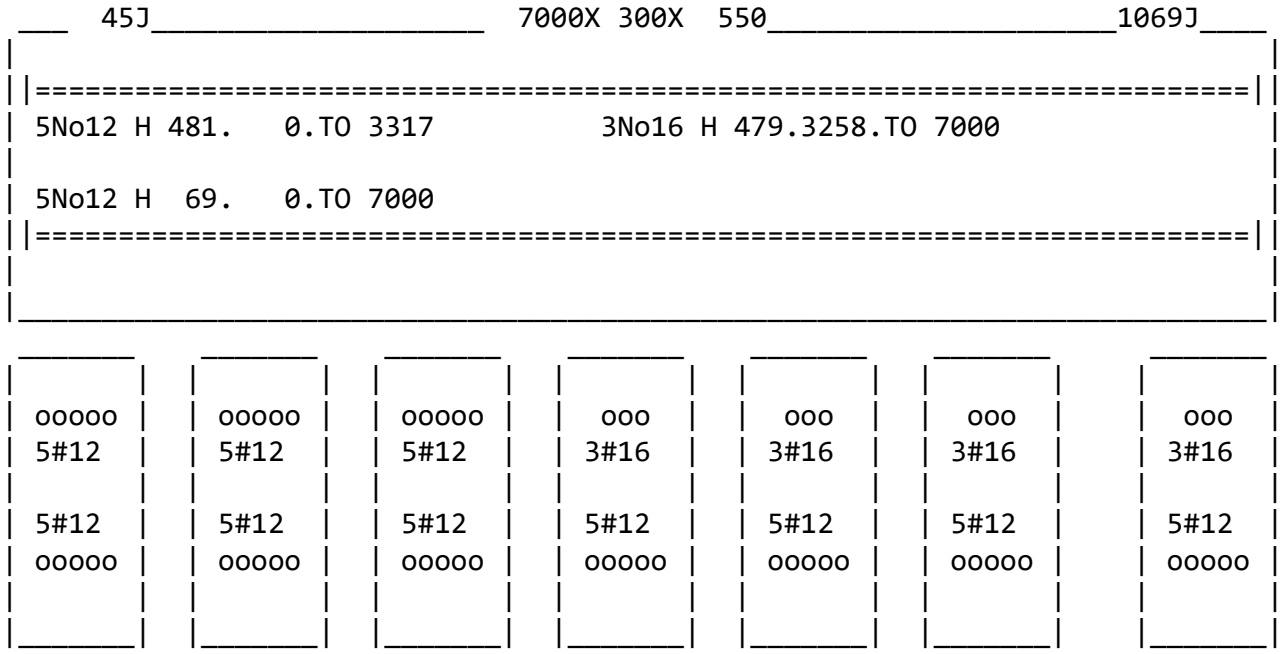
ACI 318-11 BEAM NO. 1504 DESIGN RESULTS

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7000.	YES	YES
2	481.	5 - 12MM	0.	3317.	YES	NO
3	479.	3 - 16MM	3258.	7000.	NO	YES

B E A M N O. 1504 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.09 KNS Vc= 257.13 KNS Vs= 0.00 KNS
Tu= 0.64 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 21.84 KNS Vc= 139.64 KNS Vs= 0.00 KNS
Tu= 0.64 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



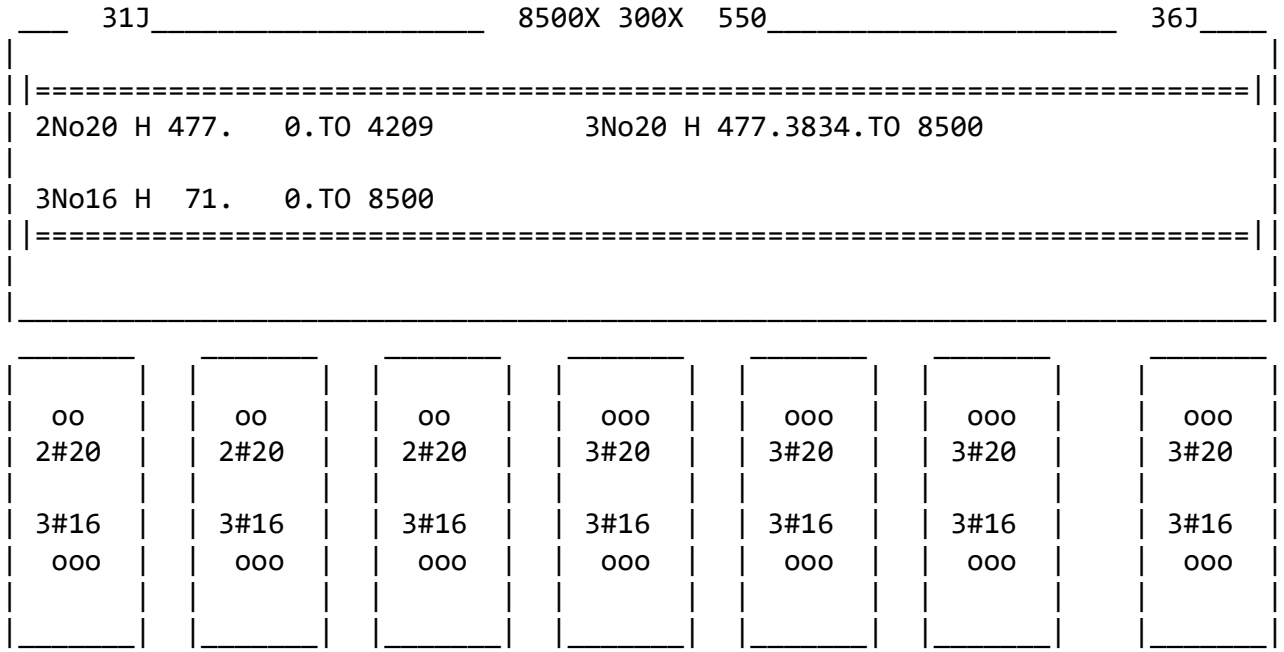
ACI 318-11 BEAM NO. 1505 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	8500.	YES	YES
2	477.	2 - 20MM	0.	4209.	YES	NO
3	477.	3 - 20MM	3834.	8500.	NO	YES

B E A M N O. 1505 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.03 KNS Vc= 130.55 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.07 KNS Vc= 130.55 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1506 DESIGN RESULTS

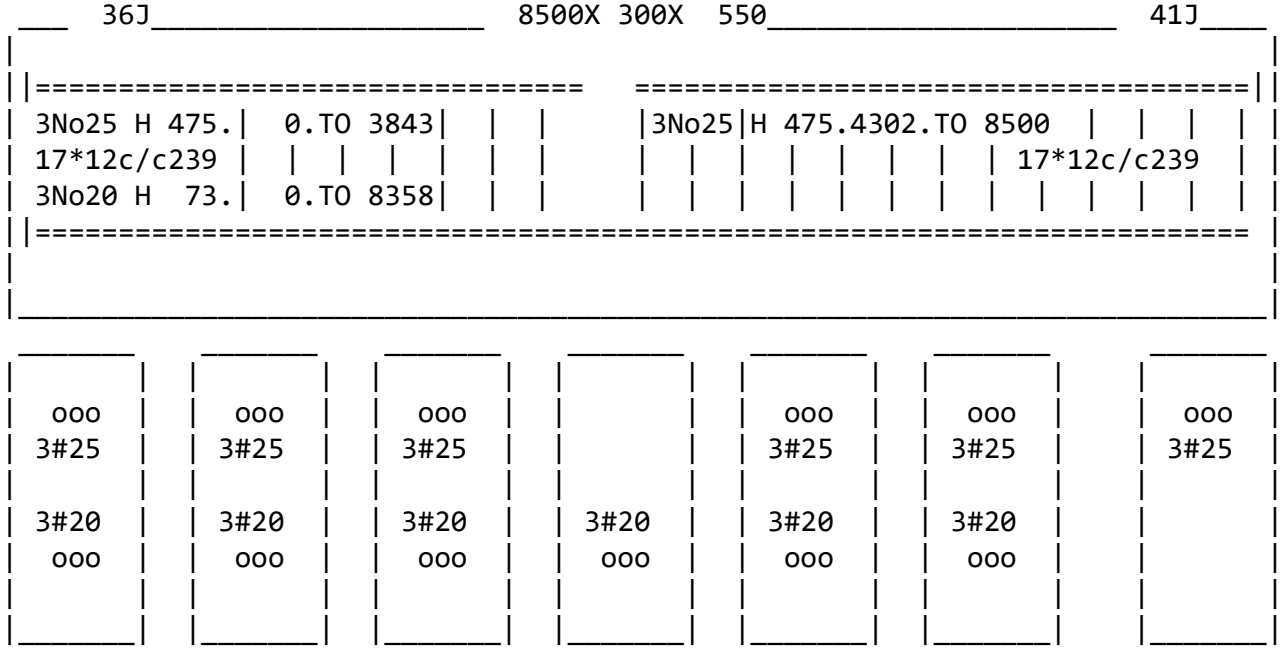
LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	8358.	YES	NO
2	475.	3 - 25MM	0.	3843.	YES	NO
3	475.	3 - 25MM	4302.	8500.	NO	YES

B E A M N O. 1506 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 84.81 KNS Vc= 145.34 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 3775. MM

AT END SUPPORT - Vu= 77.32 KNS Vc= 139.31 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 3775. MM



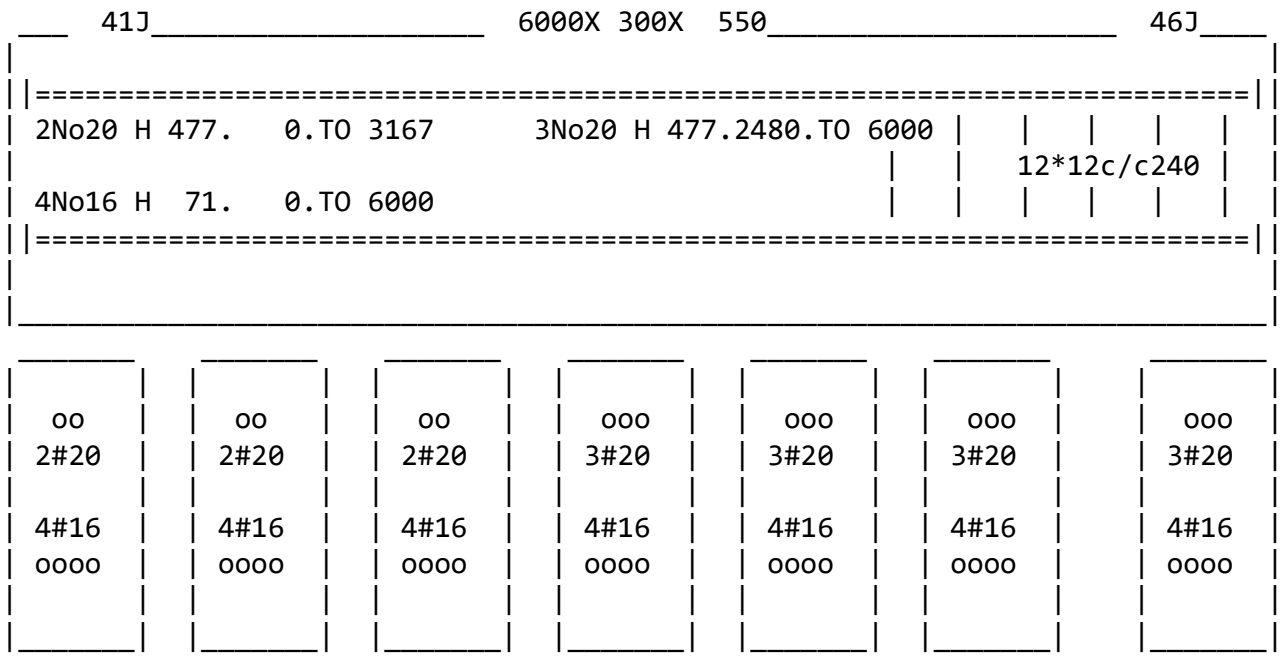
ACI 318-11 BEAM NO. 1507 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	6000.	YES	YES
2	477.	2 - 20MM	0.	3167.	YES	NO
3	477.	3 - 20MM	2480.	6000.	NO	YES

B E A M N O. 1507 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.79 KNS Vc= 164.07 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 6.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 54.85 KNS Vc= 137.90 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 7.1 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 2525. MM



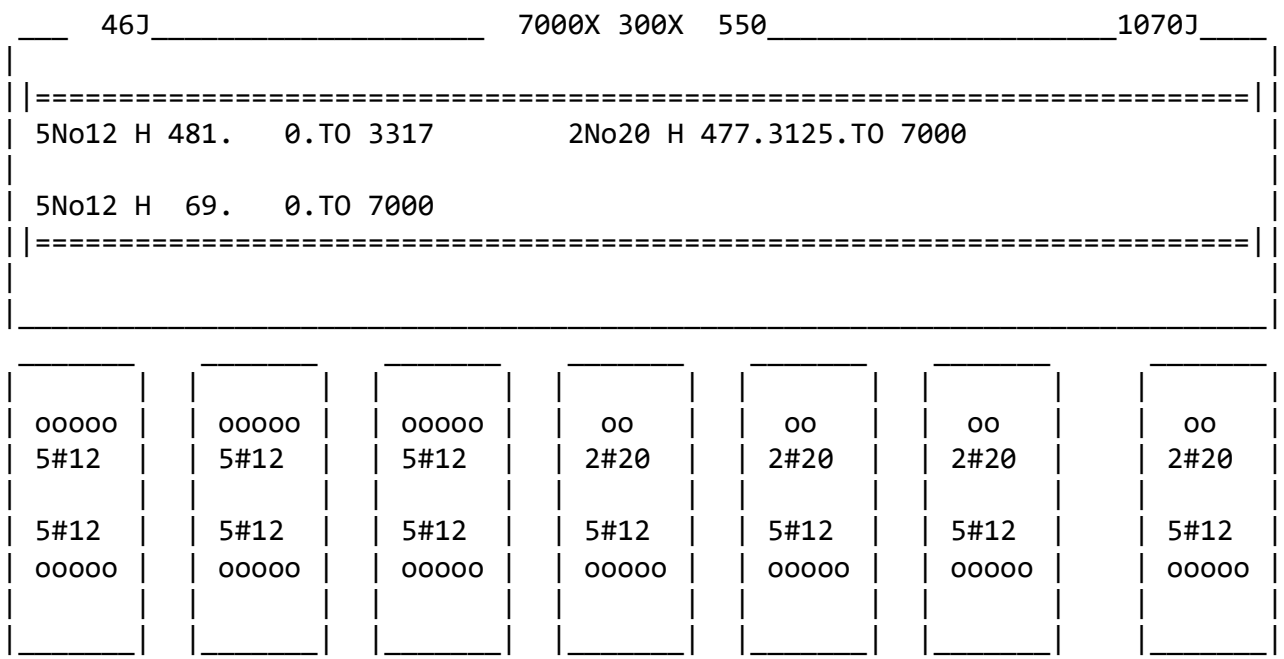
ACI 318-11 BEAM NO. 1508 DESIGN RESULTS

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7000.	YES	YES
2	481.	5 - 12MM	0.	3317.	YES	NO
3	477.	2 - 20MM	3125.	7000.	NO	YES

B E A M N O. 1508 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.85 KNS Vc= 258.12 KNS Vs= 0.00 KNS
Tu= 0.37 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.08 KNS Vc= 139.89 KNS Vs= 0.00 KNS
Tu= 0.37 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



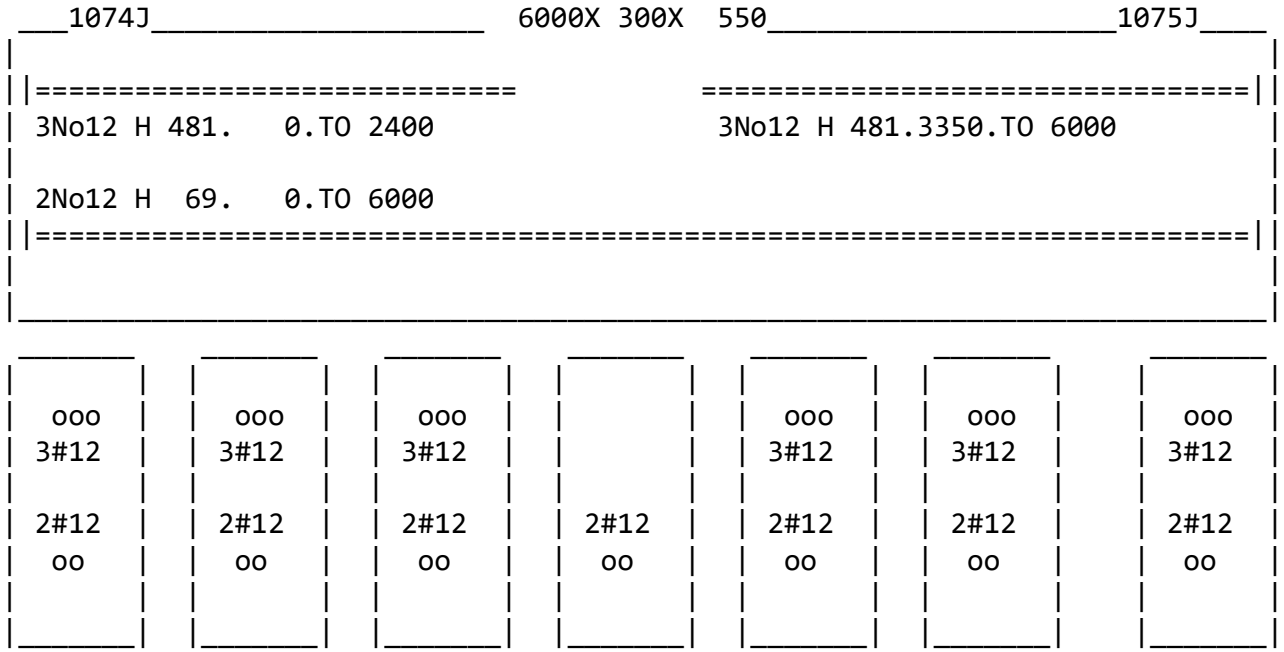
ACI 318-11 BEAM NO. 1509 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	6000.	YES	YES
2	481.	3 - 12MM	0.	2400.	YES	NO
3	481.	3 - 12MM	3350.	6000.	NO	YES

B E A M N O. 1509 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.21 KNS Vc= 136.87 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.28 KNS Vc= 136.44 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1510 DESIGN RESULTS

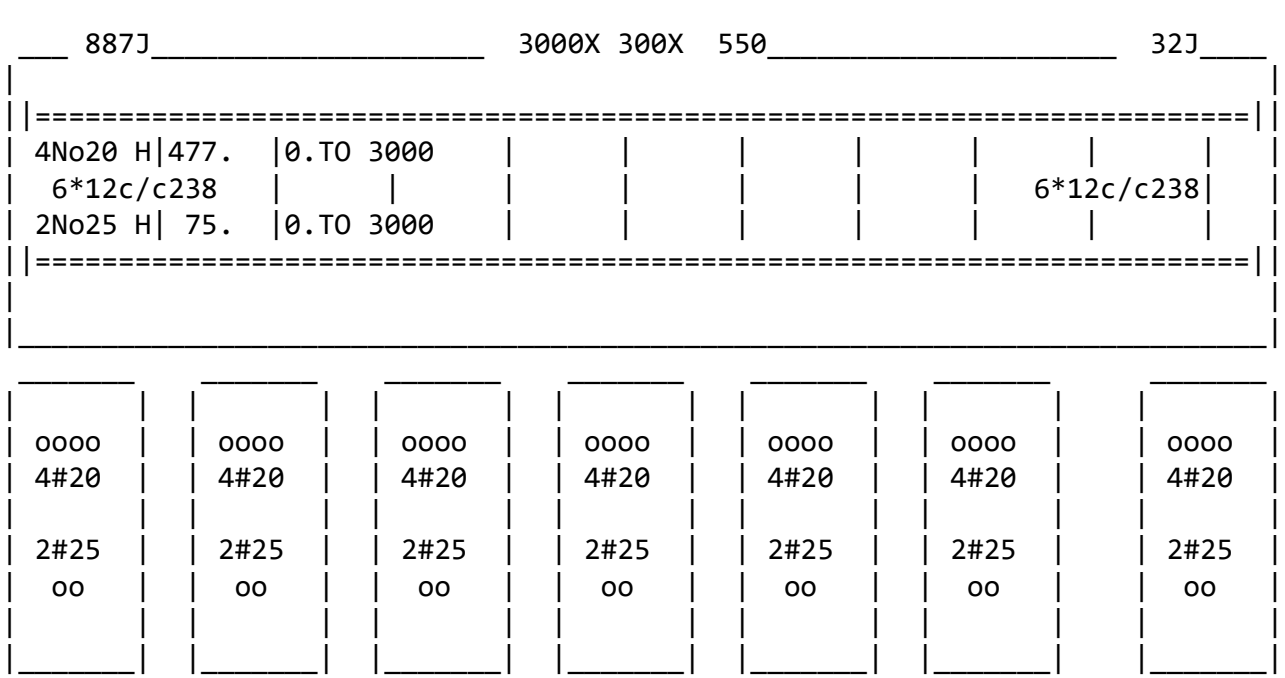
LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	3000.	YES	YES
2	477.	4 - 20MM	0.	3000.	YES	YES

B E A M N O. 1510 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 115.66 KNS Vc= 141.66 KNS Vs= 12.54 KNS
Tu= 1.15 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 1025. MM

AT END SUPPORT - Vu= 125.22 KNS Vc= 140.55 KNS Vs= 26.41 KNS
Tu= 1.15 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 1025. MM



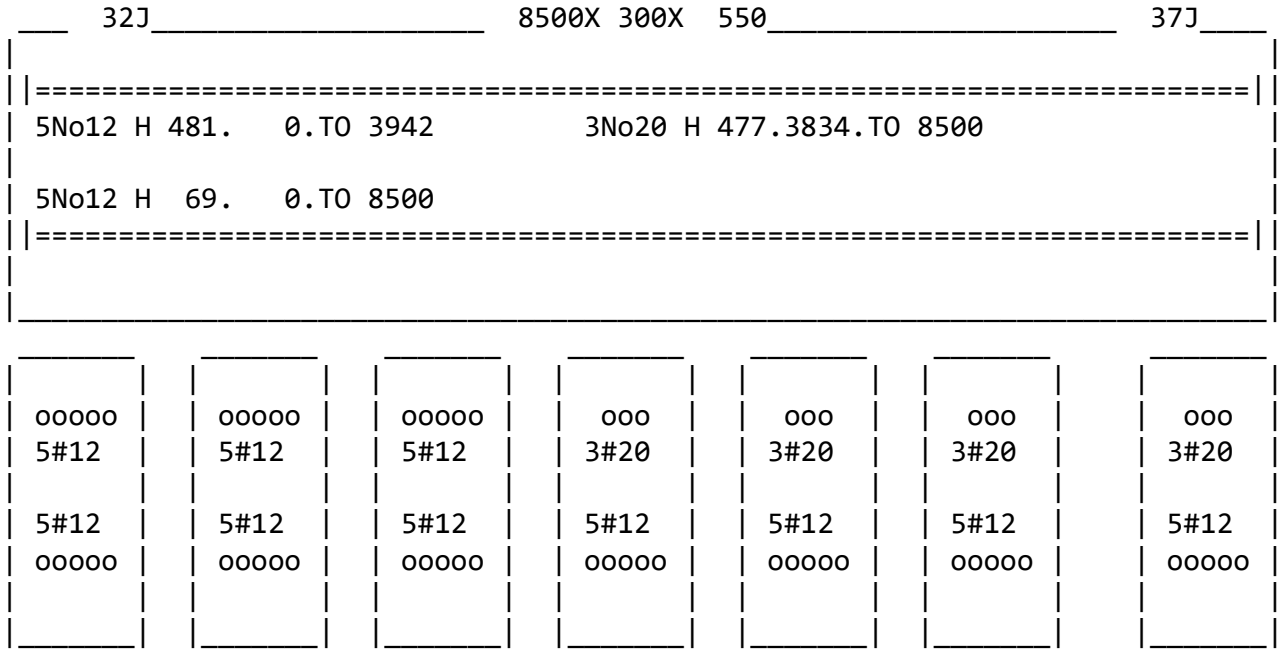
ACI 318-11 BEAM NO. 1511 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	8500.	YES	YES
2	481.	5 - 12MM	0.	3942.	YES	NO
3	477.	3 - 20MM	3834.	8500.	NO	YES

B E A M N O. 1511 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.75 KNS Vc= 131.60 KNS Vs= 0.00 KNS
Tu= 0.89 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 29.34 KNS Vc= 131.60 KNS Vs= 0.00 KNS
Tu= 0.89 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1512 DESIGN RESULTS

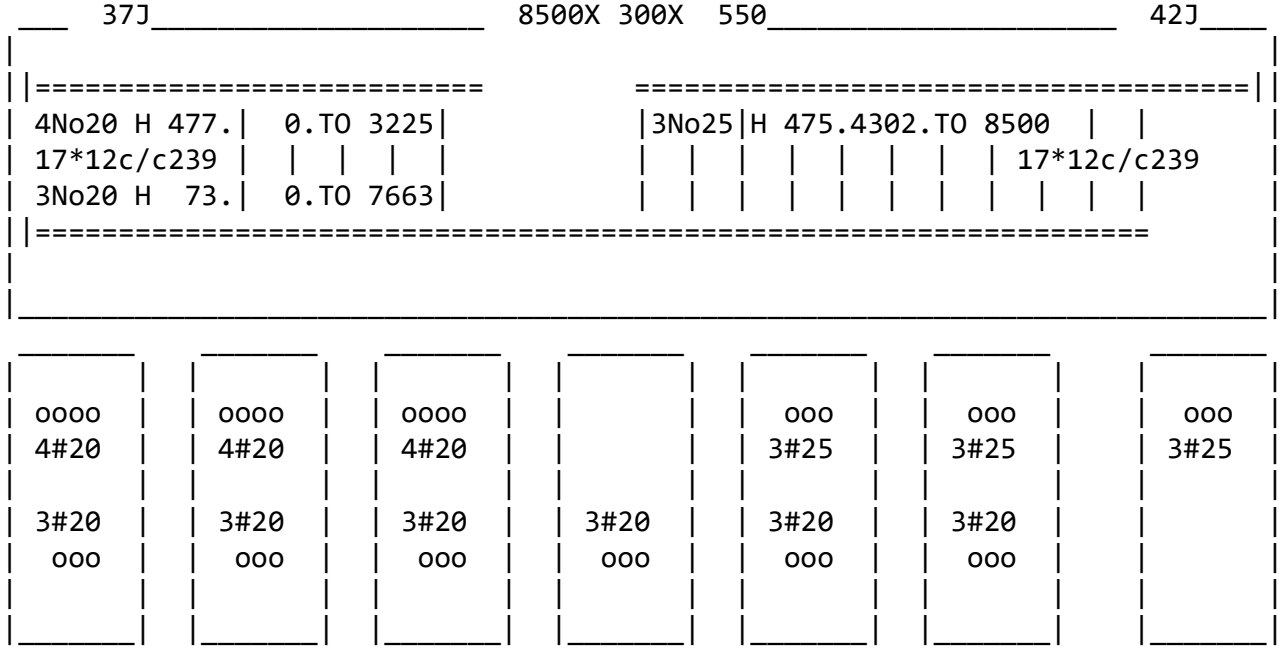
LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	7663.	YES	NO
2	477.	4 - 20MM	0.	3225.	YES	NO
3	475.	3 - 25MM	4302.	8500.	NO	YES

B E A M N O. 1512 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 83.42 KNS Vc= 148.19 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 3775. MM

AT END SUPPORT - Vu= 78.70 KNS Vc= 139.54 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 3775. MM



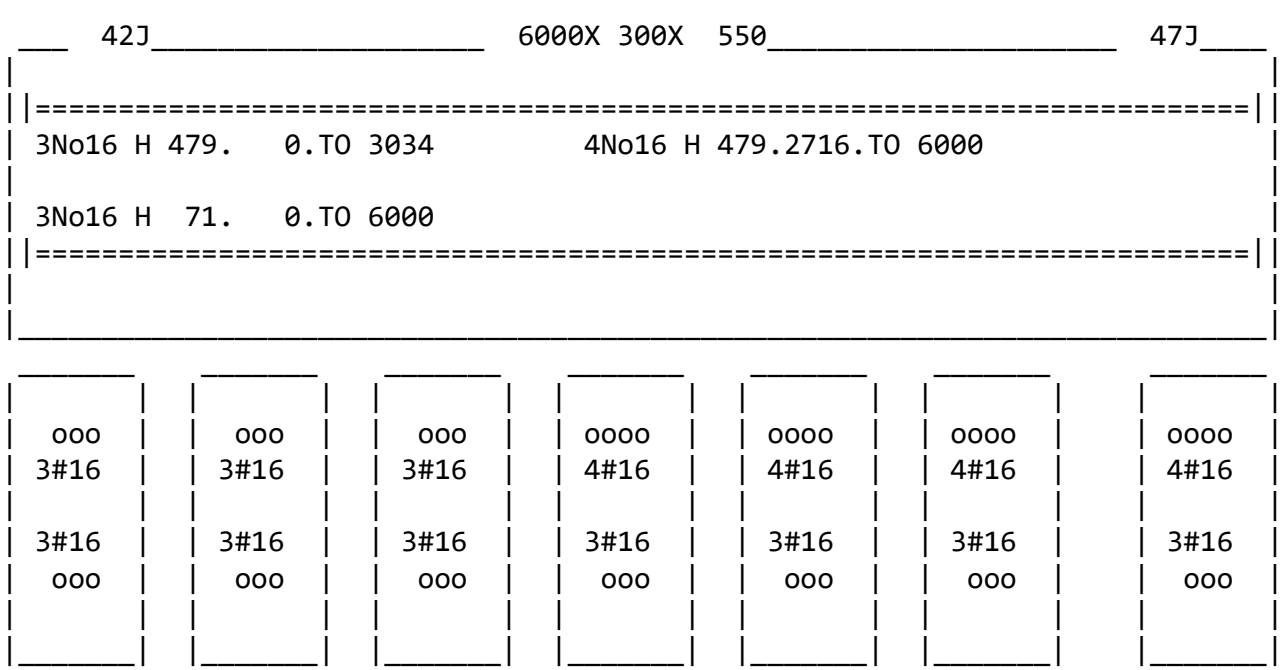
ACI 318-11 BEAM NO. 1513 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	6000.	YES	YES
2	479.	3 - 16MM	0.	3034.	YES	NO
3	479.	4 - 16MM	2716.	6000.	NO	YES

B E A M N O. 1513 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.37 KNS Vc= 256.74 KNS Vs= 0.00 KNS
Tu= 0.25 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 20.12 KNS Vc= 139.62 KNS Vs= 0.00 KNS
Tu= 0.25 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



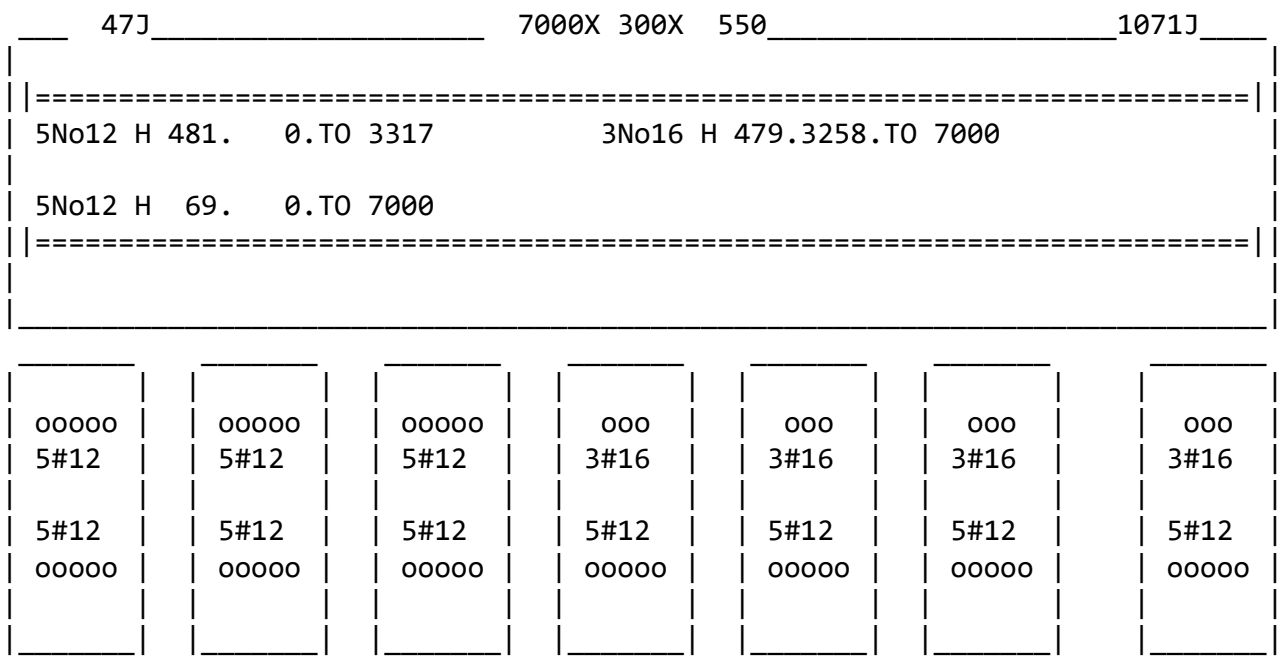
ACI 318-11 BEAM NO. 1514 DESIGN RESULTS

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7000.	YES	YES
2	481.	5 - 12MM	0.	3317.	YES	NO
3	479.	3 - 16MM	3258.	7000.	NO	YES

B E A M N O. 1514 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.82 KNS Vc= 258.77 KNS Vs= 0.00 KNS
Tu= 0.54 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.12 KNS Vc= 140.08 KNS Vs= 0.00 KNS
Tu= 0.54 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



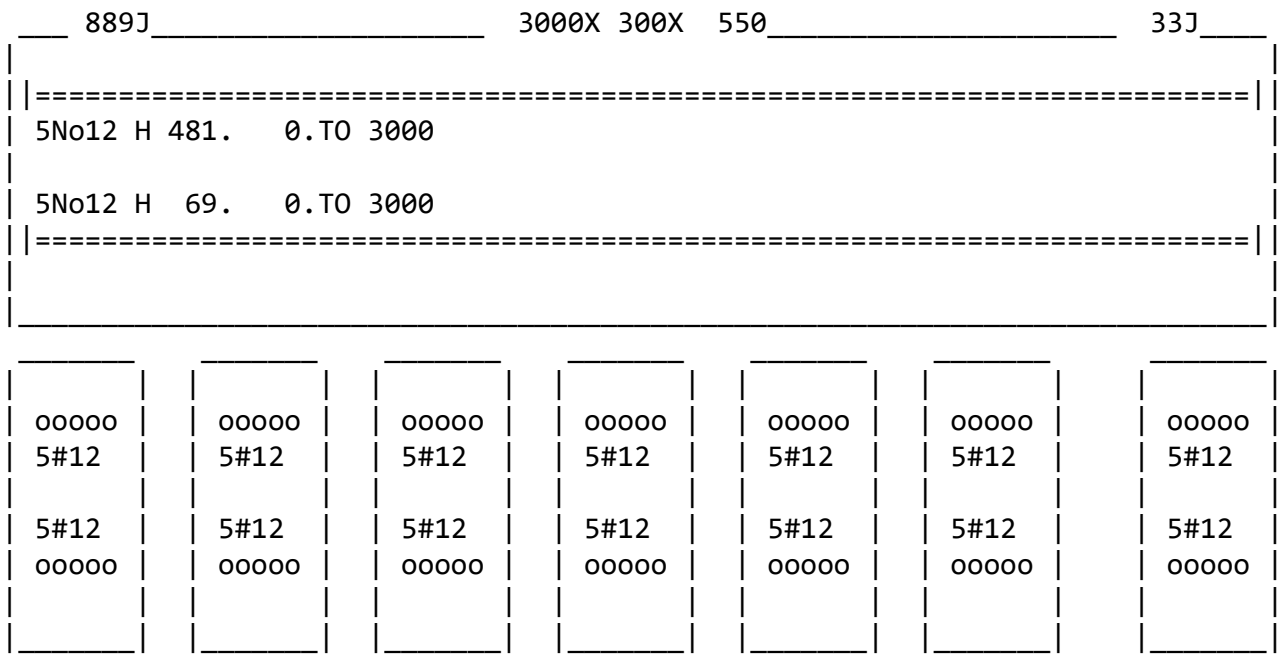
ACI 318-11 BEAM NO. 1515 DESIGN RESULTS

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3000.	YES	YES
2	481.	5 - 12MM	0.	3000.	YES	YES

B E A M N O. 1515 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.73 KNS Vc= 135.12 KNS Vs= 0.00 KNS
Tu= 1.18 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.89 KNS Vc= 143.31 KNS Vs= 0.00 KNS
Tu= 1.18 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



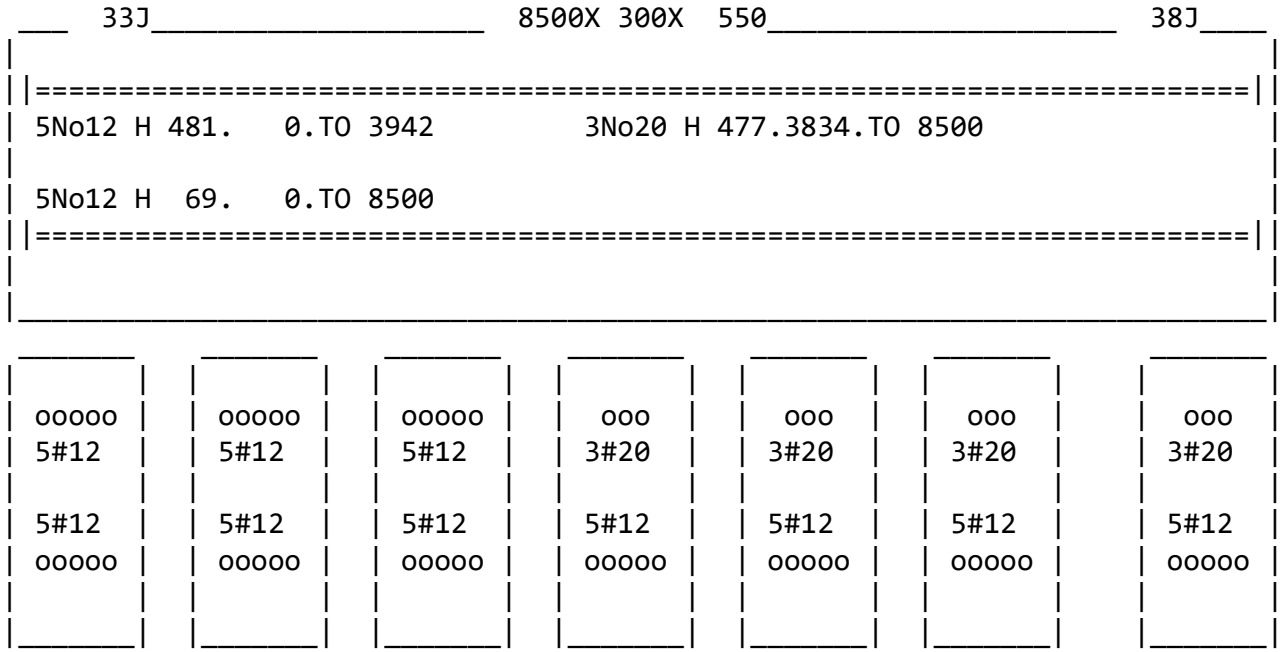
ACI 318-11 BEAM NO. 1516 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	8500.	YES	YES
2	481.	5 - 12MM	0.	3942.	YES	NO
3	477.	3 - 20MM	3834.	8500.	NO	YES

B E A M N O. 1516 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.54 KNS Vc= 131.98 KNS Vs= 0.00 KNS
Tu= 0.79 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 29.55 KNS Vc= 131.98 KNS Vs= 0.00 KNS
Tu= 0.79 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1517 DESIGN RESULTS

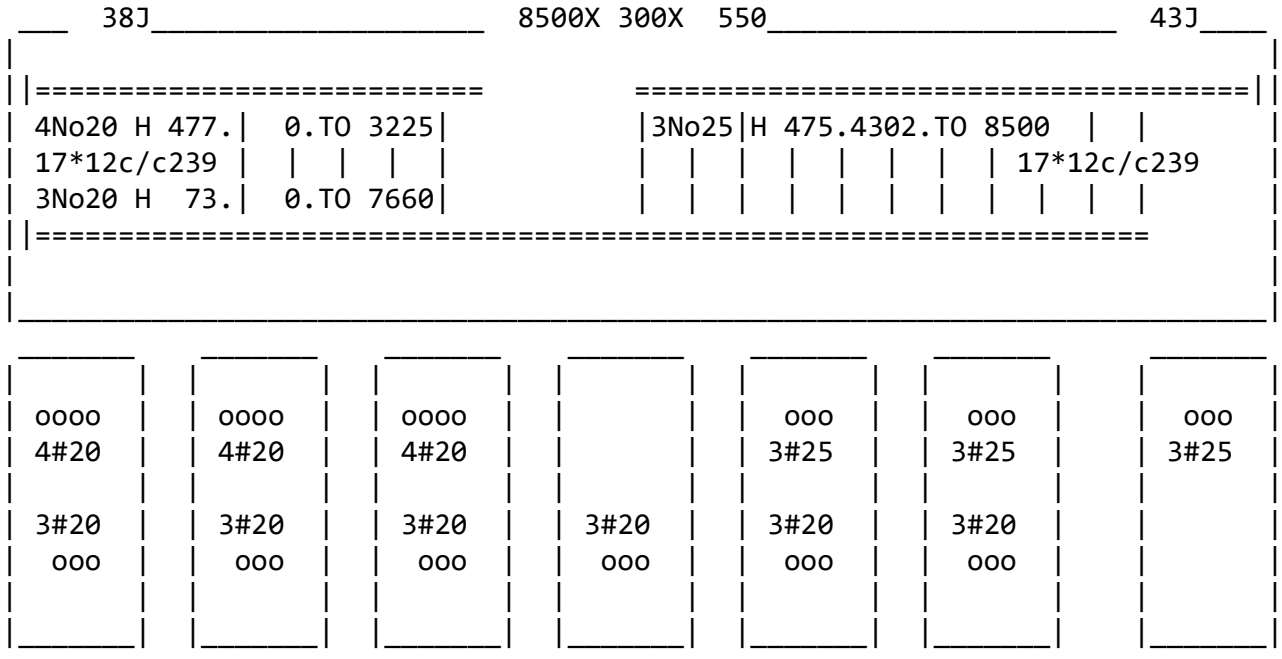
LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	7660.	YES	NO
2	477.	4 - 20MM	0.	3225.	YES	NO
3	475.	3 - 25MM	4302.	8500.	NO	YES

B E A M N O. 1517 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 83.34 KNS Vc= 147.72 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 3775. MM

AT END SUPPORT - Vu= 78.79 KNS Vc= 139.42 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 3775. MM



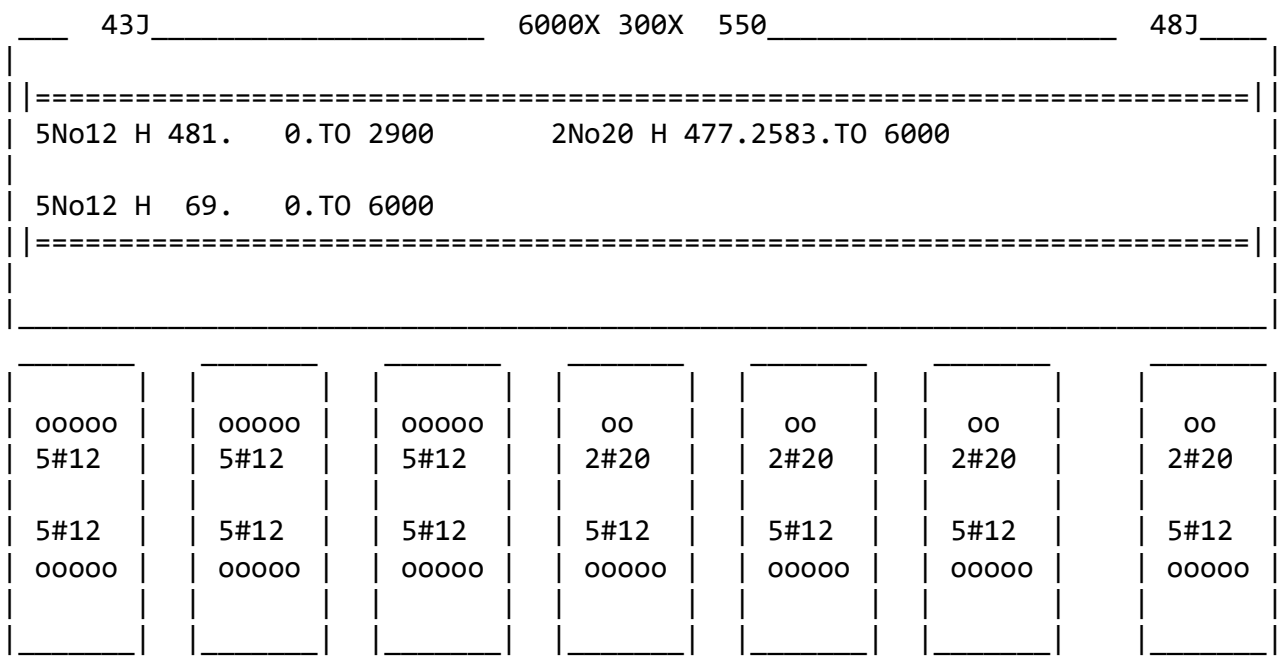
ACI 318-11 BEAM NO. 1518 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	6000.	YES	YES
2	481.	5 - 12MM	0.	2900.	YES	NO
3	477.	2 - 20MM	2583.	6000.	NO	YES

B E A M N O. 1518 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.16 KNS Vc= 257.21 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 20.33 KNS Vc= 139.43 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



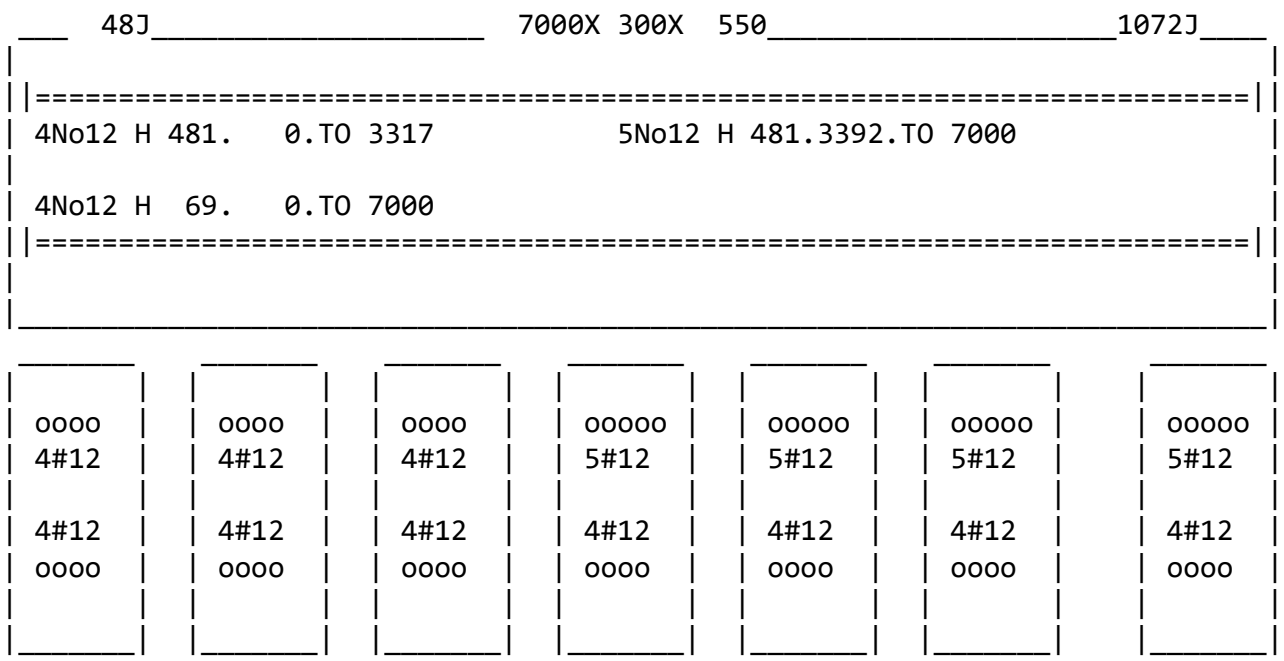
ACI 318-11 BEAM NO. 1519 DESIGN RESULTS

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	7000.	YES	YES
2	481.	4 - 12MM	0.	3317.	YES	NO
3	481.	5 - 12MM	3392.	7000.	NO	YES

B E A M N O. 1519 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.09 KNS Vc= 258.39 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 21.84 KNS Vc= 139.12 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1520 DESIGN RESULTS

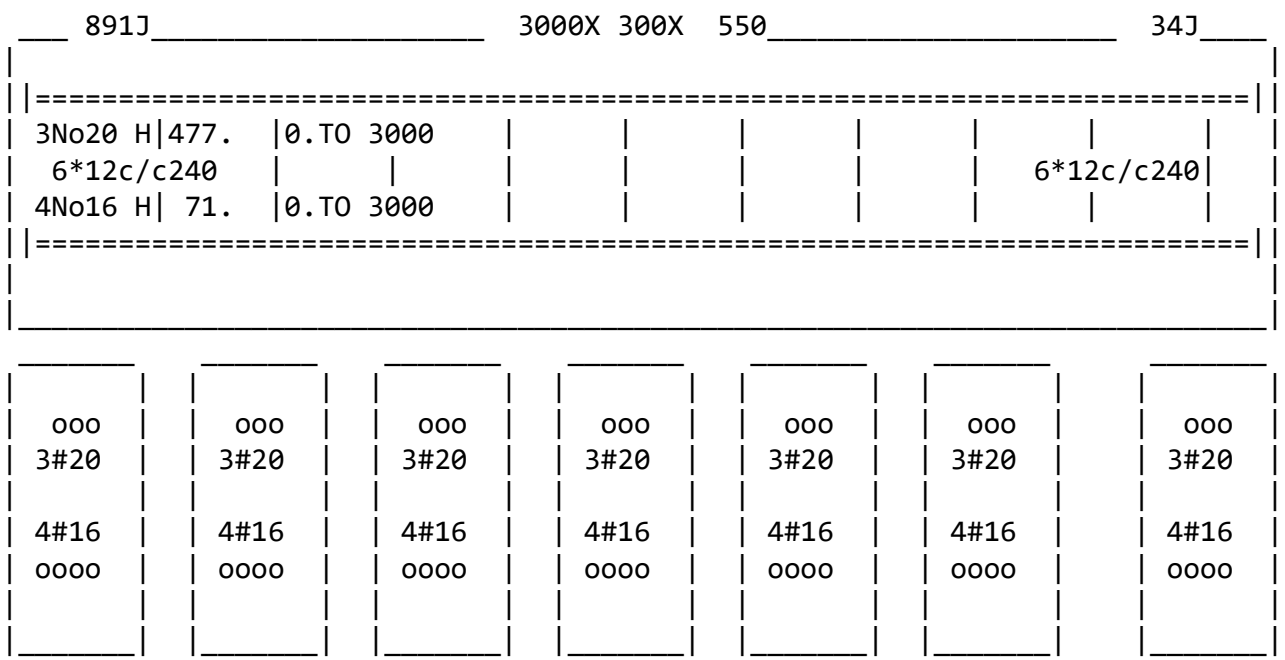
LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	3000.	YES	YES
2	477.	3 - 20MM	0.	3000.	YES	YES

B E A M N O. 1520 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 88.53 KNS Vc= 141.30 KNS Vs= 0.00 KNS
Tu= 1.70 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 1025. MM

AT END SUPPORT - Vu= 98.10 KNS Vc= 140.63 KNS Vs= 0.00 KNS
Tu= 1.70 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 1025. MM



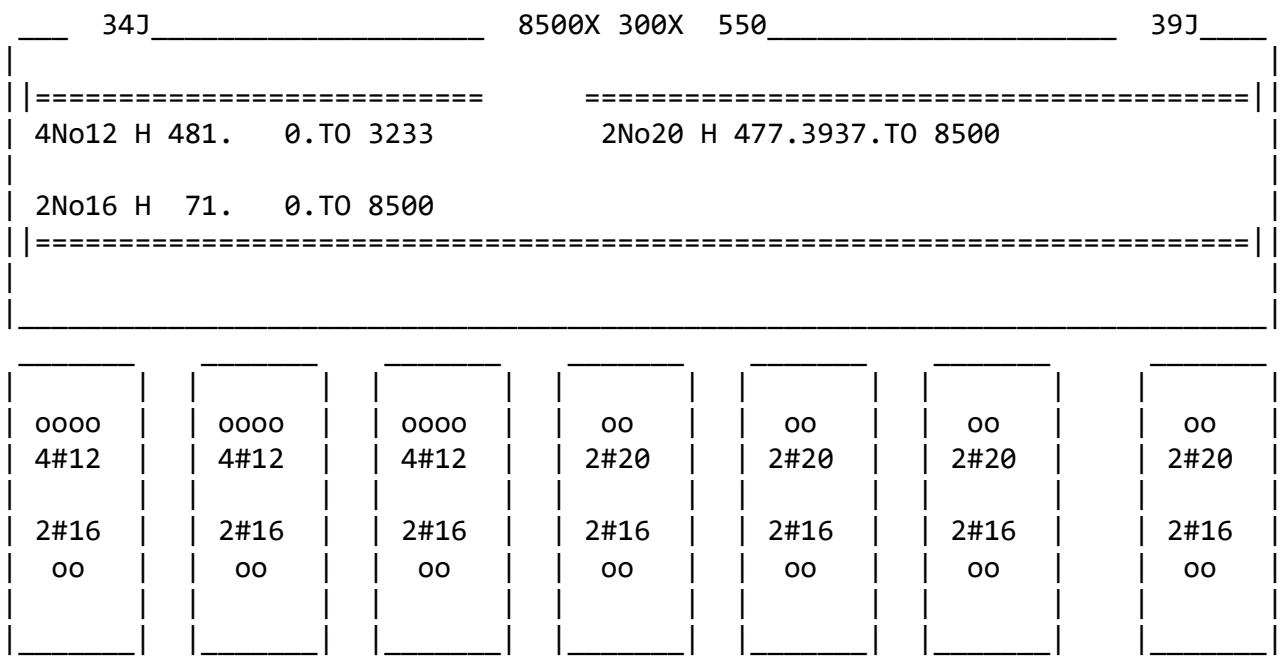
ACI 318-11 BEAM NO. 1521 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	8500.	YES	YES
2	481.	4 - 12MM	0.	3233.	YES	NO
3	477.	2 - 20MM	3937.	8500.	NO	YES

B E A M N O. 1521 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.75 KNS Vc= 170.48 KNS Vs= 0.00 KNS
Tu= 0.89 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 27.35 KNS Vc= 136.11 KNS Vs= 0.00 KNS
Tu= 0.89 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



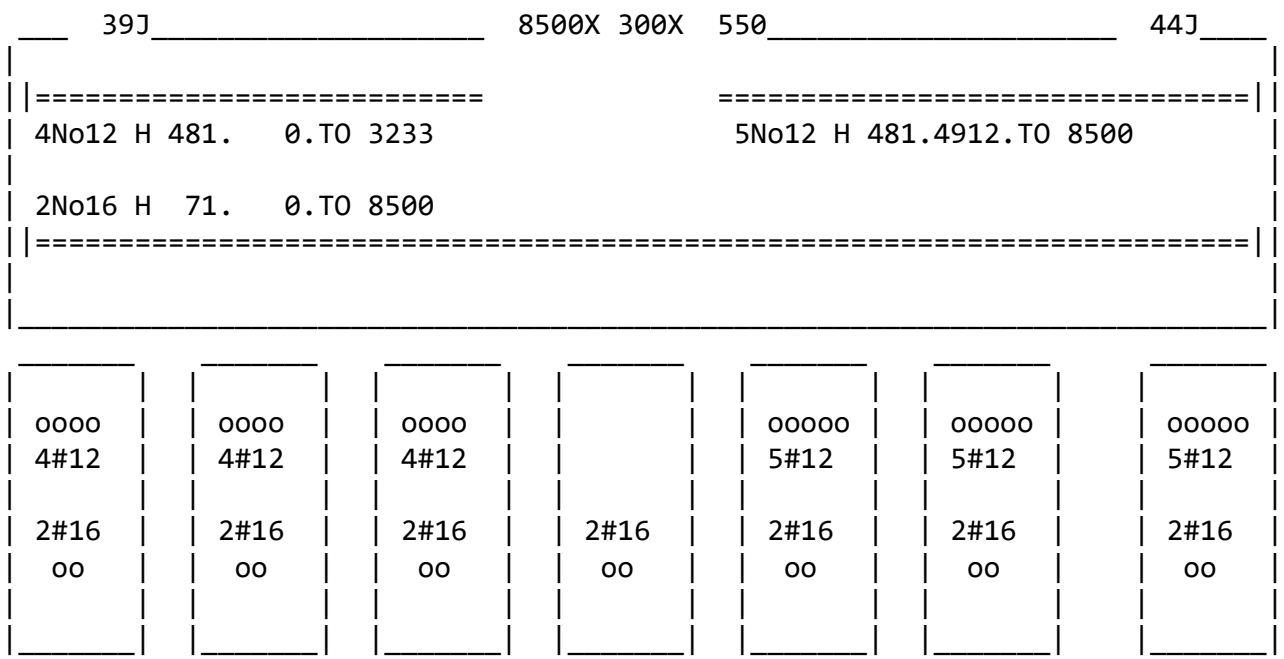
ACI 318-11 BEAM NO. 1522 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	8500.	YES	YES
2	481.	4 - 12MM	0.	3233.	YES	NO
3	481.	5 - 12MM	4912.	8500.	NO	YES

B E A M N O. 1522 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.50 KNS Vc= 260.55 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 6.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 26.59 KNS Vc= 137.09 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 6.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



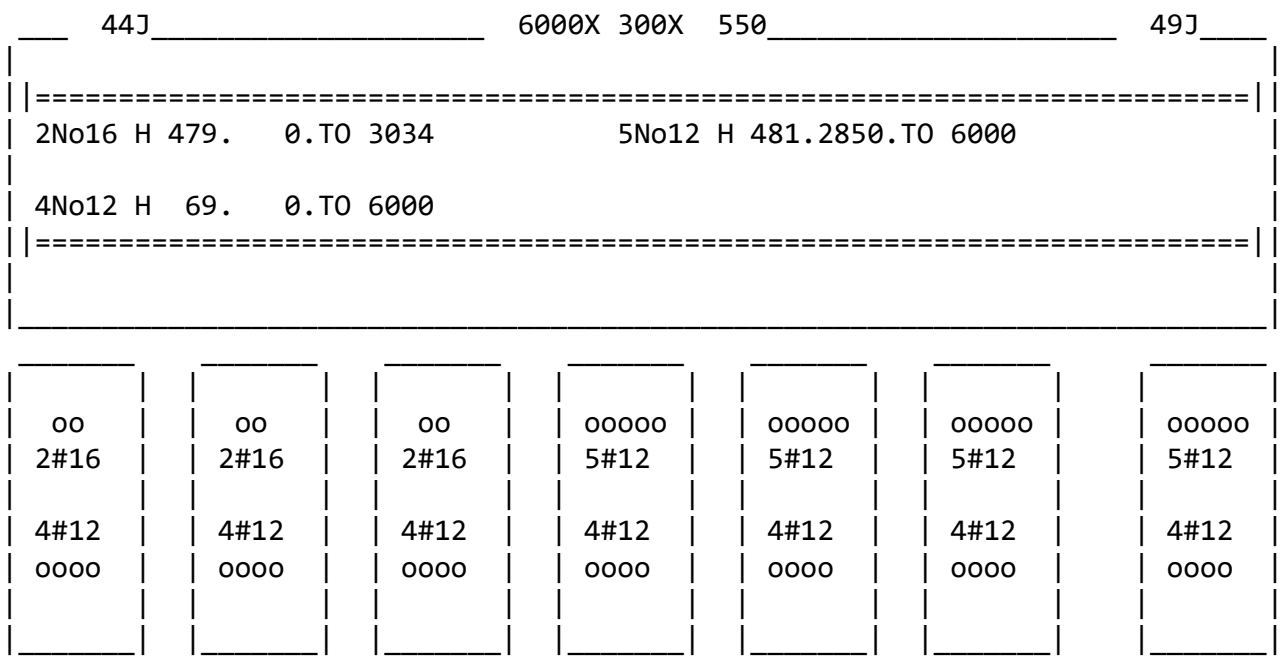
ACI 318-11 BEAM NO. 1523 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	6000.	YES	YES
2	479.	2 - 16MM	0.	3034.	YES	NO
3	481.	5 - 12MM	2850.	6000.	NO	YES

B E A M N O. 1523 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.55 KNS Vc= 259.24 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 20.94 KNS Vc= 139.52 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



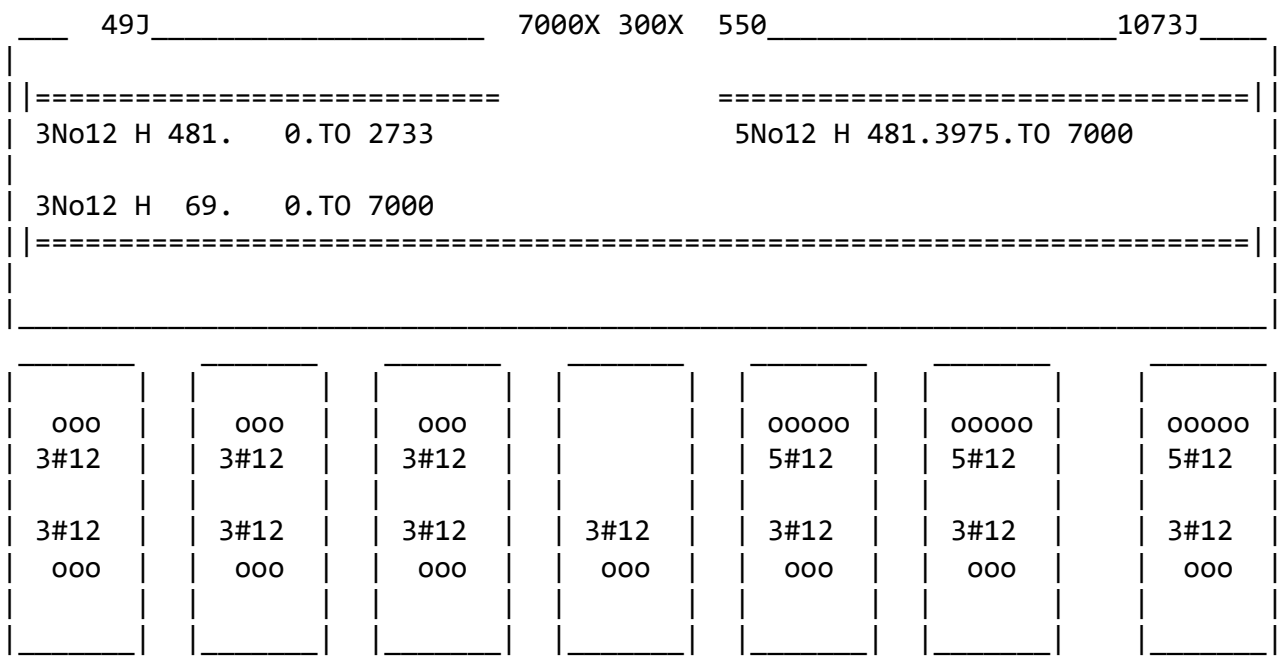
ACI 318-11 BEAM NO. 1524 DESIGN RESULTS

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	7000.	YES	YES
2	481.	3 - 12MM	0.	2733.	YES	NO
3	481.	5 - 12MM	3975.	7000.	NO	YES

B E A M N O. 1524 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.70 KNS Vc= 259.06 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.23 KNS Vc= 138.21 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



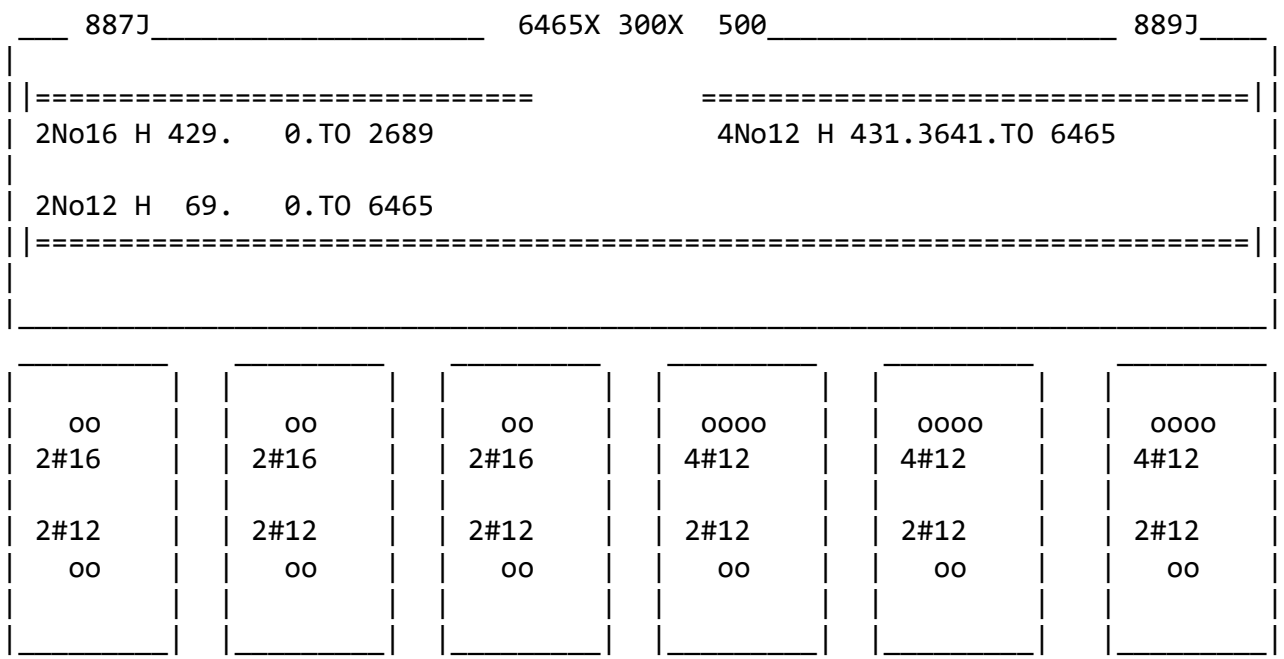
ACI 318-11 BEAM NO. 1531 DESIGN RESULTS

LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	6465.	YES	YES
2	429.	2 - 16MM	0.	2689.	YES	NO
3	431.	4 - 12MM	3641.	6465.	NO	YES

B E A M N O. 1531 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.14 KNS Vc= 125.76 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 15.64 KNS Vc= 125.76 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



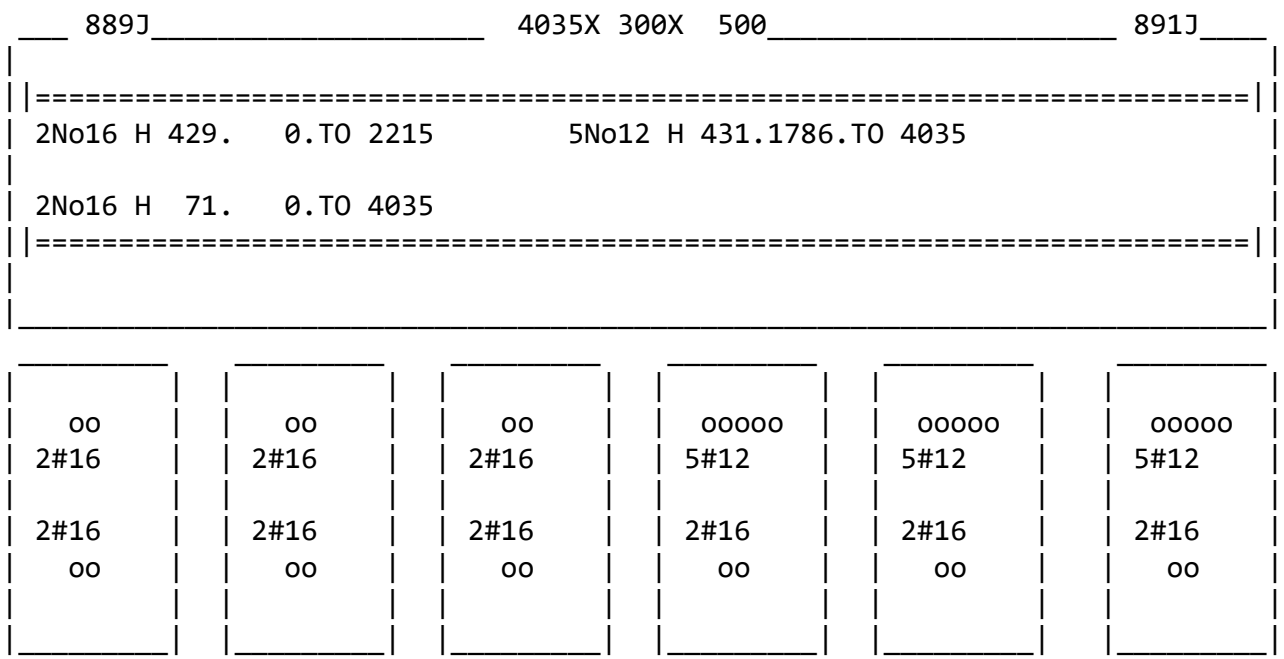
ACI 318-11 BEAM NO. 1532 DESIGN RESULTS

LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	4035.	YES	YES
2	429.	2 - 16MM	0.	2215.	YES	NO
3	431.	5 - 12MM	1786.	4035.	NO	YES

B E A M N O. 1532 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.23 KNS Vc= 121.52 KNS Vs= 0.00 KNS
Tu= 0.44 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.53 KNS Vc= 123.76 KNS Vs= 0.00 KNS
Tu= 0.44 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



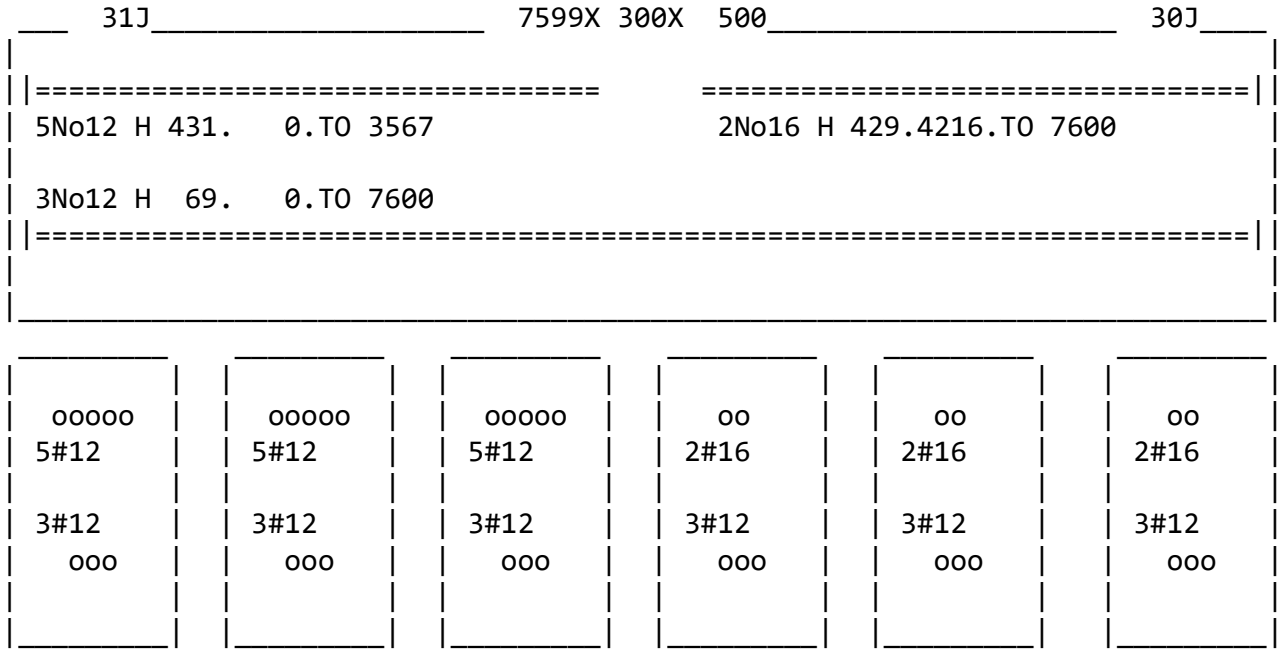
ACI 318-11 BEAM NO. 1533 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	7600.	YES	YES
2	431.	5 - 12MM	0.	3567.	YES	NO
3	429.	2 - 16MM	4216.	7600.	NO	YES

B E A M N O. 1533 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 20.06 KNS Vc= 127.25 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.34 KNS Vc= 127.25 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



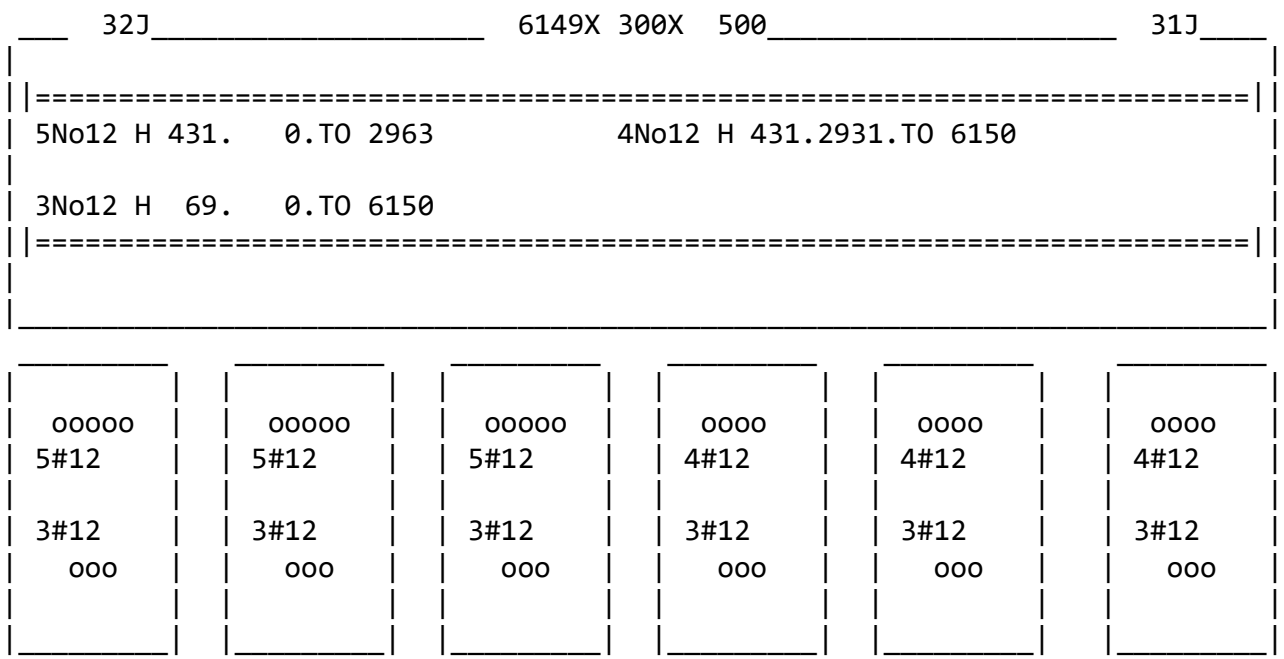
ACI 318-11 BEAM NO. 1534 DESIGN RESULTS

LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	6150.	YES	YES
2	431.	5 - 12MM	0.	2963.	YES	NO
3	431.	4 - 12MM	2931.	6150.	NO	YES

B E A M N O. 1534 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.62 KNS Vc= 123.20 KNS Vs= 0.00 KNS
Tu= 0.39 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 9.61 KNS Vc= 222.85 KNS Vs= 0.00 KNS
Tu= 0.39 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



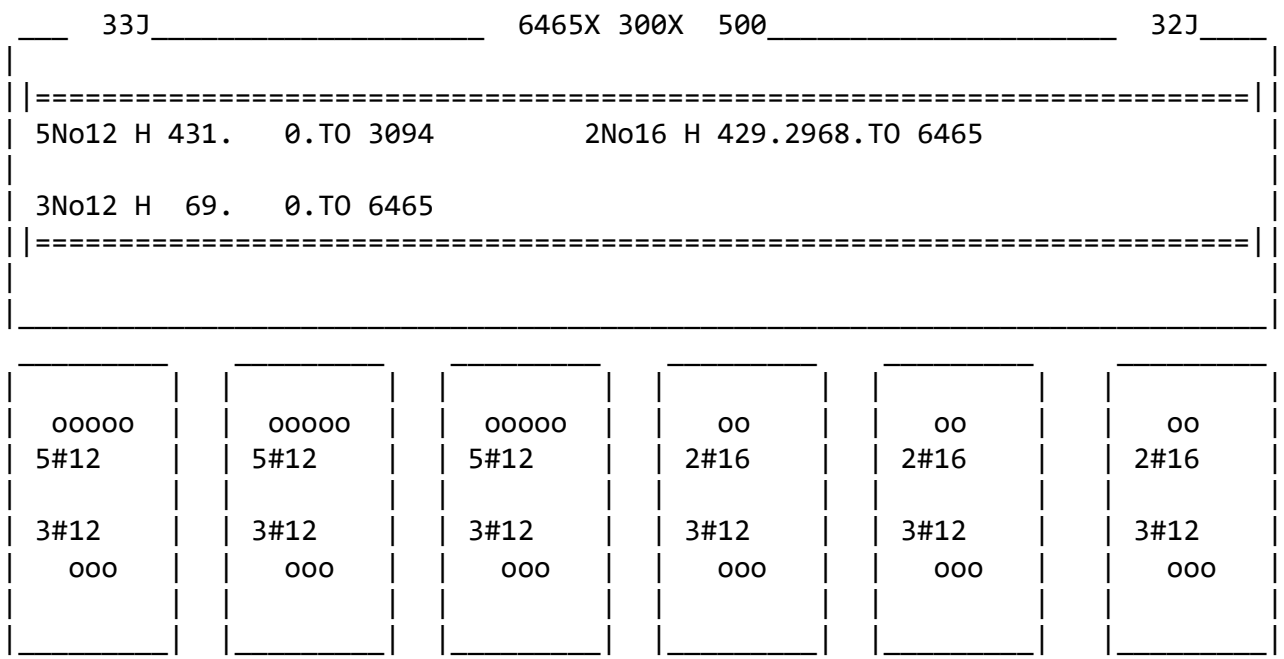
ACI 318-11 BEAM NO. 1535 DESIGN RESULTS

LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	6465.	YES	YES
2	431.	5 - 12MM	0.	3094.	YES	NO
3	429.	2 - 16MM	2968.	6465.	NO	YES

B E A M N O. 1535 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.86 KNS Vc= 124.83 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 9.93 KNS Vc= 124.83 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



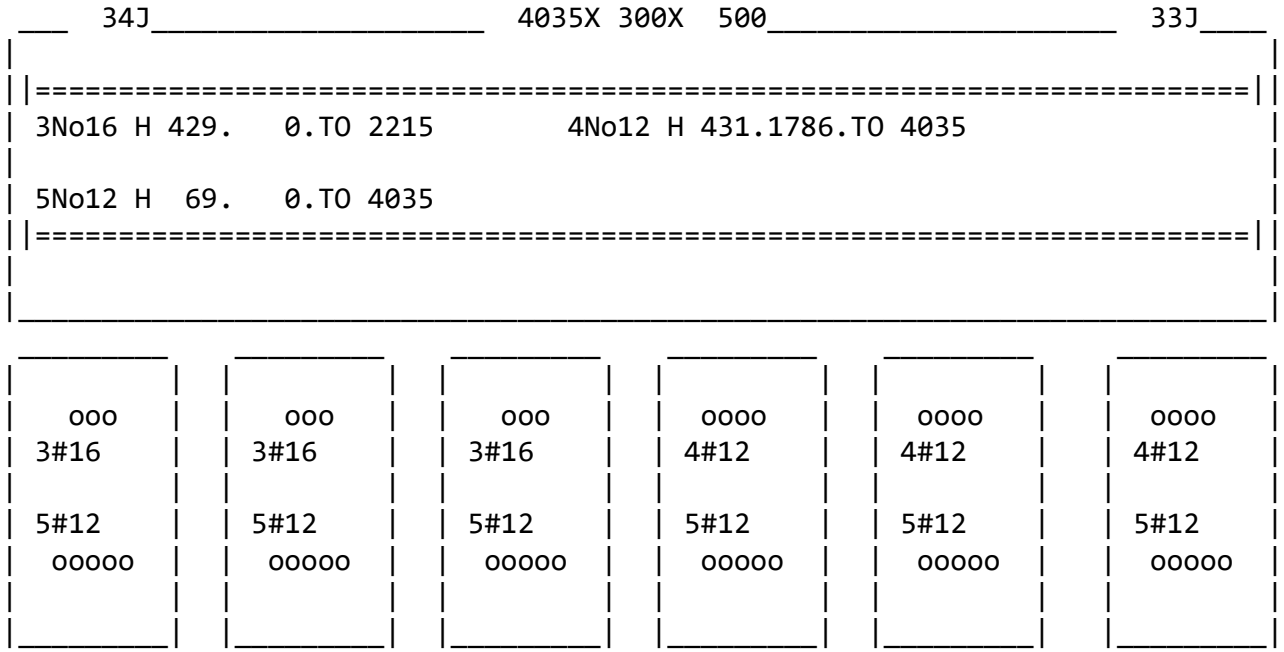
ACI 318-11 BEAM NO. 1536 DESIGN RESULTS

LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	4035.	YES	YES
2	429.	3 - 16MM	0.	2215.	YES	NO
3	431.	4 - 12MM	1786.	4035.	NO	YES

B E A M N O. 1536 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 18.85 KNS Vc= 124.24 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 3.09 KNS Vc= 121.73 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



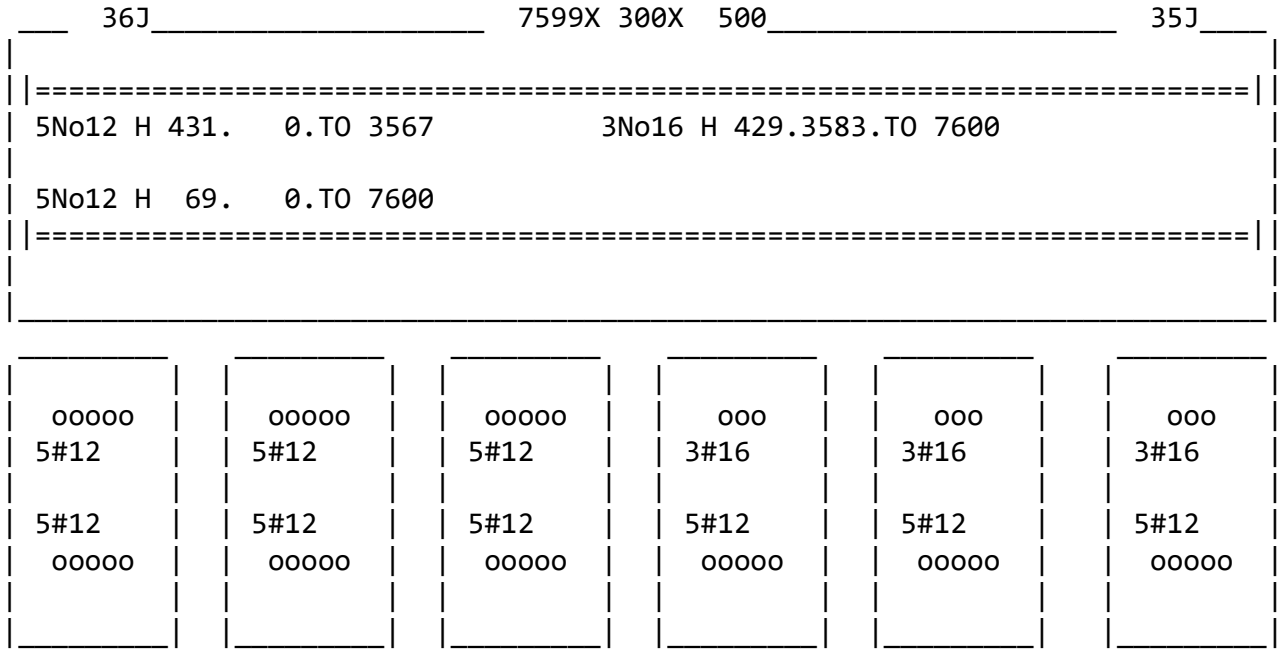
ACI 318-11 BEAM NO. 1537 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7600.	YES	YES
2	431.	5 - 12MM	0.	3567.	YES	NO
3	429.	3 - 16MM	3583.	7600.	NO	YES

B E A M N O. 1537 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.77 KNS Vc= 126.13 KNS Vs= 0.00 KNS
Tu= 1.07 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.63 KNS Vc= 126.13 KNS Vs= 0.00 KNS
Tu= 1.07 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



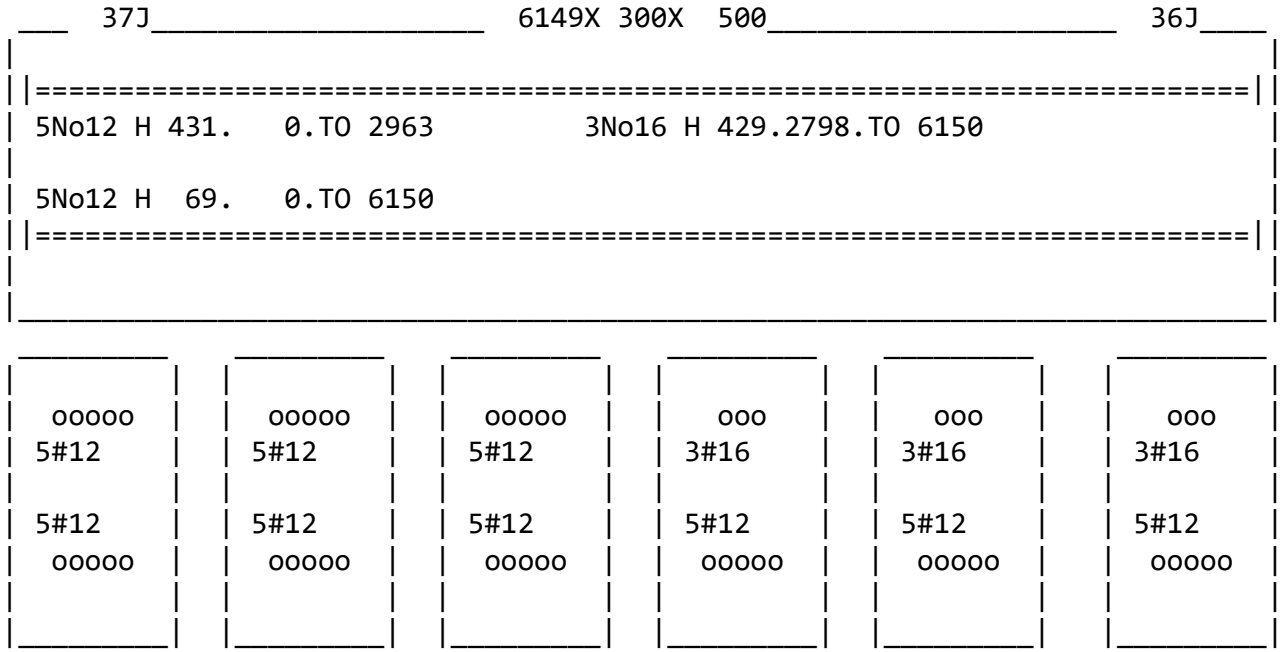
ACI 318-11 BEAM NO. 1538 DESIGN RESULTS

LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	6150.	YES	YES
2	431.	5 - 12MM	0.	2963.	YES	NO
3	429.	3 - 16MM	2798.	6150.	NO	YES

B E A M N O. 1538 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.27 KNS Vc= 127.05 KNS Vs= 0.00 KNS
Tu= 0.84 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.96 KNS Vc= 127.05 KNS Vs= 0.00 KNS
Tu= 0.84 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1539 DESIGN RESULTS

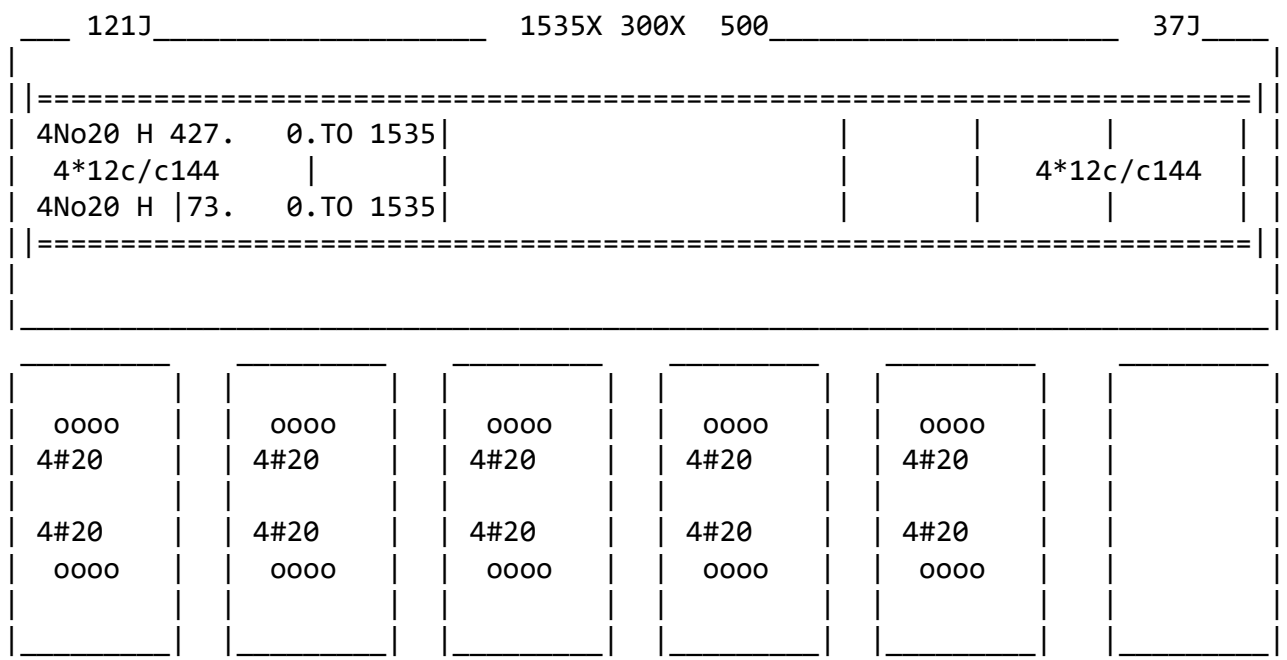
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	1535.	YES	YES
2	427.	4 - 20MM	0.	1535.	YES	YES

B E A M N O. 1539 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 217.43 KNS Vc= 120.06 KNS Vs= 169.85 KNS
Tu= 7.02 KN-MET Tc= 5.0 KN-MET Ts= 9.4 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 343. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.07 SQ.CM.

AT END SUPPORT - Vu= 214.52 KNS Vc= 120.06 KNS Vs= 165.97 KNS
Tu= 7.02 KN-MET Tc= 5.0 KN-MET Ts= 9.4 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 343. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.07 SQ.CM.



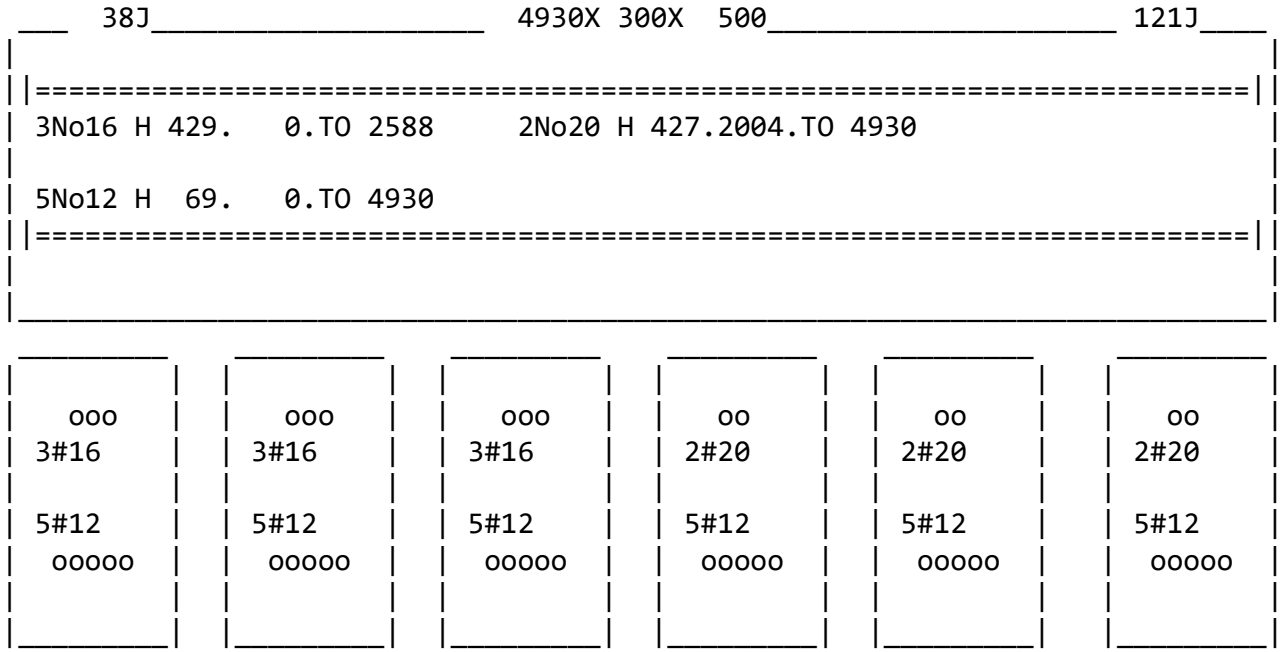
ACI 318-11 BEAM NO. 1540 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	4930.	YES	YES
2	429.	3 - 16MM	0.	2588.	YES	NO
3	427.	2 - 20MM	2004.	4930.	NO	YES

B E A M N O. 1540 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.17 KNS Vc= 126.68 KNS Vs= 0.00 KNS
Tu= 1.67 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.02 KNS Vc= 126.68 KNS Vs= 0.00 KNS
Tu= 1.67 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1541 DESIGN RESULTS

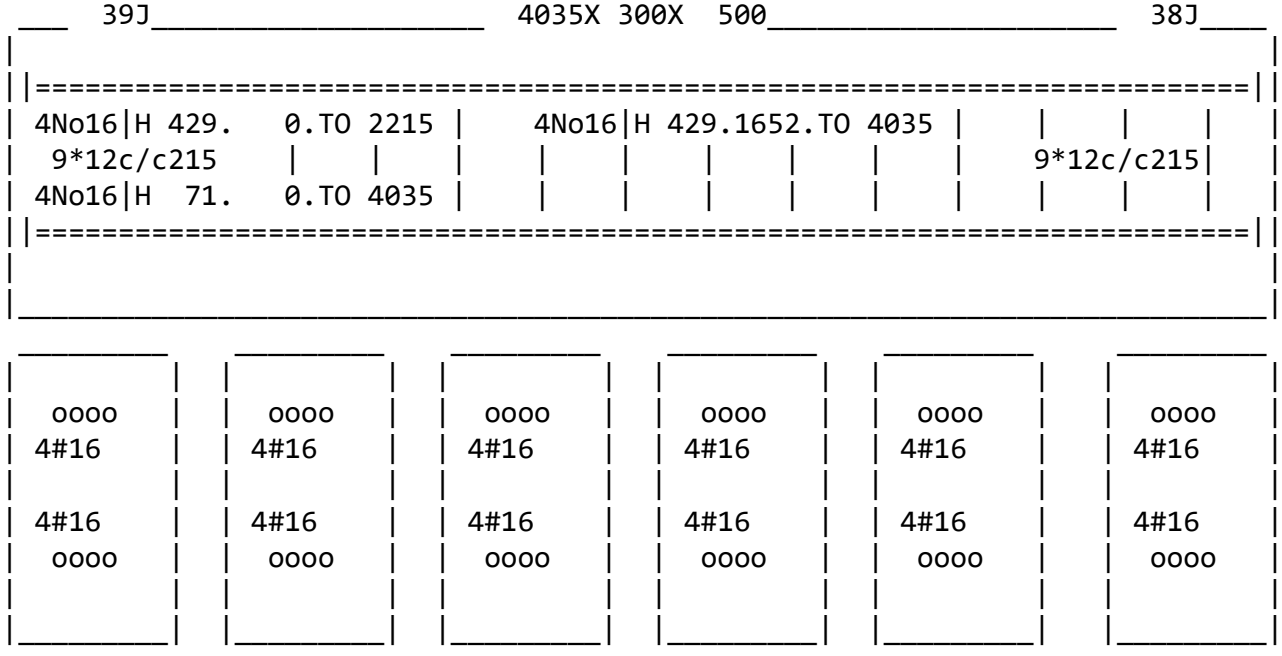
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	4035.	YES	YES
2	429.	4 - 16MM	0.	2215.	YES	NO
3	429.	4 - 16MM	1652.	4035.	NO	YES

B E A M N O. 1541 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 62.55 KNS Vc= 123.91 KNS Vs= 0.00 KNS
Tu= 2.80 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 1593. MM

AT END SUPPORT - Vu= 49.04 KNS Vc= 123.43 KNS Vs= 0.00 KNS
Tu= 2.80 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 1593. MM



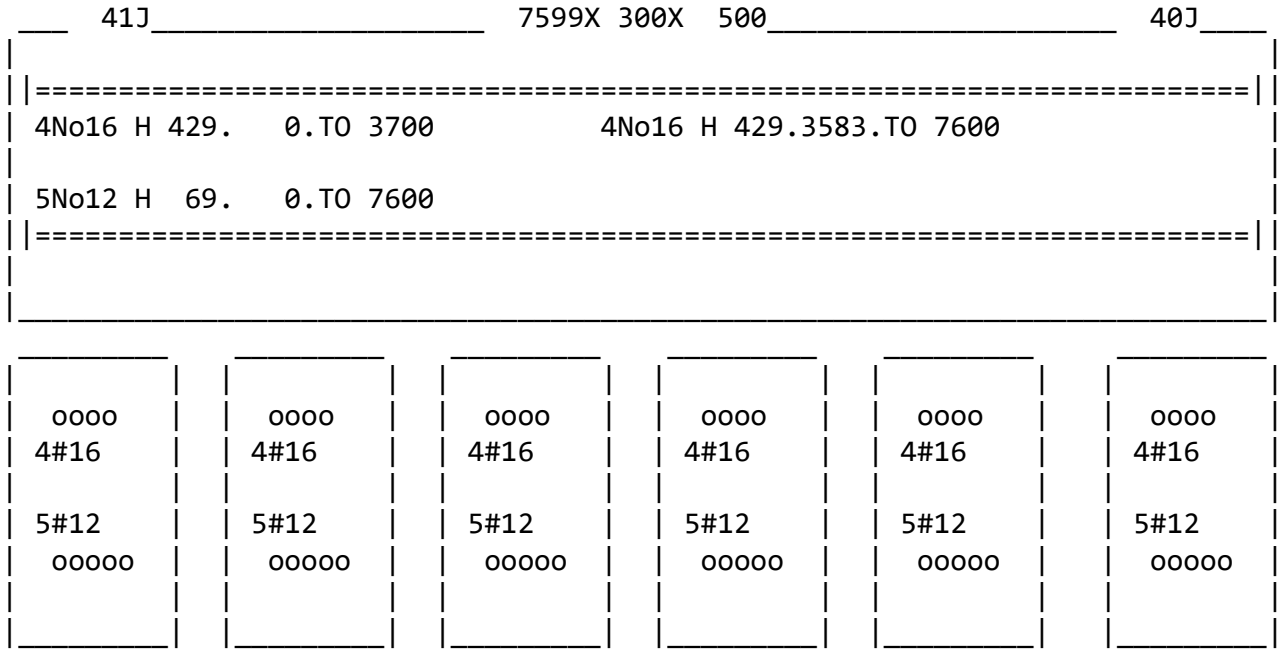
ACI 318-11 BEAM NO. 1543 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7600.	YES	YES
2	429.	4 - 16MM	0.	3700.	YES	NO
3	429.	4 - 16MM	3583.	7600.	NO	YES

B E A M N O. 1543 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.26 KNS Vc= 125.35 KNS Vs= 0.00 KNS
Tu= 0.38 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 16.15 KNS Vc= 125.35 KNS Vs= 0.00 KNS
Tu= 0.38 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1544 DESIGN RESULTS

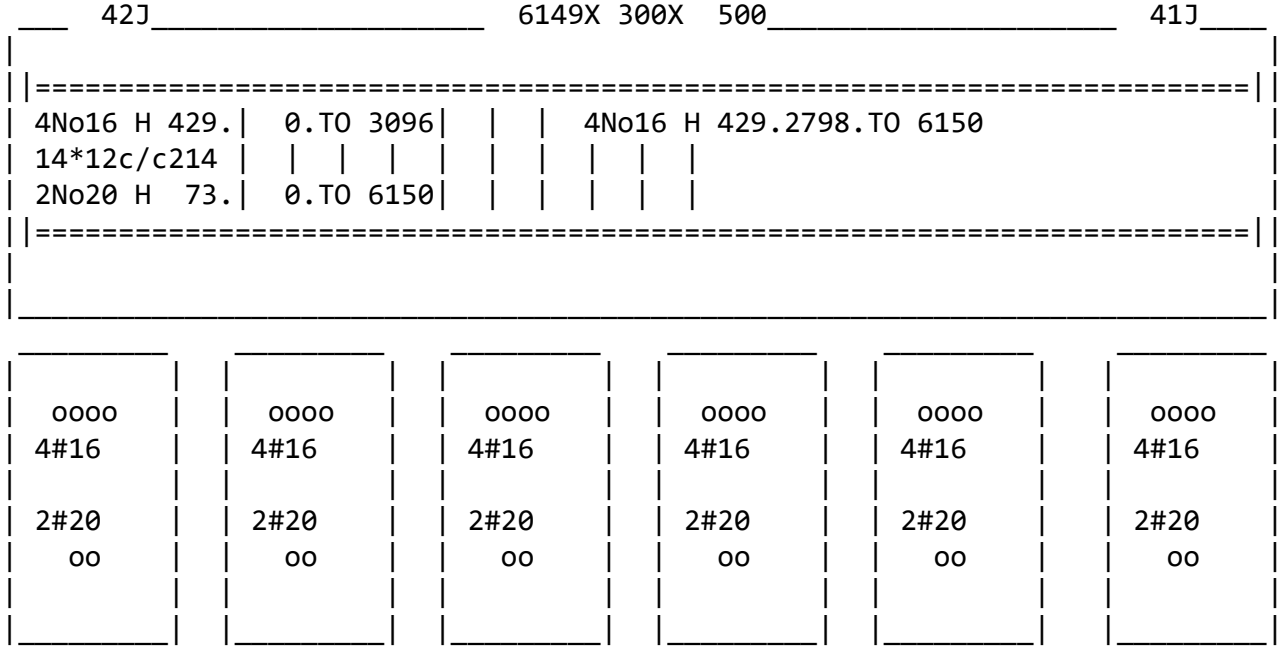
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	2 - 20MM	0.	6150.	YES	YES
2	429.	4 - 16MM	0.	3096.	YES	NO
3	429.	4 - 16MM	2798.	6150.	NO	YES

B E A M N O. 1544 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 46.41 KNS Vc= 119.57 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 2650. MM

AT END SUPPORT - Vu= 11.48 KNS Vc= 125.78 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



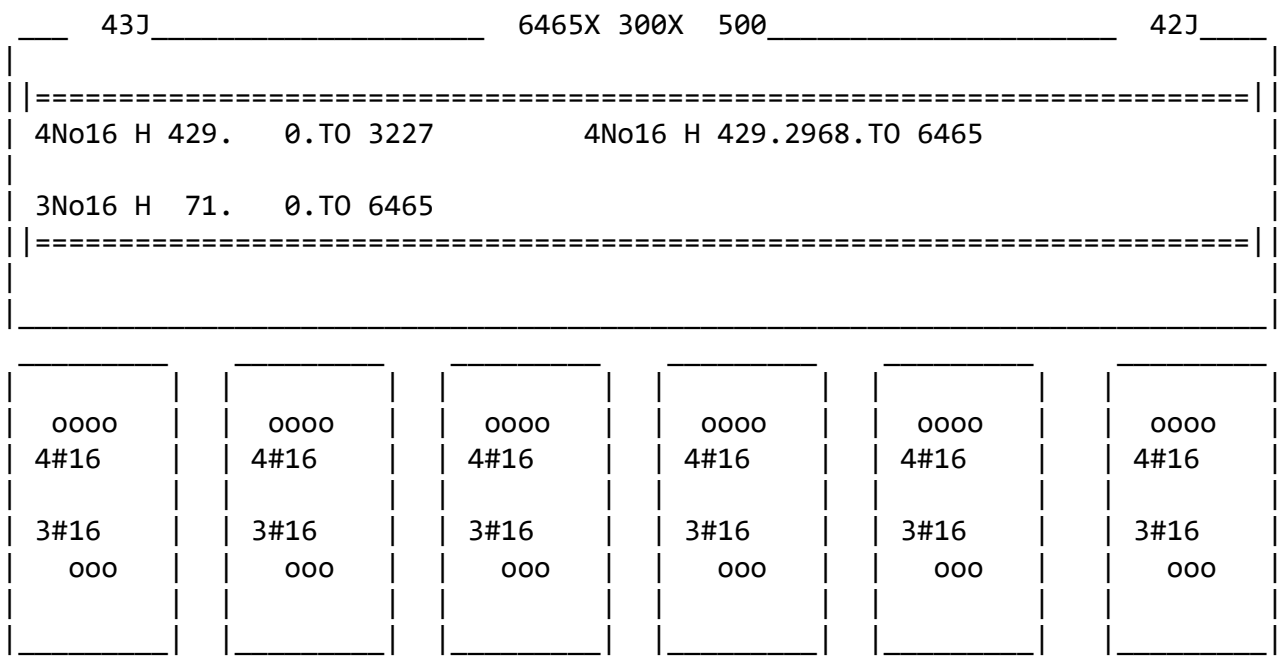
ACI 318-11 BEAM NO. 1545 DESIGN RESULTS

LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	6465.	YES	YES
2	429.	4 - 16MM	0.	3227.	YES	NO
3	429.	4 - 16MM	2968.	6465.	NO	YES

B E A M N O. 1545 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.29 KNS Vc= 126.64 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.50 KNS Vc= 126.64 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1546 DESIGN RESULTS

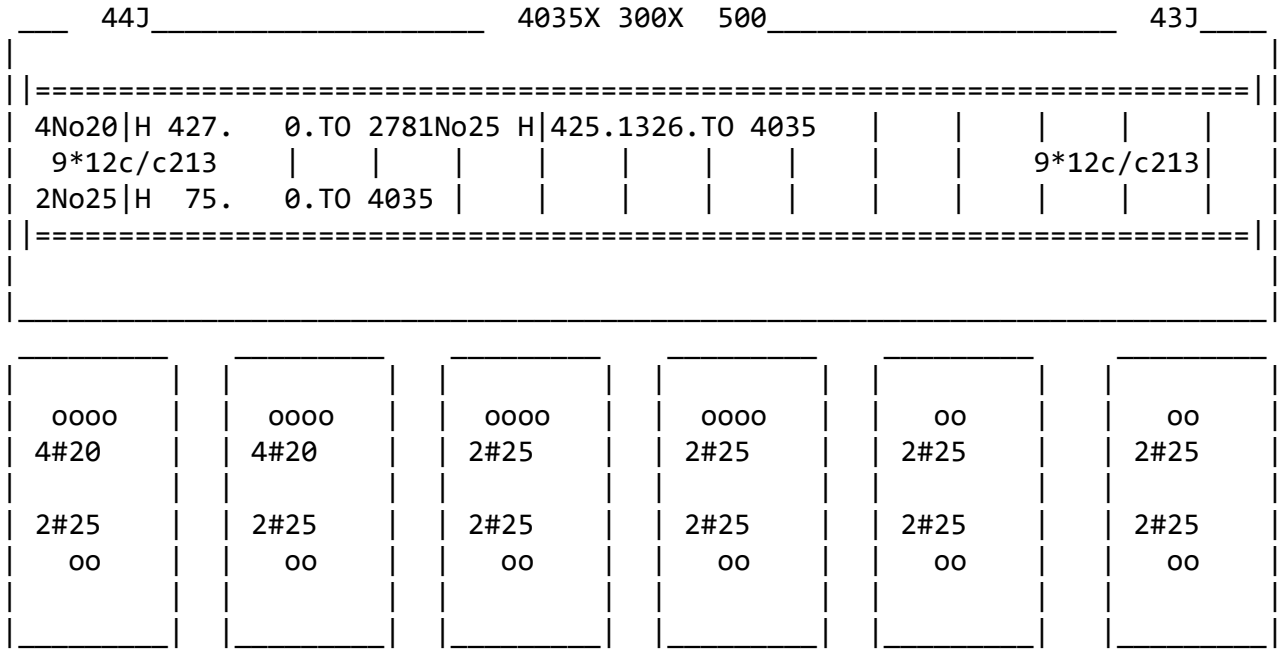
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	4035.	YES	YES
2	427.	4 - 20MM	0.	2781.	YES	NO
3	425.	2 - 25MM	1326.	4035.	NO	YES

B E A M N O. 1546 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 83.00 KNS Vc= 123.31 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 213. MM C/C FOR 1593. MM

AT END SUPPORT - Vu= 69.49 KNS Vc= 123.31 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 213. MM C/C FOR 1593. MM



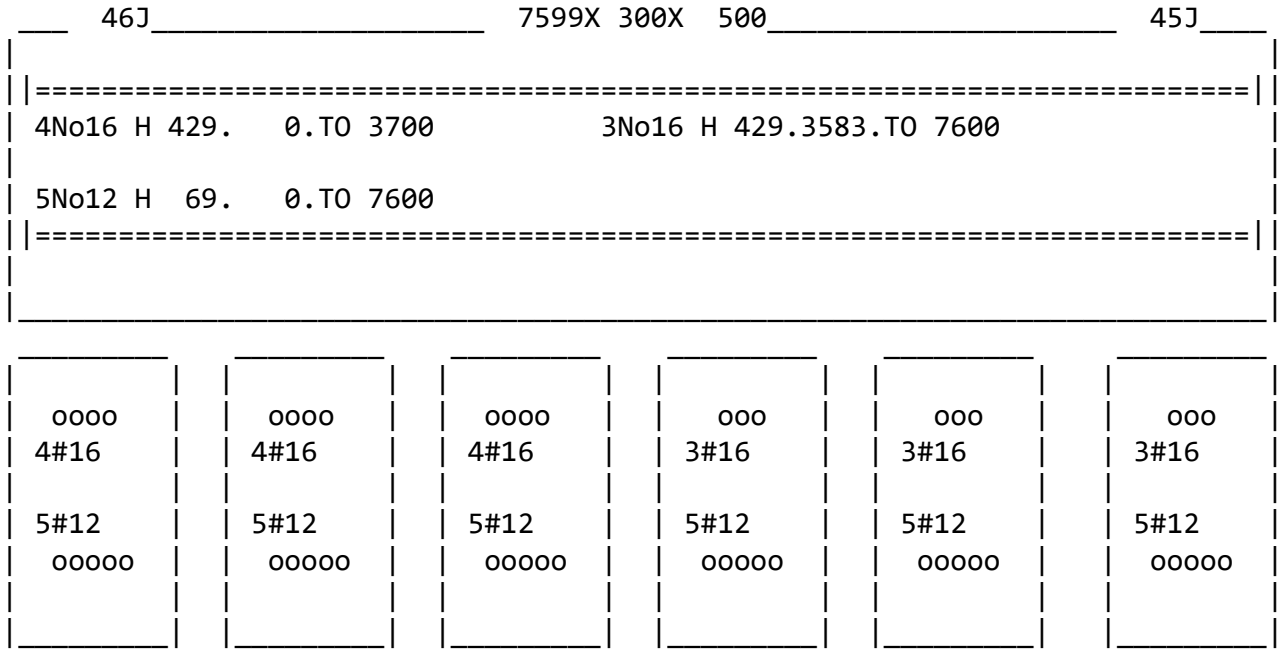
ACI 318-11 BEAM NO. 1548 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7600.	YES	YES
2	429.	4 - 16MM	0.	3700.	YES	NO
3	429.	3 - 16MM	3583.	7600.	NO	YES

B E A M N O. 1548 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.36 KNS Vc= 126.93 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 16.04 KNS Vc= 126.93 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



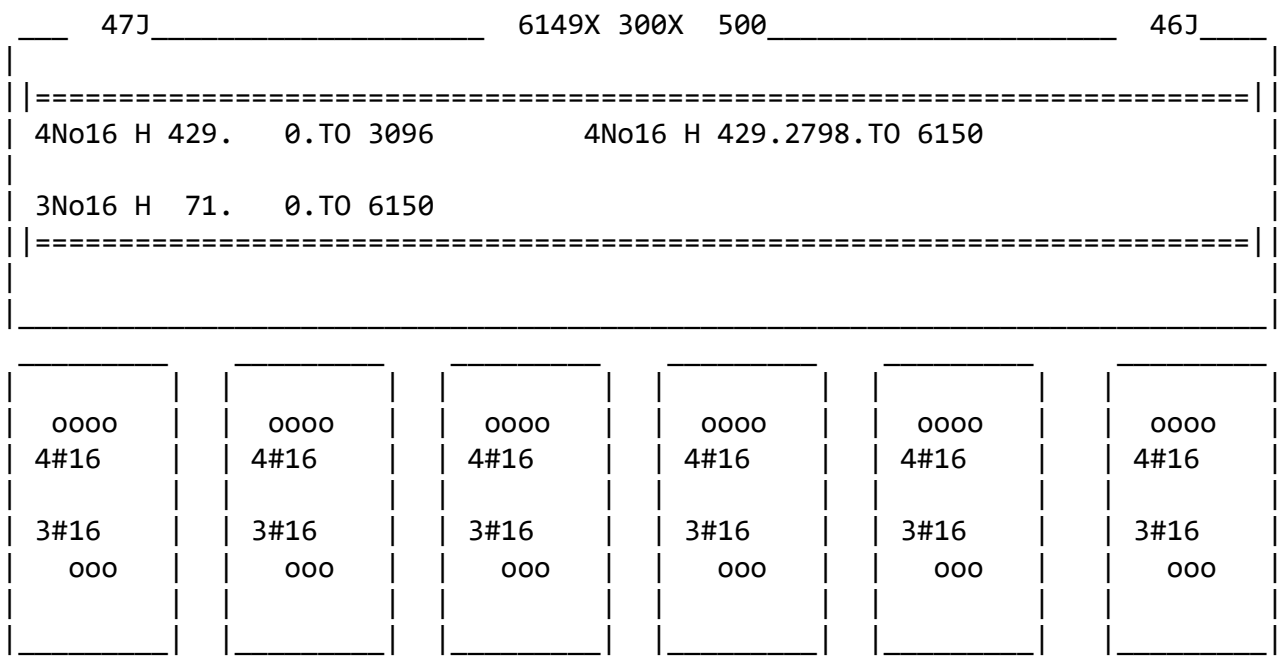
ACI 318-11 BEAM NO. 1549 DESIGN RESULTS

LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	6150.	YES	YES
2	429.	4 - 16MM	0.	3096.	YES	NO
3	429.	4 - 16MM	2798.	6150.	NO	YES

B E A M N O. 1549 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.95 KNS Vc= 127.08 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.28 KNS Vc= 126.61 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



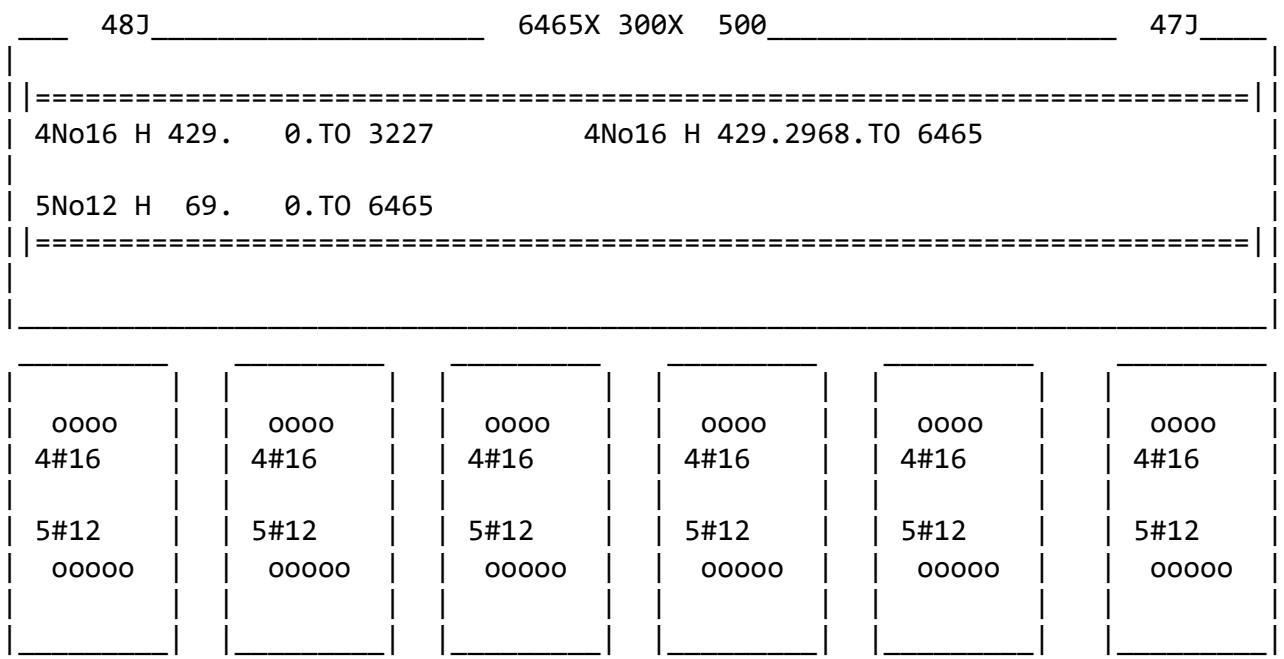
ACI 318-11 BEAM NO. 1550 DESIGN RESULTS

LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	6465.	YES	YES
2	429.	4 - 16MM	0.	3227.	YES	NO
3	429.	4 - 16MM	2968.	6465.	NO	YES

B E A M N O. 1550 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.43 KNS Vc= 125.72 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.36 KNS Vc= 126.82 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1551 DESIGN RESULTS

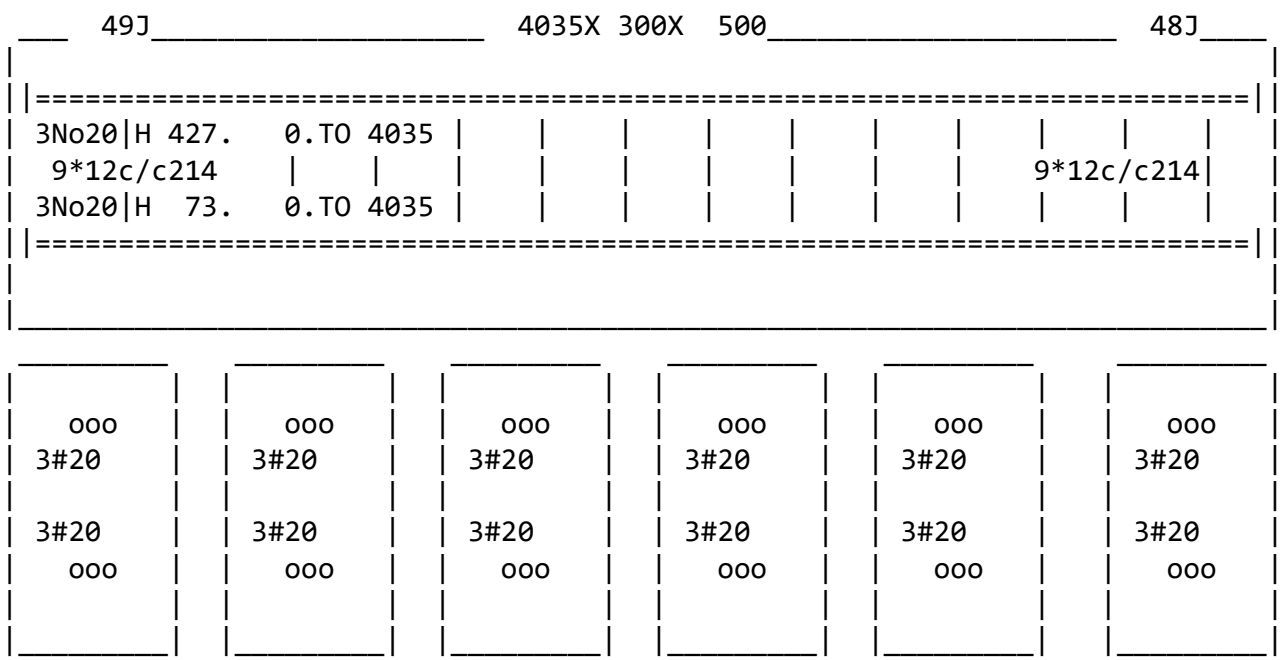
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	4035.	YES	YES
2	427.	3 - 20MM	0.	4035.	YES	YES

B E A M N O. 1551 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 71.60 KNS Vc= 124.44 KNS Vs= 0.00 KNS
Tu= 0.43 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 1593. MM

AT END SUPPORT - Vu= 58.09 KNS Vc= 124.44 KNS Vs= 0.00 KNS
Tu= 0.43 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 1593. MM



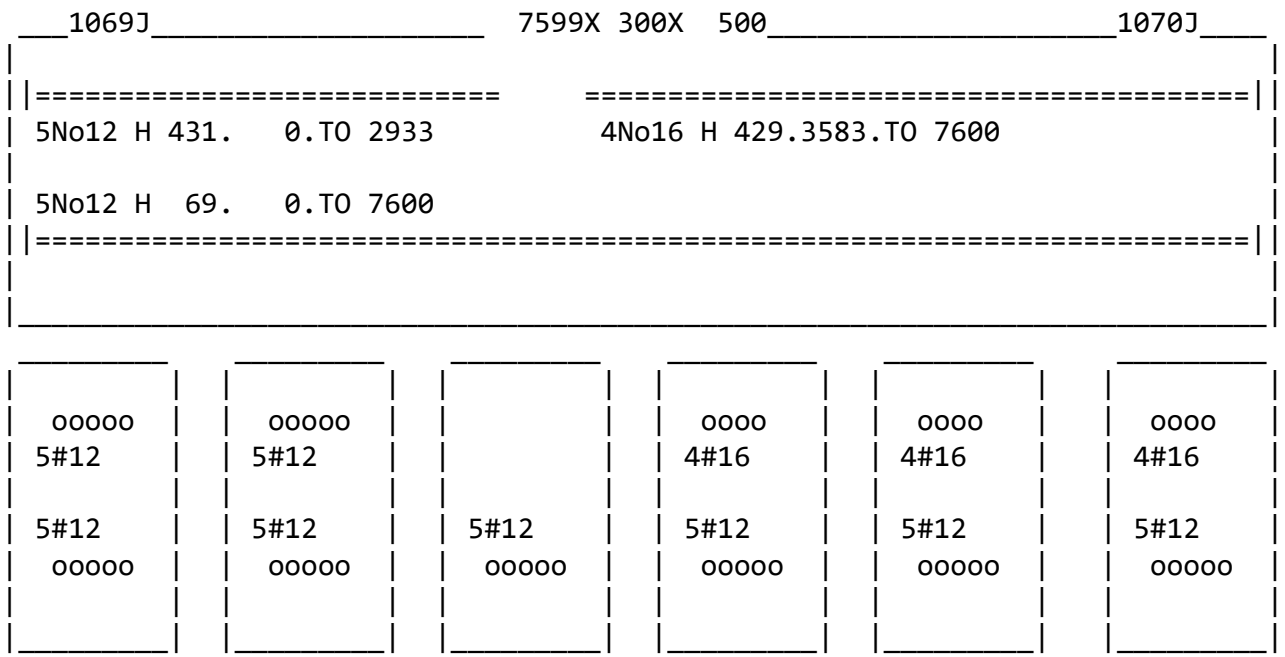
ACI 318-11 BEAM NO. 1552 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7600.	YES	YES
2	431.	5 - 12MM	0.	2933.	YES	NO
3	429.	4 - 16MM	3583.	7600.	NO	YES

B E A M N O. 1552 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.65 KNS Vc= 126.66 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 18.76 KNS Vc= 123.77 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



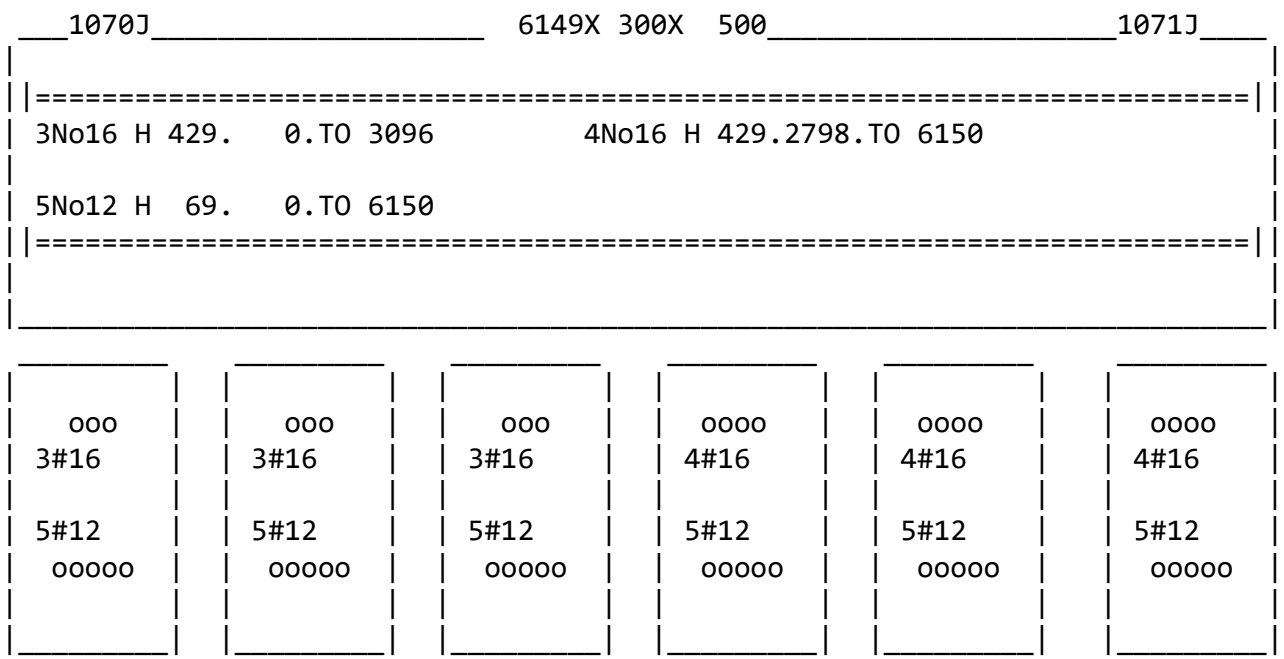
ACI 318-11 BEAM NO. 1553 DESIGN RESULTS

LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	6150.	YES	YES
2	429.	3 - 16MM	0.	3096.	YES	NO
3	429.	4 - 16MM	2798.	6150.	NO	YES

B E A M N O. 1553 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.56 KNS Vc= 148.18 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 15.67 KNS Vc= 124.86 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1554 DESIGN RESULTS

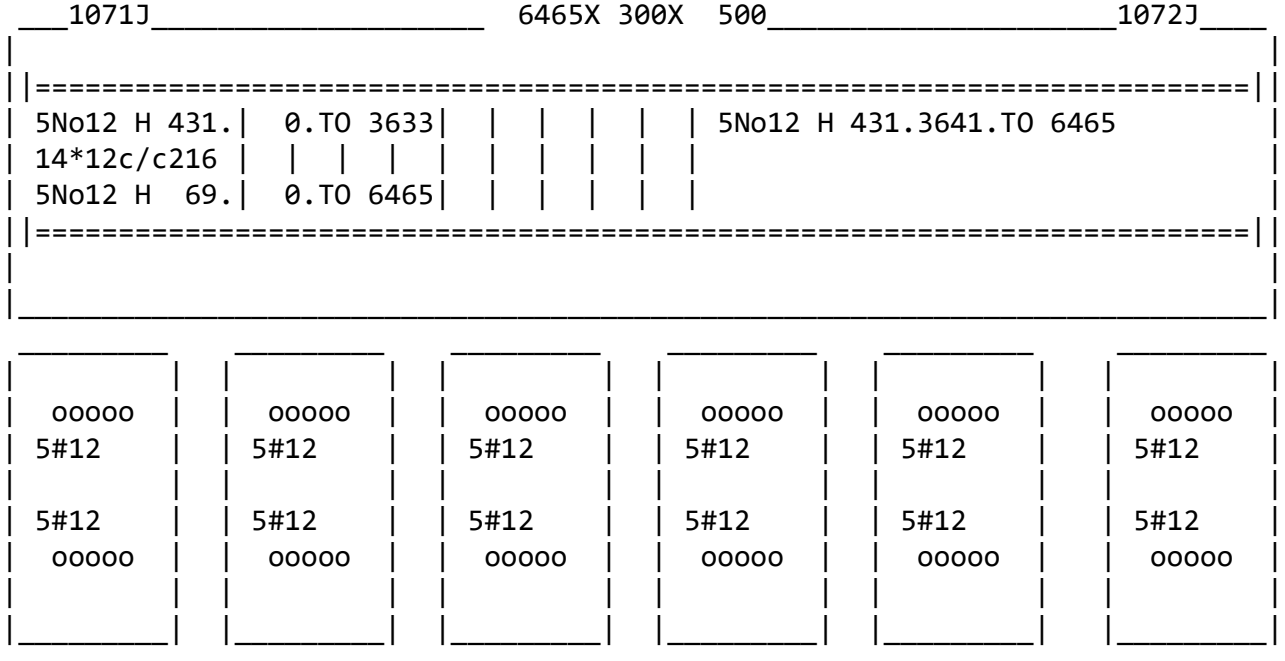
LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	6465.	YES	YES
2	431.	5 - 12MM	0.	3633.	YES	NO
3	431.	5 - 12MM	3641.	6465.	NO	YES

B E A M N O. 1554 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 30.58 KNS Vc= 77.75 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 2.9 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2808. MM

AT END SUPPORT - Vu= 15.62 KNS Vc= 127.21 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



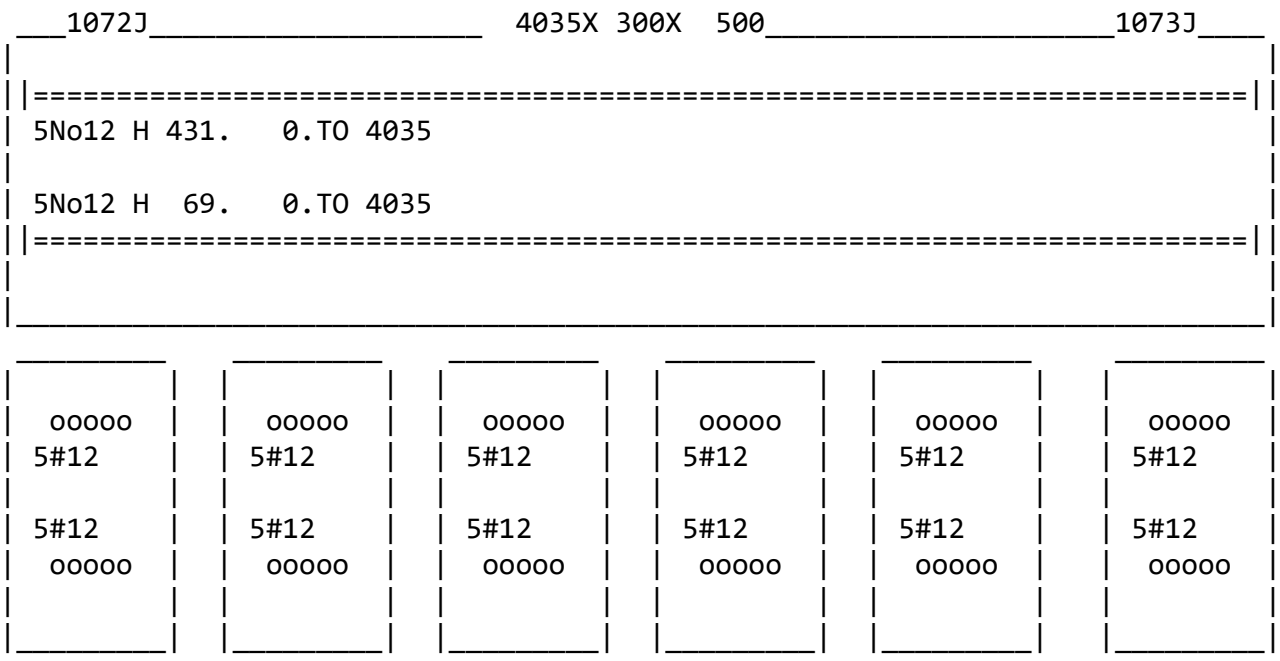
ACI 318-11 BEAM NO. 1555 DESIGN RESULTS

LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	4035.	YES	YES
2	431.	5 - 12MM	0.	4035.	YES	YES

B E A M N O. 1555 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.31 KNS Vc= 225.58 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.45 KNS Vc= 128.21 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

2223. DESIGN BEAM 333 946 1651 TO 1672 1675 TO 1699 1801 TO 1838 2321

ACI 318-11 BEAM NO. 1651 DESIGN RESULTS

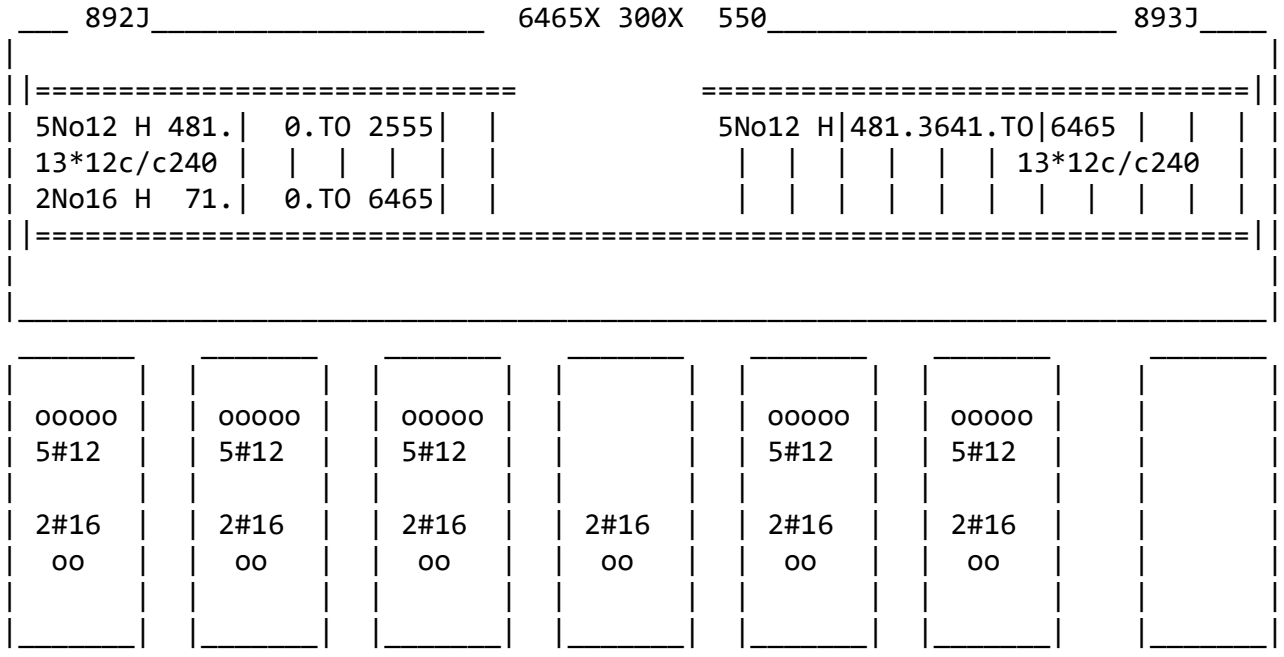
LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	6465.	YES	YES
2	481.	5 - 12MM	0.	2555.	YES	NO
3	481.	5 - 12MM	3641.	6465.	NO	YES

B E A M N O. 1651 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 64.20 KNS Vc= 139.73 KNS Vs= 0.00 KNS
Tu= 1.17 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 2758. MM

AT END SUPPORT - Vu= 68.90 KNS Vc= 138.71 KNS Vs= 0.00 KNS
Tu= 1.17 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 2758. MM



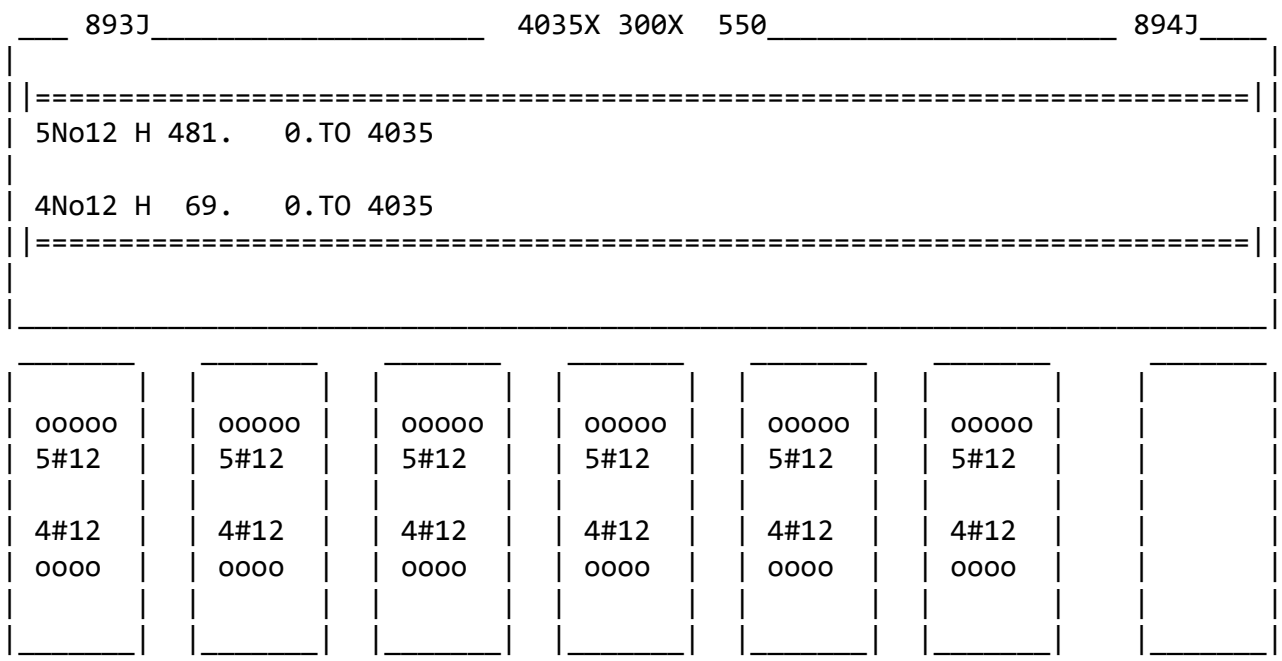
ACI 318-11 BEAM NO. 1652 DESIGN RESULTS

LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	4035.	YES	YES
2	481.	5 - 12MM	0.	4035.	YES	YES

B E A M N O. 1652 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.19 KNS Vc= 141.74 KNS Vs= 0.00 KNS
Tu= 1.86 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 15.61 KNS Vc= 141.74 KNS Vs= 0.00 KNS
Tu= 1.86 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



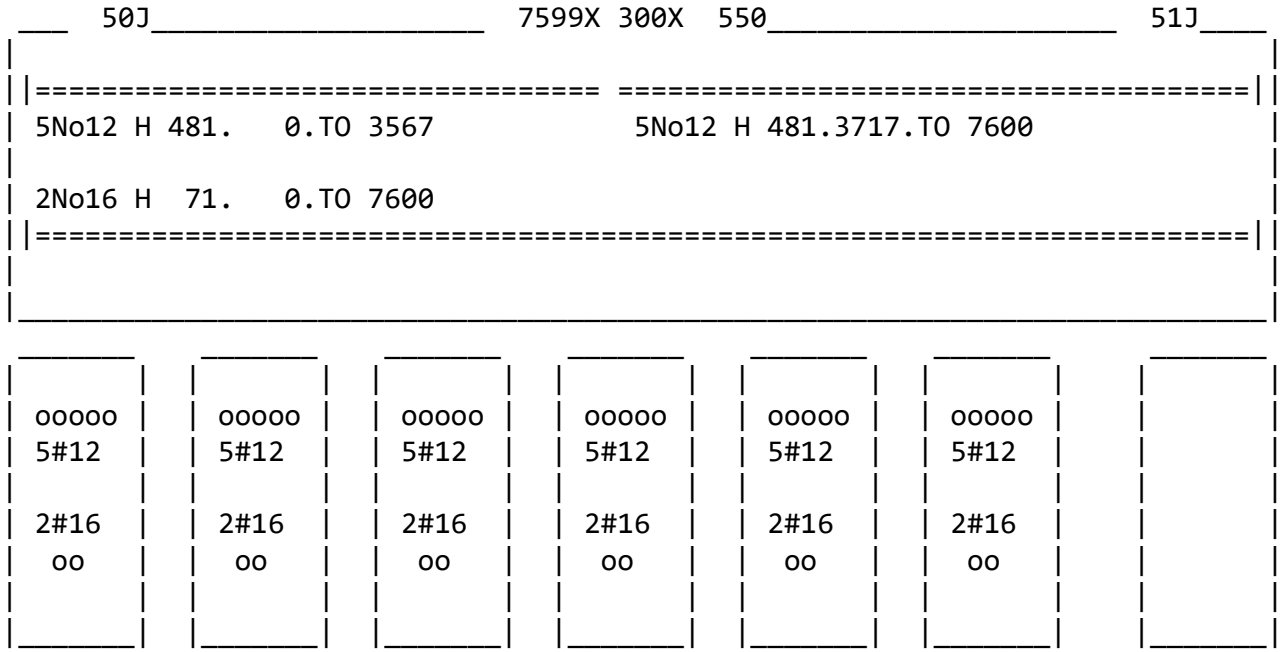
ACI 318-11 BEAM NO. 1653 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	7600.	YES	YES
2	481.	5 - 12MM	0.	3567.	YES	NO
3	481.	5 - 12MM	3717.	7600.	NO	YES

B E A M N O. 1653 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.92 KNS Vc= 140.29 KNS Vs= 0.00 KNS
Tu= 0.79 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 21.28 KNS Vc= 140.29 KNS Vs= 0.00 KNS
Tu= 0.79 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



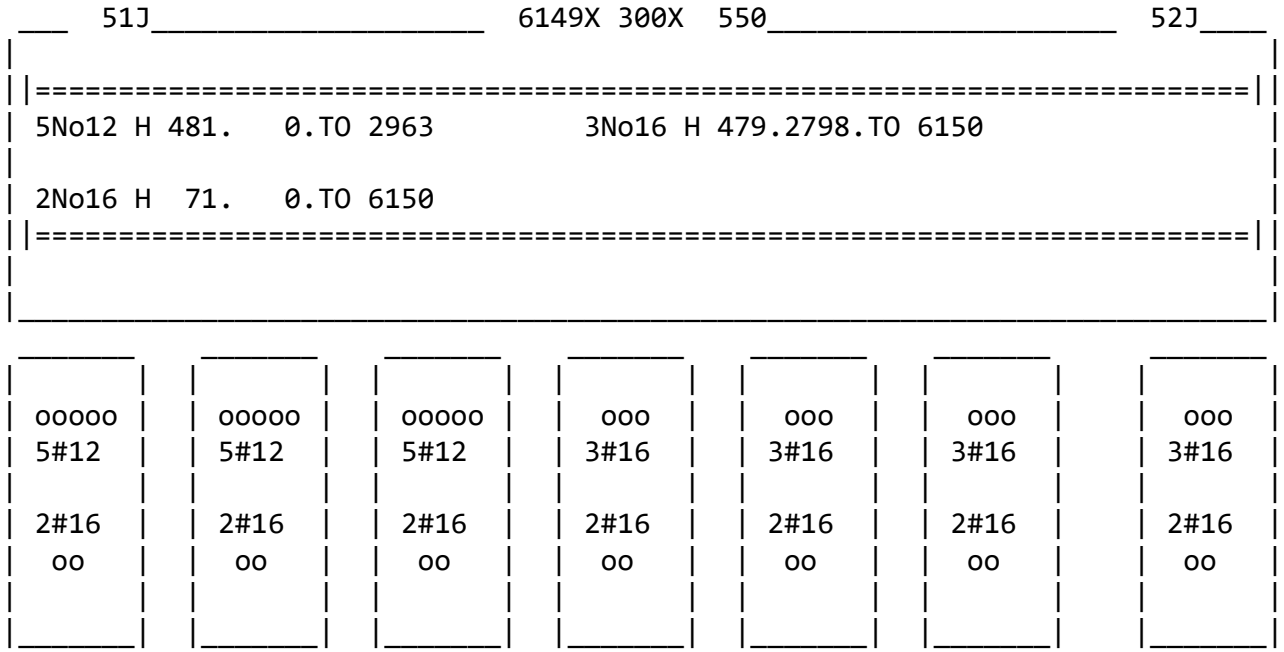
ACI 318-11 BEAM NO. 1654 DESIGN RESULTS

LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	6150.	YES	YES
2	481.	5 - 12MM	0.	2963.	YES	NO
3	479.	3 - 16MM	2798.	6150.	NO	YES

B E A M N O. 1654 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 20.48 KNS Vc= 136.88 KNS Vs= 0.00 KNS
Tu= 2.09 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 27.21 KNS Vc= 136.88 KNS Vs= 0.00 KNS
Tu= 2.09 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1655 DESIGN RESULTS

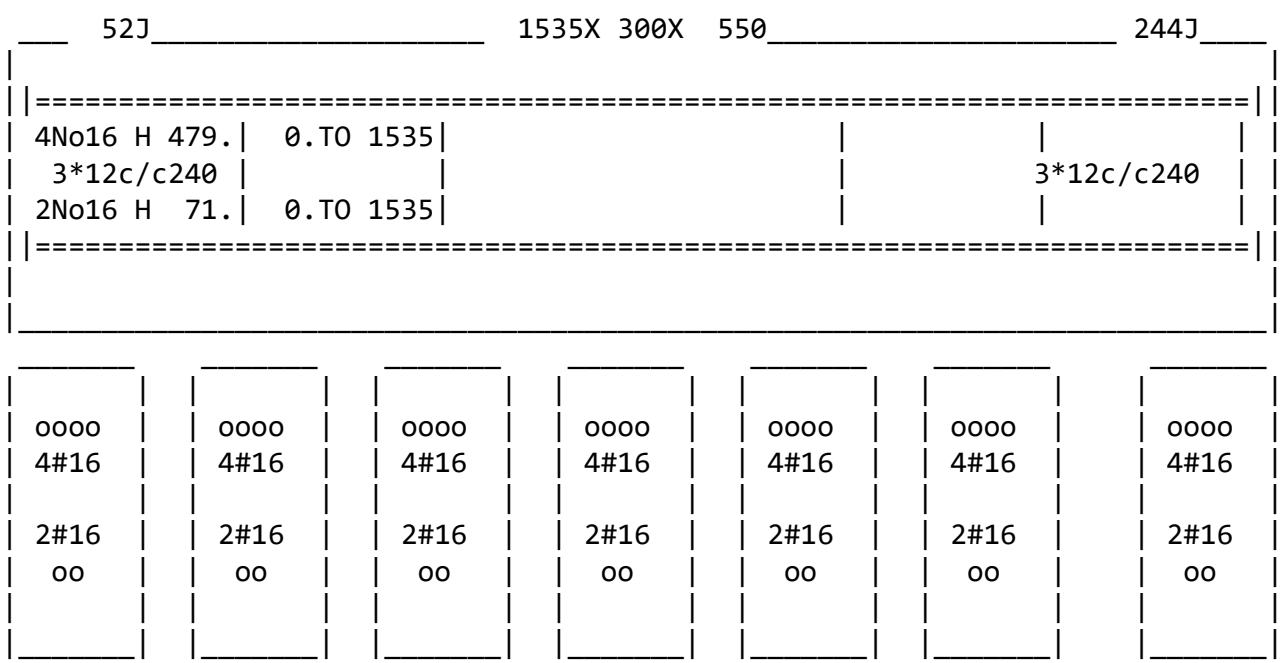
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	1535.	YES	YES
2	479.	4 - 16MM	0.	1535.	YES	YES

B E A M N O. 1655 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 63.97 KNS Vc= 140.86 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 293. MM

AT END SUPPORT - Vu= 56.68 KNS Vc= 140.86 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 293. MM



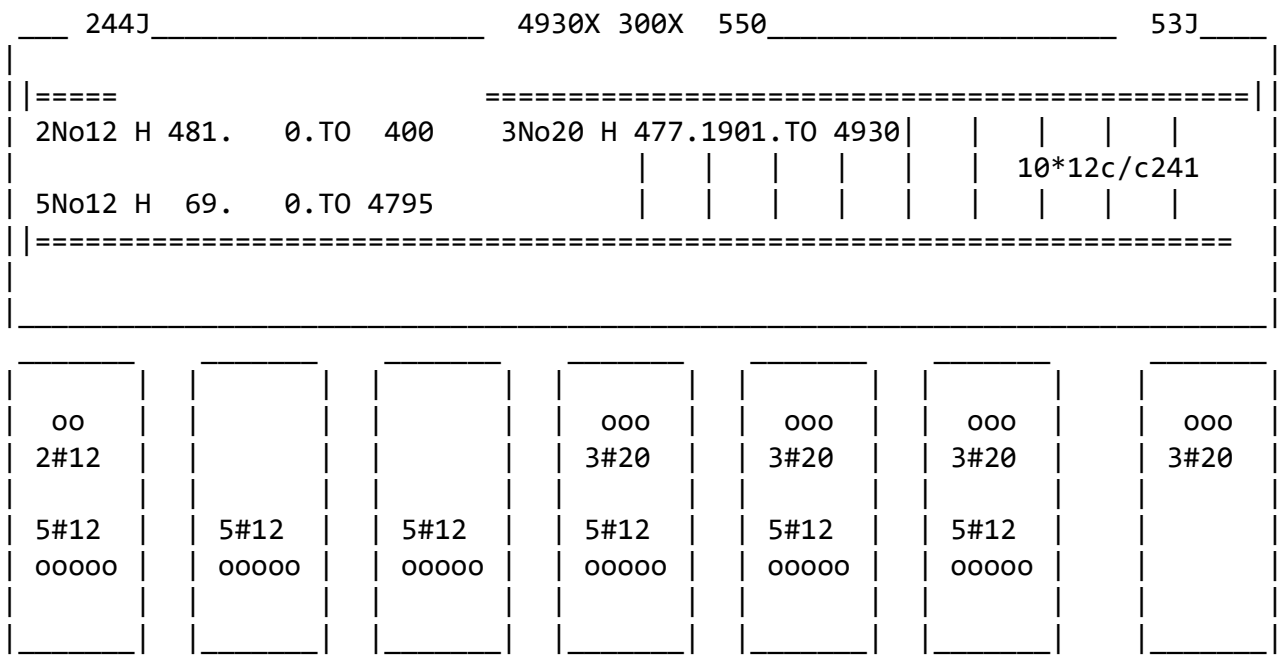
ACI 318-11 BEAM NO. 1656 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	4795.	YES	NO
2	481.	2 - 12MM	0.	400.	YES	NO
3	477.	3 - 20MM	1901.	4930.	NO	YES

B E A M N O. 1656 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.15 KNS Vc= 136.75 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 56.35 KNS Vc= 139.84 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 1990. MM



ACI 318-11 BEAM NO. 1657 DESIGN RESULTS

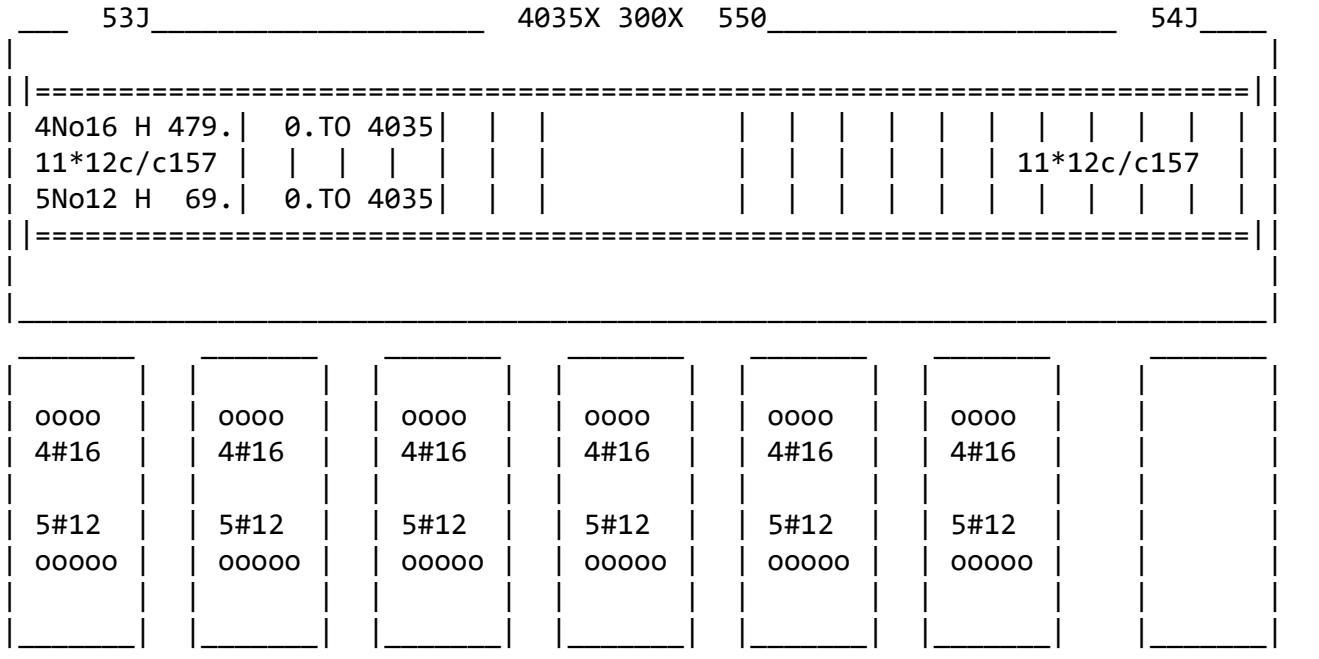
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	4035.	YES	YES
2	479.	4 - 16MM	0.	4035.	YES	YES

B E A M N O. 1657 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 40.32 KNS Vc= 156.39 KNS Vs= 0.00 KNS
Tu= 8.14 KN-MET Tc= 5.9 KN-MET Ts= 10.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1543. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.31 SQ.CM.

AT END SUPPORT - Vu= 56.24 KNS Vc= 140.79 KNS Vs= 0.00 KNS
Tu= 8.14 KN-MET Tc= 5.9 KN-MET Ts= 10.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1543. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.31 SQ.CM.



ACI 318-11 BEAM NO. 1658 DESIGN RESULTS

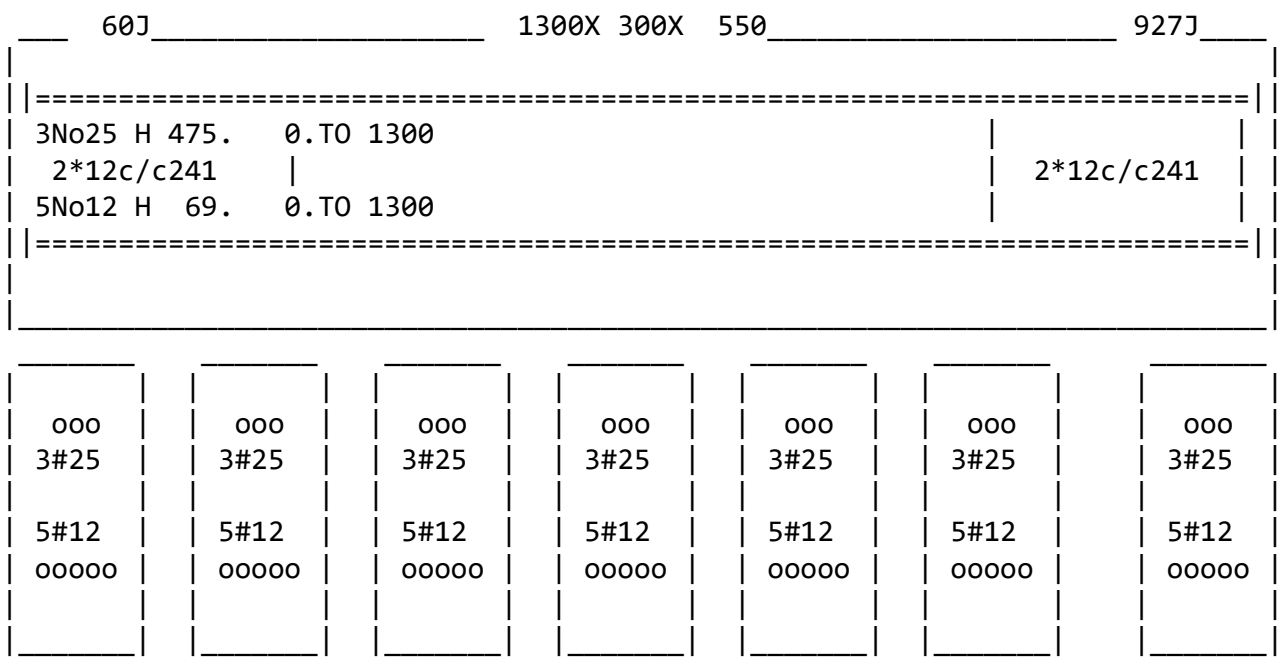
LEN - 1300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1300.	YES	YES
2	475.	3 - 25MM	0.	1300.	YES	YES

B E A M N O. 1658 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 59.90 KNS Vc= 137.71 KNS Vs= 0.00 KNS
Tu= 1.41 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 175. MM

AT END SUPPORT - Vu= 56.15 KNS Vc= 137.71 KNS Vs= 0.00 KNS
Tu= 1.41 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 175. MM



ACI 318-11 BEAM NO. 1659 DESIGN RESULTS

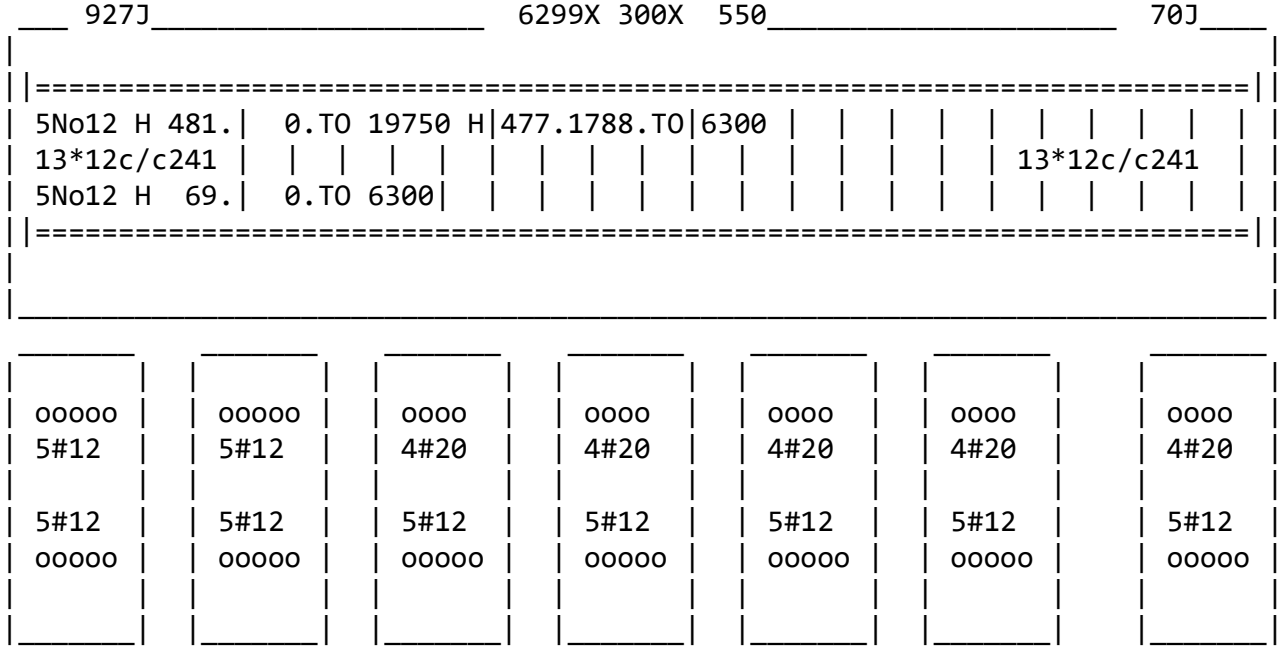
LEN - 6300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	6300.	YES	YES
2	481.	5 - 12MM	0.	1975.	YES	NO
3	477.	4 - 20MM	1788.	6300.	NO	YES

B E A M N O. 1659 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 52.99 KNS Vc= 136.70 KNS Vs= 0.00 KNS
Tu= 1.07 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 2675. MM

AT END SUPPORT - Vu= 60.06 KNS Vc= 136.39 KNS Vs= 0.00 KNS
Tu= 2.67 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 2675. MM



ACI 318-11 BEAM NO. 1660 DESIGN RESULTS

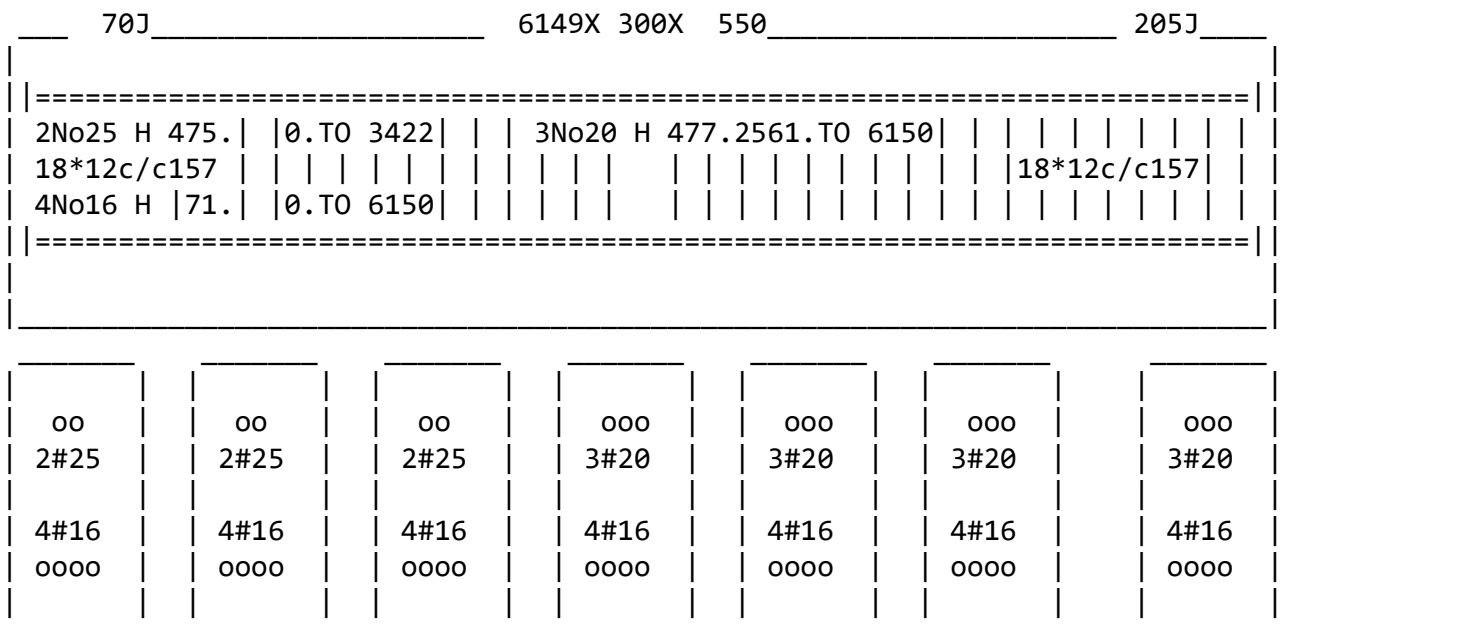
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	6150.	YES	YES
2	475.	2 - 25MM	0.	3422.	YES	NO
3	477.	3 - 20MM	2561.	6150.	NO	YES

B E A M N O. 1660 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 44.01 KNS Vc= 135.96 KNS Vs= 0.00 KNS
Tu= 5.79 KN-MET Tc= 5.7 KN-MET Ts= 7.7 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 2600. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.64 SQ.CM.

AT END SUPPORT - Vu= 47.49 KNS Vc= 135.96 KNS Vs= 0.00 KNS
Tu= 5.79 KN-MET Tc= 5.7 KN-MET Ts= 7.7 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 2600. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.64 SQ.CM.



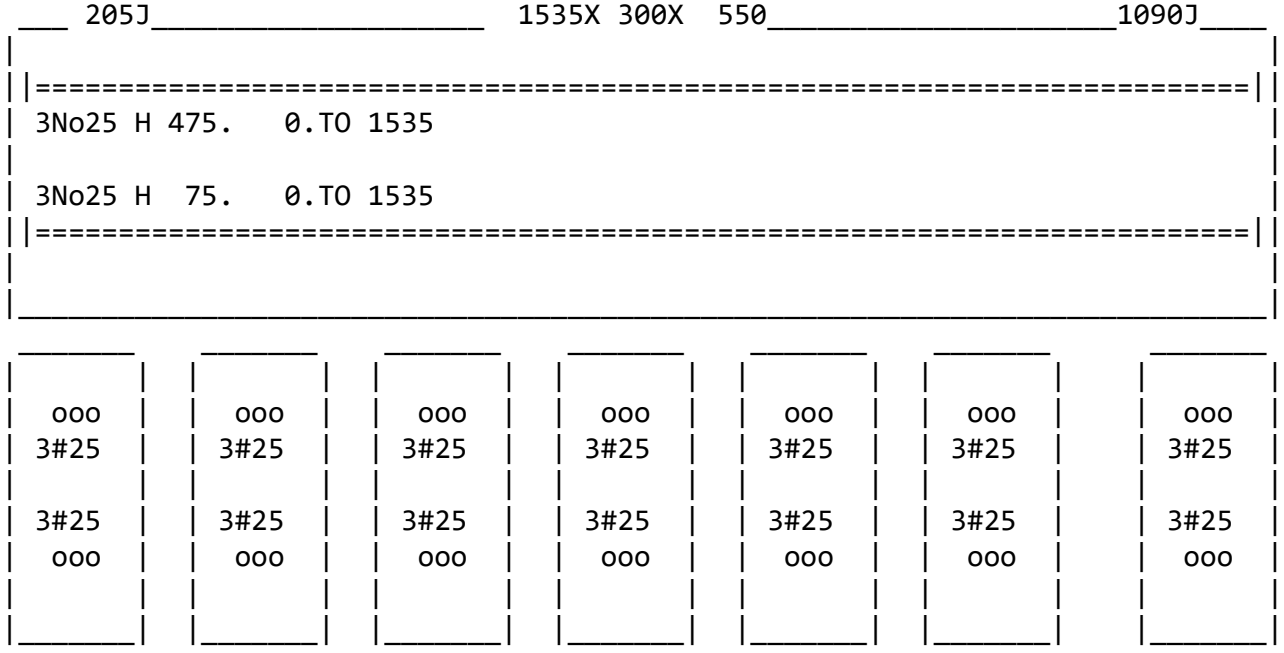
ACI 318-11 BEAM NO. 1661 DESIGN RESULTS

LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1535.	YES	YES
2	475.	3 - 25MM	0.	1535.	YES	YES

B E A M N O. 1661 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 66.19 KNS Tu= 35.6 KN-MET
Vc= 138.0 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 403 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
AT END SUPPORT - Vu= 75.60 KNS Tu= 35.6 KN-MET
Vc= 138.0 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 403 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.



ACI 318-11 BEAM NO. 1662 DESIGN RESULTS

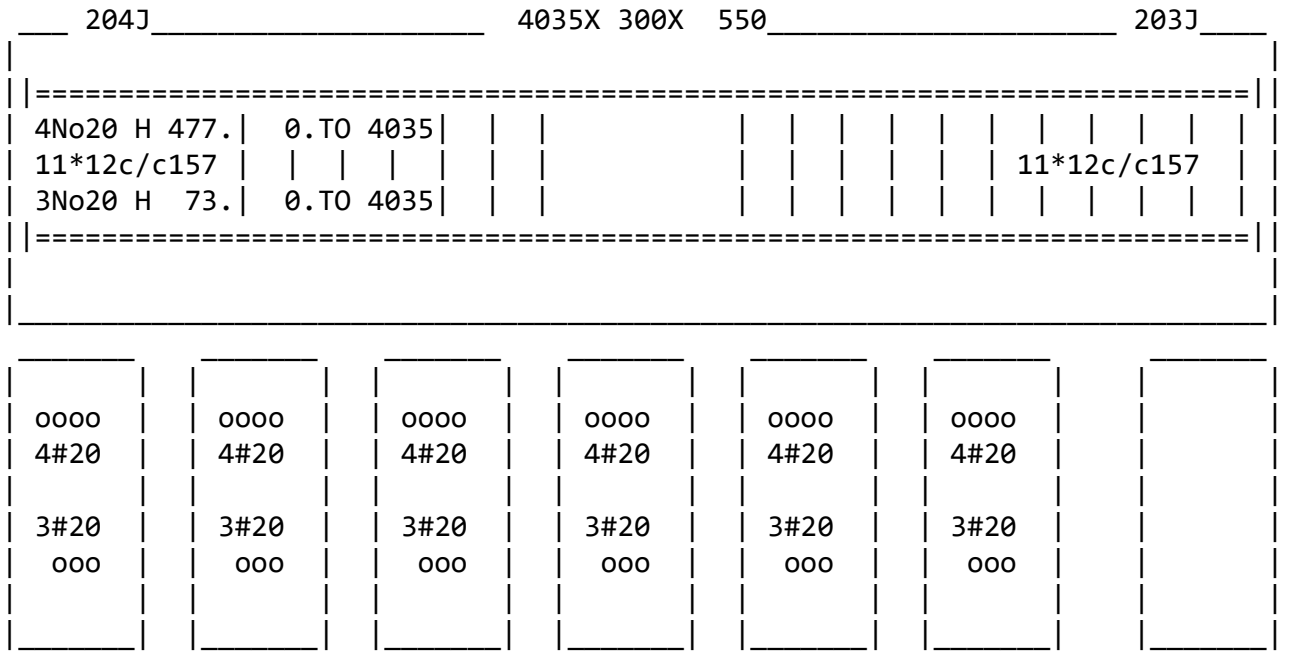
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	4035.	YES	YES
2	477.	4 - 20MM	0.	4035.	YES	YES

B E A M N O. 1662 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.36 KNS Vc= 140.28 KNS Vs= 0.00 KNS
Tu= 7.00 KN-MET Tc= 5.9 KN-MET Ts= 9.3 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1543. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.99 SQ.CM.

AT END SUPPORT - Vu= 42.03 KNS Vc= 140.28 KNS Vs= 0.00 KNS
Tu= 7.00 KN-MET Tc= 5.9 KN-MET Ts= 9.3 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1543. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.99 SQ.CM.



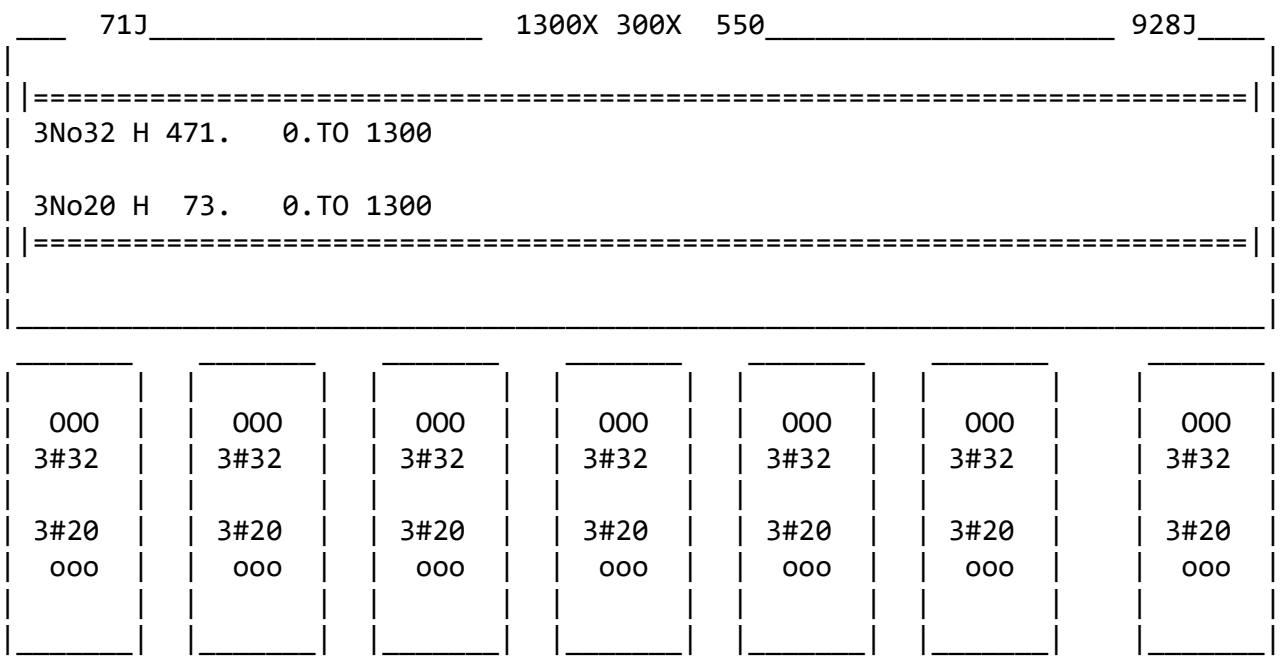
ACI 318-11 BEAM NO. 1663 DESIGN RESULTS

LEN - 1300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	1300.	YES	YES
2	471.	3 - 32MM	0.	1300.	YES	YES

B E A M N O. 1663 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 88.61 KNS Tu= 43.2 KN-MET
Vc= 139.7 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 407 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
AT END SUPPORT - Vu= 84.28 KNS Tu= 43.2 KN-MET
Vc= 139.7 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 407 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.



ACI 318-11 BEAM NO. 1664 DESIGN RESULTS

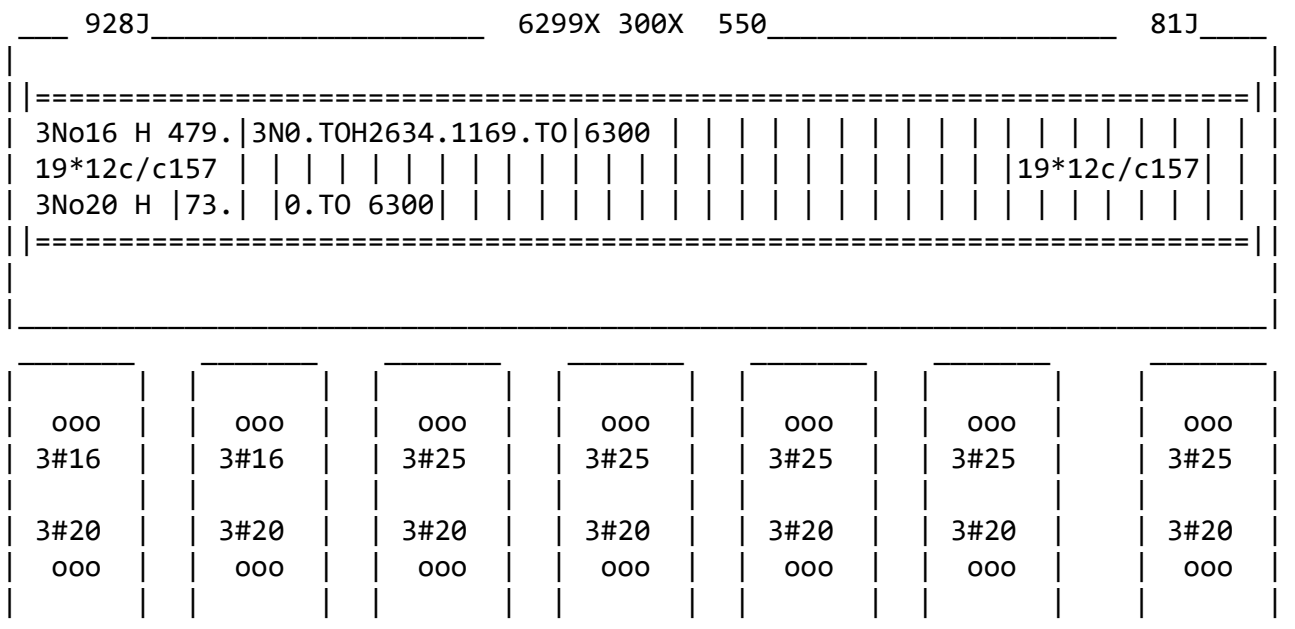
LEN - 6300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	6300.	YES	YES
2	479.	3 - 16MM	0.	2634.	YES	NO
3	475.	3 - 25MM	1169.	6300.	NO	YES

B E A M N O. 1664 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.97 KNS Vc= 139.75 KNS Vs= 0.00 KNS
Tu= 11.82 KN-MET Tc= 5.9 KN-MET Ts= 15.8 KN-MET LOAD 407
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 2675. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.35 SQ.CM.

AT END SUPPORT - Vu= 21.99 KNS Vc= 139.75 KNS Vs= 0.00 KNS
Tu= 11.82 KN-MET Tc= 5.9 KN-MET Ts= 15.8 KN-MET LOAD 407
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 2675. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.35 SQ.CM.



ACI 318-11 BEAM NO. 1665 DESIGN RESULTS

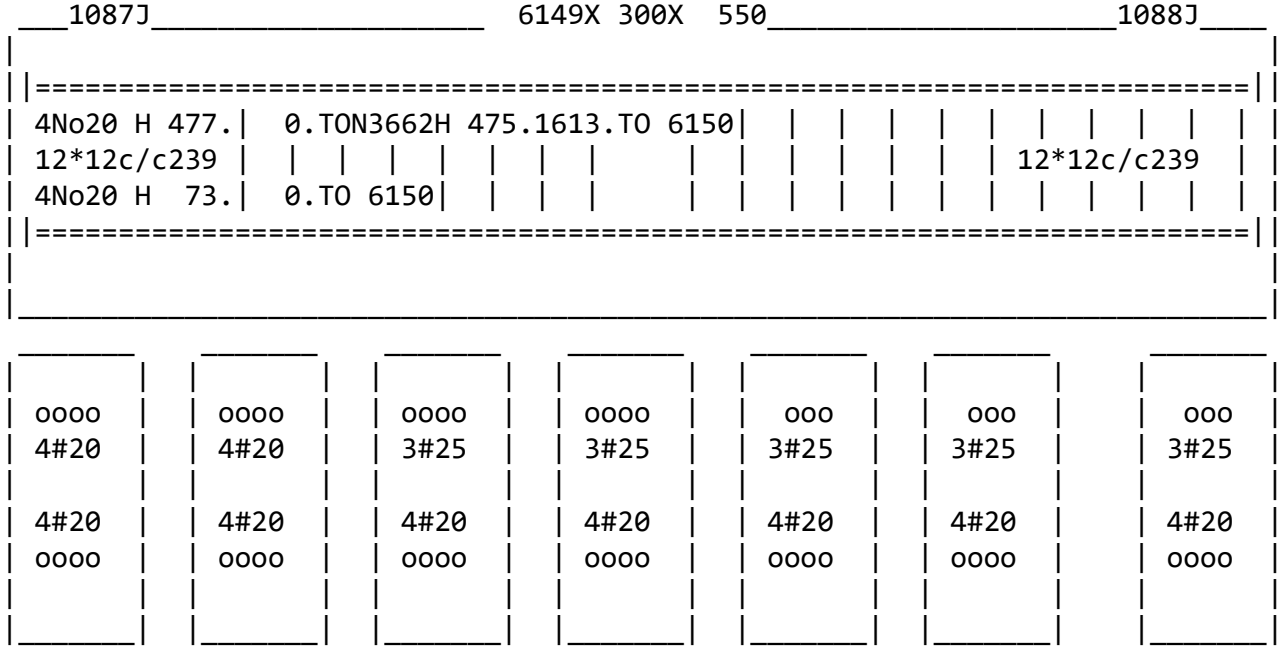
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	6150.	YES	YES
2	477.	4 - 20MM	0.	3662.	YES	NO
3	475.	3 - 25MM	1613.	6150.	NO	YES

B E A M N O. 1665 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 54.03 KNS Vc= 124.25 KNS Vs= 0.00 KNS
Tu= 0.19 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2600. MM

AT END SUPPORT - Vu= 78.29 KNS Vc= 124.25 KNS Vs= 0.00 KNS
Tu= 0.19 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2600. MM



ACI 318-11 BEAM NO. 1666 DESIGN RESULTS

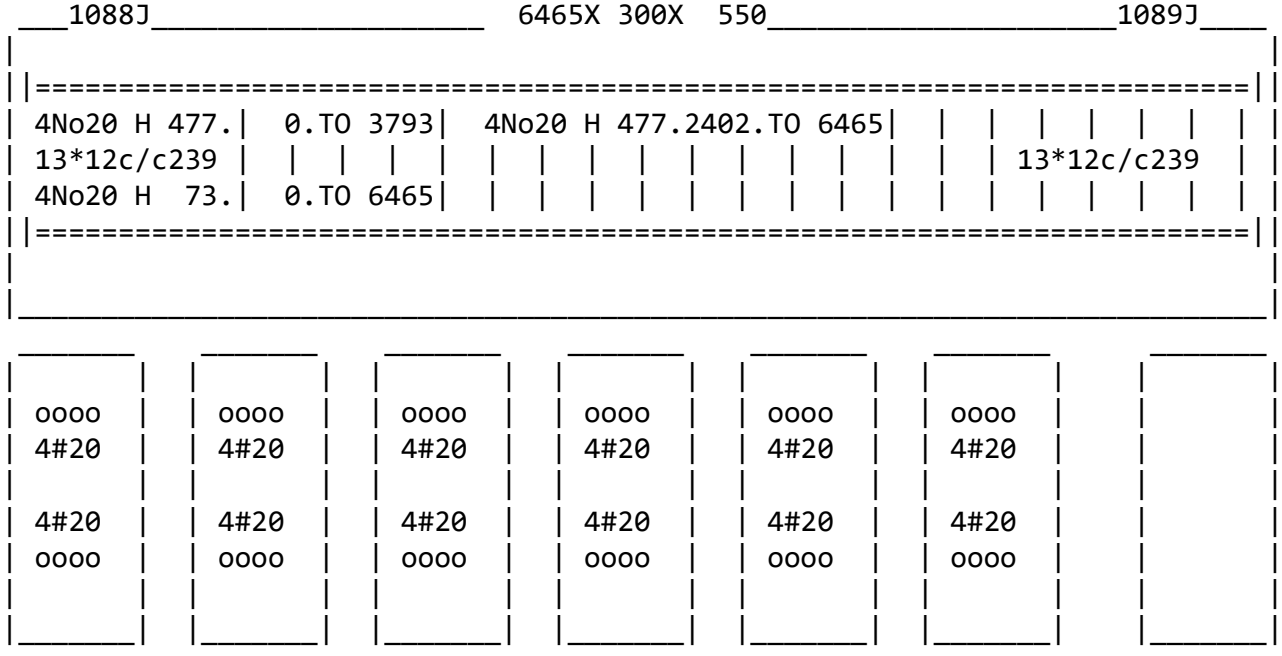
LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	6465.	YES	YES
2	477.	4 - 20MM	0.	3793.	YES	NO
3	477.	4 - 20MM	2402.	6465.	NO	YES

B E A M N O. 1666 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 65.12 KNS Vc= 140.22 KNS Vs= 0.00 KNS
Tu= 0.25 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2758. MM

AT END SUPPORT - Vu= 70.50 KNS Vc= 136.41 KNS Vs= 0.00 KNS
Tu= 0.23 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2758. MM



ACI 318-11 BEAM NO. 1667 DESIGN RESULTS

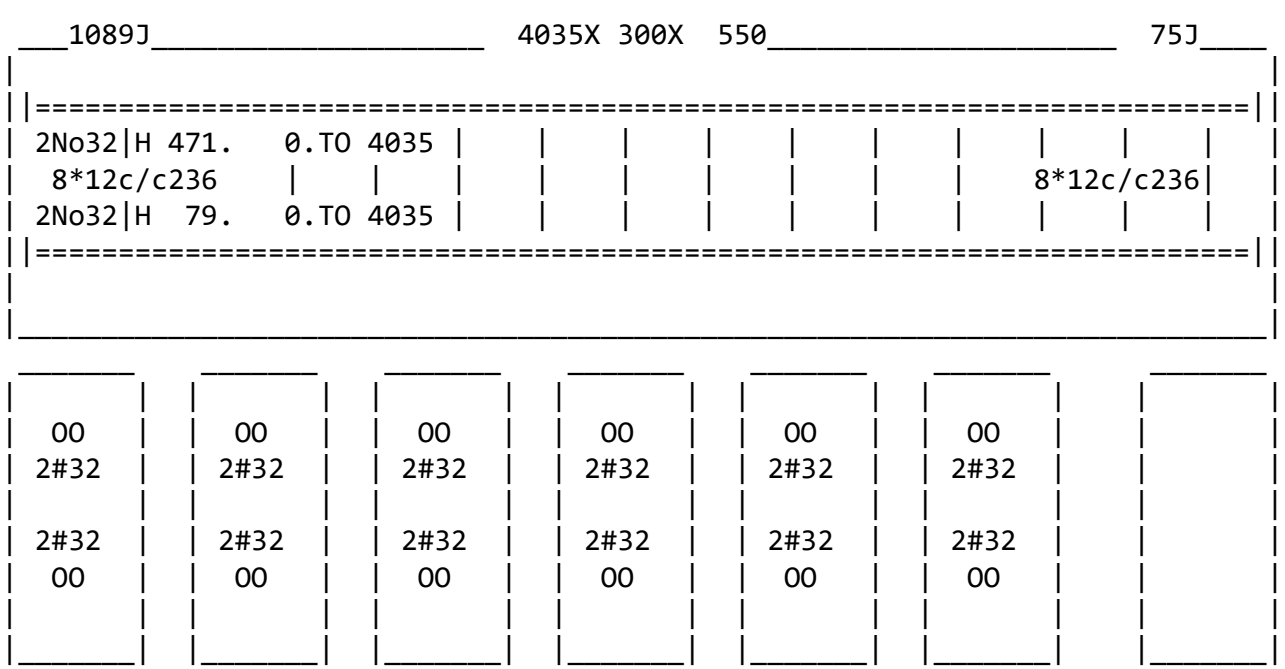
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	2 - 32MM	0.	4035.	YES	YES
2	471.	2 - 32MM	0.	4035.	YES	YES

B E A M N O. 1667 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 115.97 KNS Vc= 136.80 KNS Vs= 17.83 KNS
Tu= 1.69 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 236. MM C/C FOR 1543. MM

AT END SUPPORT - Vu= 130.37 KNS Vc= 136.80 KNS Vs= 37.02 KNS
Tu= 1.69 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 236. MM C/C FOR 1543. MM



ACI 318-11 BEAM NO. 1668 DESIGN RESULTS

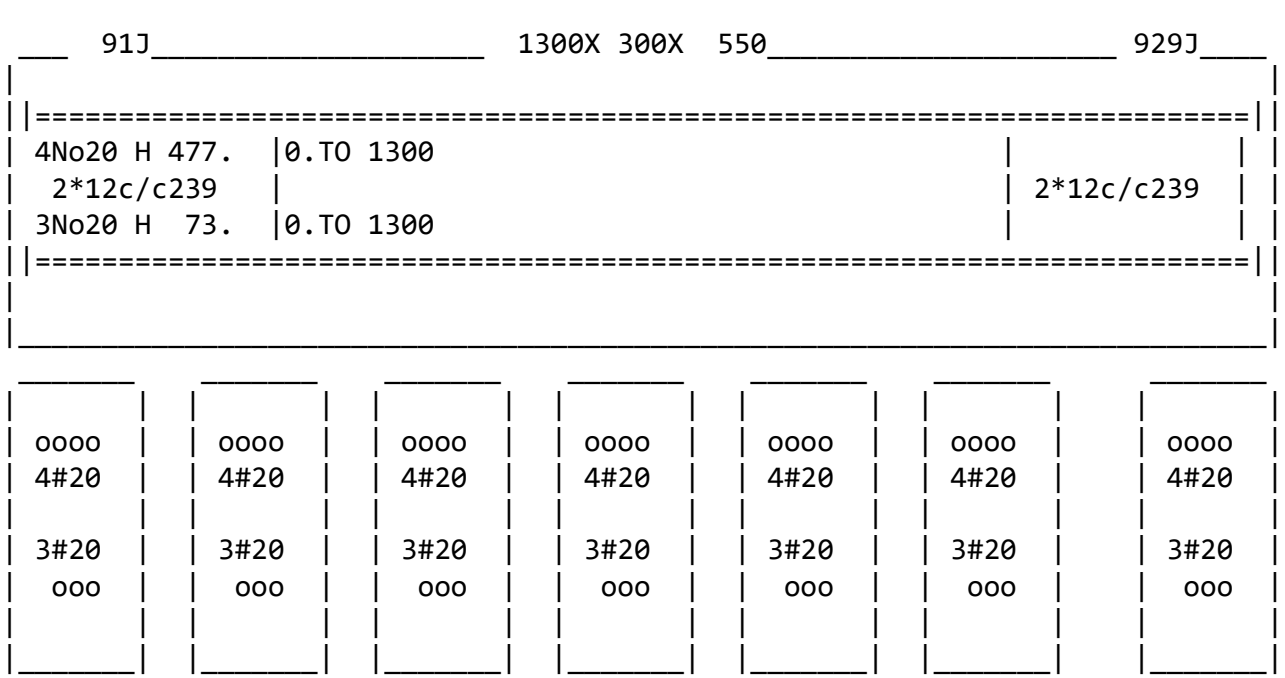
LEN - 1300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	1300.	YES	YES
2	477.	4 - 20MM	0.	1300.	YES	YES

B E A M N O. 1668 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.97 KNS Vc= 137.30 KNS Vs= 0.00 KNS
Tu= 1.22 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 175. MM

AT END SUPPORT - Vu= 74.99 KNS Vc= 137.80 KNS Vs= 0.00 KNS
Tu= 1.22 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 175. MM



ACI 318-11 BEAM NO. 1669 DESIGN RESULTS

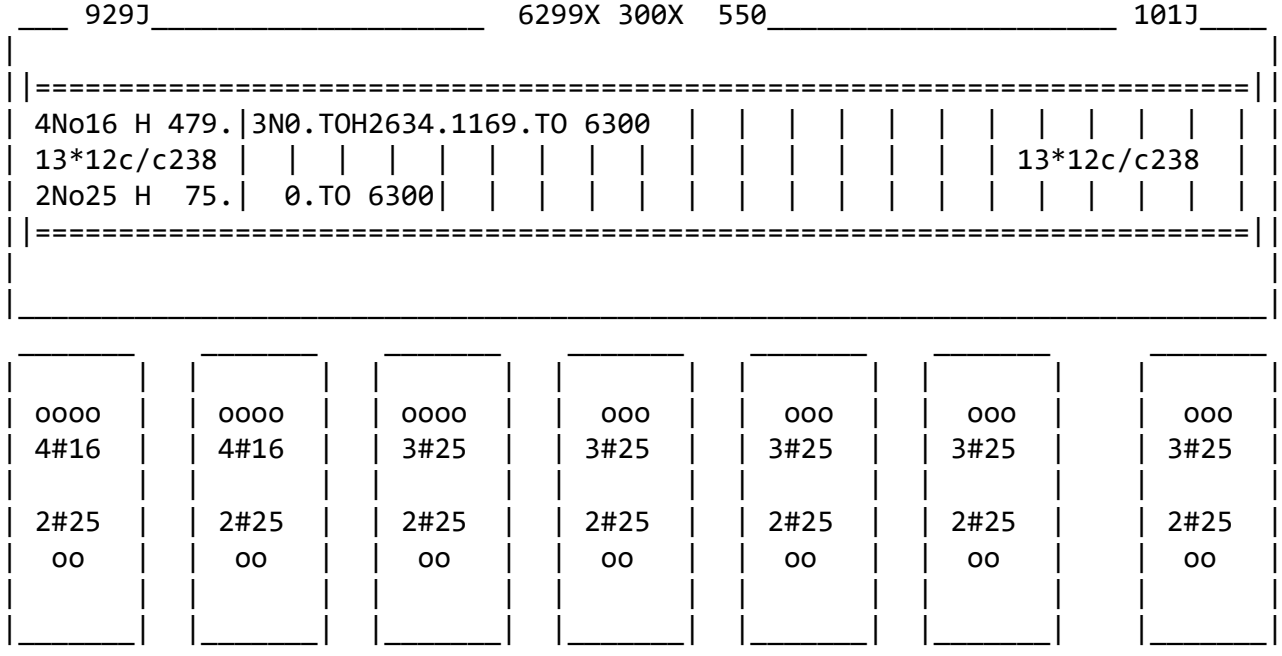
LEN - 6300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	6300.	YES	YES
2	479.	4 - 16MM	0.	2634.	YES	NO
3	475.	3 - 25MM	1169.	6300.	NO	YES

B E A M N O. 1669 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 53.88 KNS Vc= 138.36 KNS Vs= 0.00 KNS
Tu= 1.22 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 2675. MM

AT END SUPPORT - Vu= 66.16 KNS Vc= 135.98 KNS Vs= 0.00 KNS
Tu= 0.40 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 2675. MM



ACI 318-11 BEAM NO. 1670 DESIGN RESULTS

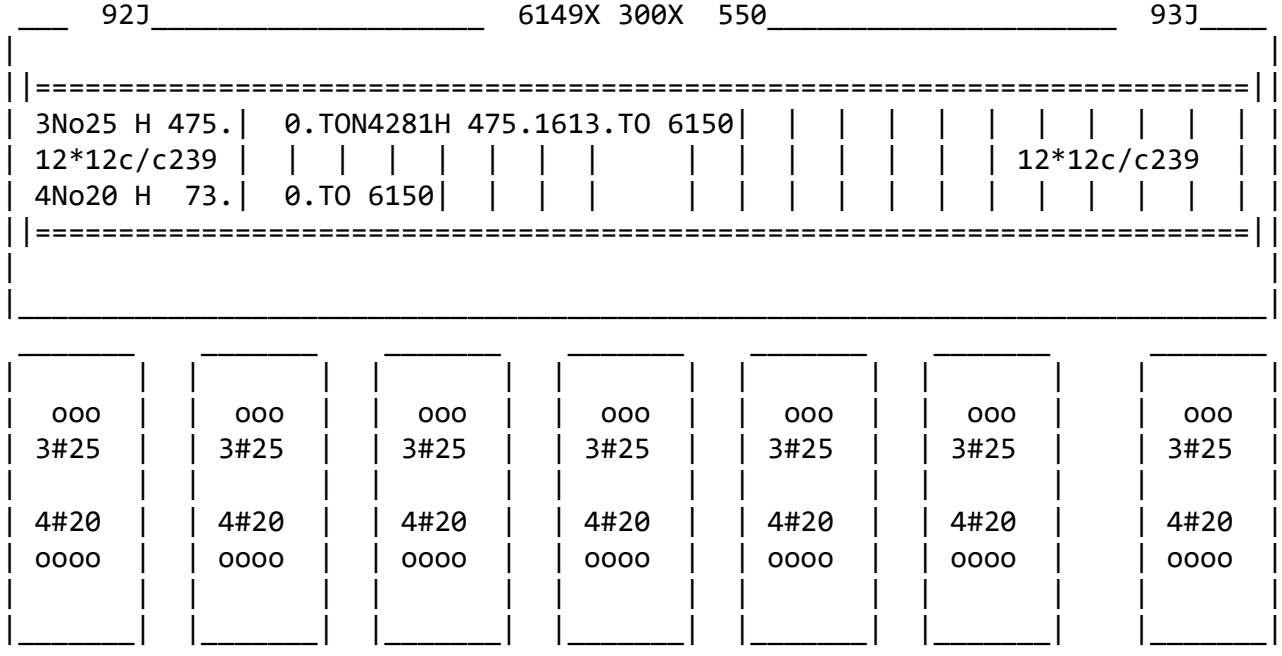
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	6150.	YES	YES
2	475.	3 - 25MM	0.	4281.	YES	NO
3	475.	3 - 25MM	1613.	6150.	NO	YES

B E A M N O. 1670 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 58.37 KNS Vc= 129.94 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2600. MM

AT END SUPPORT - Vu= 82.64 KNS Vc= 129.94 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2600. MM



ACI 318-11 BEAM NO. 1671 DESIGN RESULTS

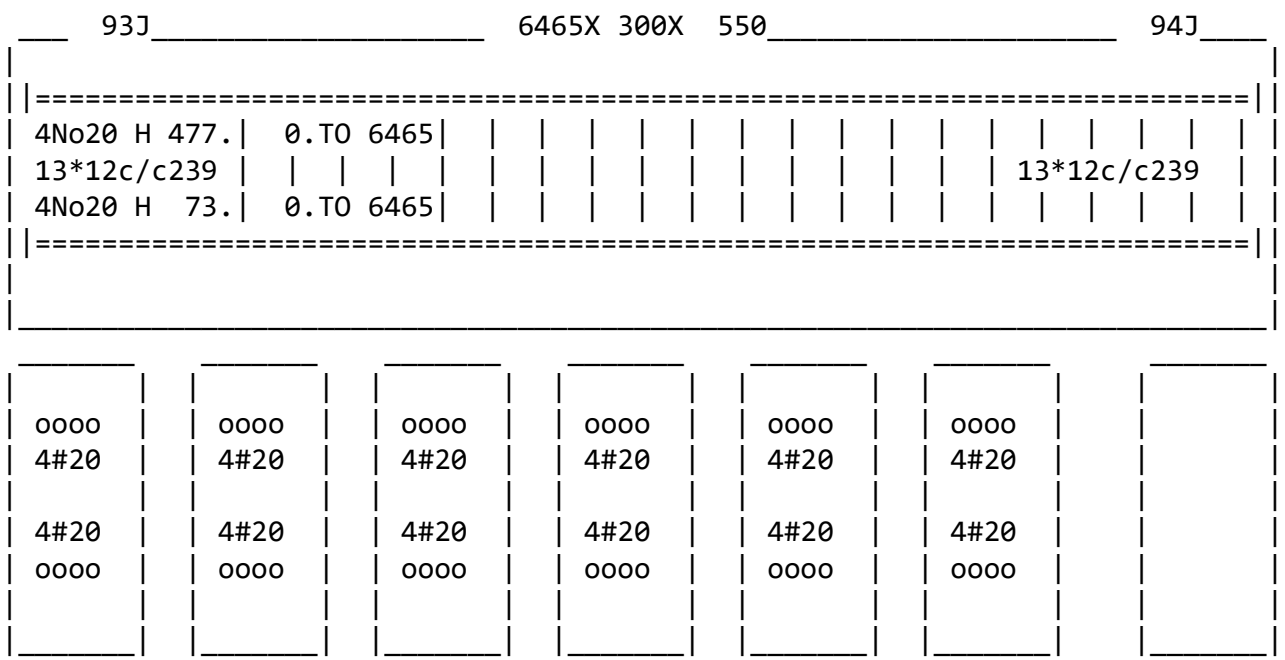
LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	6465.	YES	YES
2	477.	4 - 20MM	0.	6465.	YES	YES

B E A M N O. 1671 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 65.37 KNS Vc= 137.52 KNS Vs= 0.00 KNS
Tu= 0.35 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2758. MM

AT END SUPPORT - Vu= 68.15 KNS Vc= 131.27 KNS Vs= 0.00 KNS
Tu= 0.71 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 239. MM C/C FOR 2758. MM



ACI 318-11 BEAM NO. 1672 DESIGN RESULTS

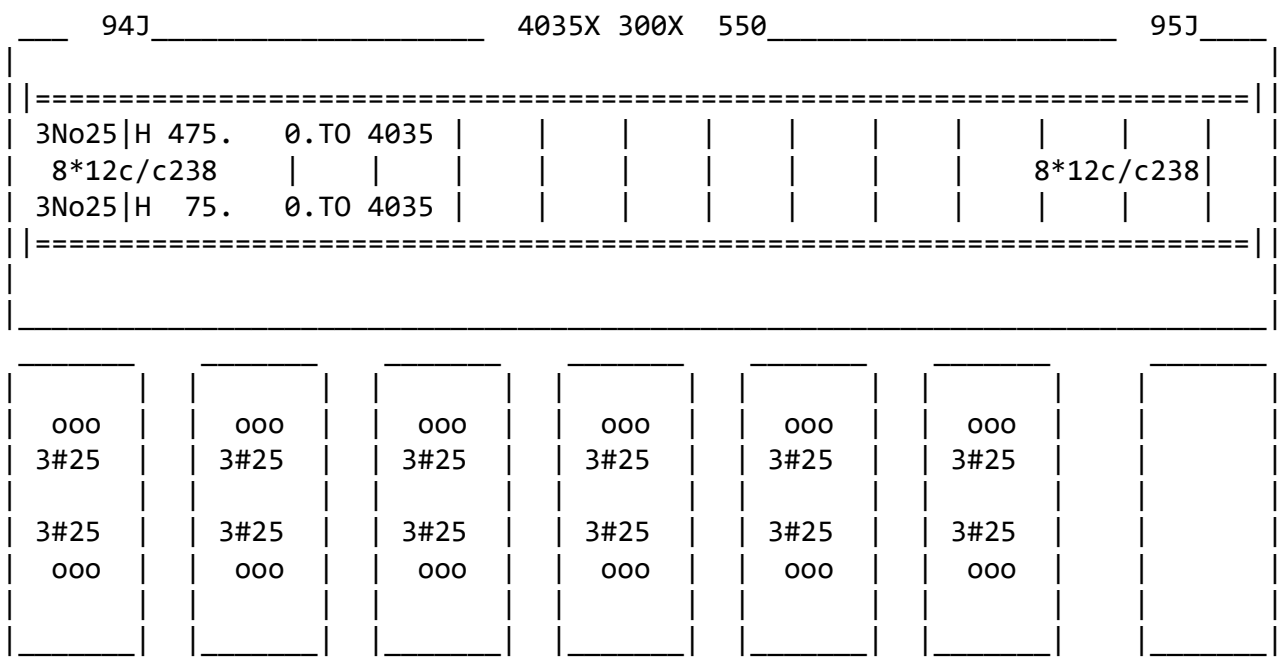
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	4035.	YES	YES
2	475.	3 - 25MM	0.	4035.	YES	YES

B E A M N O. 1672 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 100.97 KNS Vc= 134.54 KNS Vs= 0.09 KNS
Tu= 0.44 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 1543. MM

AT END SUPPORT - Vu= 115.37 KNS Vc= 134.54 KNS Vs= 19.29 KNS
Tu= 0.44 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 1543. MM



ACI 318-11 BEAM NO. 1675 DESIGN RESULTS

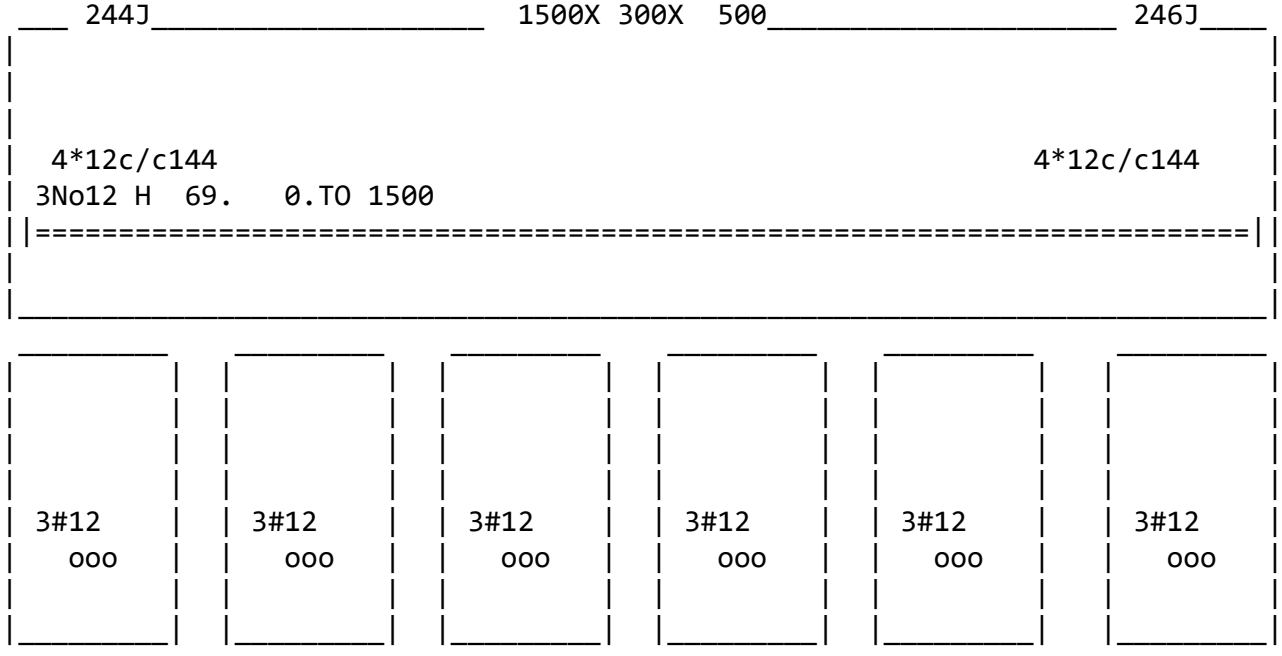
LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1500.	YES	YES

B E A M N O. 1675 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.70 KNS Vc= 123.68 KNS Vs= 0.00 KNS
Tu= 6.67 KN-MET Tc= 5.0 KN-MET Ts= 8.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 325. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.97 SQ.CM.

AT END SUPPORT - Vu= 19.61 KNS Vc= 123.68 KNS Vs= 0.00 KNS
Tu= 6.67 KN-MET Tc= 5.0 KN-MET Ts= 8.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 325. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.97 SQ.CM.



ACI 318-11 BEAM NO. 1676 DESIGN RESULTS

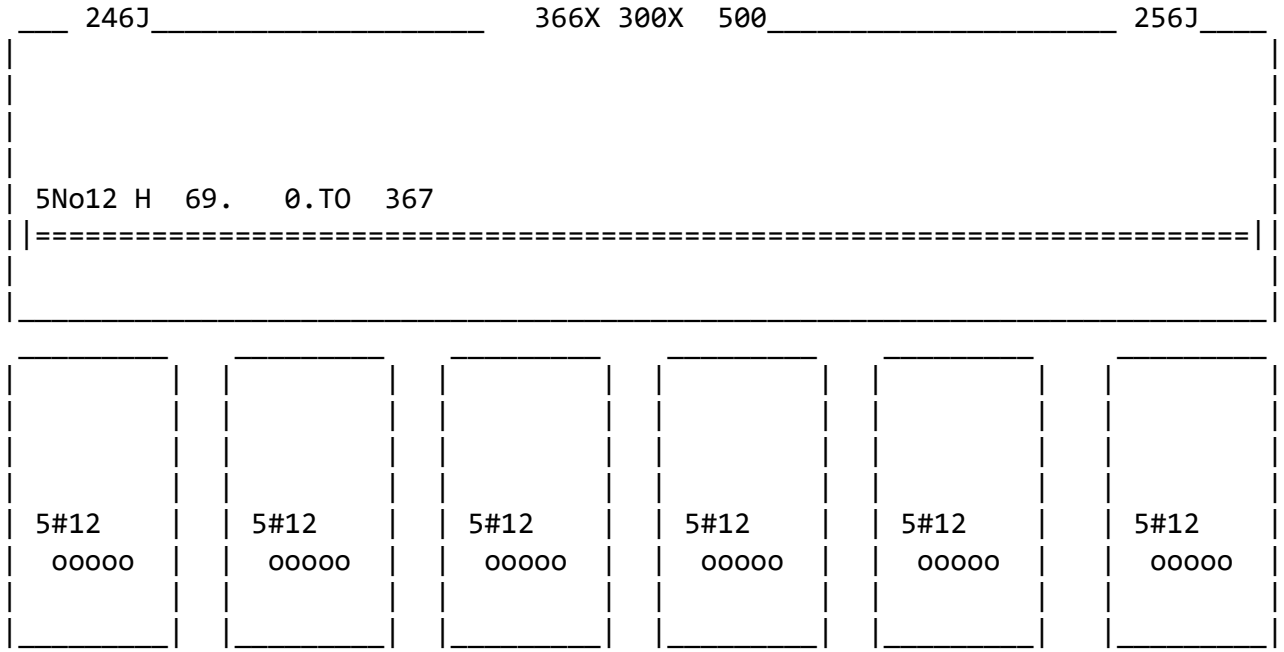
LEN - 367. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	367.	YES	YES

B E A M N O. 1676 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1676 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1676 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1677 DESIGN RESULTS

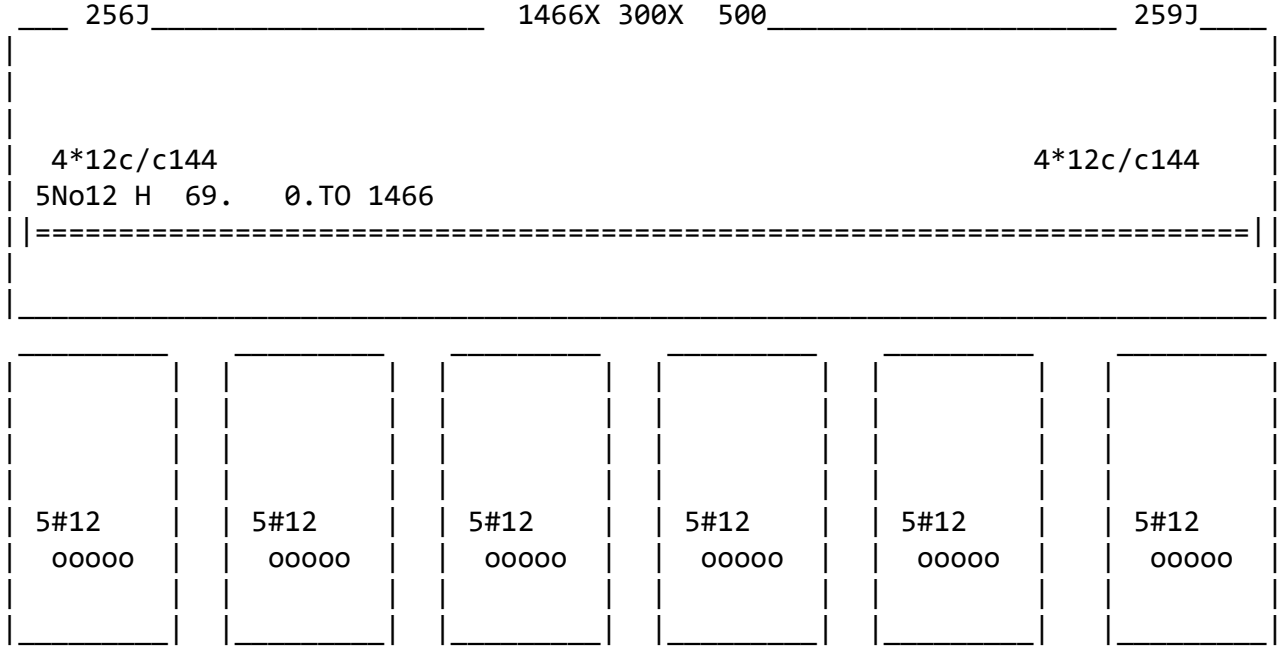
LEN - 1466. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1466.	YES	YES

B E A M N O. 1677 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.44 KNS Vc= 122.57 KNS Vs= 0.00 KNS
Tu= 6.67 KN-MET Tc= 5.0 KN-MET Ts= 8.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 309. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.97 SQ.CM.

AT END SUPPORT - Vu= 2.84 KNS Vc= 122.57 KNS Vs= 0.00 KNS
Tu= 6.67 KN-MET Tc= 5.0 KN-MET Ts= 8.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 309. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.97 SQ.CM.



ACI 318-11 BEAM NO. 1678 DESIGN RESULTS

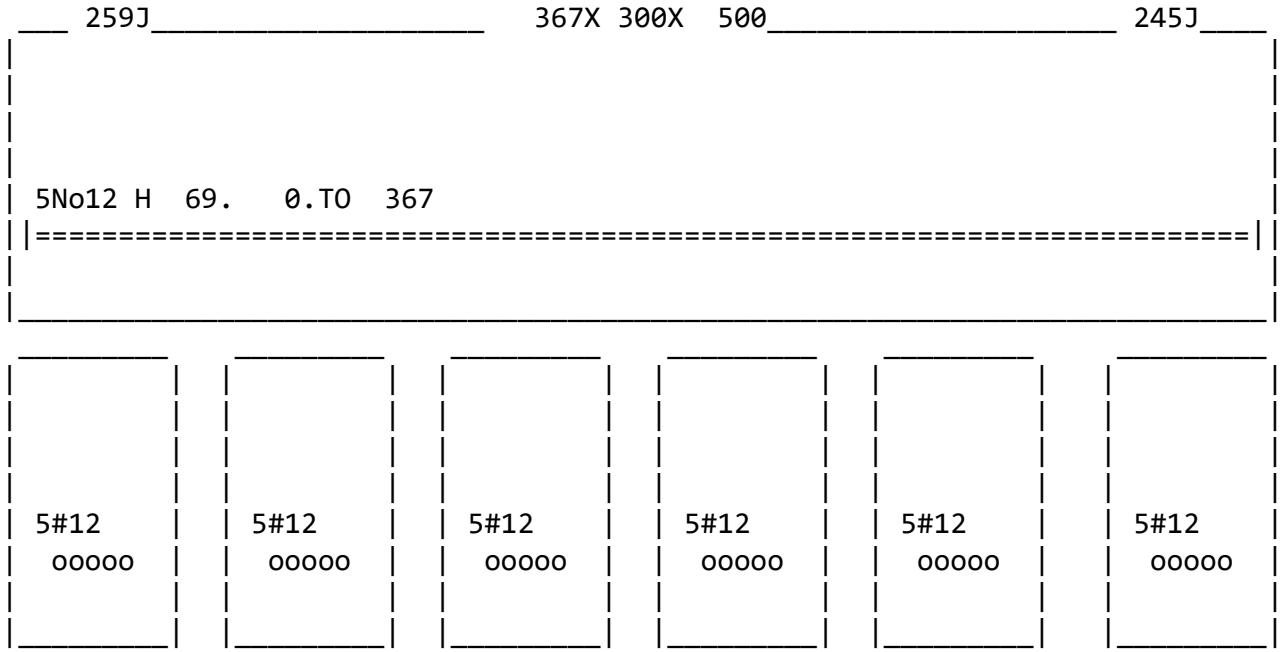
LEN - 367. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	367.	YES	YES

B E A M N O. 1678 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1678 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1678 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1679 DESIGN RESULTS

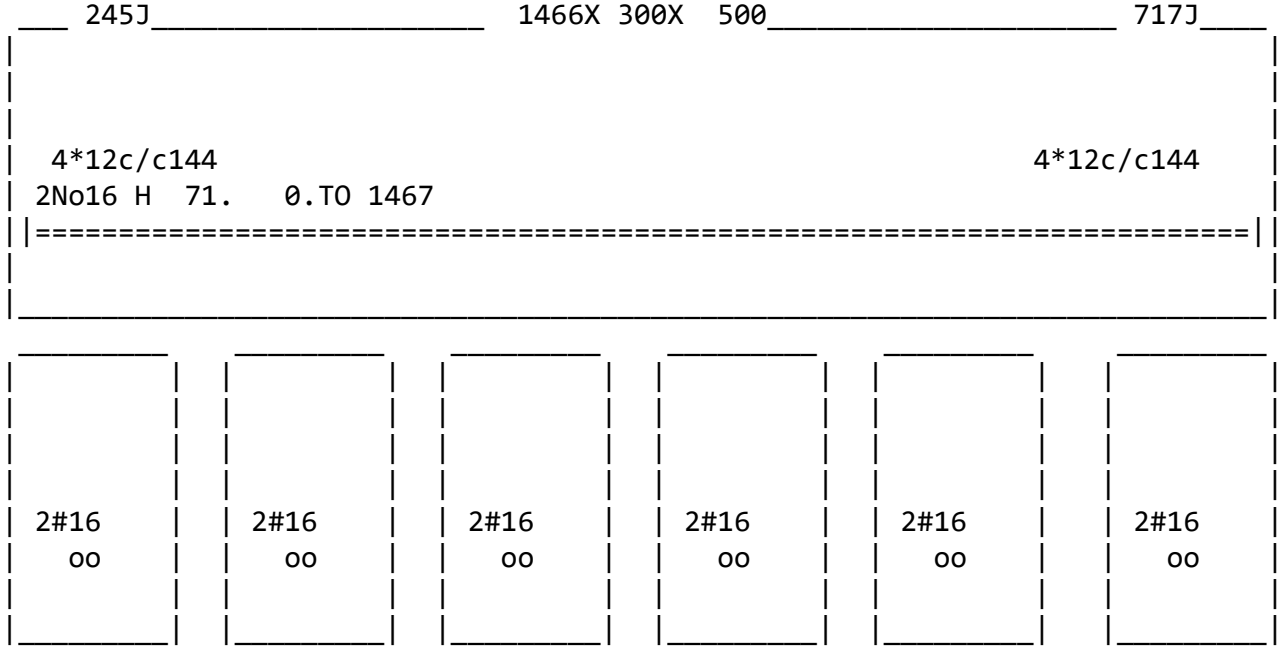
LEN - 1467. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	1467.	YES	YES

B E A M N O. 1679 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 26.05 KNS Vc= 124.26 KNS Vs= 0.00 KNS
Tu= 6.67 KN-MET Tc= 5.1 KN-MET Ts= 8.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 309. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.97 SQ.CM.

AT END SUPPORT - Vu= 40.26 KNS Vc= 124.26 KNS Vs= 0.00 KNS
Tu= 6.67 KN-MET Tc= 5.1 KN-MET Ts= 8.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 309. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.97 SQ.CM.



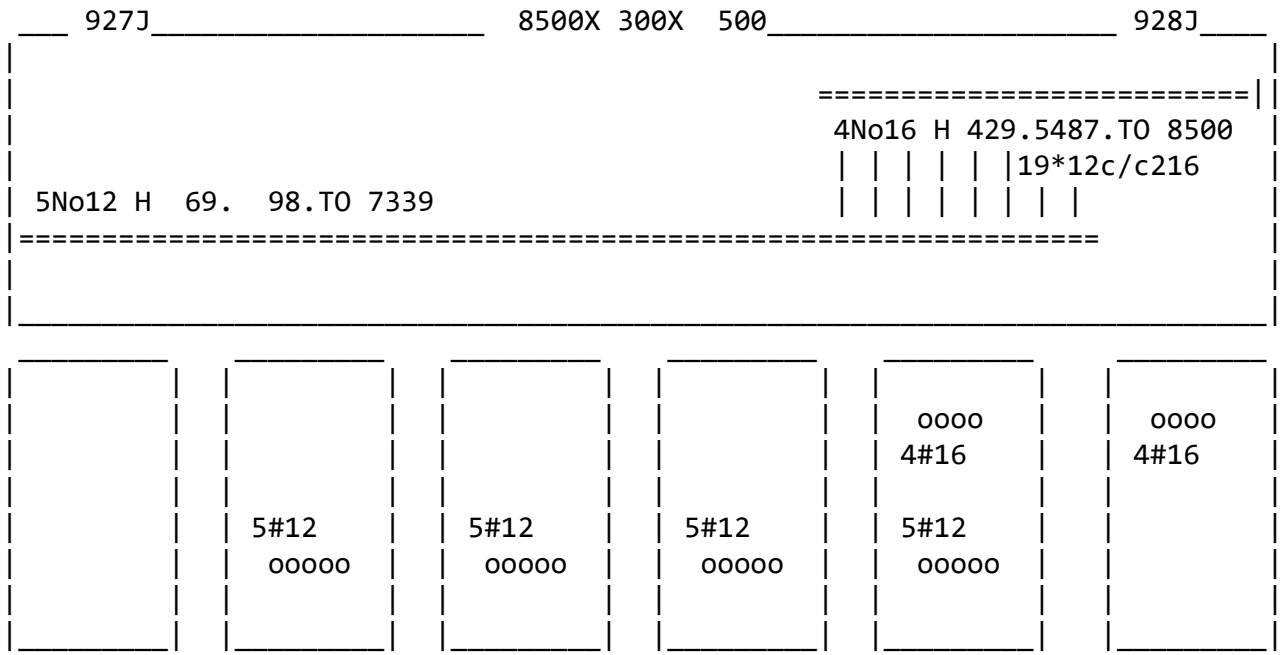
ACI 318-11 BEAM NO. 1680 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	98.	7339.	NO	NO
2	429.	4 - 16MM	5487.	8500.	NO	YES

B E A M N O. 1680 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.12 KNS Vc= 129.17 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 64.28 KNS Vc= 123.74 KNS Vs= 0.00 KNS
Tu= 0.45 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 3825. MM



ACI 318-11 BEAM NO. 1681 DESIGN RESULTS

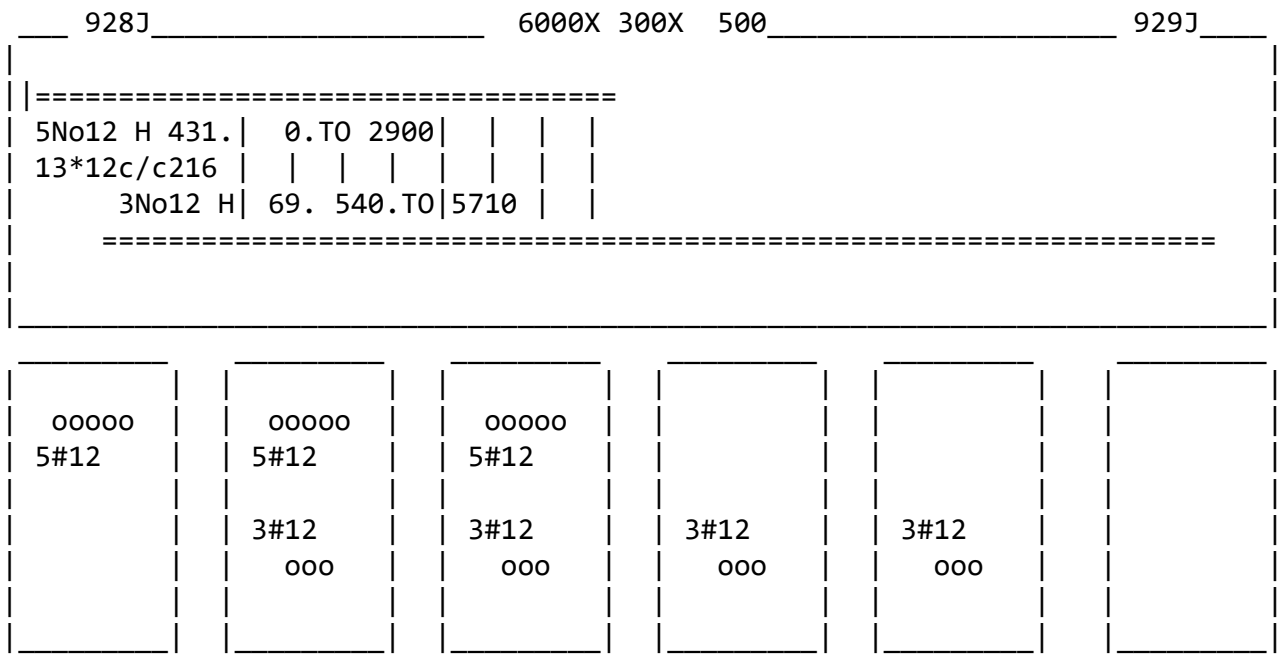
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	540.	5710.	NO	NO
2	431.	5 - 12MM	0.	2900.	YES	NO

B E A M N O. 1681 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 48.06 KNS Vc= 122.40 KNS Vs= 0.00 KNS
Tu= 1.91 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2575. MM

AT END SUPPORT - Vu= 13.44 KNS Vc= 127.17 KNS Vs= 0.00 KNS
Tu= 1.21 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1682 DESIGN RESULTS

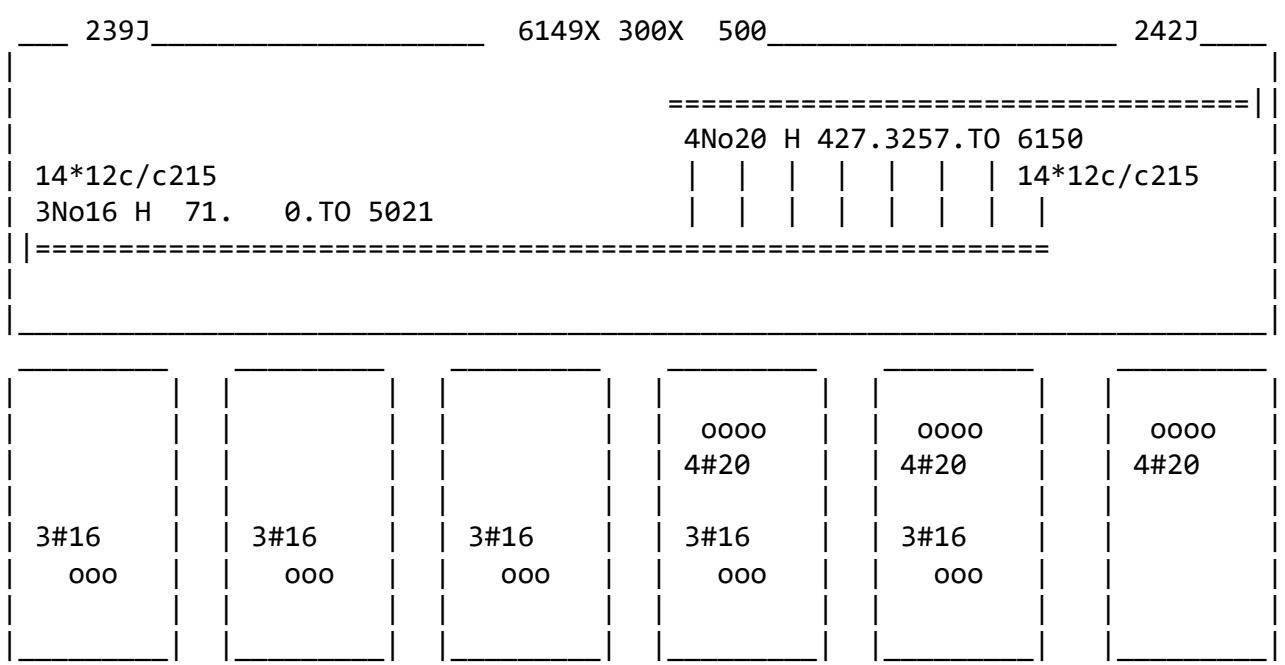
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	5021.	YES	NO
2	427.	4 - 20MM	3257.	6150.	NO	YES

B E A M N O. 1682 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 59.15 KNS Vc= 117.24 KNS Vs= 0.00 KNS
Tu= 2.15 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2650. MM

AT END SUPPORT - Vu= 73.37 KNS Vc= 125.60 KNS Vs= 0.00 KNS
Tu= 1.86 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2650. MM



ACI 318-11 BEAM NO. 1683 DESIGN RESULTS

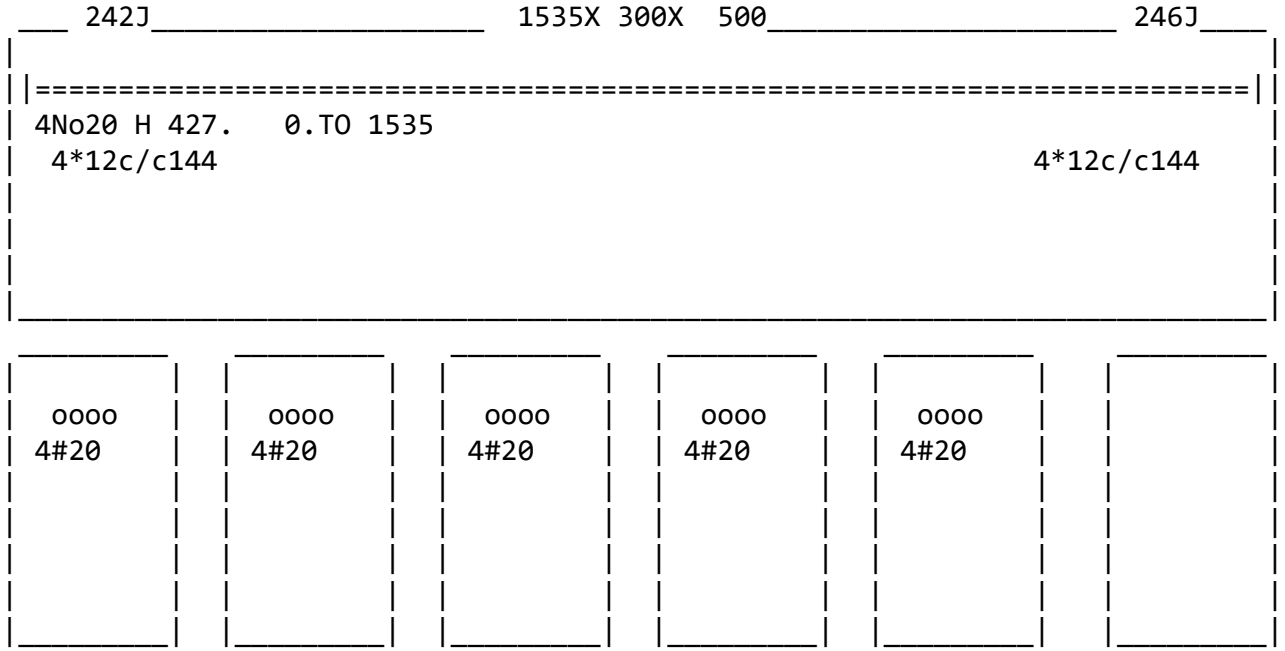
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	427.	4 - 20MM	0.	1535.	YES	YES

B E A M N O. 1683 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 115.13 KNS Vc= 120.96 KNS Vs= 32.54 KNS
Tu= 6.76 KN-MET Tc= 5.2 KN-MET Ts= 9.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 343. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.99 SQ.CM.

AT END SUPPORT - Vu= 100.39 KNS Vc= 120.96 KNS Vs= 12.89 KNS
Tu= 6.76 KN-MET Tc= 5.4 KN-MET Ts= 9.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 343. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.99 SQ.CM.



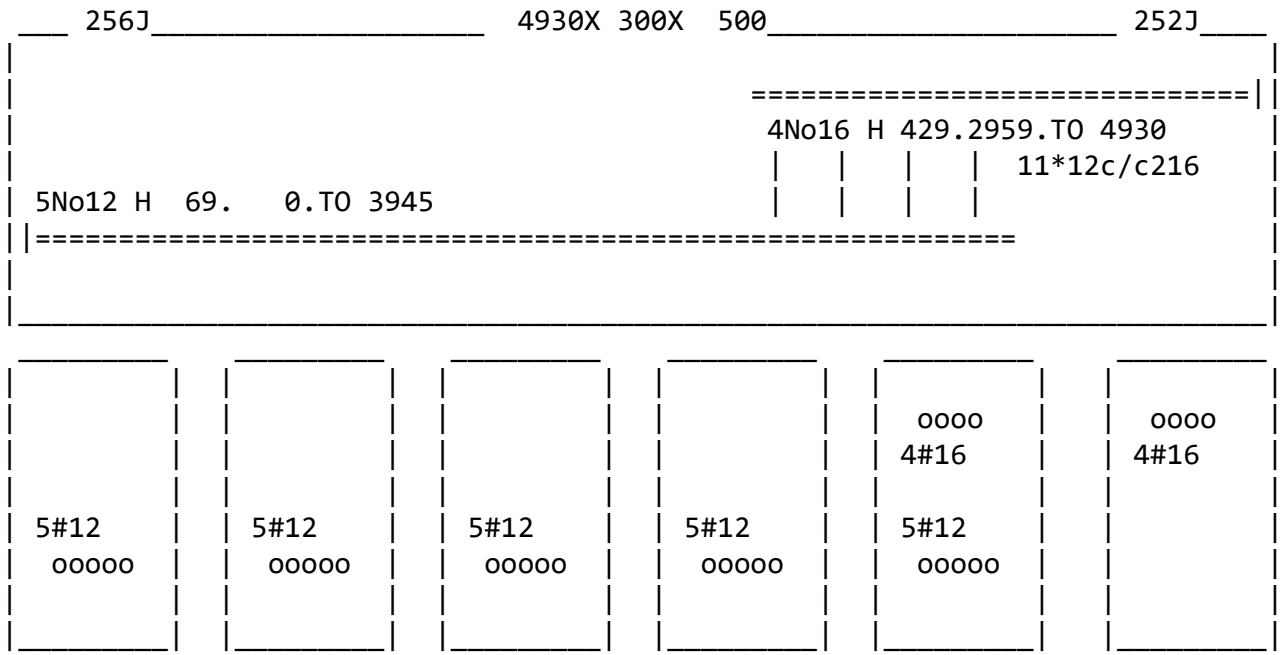
ACI 318-11 BEAM NO. 1684 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3945.	YES	NO
2	429.	4 - 16MM	2959.	4930.	NO	YES

B E A M N O. 1684 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.19 KNS Vc= 137.36 KNS Vs= 0.00 KNS
Tu= 1.79 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 50.29 KNS Vc= 126.42 KNS Vs= 0.00 KNS
Tu= 2.17 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2040. MM



ACI 318-11 BEAM NO. 1685 DESIGN RESULTS

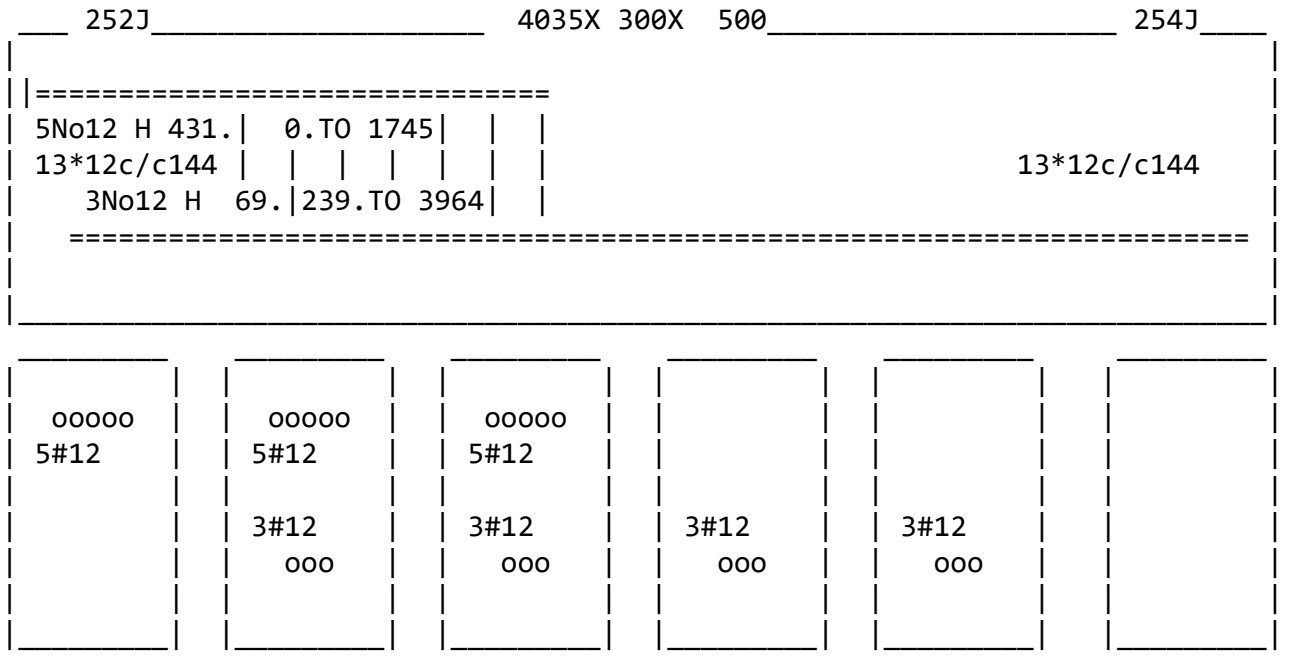
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	239.	3964.	NO	NO
2	431.	5 - 12MM	0.	1745.	YES	NO

B E A M N O. 1685 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 74.98 KNS Vc= 125.82 KNS Vs= 0.00 KNS
Tu= 5.66 KN-MET Tc= 5.2 KN-MET Ts= 7.6 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 1593. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.67 SQ.CM.

AT END SUPPORT - Vu= 41.46 KNS Vc= 126.03 KNS Vs= 0.00 KNS
Tu= 5.66 KN-MET Tc= 5.3 KN-MET Ts= 7.6 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 144. MM C/C FOR 1593. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.67 SQ.CM.



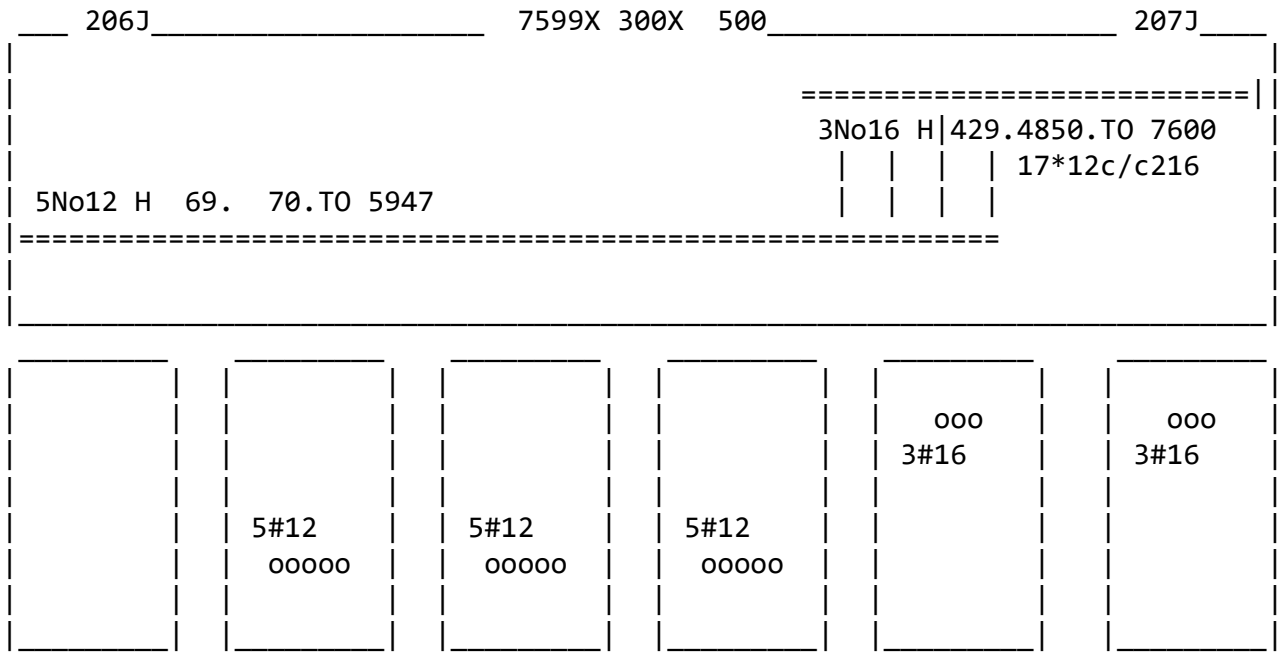
ACI 318-11 BEAM NO. 1686 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	70.	5947.	NO	NO
2	429.	3 - 16MM	4850.	7600.	NO	YES

B E A M N O. 1686 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 21.61 KNS Vc= 121.37 KNS Vs= 0.00 KNS
Tu= 0.82 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 60.72 KNS Vc= 122.44 KNS Vs= 0.00 KNS
Tu= 1.28 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 3375. MM



ACI 318-11 BEAM NO. 1687 DESIGN RESULTS

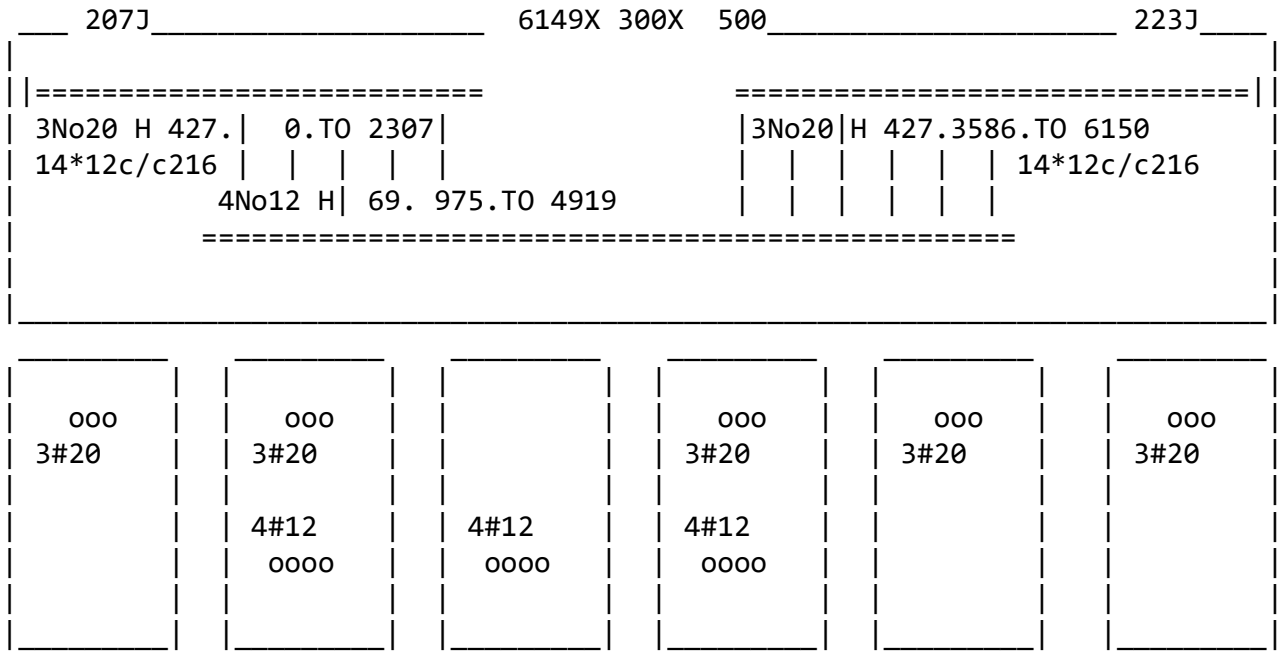
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	975.	4919.	NO	NO
2	427.	3 - 20MM	0.	2307.	YES	NO
3	427.	3 - 20MM	3586.	6150.	NO	YES

B E A M N O. 1687 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 48.71 KNS Vc= 118.26 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2650. MM

AT END SUPPORT - Vu= 55.81 KNS Vc= 125.78 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2650. MM



ACI 318-11 BEAM NO. 1688 DESIGN RESULTS

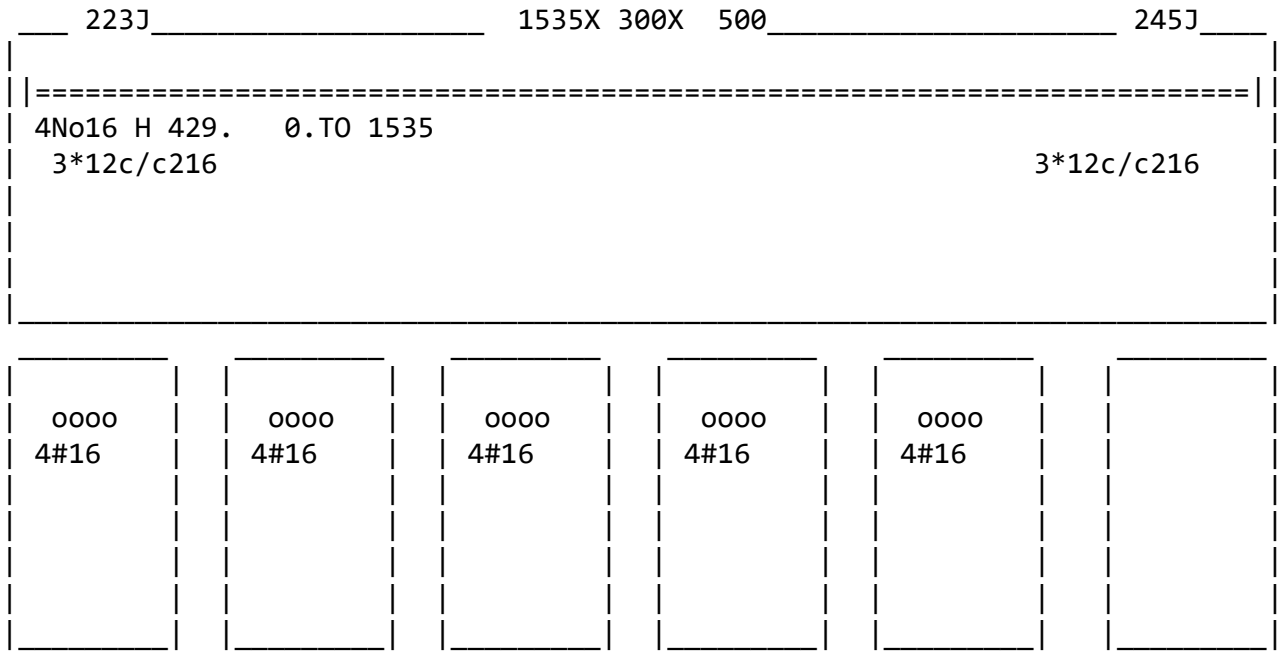
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	429.	4 - 16MM	0.	1535.	YES	YES

B E A M N O. 1688 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 53.11 KNS Vc= 120.96 KNS Vs= 0.00 KNS
Tu= 0.94 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 343. MM

AT END SUPPORT - Vu= 45.61 KNS Vc= 120.96 KNS Vs= 0.00 KNS
Tu= 0.94 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 343. MM



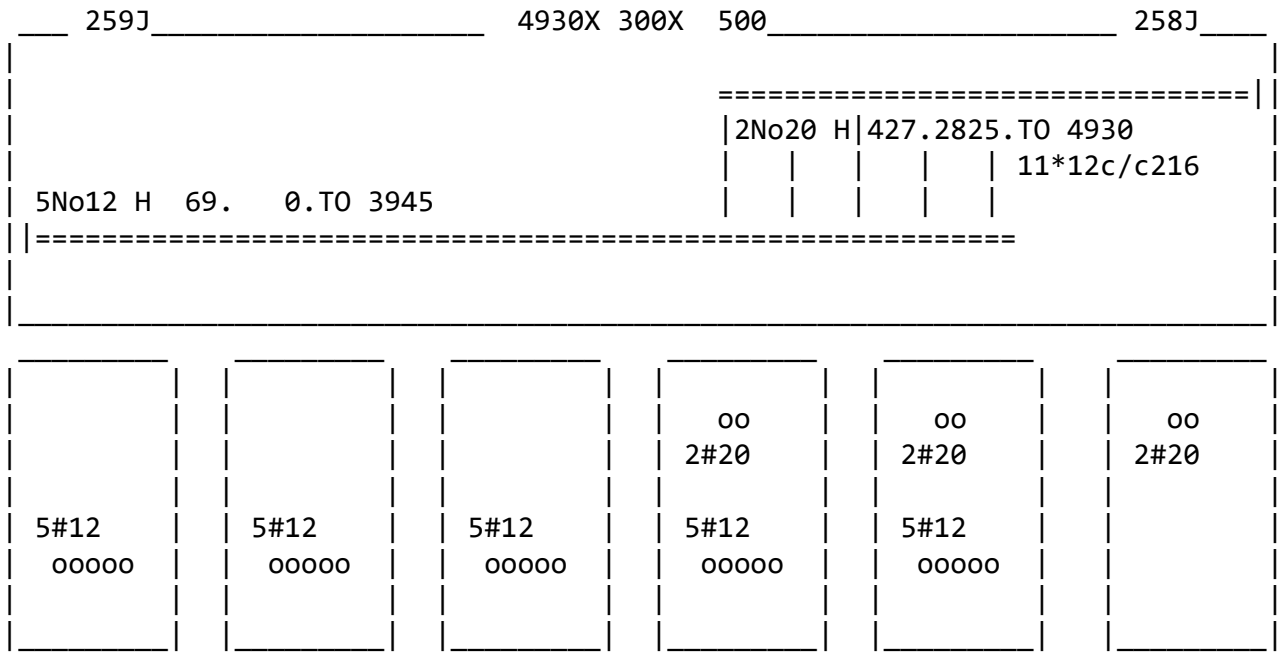
ACI 318-11 BEAM NO. 1689 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3945.	YES	NO
2	427.	2 - 20MM	2825.	4930.	NO	YES

B E A M N O. 1689 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.13 KNS Vc= 143.48 KNS Vs= 0.00 KNS
Tu= 0.57 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 49.68 KNS Vc= 126.91 KNS Vs= 0.00 KNS
Tu= 0.90 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2040. MM



ACI 318-11 BEAM NO. 1690 DESIGN RESULTS

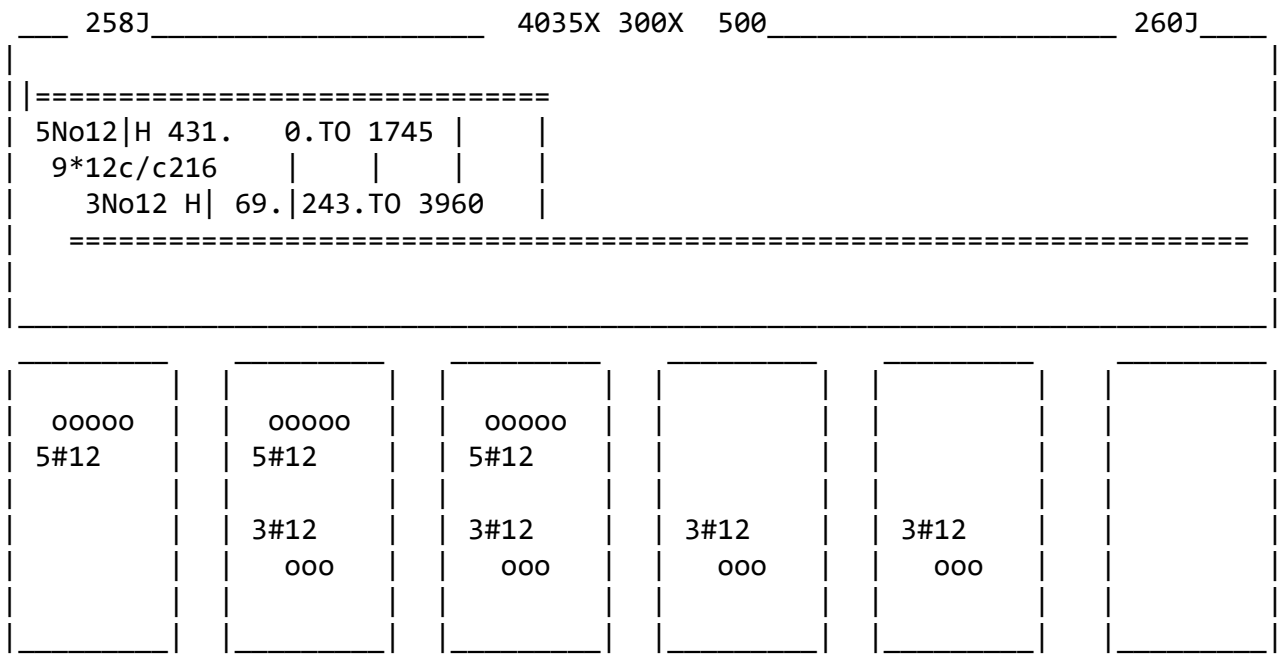
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	243.	3960.	NO	NO
2	431.	5 - 12MM	0.	1745.	YES	NO

B E A M N O. 1690 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.22 KNS Vc= 124.99 KNS Vs= 0.00 KNS
Tu= 1.78 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 1593. MM

AT END SUPPORT - Vu= 21.74 KNS Vc= 127.33 KNS Vs= 0.00 KNS
Tu= 1.20 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1691 DESIGN RESULTS

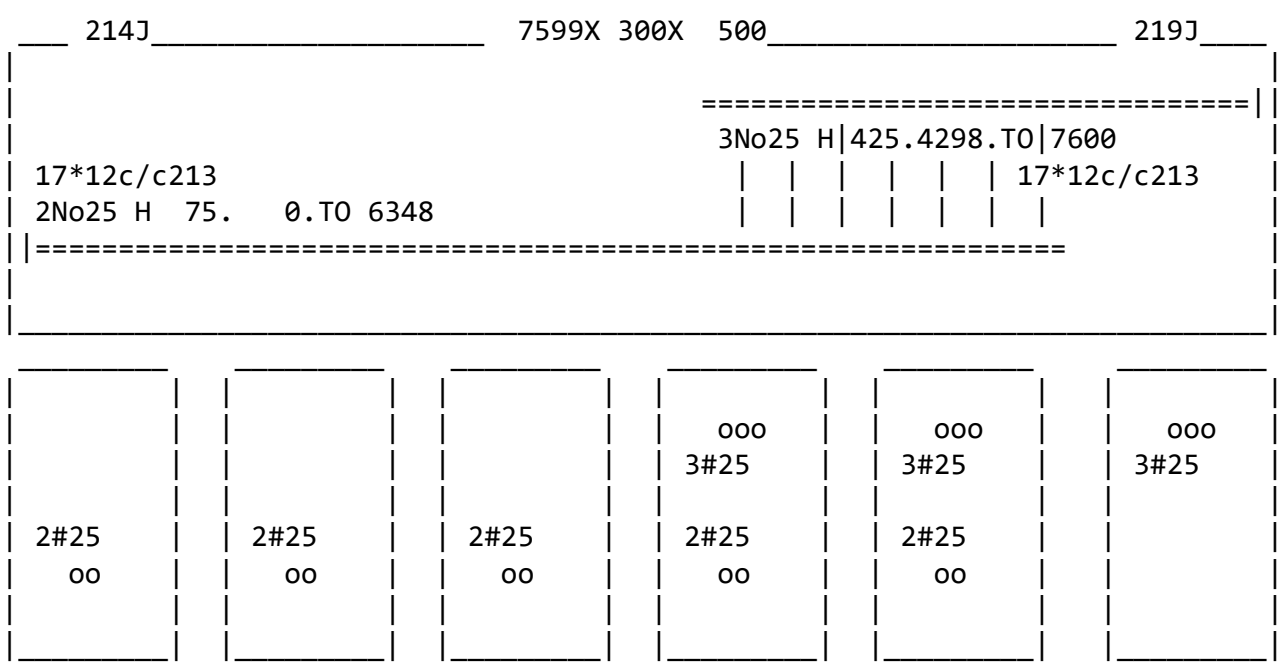
LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	6348.	YES	NO
2	425.	3 - 25MM	4298.	7600.	NO	YES

B E A M N O. 1691 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 71.75 KNS Vc= 135.42 KNS Vs= 0.00 KNS
Tu= 0.52 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 213. MM C/C FOR 3375. MM

AT END SUPPORT - Vu= 135.21 KNS Vc= 126.37 KNS Vs= 53.90 KNS
Tu= 0.17 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 213. MM C/C FOR 3375. MM



ACI 318-11 BEAM NO. 1692 DESIGN RESULTS

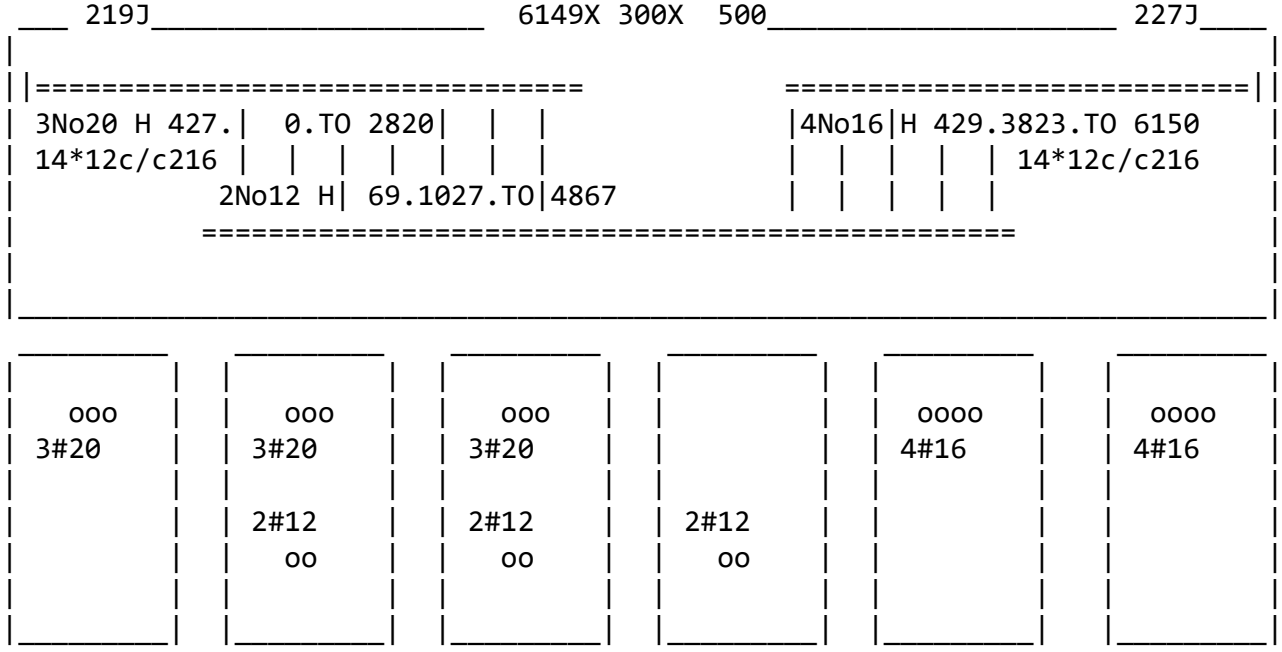
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	1027.	4867.	NO	NO
2	427.	3 - 20MM	0.	2820.	YES	NO
3	429.	4 - 16MM	3823.	6150.	NO	YES

B E A M N O. 1692 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 47.28 KNS Vc= 122.24 KNS Vs= 0.00 KNS
Tu= 1.39 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2650. MM

AT END SUPPORT - Vu= 51.98 KNS Vc= 122.59 KNS Vs= 0.00 KNS
Tu= 1.78 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2650. MM



ACI 318-11 BEAM NO. 1693 DESIGN RESULTS

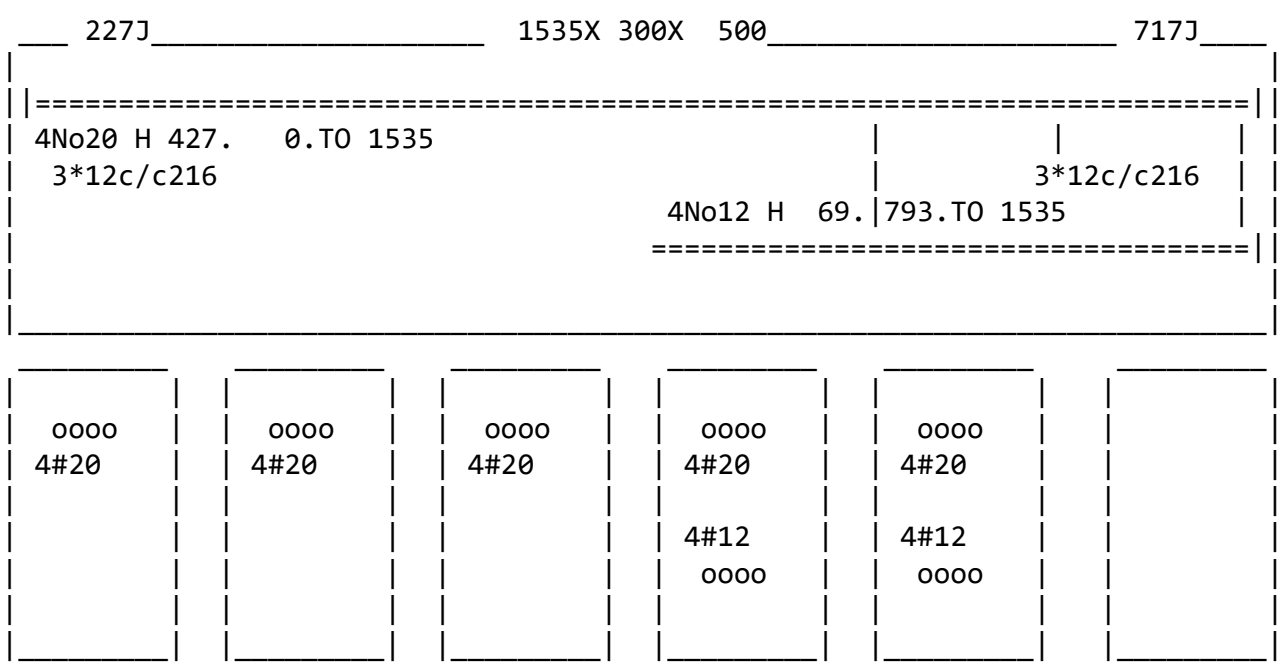
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	793.	1535.	NO	YES
2	427.	4 - 20MM	0.	1535.	YES	YES

B E A M N O. 1693 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 148.67 KNS Vc= 126.21 KNS Vs= 72.02 KNS
Tu= 0.41 KN-MET Tc= 5.6 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 343. MM

AT END SUPPORT - Vu= 87.81 KNS Vc= 232.25 KNS Vs= 0.00 KNS
Tu= 0.39 KN-MET Tc= 5.6 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 343. MM



ACI 318-11 BEAM NO. 1694 DESIGN RESULTS

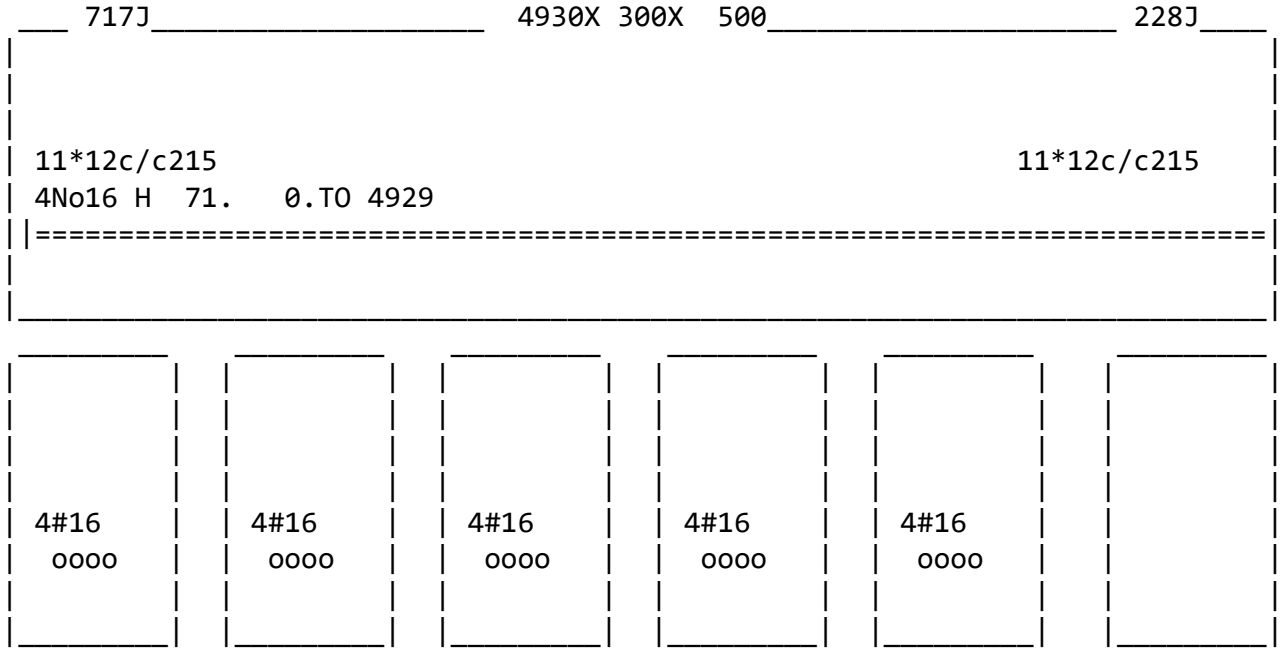
LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	4929.	YES	NO

B E A M N O. 1694 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 50.64 KNS Vc= 126.10 KNS Vs= 0.00 KNS
Tu= 0.74 KN-MET Tc= 5.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2040. MM

AT END SUPPORT - Vu= 70.11 KNS Vc= 140.15 KNS Vs= 0.00 KNS
Tu= 0.74 KN-MET Tc= 5.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2040. MM



ACI 318-11 BEAM NO. 1695 DESIGN RESULTS

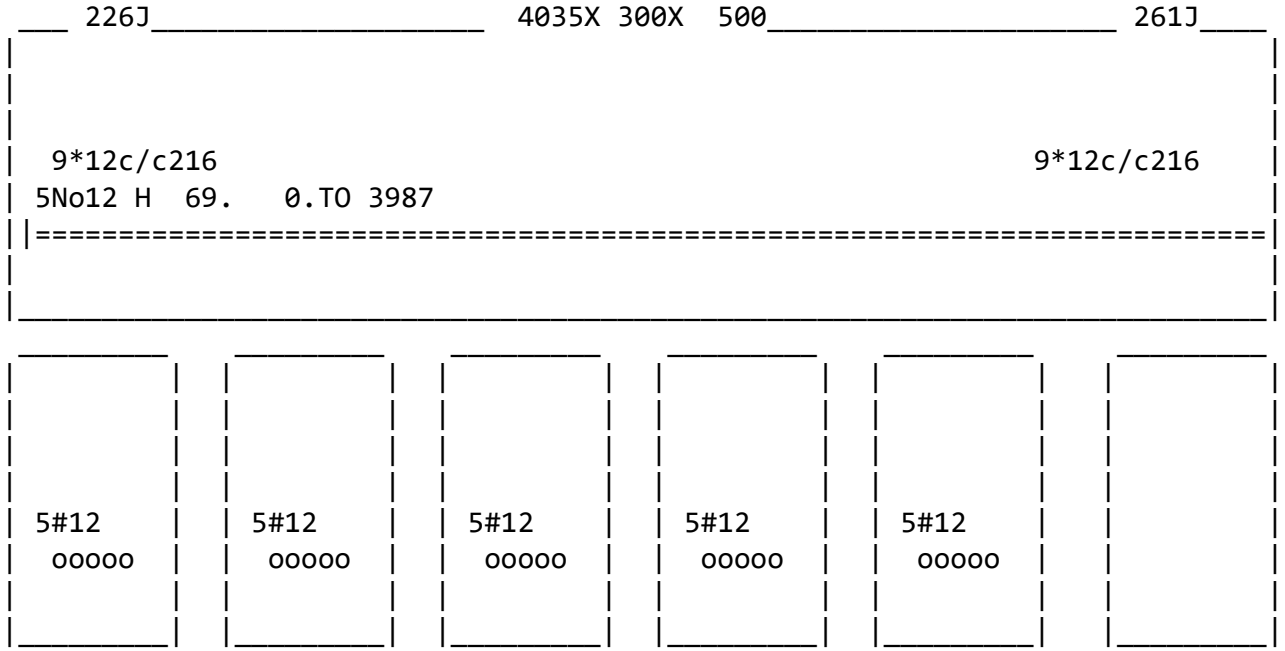
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3987.	YES	NO

B E A M N O. 1695 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 56.89 KNS Vc= 134.04 KNS Vs= 0.00 KNS
Tu= 0.62 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 1593. MM

AT END SUPPORT - Vu= 57.14 KNS Vc= 131.40 KNS Vs= 0.00 KNS
Tu= 0.62 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 1593. MM



ACI 318-11 BEAM NO. 1696 DESIGN RESULTS

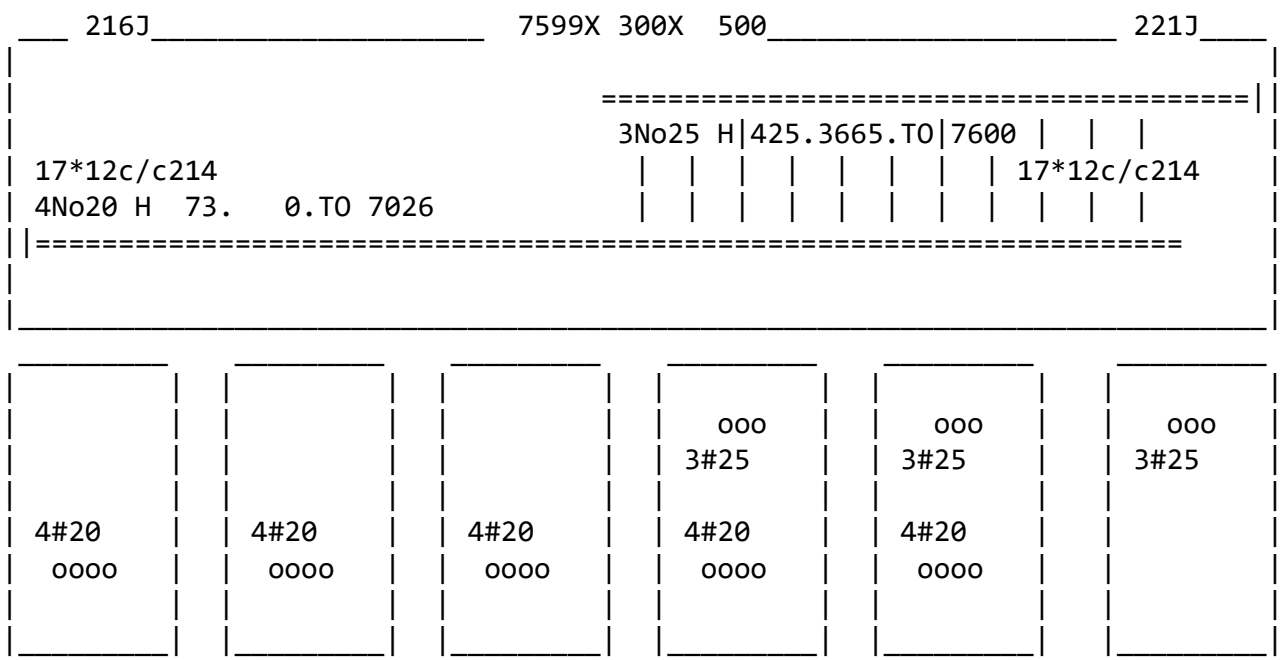
LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	7026.	YES	NO
2	425.	3 - 25MM	3665.	7600.	NO	YES

B E A M N O. 1696 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.63 KNS Vc= 137.23 KNS Vs= 0.00 KNS
Tu= 2.05 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 3375. MM

AT END SUPPORT - Vu= 145.93 KNS Vc= 125.68 KNS Vs= 68.90 KNS
Tu= 1.68 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 3375. MM



ACI 318-11 BEAM NO. 1697 DESIGN RESULTS

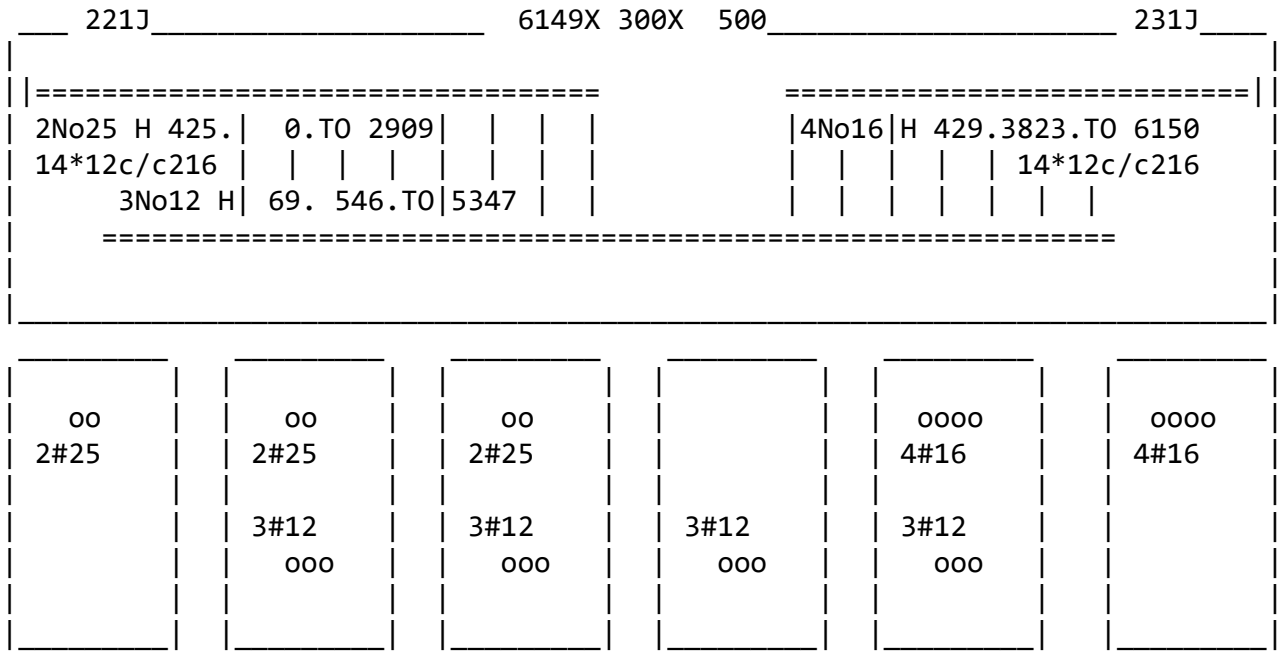
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	546.	5347.	NO	NO
2	425.	2 - 25MM	0.	2909.	YES	NO
3	429.	4 - 16MM	3823.	6150.	NO	YES

B E A M N O. 1697 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 50.01 KNS Vc= 122.73 KNS Vs= 0.00 KNS
Tu= 2.16 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2650. MM

AT END SUPPORT - Vu= 54.48 KNS Vc= 123.24 KNS Vs= 0.00 KNS
Tu= 2.63 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2650. MM



ACI 318-11 BEAM NO. 1698 DESIGN RESULTS

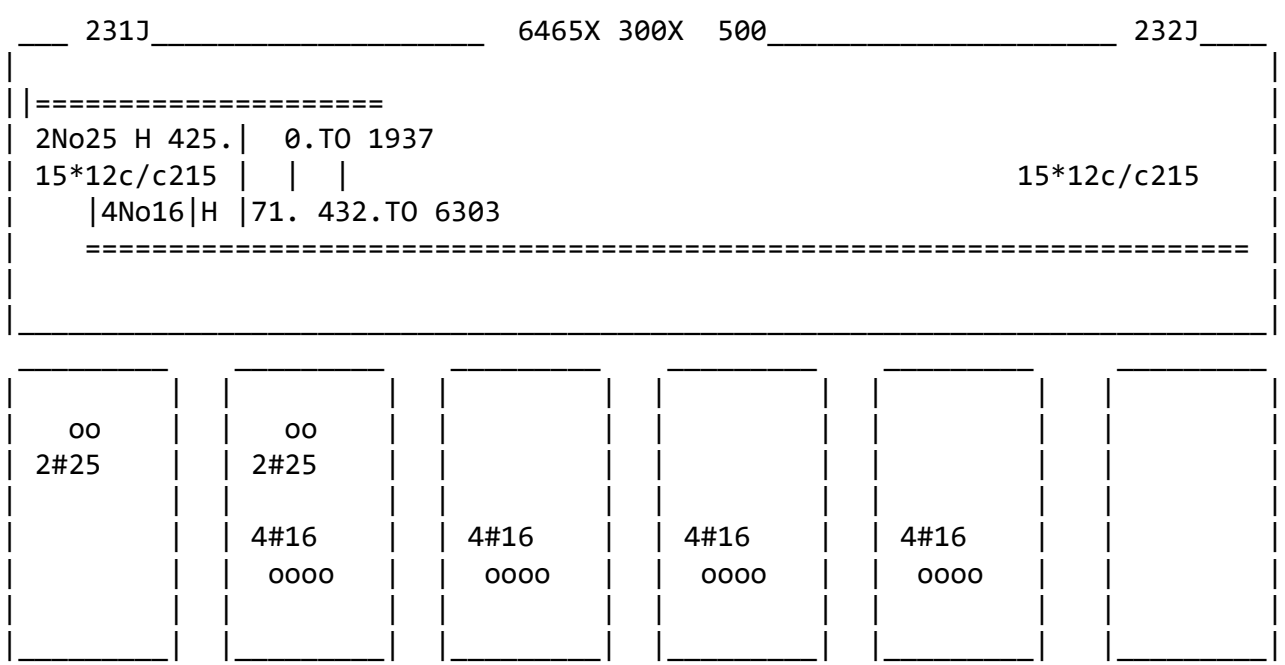
LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	432.	6303.	NO	NO
2	425.	2 - 25MM	0.	1937.	YES	NO

B E A M N O. 1698 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 62.40 KNS Vc= 128.96 KNS Vs= 0.00 KNS
Tu= 0.98 KN-MET Tc= 5.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2808. MM

AT END SUPPORT - Vu= 65.94 KNS Vc= 136.83 KNS Vs= 0.00 KNS
Tu= 1.83 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2808. MM



ACI 318-11 BEAM NO. 1699 DESIGN RESULTS

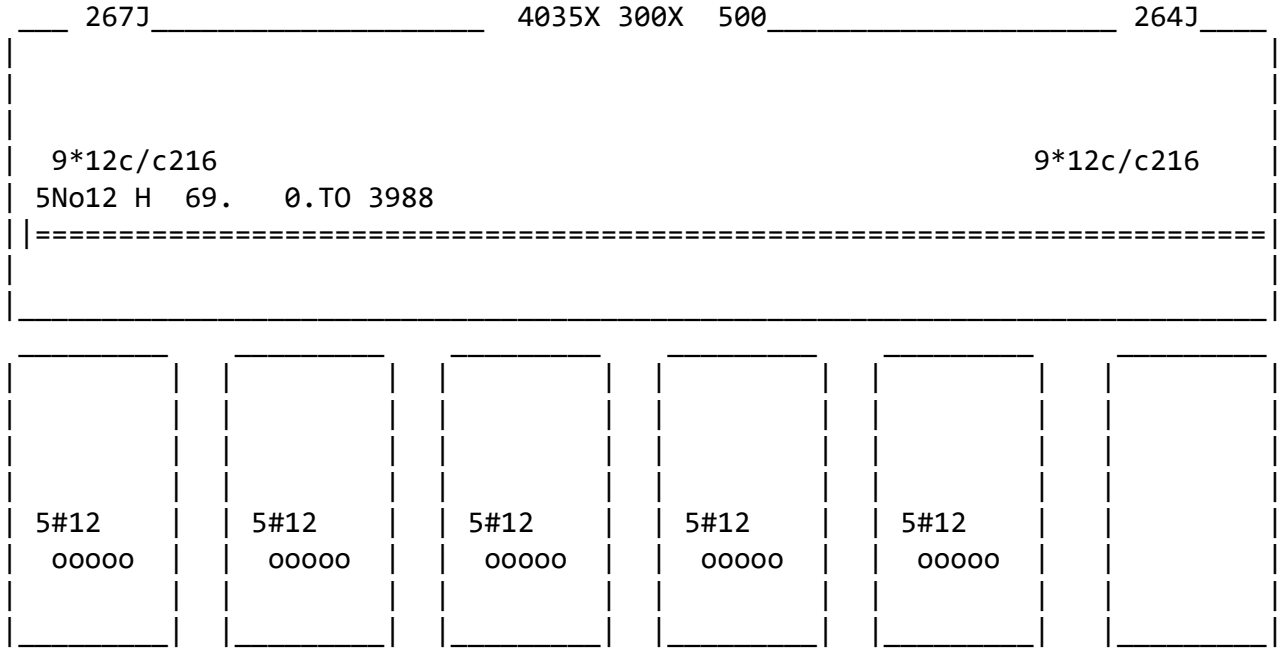
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3988.	YES	NO

B E A M N O. 1699 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 57.11 KNS Vc= 133.10 KNS Vs= 0.00 KNS
Tu= 2.90 KN-MET Tc= 5.5 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 1593. MM

AT END SUPPORT - Vu= 57.35 KNS Vc= 130.81 KNS Vs= 0.00 KNS
Tu= 2.90 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 1593. MM



ACI 318-11 BEAM NO. 1801 DESIGN RESULTS

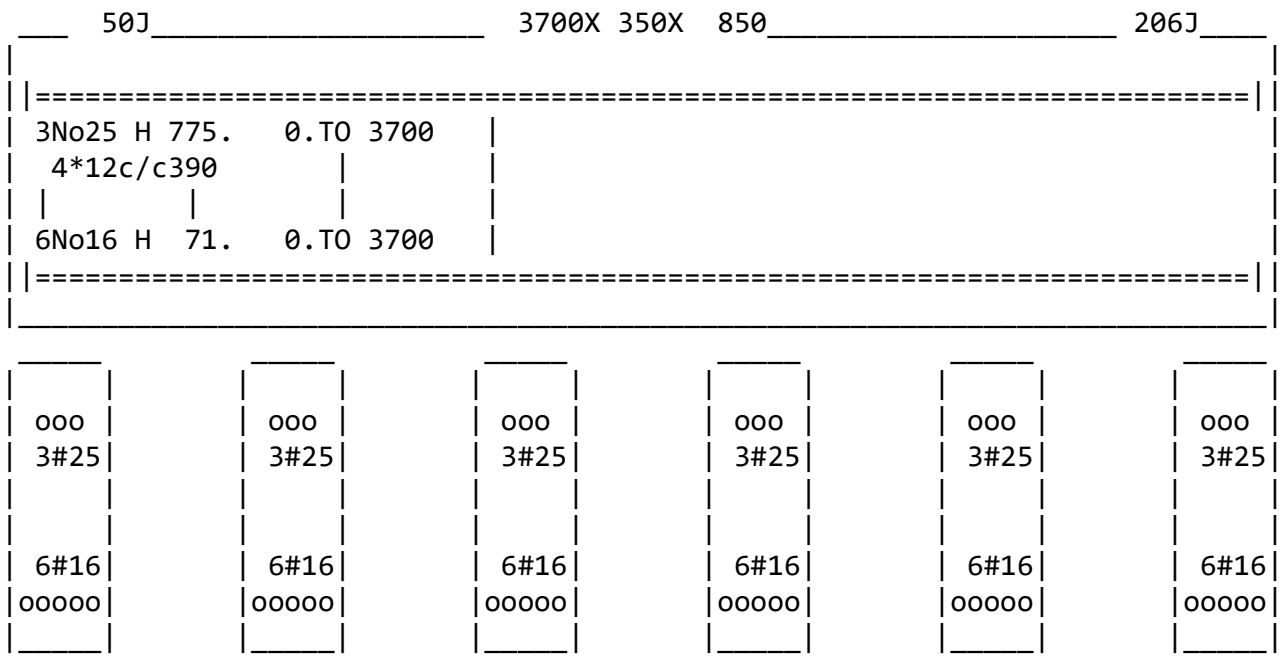
LEN - 3700. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	3700.	YES	YES
2	775.	3 - 25MM	0.	3700.	YES	YES

B E A M N O. 1801 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 137.80 KNS Vc= 262.82 KNS Vs= 0.00 KNS
Tu= 1.64 KN-MET Tc= 13.7 KN-MET Ts= 0.0 KN-MET LOAD 408
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 1075. MM

AT END SUPPORT - Vu= 28.80 KNS Vc= 259.28 KNS Vs= 0.00 KNS
Tu= 1.64 KN-MET Tc= 13.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1802 DESIGN RESULTS

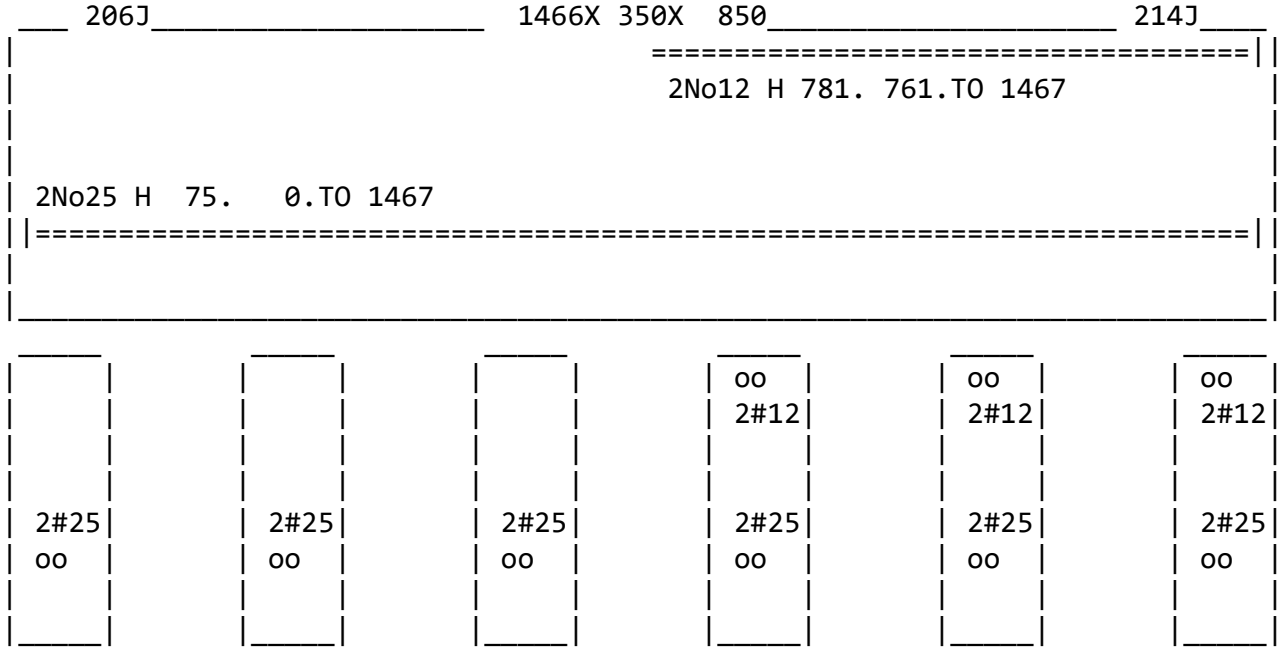
LEN - 1467. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1467.	YES	YES
2	781.	2 - 12MM	761.	1467.	NO	YES

B E A M N O. 1802 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1802 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1802 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1803 DESIGN RESULTS

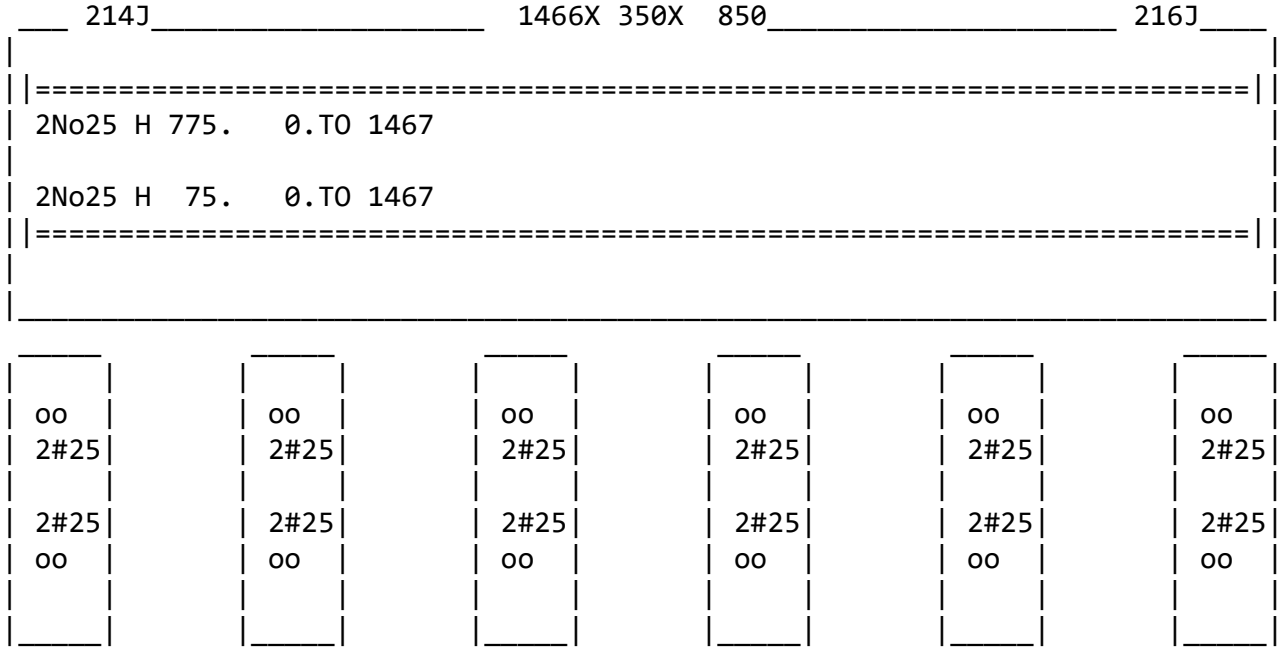
LEN - 1467. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1467.	YES	YES
2	775.	2 - 25MM	0.	1467.	YES	YES

B E A M N O. 1803 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1803 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1803 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1804 DESIGN RESULTS

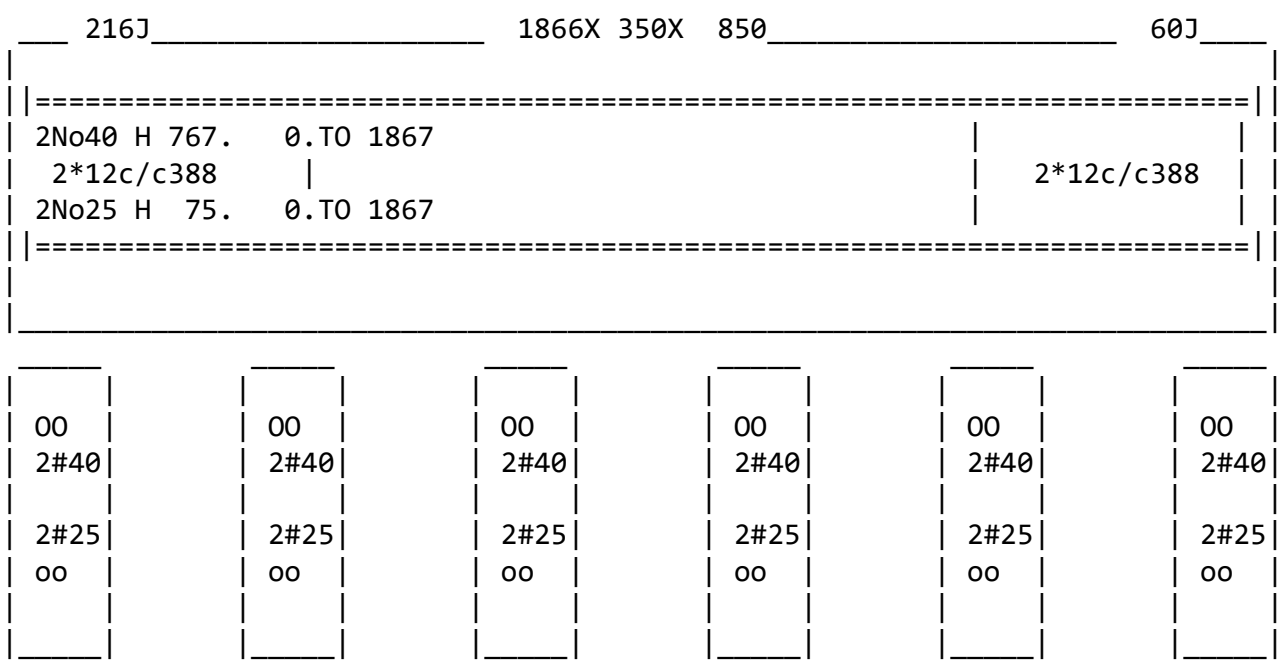
LEN - 1867. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1867.	YES	YES
2	767.	2 - 40MM	0.	1867.	YES	YES

B E A M N O. 1804 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 155.59 KNS Vc= 268.57 KNS Vs= 0.00 KNS
Tu= 1.64 KN-MET Tc= 13.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 159. MM

AT END SUPPORT - Vu= 159.77 KNS Vc= 264.63 KNS Vs= 0.00 KNS
Tu= 1.64 KN-MET Tc= 13.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 159. MM



ACI 318-11 BEAM NO. 1805 DESIGN RESULTS

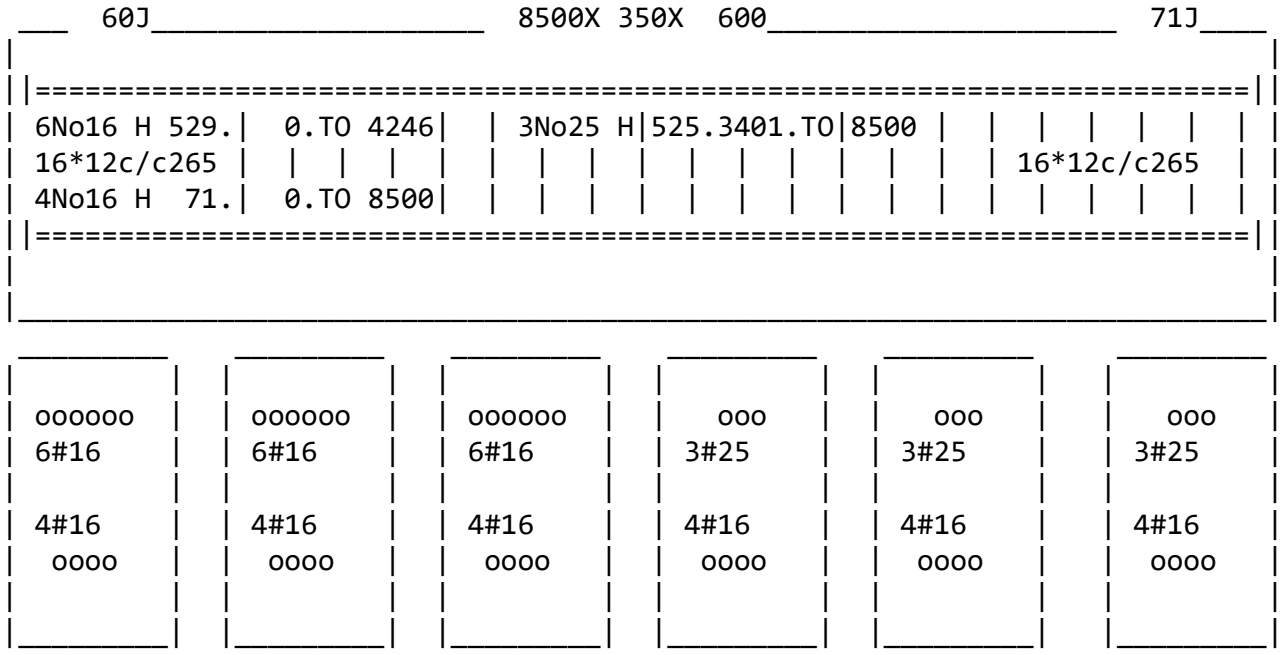
LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	8500.	YES	YES
2	529.	6 - 16MM	0.	4246.	YES	NO
3	525.	3 - 25MM	3401.	8500.	NO	YES

B E A M N O. 1805 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 70.16 KNS Vc= 177.62 KNS Vs= 0.00 KNS
Tu= 0.65 KN-MET Tc= 8.4 KN-MET Ts= 0.0 KN-MET LOAD 408
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 3725. MM

AT END SUPPORT - Vu= 82.84 KNS Vc= 175.70 KNS Vs= 0.00 KNS
Tu= 0.82 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 3725. MM



ACI 318-11 BEAM NO. 1806 DESIGN RESULTS

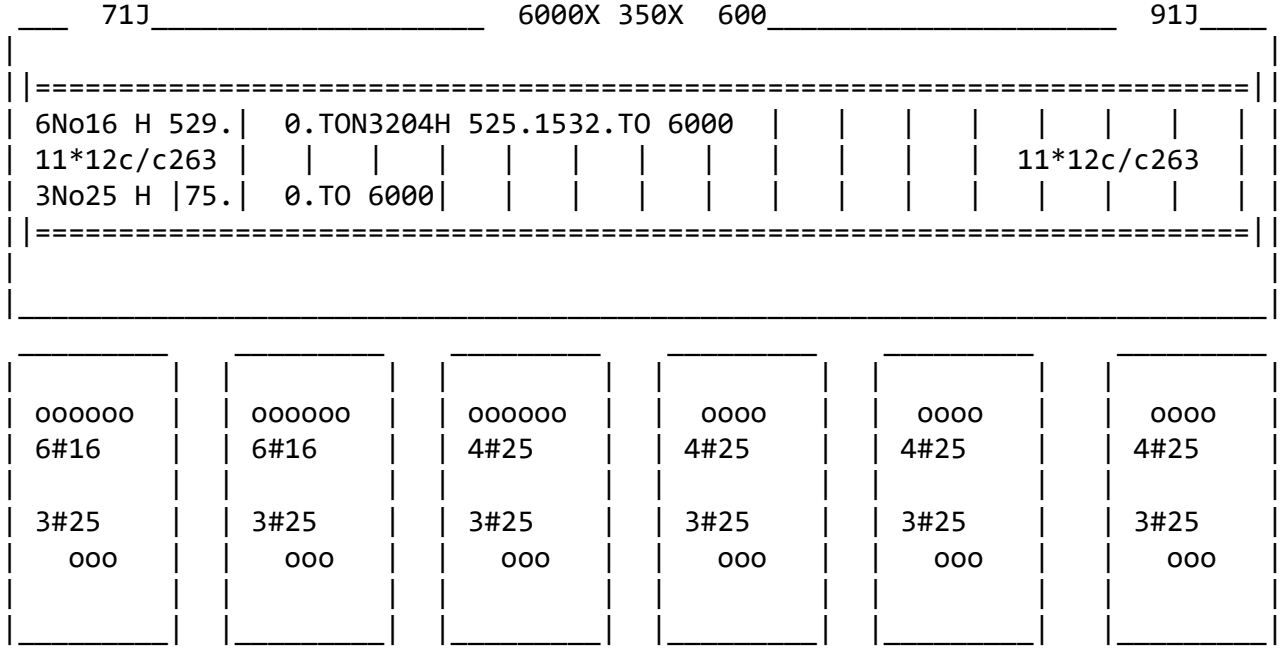
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	6000.	YES	YES
2	529.	6 - 16MM	0.	3204.	YES	NO
3	525.	4 - 25MM	1532.	6000.	NO	YES

B E A M N O. 1806 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 87.66 KNS Vc= 177.87 KNS Vs= 0.00 KNS
Tu= 1.06 KN-MET Tc= 8.6 KN-MET Ts= 0.0 KN-MET LOAD 408
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 263. MM C/C FOR 2475. MM

AT END SUPPORT - Vu= 115.64 KNS Vc= 177.34 KNS Vs= 0.00 KNS
Tu= 0.59 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 263. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 1807 DESIGN RESULTS

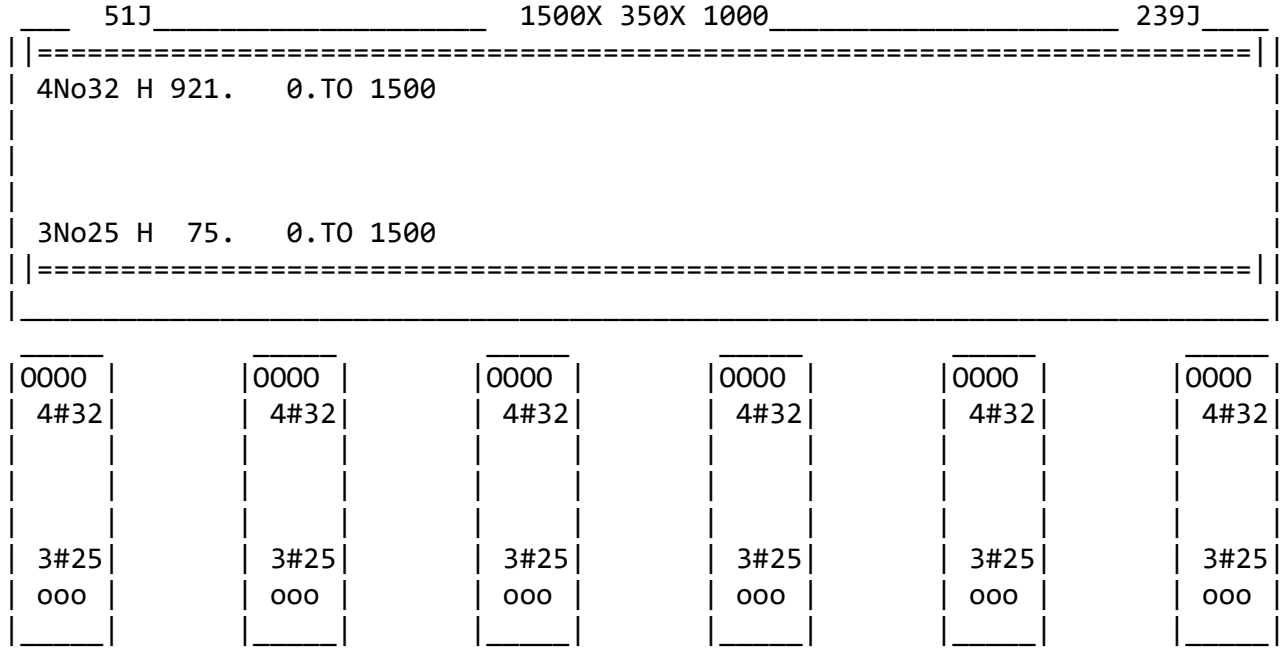
LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1500.	YES	YES
2	921.	4 - 32MM	0.	1500.	YES	YES

B E A M N O. 1807 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1807 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1807 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1808 DESIGN RESULTS

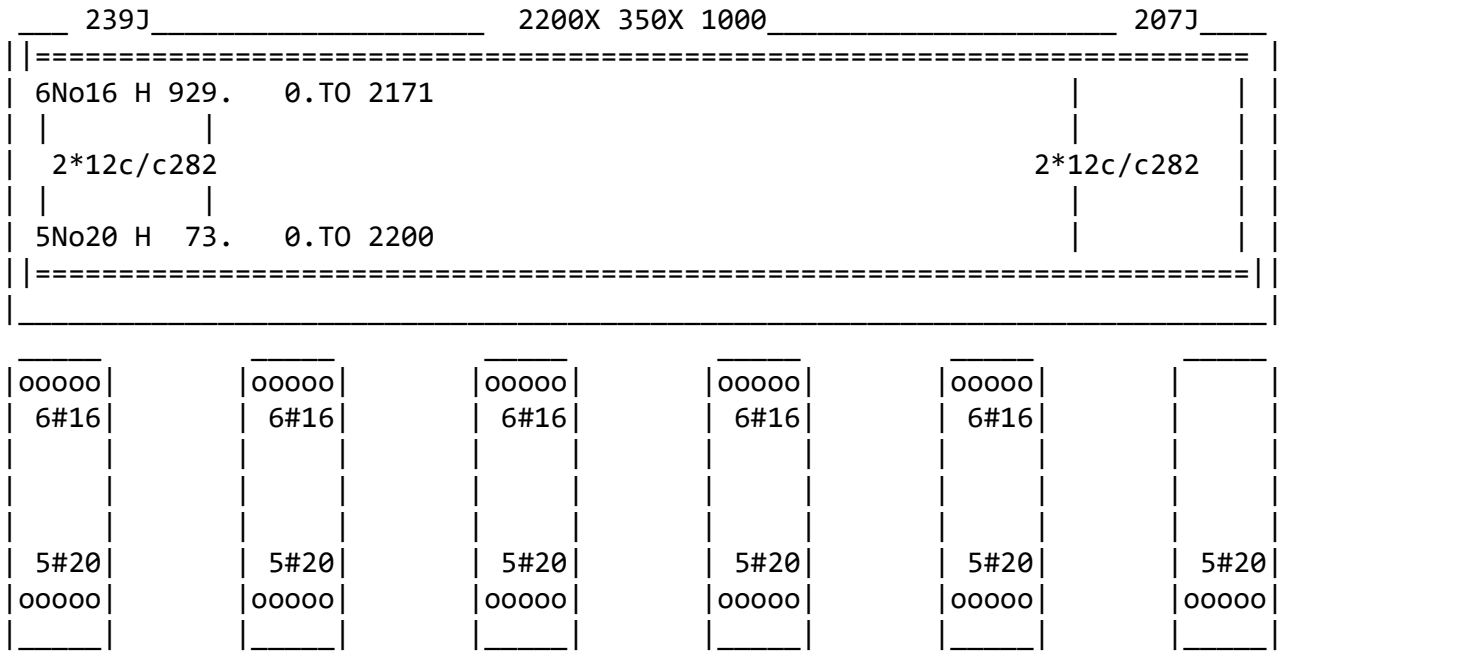
LEN - 2200. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	5 - 20MM	0.	2200.	YES	YES
2	929.	6 - 16MM	0.	2171.	YES	NO

B E A M N O. 1808 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 129.36 KNS Vc= 309.39 KNS Vs= 0.00 KNS
Tu= 24.81 KN-MET Tc= 16.3 KN-MET Ts= 33.1 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 282. MM C/C FOR 175. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.94 SQ.CM.

AT END SUPPORT - Vu= 124.04 KNS Vc= 309.39 KNS Vs= 0.00 KNS
Tu= 24.81 KN-MET Tc= 16.3 KN-MET Ts= 33.1 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 282. MM C/C FOR 175. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.94 SQ.CM.



ACI 318-11 BEAM NO. 1809 DESIGN RESULTS

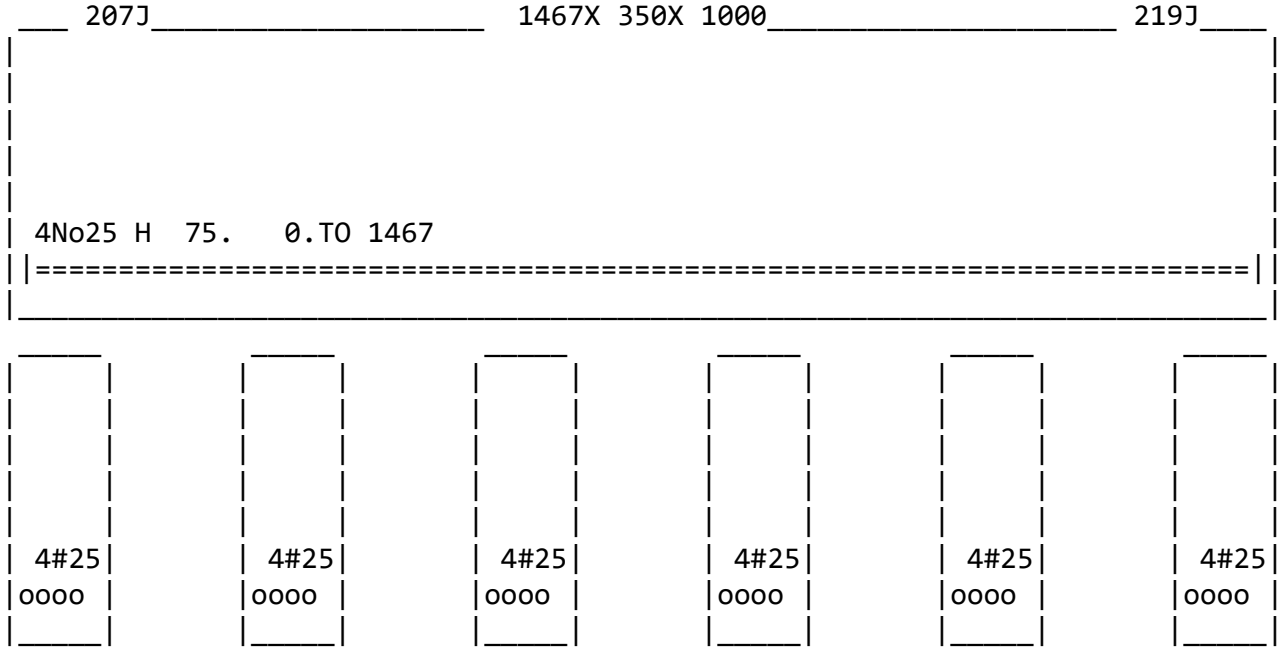
LEN - 1467. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	1467.	YES	YES

B E A M N O. 1809 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1809 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1809 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1810 DESIGN RESULTS

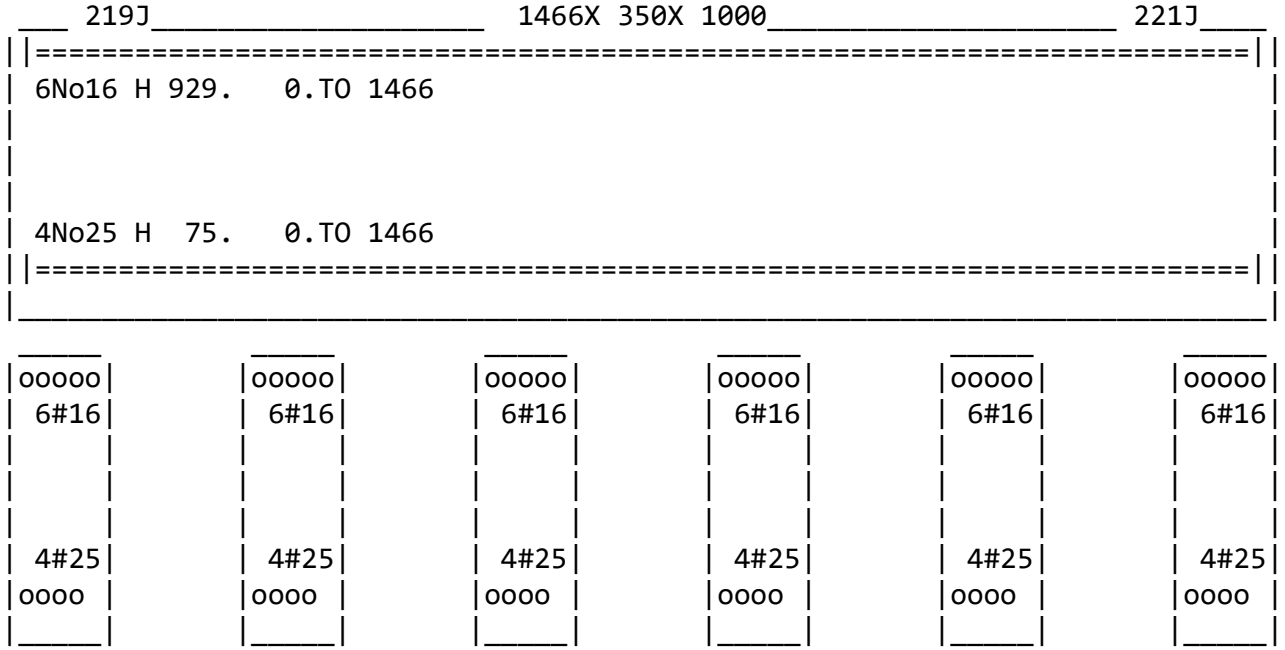
LEN - 1466. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	1466.	YES	YES
2	929.	6 - 16MM	0.	1466.	YES	YES

B E A M N O. 1810 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1810 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1810 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1811 DESIGN RESULTS

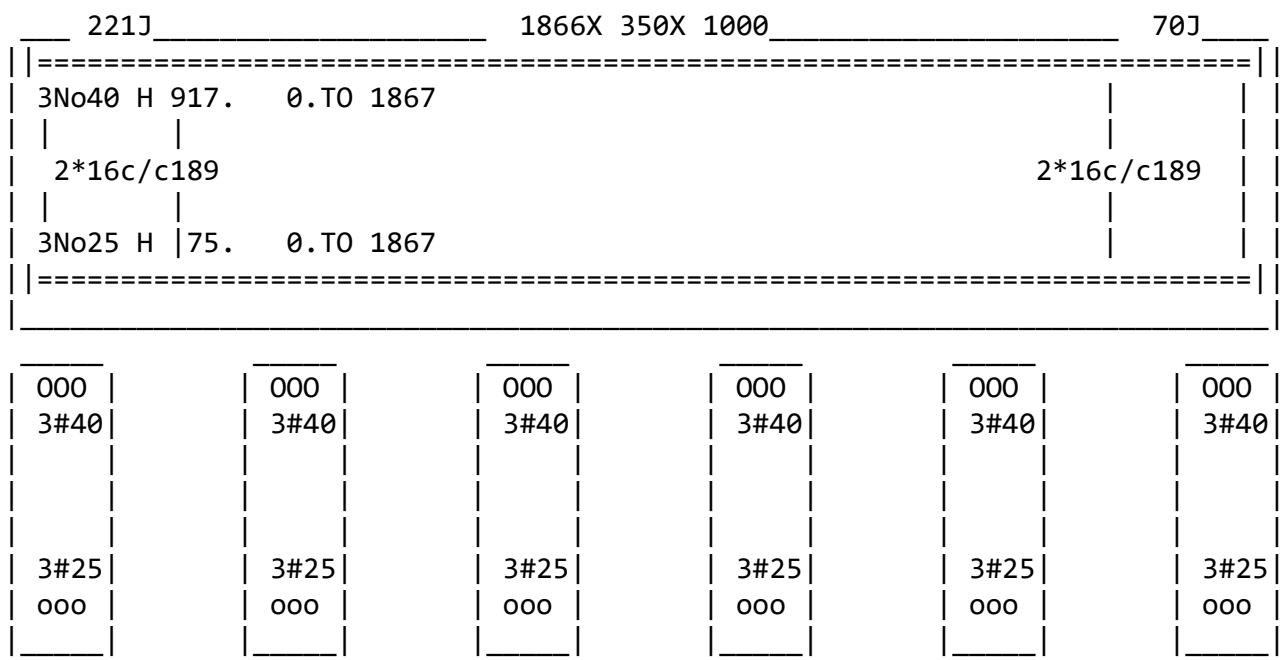
LEN - 1867. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1867.	YES	YES
2	917.	3 - 40MM	0.	1867.	YES	YES

B E A M N O. 1811 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 344.48 KNS Vc= 310.62 KNS Vs= 148.68 KNS
Tu= 98.83 KN-MET Tc= 16.4 KN-MET Ts= 131.8 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 16 MM 2-LEGGED STIRRUPS AT 189. MM C/C FOR 9. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 19.67 SQ.CM.

AT END SUPPORT - Vu= 344.81 KNS Vc= 310.62 KNS Vs= 149.12 KNS
Tu= 98.83 KN-MET Tc= 16.4 KN-MET Ts= 131.8 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 16 MM 2-LEGGED STIRRUPS AT 189. MM C/C FOR 9. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 19.67 SQ.CM.



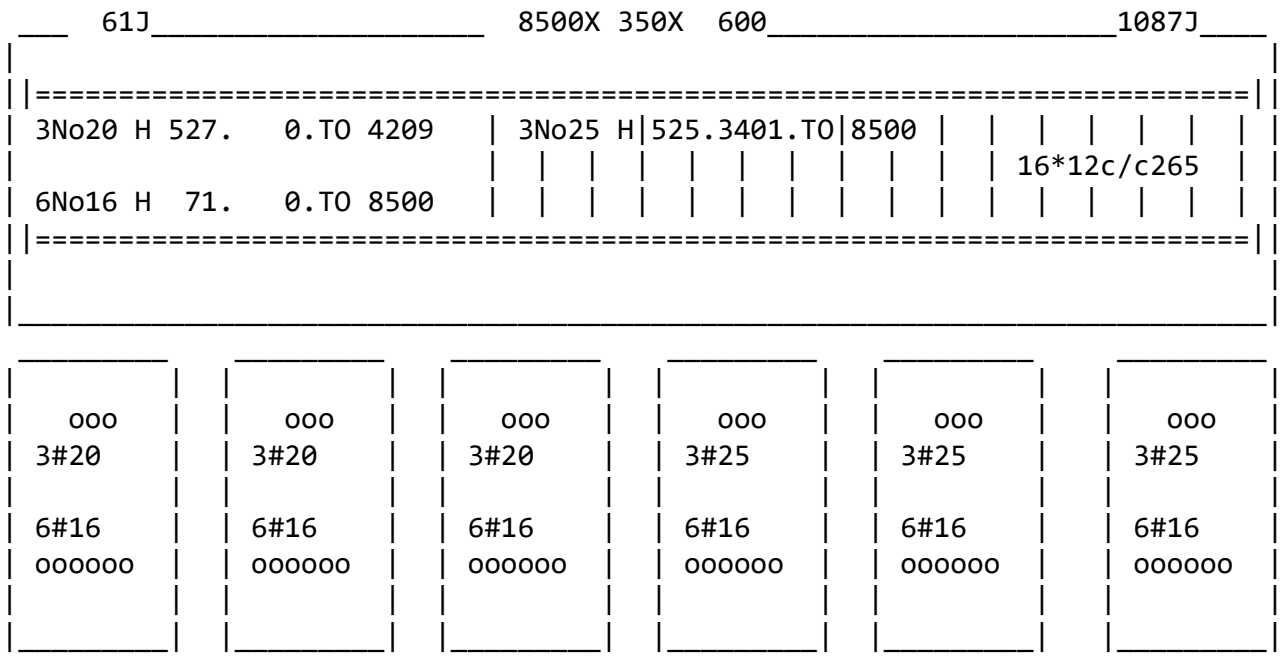
ACI 318-11 BEAM NO. 1812 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	8500.	YES	YES
2	527.	3 - 20MM	0.	4209.	YES	NO
3	525.	3 - 25MM	3401.	8500.	NO	YES

B E A M N O. 1812 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.05 KNS Vc= 174.42 KNS Vs= 0.00 KNS
Tu= 0.19 KN-MET Tc= 8.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 76.63 KNS Vc= 158.41 KNS Vs= 0.00 KNS
Tu= 0.27 KN-MET Tc= 7.5 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 3725. MM



ACI 318-11 BEAM NO. 1813 DESIGN RESULTS

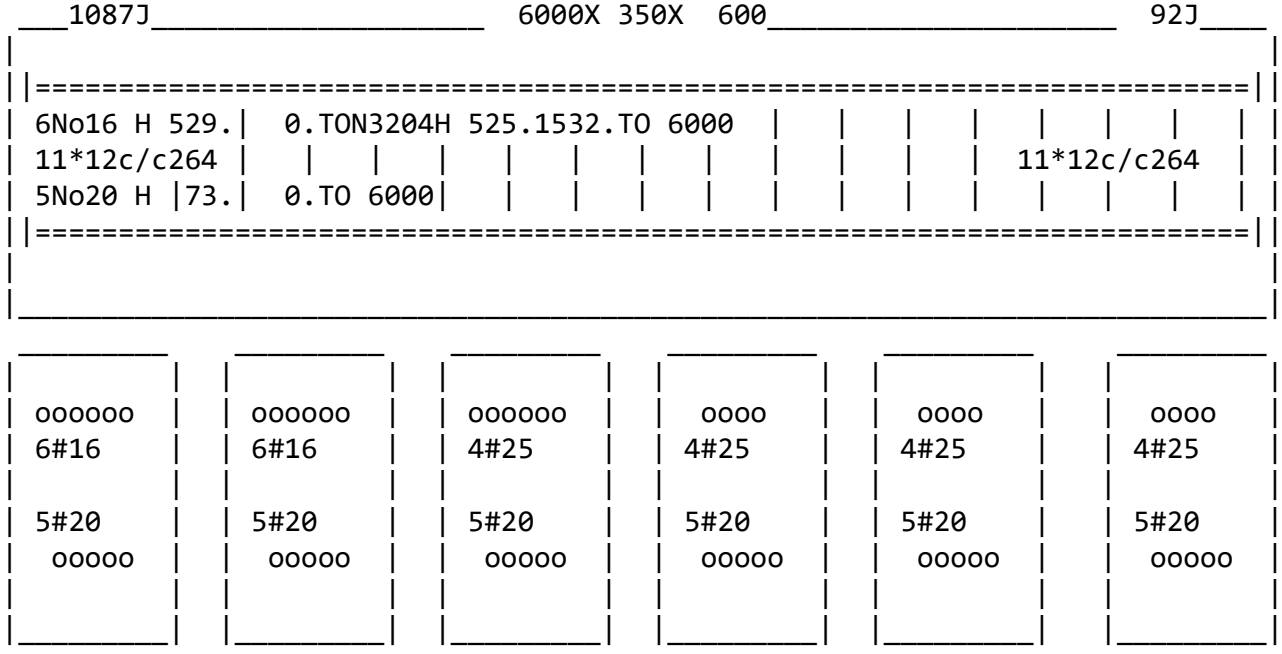
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	5 - 20MM	0.	6000.	YES	YES
2	529.	6 - 16MM	0.	3204.	YES	NO
3	525.	4 - 25MM	1532.	6000.	NO	YES

B E A M N O. 1813 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 87.54 KNS Vc= 178.50 KNS Vs= 0.00 KNS
Tu= 0.27 KN-MET Tc= 8.4 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 2475. MM

AT END SUPPORT - Vu= 116.94 KNS Vc= 178.50 KNS Vs= 0.00 KNS
Tu= 0.27 KN-MET Tc= 8.4 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 1814 DESIGN RESULTS

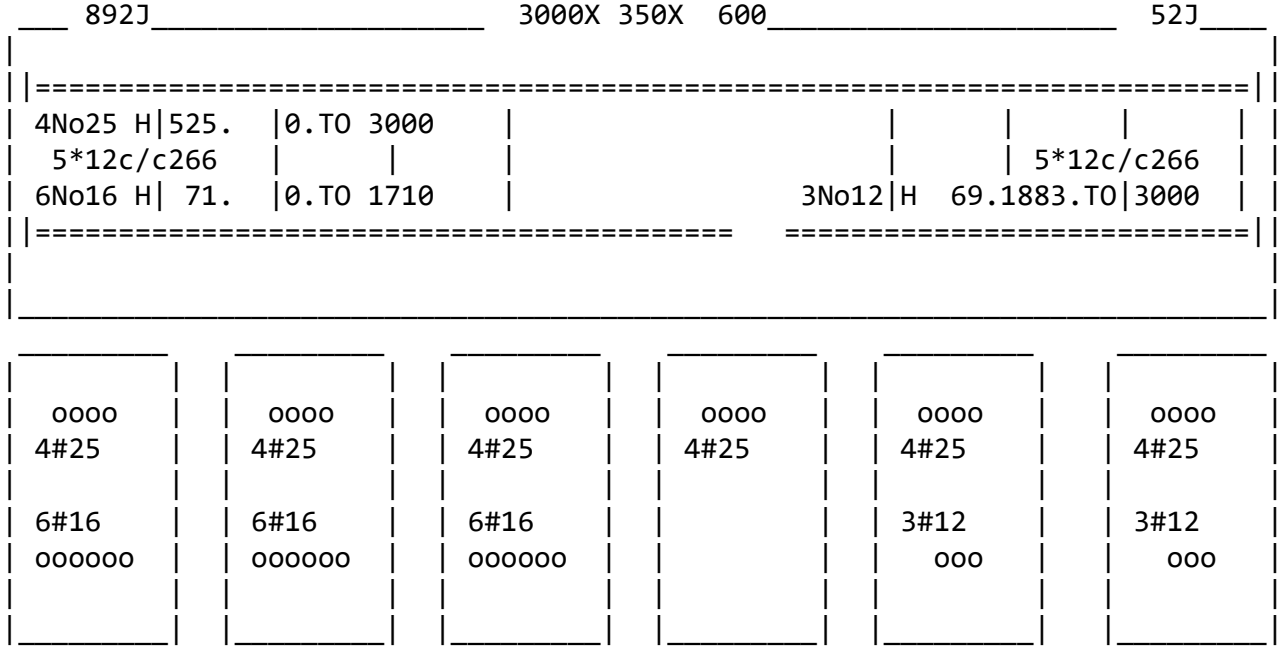
LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	1710.	YES	NO
2	69.	3 - 12MM	1883.	3000.	NO	YES
3	525.	4 - 25MM	0.	3000.	YES	YES

B E A M N O. 1814 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 94.11 KNS Vc= 178.69 KNS Vs= 0.00 KNS
Tu= 0.18 KN-MET Tc= 8.4 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 975. MM

AT END SUPPORT - Vu= 84.23 KNS Vc= 179.95 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 8.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 975. MM



ACI 318-11 BEAM NO. 1815 DESIGN RESULTS

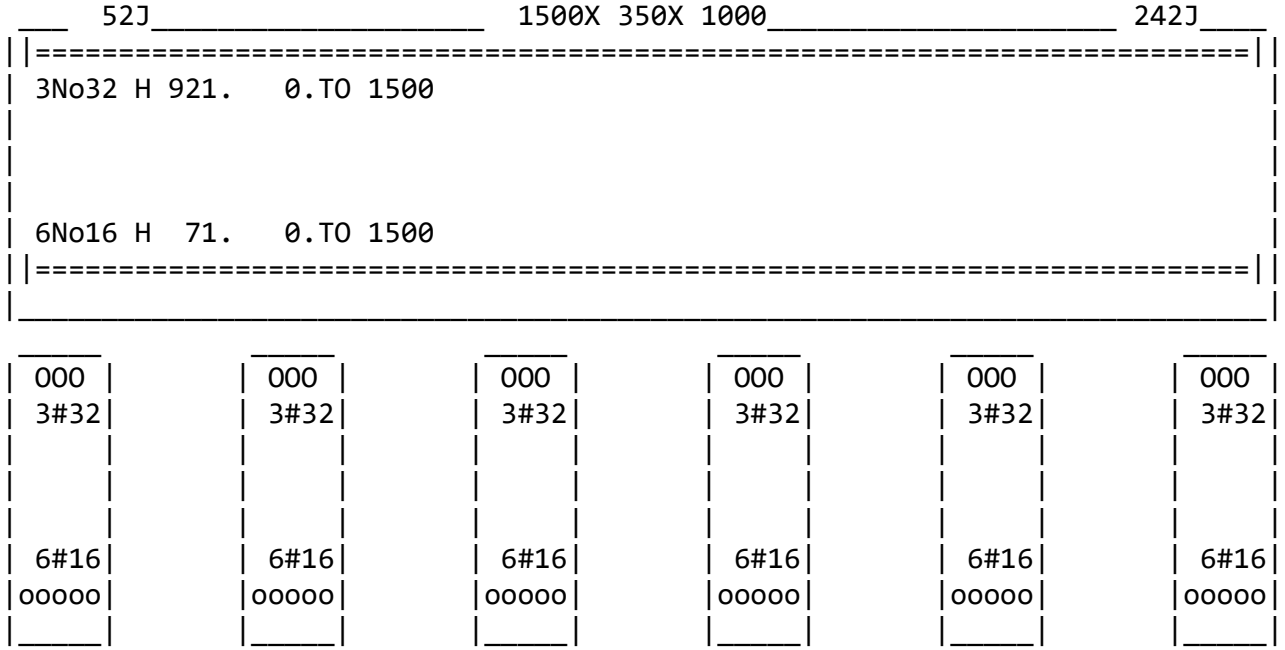
LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	1500.	YES	YES
2	921.	3 - 32MM	0.	1500.	YES	YES

B E A M N O. 1815 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1815 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1815 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1816 DESIGN RESULTS

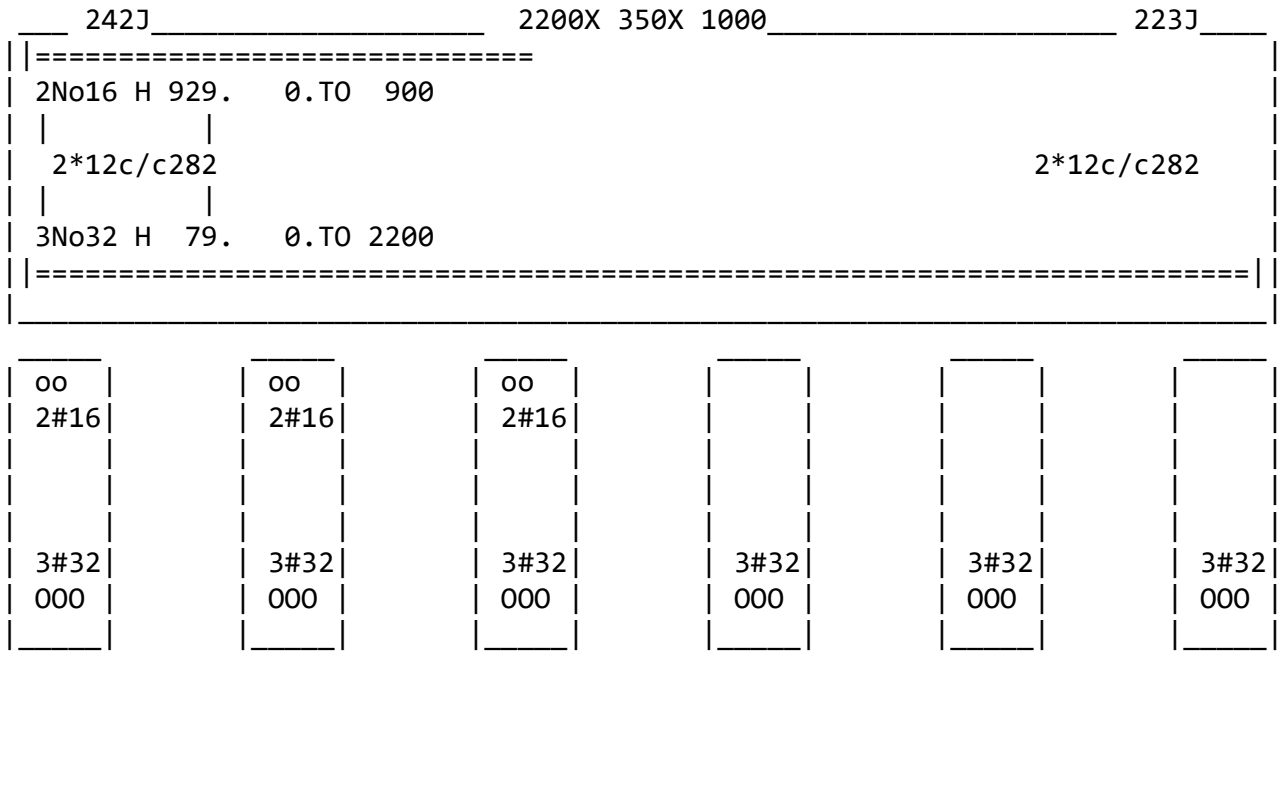
LEN - 2200. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	2200.	YES	YES
2	929.	2 - 16MM	0.	900.	YES	NO

B E A M N O. 1816 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 244.25 KNS Vc= 297.29 KNS Vs= 28.38 KNS
Tu= 20.59 KN-MET Tc= 15.8 KN-MET Ts= 27.4 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 282. MM C/C FOR 175. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.10 SQ.CM.

AT END SUPPORT - Vu= 222.11 KNS Vc= 316.40 KNS Vs= 0.00 KNS
Tu= 23.18 KN-MET Tc= 17.9 KN-MET Ts= 30.9 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 282. MM C/C FOR 175. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.61 SQ.CM.



ACI 318-11 BEAM NO. 1817 DESIGN RESULTS

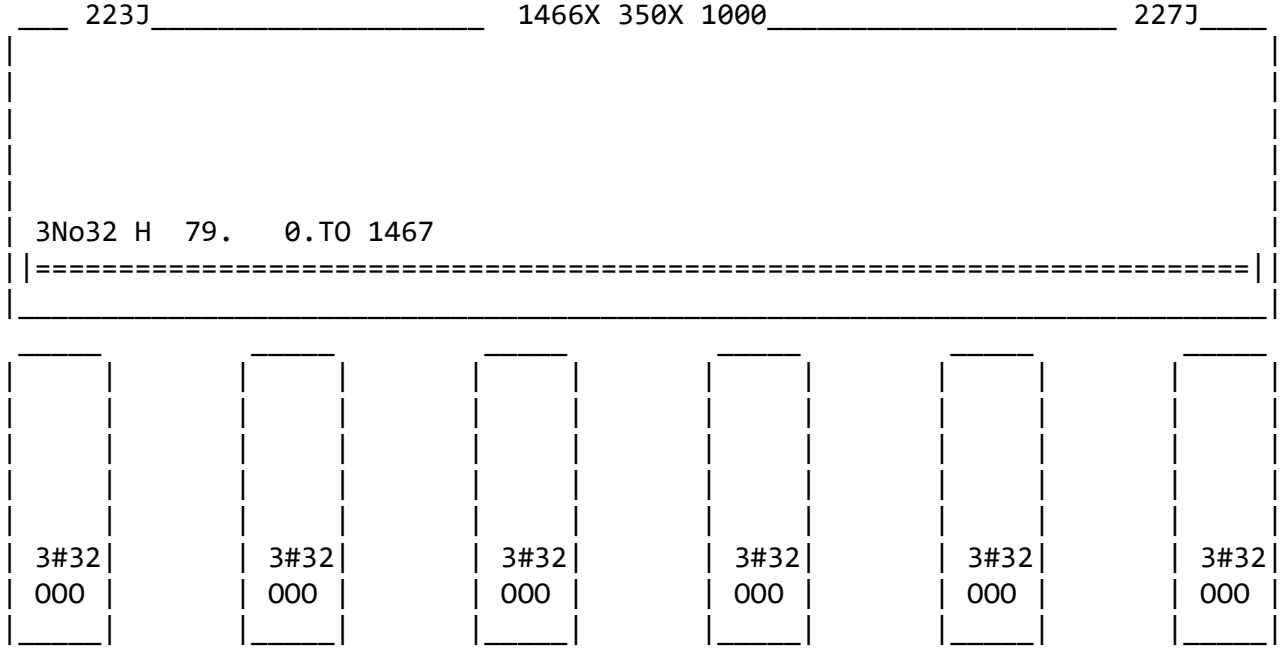
LEN - 1467. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	1467.	YES	YES

B E A M N O. 1817 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1817 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1817 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1818 DESIGN RESULTS

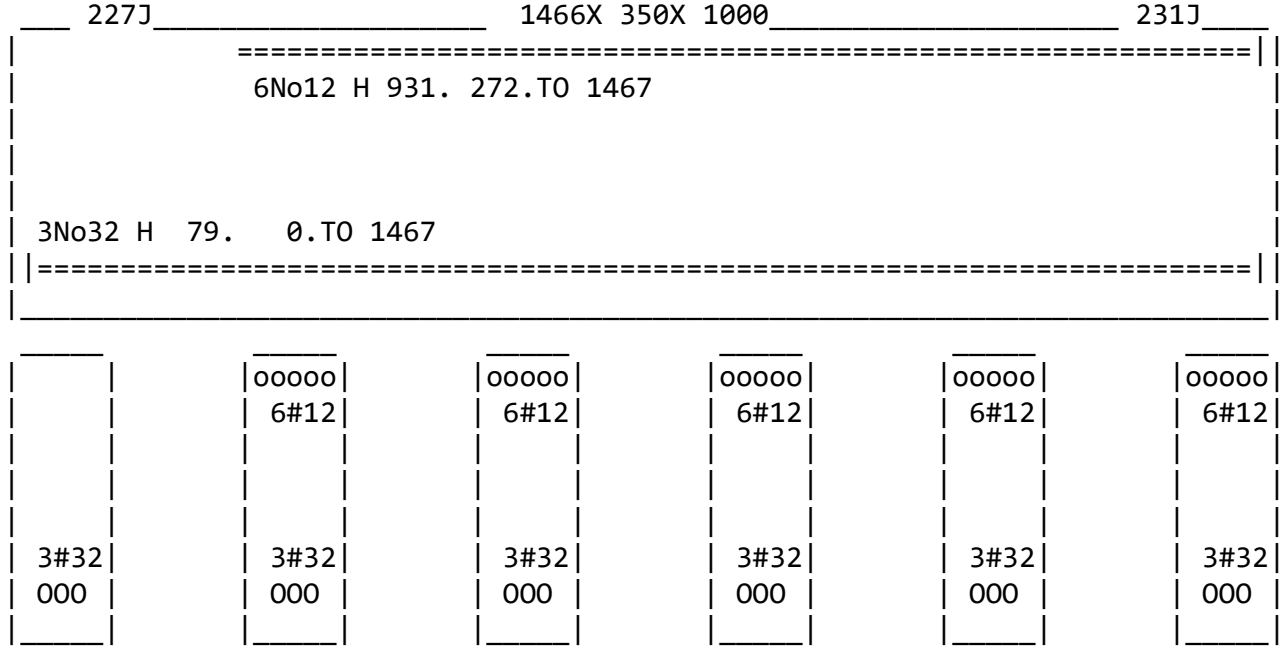
LEN - 1467. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	1467.	YES	YES
2	931.	6 - 12MM	272.	1467.	NO	YES

B E A M N O. 1818 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1818 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1818 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1819 DESIGN RESULTS

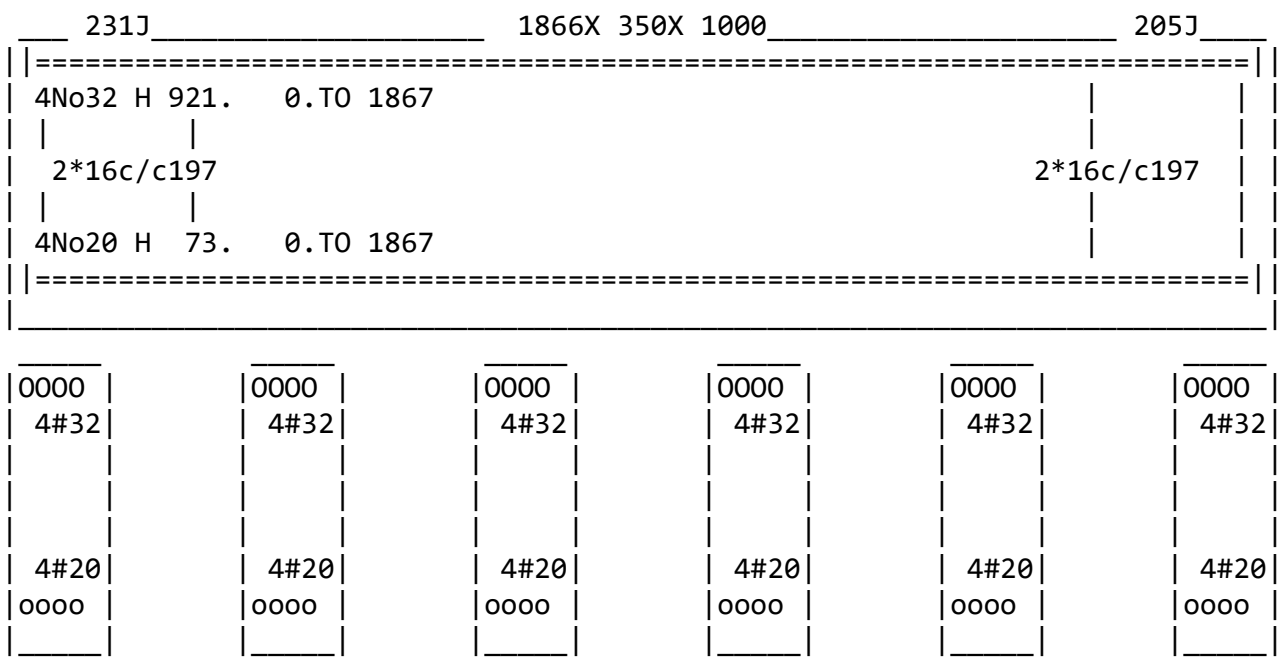
LEN - 1867. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	1867.	YES	YES
2	921.	4 - 32MM	0.	1867.	YES	YES

B E A M N O. 1819 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 545.83 KNS Vc= 298.78 KNS Vs= 429.00 KNS
Tu= 53.39 KN-MET Tc= 15.8 KN-MET Ts= 71.2 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 16 MM 2-LEGGED STIRRUPS AT 197. MM C/C FOR 9. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 10.62 SQ.CM.

AT END SUPPORT - Vu= 546.35 KNS Vc= 298.78 KNS Vs= 429.68 KNS
Tu= 53.39 KN-MET Tc= 15.8 KN-MET Ts= 71.2 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 16 MM 2-LEGGED STIRRUPS AT 197. MM C/C FOR 9. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 10.62 SQ.CM.



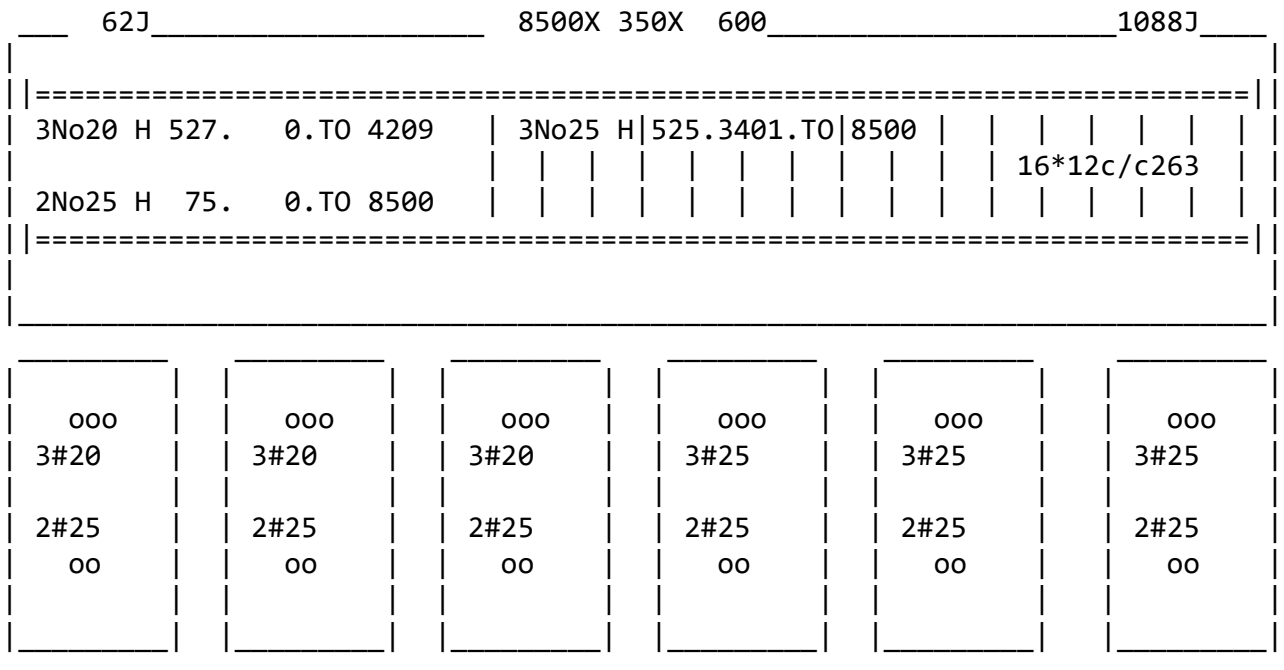
ACI 318-11 BEAM NO. 1820 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	8500.	YES	YES
2	527.	3 - 20MM	0.	4209.	YES	NO
3	525.	3 - 25MM	3401.	8500.	NO	YES

B E A M N O. 1820 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.21 KNS Vc= 164.95 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 74.69 KNS Vc= 146.42 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 7.0 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 263. MM C/C FOR 3725. MM



ACI 318-11 BEAM NO. 1821 DESIGN RESULTS

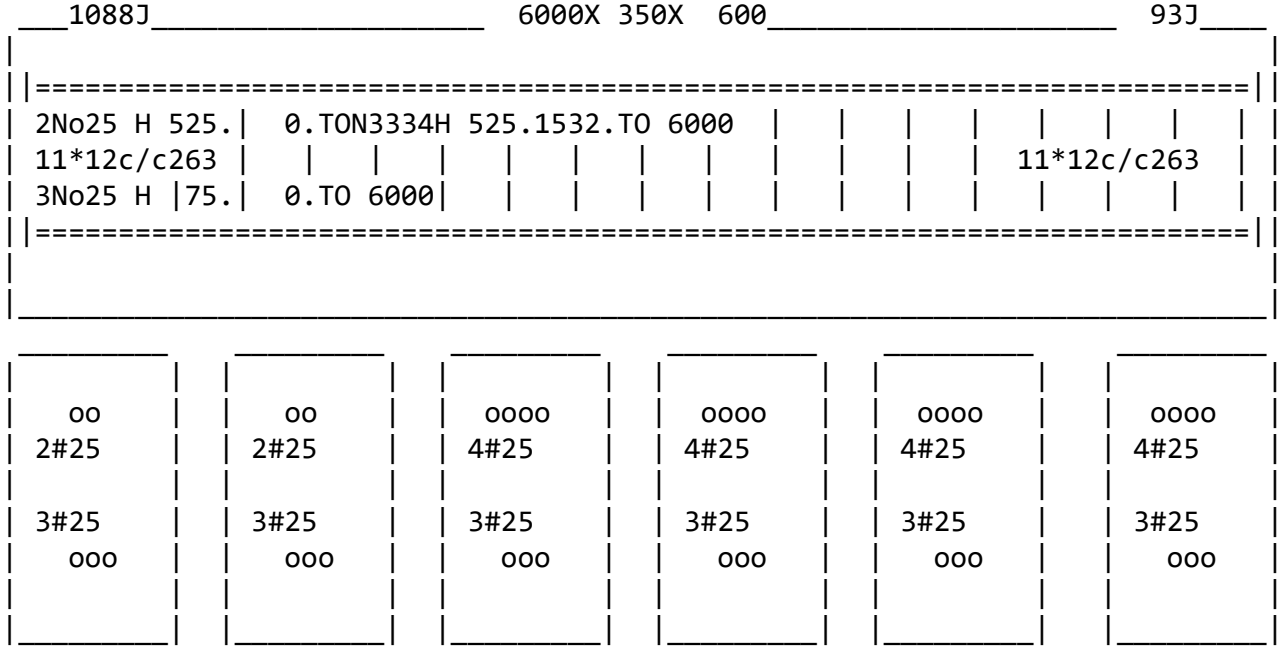
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	6000.	YES	YES
2	525.	2 - 25MM	0.	3334.	YES	NO
3	525.	4 - 25MM	1532.	6000.	NO	YES

B E A M N O. 1821 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 82.11 KNS Vc= 171.36 KNS Vs= 0.00 KNS
Tu= 0.37 KN-MET Tc= 8.2 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 263. MM C/C FOR 2475. MM

AT END SUPPORT - Vu= 111.50 KNS Vc= 171.36 KNS Vs= 0.00 KNS
Tu= 0.37 KN-MET Tc= 8.2 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 263. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 1822 DESIGN RESULTS

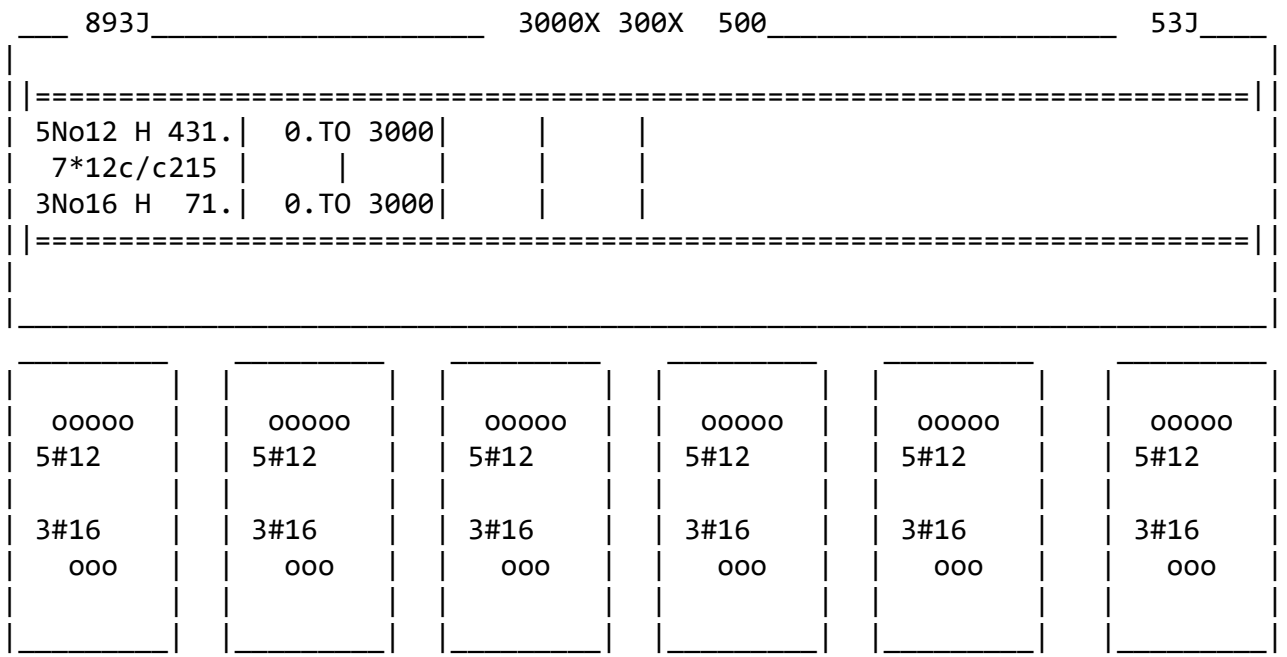
LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	3000.	YES	YES
2	431.	5 - 12MM	0.	3000.	YES	YES

B E A M N O. 1822 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 48.13 KNS Vc= 117.12 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 1075. MM

AT END SUPPORT - Vu= 20.73 KNS Vc= 129.23 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1823 DESIGN RESULTS

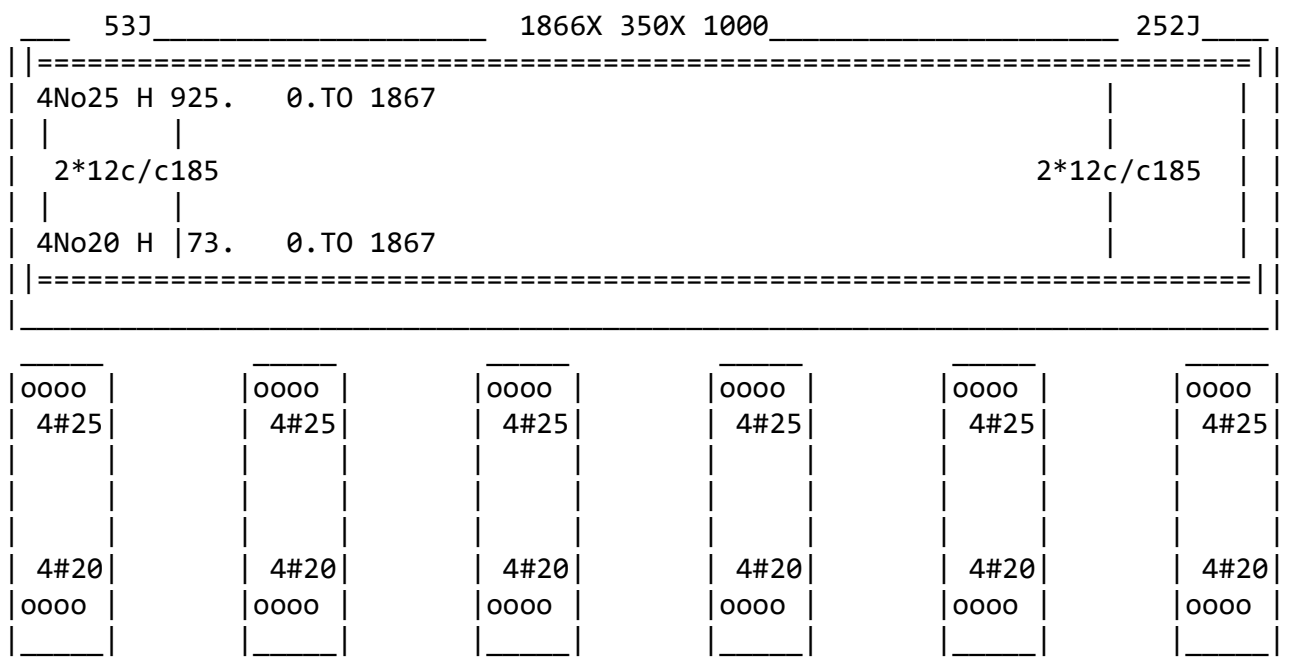
LEN - 1867. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	1867.	YES	YES
2	925.	4 - 25MM	0.	1867.	YES	YES

B E A M N O. 1823 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 405.80 KNS Vc= 282.83 KNS Vs= 258.23 KNS
Tu= 31.75 KN-MET Tc= 15.0 KN-MET Ts= 42.3 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 185. MM C/C FOR 9. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.32 SQ.CM.

AT END SUPPORT - Vu= 405.29 KNS Vc= 282.83 KNS Vs= 257.55 KNS
Tu= 31.75 KN-MET Tc= 15.0 KN-MET Ts= 42.3 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 185. MM C/C FOR 9. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.32 SQ.CM.



ACI 318-11 BEAM NO. 1824 DESIGN RESULTS

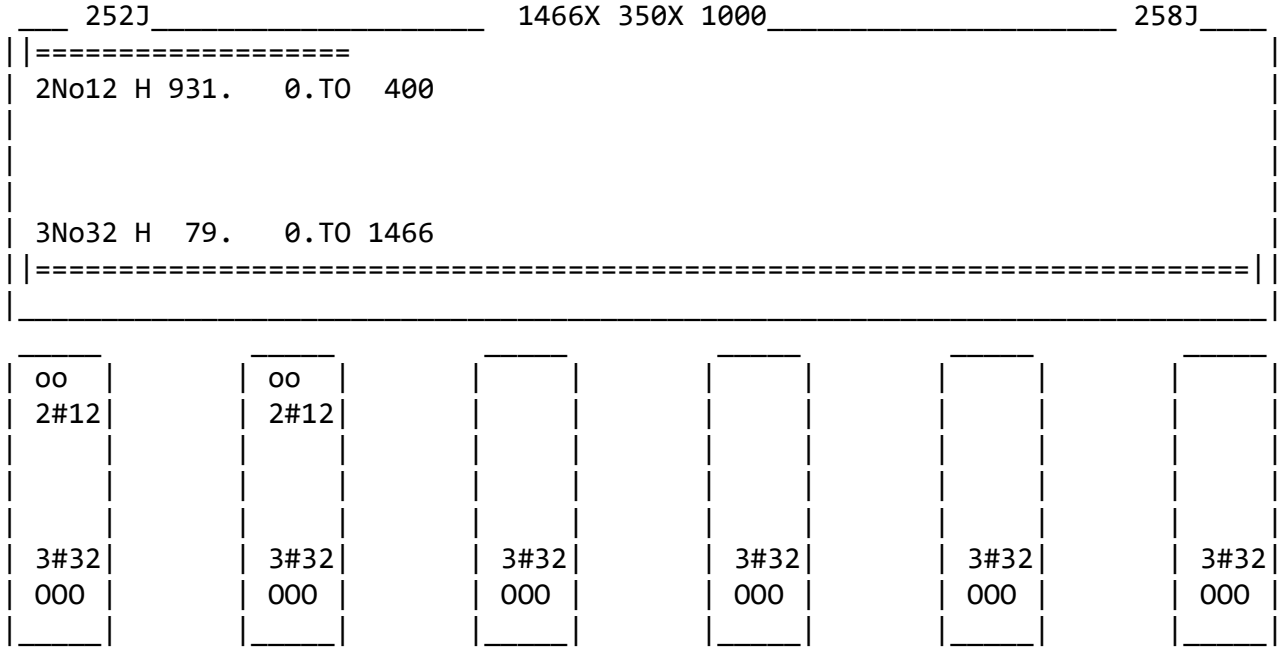
LEN - 1466. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	1466.	YES	YES
2	931.	2 - 12MM	0.	400.	YES	NO

B E A M N O. 1824 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1824 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1824 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1825 DESIGN RESULTS

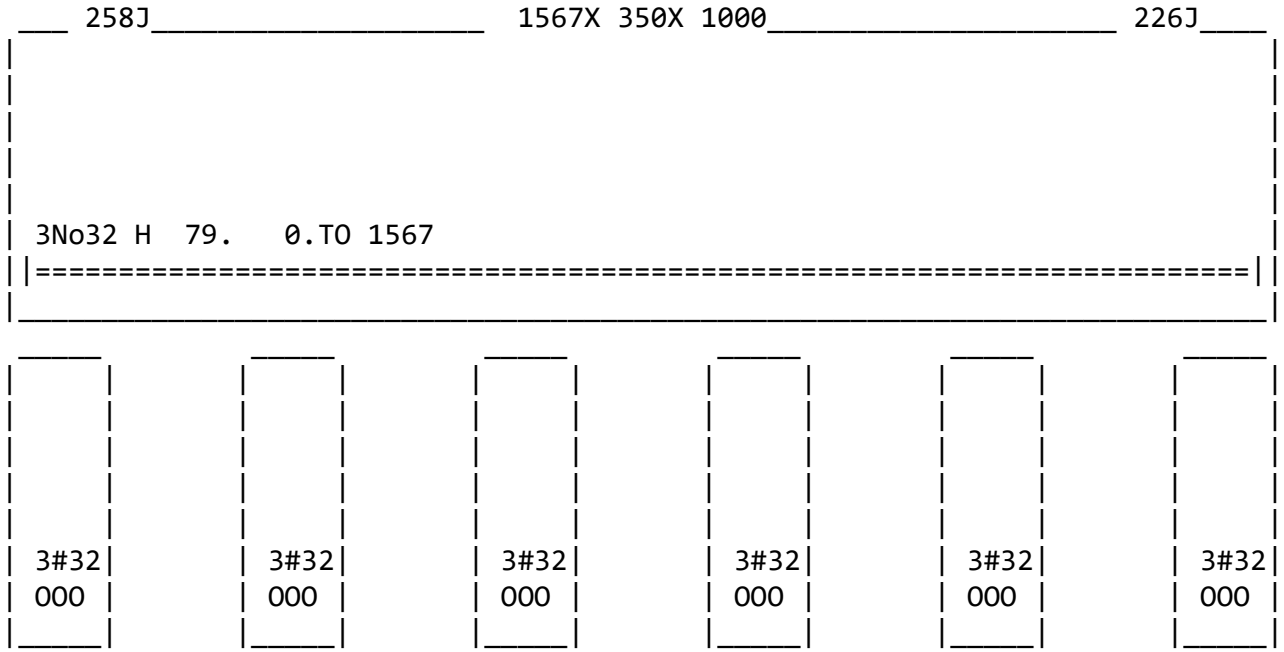
LEN - 1567. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	1567.	YES	YES

B E A M N O. 1825 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1825 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1825 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1826 DESIGN RESULTS

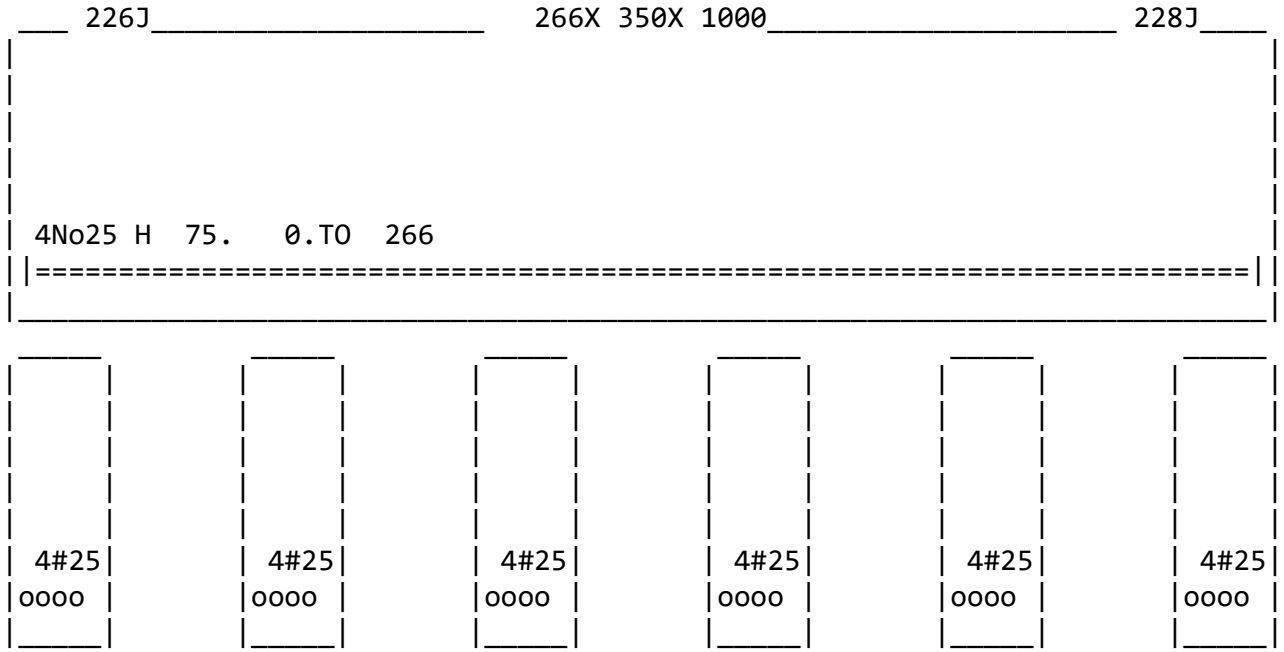
LEN - 266. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	266.	YES	YES

B E A M N O. 1826 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1826 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1826 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1827 DESIGN RESULTS

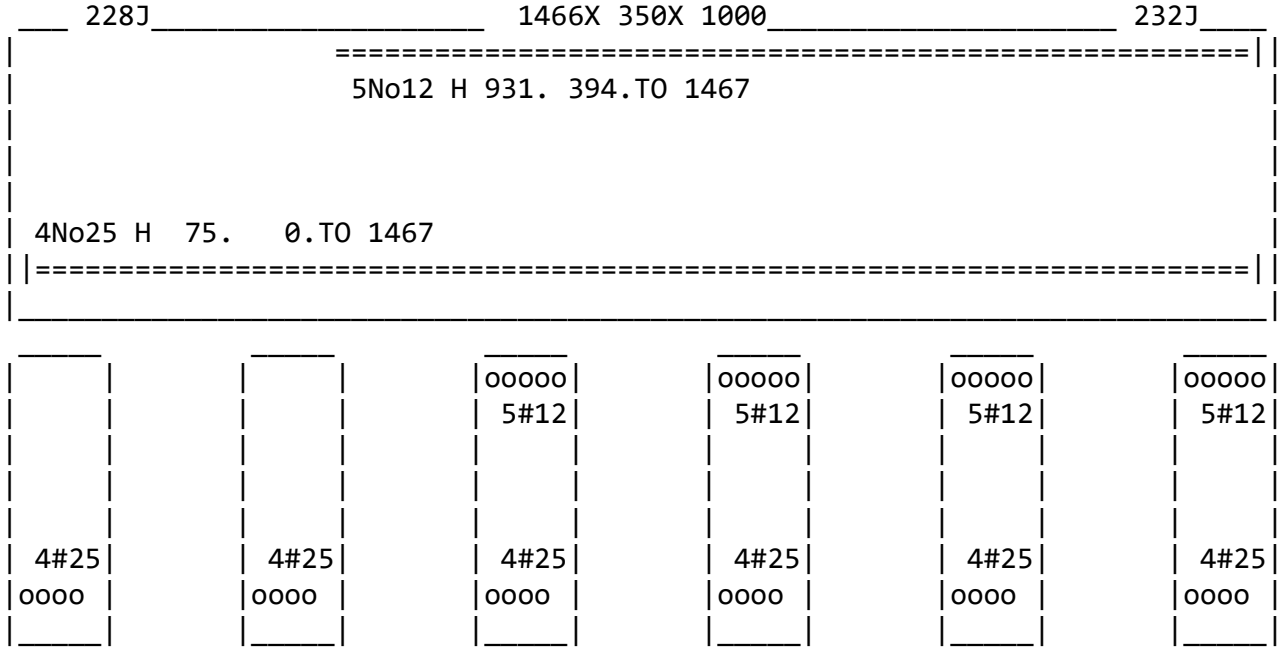
LEN - 1467. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	1467.	YES	YES
2	931.	5 - 12MM	394.	1467.	NO	YES

B E A M N O. 1827 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1827 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1827 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1828 DESIGN RESULTS

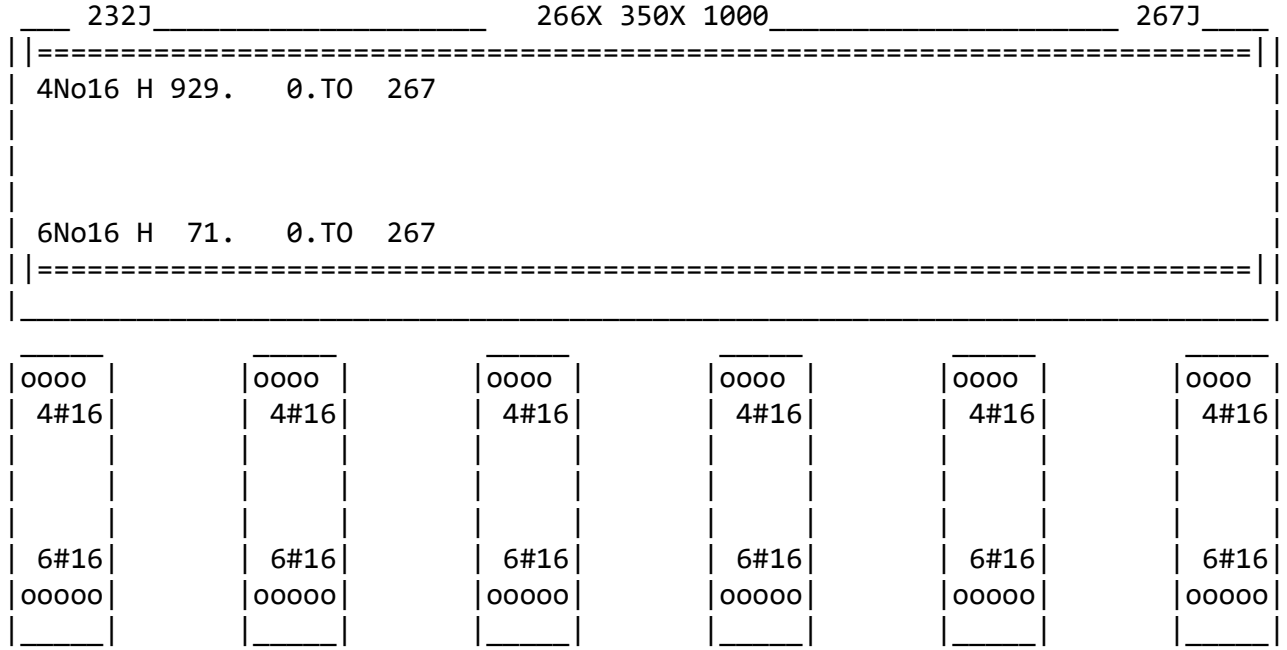
LEN - 267. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	267.	YES	YES
2	929.	4 - 16MM	0.	267.	YES	YES

B E A M N O. 1828 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1828 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1828 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1829 DESIGN RESULTS

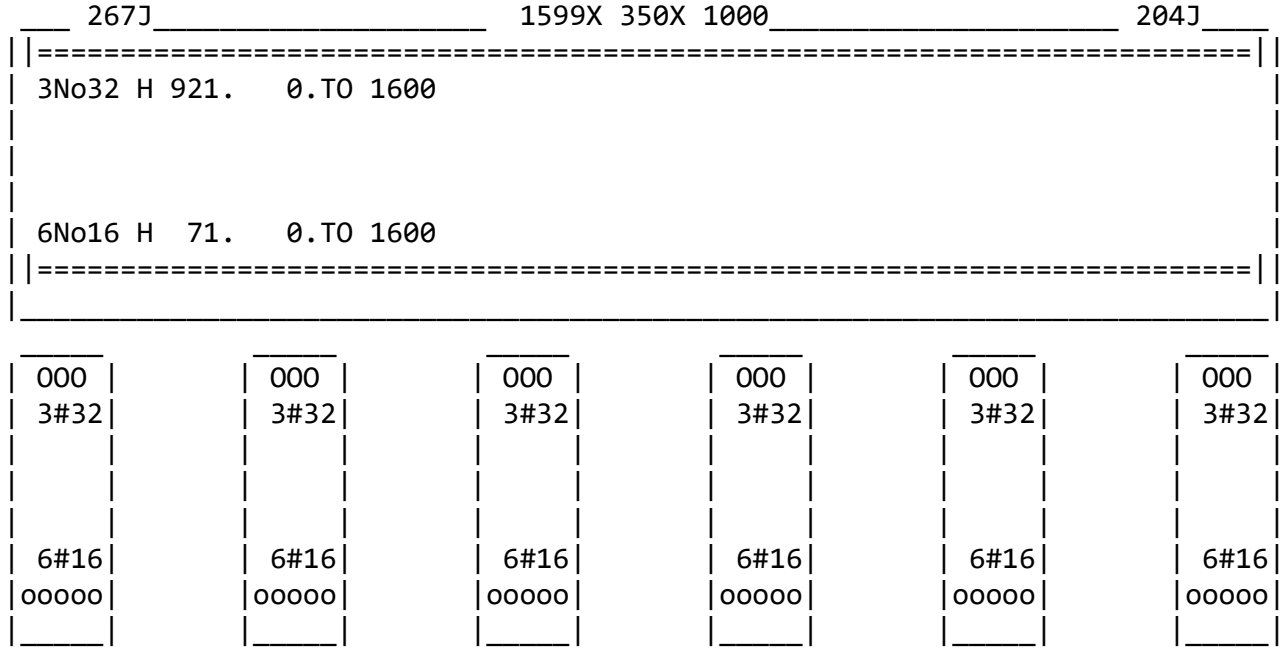
LEN - 1600. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	1600.	YES	YES
2	921.	3 - 32MM	0.	1600.	YES	YES

B E A M N O. 1829 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1829 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1829 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



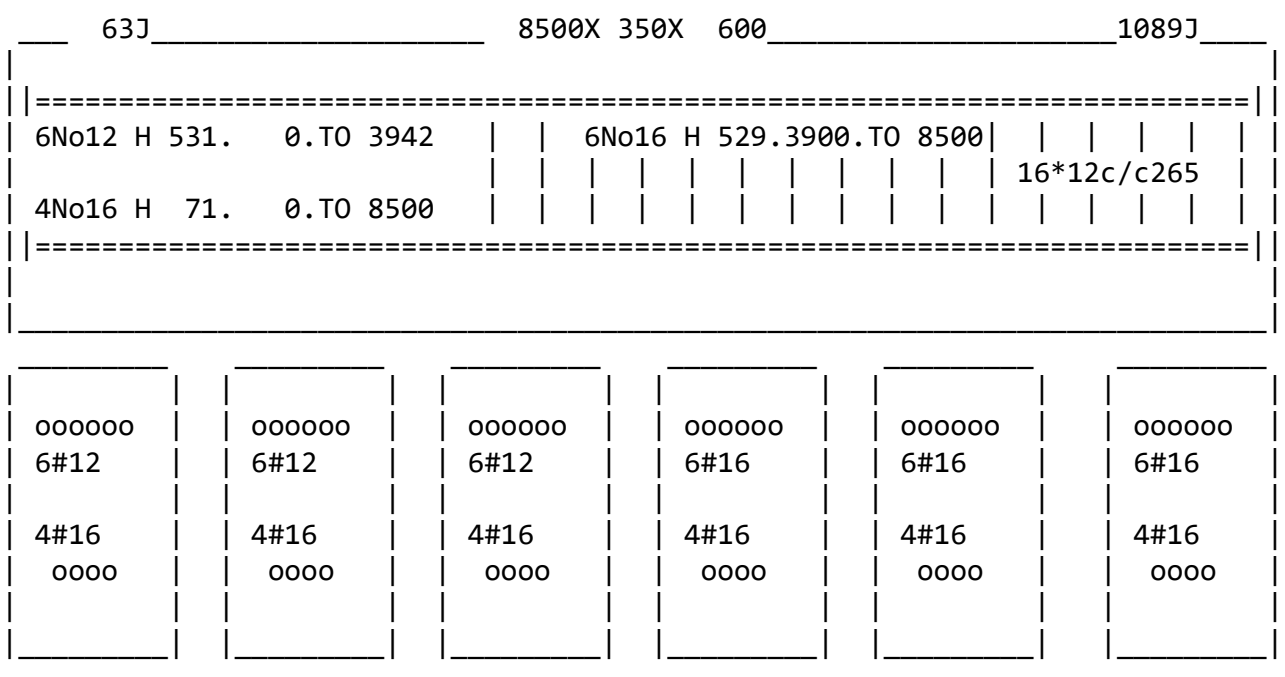
ACI 318-11 BEAM NO. 1830 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	8500.	YES	YES
2	531.	6 - 12MM	0.	3942.	YES	NO
3	529.	6 - 16MM	3900.	8500.	NO	YES

B E A M N O. 1830 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.17 KNS Vc= 169.06 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 8.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 47.83 KNS Vc= 125.57 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 3725. MM



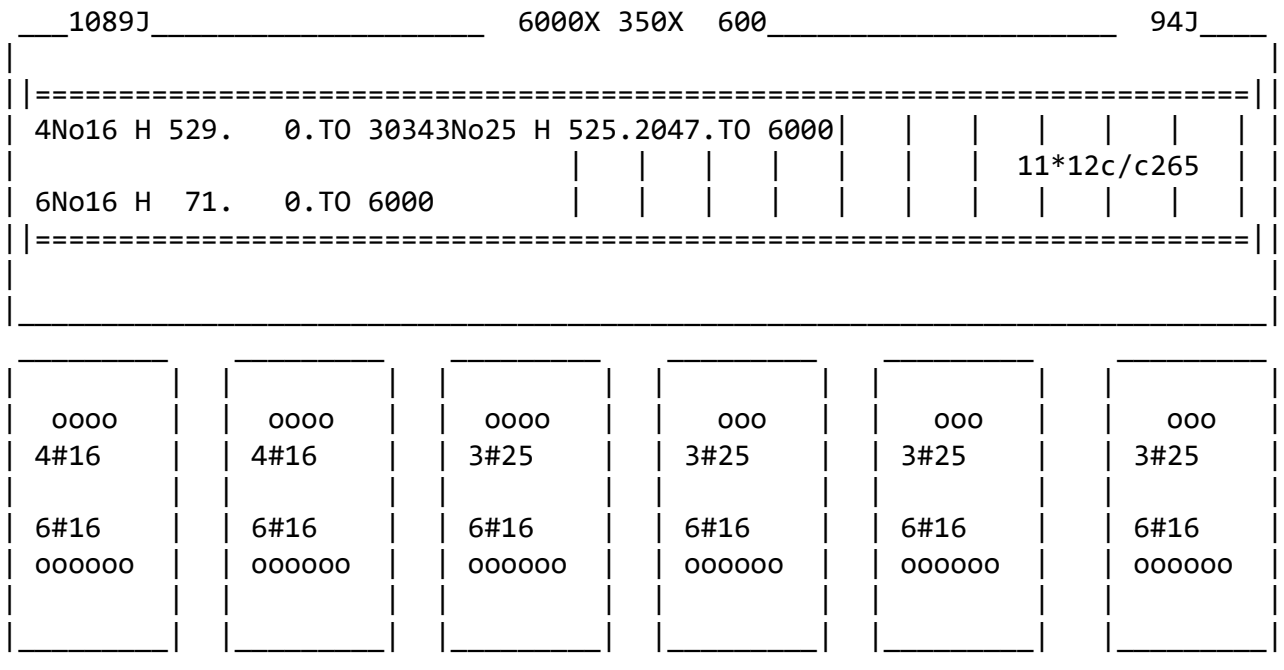
ACI 318-11 BEAM NO. 1831 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	6000.	YES	YES
2	529.	4 - 16MM	0.	3034.	YES	NO
3	525.	3 - 25MM	2047.	6000.	NO	YES

B E A M N O. 1831 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.24 KNS Vc= 179.86 KNS Vs= 0.00 KNS
Tu= 1.02 KN-MET Tc= 8.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 92.54 KNS Vc= 174.93 KNS Vs= 0.00 KNS
Tu= 0.25 KN-MET Tc= 8.2 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 1832 DESIGN RESULTS

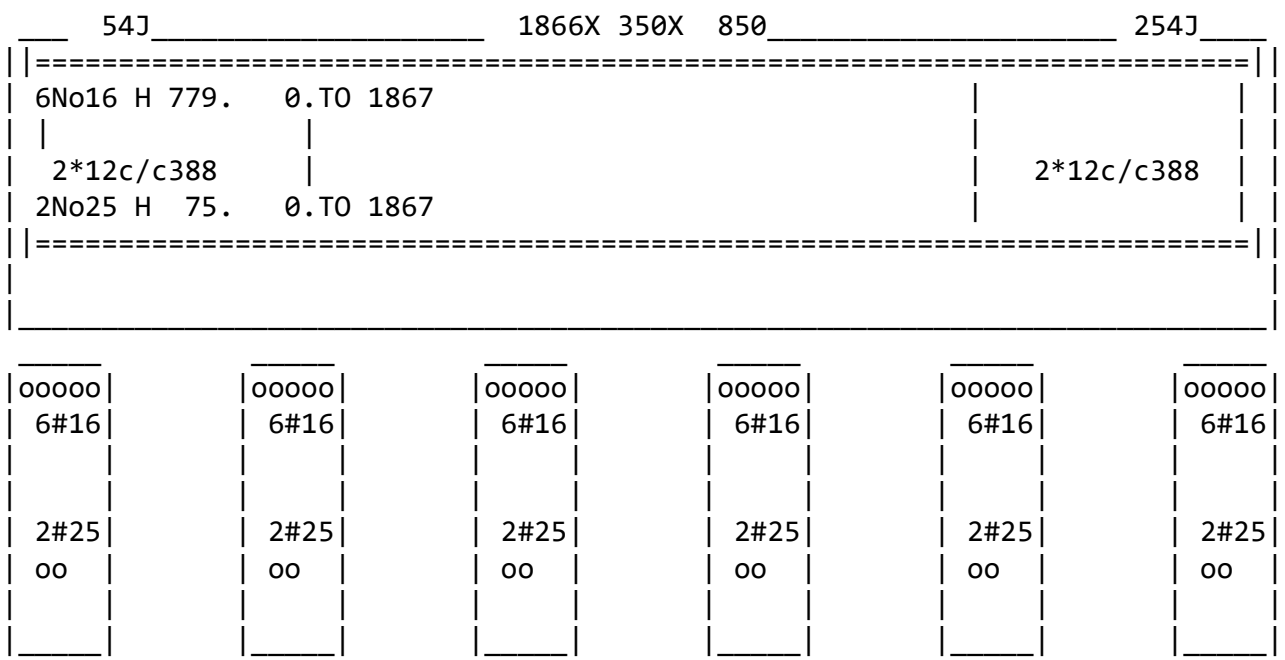
LEN - 1867. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1867.	YES	YES
2	779.	6 - 16MM	0.	1867.	YES	YES

B E A M N O. 1832 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 108.73 KNS Vc= 265.60 KNS Vs= 0.00 KNS
Tu= 1.43 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 159. MM

AT END SUPPORT - Vu= 104.55 KNS Vc= 265.60 KNS Vs= 0.00 KNS
Tu= 1.43 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 159. MM



ACI 318-11 BEAM NO. 1833 DESIGN RESULTS

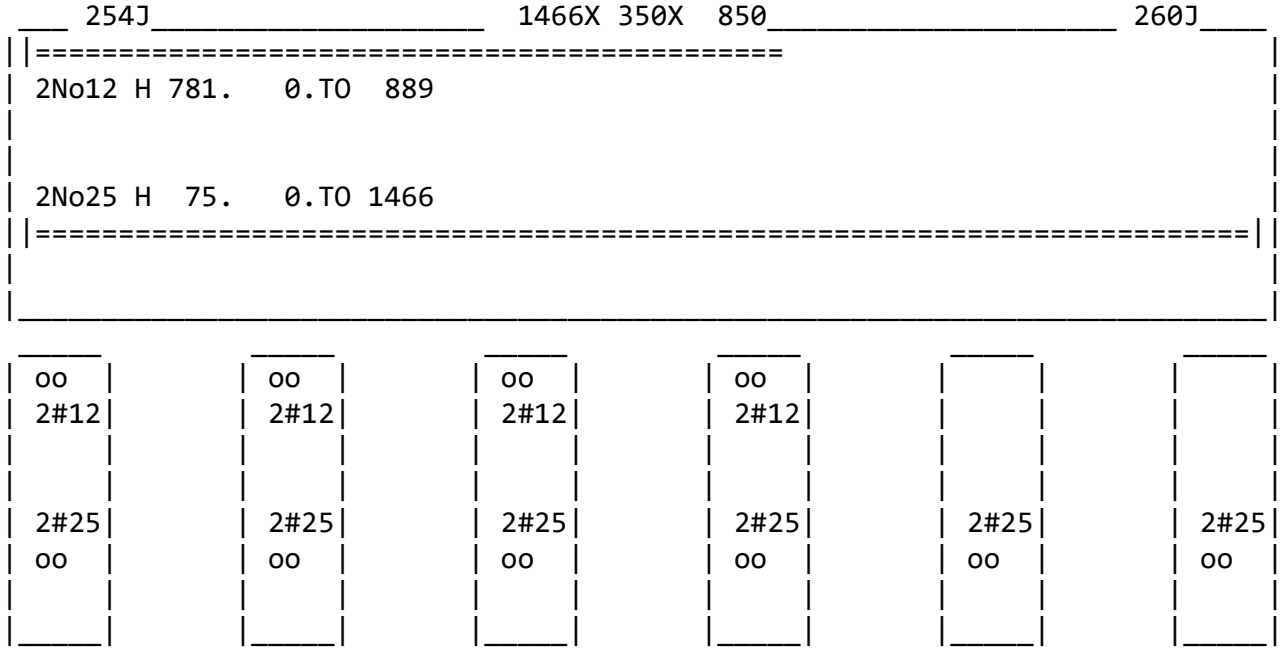
LEN - 1466. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1466.	YES	YES
2	781.	2 - 12MM	0.	889.	YES	NO

B E A M N O. 1833 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1833 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1833 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



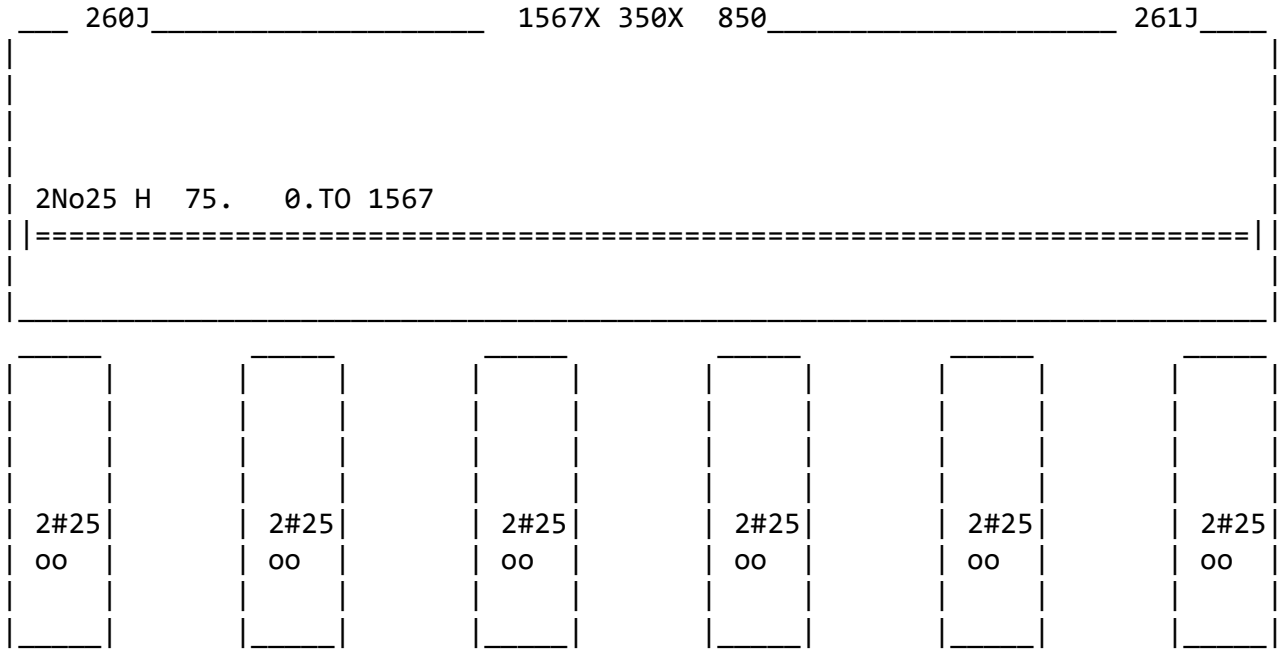
ACI 318-11 BEAM NO. 1834 DESIGN RESULTS

LEN - 1567. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1567.	YES	YES

B E A M N O. 1834 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 4.62 KNS Vc= 253.67 KNS Vs= 0.00 KNS
Tu= 1.28 KN-MET Tc= 13.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 4.85 KNS Vc= 253.69 KNS Vs= 0.00 KNS
Tu= 1.28 KN-MET Tc= 13.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1835 DESIGN RESULTS

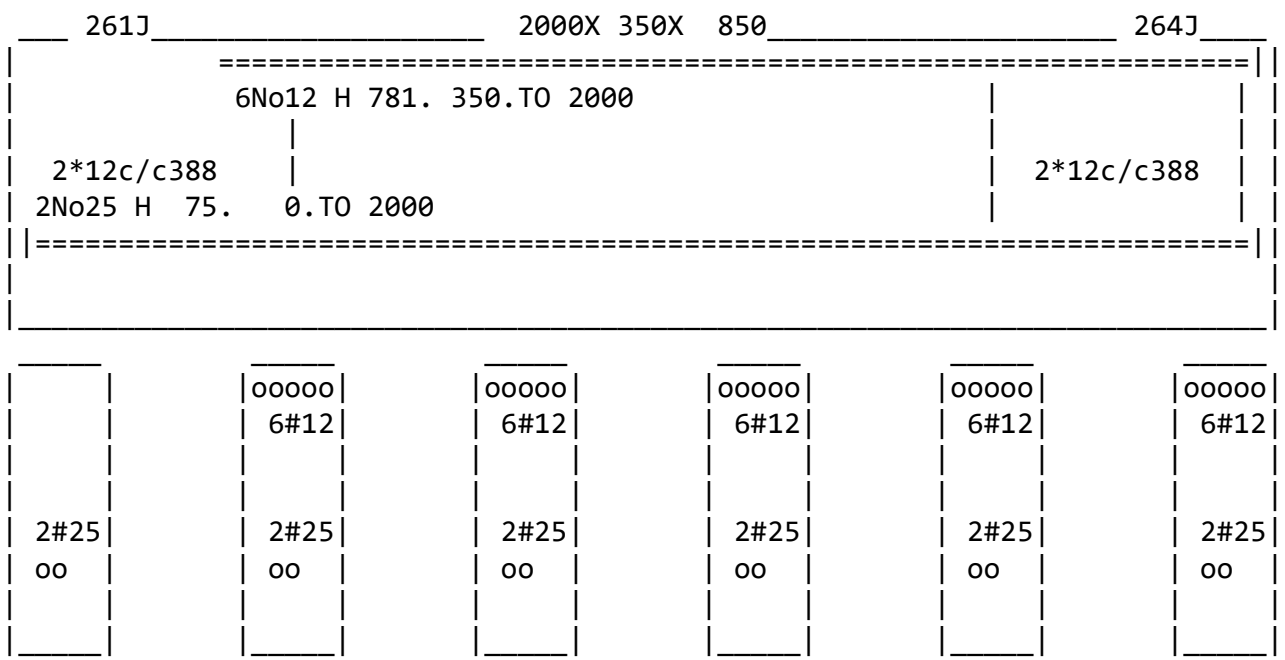
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	2000.	YES	YES
2	781.	6 - 12MM	350.	2000.	NO	YES

B E A M N O. 1835 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 112.18 KNS Vc= 262.63 KNS Vs= 0.00 KNS
Tu= 1.04 KN-MET Tc= 13.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 225. MM

AT END SUPPORT - Vu= 121.57 KNS Vc= 268.66 KNS Vs= 0.00 KNS
Tu= 1.04 KN-MET Tc= 13.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 225. MM



ACI 318-11 BEAM NO. 1836 DESIGN RESULTS

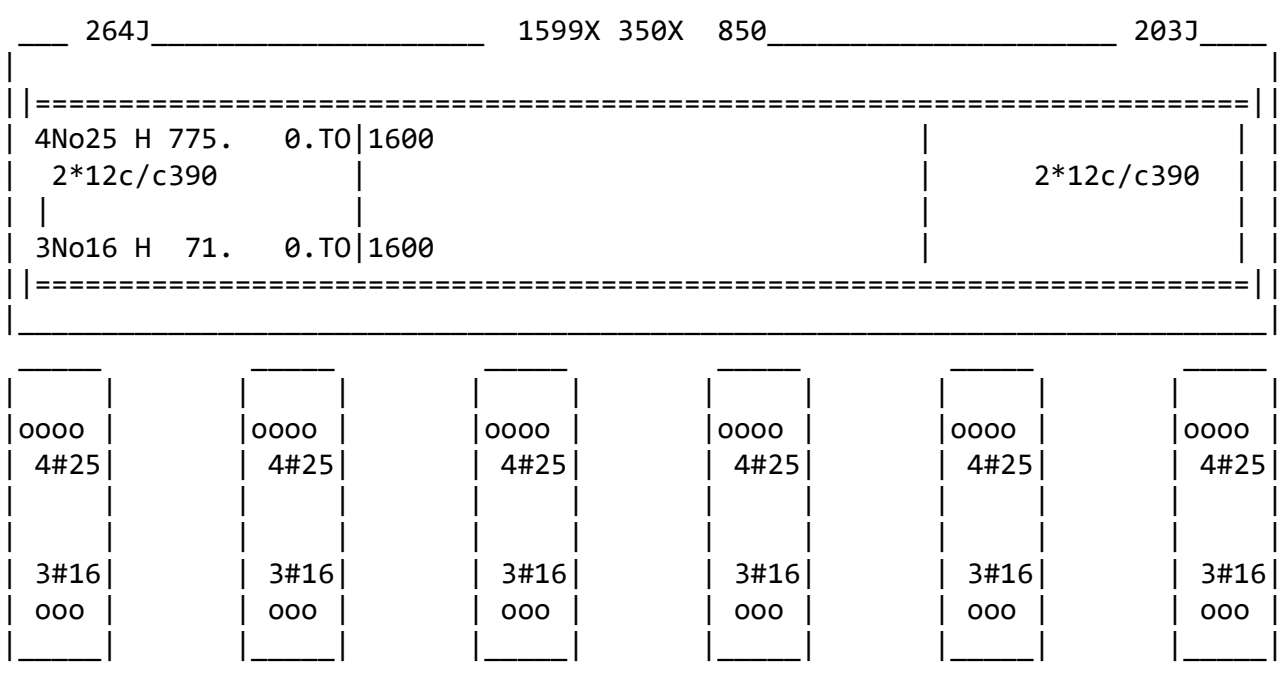
LEN - 1600. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	1600.	YES	YES
2	775.	4 - 25MM	0.	1600.	YES	YES

B E A M N O. 1836 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 180.23 KNS Vc= 267.05 KNS Vs= 0.00 KNS
Tu= 1.28 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 25. MM

AT END SUPPORT - Vu= 180.87 KNS Vc= 267.05 KNS Vs= 0.00 KNS
Tu= 1.28 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 25. MM



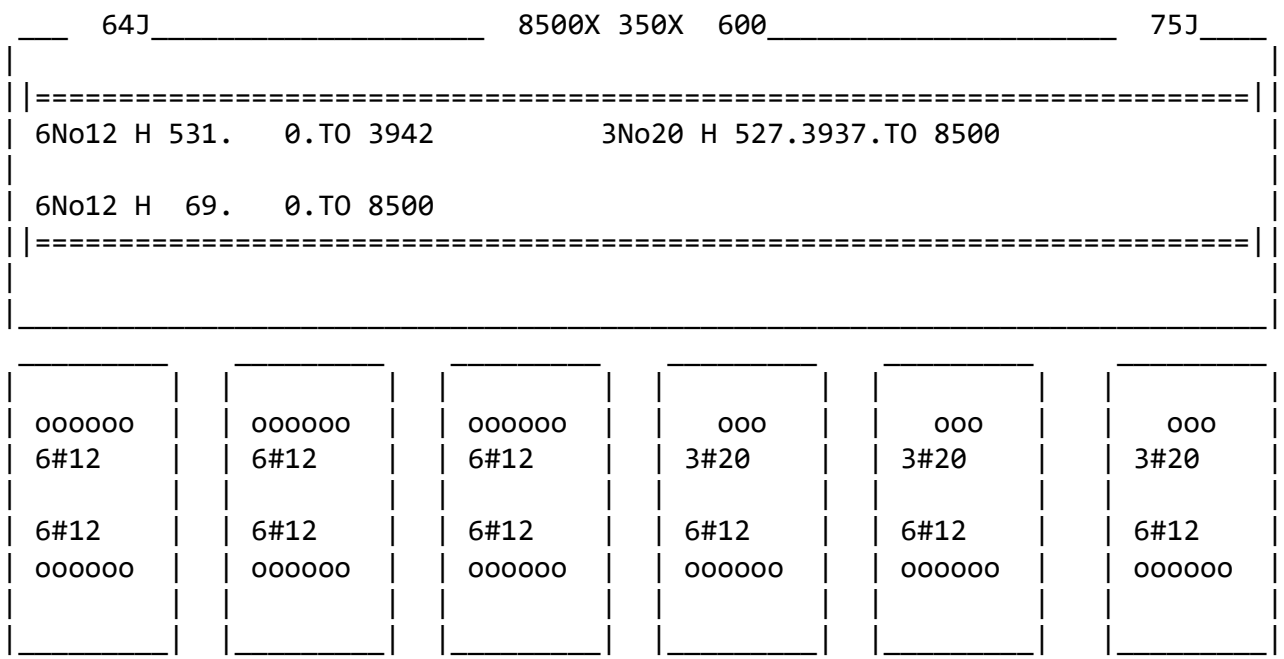
ACI 318-11 BEAM NO. 1837 DESIGN RESULTS

LEN - 8500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	6 - 12MM	0.	8500.	YES	YES
2	531.	6 - 12MM	0.	3942.	YES	NO
3	527.	3 - 20MM	3937.	8500.	NO	YES

B E A M N O. 1837 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.94 KNS Vc= 175.19 KNS Vs= 0.00 KNS
Tu= 0.49 KN-MET Tc= 8.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 39.67 KNS Vc= 175.19 KNS Vs= 0.00 KNS
Tu= 0.49 KN-MET Tc= 8.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



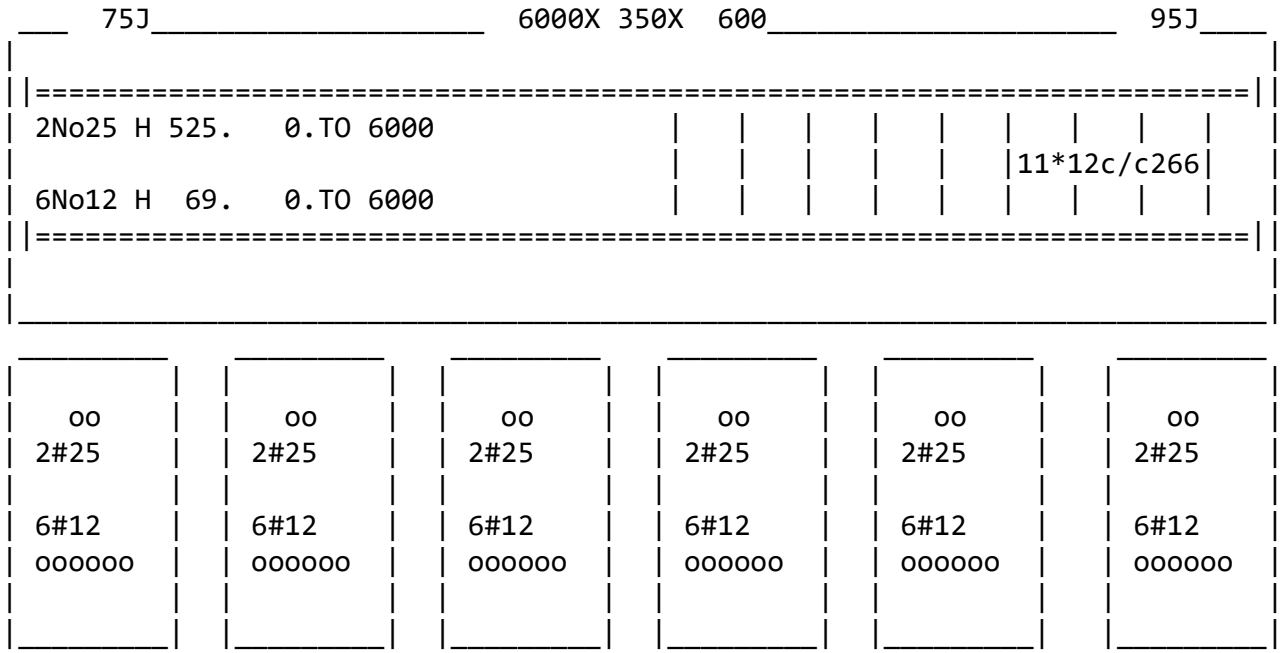
ACI 318-11 BEAM NO. 1838 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	6 - 12MM	0.	6000.	YES	YES
2	525.	2 - 25MM	0.	6000.	YES	YES

B E A M N O. 1838 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.73 KNS Vc= 180.83 KNS Vs= 0.00 KNS
Tu= 0.36 KN-MET Tc= 8.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 66.91 KNS Vc= 178.27 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 8.4 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 2321 DESIGN RESULTS

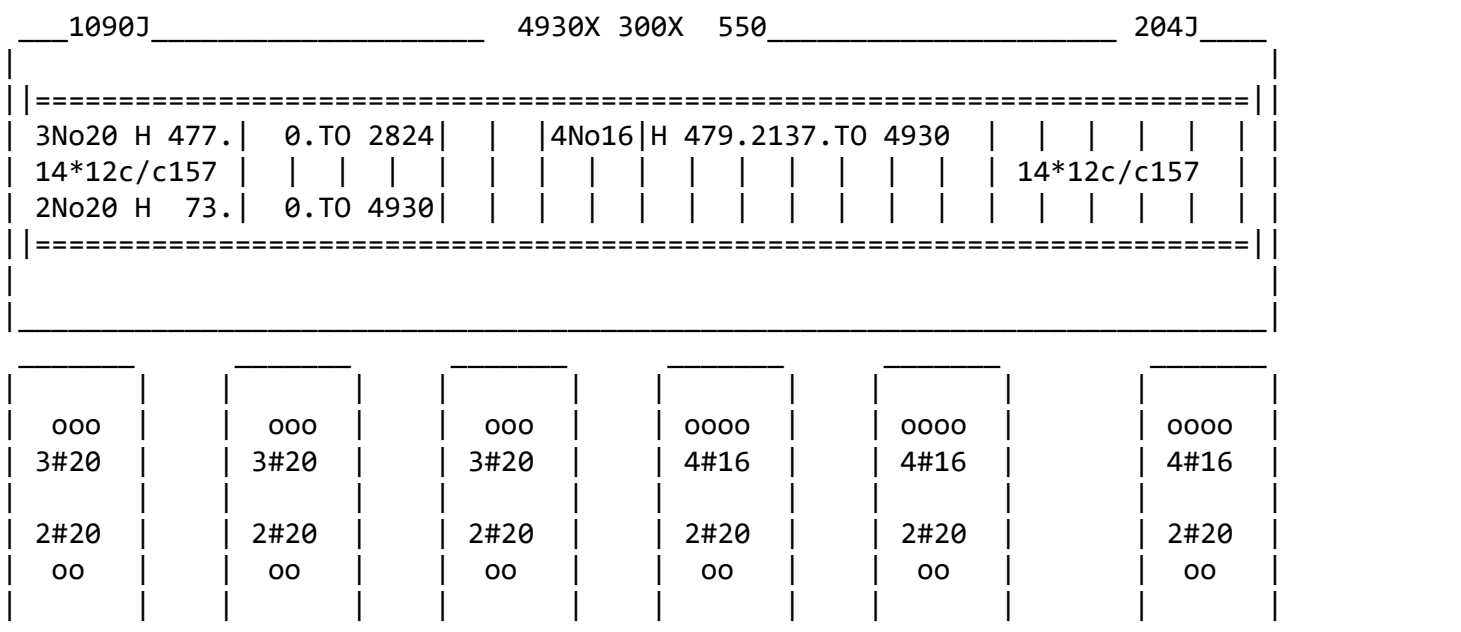
LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	2 - 20MM	0.	4930.	YES	YES
2	477.	3 - 20MM	0.	2824.	YES	NO
3	479.	4 - 16MM	2137.	4930.	NO	YES

B E A M N O. 2321 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 26.93 KNS Vc= 141.55 KNS Vs= 0.00 KNS
Tu= 15.06 KN-MET Tc= 5.9 KN-MET Ts= 20.1 KN-MET LOAD 411
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1990. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.28 SQ.CM.

AT END SUPPORT - Vu= 16.33 KNS Vc= 247.60 KNS Vs= 0.00 KNS
Tu= 15.06 KN-MET Tc= 5.9 KN-MET Ts= 20.1 KN-MET LOAD 411
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1990. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.28 SQ.CM.



**********END OF BEAM DESIGN**********

2224. DESIGN BEAM 1701 TO 1706 1710 TO 1716 1719 TO 1726 2104 2105

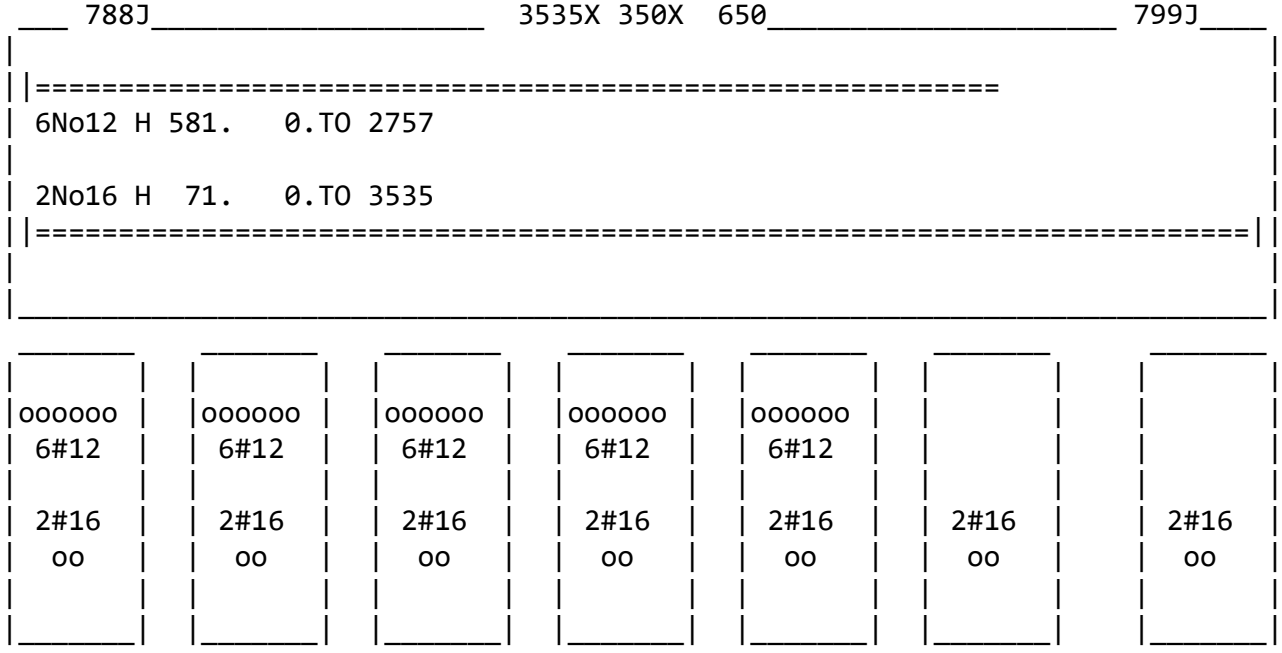
ACI 318-11 BEAM NO. 1701 DESIGN RESULTS

LEN - 3535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	3535.	YES	YES
2	581.	6 - 12MM	0.	2757.	YES	NO

B E A M N O. 1701 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 32.31 KNS Vc= 182.76 KNS Vs= 0.00 KNS
Tu= 1.62 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.41 KNS Vc= 182.76 KNS Vs= 0.00 KNS
Tu= 1.62 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



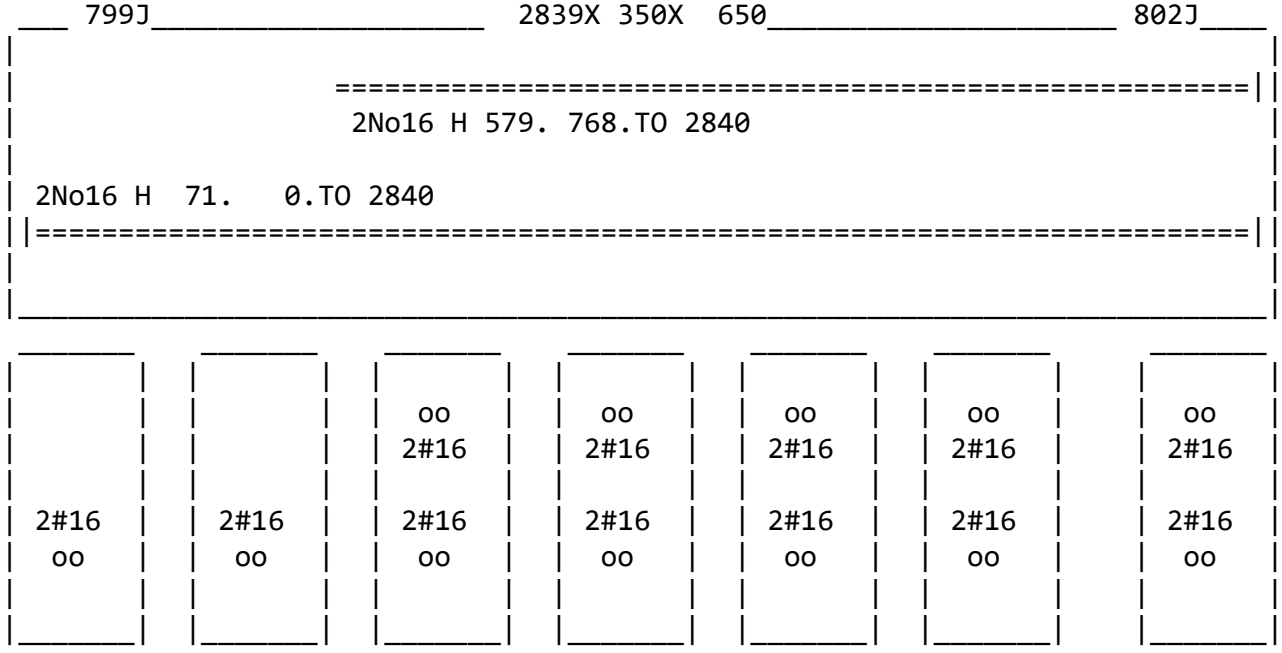
ACI 318-11 BEAM NO. 1702 DESIGN RESULTS

LEN - 2840. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	2840.	YES	YES
2	579.	2 - 16MM	768.	2840.	NO	YES

B E A M N O. 1702 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.07 KNS Vc= 182.72 KNS Vs= 0.00 KNS
Tu= 1.62 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.76 KNS Vc= 182.72 KNS Vs= 0.00 KNS
Tu= 1.62 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



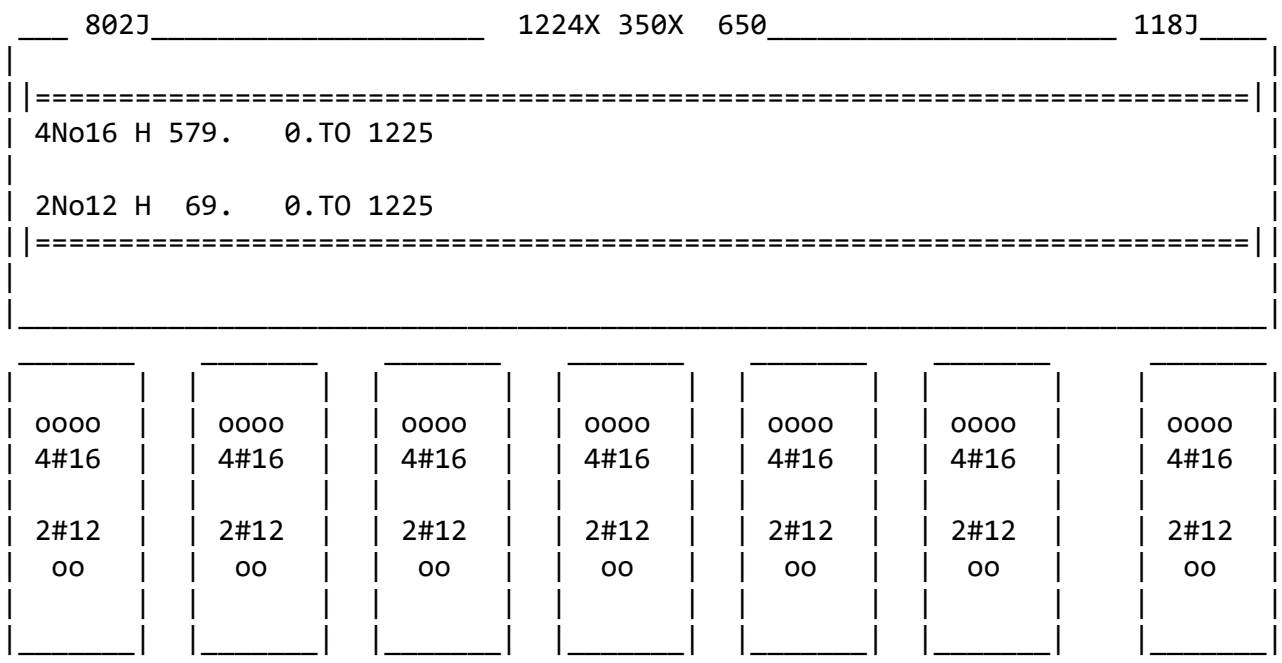
ACI 318-11 BEAM NO. 1703 DESIGN RESULTS

LEN - 1225. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1225.	YES	YES
2	579.	4 - 16MM	0.	1225.	YES	YES

B E A M N O. 1703 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 31.38 KNS Vc= 183.35 KNS Vs= 0.00 KNS
Tu= 1.62 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 31.95 KNS Vc= 183.35 KNS Vs= 0.00 KNS
Tu= 1.62 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



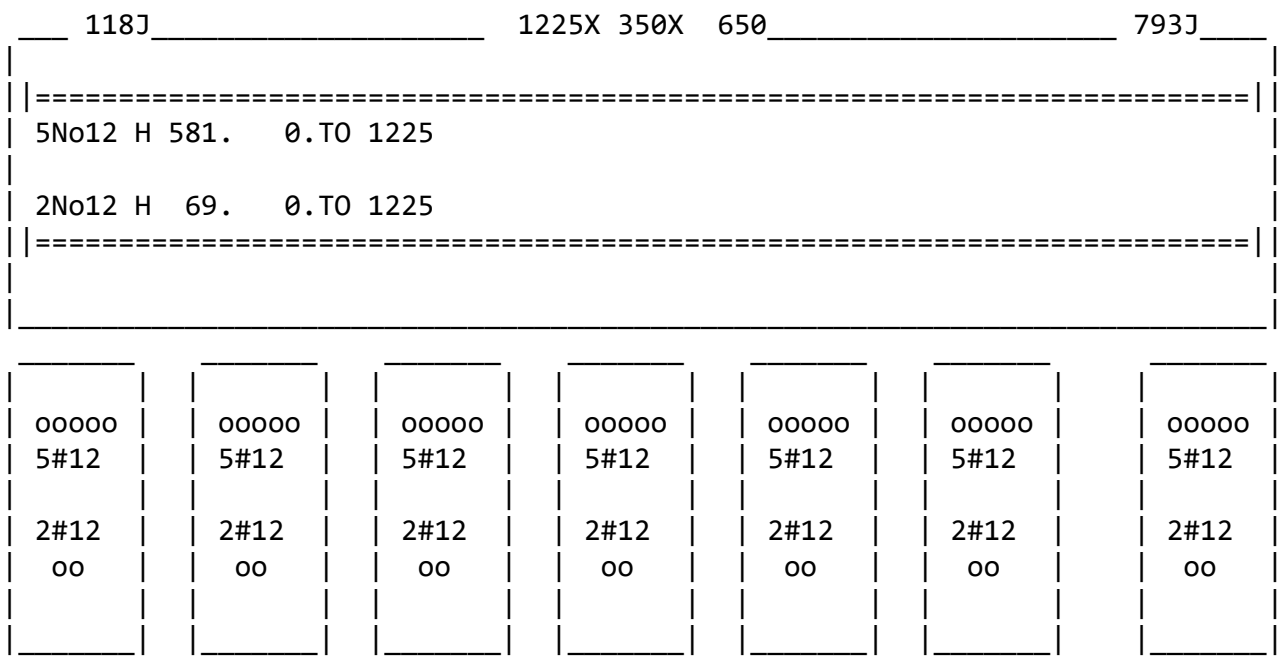
ACI 318-11 BEAM NO. 1704 DESIGN RESULTS

LEN - 1225. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1225.	YES	YES
2	581.	5 - 12MM	0.	1225.	YES	YES

B E A M N O. 1704 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 22.19 KNS Vc= 192.51 KNS Vs= 0.00 KNS
Tu= 0.74 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 21.62 KNS Vc= 192.65 KNS Vs= 0.00 KNS
Tu= 0.74 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



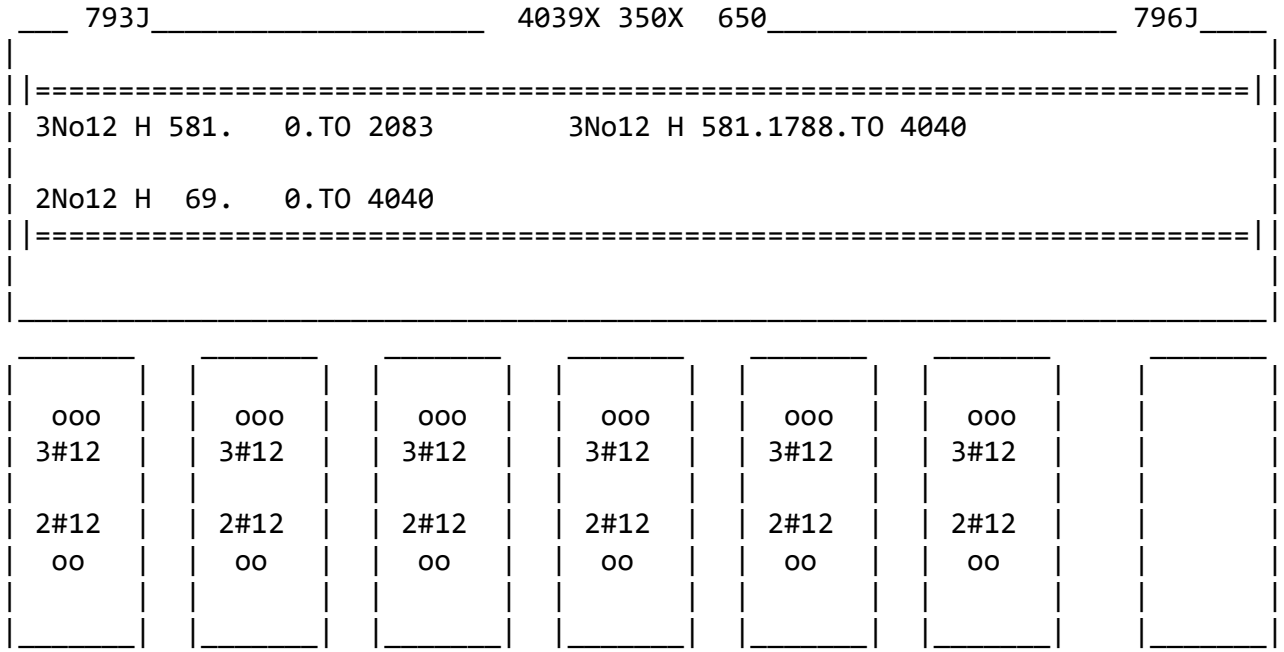
ACI 318-11 BEAM NO. 1705 DESIGN RESULTS

LEN - 4040. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	4040.	YES	YES
2	581.	3 - 12MM	0.	2083.	YES	NO
3	581.	3 - 12MM	1788.	4040.	NO	YES

B E A M N O. 1705 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.00 KNS Vc= 351.24 KNS Vs= 0.00 KNS
Tu= 0.74 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 8.69 KNS Vc= 194.09 KNS Vs= 0.00 KNS
Tu= 0.74 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1706 DESIGN RESULTS

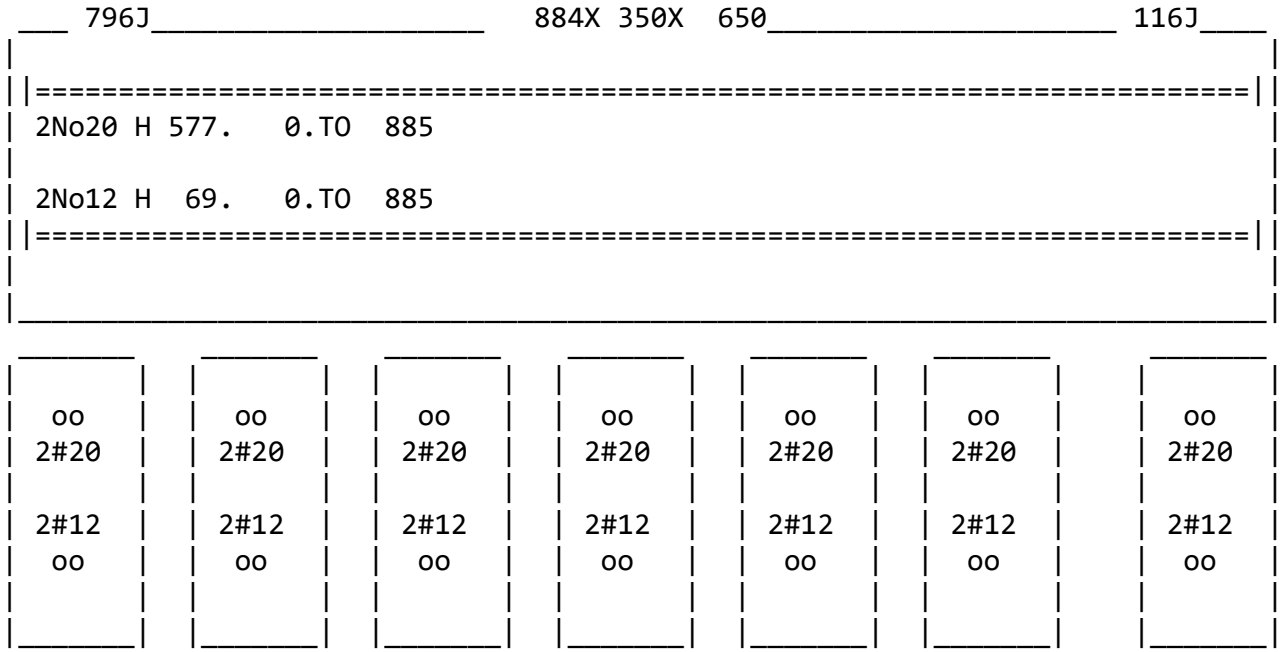
LEN - 885. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	885.	YES	YES
2	577.	2 - 20MM	0.	885.	YES	YES

B E A M N O. 1706 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1706 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1706 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1710 DESIGN RESULTS

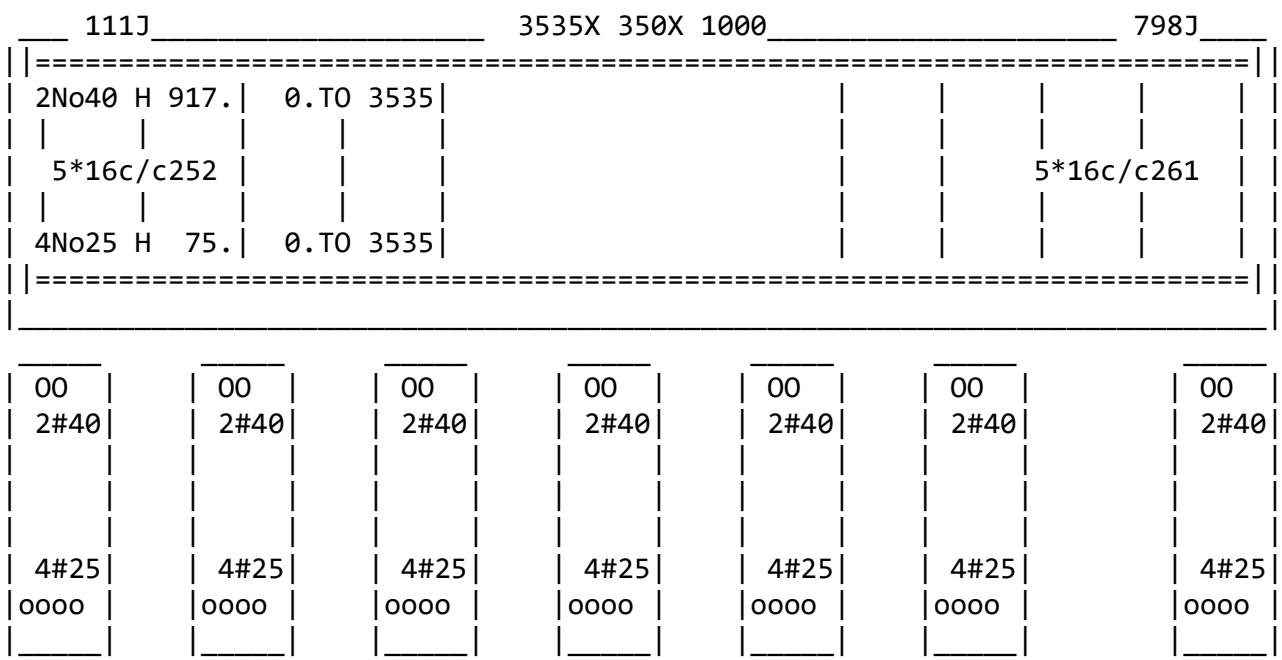
LEN - 3535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	3535.	YES	YES
2	917.	2 - 40MM	0.	3535.	YES	YES

B E A M N O. 1710 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 294.77 KNS Vc= 237.34 KNS Vs= 155.68 KNS
Tu= 67.62 KN-MET Tc= 12.4 KN-MET Ts= 90.2 KN-MET LOAD 411
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 16 MM 2-LEGGED STIRRUPS AT 252. MM C/C FOR 843. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 13.46 SQ.CM.

AT END SUPPORT - Vu= 278.09 KNS Vc= 237.34 KNS Vs= 133.44 KNS
Tu= 67.62 KN-MET Tc= 12.4 KN-MET Ts= 90.2 KN-MET LOAD 411
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 16 MM 2-LEGGED STIRRUPS AT 261. MM C/C FOR 843. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 13.46 SQ.CM.



ACI 318-11 BEAM NO. 1711 DESIGN RESULTS

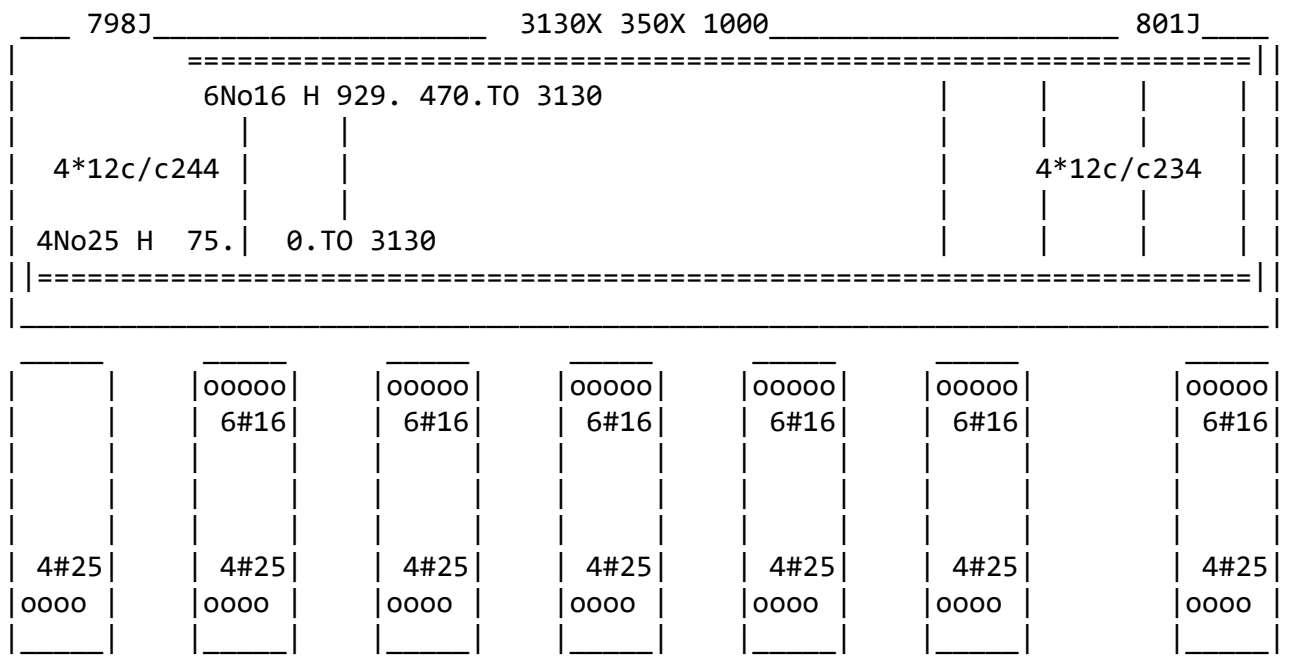
LEN - 3130. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	3130.	YES	YES
2	929.	6 - 16MM	470.	3130.	NO	YES

B E A M N O. 1711 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 269.72 KNS Vc= 228.33 KNS Vs= 131.29 KNS
Tu= 33.24 KN-MET Tc= 11.9 KN-MET Ts= 44.3 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 244. MM C/C FOR 640. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.61 SQ.CM.

AT END SUPPORT - Vu= 282.39 KNS Vc= 228.33 KNS Vs= 148.19 KNS
Tu= 33.24 KN-MET Tc= 11.9 KN-MET Ts= 44.3 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 234. MM C/C FOR 640. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.61 SQ.CM.



ACI 318-11 BEAM NO. 1712 DESIGN RESULTS

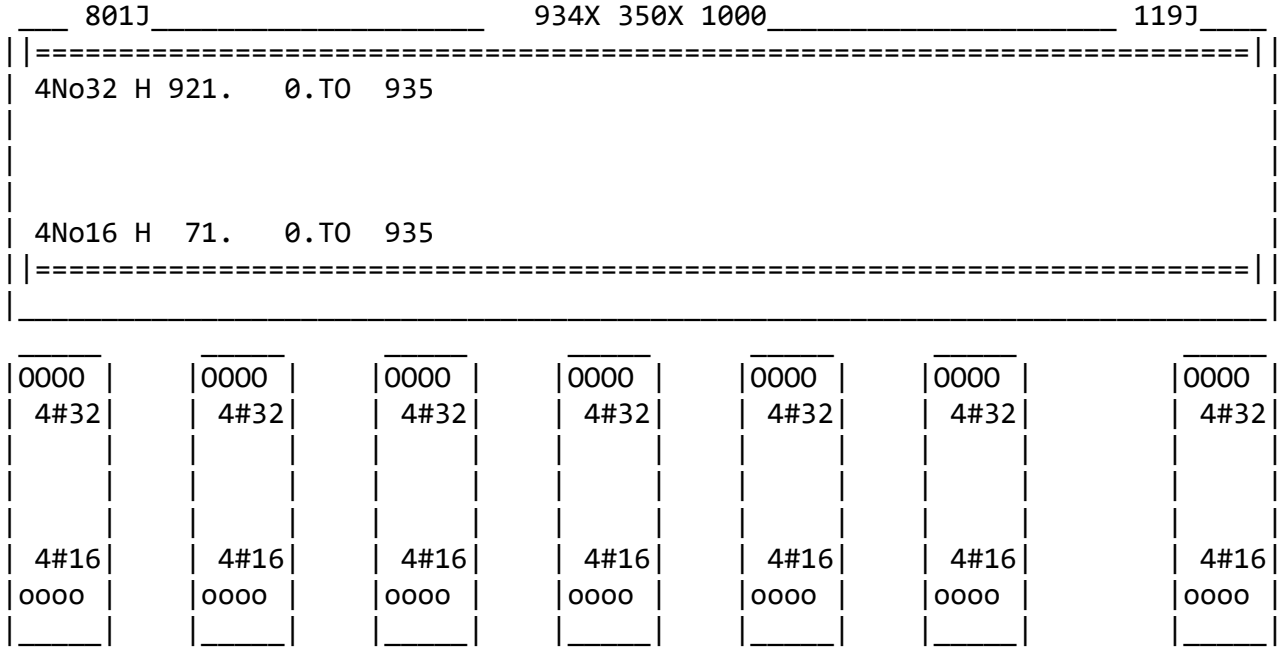
LEN - 935. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	935.	YES	YES
2	921.	4 - 32MM	0.	935.	YES	YES

B E A M N O. 1712 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1712 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1712 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1713 DESIGN RESULTS

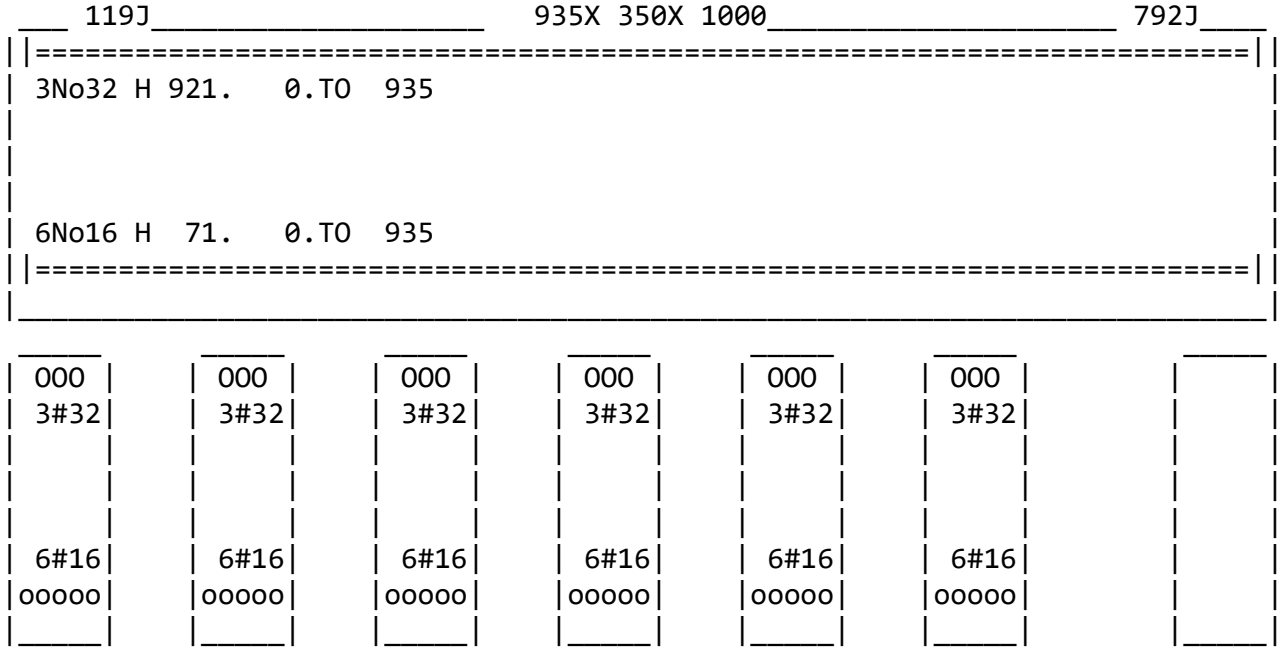
LEN - 935. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	935.	YES	YES
2	921.	3 - 32MM	0.	935.	YES	YES

B E A M N O. 1713 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1713 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1713 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1714 DESIGN RESULTS

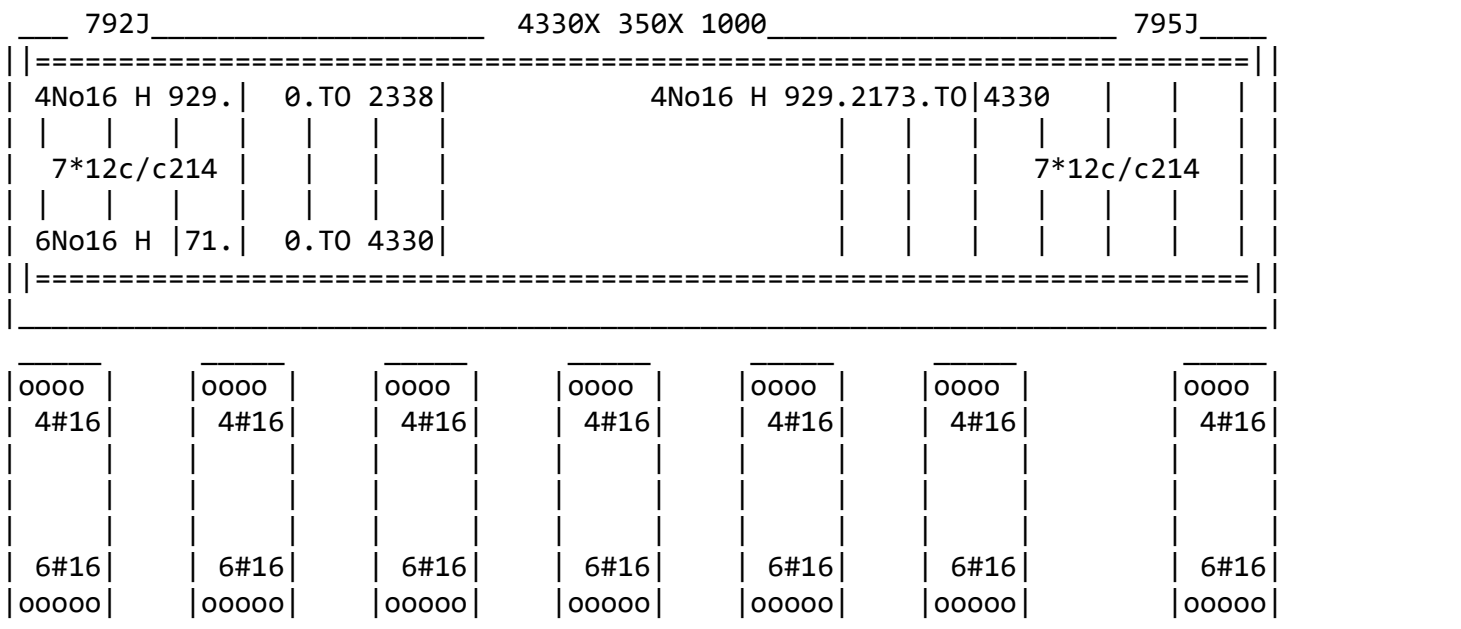
LEN - 4330. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	4330.	YES	YES
2	929.	4 - 16MM	0.	2338.	YES	NO
3	929.	4 - 16MM	2173.	4330.	NO	YES

B E A M N O. 1714 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.19 KNS Vc= 292.35 KNS Vs= 0.00 KNS
Tu= 59.85 KN-MET Tc= 15.4 KN-MET Ts= 79.8 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 1240. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 11.91 SQ.CM.

AT END SUPPORT - Vu= 38.83 KNS Vc= 292.35 KNS Vs= 0.00 KNS
Tu= 59.85 KN-MET Tc= 15.4 KN-MET Ts= 79.8 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 1240. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 11.91 SQ.CM.



ACI 318-11 BEAM NO. 1715 DESIGN RESULTS

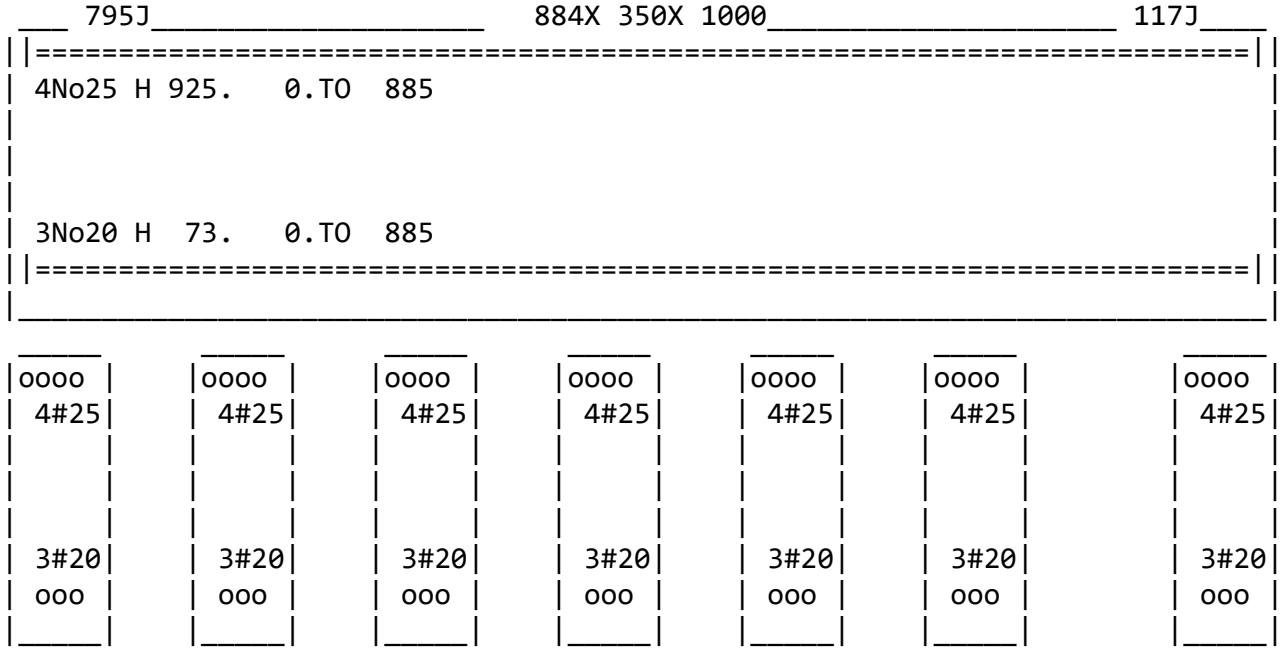
LEN - 885. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	885.	YES	YES
2	925.	4 - 25MM	0.	885.	YES	YES

B E A M N O. 1715 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1715 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1715 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1716 DESIGN RESULTS

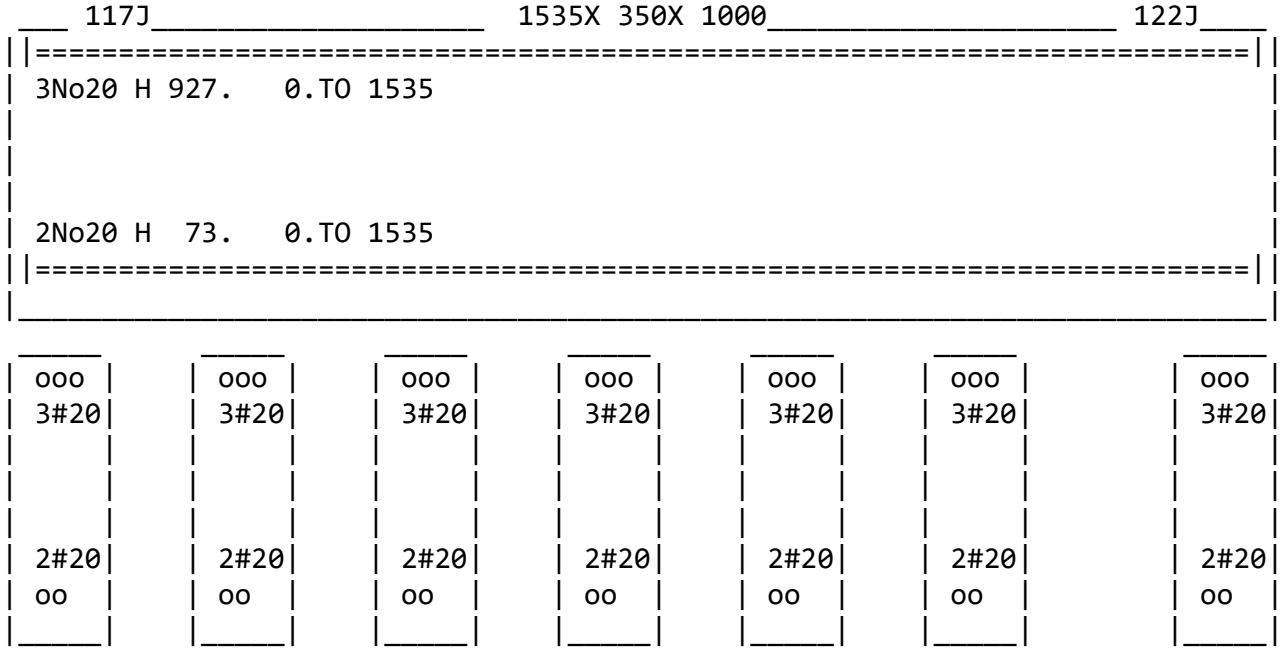
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1000. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	2 - 20MM	0.	1535.	YES	YES
2	927.	3 - 20MM	0.	1535.	YES	YES

B E A M N O. 1716 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1716 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1716 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1719 DESIGN RESULTS

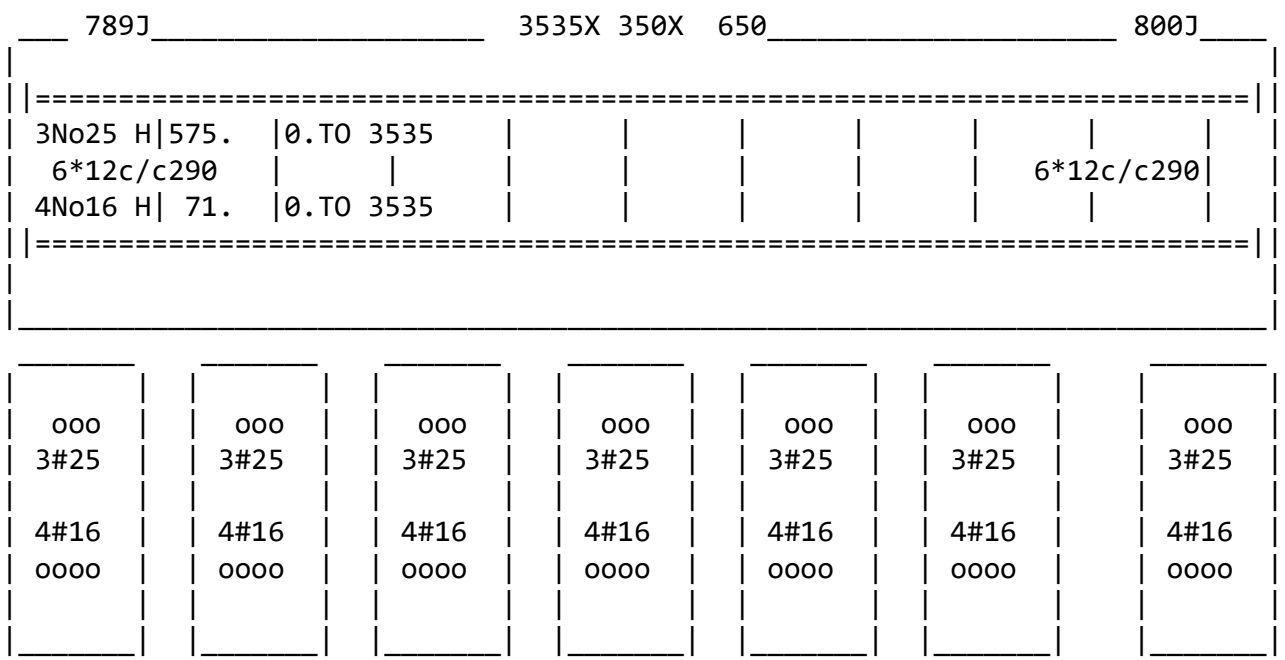
LEN - 3535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	3535.	YES	YES
2	575.	3 - 25MM	0.	3535.	YES	YES

B E A M N O. 1719 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 78.84 KNS Vc= 195.41 KNS Vs= 0.00 KNS
Tu= 1.49 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 1193. MM

AT END SUPPORT - Vu= 83.01 KNS Vc= 191.63 KNS Vs= 0.00 KNS
Tu= 1.64 KN-MET Tc= 9.2 KN-MET Ts= 0.0 KN-MET LOAD 414
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 1193. MM



ACI 318-11 BEAM NO. 1720 DESIGN RESULTS

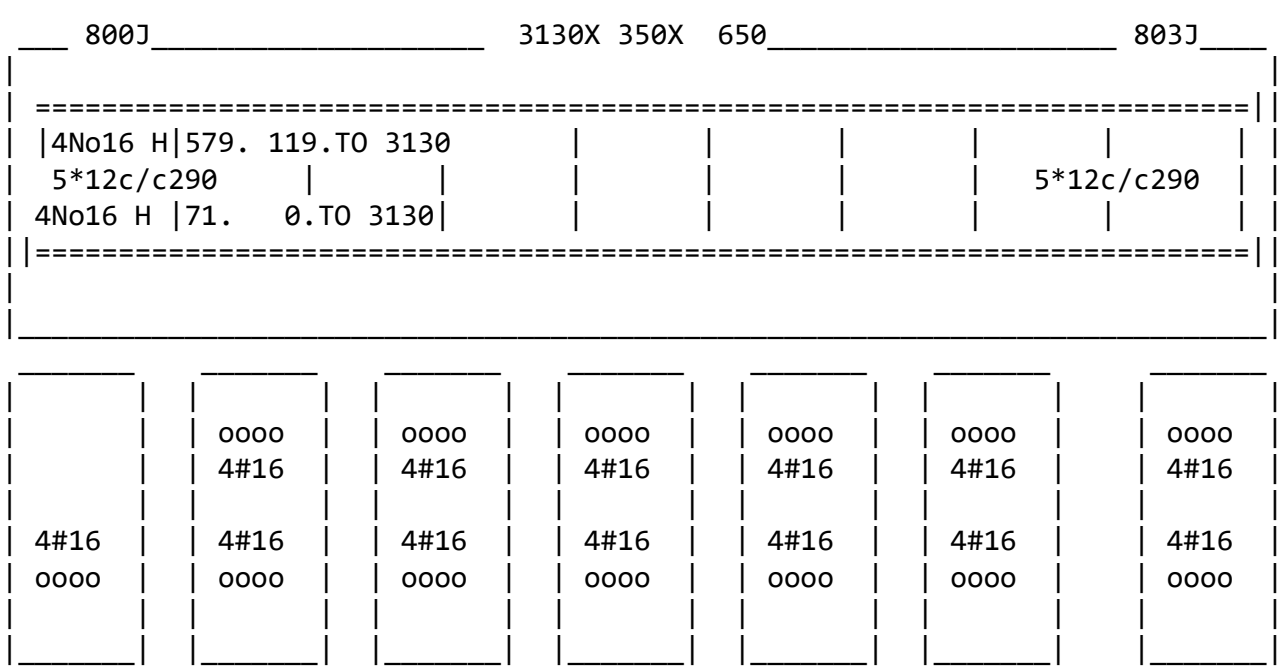
LEN - 3130. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	3130.	YES	YES
2	579.	4 - 16MM	119.	3130.	NO	YES

B E A M N O. 1720 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.71 KNS Vc= 177.70 KNS Vs= 0.00 KNS
Tu= 1.48 KN-MET Tc= 8.6 KN-MET Ts= 0.0 KN-MET LOAD 413
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 990. MM

AT END SUPPORT - Vu= 68.37 KNS Vc= 181.47 KNS Vs= 0.00 KNS
Tu= 1.63 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 990. MM



ACI 318-11 BEAM NO. 1721 DESIGN RESULTS

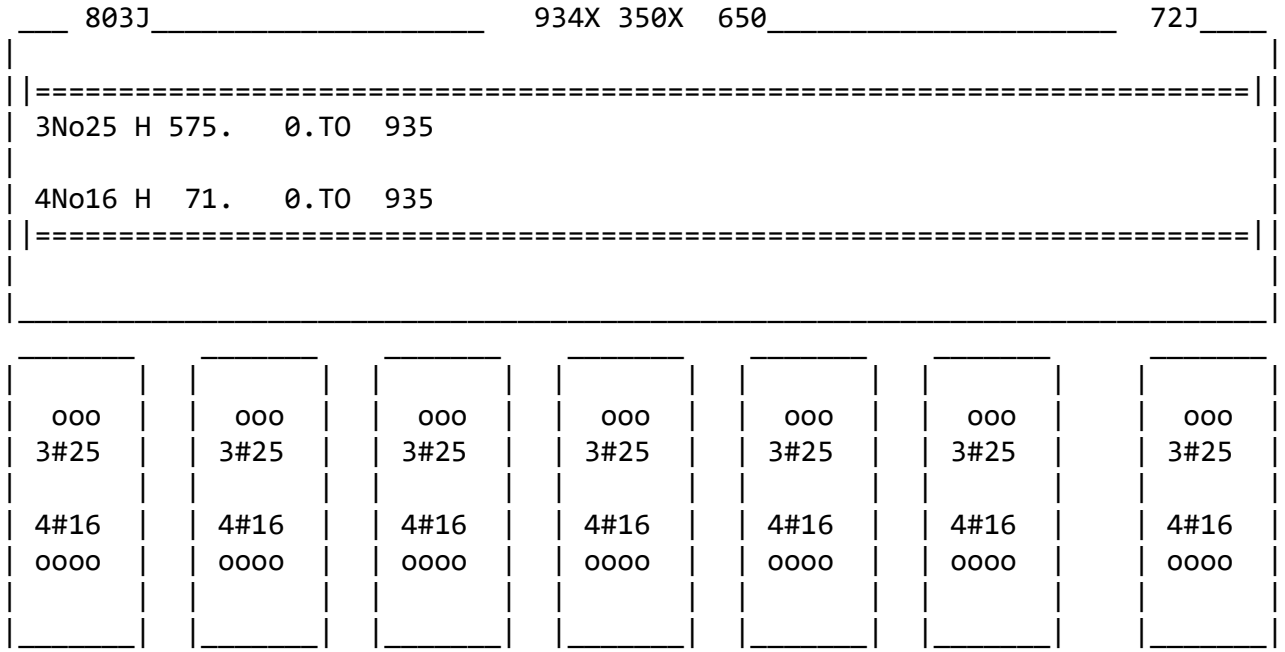
LEN - 935. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	935.	YES	YES
2	575.	3 - 25MM	0.	935.	YES	YES

B E A M N O. 1721 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1721 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1721 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1722 DESIGN RESULTS

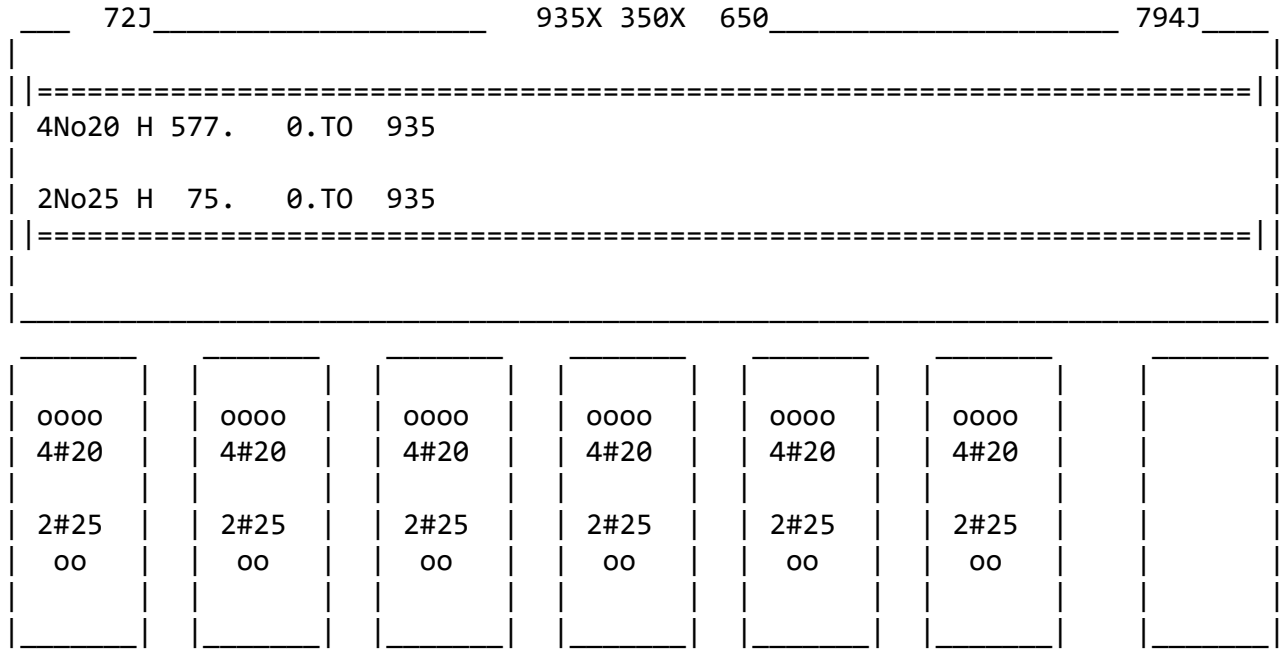
LEN - 935. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	935.	YES	YES
2	577.	4 - 20MM	0.	935.	YES	YES

B E A M N O. 1722 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1722 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1722 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1723 DESIGN RESULTS

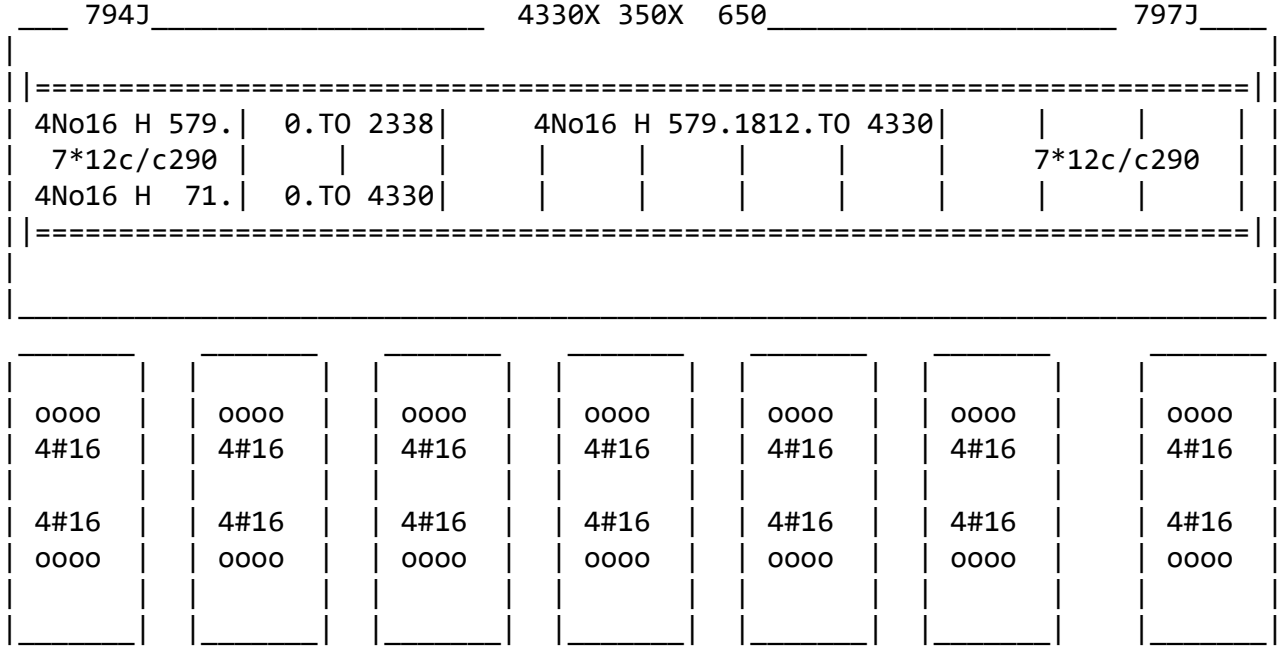
LEN - 4330. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	4330.	YES	YES
2	579.	4 - 16MM	0.	2338.	YES	NO
3	579.	4 - 16MM	1812.	4330.	NO	YES

B E A M N O. 1723 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 75.14 KNS Vc= 195.30 KNS Vs= 0.00 KNS
Tu= 3.67 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 1590. MM

AT END SUPPORT - Vu= 85.62 KNS Vc= 188.08 KNS Vs= 0.00 KNS
Tu= 0.76 KN-MET Tc= 9.0 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 1590. MM



ACI 318-11 BEAM NO. 1724 DESIGN RESULTS

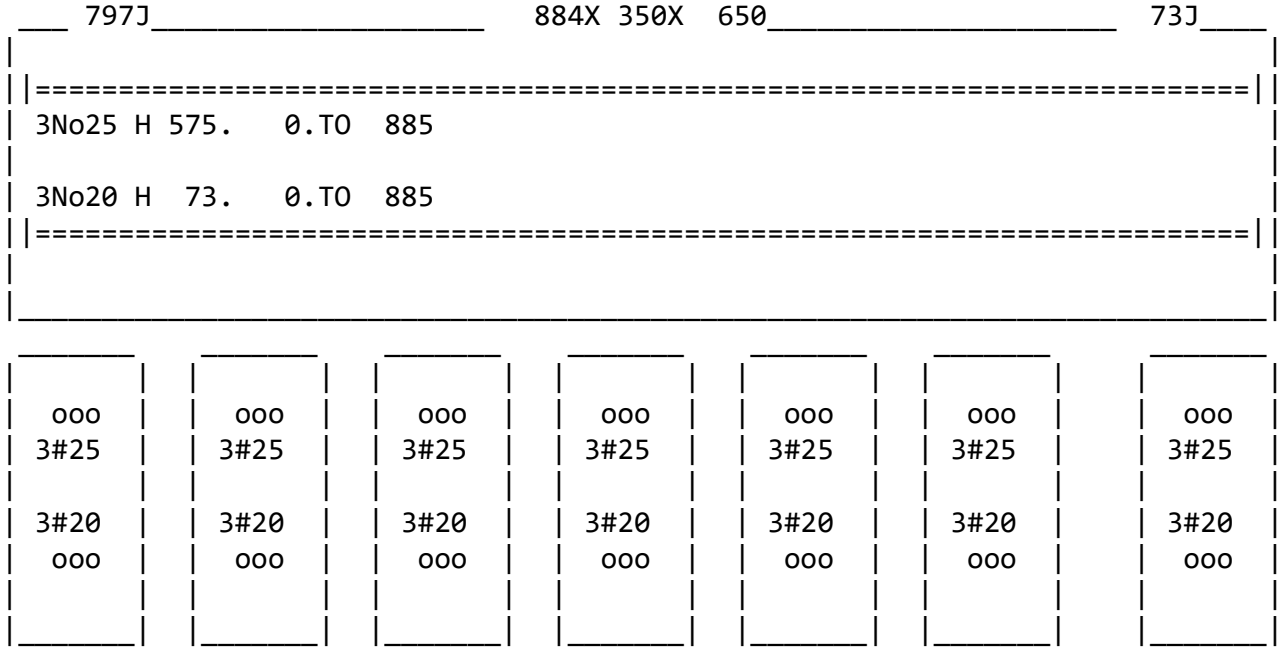
LEN - 885. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	885.	YES	YES
2	575.	3 - 25MM	0.	885.	YES	YES

B E A M N O. 1724 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1724 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1724 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1725 DESIGN RESULTS

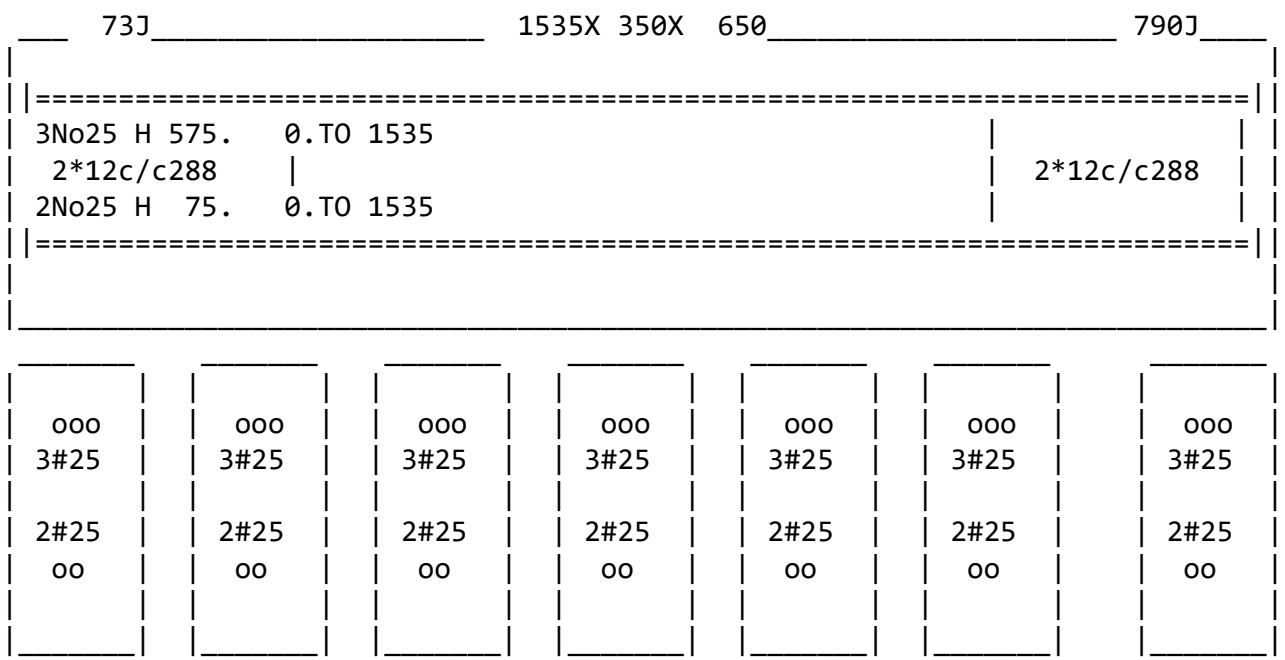
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1535.	YES	YES
2	575.	3 - 25MM	0.	1535.	YES	YES

B E A M N O. 1725 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 111.21 KNS Vc= 183.20 KNS Vs= 0.00 KNS
Tu= 0.26 KN-MET Tc= 8.9 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 288. MM C/C FOR 193. MM

AT END SUPPORT - Vu= 108.31 KNS Vc= 183.20 KNS Vs= 0.00 KNS
Tu= 0.26 KN-MET Tc= 8.9 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 288. MM C/C FOR 193. MM



ACI 318-11 BEAM NO. 1726 DESIGN RESULTS

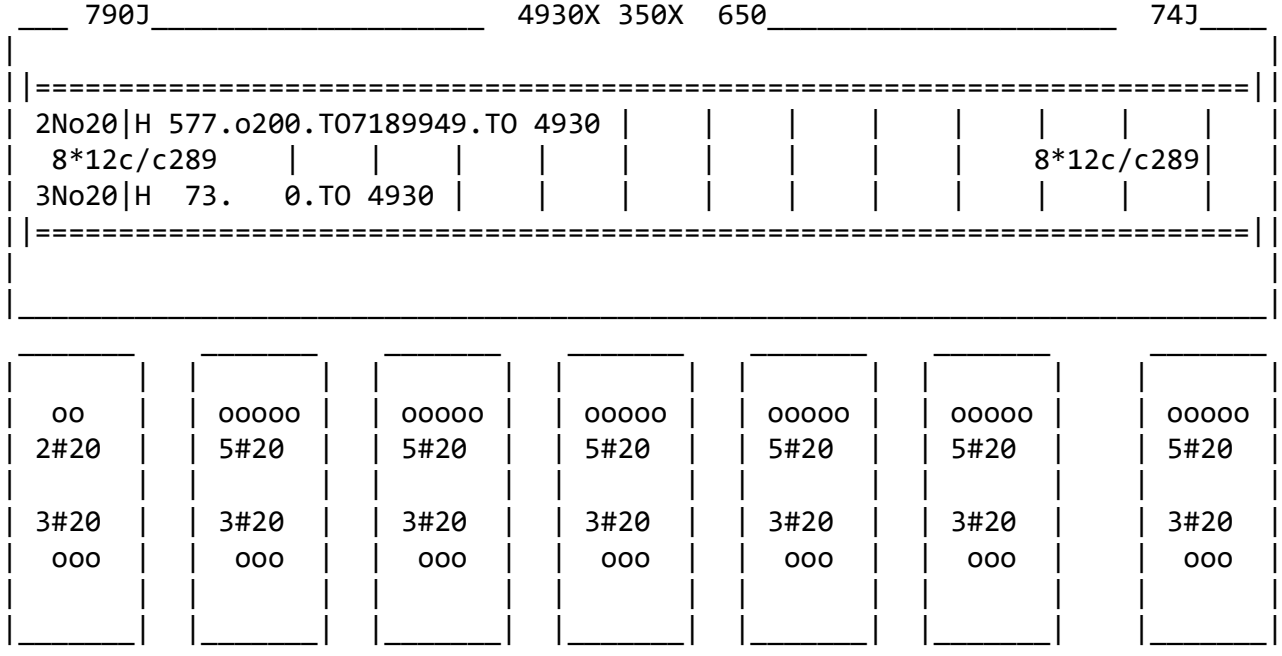
LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	4930.	YES	YES
2	577.	2 - 20MM	0.	1899.	YES	NO
3	577.	5 - 20MM	749.	4930.	NO	YES

B E A M N O. 1726 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 70.77 KNS Vc= 187.01 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 9.0 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 289. MM C/C FOR 1890. MM

AT END SUPPORT - Vu= 95.09 KNS Vc= 187.01 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 9.0 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 289. MM C/C FOR 1890. MM



ACI 318-11 BEAM NO. 2104 DESIGN RESULTS

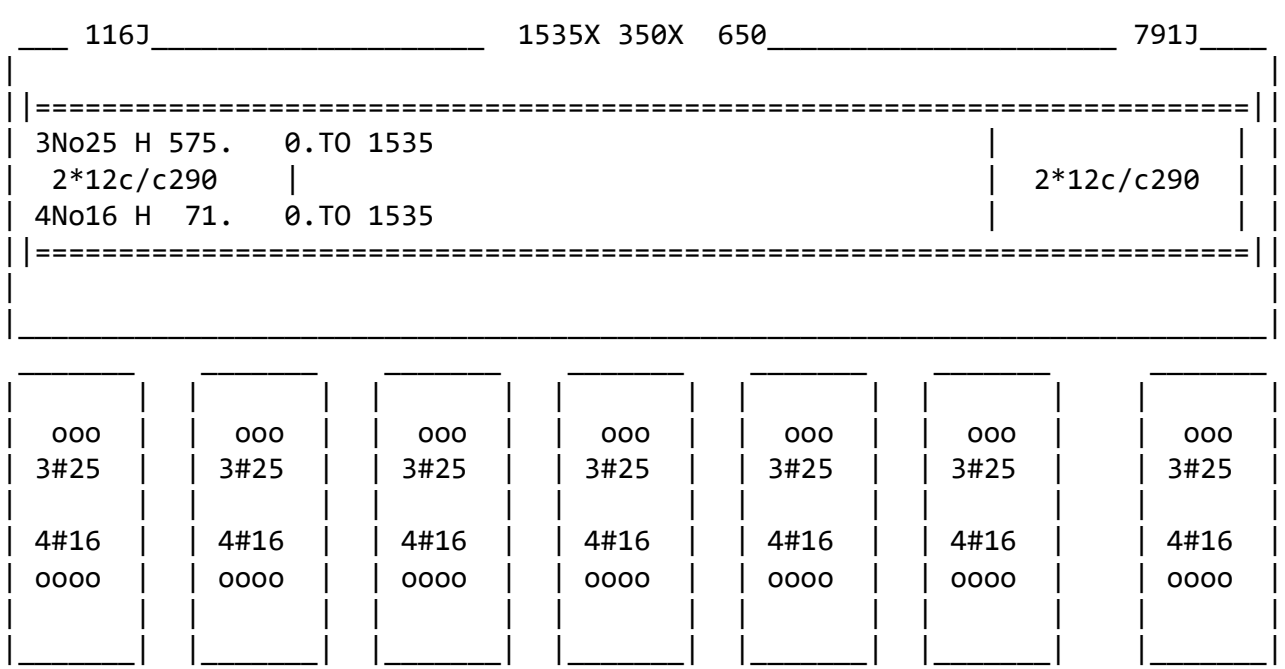
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	1535.	YES	YES
2	575.	3 - 25MM	0.	1535.	YES	YES

B E A M N O. 2104 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 275.78 KNS Vc= 180.22 KNS Vs= 187.48 KNS
Tu= 1.53 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 193. MM

AT END SUPPORT - Vu= 272.88 KNS Vc= 180.22 KNS Vs= 183.62 KNS
Tu= 1.53 KN-MET Tc= 8.7 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 193. MM



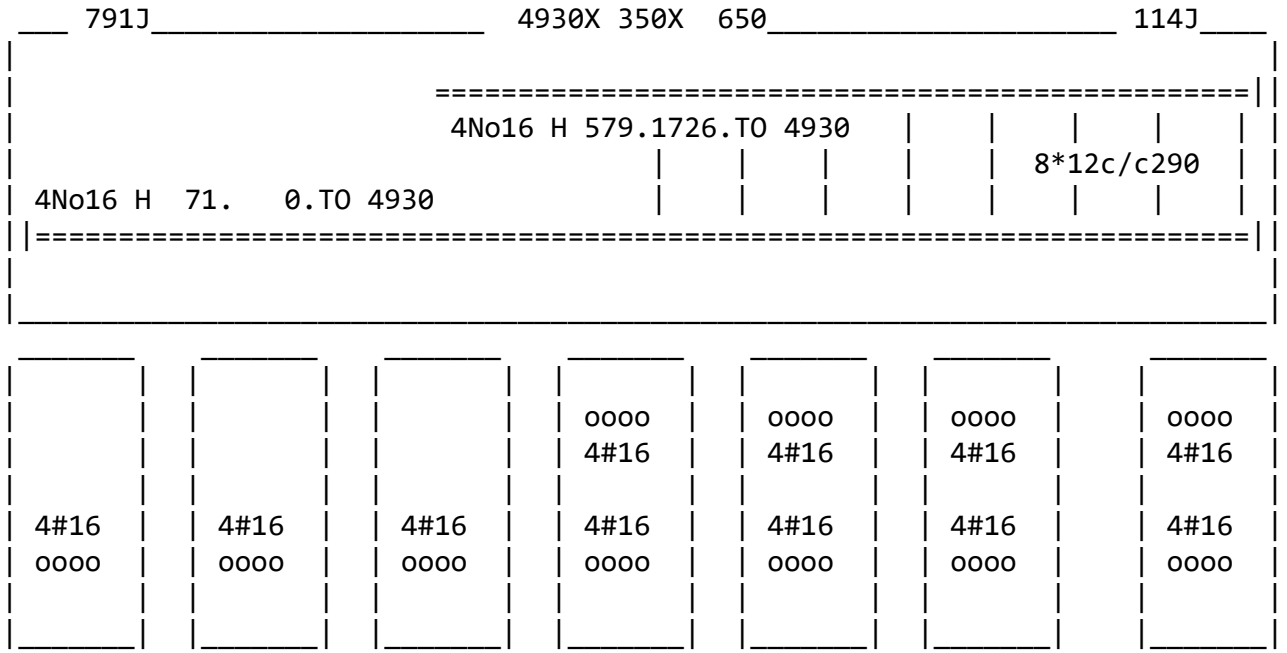
ACI 318-11 BEAM NO. 2105 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	4930.	YES	YES
2	579.	4 - 16MM	1726.	4930.	NO	YES

B E A M N O. 2105 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.91 KNS Vc= 193.29 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 9.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 72.38 KNS Vc= 182.74 KNS Vs= 0.00 KNS
Tu= 1.29 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 1890. MM



**********END OF BEAM DESIGN**********

2225. DESIGN BEAM 318 1024 1322 1892 TO 1917 1974 TO 1978 1980 TO 1984 2151 TO 2191 -
2226. 2201 TO 2312 2328 TO 2330

ACI 318-11 BEAM NO. 1322 DESIGN RESULTS

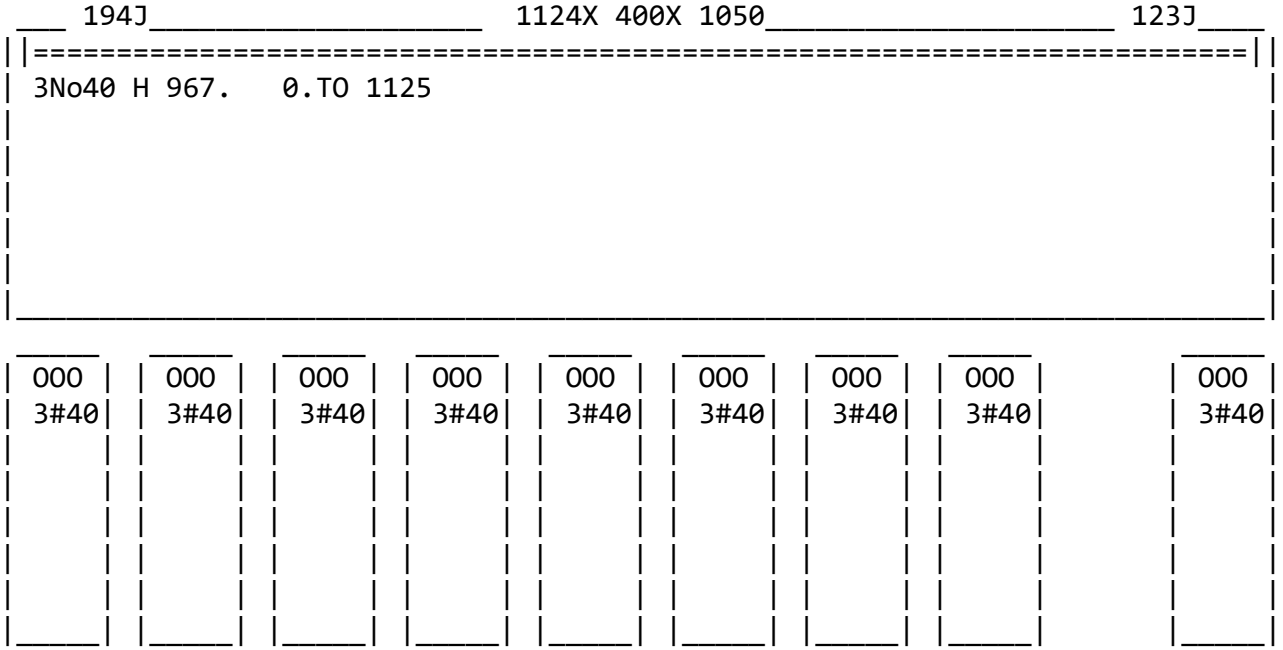
LEN - 1125. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	967.	3 - 40MM	0.	1125.	YES	YES

B E A M N O. 1322 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1322 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1322 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1892 DESIGN RESULTS

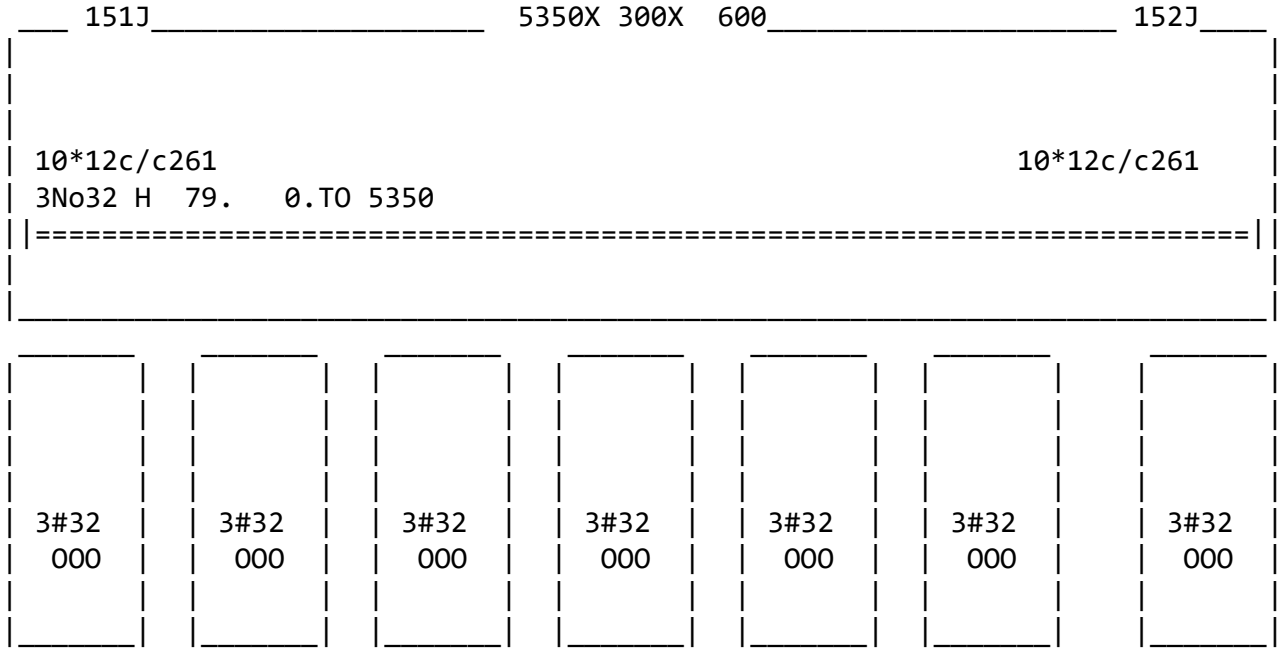
LEN - 5350. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	5350.	YES	YES

B E A M N O. 1892 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 169.65 KNS Vc= 152.67 KNS Vs= 73.53 KNS
Tu= 1.44 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 261. MM C/C FOR 2150. MM

AT END SUPPORT - Vu= 163.53 KNS Vc= 152.67 KNS Vs= 65.37 KNS
Tu= 1.44 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 261. MM C/C FOR 2150. MM



ACI 318-11 BEAM NO. 1893 DESIGN RESULTS

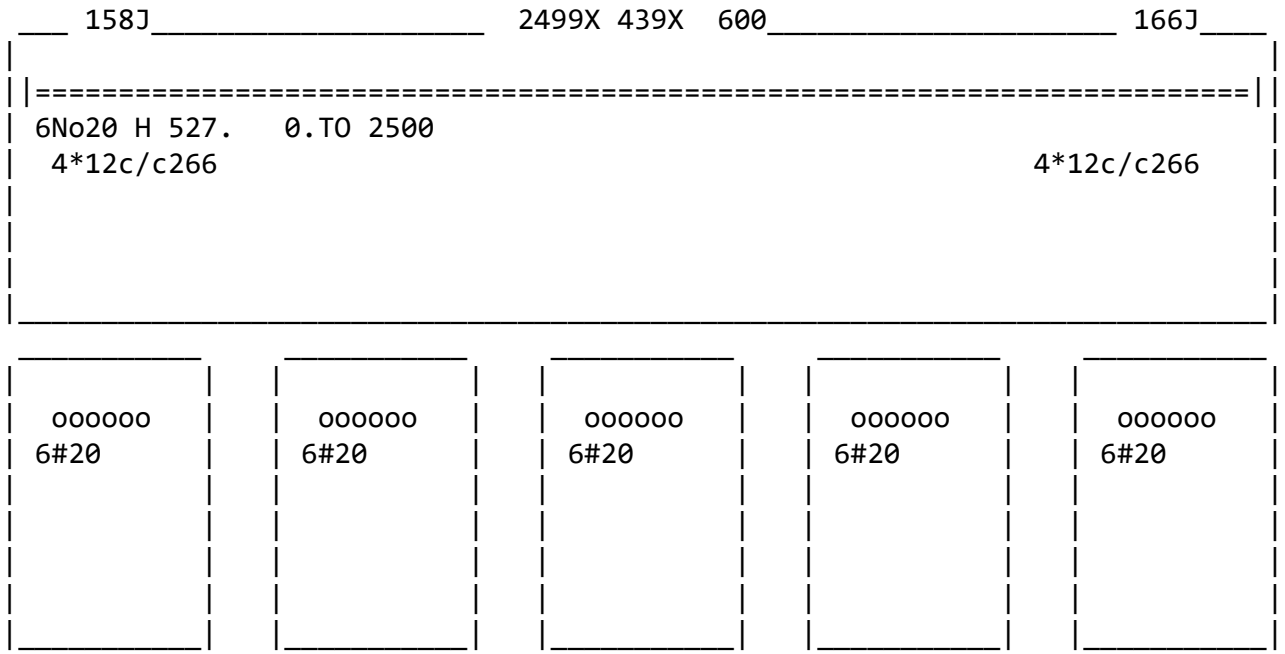
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	527.	6 - 20MM	0.	2500.	YES	YES

B E A M N O. 1893 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 87.26 KNS Vc= 218.48 KNS Vs= 0.00 KNS
Tu= 0.43 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 725. MM

AT END SUPPORT - Vu= 91.28 KNS Vc= 218.48 KNS Vs= 0.00 KNS
Tu= 0.63 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 725. MM



ACI 318-11 BEAM NO. 1894 DESIGN RESULTS

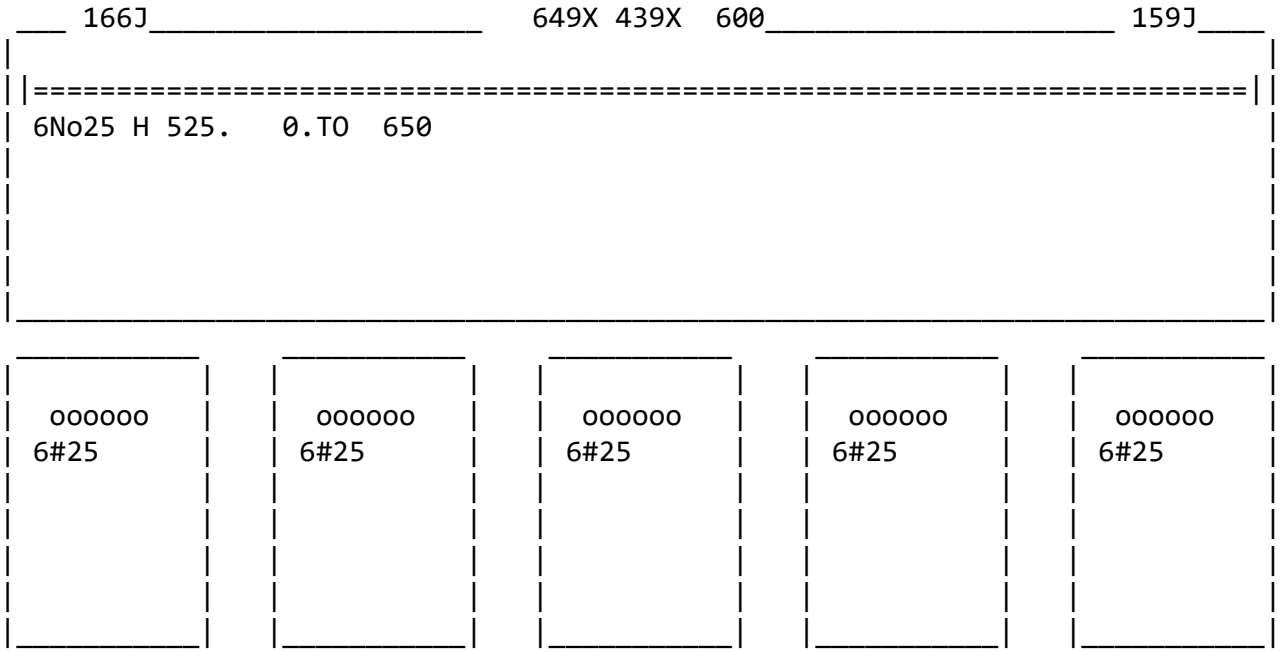
LEN - 650. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	525.	6 - 25MM	0.	650.	YES	YES

B E A M N O. 1894 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1894 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1894 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1895 DESIGN RESULTS

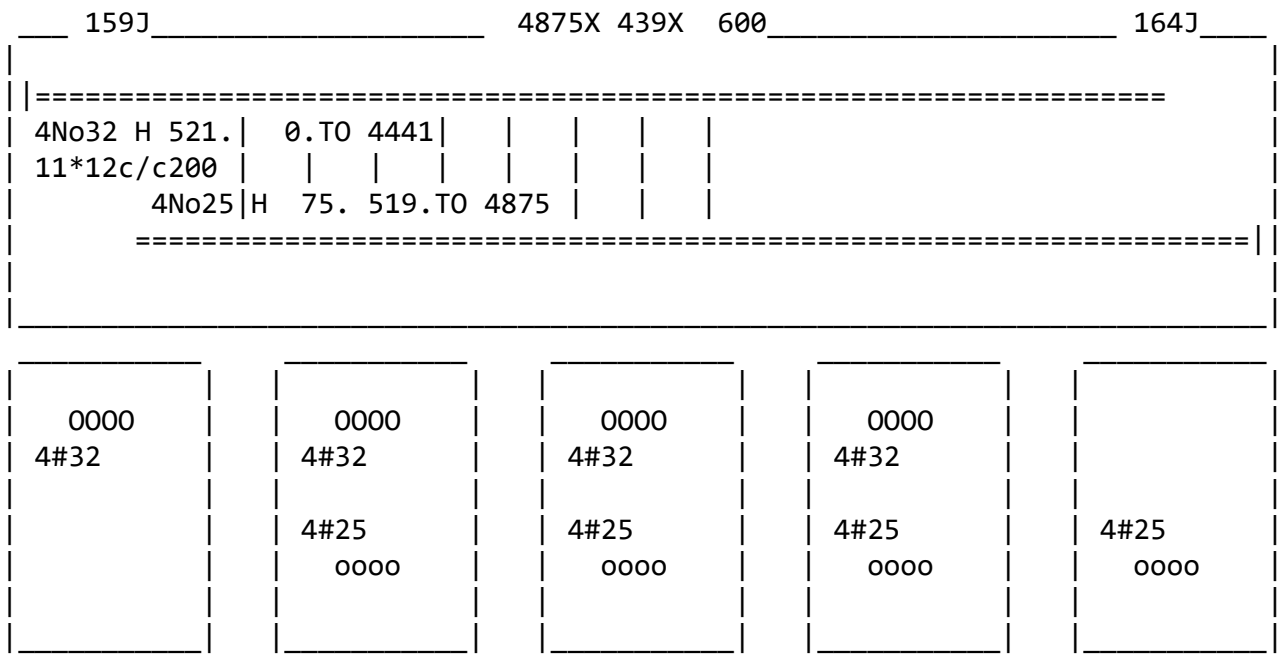
LEN - 4875. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	519.	4875.	NO	YES
2	521.	4 - 32MM	0.	4441.	YES	NO

B E A M N O. 1895 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 357.13 KNS Vc= 225.96 KNS Vs= 250.21 KNS
Tu= 4.75 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 200. MM C/C FOR 1913. MM

AT END SUPPORT - Vu= 29.84 KNS Vc= 219.02 KNS Vs= 0.00 KNS
Tu= 1.57 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1896 DESIGN RESULTS

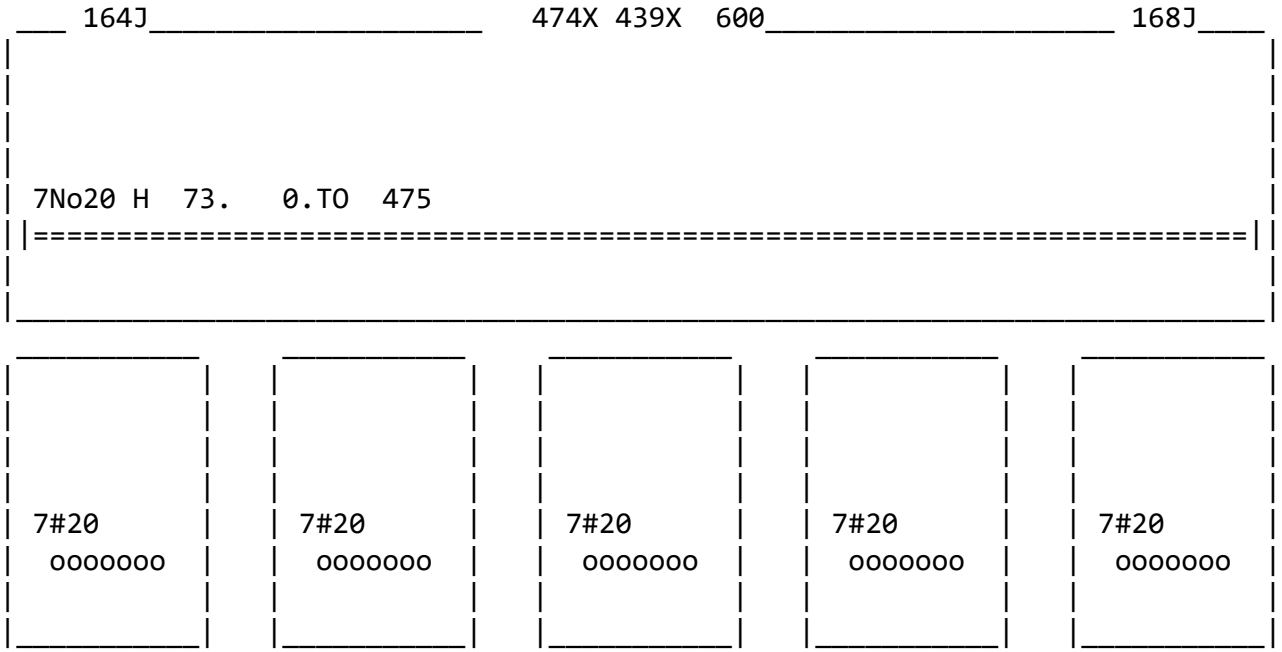
LEN - 475. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	7 - 20MM	0.	475.	YES	YES

B E A M N O. 1896 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1896 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1896 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1897 DESIGN RESULTS

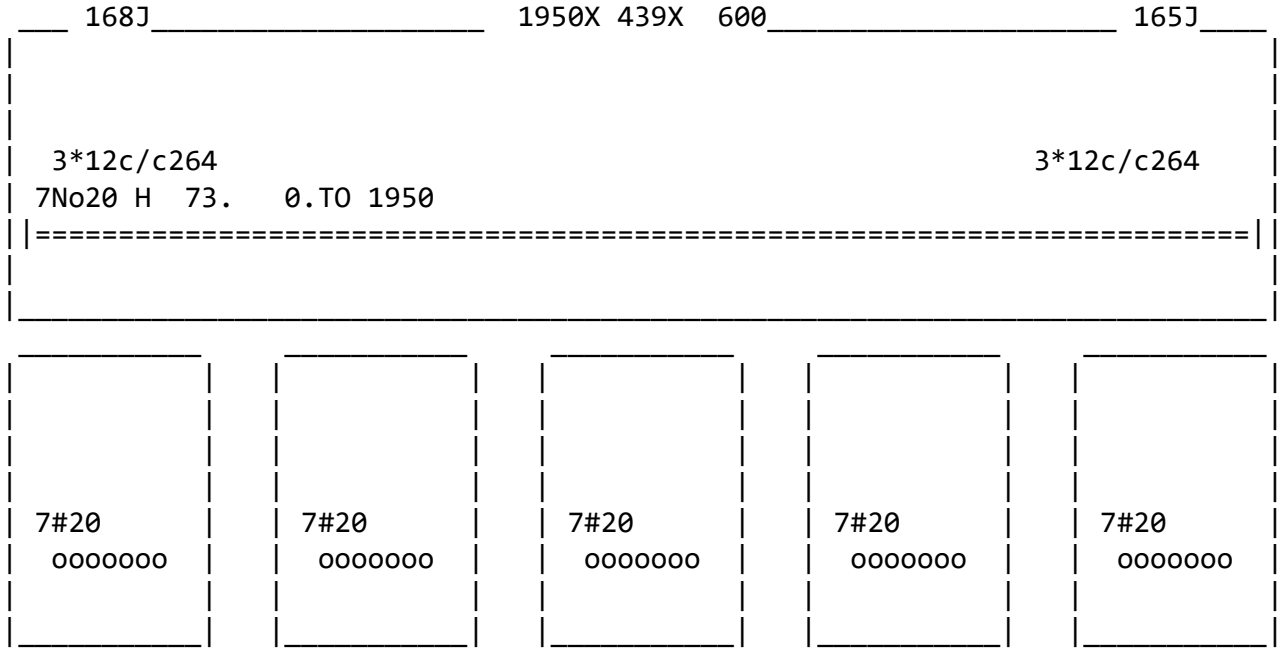
LEN - 1950. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	7 - 20MM	0.	1950.	YES	YES

B E A M N O. 1897 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 89.18 KNS Vc= 222.24 KNS Vs= 0.00 KNS
Tu= 3.95 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 450. MM

AT END SUPPORT - Vu= 94.06 KNS Vc= 227.17 KNS Vs= 0.00 KNS
Tu= 4.35 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 450. MM



ACI 318-11 BEAM NO. 1898 DESIGN RESULTS

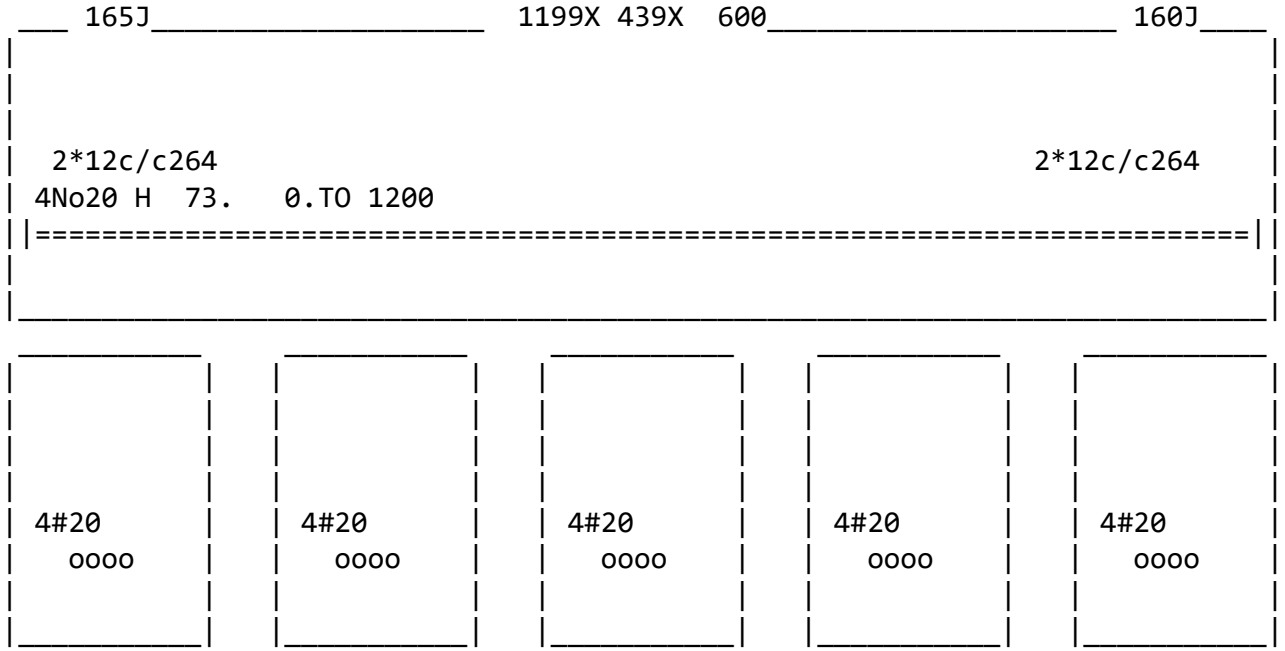
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	1200.	YES	YES

B E A M N O. 1898 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 95.24 KNS Vc= 232.80 KNS Vs= 0.00 KNS
Tu= 1.57 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 75. MM

AT END SUPPORT - Vu= 97.72 KNS Vc= 237.72 KNS Vs= 0.00 KNS
Tu= 1.57 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 75. MM



ACI 318-11 BEAM NO. 1899 DESIGN RESULTS

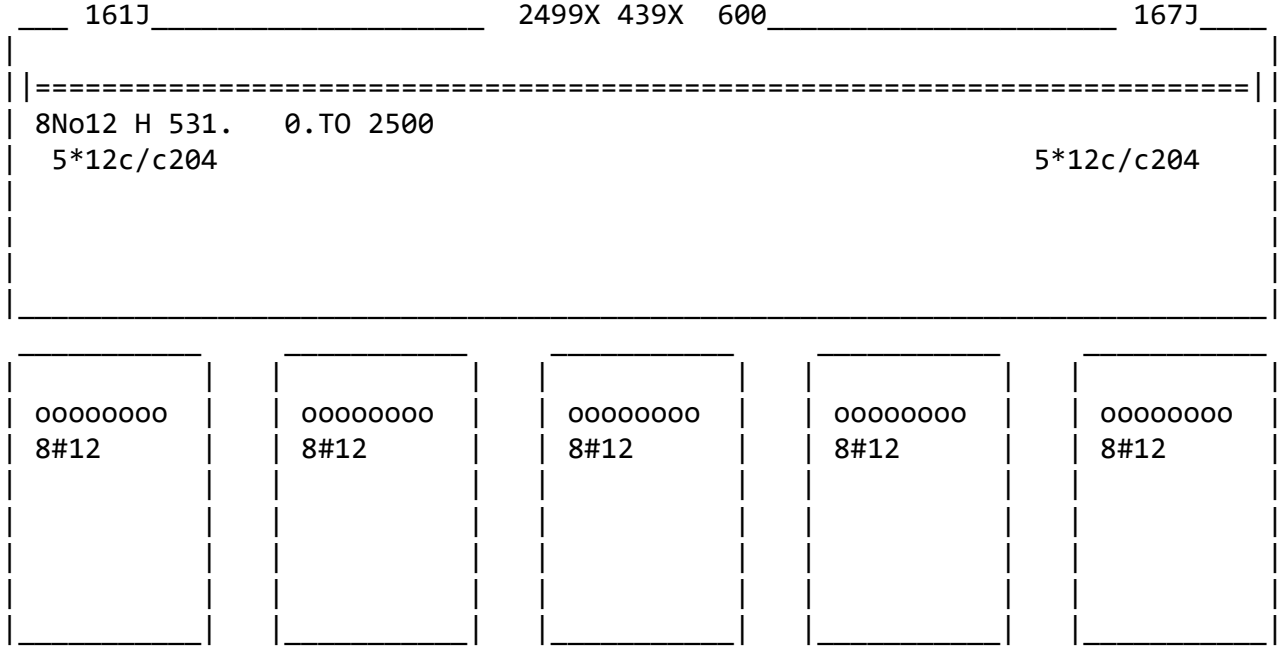
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	531.	8 - 12MM	0.	2500.	YES	YES

B E A M N O. 1899 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 32.35 KNS Vc= 225.95 KNS Vs= 0.00 KNS
Tu= 23.08 KN-MET Tc= 12.2 KN-MET Ts= 30.8 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 725. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.39 SQ.CM.

AT END SUPPORT - Vu= 67.33 KNS Vc= 225.95 KNS Vs= 0.00 KNS
Tu= 23.08 KN-MET Tc= 12.2 KN-MET Ts= 30.8 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 725. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.39 SQ.CM.



ACI 318-11 BEAM NO. 1900 DESIGN RESULTS

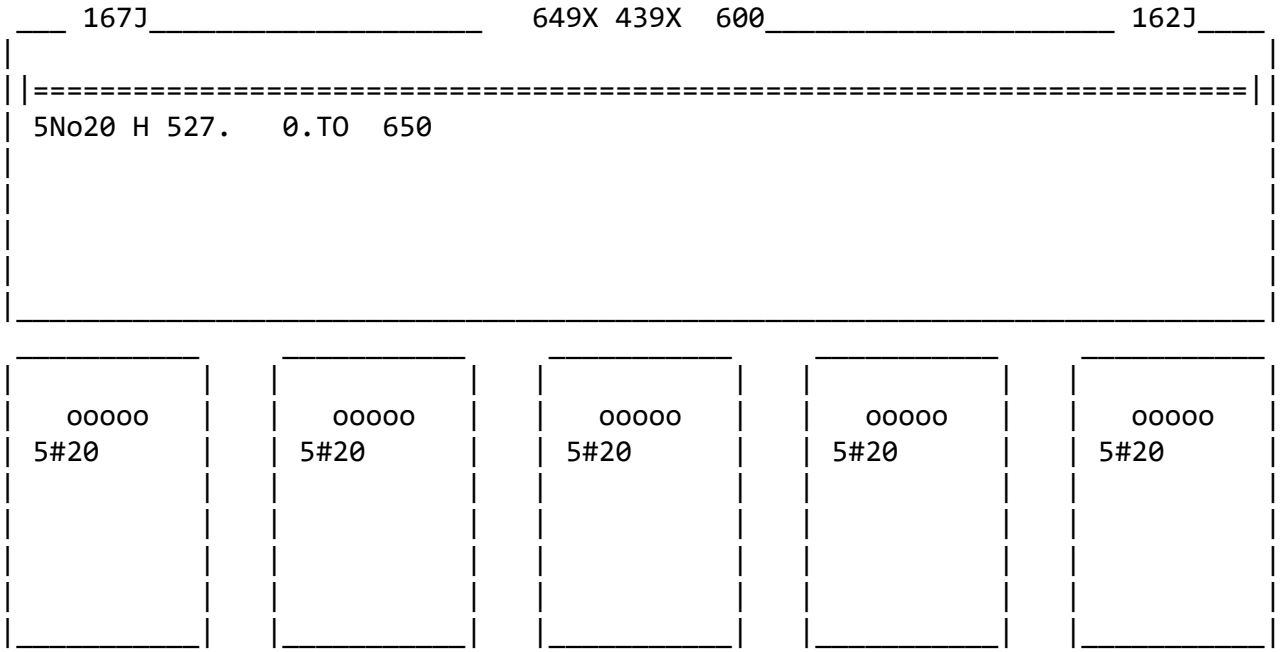
LEN - 650. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	527.	5 - 20MM	0.	650.	YES	YES

B E A M N O. 1900 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1900 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1900 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1901 DESIGN RESULTS

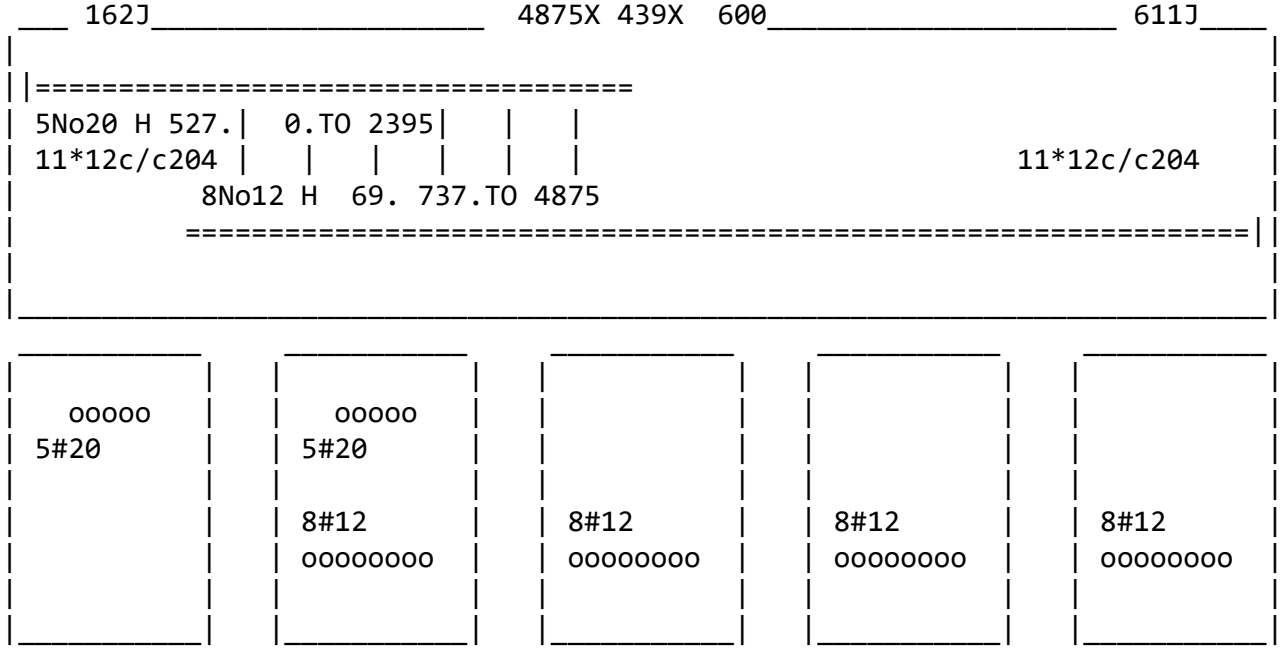
LEN - 4875. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	737.	4875.	NO	YES
2	527.	5 - 20MM	0.	2395.	YES	NO

B E A M N O. 1901 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 217.26 KNS Vc= 215.21 KNS Vs= 74.47 KNS
Tu= 12.76 KN-MET Tc= 11.6 KN-MET Ts= 17.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 1913. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.

AT END SUPPORT - Vu= 44.20 KNS Vc= 216.91 KNS Vs= 0.00 KNS
Tu= 12.76 KN-MET Tc= 11.7 KN-MET Ts= 17.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 1913. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.



ACI 318-11 BEAM NO. 1902 DESIGN RESULTS

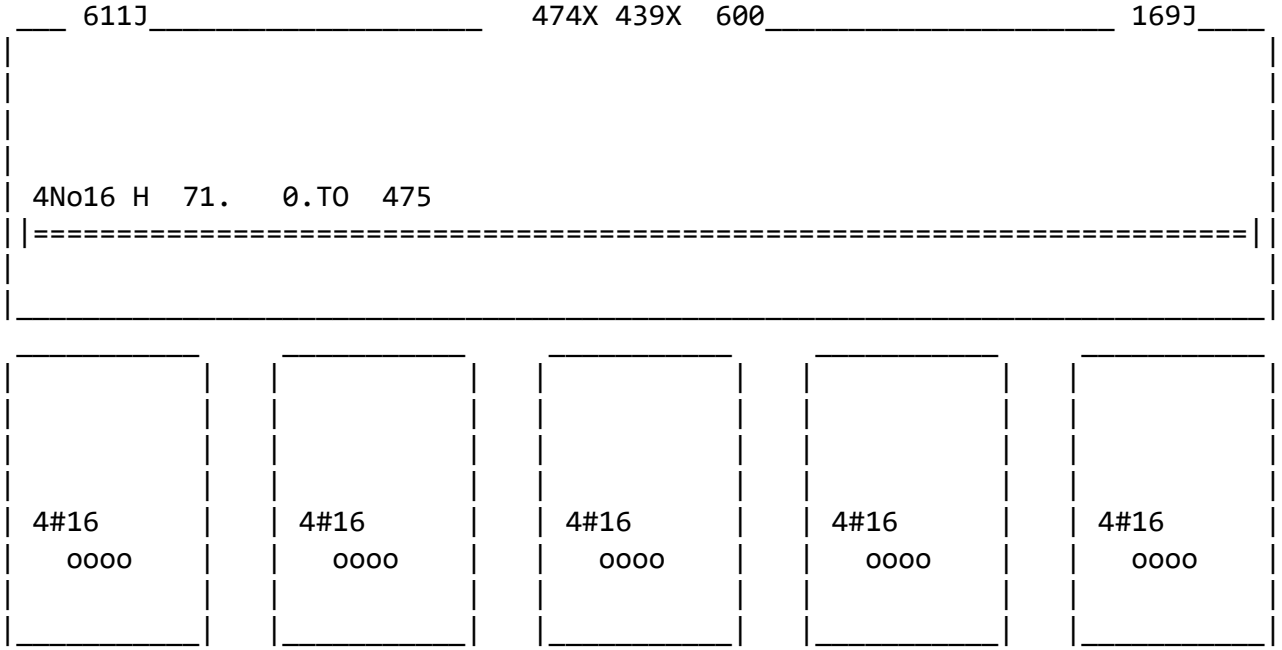
LEN - 475. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	475.	YES	YES

B E A M N O. 1902 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1902 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1902 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1903 DESIGN RESULTS

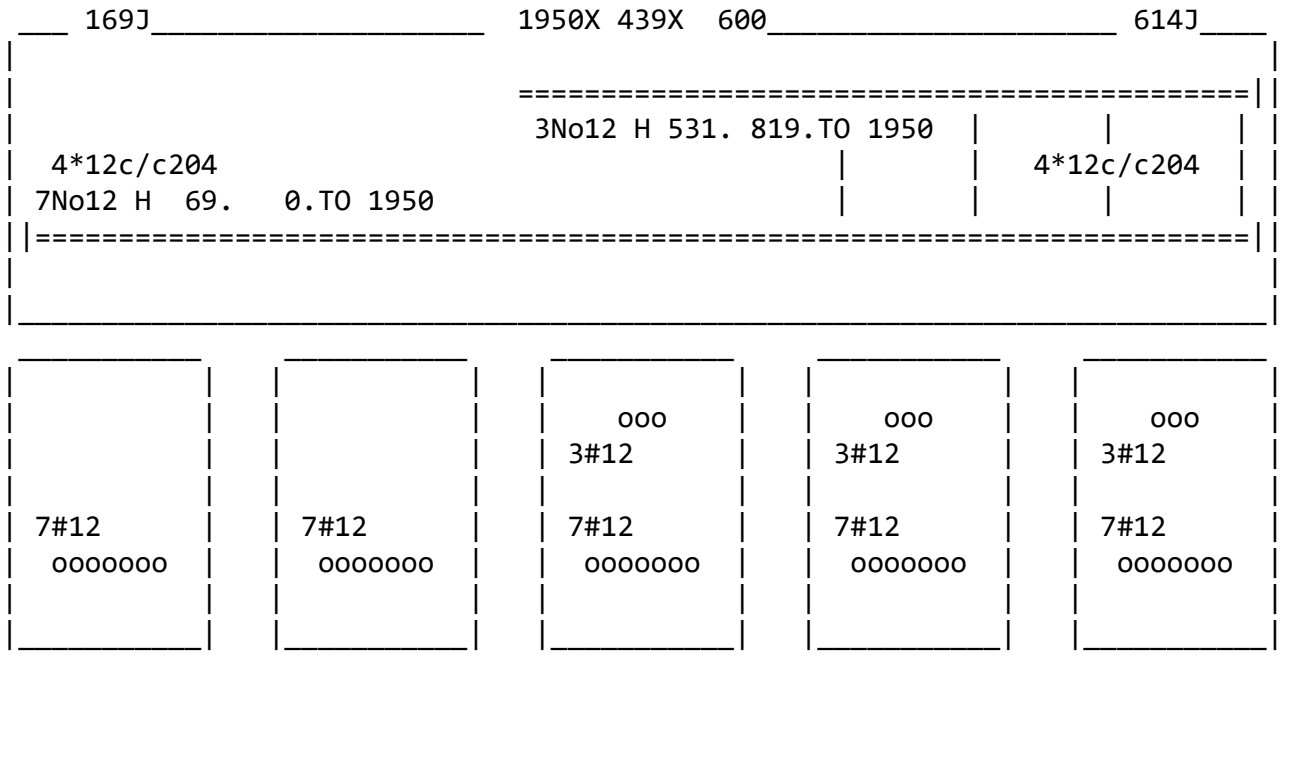
LEN - 1950. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	7 - 12MM	0.	1950.	YES	YES
2	531.	3 - 12MM	819.	1950.	NO	YES

B E A M N O. 1903 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 38.69 KNS Vc= 221.13 KNS Vs= 0.00 KNS
Tu= 12.76 KN-MET Tc= 11.9 KN-MET Ts= 17.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 450. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.

AT END SUPPORT - Vu= 63.94 KNS Vc= 221.13 KNS Vs= 0.00 KNS
Tu= 12.76 KN-MET Tc= 11.9 KN-MET Ts= 17.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 450. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.



ACI 318-11 BEAM NO. 1904 DESIGN RESULTS

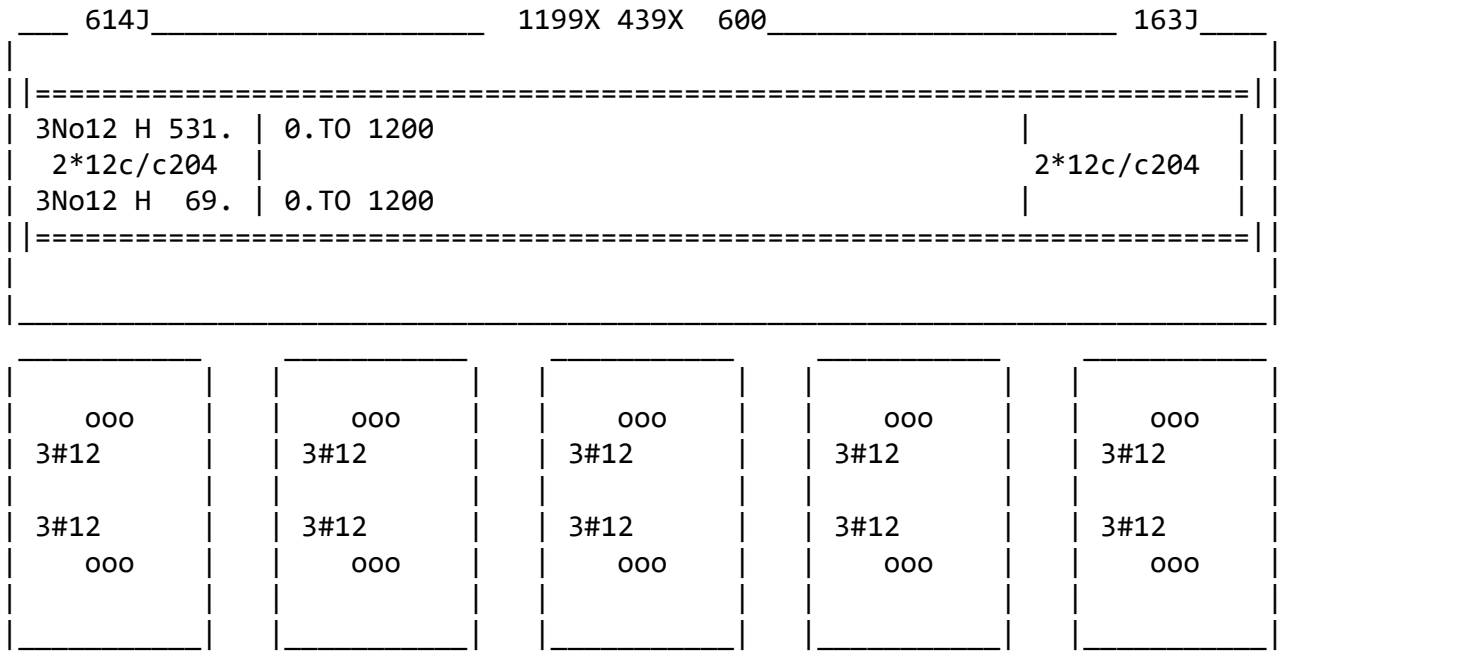
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1200.	YES	YES
2	531.	3 - 12MM	0.	1200.	YES	YES

B E A M N O. 1904 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.08 KNS Vc= 224.44 KNS Vs= 0.00 KNS
Tu= 12.76 KN-MET Tc= 12.1 KN-MET Ts= 17.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 75. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.

AT END SUPPORT - Vu= 11.21 KNS Vc= 224.44 KNS Vs= 0.00 KNS
Tu= 12.76 KN-MET Tc= 12.1 KN-MET Ts= 17.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 75. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.



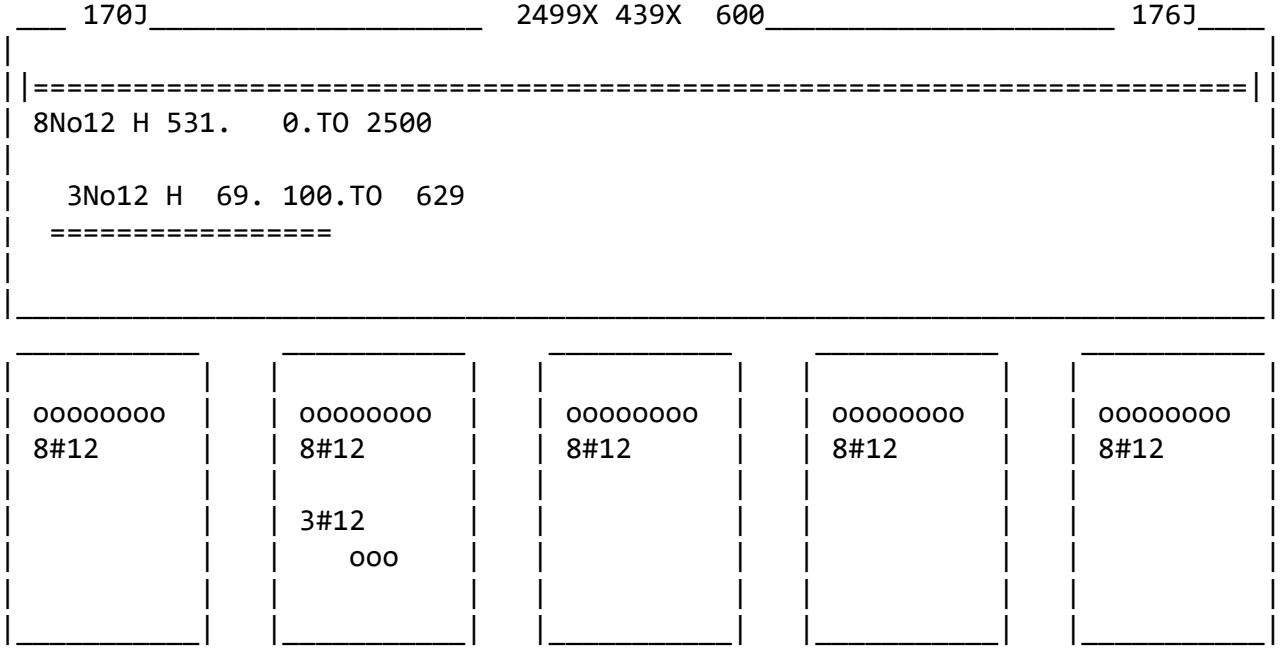
ACI 318-11 BEAM NO. 1905 DESIGN RESULTS

LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	100.	629.	NO	NO
2	531.	8 - 12MM	0.	2500.	YES	YES

B E A M N O. 1905 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.93 KNS Vc= 218.68 KNS Vs= 0.00 KNS
Tu= 0.59 KN-MET Tc= 12.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 45.97 KNS Vc= 218.51 KNS Vs= 0.00 KNS
Tu= 0.59 KN-MET Tc= 12.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1906 DESIGN RESULTS

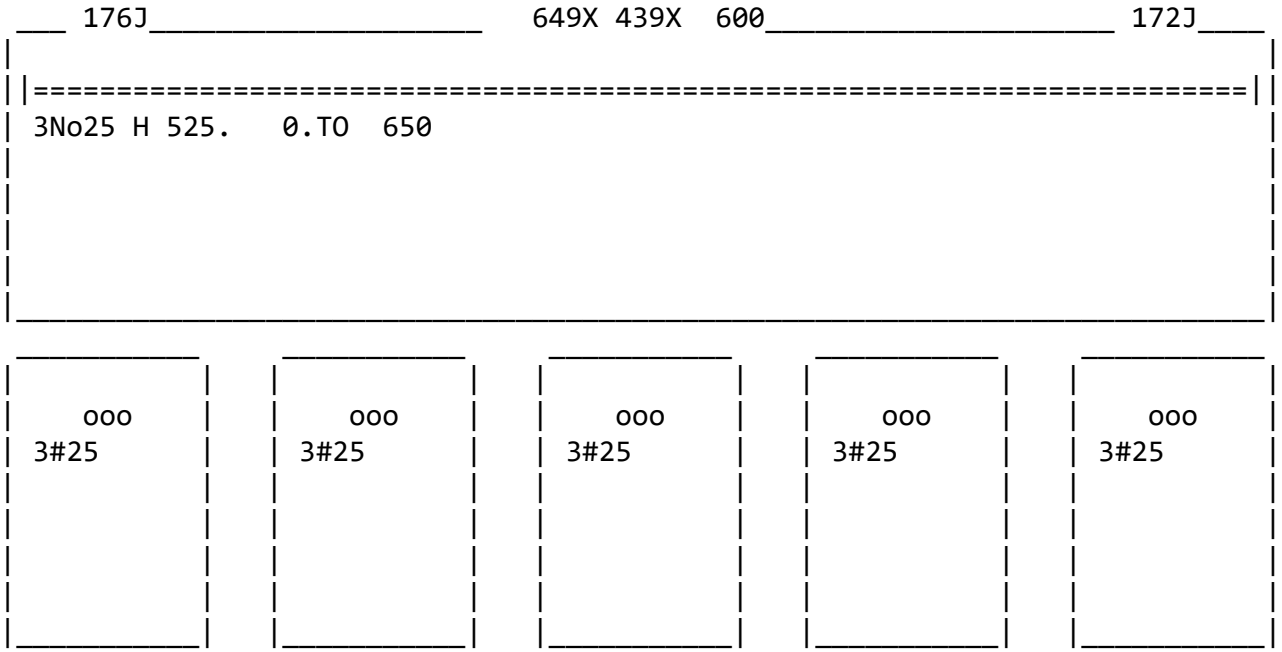
LEN - 650. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	525.	3 - 25MM	0.	650.	YES	YES

B E A M N O. 1906 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1906 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1906 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1907 DESIGN RESULTS

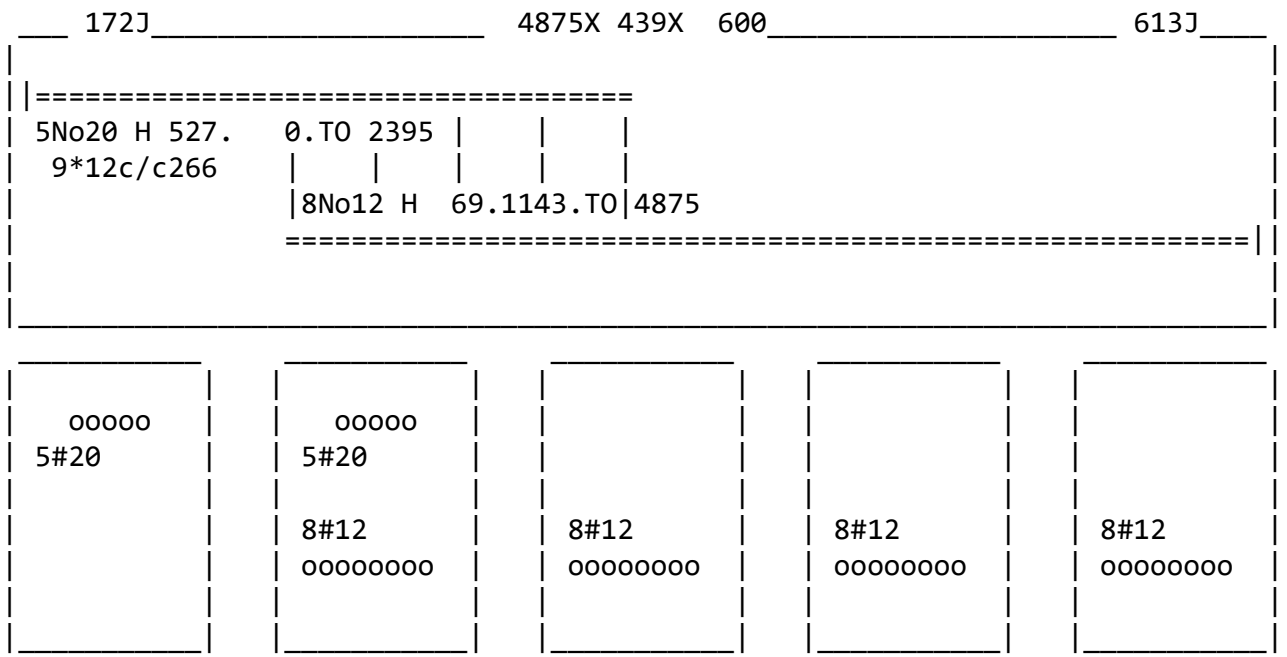
LEN - 4875. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	1143.	4875.	NO	YES
2	527.	5 - 20MM	0.	2395.	YES	NO

B E A M N O. 1907 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 118.86 KNS Vc= 227.77 KNS Vs= 0.00 KNS
Tu= 2.71 KN-MET Tc= 12.5 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 1913. MM

AT END SUPPORT - Vu= 14.46 KNS Vc= 220.92 KNS Vs= 0.00 KNS
Tu= 2.71 KN-MET Tc= 12.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1908 DESIGN RESULTS

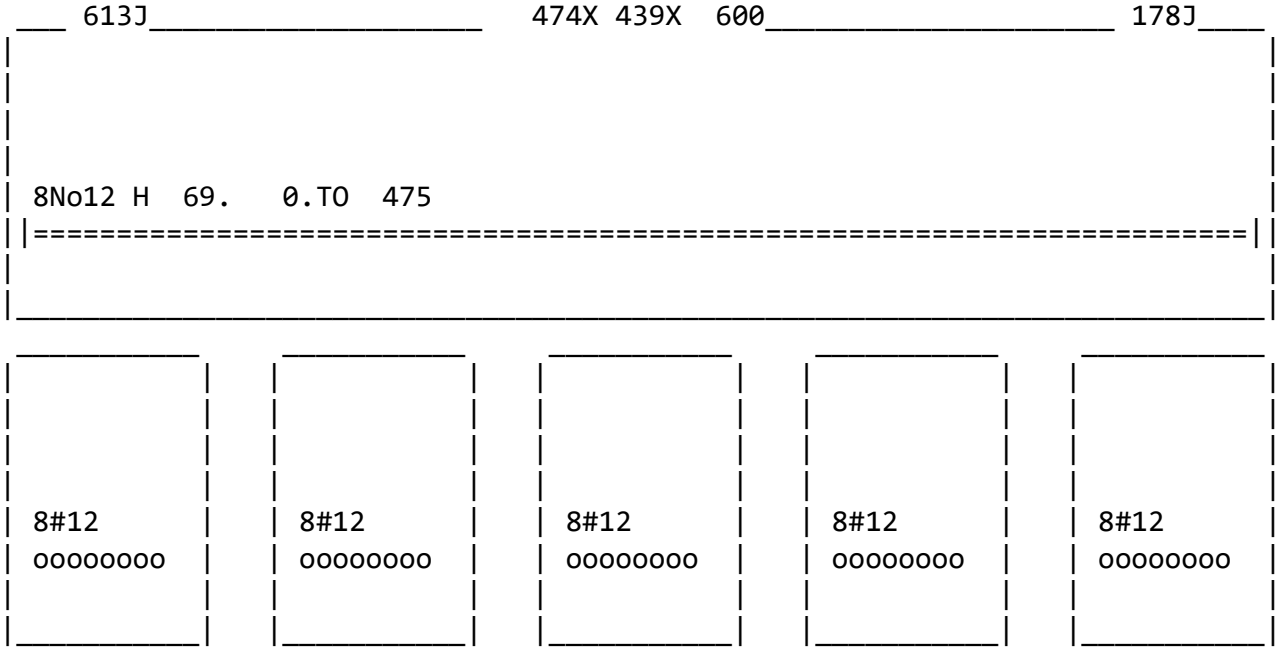
LEN - 475. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	0.	475.	YES	YES

B E A M N O. 1908 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1908 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1908 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



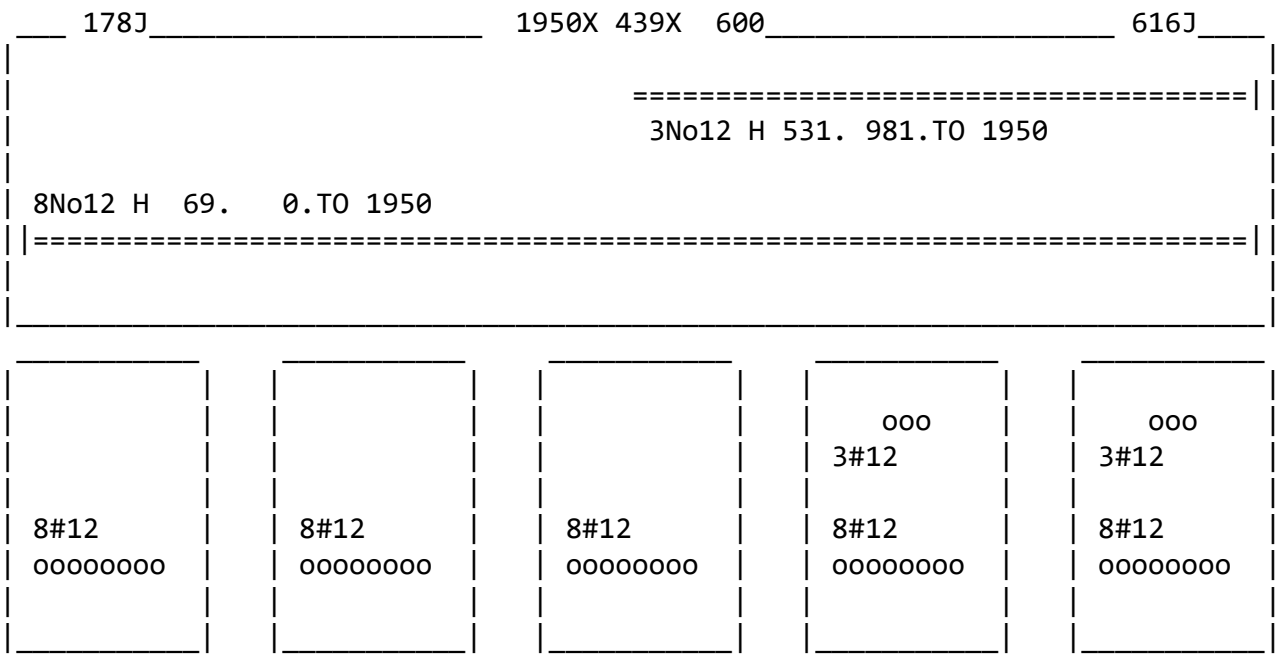
ACI 318-11 BEAM NO. 1909 DESIGN RESULTS

LEN - 1950. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	0.	1950.	YES	YES
2	531.	3 - 12MM	981.	1950.	NO	YES

B E A M N O. 1909 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.00 KNS Vc= 221.88 KNS Vs= 0.00 KNS
Tu= 2.71 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 37.96 KNS Vc= 230.59 KNS Vs= 0.00 KNS
Tu= 2.71 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



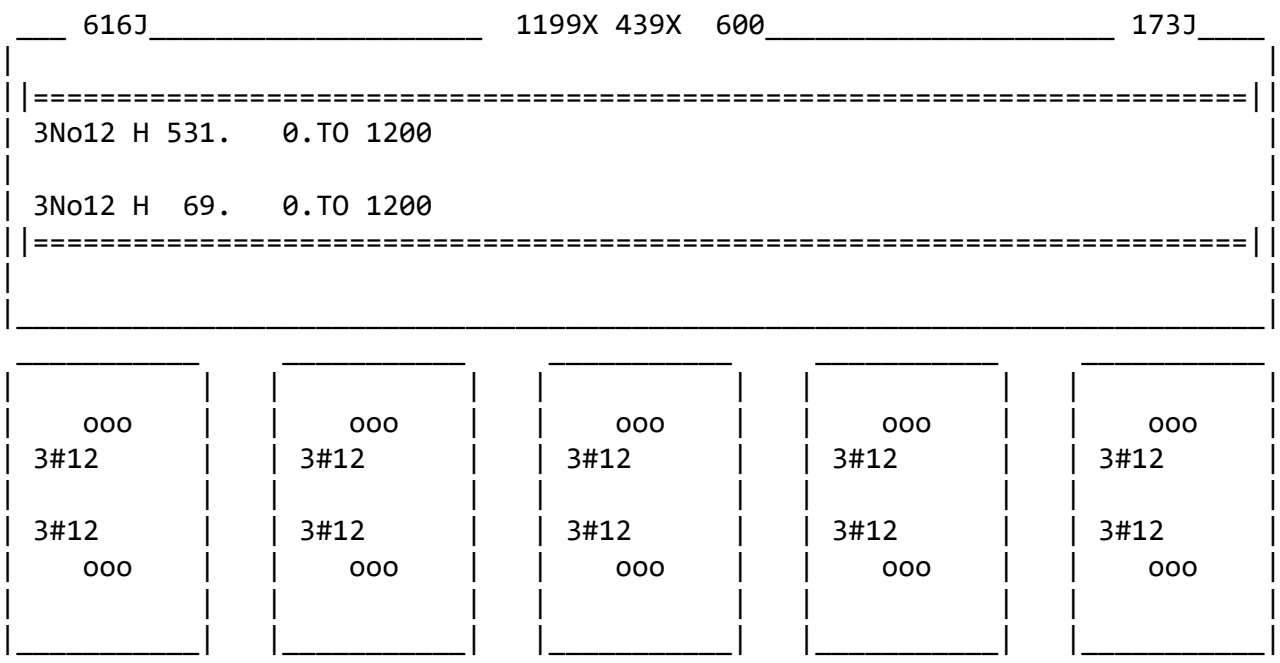
ACI 318-11 BEAM NO. 1910 DESIGN RESULTS

LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1200.	YES	YES
2	531.	3 - 12MM	0.	1200.	YES	YES

B E A M N O. 1910 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.66 KNS Vc= 226.88 KNS Vs= 0.00 KNS
Tu= 2.71 KN-MET Tc= 12.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.13 KNS Vc= 226.88 KNS Vs= 0.00 KNS
Tu= 2.71 KN-MET Tc= 12.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1911 DESIGN RESULTS

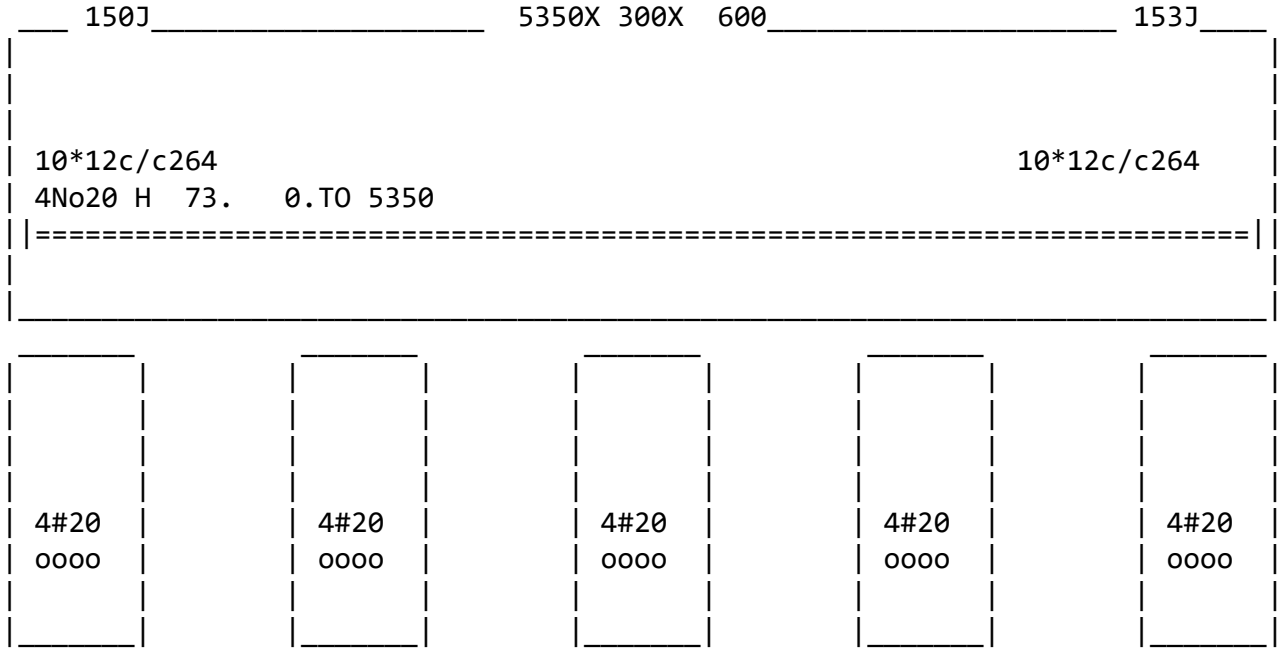
LEN - 5350. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	5350.	YES	YES

B E A M N O. 1911 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 78.37 KNS Vc= 168.85 KNS Vs= 0.00 KNS
Tu= 0.49 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 2150. MM

AT END SUPPORT - Vu= 72.25 KNS Vc= 168.81 KNS Vs= 0.00 KNS
Tu= 0.49 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 2150. MM



ACI 318-11 BEAM NO. 1912 DESIGN RESULTS

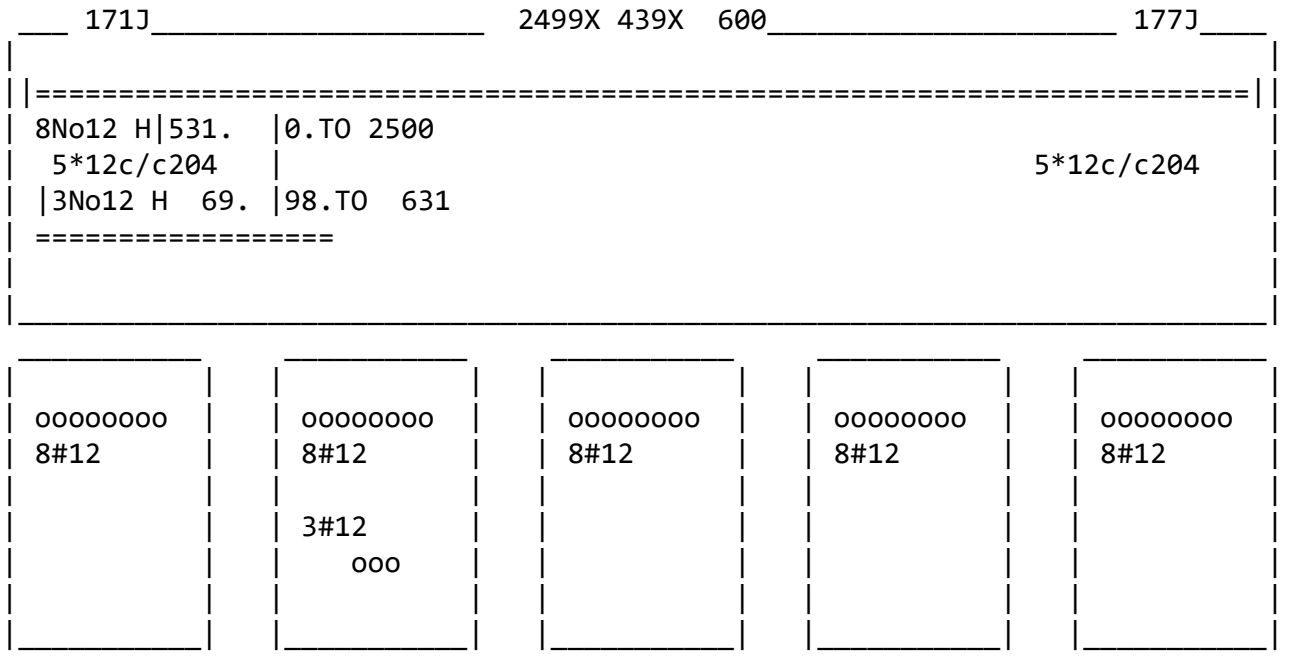
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	98.	631.	NO	NO
2	531.	8 - 12MM	0.	2500.	YES	YES

B E A M N O. 1912 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 24.41 KNS Vc= 227.51 KNS Vs= 0.00 KNS
Tu= 12.75 KN-MET Tc= 12.2 KN-MET Ts= 17.0 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 725. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.

AT END SUPPORT - Vu= 70.03 KNS Vc= 227.51 KNS Vs= 0.00 KNS
Tu= 12.75 KN-MET Tc= 12.2 KN-MET Ts= 17.0 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 204. MM C/C FOR 725. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.



ACI 318-11 BEAM NO. 1913 DESIGN RESULTS

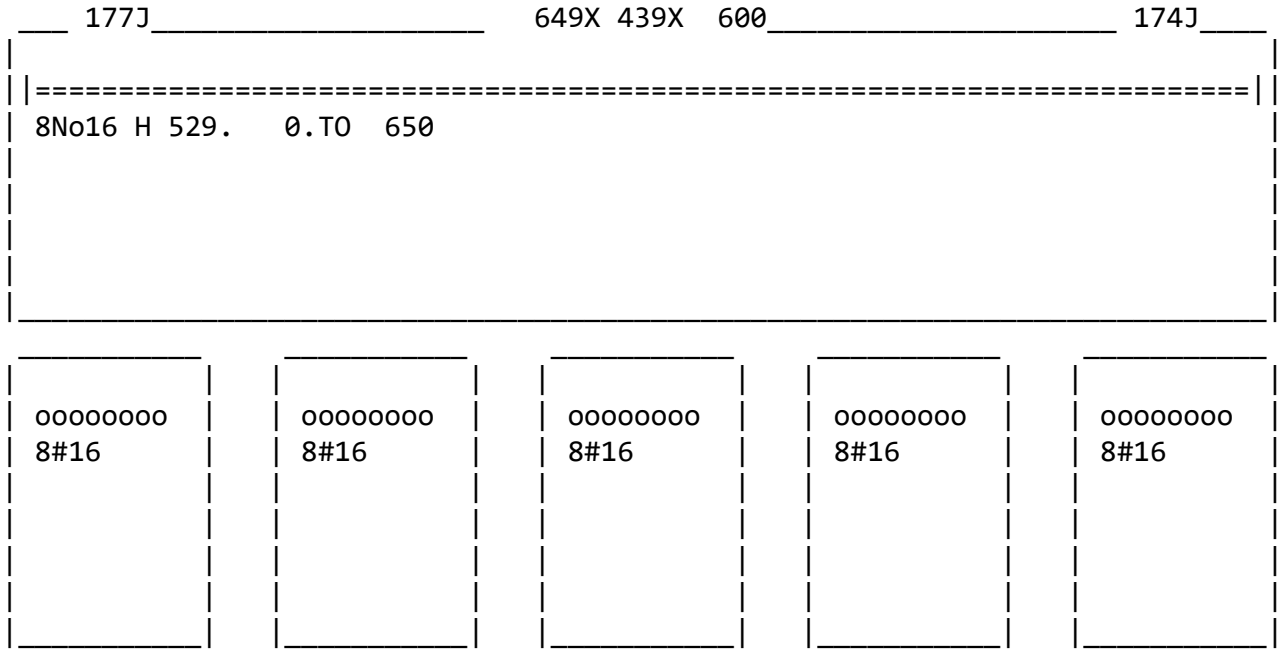
LEN - 650. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	529.	8 - 16MM	0.	650.	YES	YES

B E A M N O. 1913 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1913 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1913 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1914 DESIGN RESULTS

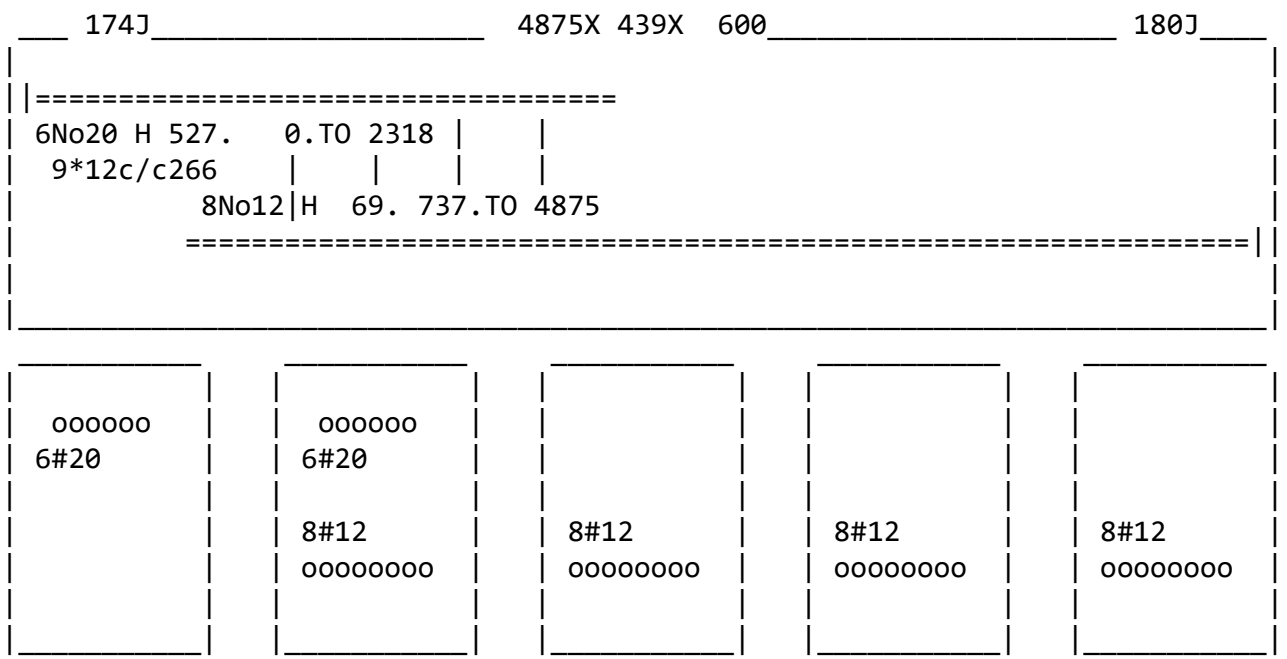
LEN - 4875. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	737.	4875.	NO	YES
2	527.	6 - 20MM	0.	2318.	YES	NO

B E A M N O. 1914 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 238.83 KNS Vc= 227.92 KNS Vs= 90.52 KNS
Tu= 7.07 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 1913. MM

AT END SUPPORT - Vu= 21.11 KNS Vc= 221.86 KNS Vs= 0.00 KNS
Tu= 2.63 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1915 DESIGN RESULTS

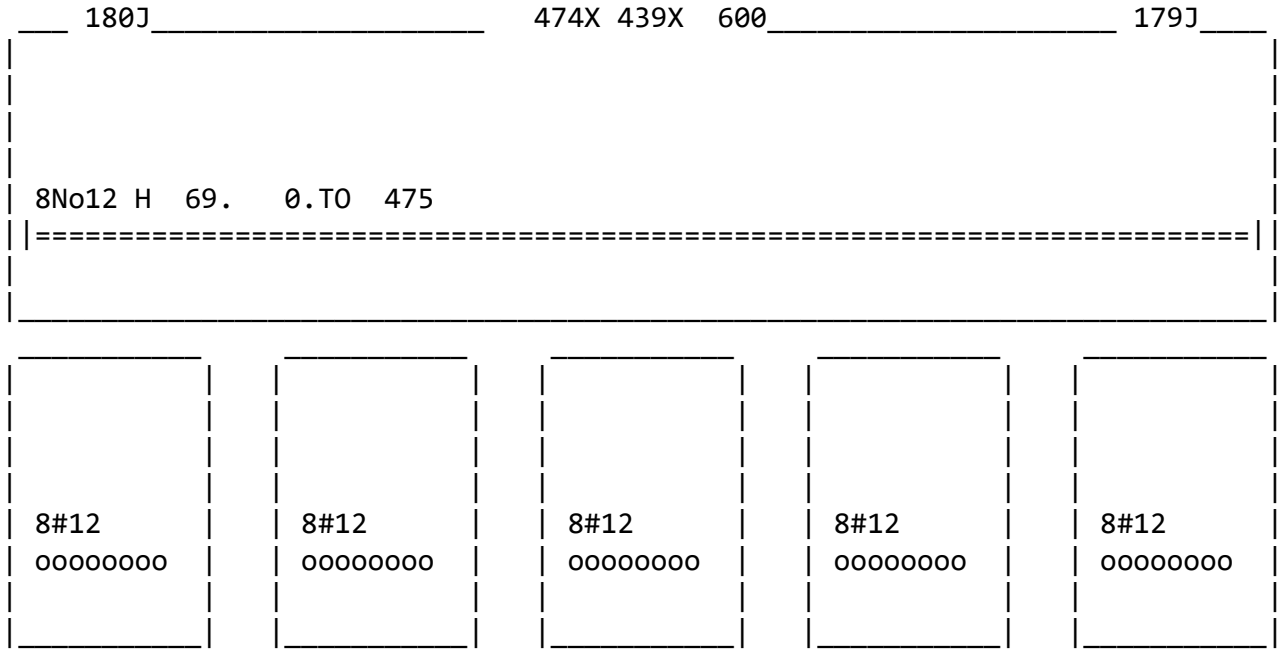
LEN - 475. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	0.	475.	YES	YES

B E A M N O. 1915 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1915 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1915 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



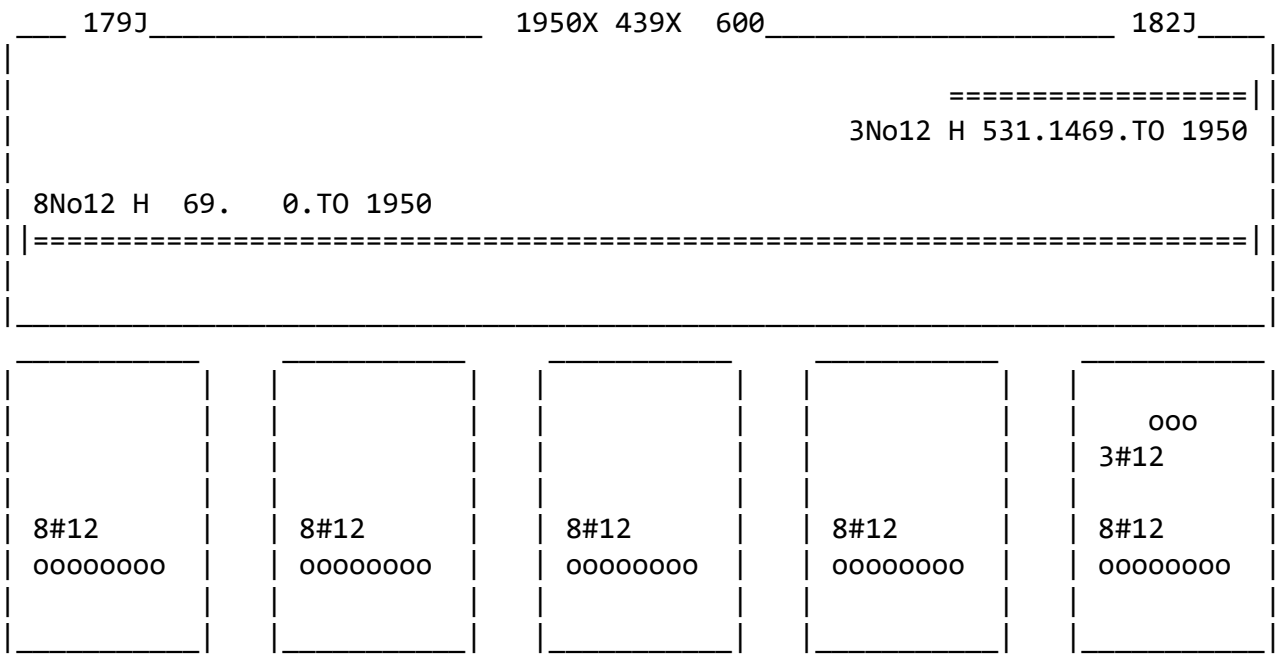
ACI 318-11 BEAM NO. 1916 DESIGN RESULTS

LEN - 1950. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	0.	1950.	YES	YES
2	531.	3 - 12MM	1469.	1950.	NO	YES

B E A M N O. 1916 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.04 KNS Vc= 220.75 KNS Vs= 0.00 KNS
Tu= 2.63 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 33.42 KNS Vc= 226.63 KNS Vs= 0.00 KNS
Tu= 2.63 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



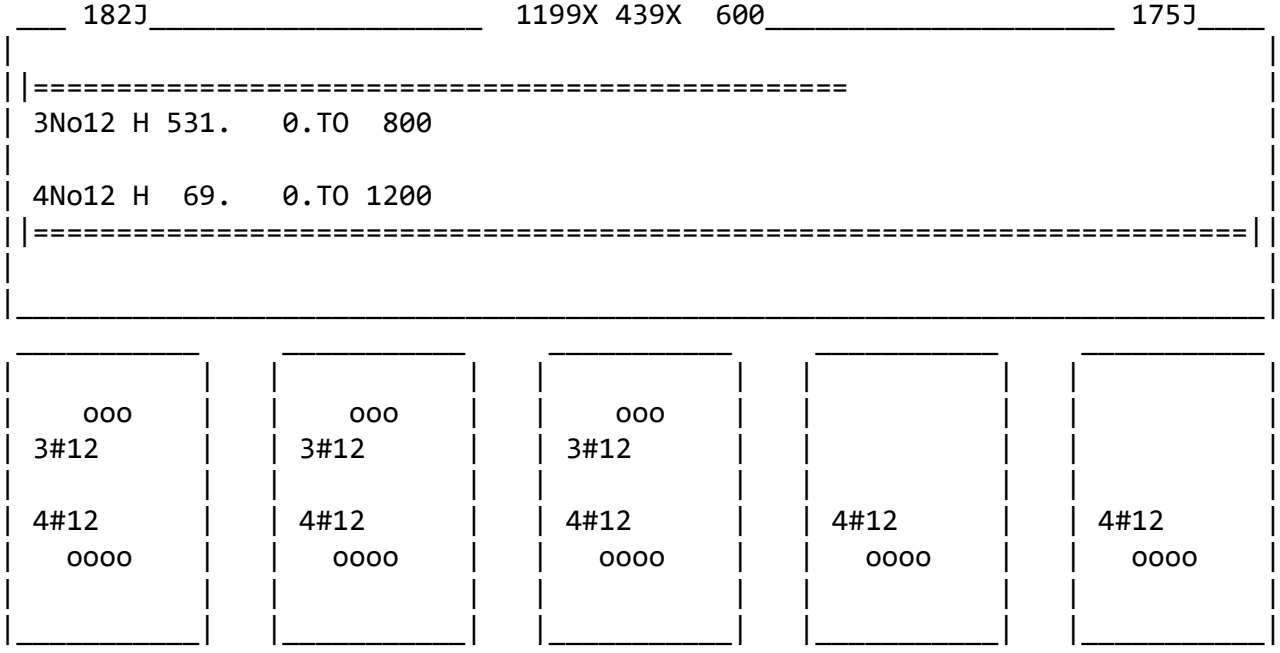
ACI 318-11 BEAM NO. 1917 DESIGN RESULTS

LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 440. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	1200.	YES	YES
2	531.	3 - 12MM	0.	800.	YES	NO

B E A M N O. 1917 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.63 KNS Vc= 222.57 KNS Vs= 0.00 KNS
Tu= 2.63 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.12 KNS Vc= 224.34 KNS Vs= 0.00 KNS
Tu= 2.63 KN-MET Tc= 12.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1974 DESIGN RESULTS

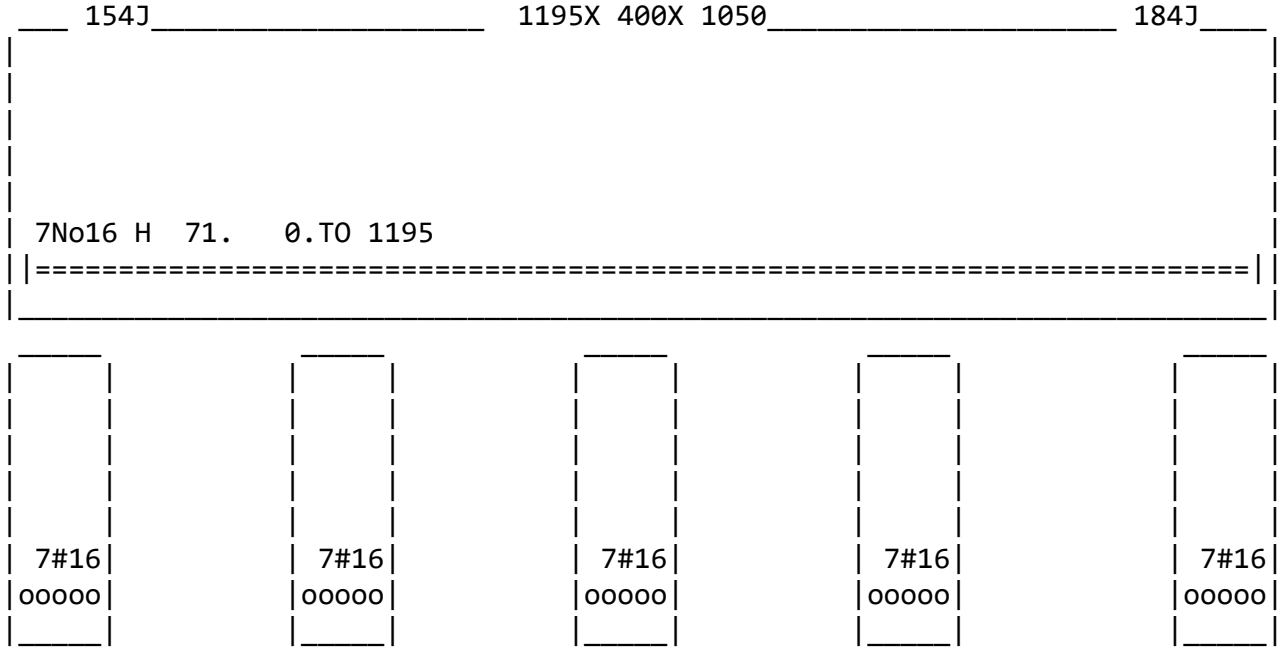
LEN - 1195. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	7 - 16MM	0.	1195.	YES	YES

B E A M N O. 1974 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1974 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1974 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1975 DESIGN RESULTS

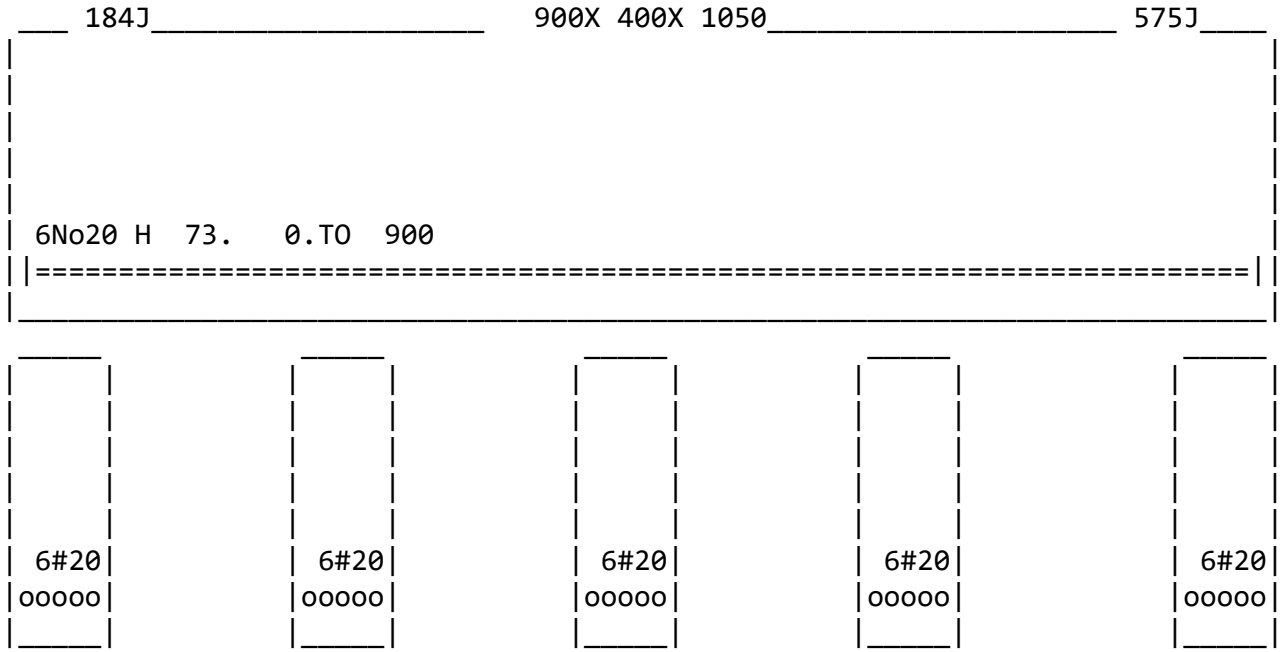
LEN - 900. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	6 - 20MM	0.	900.	YES	YES

B E A M N O. 1975 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1975 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1975 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



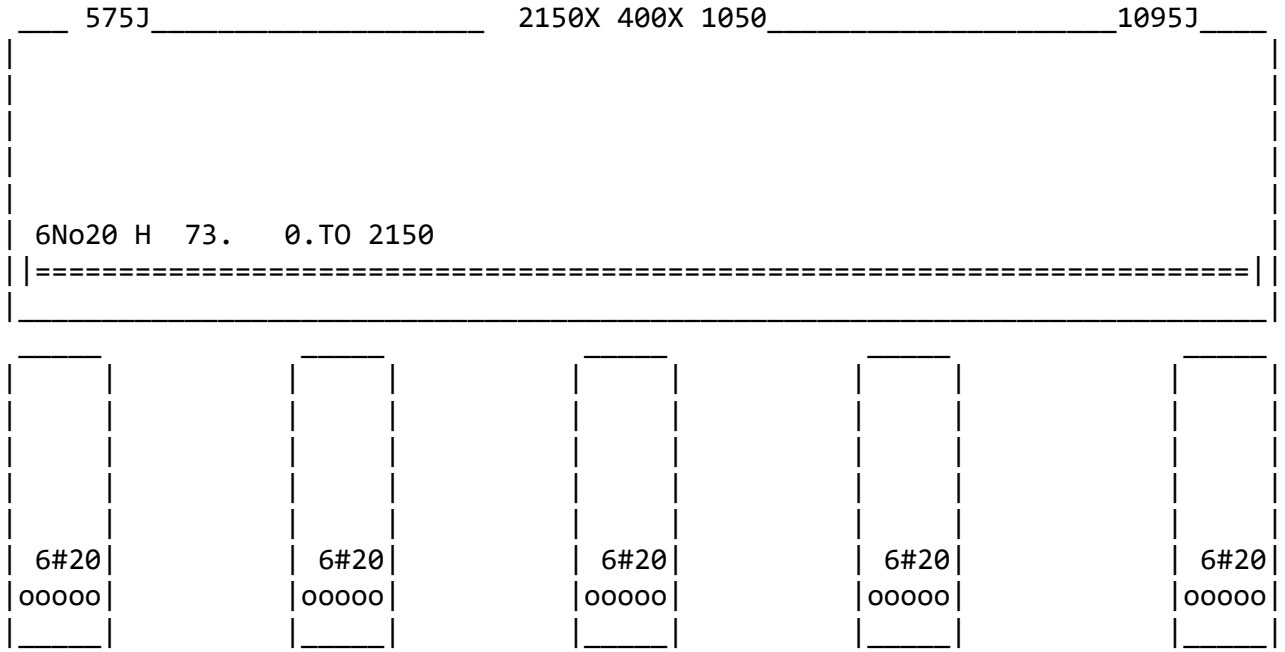
ACI 318-11 BEAM NO. 1976 DESIGN RESULTS

LEN - 2150. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	6 - 20MM	0.	2150.	YES	YES

B E A M N O. 1976 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 30.47 KNS Vc= 383.42 KNS Vs= 0.00 KNS
Tu= 2.97 KN-MET Tc= 22.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 34.16 KNS Vc= 383.42 KNS Vs= 0.00 KNS
Tu= 2.97 KN-MET Tc= 22.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 1977 DESIGN RESULTS

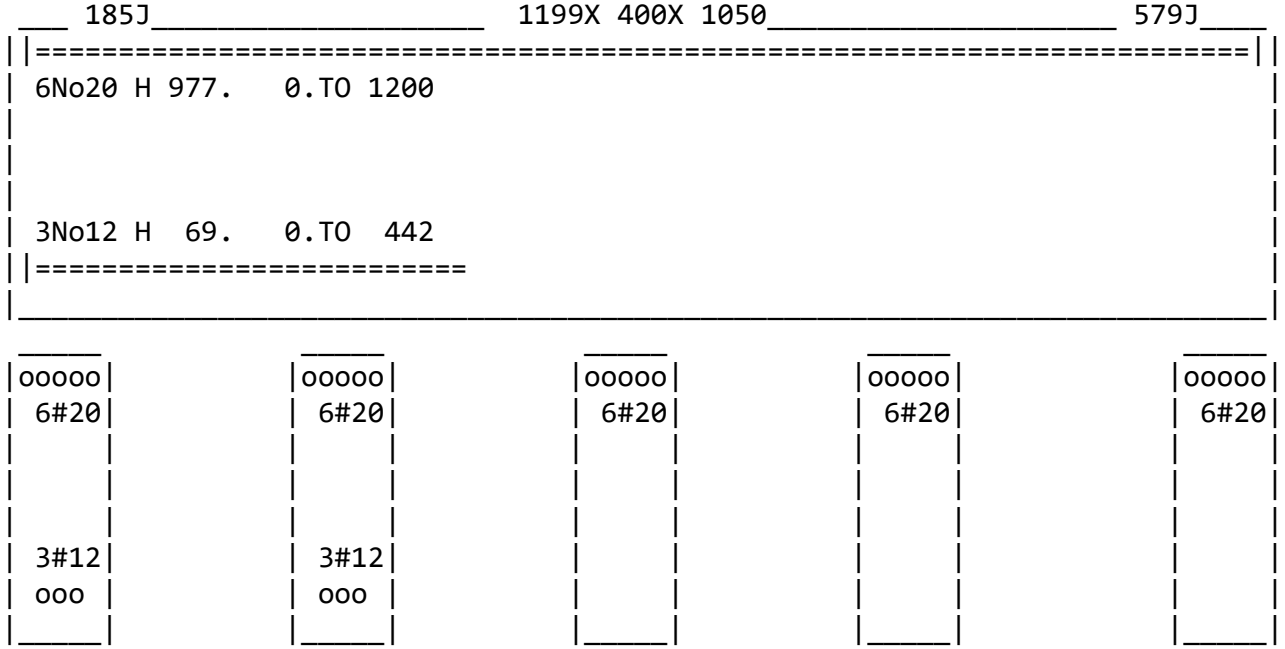
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	442.	YES	NO
2	977.	6 - 20MM	0.	1200.	YES	YES

B E A M N O. 1977 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1977 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1977 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1978 DESIGN RESULTS

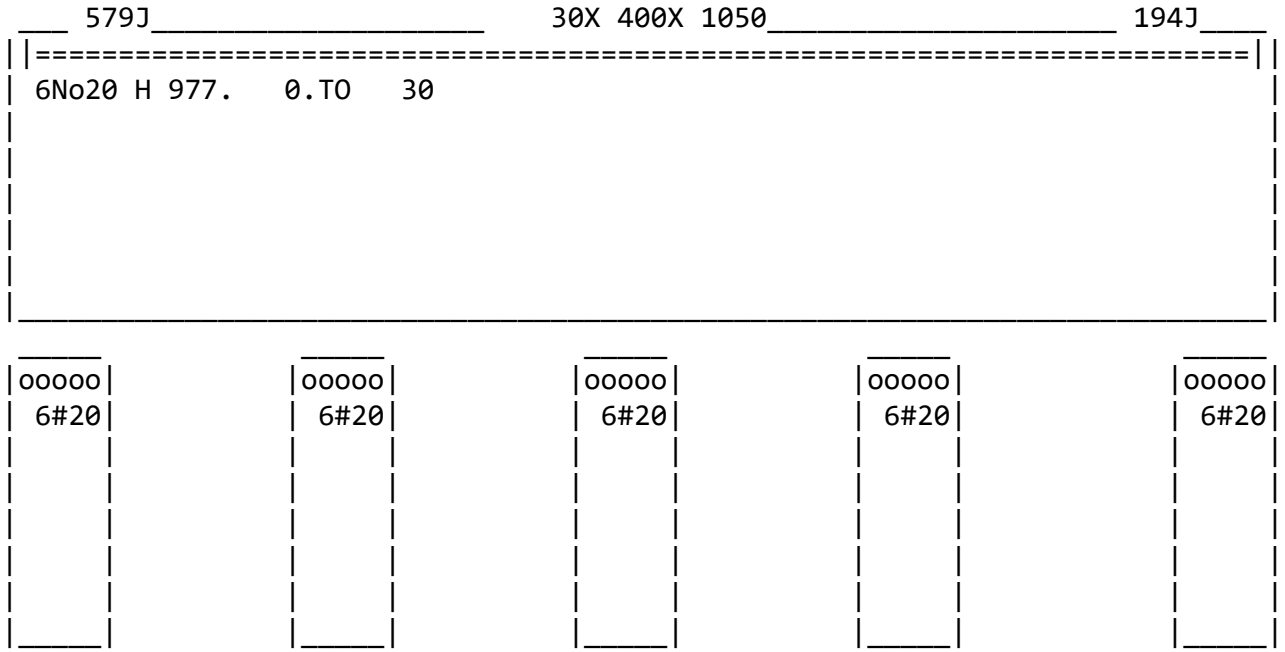
LEN - 30. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	977.	6 - 20MM	0.	30.	YES	YES

B E A M N O. 1978 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1978 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1978 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1980 DESIGN RESULTS

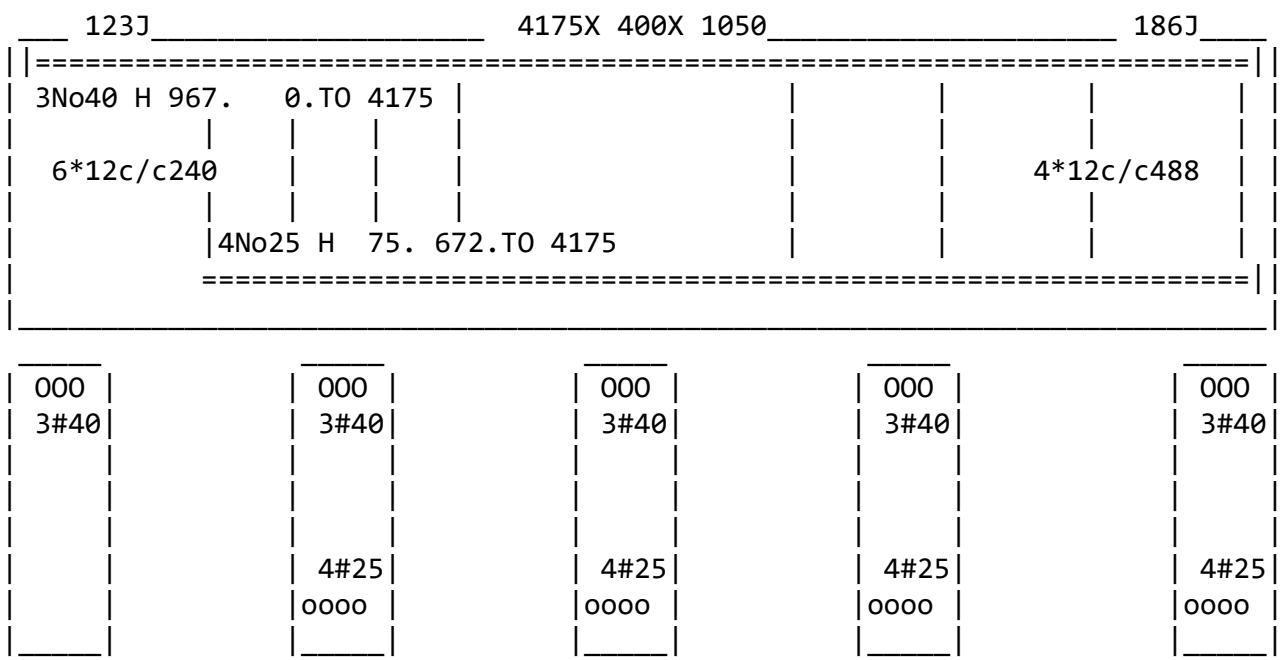
LEN - 4175. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	672.	4175.	NO	YES
2	967.	3 - 40MM	0.	4175.	YES	YES

B E A M N O. 1980 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 570.64 KNS Vc= 375.36 KNS Vs= 385.49 KNS
Tu= 1.16 KN-MET Tc= 22.0 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 1113. MM

AT END SUPPORT - Vu= 217.19 KNS Vc= 382.83 KNS Vs= 0.00 KNS
Tu= 3.61 KN-MET Tc= 22.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 488. MM C/C FOR 1113. MM



ACI 318-11 BEAM NO. 1981 DESIGN RESULTS

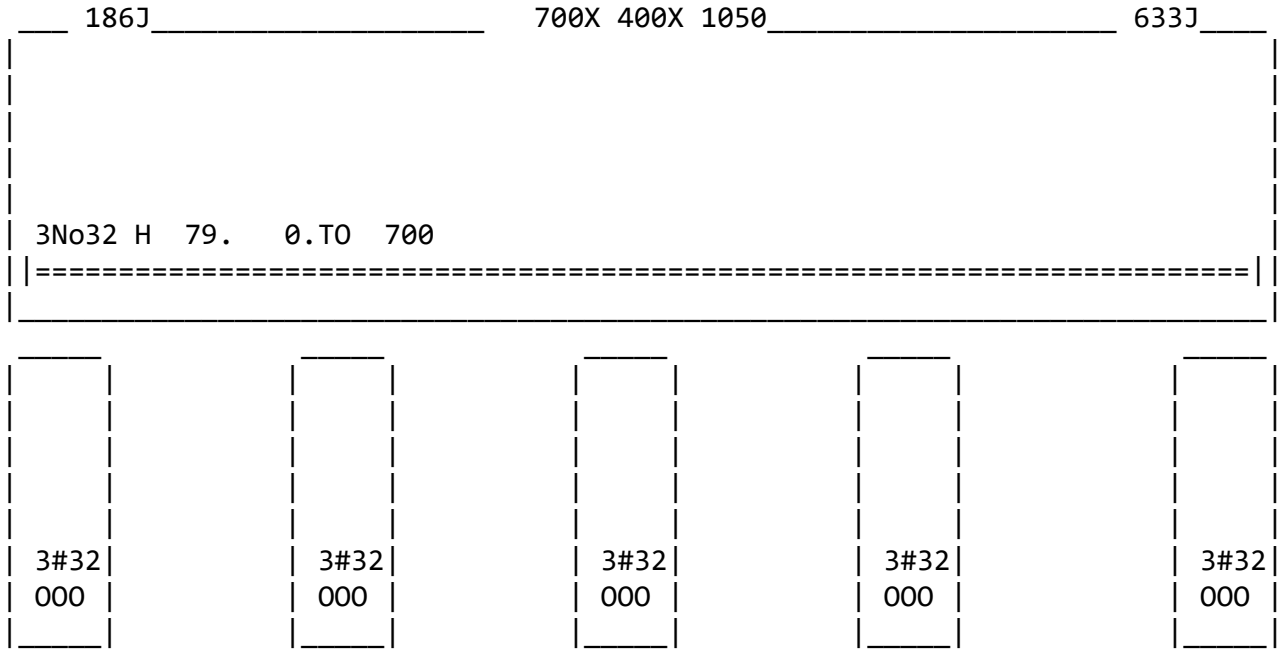
LEN - 700. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	700.	YES	YES

B E A M N O. 1981 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1981 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1981 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1982 DESIGN RESULTS

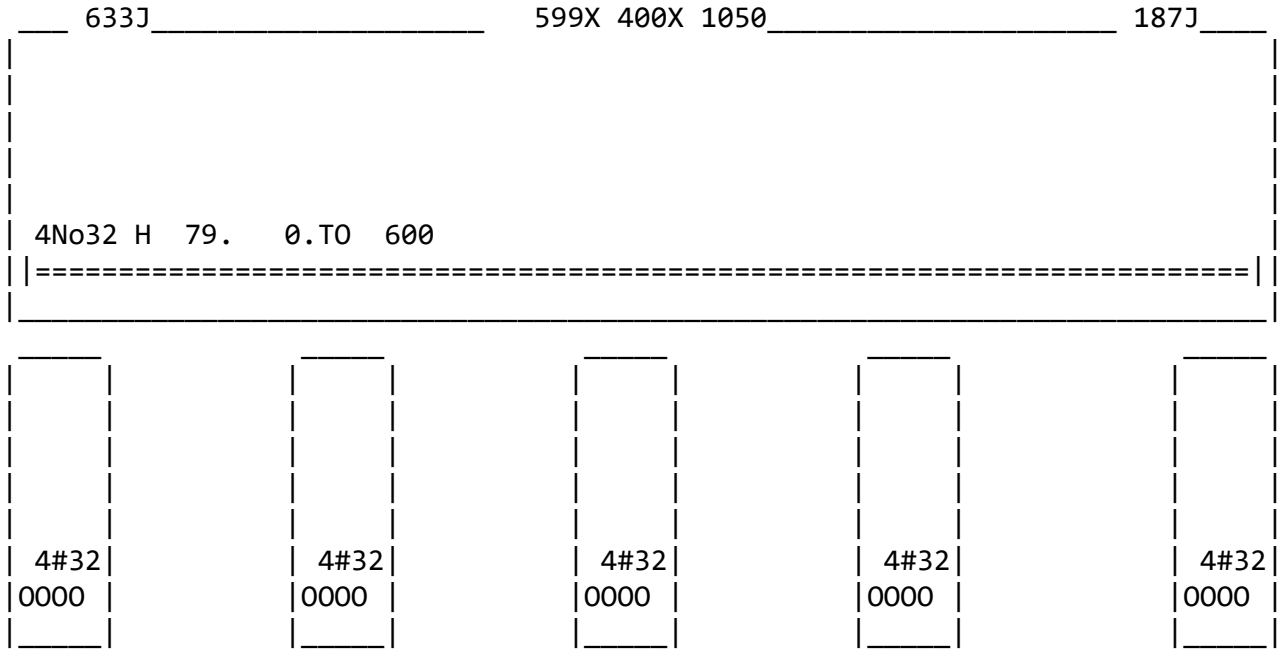
LEN - 600. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	4 - 32MM	0.	600.	YES	YES

B E A M N O. 1982 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1982 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1982 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1983 DESIGN RESULTS

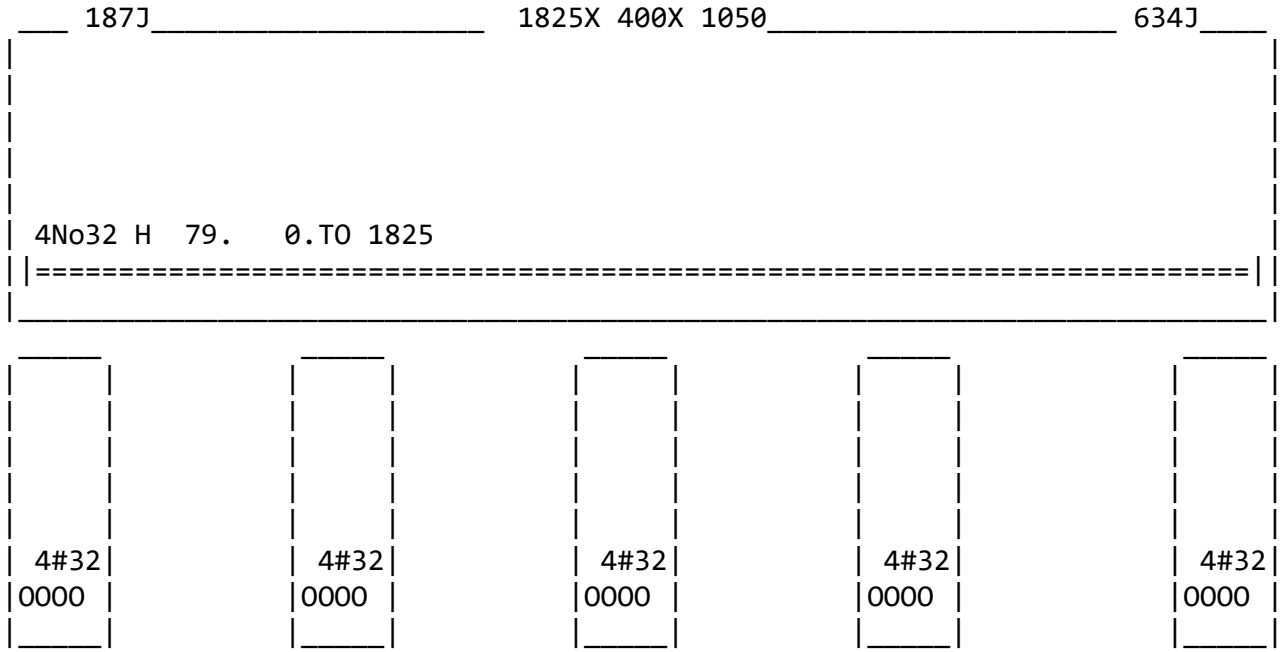
LEN - 1825. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	4 - 32MM	0.	1825.	YES	YES

B E A M N O. 1983 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1983 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1983 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 1984 DESIGN RESULTS

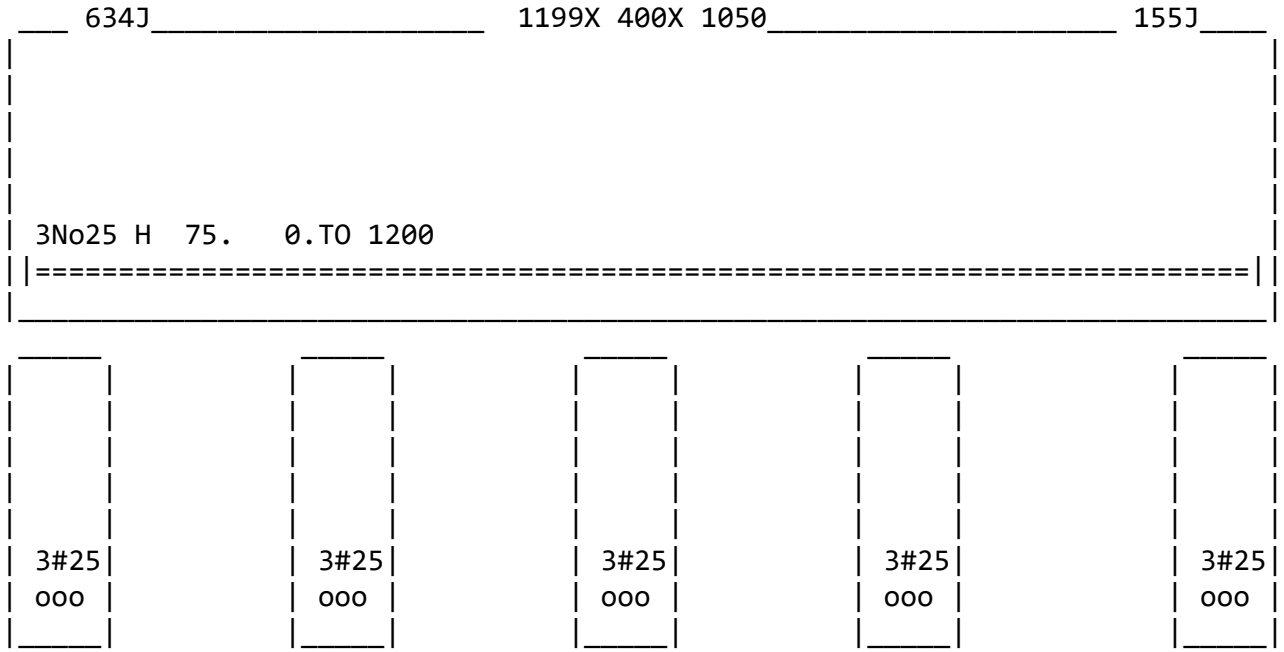
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1200.	YES	YES

B E A M N O. 1984 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 1984 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 1984 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



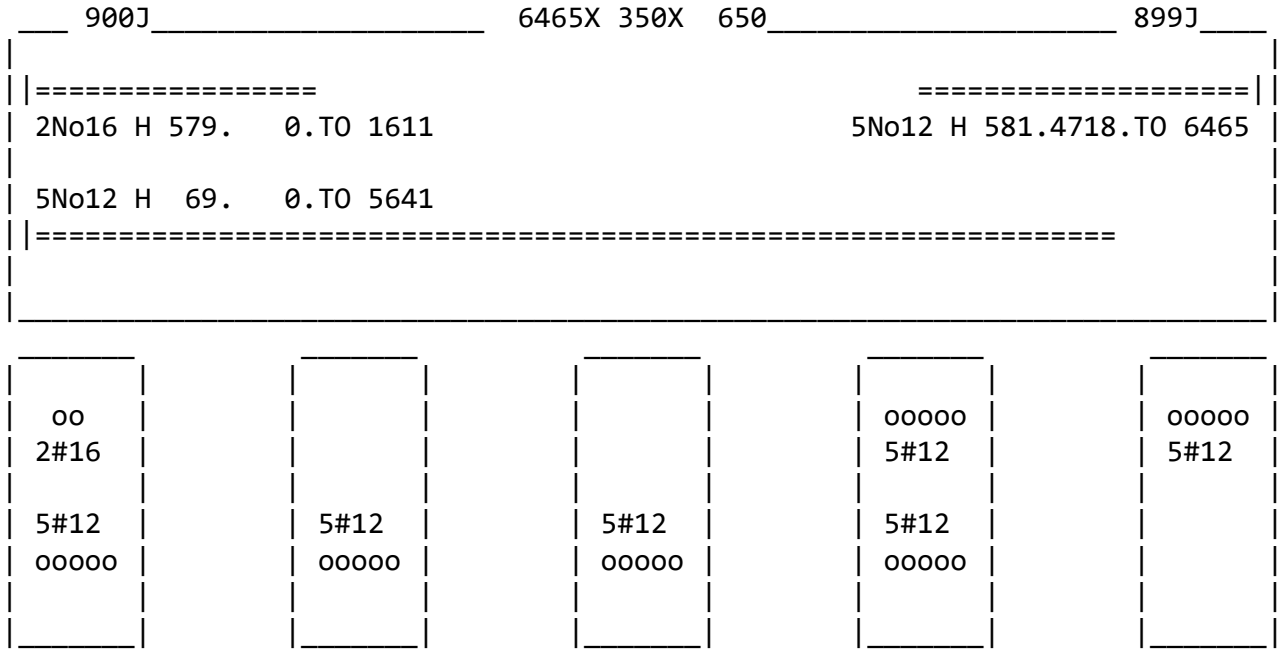
ACI 318-11 BEAM NO. 2151 DESIGN RESULTS

LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	5641.	YES	NO
2	579.	2 - 16MM	0.	1611.	YES	NO
3	581.	5 - 12MM	4718.	6465.	NO	YES

B E A M N O. 2151 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 38.71 KNS Vc= 218.97 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 9.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 42.74 KNS Vc= 201.89 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 9.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2152 DESIGN RESULTS

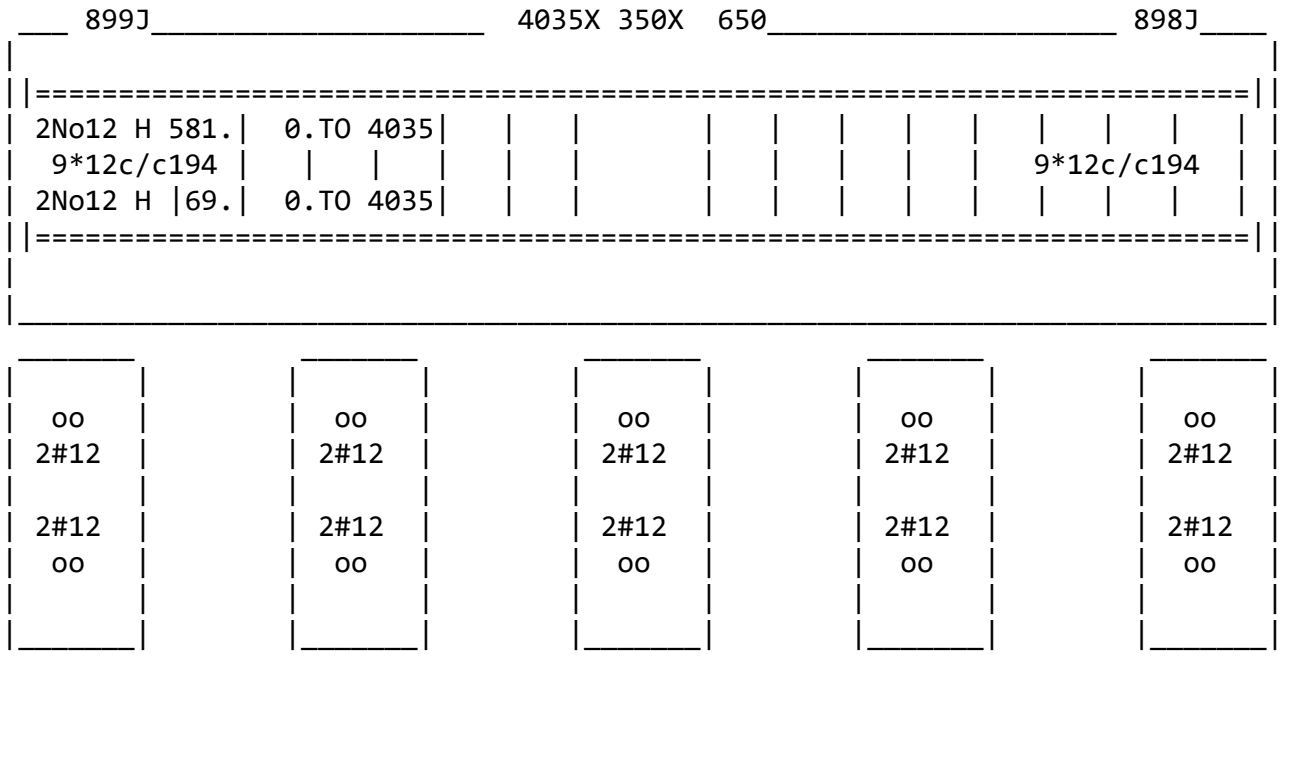
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	4035.	YES	YES
2	581.	2 - 12MM	0.	4035.	YES	YES

B E A M N O. 2152 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.79 KNS Vc= 190.76 KNS Vs= 0.00 KNS
Tu= 10.51 KN-MET Tc= 9.6 KN-MET Ts= 14.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 194. MM C/C FOR 1443. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.38 SQ.CM.

AT END SUPPORT - Vu= 7.87 KNS Vc= 190.88 KNS Vs= 0.00 KNS
Tu= 10.51 KN-MET Tc= 9.6 KN-MET Ts= 14.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 194. MM C/C FOR 1443. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.38 SQ.CM.



ACI 318-11 BEAM NO. 2153 DESIGN RESULTS

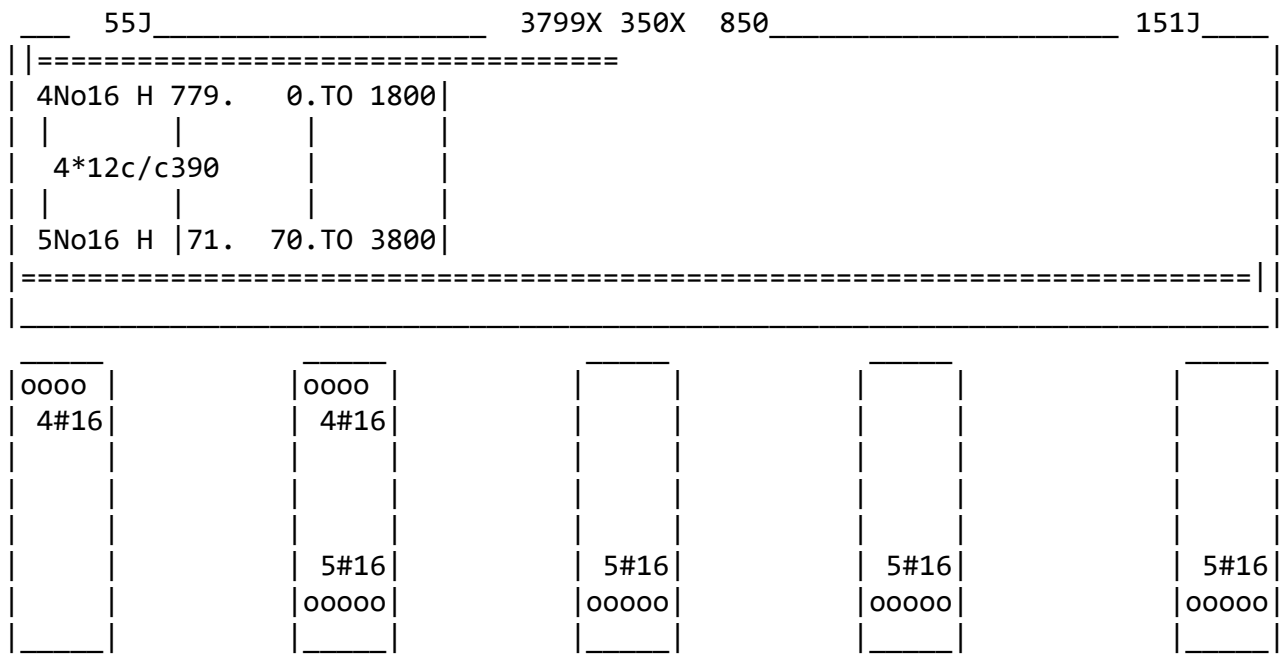
LEN - 3800. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	5 - 16MM	70.	3800.	NO	YES
2	779.	4 - 16MM	0.	1800.	YES	NO

B E A M N O. 2153 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 119.66 KNS Vc= 306.76 KNS Vs= 0.00 KNS
Tu= 4.84 KN-MET Tc= 14.4 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 1125. MM

AT END SUPPORT - Vu= 56.17 KNS Vc= 260.26 KNS Vs= 0.00 KNS
Tu= 4.78 KN-MET Tc= 14.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2154 DESIGN RESULTS

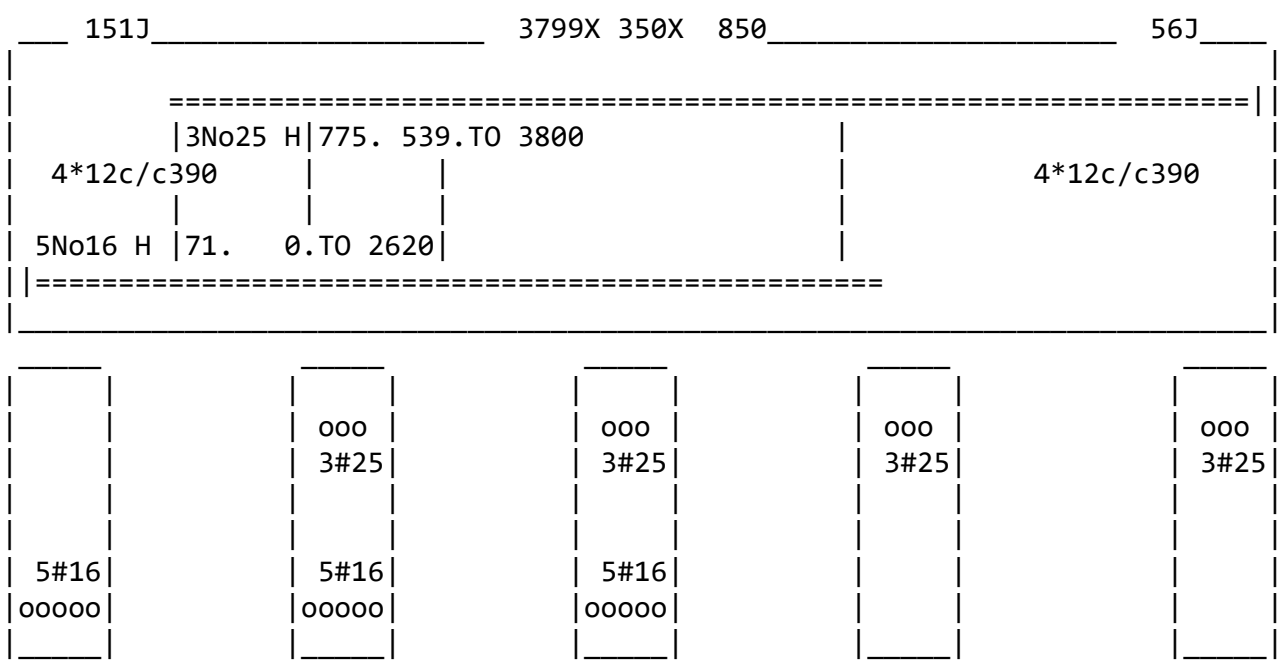
LEN - 3800. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	5 - 16MM	0.	2620.	YES	NO
2	775.	3 - 25MM	539.	3800.	NO	YES

B E A M N O. 2154 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 129.41 KNS Vc= 265.68 KNS Vs= 0.00 KNS
Tu= 5.60 KN-MET Tc= 14.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 1125. MM

AT END SUPPORT - Vu= 146.51 KNS Vc= 272.46 KNS Vs= 0.00 KNS
Tu= 4.78 KN-MET Tc= 14.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 1125. MM



ACI 318-11 BEAM NO. 2155 DESIGN RESULTS

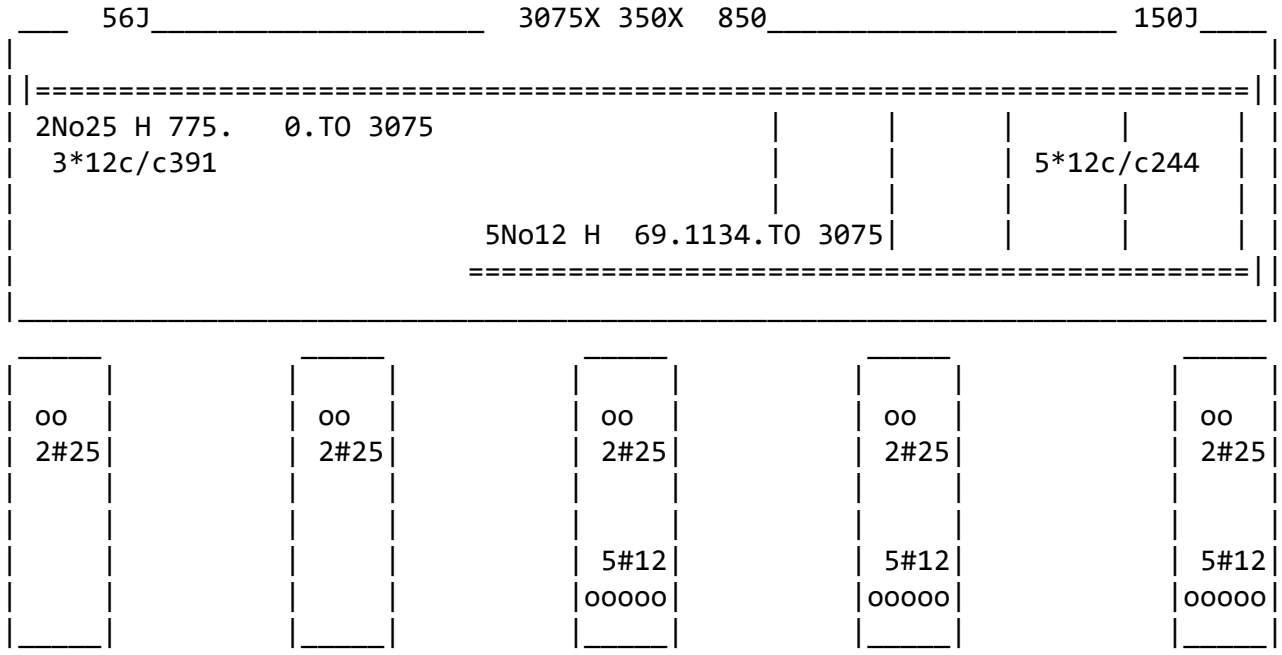
LEN - 3075. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	1134.	3075.	NO	YES
2	775.	2 - 25MM	0.	3075.	YES	YES

B E A M N O. 2155 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 135.46 KNS Vc= 262.71 KNS Vs= 0.00 KNS
Tu= 7.33 KN-MET Tc= 13.8 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 391. MM C/C FOR 763. MM

AT END SUPPORT - Vu= 93.11 KNS Vc= 272.34 KNS Vs= 0.00 KNS
Tu= 13.87 KN-MET Tc= 13.8 KN-MET Ts= 18.5 KN-MET LOAD 404
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 244. MM C/C FOR 763. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.88 SQ.CM.



ACI 318-11 BEAM NO. 2156 DESIGN RESULTS

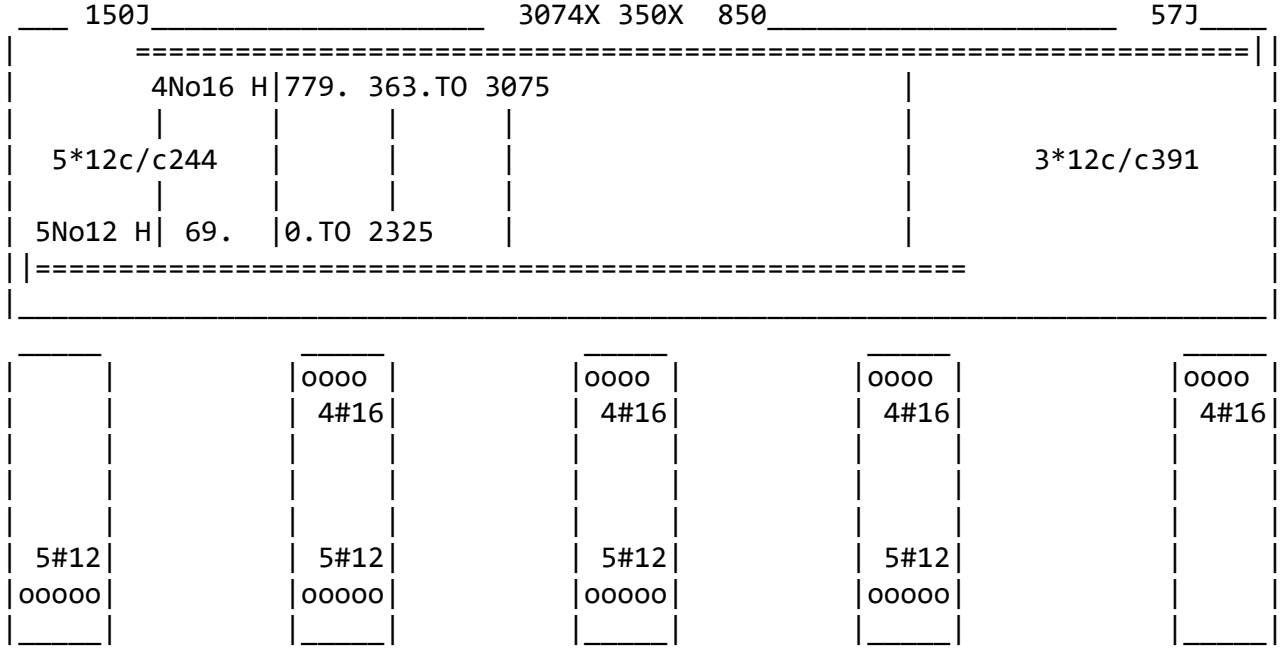
LEN - 3075. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	2325.	YES	NO
2	779.	4 - 16MM	363.	3075.	NO	YES

B E A M N O. 2156 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 63.68 KNS Vc= 262.76 KNS Vs= 0.00 KNS
Tu= 13.87 KN-MET Tc= 13.8 KN-MET Ts= 18.5 KN-MET LOAD 404
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 244. MM C/C FOR 763. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.88 SQ.CM.

AT END SUPPORT - Vu= 102.06 KNS Vc= 269.26 KNS Vs= 0.00 KNS
Tu= 7.33 KN-MET Tc= 13.8 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 391. MM C/C FOR 763. MM



ACI 318-11 BEAM NO. 2157 DESIGN RESULTS

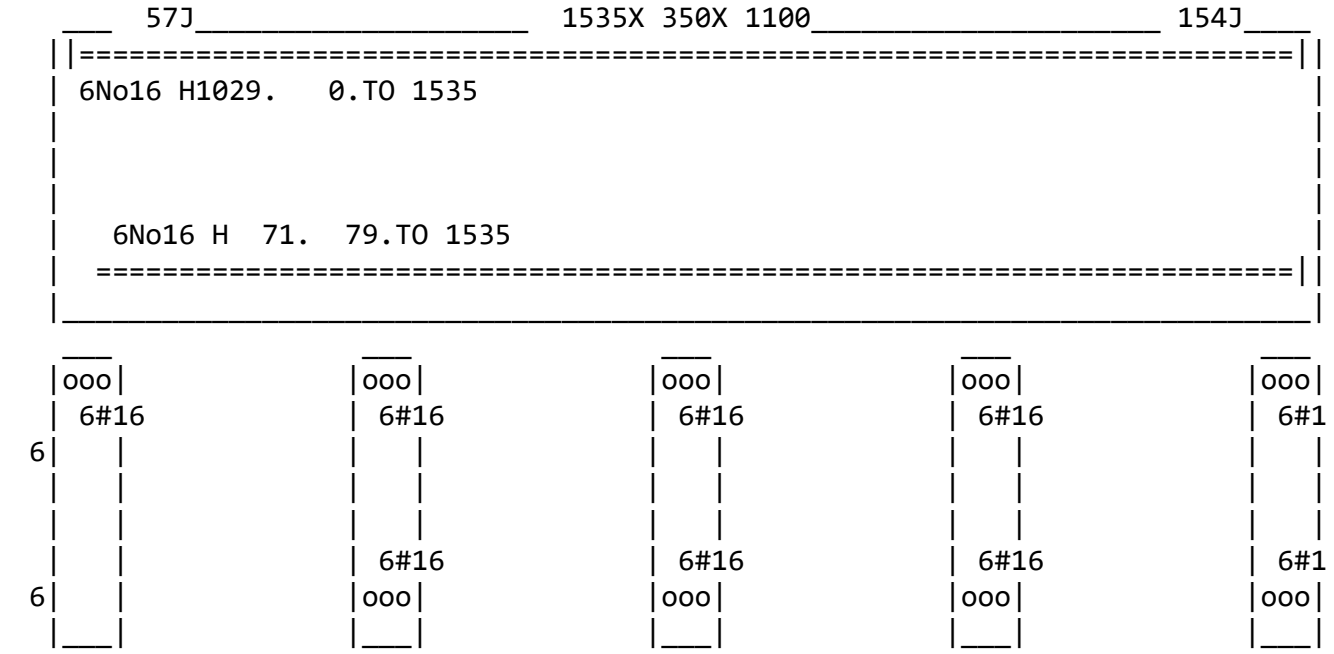
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1100. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	79.	1535.	NO	YES
2	1029.	6 - 16MM	0.	1535.	YES	YES

B E A M N O. 2157 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2157 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2157 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2158 DESIGN RESULTS

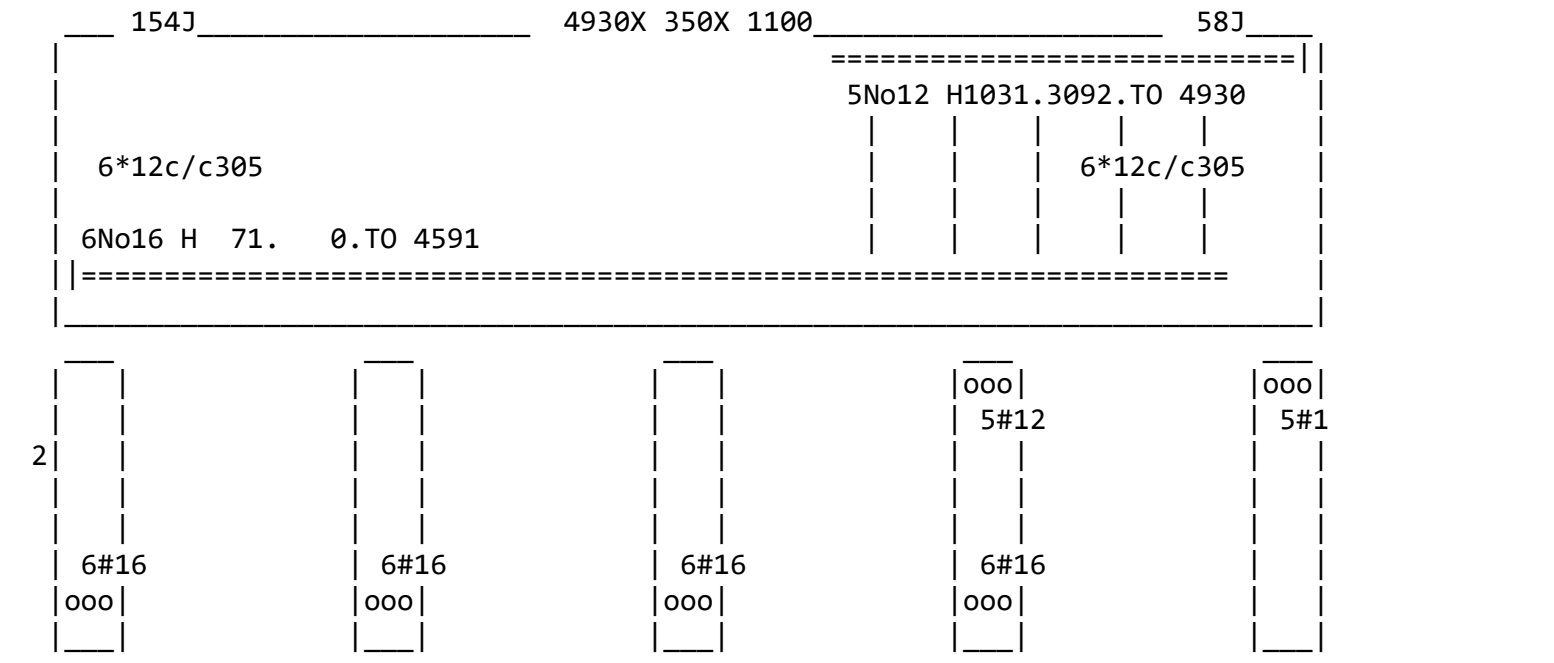
LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1100. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	4591.	YES	NO
2	1031.	5 - 12MM	3092.	4930.	NO	YES

B E A M N O. 2158 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 35.22 KNS Vc= 339.19 KNS Vs= 0.00 KNS
Tu= 25.67 KN-MET Tc= 20.2 KN-MET Ts= 34.2 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1440. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.00 SQ.CM.

AT END SUPPORT - Vu= 156.46 KNS Vc= 645.17 KNS Vs= 0.00 KNS
Tu= 25.67 KN-MET Tc= 20.2 KN-MET Ts= 34.2 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1440. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.00 SQ.CM.



ACI 318-11 BEAM NO. 2159 DESIGN RESULTS

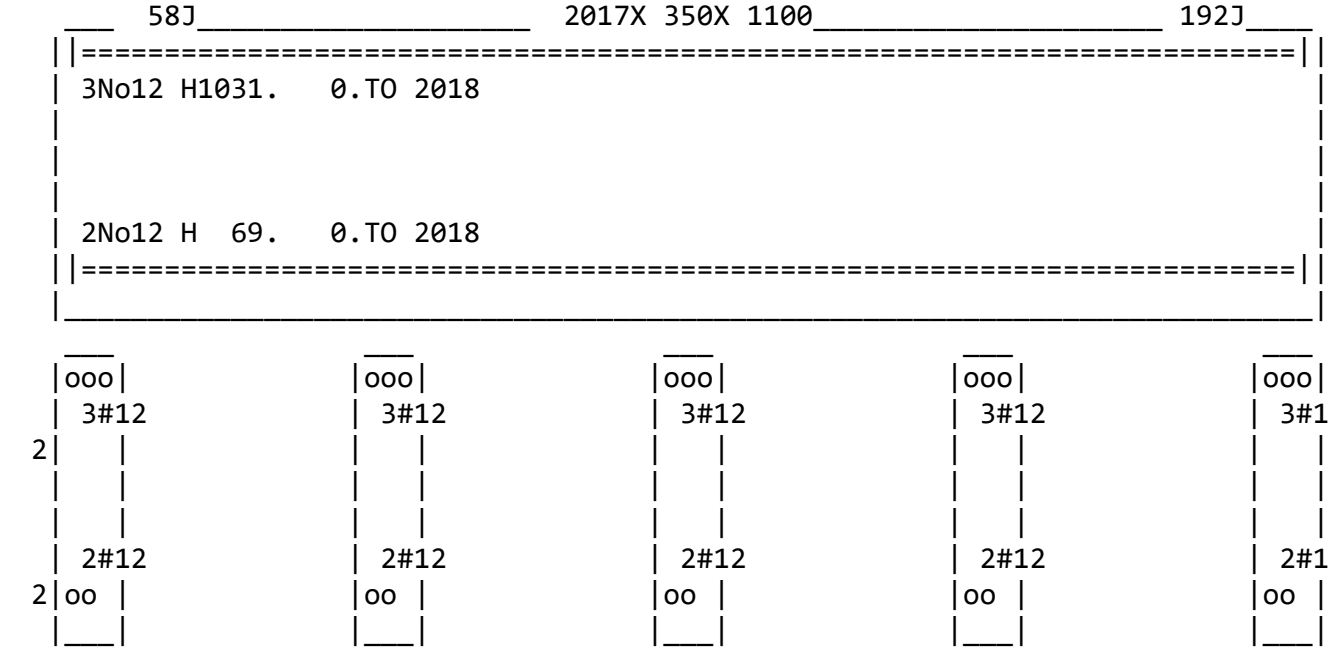
LEN - 2018. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1100. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	2018.	YES	YES
2	1031.	3 - 12MM	0.	2018.	YES	YES

B E A M N O. 2159 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2159 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2159 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2160 DESIGN RESULTS

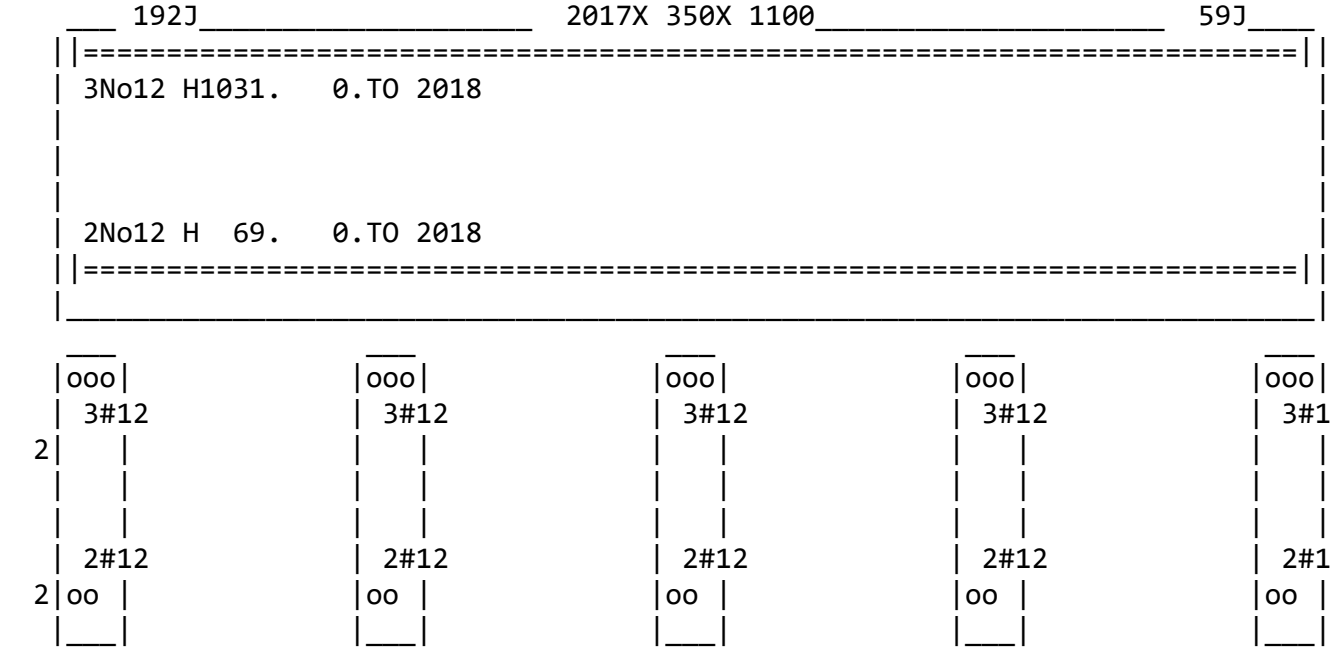
LEN - 2018. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1100. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	2018.	YES	YES
2	1031.	3 - 12MM	0.	2018.	YES	YES

B E A M N O. 2160 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2160 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2160 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2161 DESIGN RESULTS

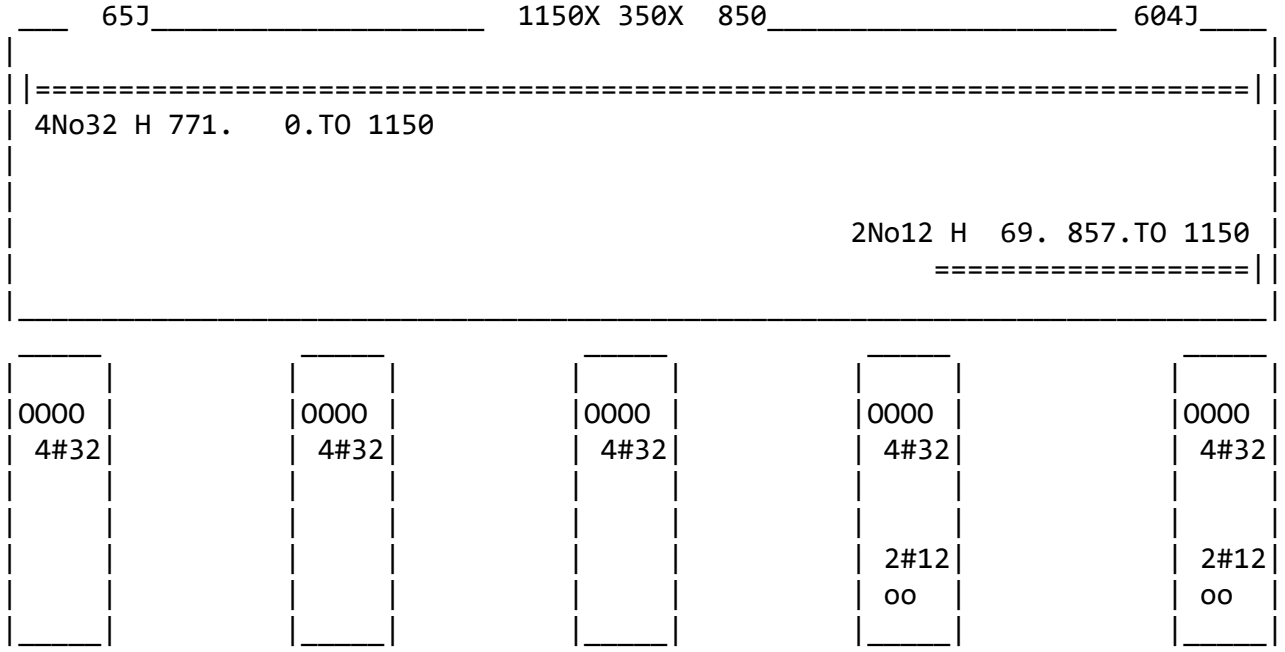
LEN - 1150. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	857.	1150.	NO	YES
2	771.	4 - 32MM	0.	1150.	YES	YES

B E A M N O. 2161 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2161 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2161 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2162 DESIGN RESULTS

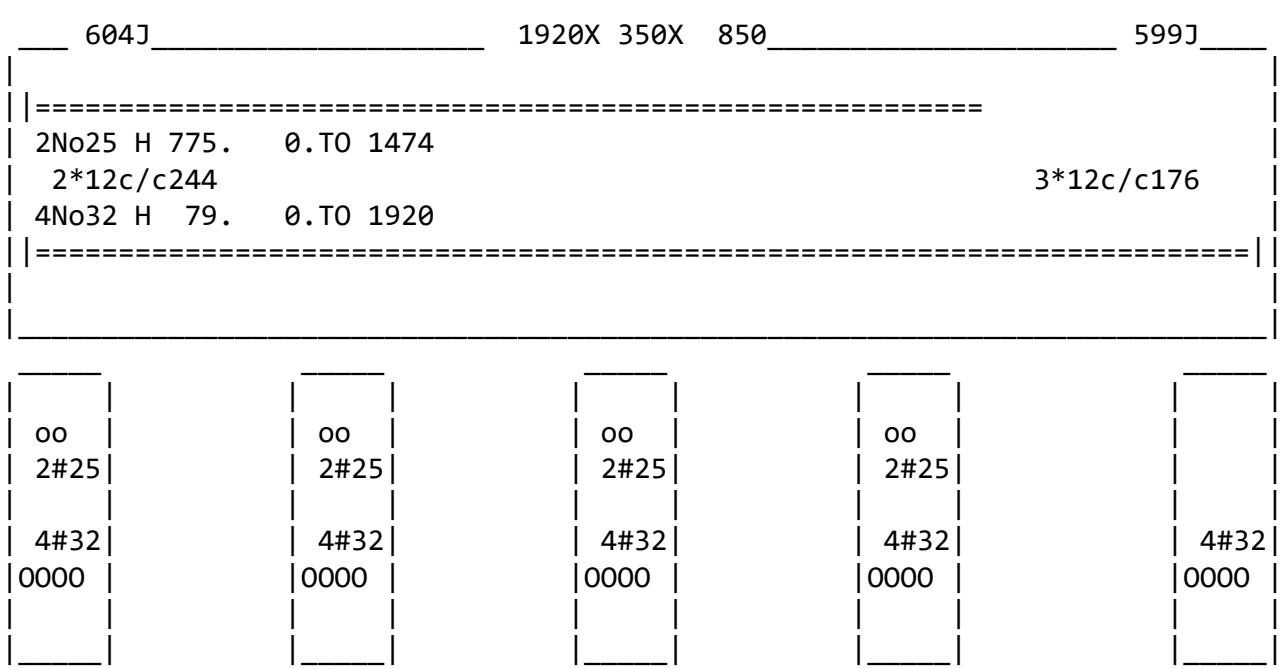
LEN - 1920. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	4 - 32MM	0.	1920.	YES	YES
2	775.	2 - 25MM	0.	1474.	YES	NO

B E A M N O. 2162 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 478.55 KNS Vc= 566.41 KNS Vs= 71.65 KNS
Tu= 19.21 KN-MET Tc= 18.5 KN-MET Ts= 25.6 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 244. MM C/C FOR 185. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.99 SQ.CM.

AT END SUPPORT - Vu= 461.11 KNS Vc= 330.04 KNS Vs= 284.77 KNS
Tu= 19.21 KN-MET Tc= 18.5 KN-MET Ts= 25.6 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 176. MM C/C FOR 185. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.99 SQ.CM.



ACI 318-11 BEAM NO. 2163 DESIGN RESULTS

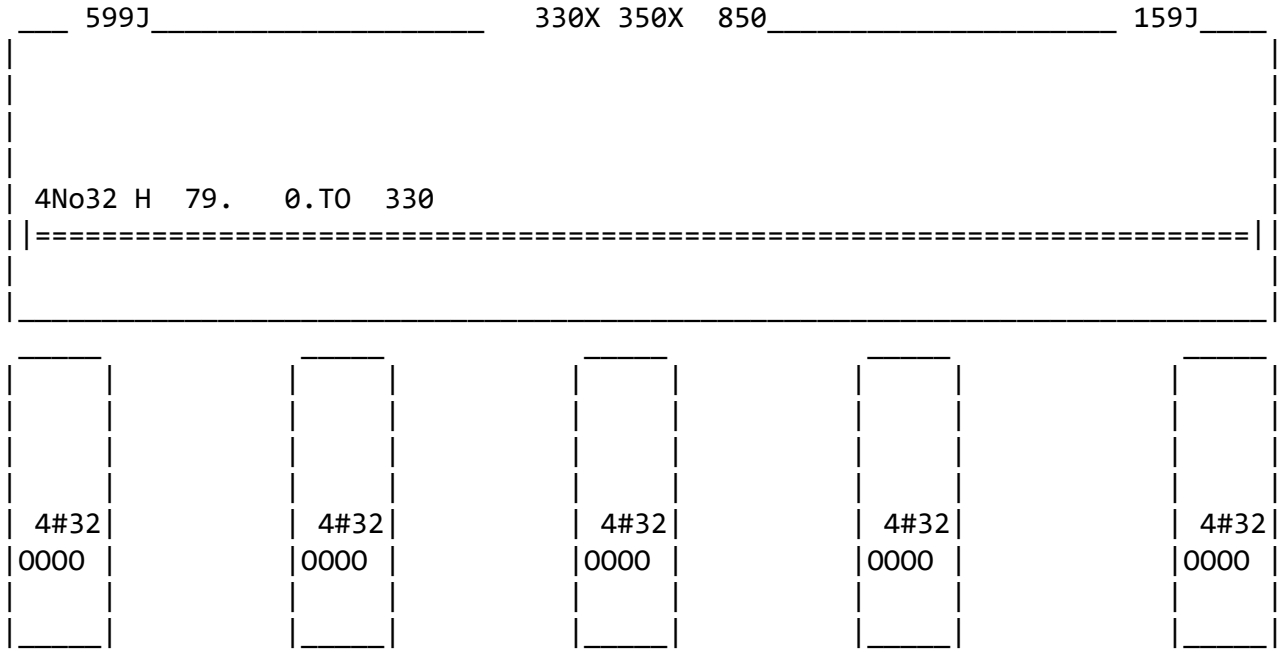
LEN - 330. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	4 - 32MM	0.	330.	YES	YES

B E A M N O. 2163 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2163 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2163 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2164 DESIGN RESULTS

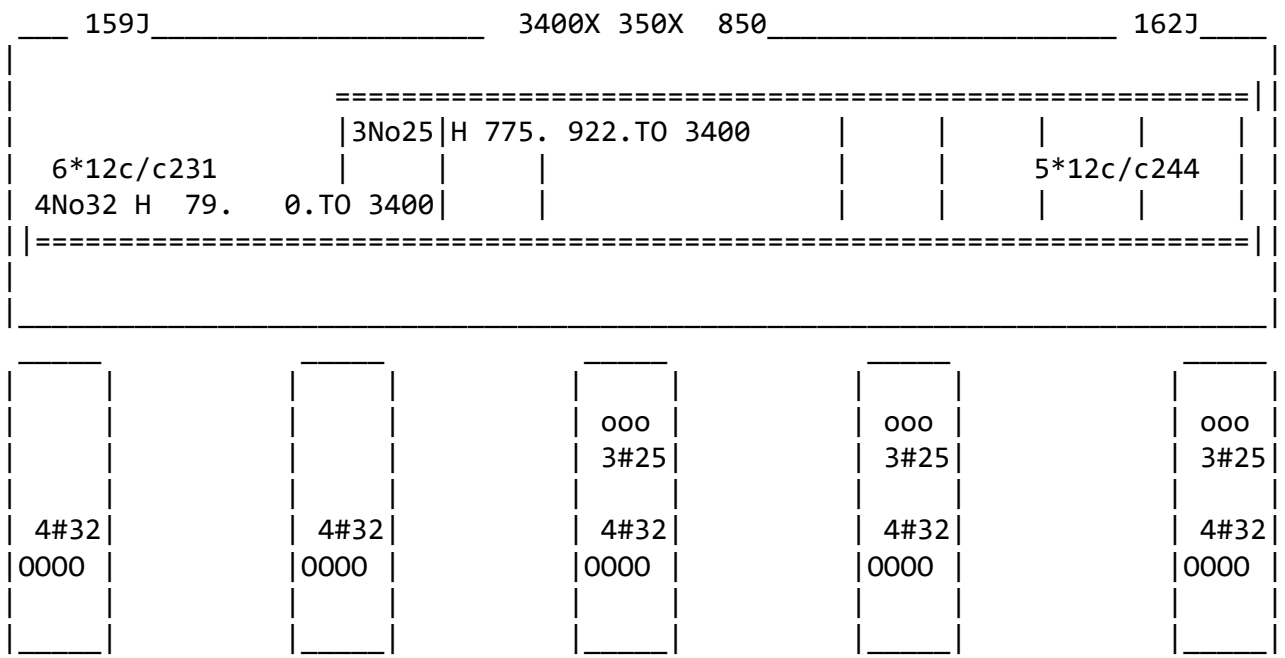
LEN - 3400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	4 - 32MM	0.	3400.	YES	YES
2	775.	3 - 25MM	922.	3400.	NO	YES

B E A M N O. 2164 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 318.46 KNS Vc= 279.25 KNS Vs= 145.36 KNS
Tu= 24.94 KN-MET Tc= 18.0 KN-MET Ts= 33.3 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 231. MM C/C FOR 925. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.18 SQ.CM.

AT END SUPPORT - Vu= 433.90 KNS Vc= 554.85 KNS Vs= 23.69 KNS
Tu= 24.94 KN-MET Tc= 18.0 KN-MET Ts= 33.3 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 244. MM C/C FOR 925. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.18 SQ.CM.



ACI 318-11 BEAM NO. 2165 DESIGN RESULTS

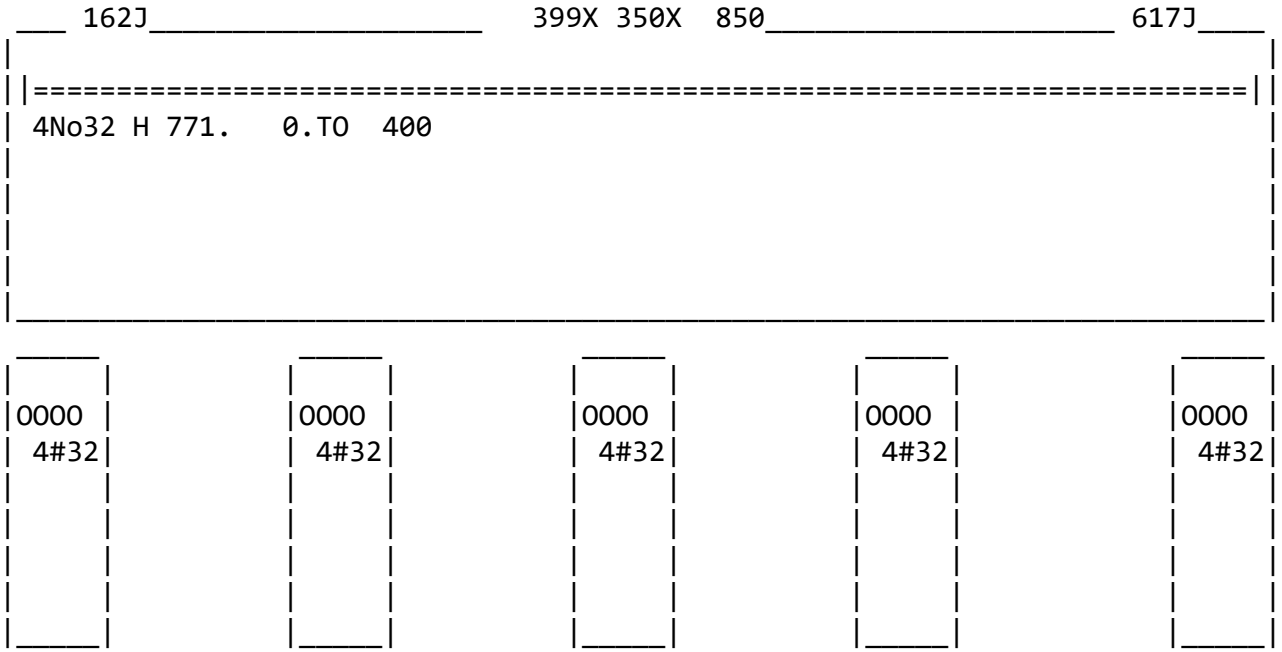
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	771.	4 - 32MM	0.	400.	YES	YES

B E A M N O. 2165 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2165 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2165 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

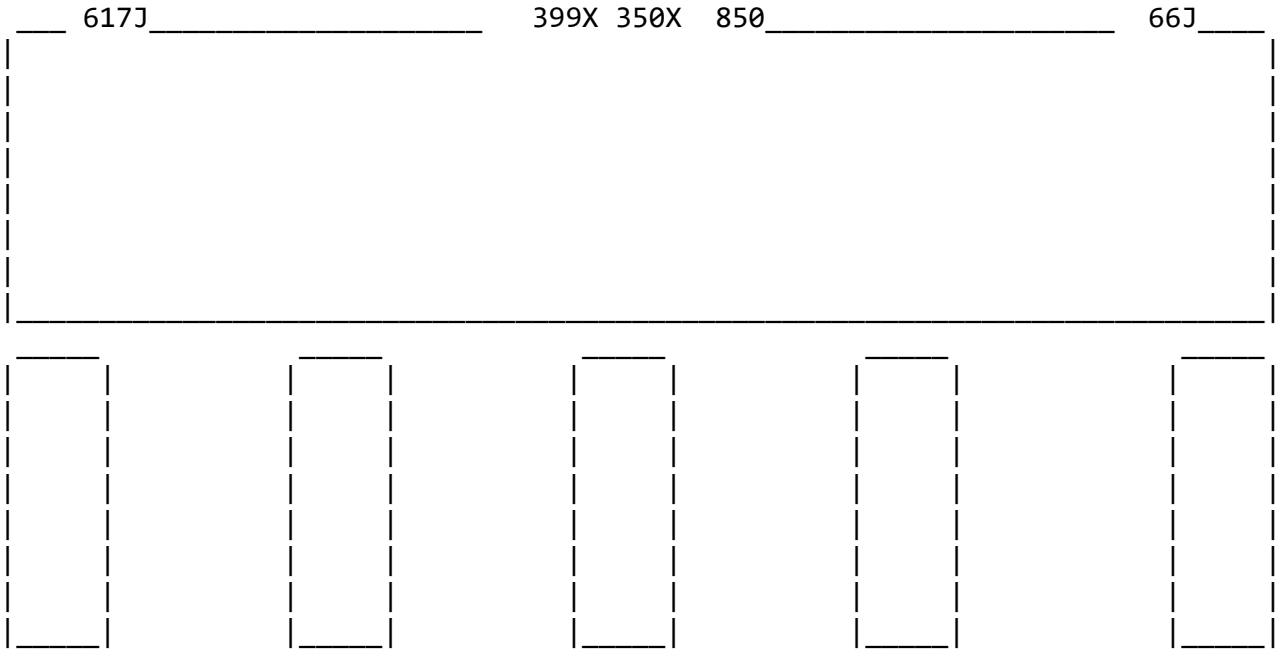


ACI 318-11 BEAM NO. 2166 DESIGN RESULTS

LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 4131. MM2, MAX. STEEL PERMISSIBLE = 6731. MM2
MAX NEG MOMENT = 1077.09 KN-MET, LOADING 406



ACI 318-11 BEAM NO. 2167 DESIGN RESULTS

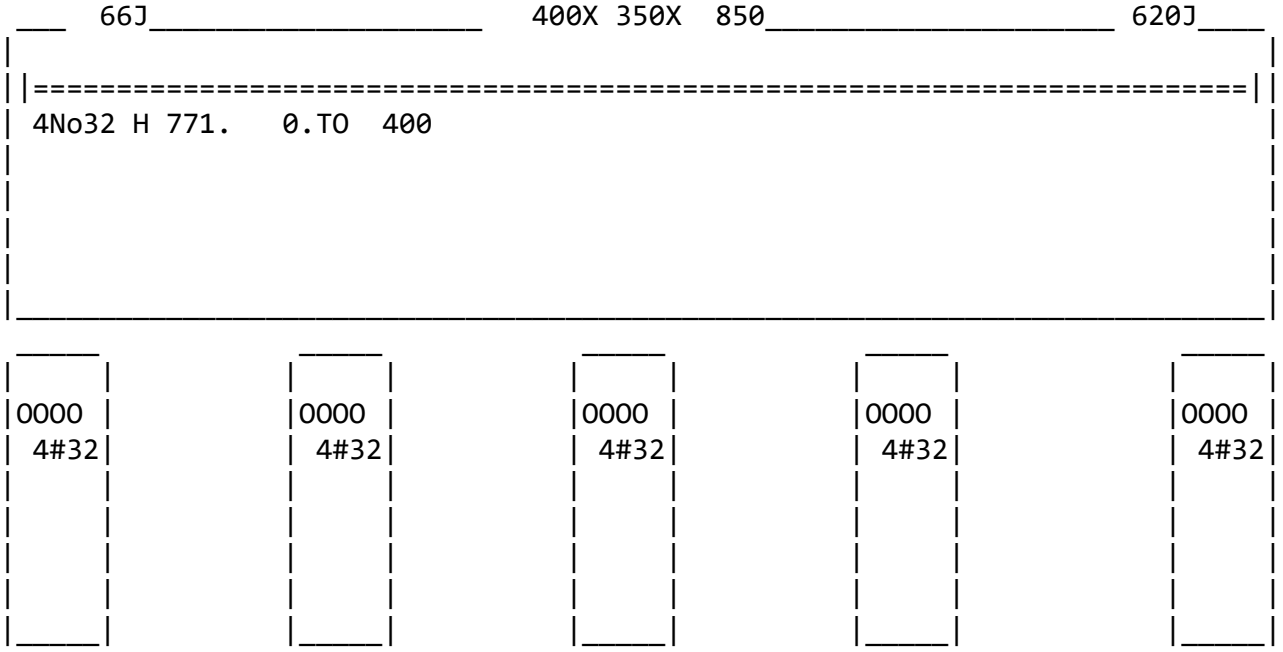
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	771.	4 - 32MM	0.	400.	YES	YES

B E A M N O. 2167 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2167 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2167 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2168 DESIGN RESULTS

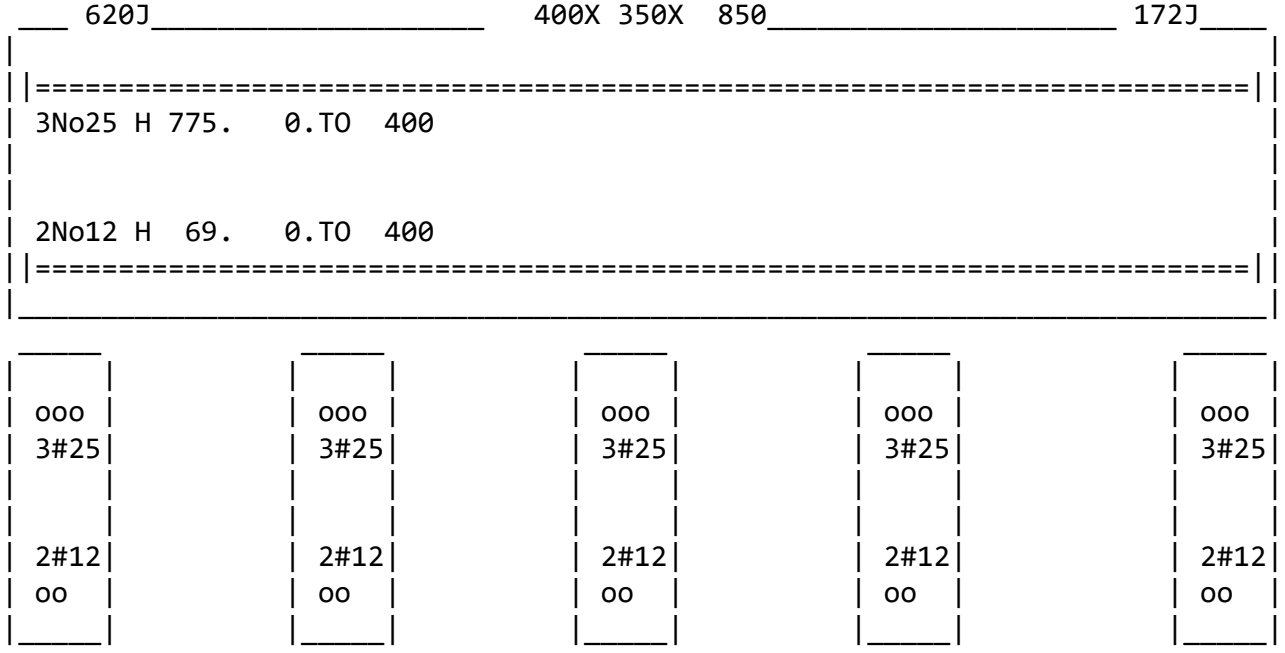
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	400.	YES	YES
2	775.	3 - 25MM	0.	400.	YES	YES

B E A M N O. 2168 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2168 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2168 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2169 DESIGN RESULTS

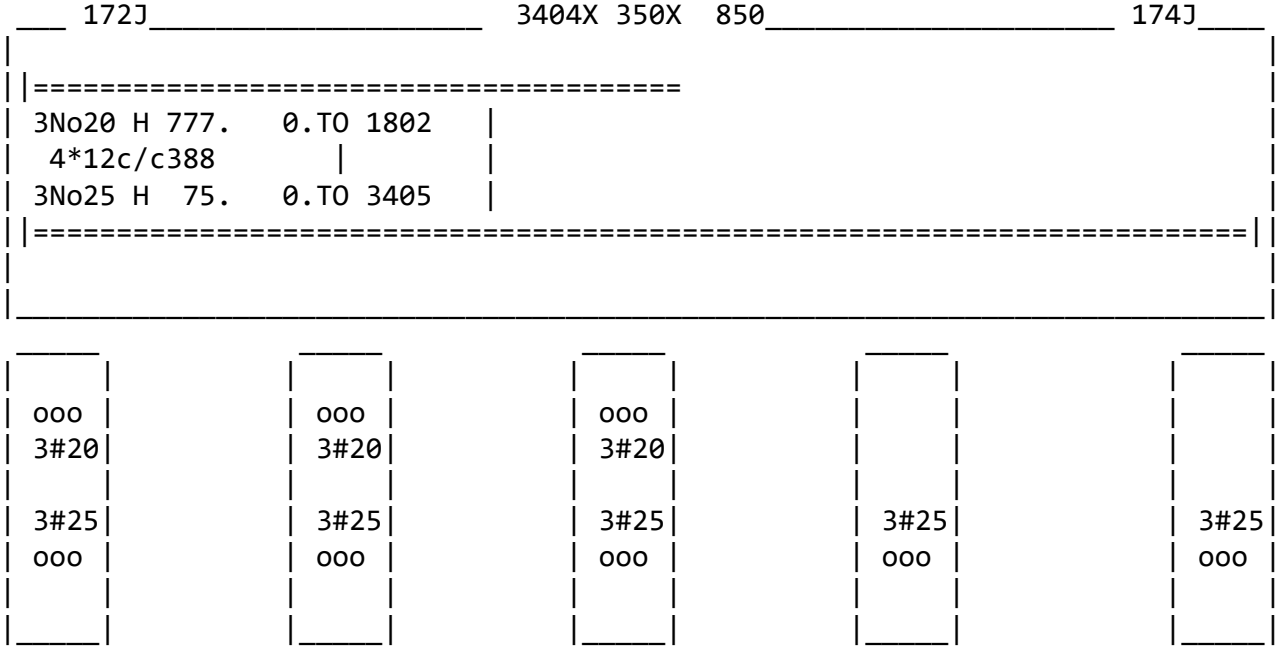
LEN - 3405. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	3405.	YES	YES
2	777.	3 - 20MM	0.	1802.	YES	NO

B E A M N O. 2169 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 192.02 KNS Vc= 504.93 KNS Vs= 0.00 KNS
Tu= 5.43 KN-MET Tc= 15.5 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 928. MM

AT END SUPPORT - Vu= 49.56 KNS Vc= 259.97 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 14.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2170 DESIGN RESULTS

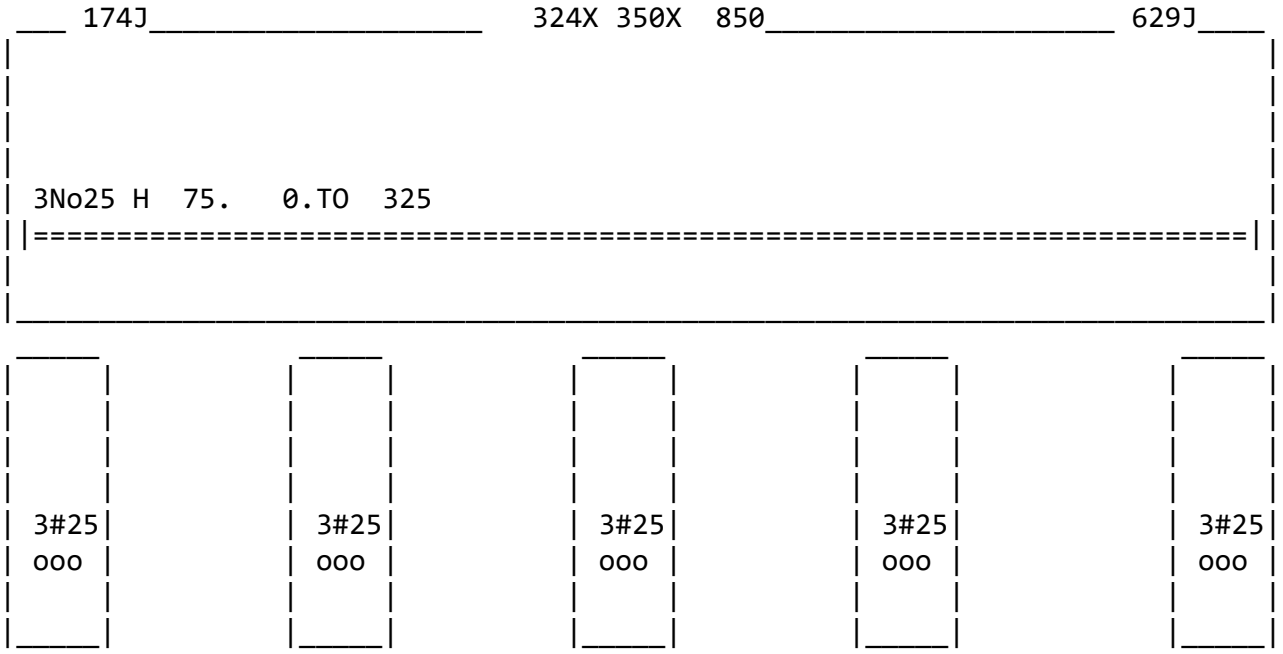
LEN - 325. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	325.	YES	YES

B E A M N O. 2170 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2170 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2170 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2171 DESIGN RESULTS

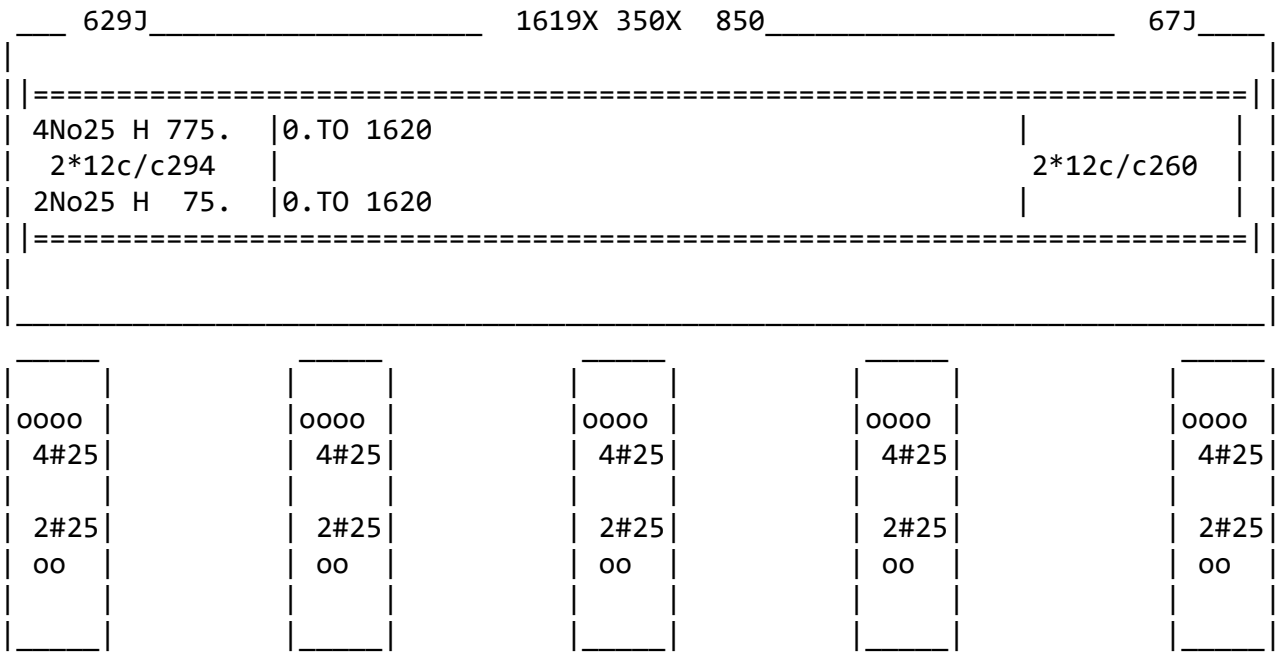
LEN - 1620. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	1620.	YES	YES
2	775.	4 - 25MM	0.	1620.	YES	YES

B E A M N O. 2171 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 405.50 KNS Vc= 290.45 KNS Vs= 250.21 KNS
Tu= 5.92 KN-MET Tc= 15.1 KN-MET Ts= 0.0 KN-MET LOAD 413
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 294. MM C/C FOR 35. MM

AT END SUPPORT - Vu= 462.55 KNS Vc= 333.62 KNS Vs= 283.11 KNS
Tu= 9.00 KN-MET Tc= 15.8 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 260. MM C/C FOR 35. MM



ACI 318-11 BEAM NO. 2172 DESIGN RESULTS

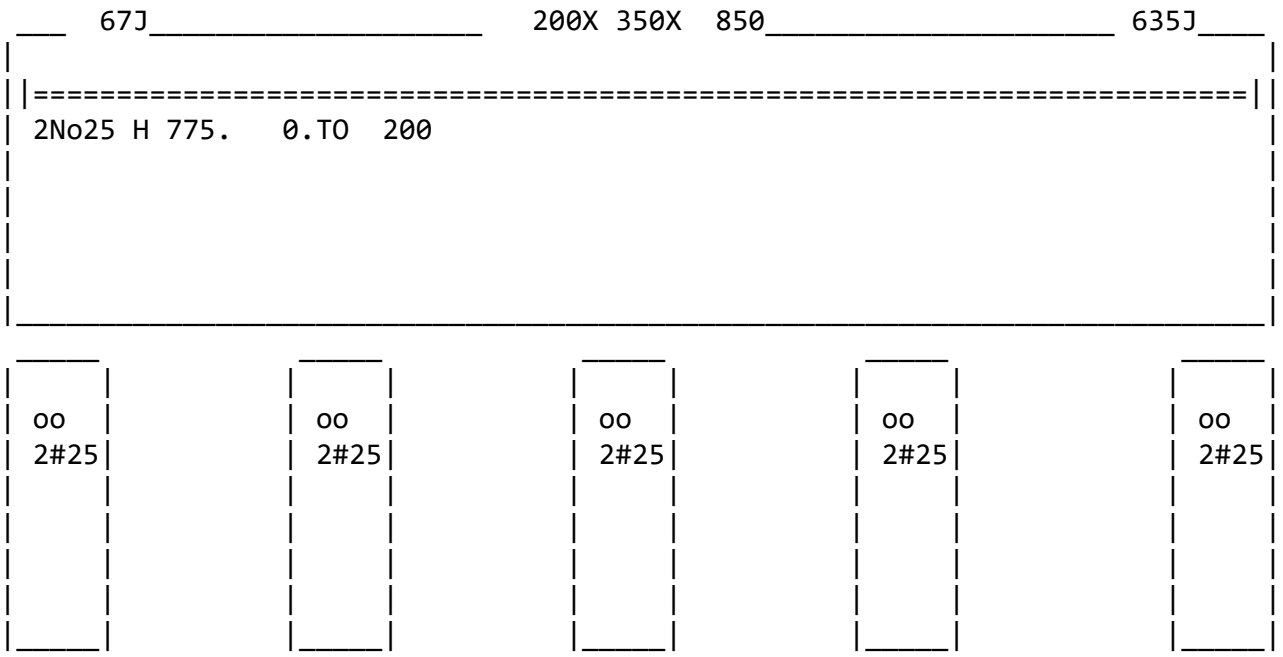
LEN - 200. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	775.	2 - 25MM	0.	200.	YES	YES

B E A M N O. 2172 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2172 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2172 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2173 DESIGN RESULTS

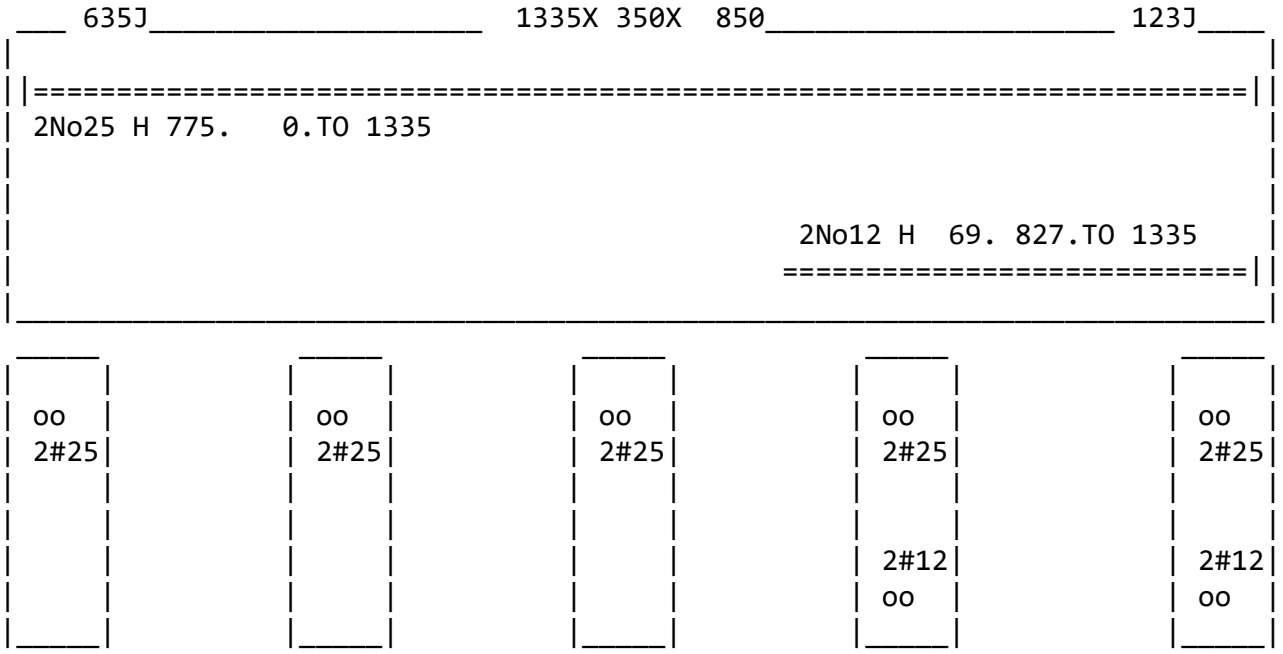
LEN - 1335. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	827.	1335.	NO	YES
2	775.	2 - 25MM	0.	1335.	YES	YES

B E A M N O. 2173 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2173 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2173 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



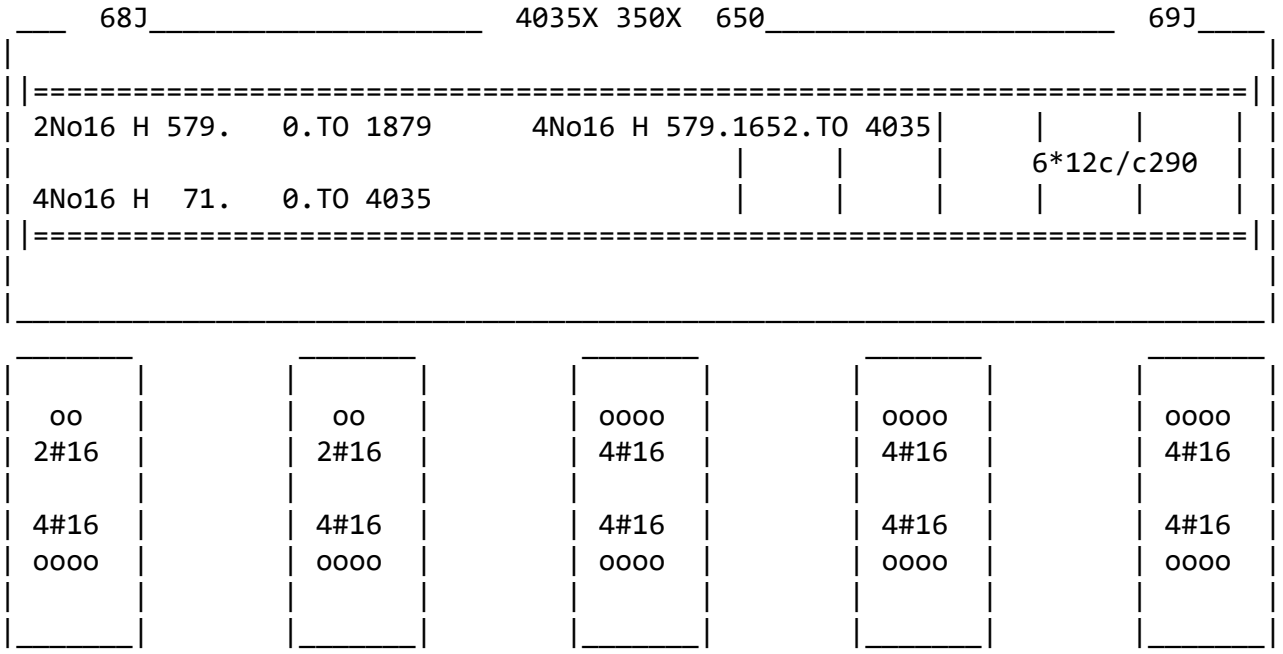
ACI 318-11 BEAM NO. 2174 DESIGN RESULTS

LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	4035.	YES	YES
2	579.	2 - 16MM	0.	1879.	YES	NO
3	579.	4 - 16MM	1652.	4035.	NO	YES

B E A M N O. 2174 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.89 KNS Vc= 191.65 KNS Vs= 0.00 KNS
Tu= 4.03 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 78.60 KNS Vc= 203.99 KNS Vs= 0.00 KNS
Tu= 1.12 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 290. MM C/C FOR 1443. MM



ACI 318-11 BEAM NO. 2175 DESIGN RESULTS

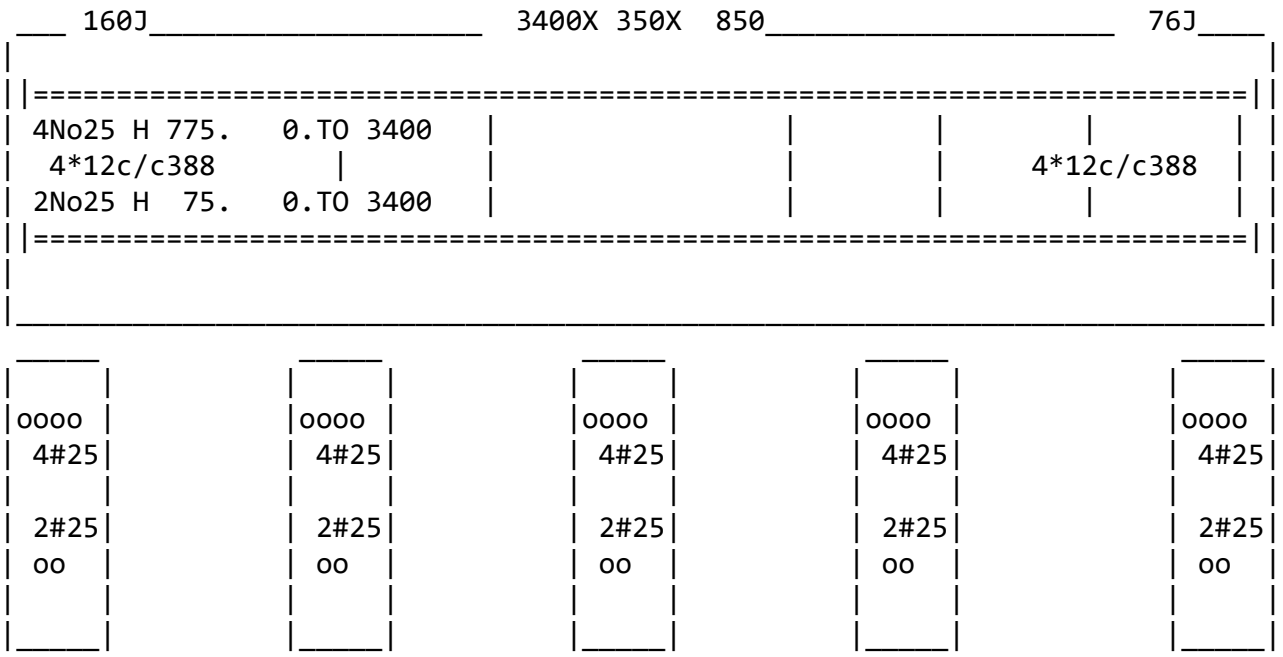
LEN - 3400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	3400.	YES	YES
2	775.	4 - 25MM	0.	3400.	YES	YES

B E A M N O. 2175 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 109.74 KNS Vc= 257.49 KNS Vs= 0.00 KNS
Tu= 5.39 KN-MET Tc= 13.2 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 925. MM

AT END SUPPORT - Vu= 118.29 KNS Vc= 256.27 KNS Vs= 0.00 KNS
Tu= 3.66 KN-MET Tc= 13.1 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 925. MM



ACI 318-11 BEAM NO. 2176 DESIGN RESULTS

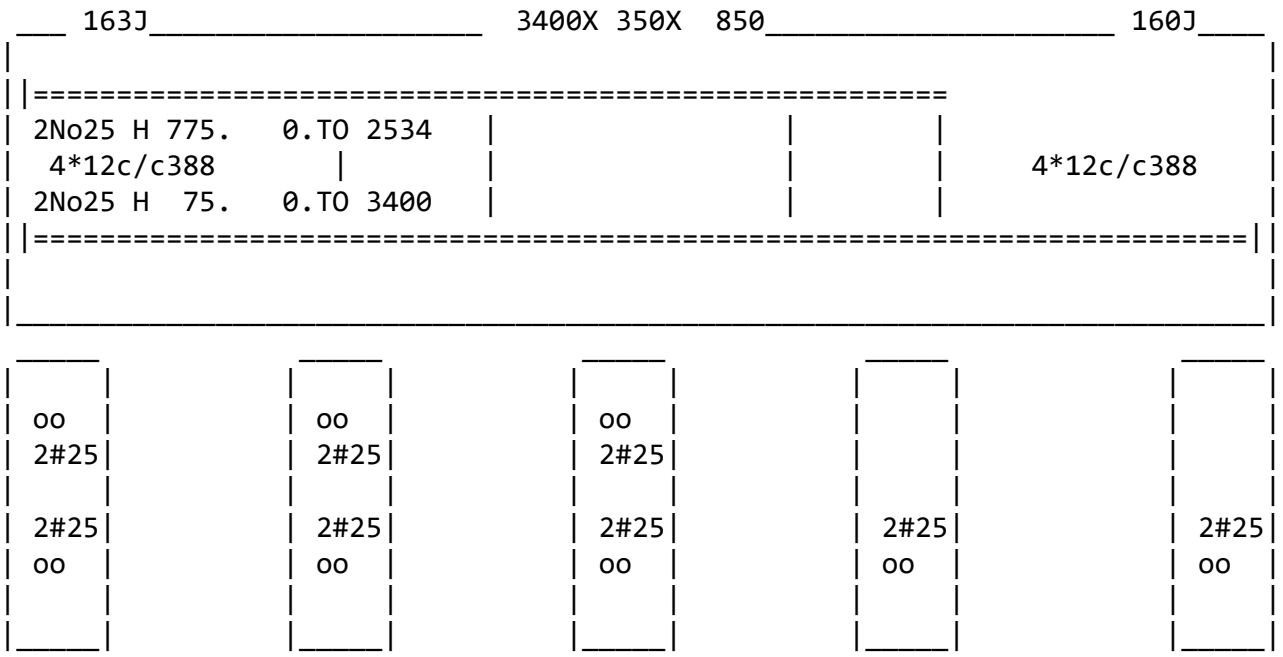
LEN - 3400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	3400.	YES	YES
2	775.	2 - 25MM	0.	2534.	YES	NO

B E A M N O. 2176 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 96.53 KNS Vc= 252.07 KNS Vs= 0.00 KNS
Tu= 5.39 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 925. MM

AT END SUPPORT - Vu= 105.41 KNS Vc= 253.90 KNS Vs= 0.00 KNS
Tu= 3.82 KN-MET Tc= 13.0 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 388. MM C/C FOR 925. MM



ACI 318-11 BEAM NO. 2177 DESIGN RESULTS

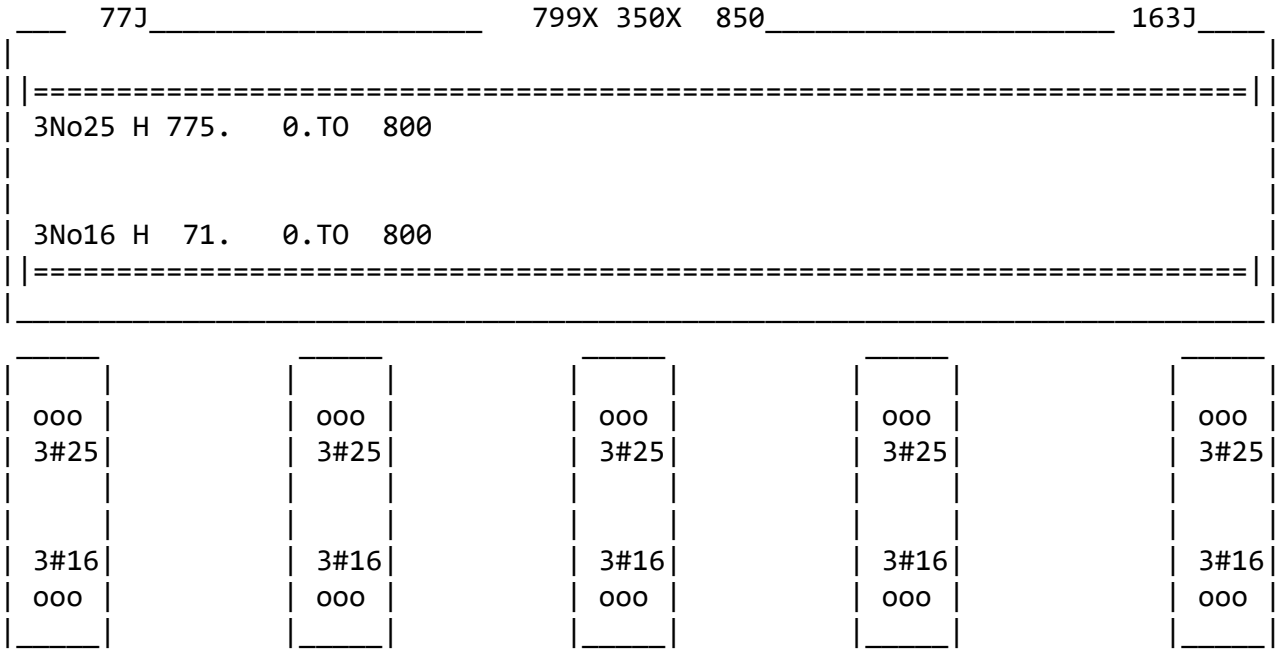
LEN - 800. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	800.	YES	YES
2	775.	3 - 25MM	0.	800.	YES	YES

B E A M N O. 2177 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2177 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2177 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2178 DESIGN RESULTS

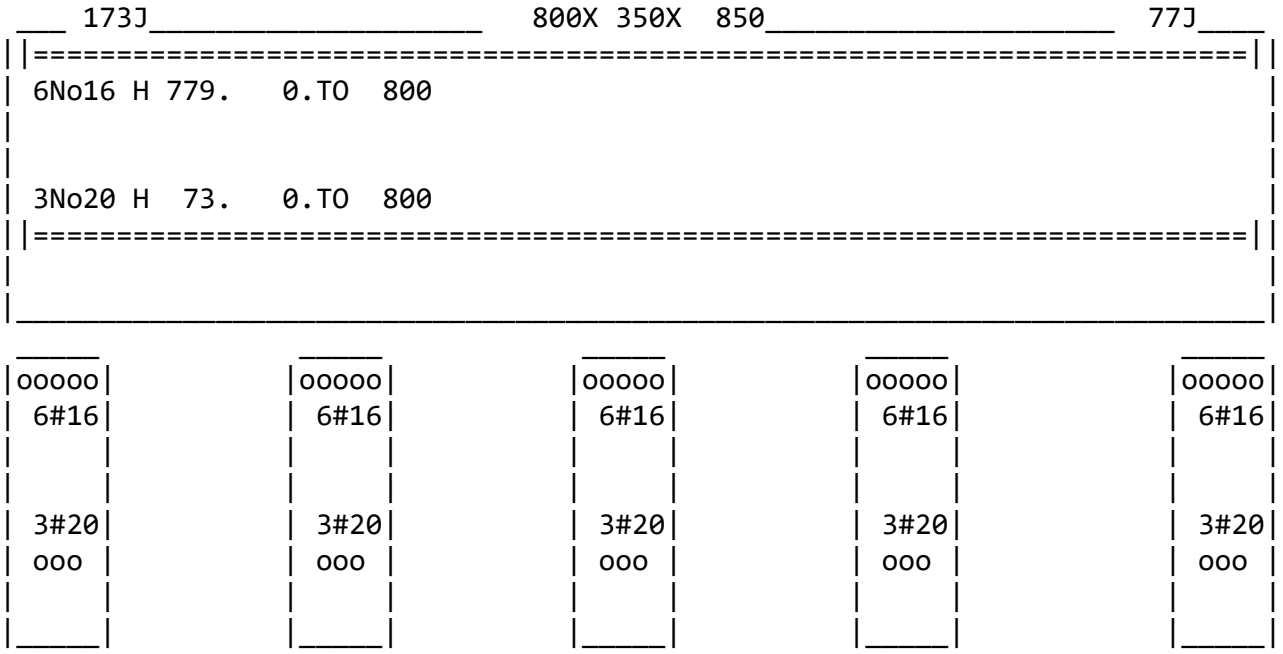
LEN - 800. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	800.	YES	YES
2	779.	6 - 16MM	0.	800.	YES	YES

B E A M N O. 2178 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2178 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2178 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



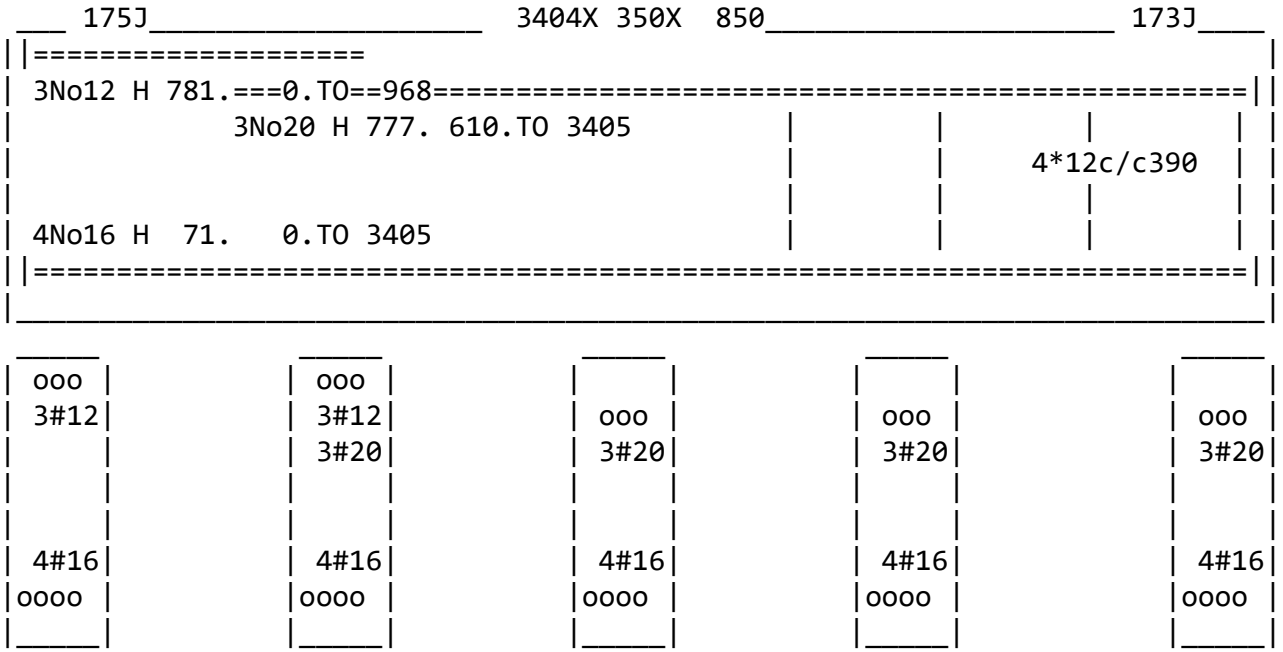
ACI 318-11 BEAM NO. 2179 DESIGN RESULTS

LEN - 3405. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	3405.	YES	YES
2	781.	3 - 12MM	0.	968.	YES	NO
3	777.	3 - 20MM	610.	3405.	NO	YES

B E A M N O. 2179 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.46 KNS Vc= 253.72 KNS Vs= 0.00 KNS
Tu= 4.76 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 98.25 KNS Vc= 254.04 KNS Vs= 0.00 KNS
Tu= 4.14 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 928. MM



ACI 318-11 BEAM NO. 2180 DESIGN RESULTS

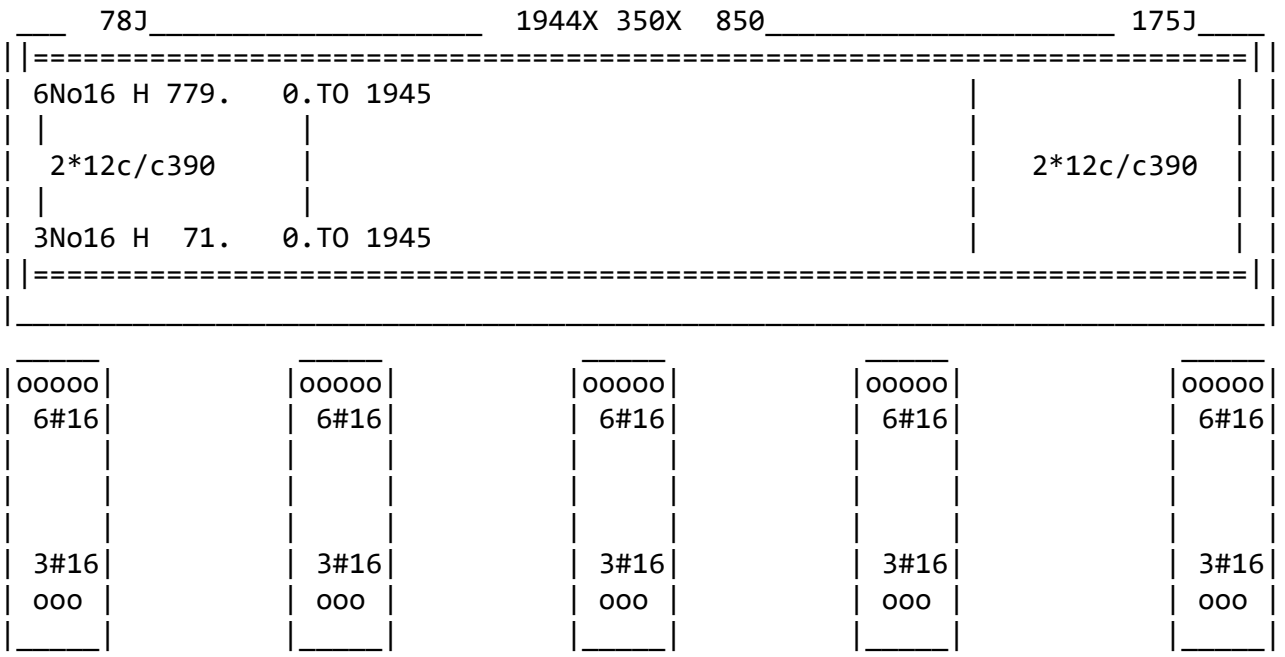
LEN - 1945. MM FY - 420. FC - 35. MPA, SIZE - 350. X 850. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	3 - 16MM	0.	1945.	YES	YES
2	779.	6 - 16MM	0.	1945.	YES	YES

B E A M N O. 2180 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 127.75 KNS Vc= 256.94 KNS Vs= 0.00 KNS
Tu= 5.12 KN-MET Tc= 13.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 198. MM

AT END SUPPORT - Vu= 116.31 KNS Vc= 256.94 KNS Vs= 0.00 KNS
Tu= 5.12 KN-MET Tc= 13.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 390. MM C/C FOR 198. MM



ACI 318-11 BEAM NO. 2181 DESIGN RESULTS

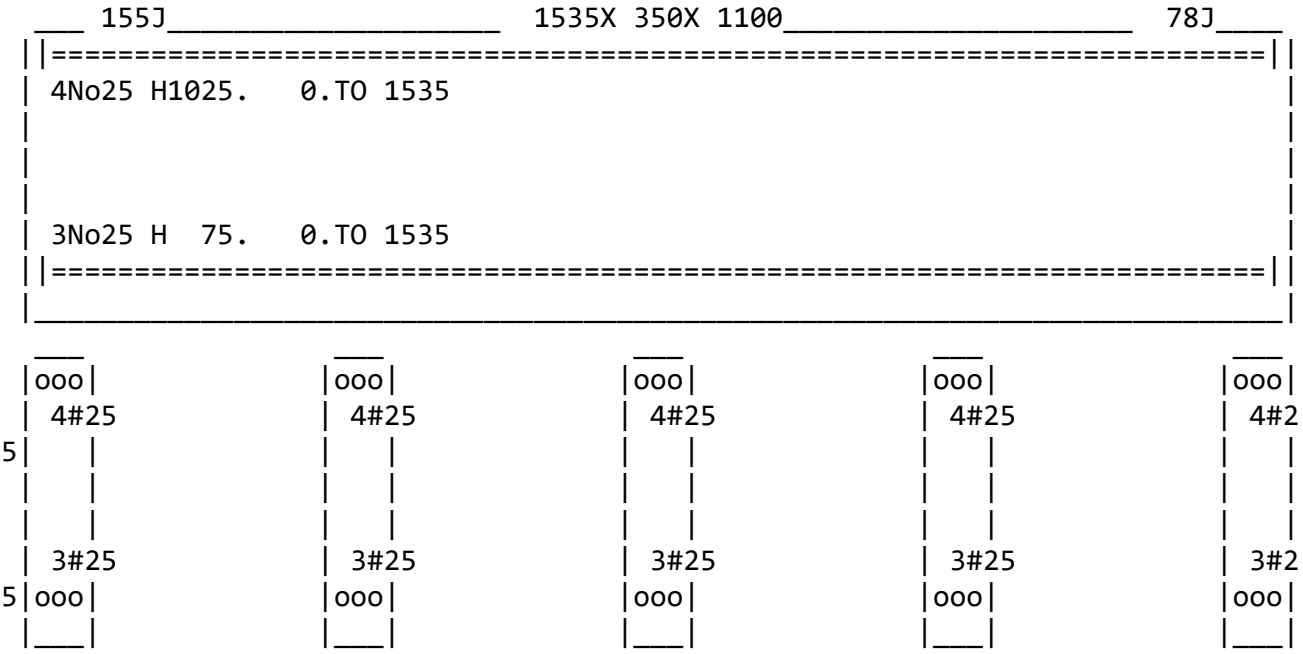
LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1100. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1535.	YES	YES
2	1025.	4 - 25MM	0.	1535.	YES	YES

B E A M N O. 2181 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2181 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2181 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2182 DESIGN RESULTS

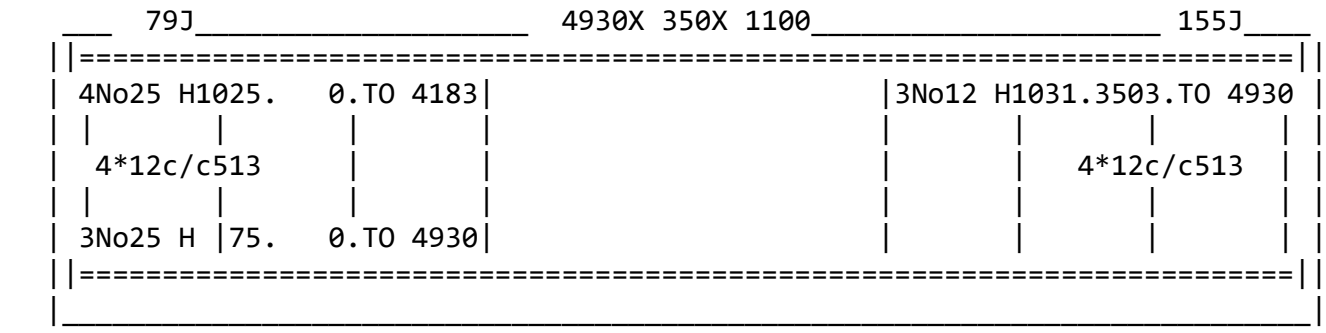
LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1100. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	4930.	YES	YES
2	1025.	4 - 25MM	0.	4183.	YES	NO
3	1031.	3 - 12MM	3503.	4930.	NO	YES

B E A M N O. 2182 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 165.28 KNS Vc= 347.27 KNS Vs= 0.00 KNS
Tu= 4.57 KN-MET Tc= 18.6 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 513. MM C/C FOR 1440. MM

AT END SUPPORT - Vu= 161.29 KNS Vc= 326.06 KNS Vs= 0.00 KNS
Tu= 5.30 KN-MET Tc= 17.5 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 513. MM C/C FOR 1440. MM



ACI 318-11 BEAM NO. 2183 DESIGN RESULTS

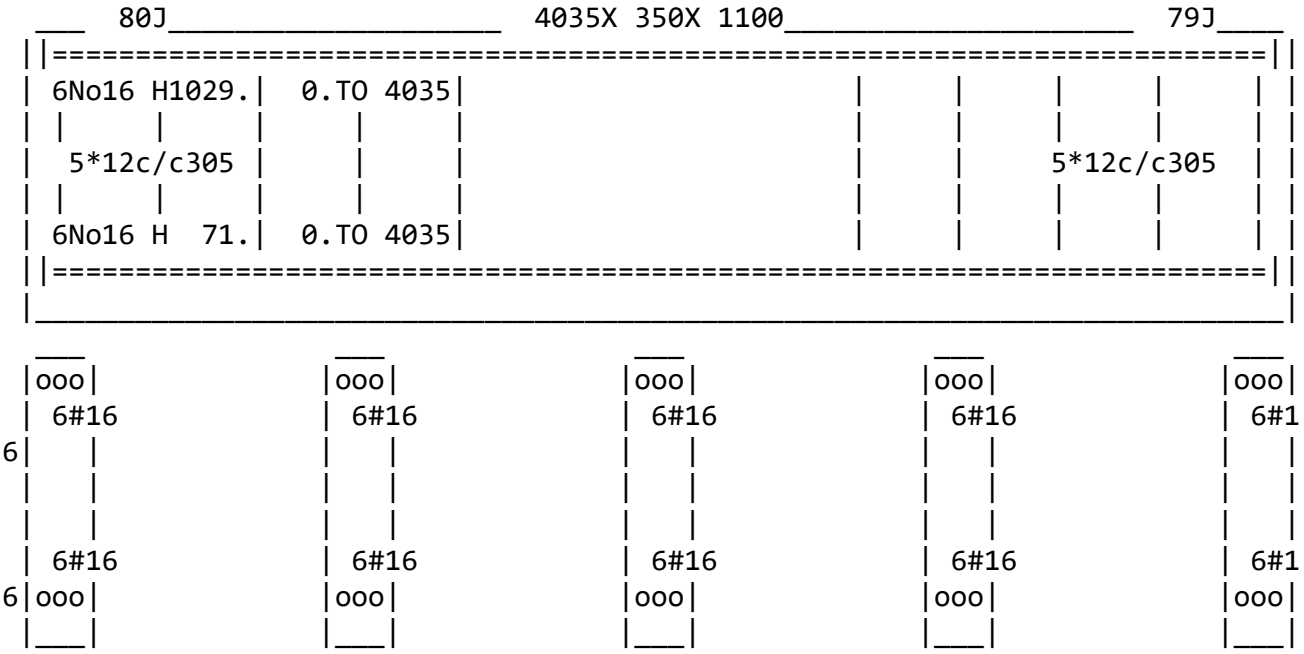
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 350. X 1100. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	4035.	YES	YES
2	1029.	6 - 16MM	0.	4035.	YES	YES

B E A M N O. 2183 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 4.30 KNS Vc= 342.92 KNS Vs= 0.00 KNS
Tu= 26.28 KN-MET Tc= 19.1 KN-MET Ts= 35.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 993. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.12 SQ.CM.

AT END SUPPORT - Vu= 67.36 KNS Vc= 370.64 KNS Vs= 0.00 KNS
Tu= 26.28 KN-MET Tc= 19.1 KN-MET Ts= 35.0 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 993. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.12 SQ.CM.



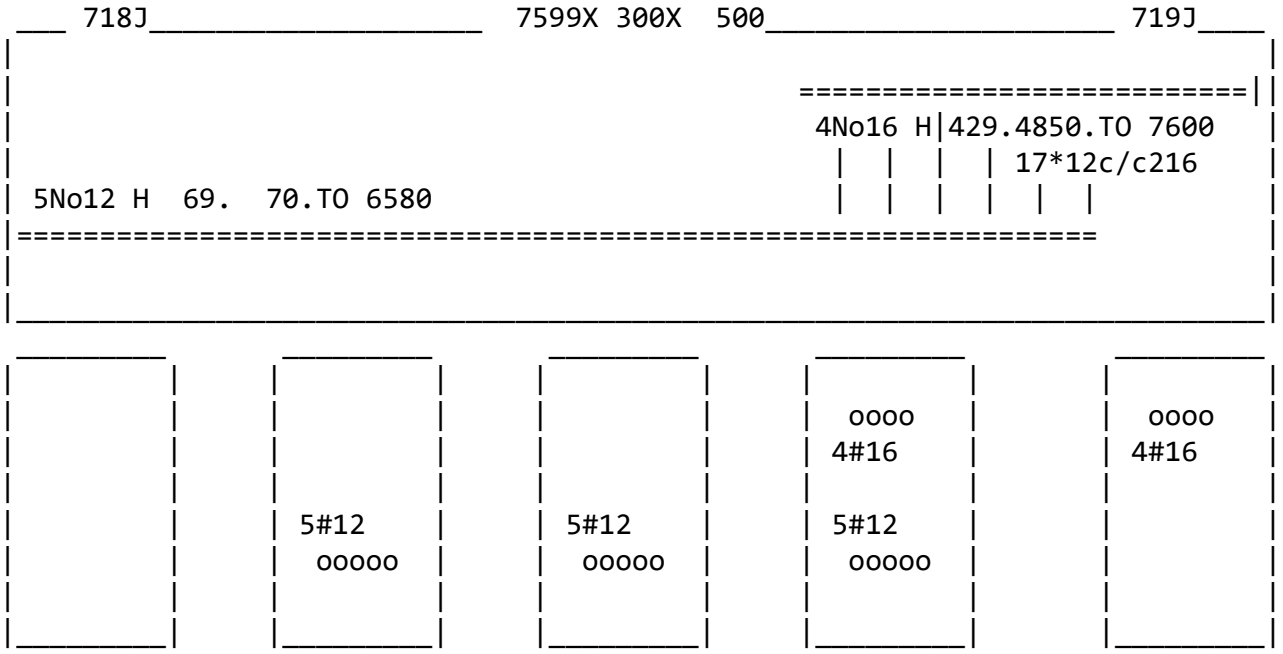
ACI 318-11 BEAM NO. 2184 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	70.	6580.	NO	NO
2	429.	4 - 16MM	4850.	7600.	NO	YES

B E A M N O. 2184 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 22.74 KNS Vc= 128.68 KNS Vs= 0.00 KNS
Tu= 0.88 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 63.54 KNS Vc= 125.11 KNS Vs= 0.00 KNS
Tu= 0.96 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 3375. MM



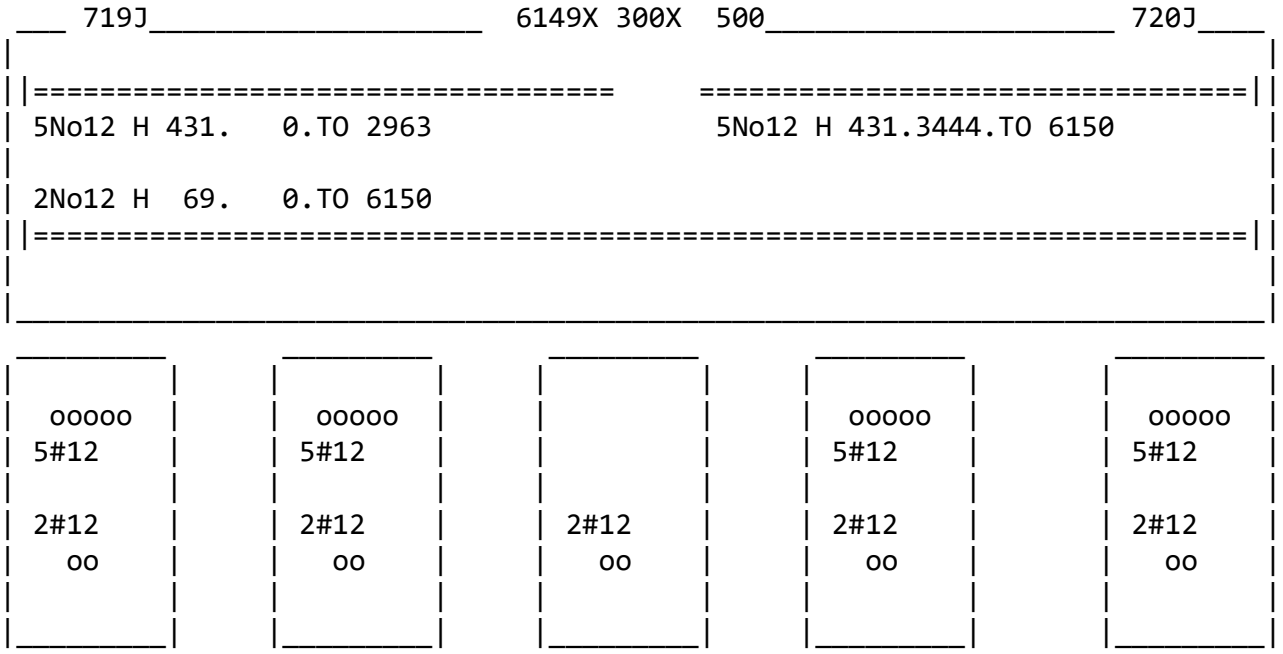
ACI 318-11 BEAM NO. 2185 DESIGN RESULTS

LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	6150.	YES	YES
2	431.	5 - 12MM	0.	2963.	YES	NO
3	431.	5 - 12MM	3444.	6150.	NO	YES

B E A M N O. 2185 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 25.05 KNS Vc= 126.61 KNS Vs= 0.00 KNS
Tu= 0.89 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.86 KNS Vc= 126.61 KNS Vs= 0.00 KNS
Tu= 0.89 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



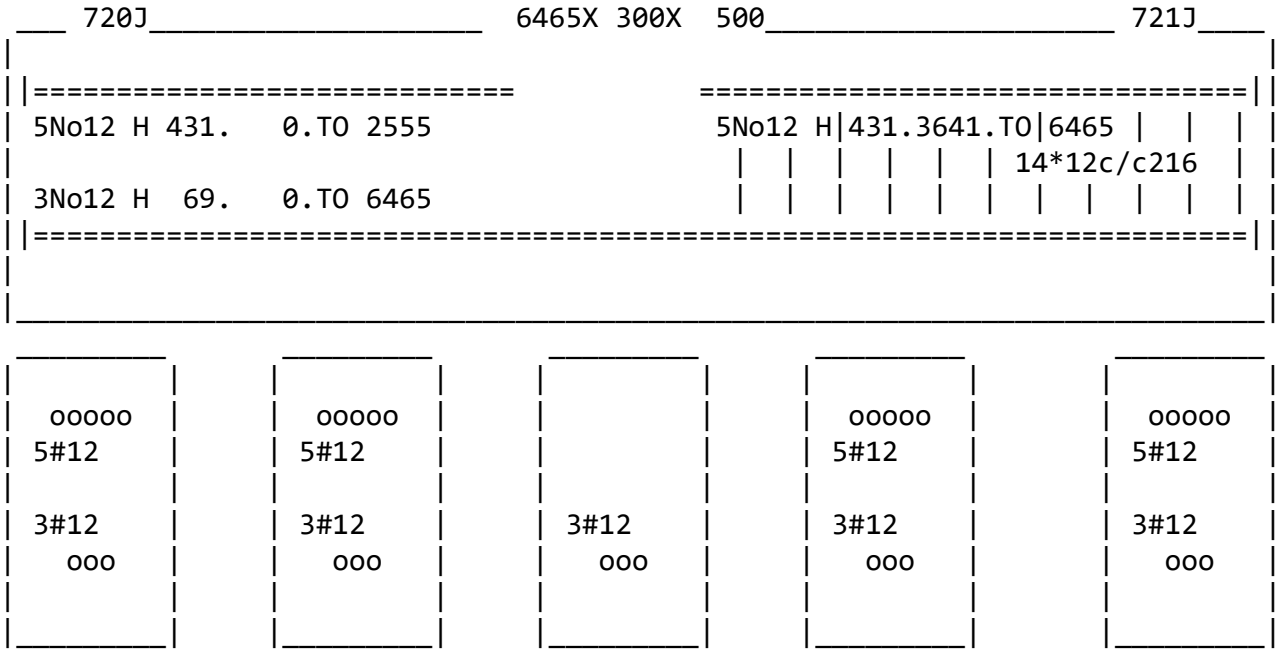
ACI 318-11 BEAM NO. 2186 DESIGN RESULTS

LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	6465.	YES	YES
2	431.	5 - 12MM	0.	2555.	YES	NO
3	431.	5 - 12MM	3641.	6465.	NO	YES

B E A M N O. 2186 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 24.46 KNS Vc= 124.97 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 47.51 KNS Vc= 124.66 KNS Vs= 0.00 KNS
Tu= 0.33 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 216. MM C/C FOR 2808. MM



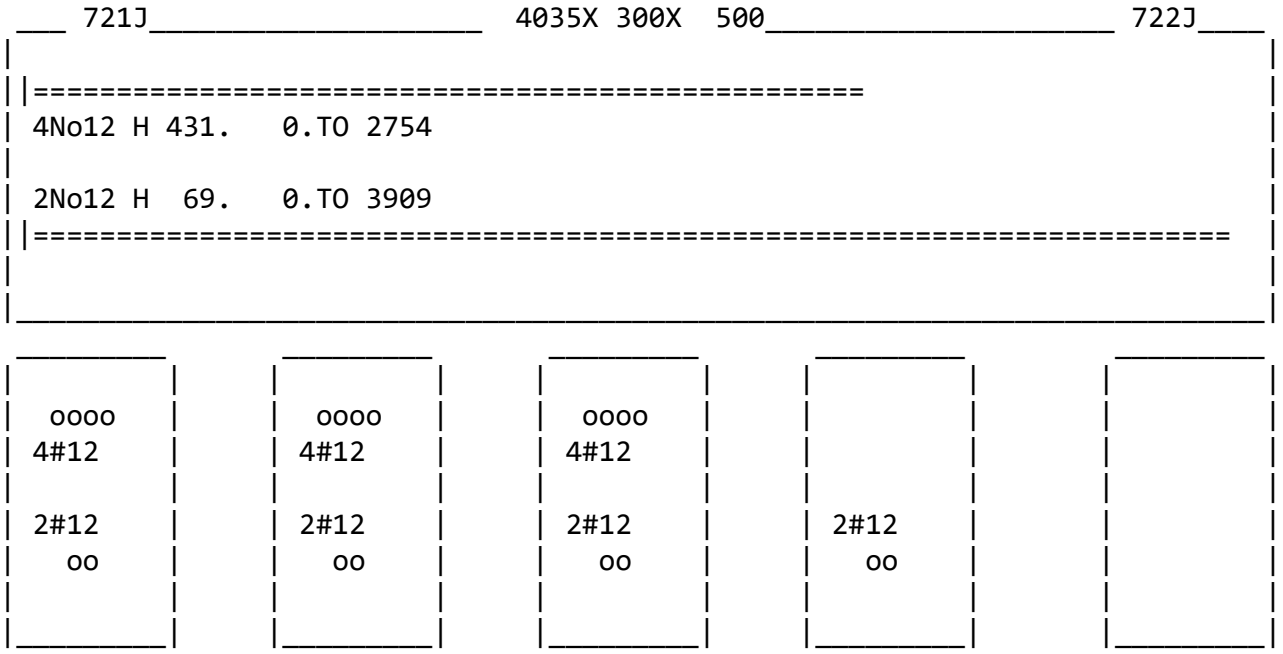
ACI 318-11 BEAM NO. 2187 DESIGN RESULTS

LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	3909.	YES	NO
2	431.	4 - 12MM	0.	2754.	YES	NO

B E A M N O. 2187 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 18.63 KNS Vc= 126.58 KNS Vs= 0.00 KNS
Tu= 1.21 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.05 KNS Vc= 126.58 KNS Vs= 0.00 KNS
Tu= 1.21 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



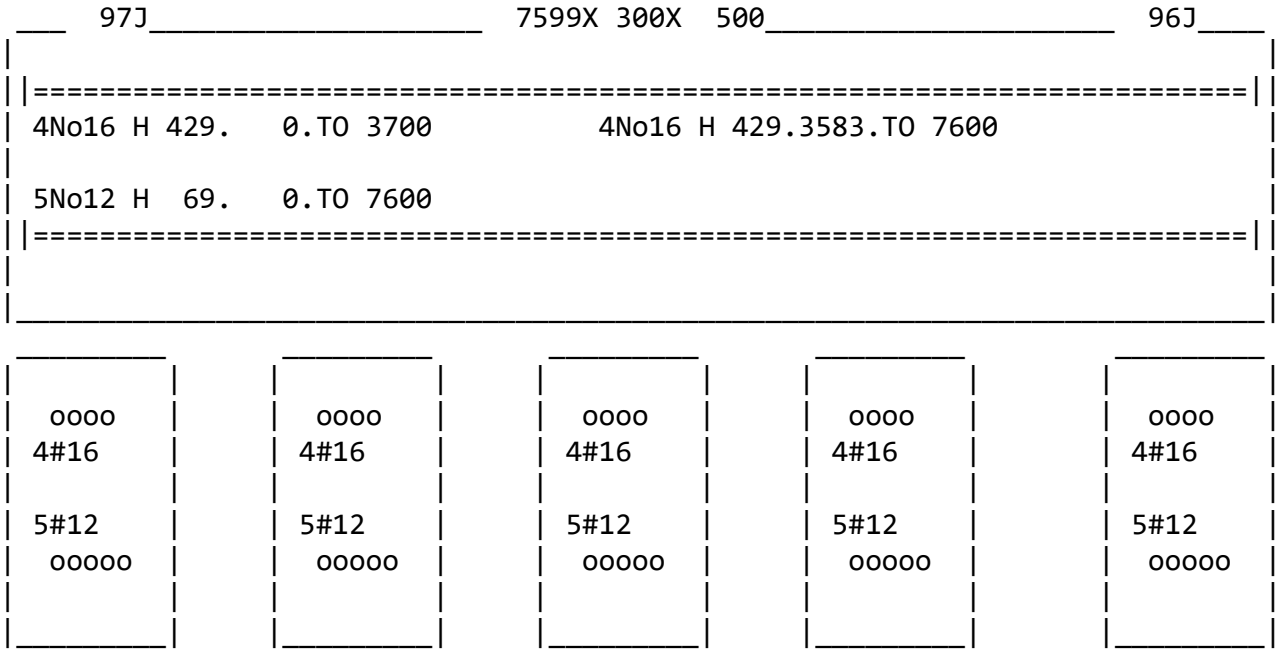
ACI 318-11 BEAM NO. 2188 DESIGN RESULTS

LEN - 7600. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	7600.	YES	YES
2	429.	4 - 16MM	0.	3700.	YES	NO
3	429.	4 - 16MM	3583.	7600.	NO	YES

B E A M N O. 2188 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.49 KNS Vc= 121.74 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 15.91 KNS Vc= 121.74 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2189 DESIGN RESULTS

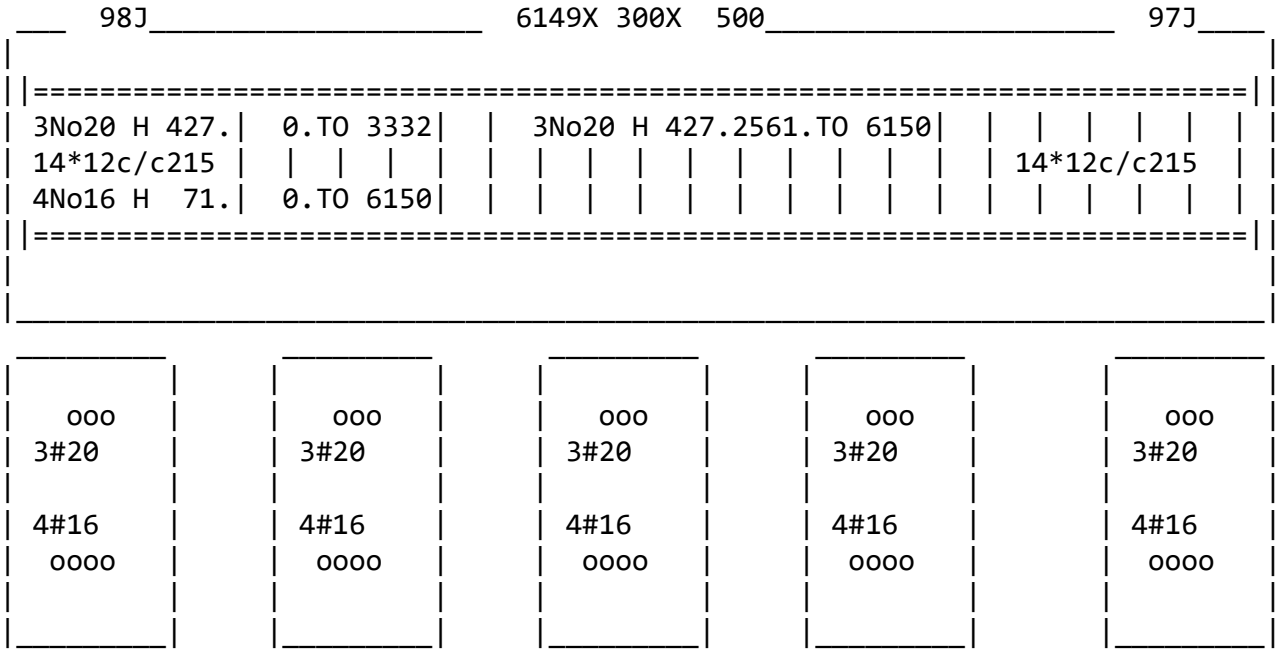
LEN - 6150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	6150.	YES	YES
2	427.	3 - 20MM	0.	3332.	YES	NO
3	427.	3 - 20MM	2561.	6150.	NO	YES

B E A M N O. 2189 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 48.19 KNS Vc= 123.20 KNS Vs= 0.00 KNS
Tu= 1.02 KN-MET Tc= 5.6 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2650. MM

AT END SUPPORT - Vu= 52.96 KNS Vc= 121.24 KNS Vs= 0.00 KNS
Tu= 1.34 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2650. MM



ACI 318-11 BEAM NO. 2190 DESIGN RESULTS

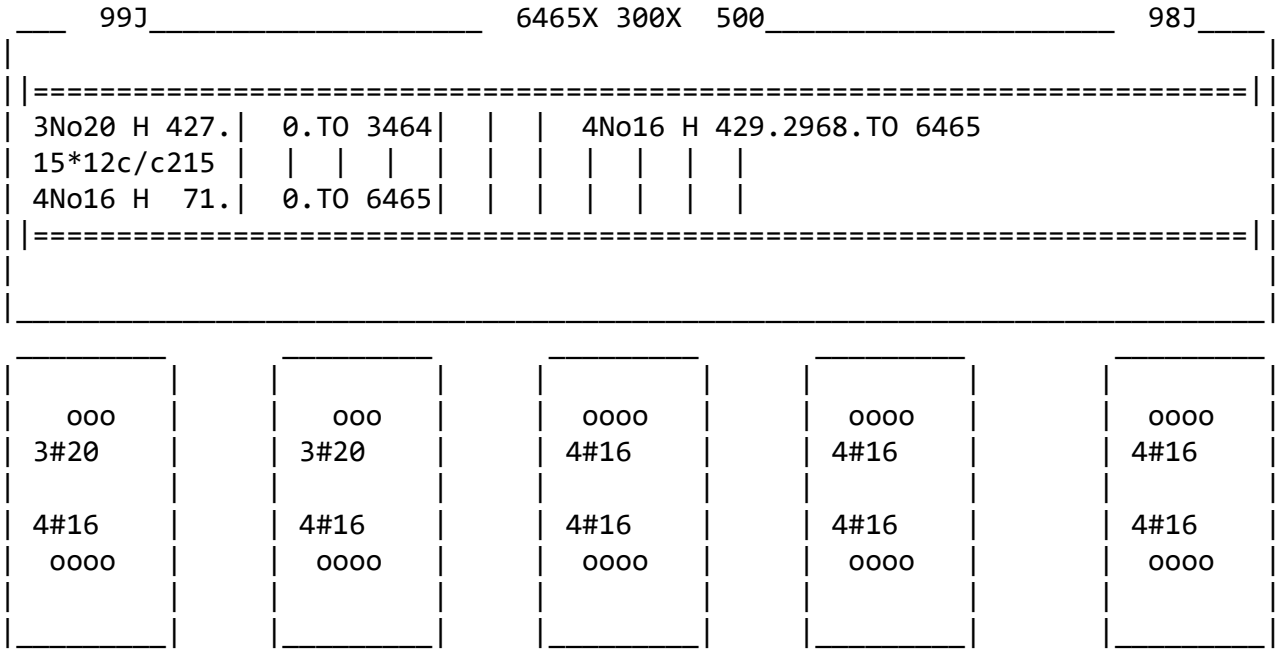
LEN - 6465. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	6465.	YES	YES
2	427.	3 - 20MM	0.	3464.	YES	NO
3	429.	4 - 16MM	2968.	6465.	NO	YES

B E A M N O. 2190 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 48.62 KNS Vc= 116.57 KNS Vs= 0.00 KNS
Tu= 1.15 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 215. MM C/C FOR 2808. MM

AT END SUPPORT - Vu= 10.64 KNS Vc= 116.76 KNS Vs= 0.00 KNS
Tu= 1.00 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2191 DESIGN RESULTS

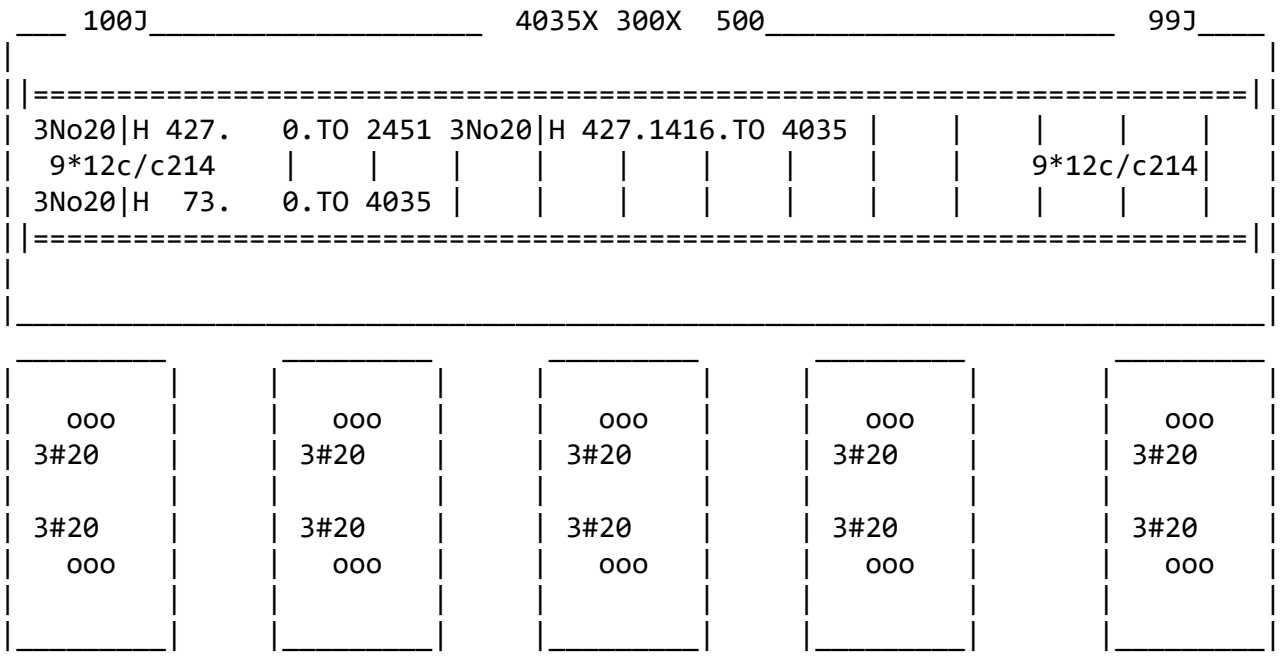
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	4035.	YES	YES
2	427.	3 - 20MM	0.	2451.	YES	NO
3	427.	3 - 20MM	1416.	4035.	NO	YES

B E A M N O. 2191 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 74.51 KNS Vc= 124.32 KNS Vs= 0.00 KNS
Tu= 1.34 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 1593. MM

AT END SUPPORT - Vu= 61.00 KNS Vc= 123.69 KNS Vs= 0.00 KNS
Tu= 1.34 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 409
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 214. MM C/C FOR 1593. MM



ACI 318-11 BEAM NO. 2201 DESIGN RESULTS

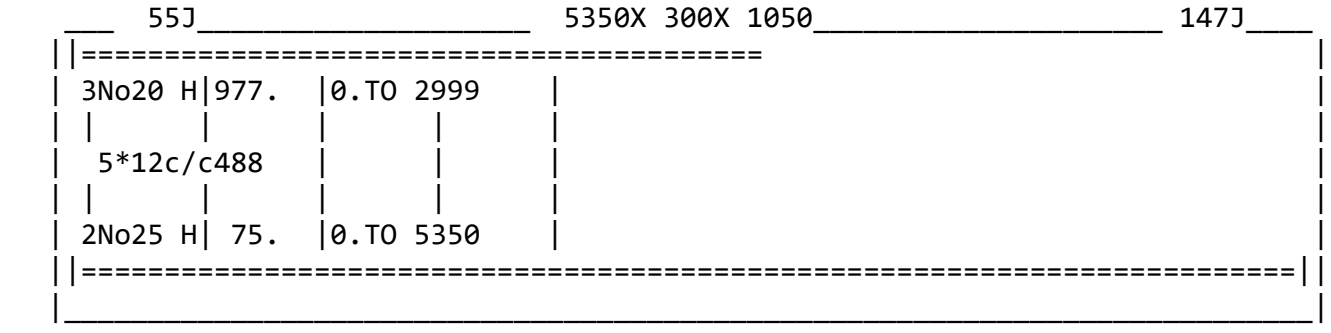
LEN - 5350. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

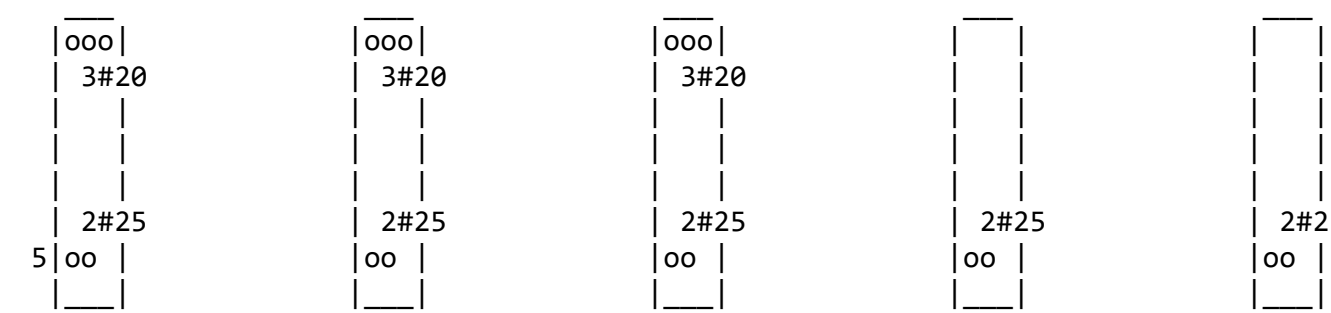
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	2 - 25MM	0.	5350.	YES	YES
2	977.	3 - 20MM	0.	2999.	YES	NO

B E A M N O. 2201 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 114.99 KNS Vc= 287.07 KNS Vs= 0.00 KNS
Tu= 1.50 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 488. MM C/C FOR 1700. MM

AT END SUPPORT - Vu= 23.62 KNS Vc= 275.46 KNS Vs= 0.00 KNS
Tu= 1.42 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.





ACI 318-11 BEAM NO. 2202 DESIGN RESULTS

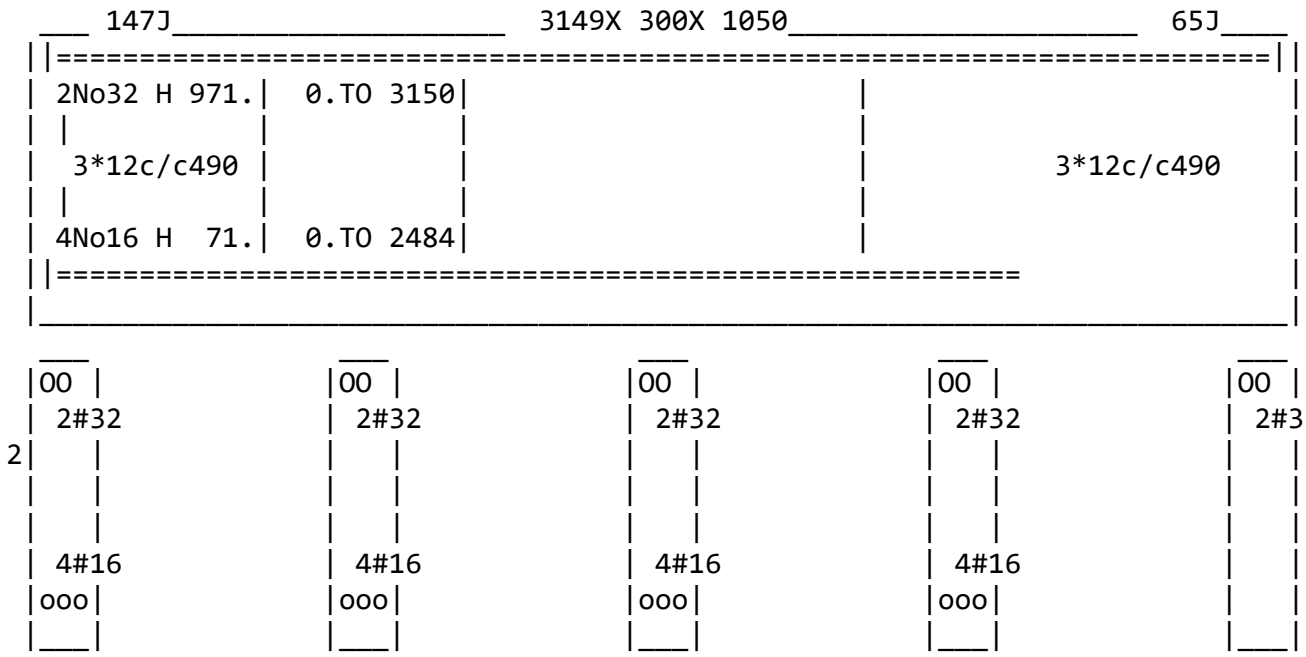
LEN - 3150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	2484.	YES	NO
2	971.	2 - 32MM	0.	3150.	YES	YES

B E A M N O. 2202 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 210.43 KNS Vc= 288.38 KNS Vs= 0.00 KNS
Tu= 1.50 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 600. MM

AT END SUPPORT - Vu= 167.02 KNS Vc= 285.60 KNS Vs= 0.00 KNS
Tu= 1.42 KN-MET Tc= 13.5 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 600. MM



ACI 318-11 BEAM NO. 2203 DESIGN RESULTS

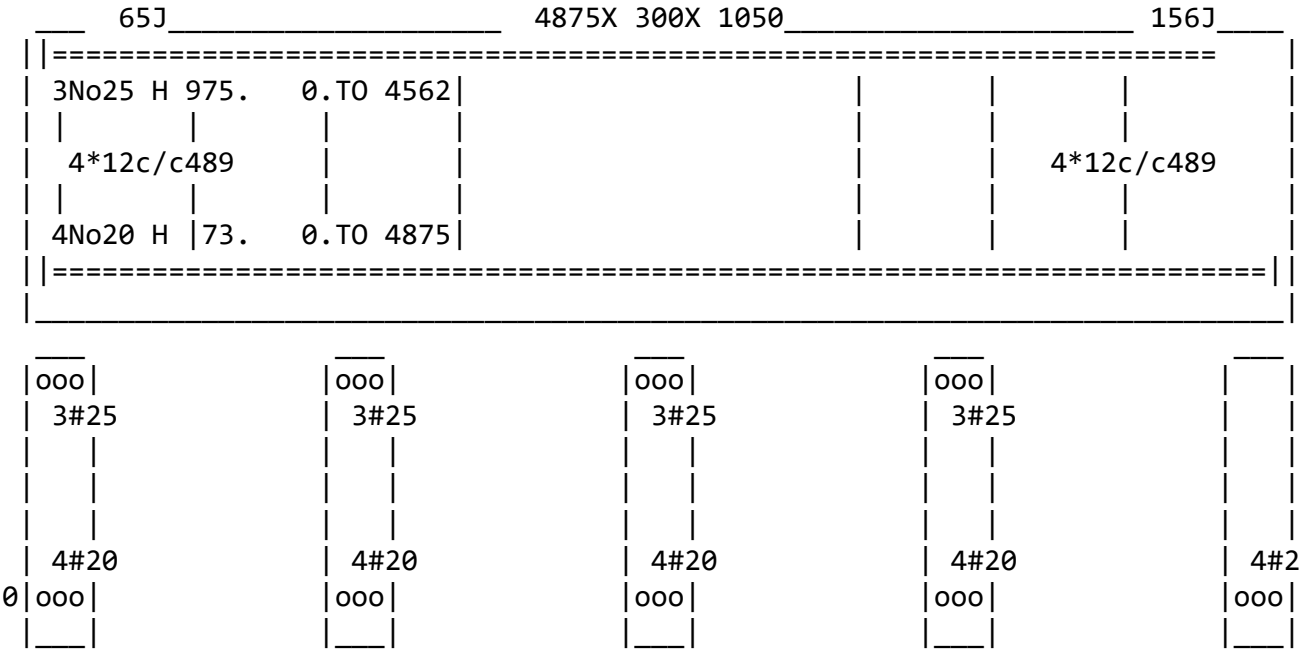
LEN - 4875. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	4875.	YES	YES
2	975.	3 - 25MM	0.	4562.	YES	NO

B E A M N O. 2203 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 105.92 KNS Vc= 273.17 KNS Vs= 0.00 KNS
Tu= 4.94 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 1463. MM

AT END SUPPORT - Vu= 123.40 KNS Vc= 280.25 KNS Vs= 0.00 KNS
Tu= 10.13 KN-MET Tc= 13.2 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 1463. MM



ACI 318-11 BEAM NO. 2204 DESIGN RESULTS

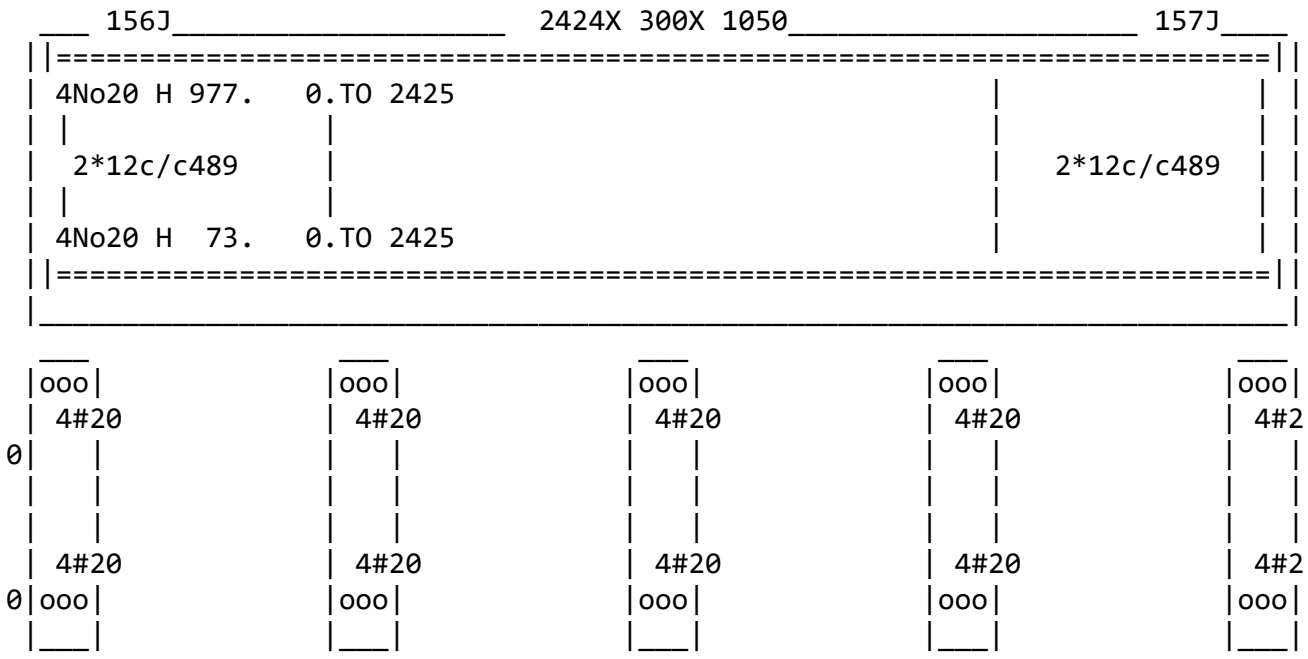
LEN - 2425. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	2425.	YES	YES
2	977.	4 - 20MM	0.	2425.	YES	YES

B E A M N O. 2204 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 107.42 KNS Vc= 273.09 KNS Vs= 0.00 KNS
Tu= 4.94 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 238. MM

AT END SUPPORT - Vu= 114.97 KNS Vc= 273.09 KNS Vs= 0.00 KNS
Tu= 4.94 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 238. MM



ACI 318-11 BEAM NO. 2205 DESIGN RESULTS

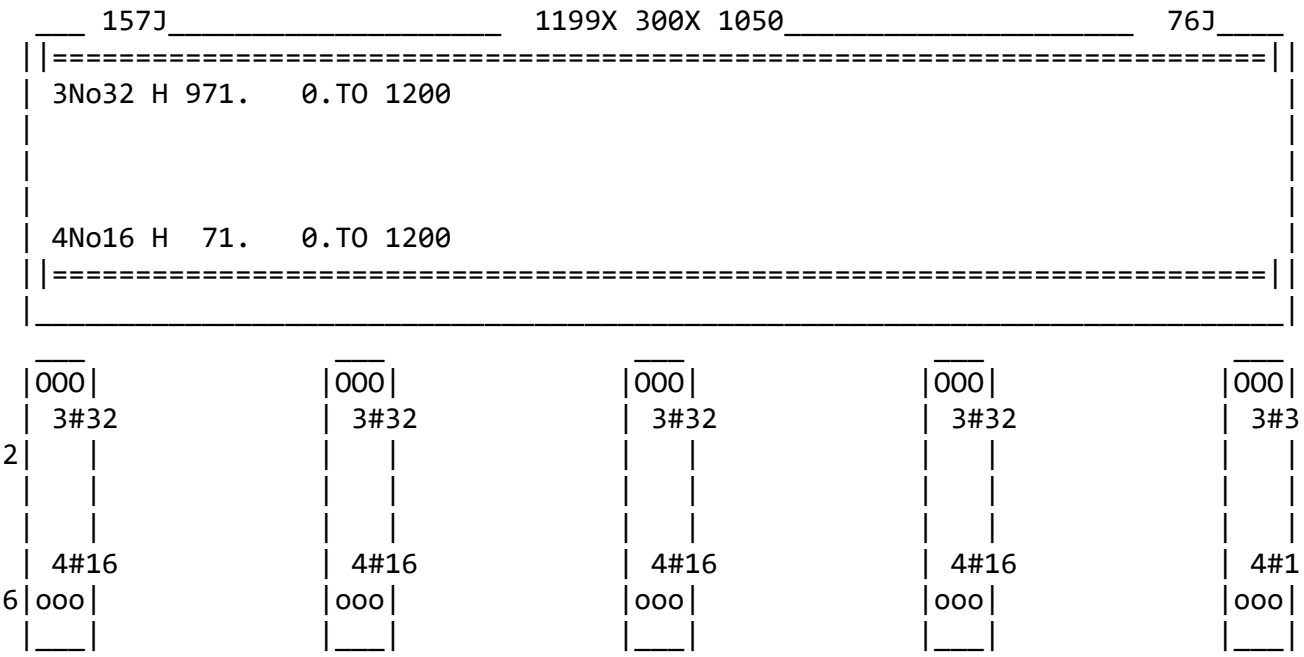
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	1200.	YES	YES
2	971.	3 - 32MM	0.	1200.	YES	YES

B E A M N O. 2205 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2205 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2205 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2206 DESIGN RESULTS

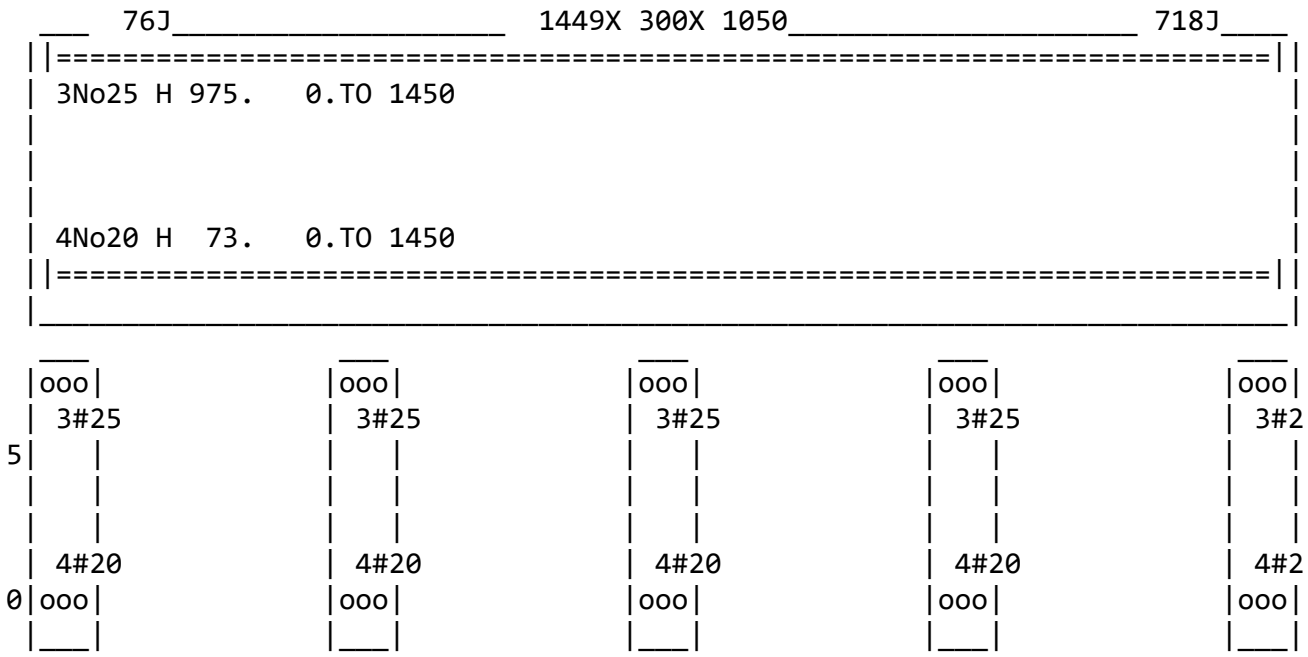
LEN - 1450. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	1450.	YES	YES
2	975.	3 - 25MM	0.	1450.	YES	YES

B E A M N O. 2206 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2206 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2206 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2207 DESIGN RESULTS

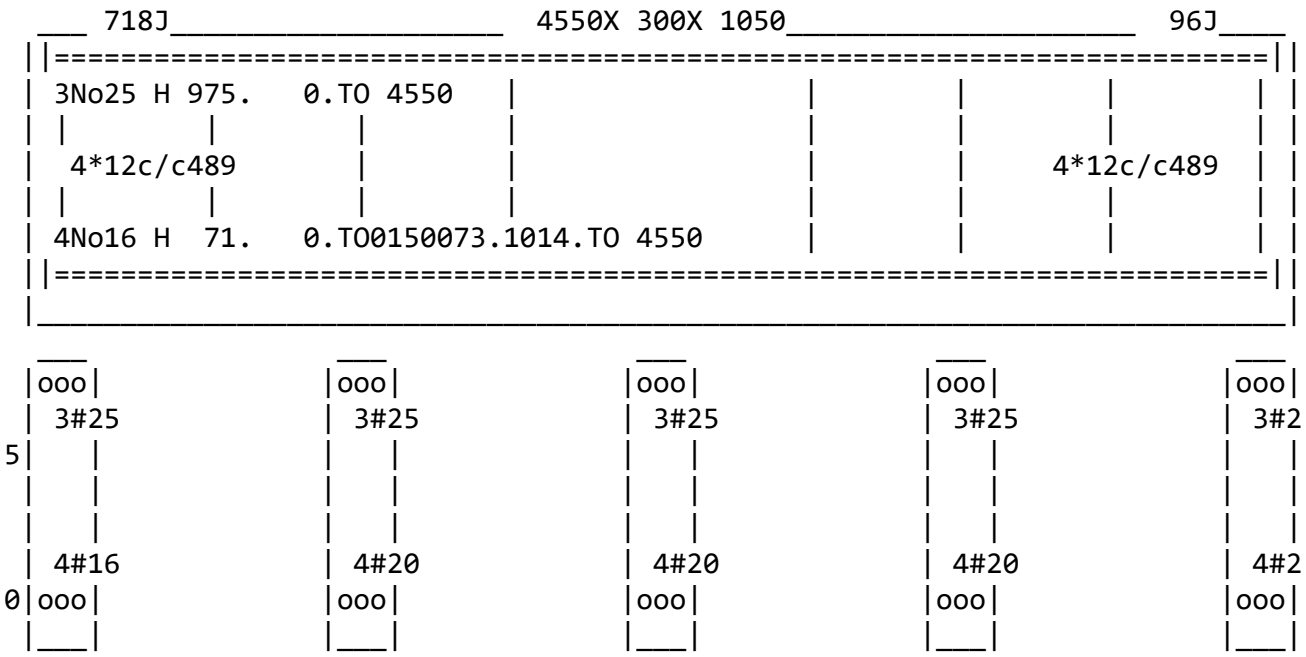
LEN - 4550. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	1500.	YES	NO
2	73.	4 - 20MM	1014.	4550.	NO	YES
3	975.	3 - 25MM	0.	4550.	YES	YES

B E A M N O. 2207 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 135.40 KNS Vc= 261.37 KNS Vs= 0.00 KNS
Tu= 1.96 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 1300. MM

AT END SUPPORT - Vu= 158.57 KNS Vc= 261.37 KNS Vs= 0.00 KNS
Tu= 1.96 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 1300. MM



ACI 318-11 BEAM NO. 2208 DESIGN RESULTS

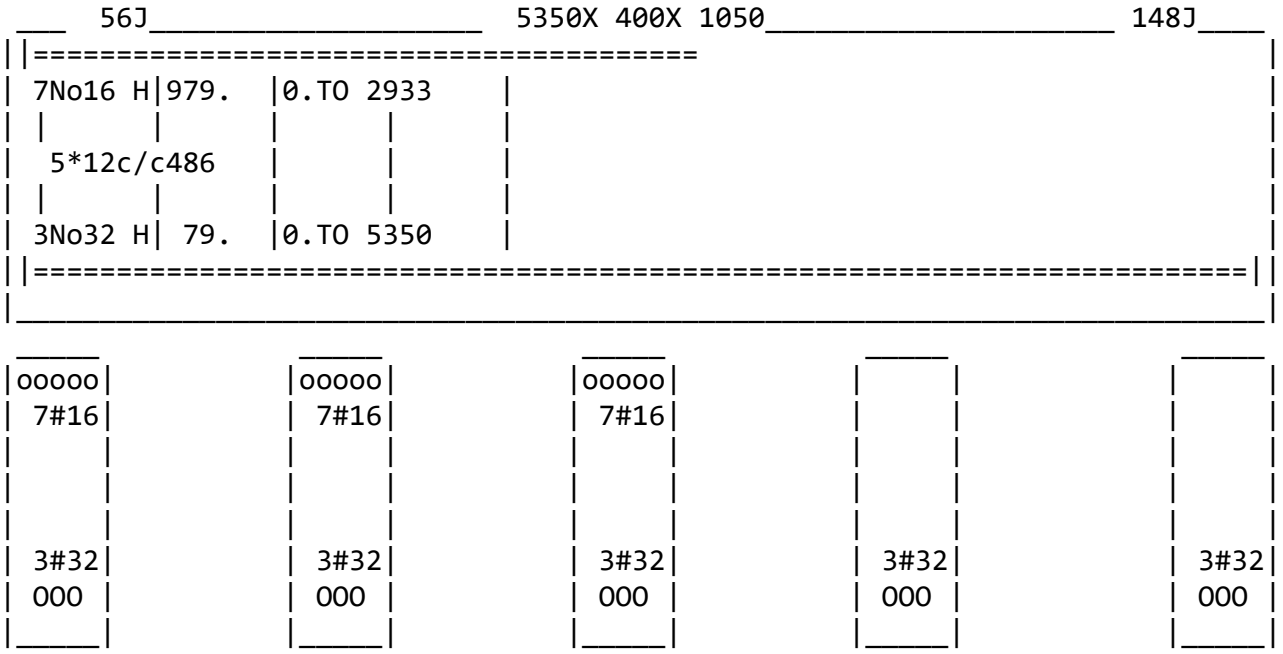
LEN - 5350. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	5350.	YES	YES
2	979.	7 - 16MM	0.	2933.	YES	NO

B E A M N O. 2208 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 301.56 KNS Vc= 627.09 KNS Vs= 0.00 KNS
Tu= 17.50 KN-MET Tc= 23.2 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 486. MM C/C FOR 1700. MM

AT END SUPPORT - Vu= 16.42 KNS Vc= 364.35 KNS Vs= 0.00 KNS
Tu= 12.53 KN-MET Tc= 23.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2209 DESIGN RESULTS

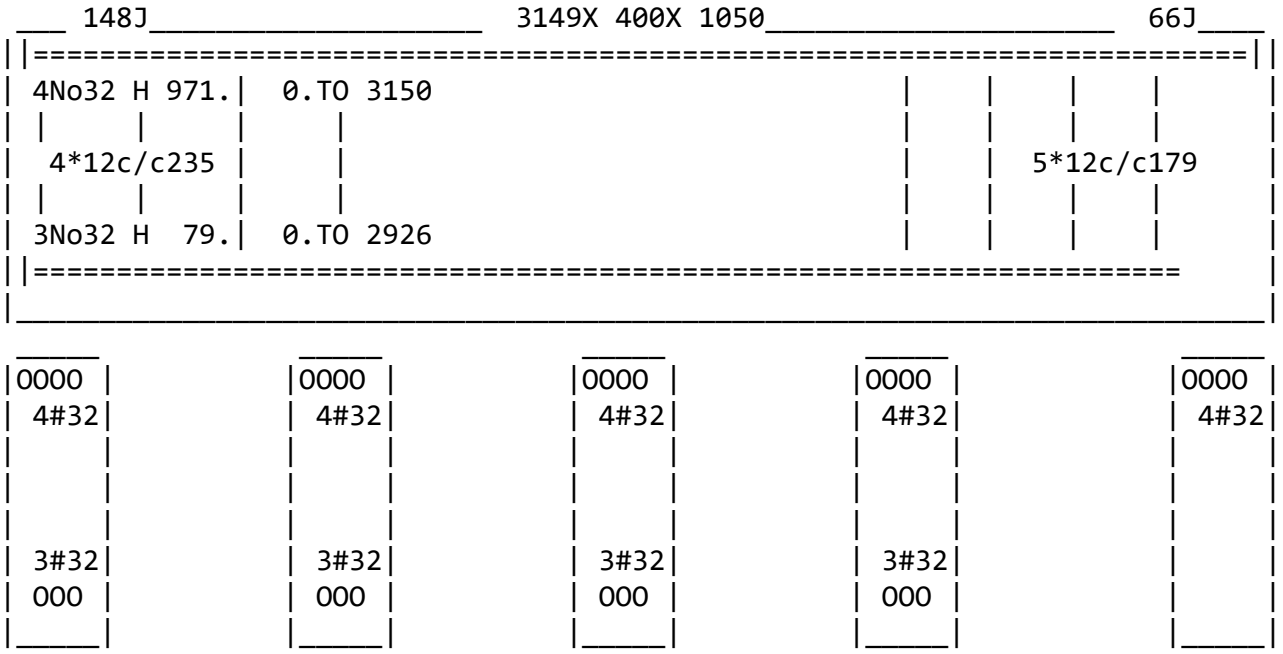
LEN - 3150. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	2926.	YES	NO
2	971.	4 - 32MM	0.	3150.	YES	YES

B E A M N O. 2209 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 515.35 KNS Vc= 444.89 KNS Vs= 242.24 KNS
Tu= 26.75 KN-MET Tc= 23.8 KN-MET Ts= 35.7 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 235. MM C/C FOR 600. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.53 SQ.CM.

AT END SUPPORT - Vu= 579.39 KNS Vc= 405.93 KNS Vs= 366.59 KNS
Tu= 26.38 KN-MET Tc= 23.4 KN-MET Ts= 35.2 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 179. MM C/C FOR 600. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.47 SQ.CM.



ACI 318-11 BEAM NO. 2210 DESIGN RESULTS

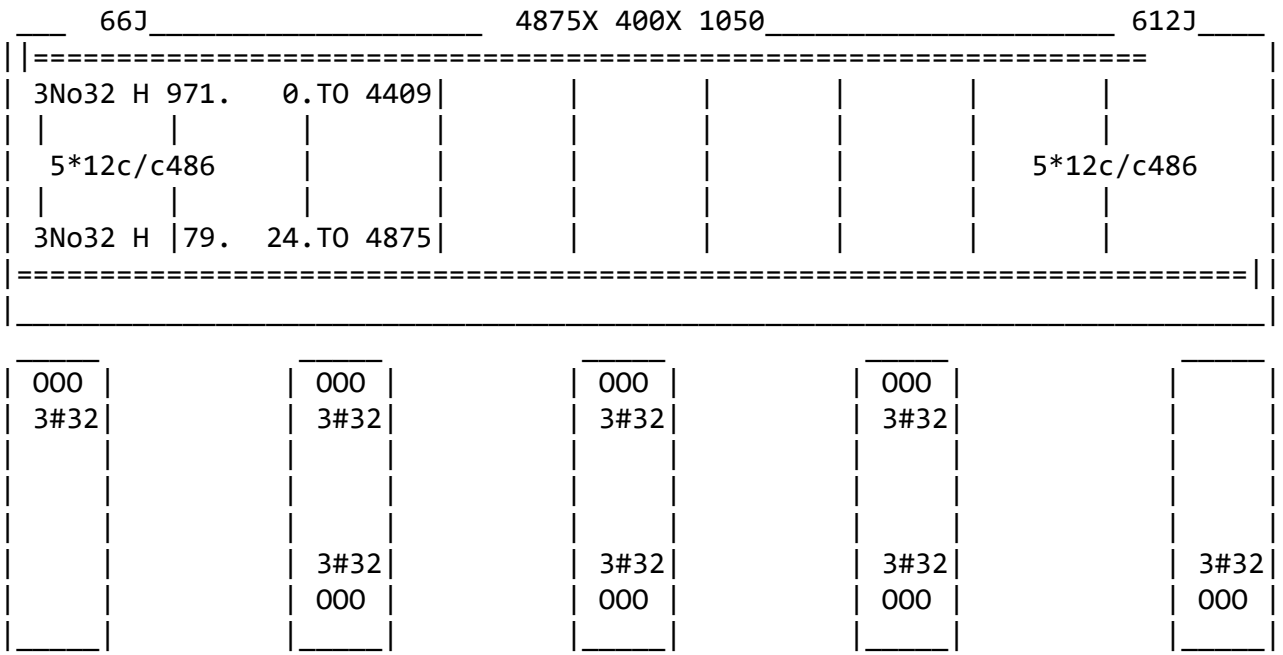
LEN - 4875. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	24.	4875.	NO	YES
2	971.	3 - 32MM	0.	4409.	YES	NO

B E A M N O. 2210 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 200.42 KNS Vc= 375.40 KNS Vs= 0.00 KNS
Tu= 2.22 KN-MET Tc= 22.1 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 486. MM C/C FOR 1463. MM

AT END SUPPORT - Vu= 146.13 KNS Vc= 375.40 KNS Vs= 0.00 KNS
Tu= 2.22 KN-MET Tc= 22.1 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 486. MM C/C FOR 1463. MM



ACI 318-11 BEAM NO. 2211 DESIGN RESULTS

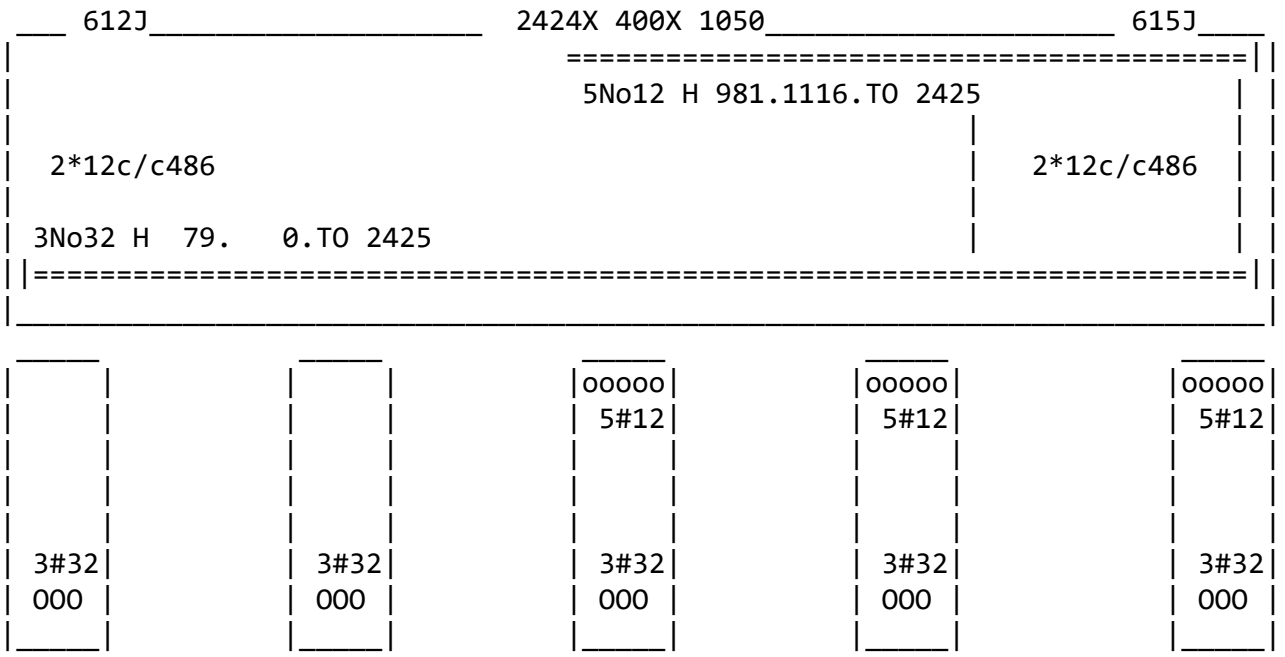
LEN - 2425. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	2425.	YES	YES
2	981.	5 - 12MM	1116.	2425.	NO	YES

B E A M N O. 2211 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 156.15 KNS Vc= 376.85 KNS Vs= 0.00 KNS
Tu= 1.41 KN-MET Tc= 22.1 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 486. MM C/C FOR 238. MM

AT END SUPPORT - Vu= 164.98 KNS Vc= 376.85 KNS Vs= 0.00 KNS
Tu= 1.41 KN-MET Tc= 22.1 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 486. MM C/C FOR 238. MM



ACI 318-11 BEAM NO. 2212 DESIGN RESULTS

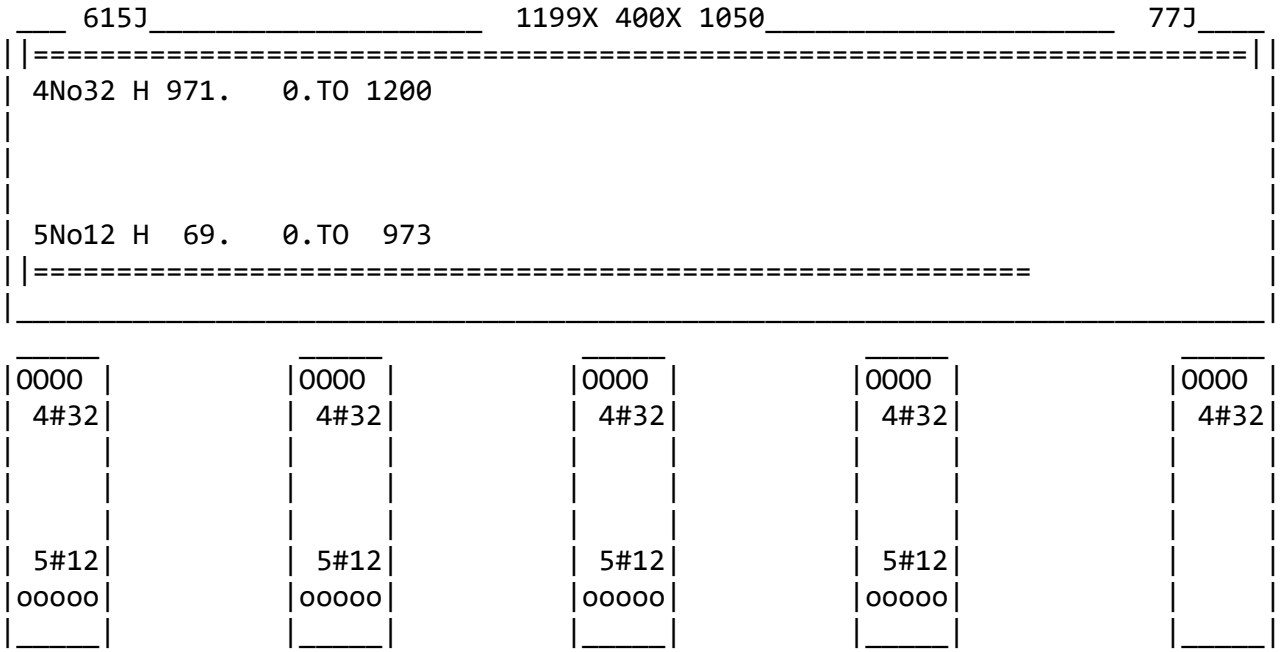
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	973.	YES	NO
2	971.	4 - 32MM	0.	1200.	YES	YES

B E A M N O. 2212 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2212 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2212 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2213 DESIGN RESULTS

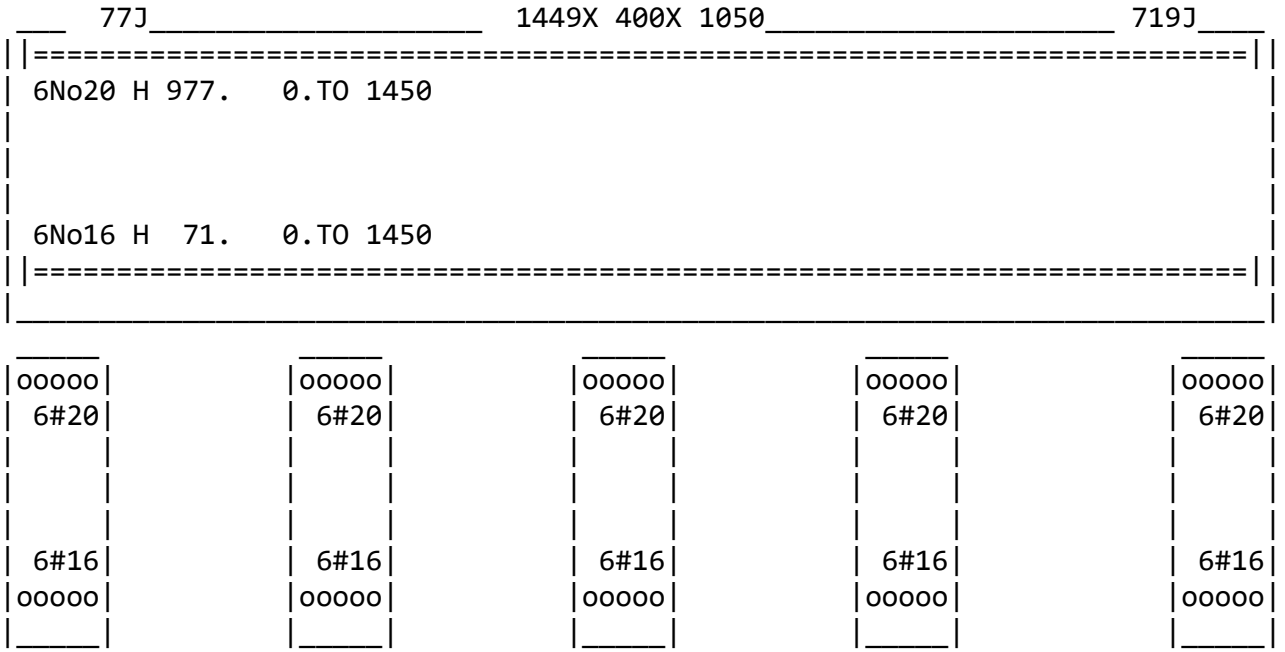
LEN - 1450. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	1450.	YES	YES
2	977.	6 - 20MM	0.	1450.	YES	YES

B E A M N O. 2213 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2213 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2213 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2214 DESIGN RESULTS

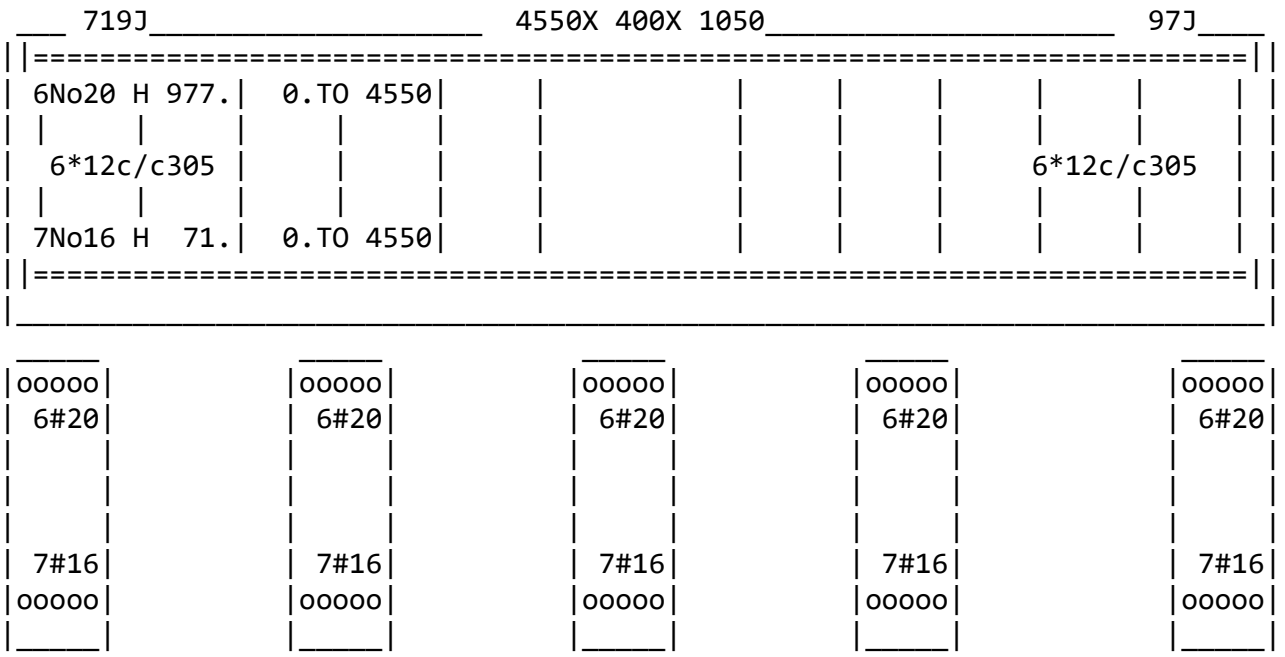
LEN - 4550. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	7 - 16MM	0.	4550.	YES	YES
2	977.	6 - 20MM	0.	4550.	YES	YES

B E A M N O. 2214 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.29 KNS Vc= 376.74 KNS Vs= 0.00 KNS
Tu= 52.77 KN-MET Tc= 22.0 KN-MET Ts= 70.4 KN-MET LOAD 409
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 8.94 SQ.CM.

AT END SUPPORT - Vu= 65.17 KNS Vc= 376.74 KNS Vs= 0.00 KNS
Tu= 52.77 KN-MET Tc= 22.0 KN-MET Ts= 70.4 KN-MET LOAD 409
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 8.94 SQ.CM.



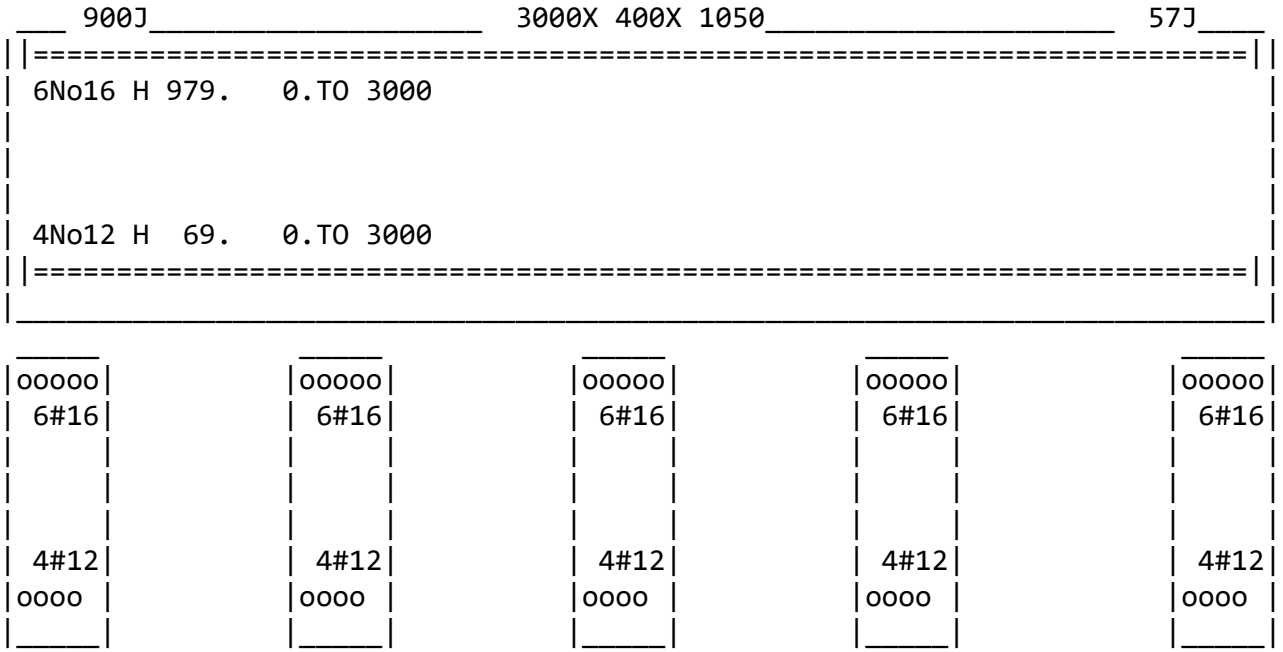
ACI 318-11 BEAM NO. 2215 DESIGN RESULTS

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	3000.	YES	YES
2	979.	6 - 16MM	0.	3000.	YES	YES

B E A M N O. 2215 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 26.11 KNS Vc= 384.84 KNS Vs= 0.00 KNS
Tu= 9.48 KN-MET Tc= 22.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 48.92 KNS Vc= 384.84 KNS Vs= 0.00 KNS
Tu= 9.48 KN-MET Tc= 22.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2216 DESIGN RESULTS

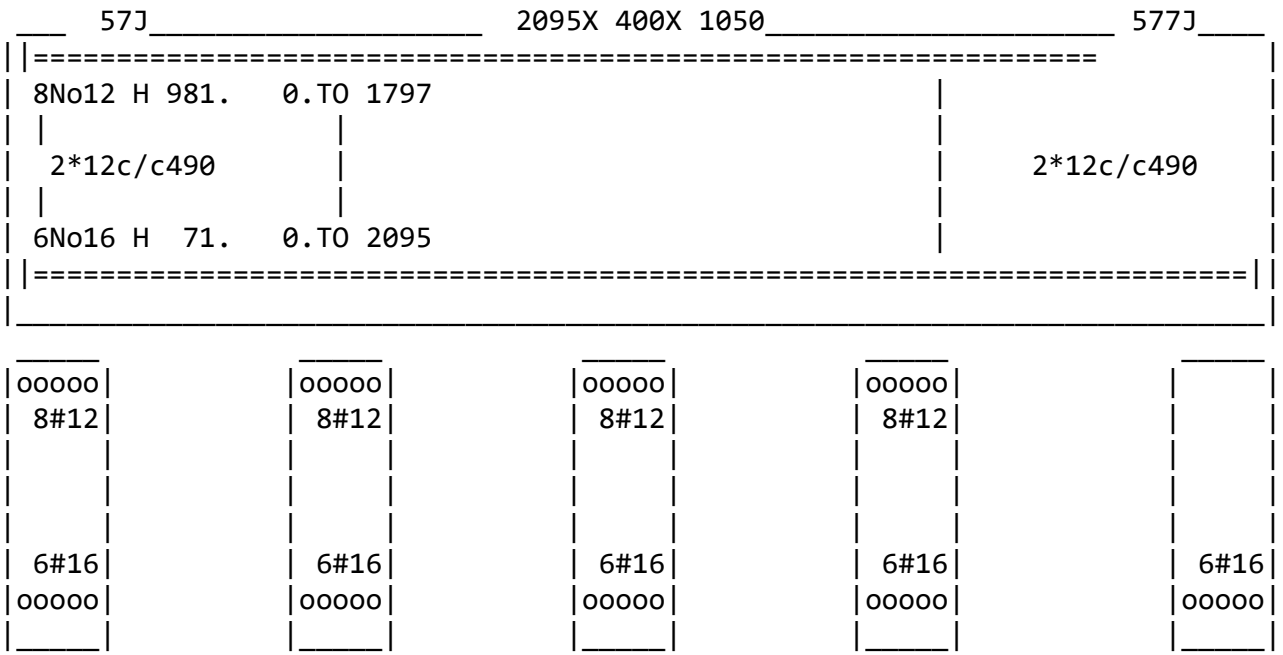
LEN - 2095. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	2095.	YES	YES
2	981.	8 - 12MM	0.	1797.	YES	NO

B E A M N O. 2216 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 176.69 KNS Vc= 333.03 KNS Vs= 0.00 KNS
Tu= 12.06 KN-MET Tc= 19.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 73. MM

AT END SUPPORT - Vu= 171.84 KNS Vc= 333.03 KNS Vs= 0.00 KNS
Tu= 12.06 KN-MET Tc= 19.6 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 73. MM



ACI 318-11 BEAM NO. 2217 DESIGN RESULTS

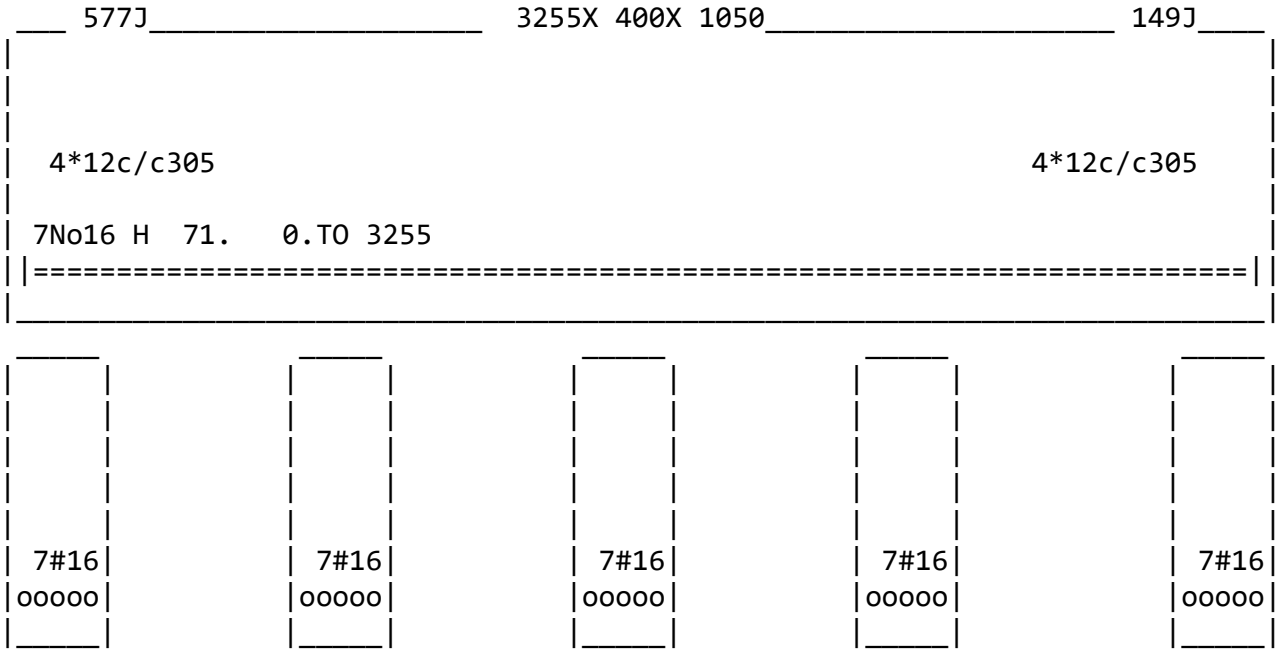
LEN - 3255. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	7 - 16MM	0.	3255.	YES	YES

B E A M N O. 2217 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.04 KNS Vc= 336.53 KNS Vs= 0.00 KNS
Tu= 19.84 KN-MET Tc= 19.8 KN-MET Ts= 26.5 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 653. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.36 SQ.CM.

AT END SUPPORT - Vu= 56.49 KNS Vc= 336.53 KNS Vs= 0.00 KNS
Tu= 19.84 KN-MET Tc= 19.8 KN-MET Ts= 26.5 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 653. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.36 SQ.CM.



ACI 318-11 BEAM NO. 2218 DESIGN RESULTS

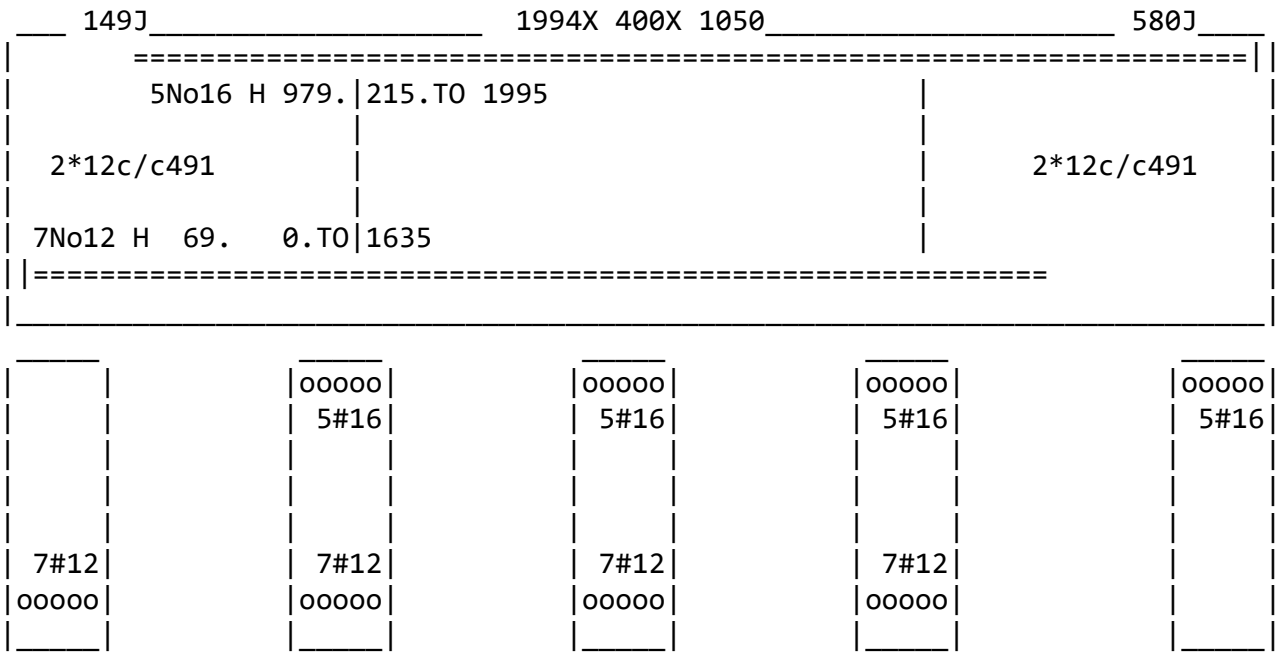
LEN - 1995. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	7 - 12MM	0.	1635.	YES	NO
2	979.	5 - 16MM	215.	1995.	NO	YES

B E A M N O. 2218 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 152.97 KNS Vc= 383.82 KNS Vs= 0.00 KNS
Tu= 8.41 KN-MET Tc= 22.3 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 491. MM C/C FOR 23. MM

AT END SUPPORT - Vu= 154.05 KNS Vc= 383.82 KNS Vs= 0.00 KNS
Tu= 8.41 KN-MET Tc= 22.3 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 491. MM C/C FOR 23. MM



ACI 318-11 BEAM NO. 2219 DESIGN RESULTS

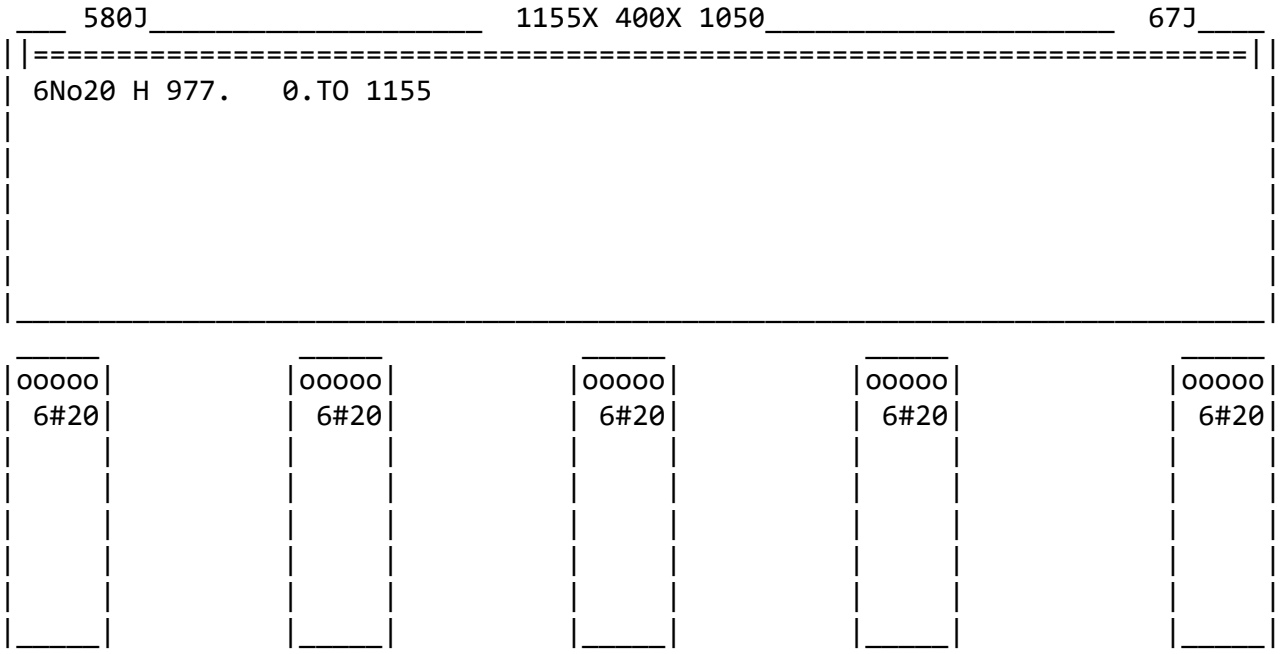
LEN - 1155. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	977.	6 - 20MM	0.	1155.	YES	YES

B E A M N O. 2219 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2219 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2219 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2220 DESIGN RESULTS

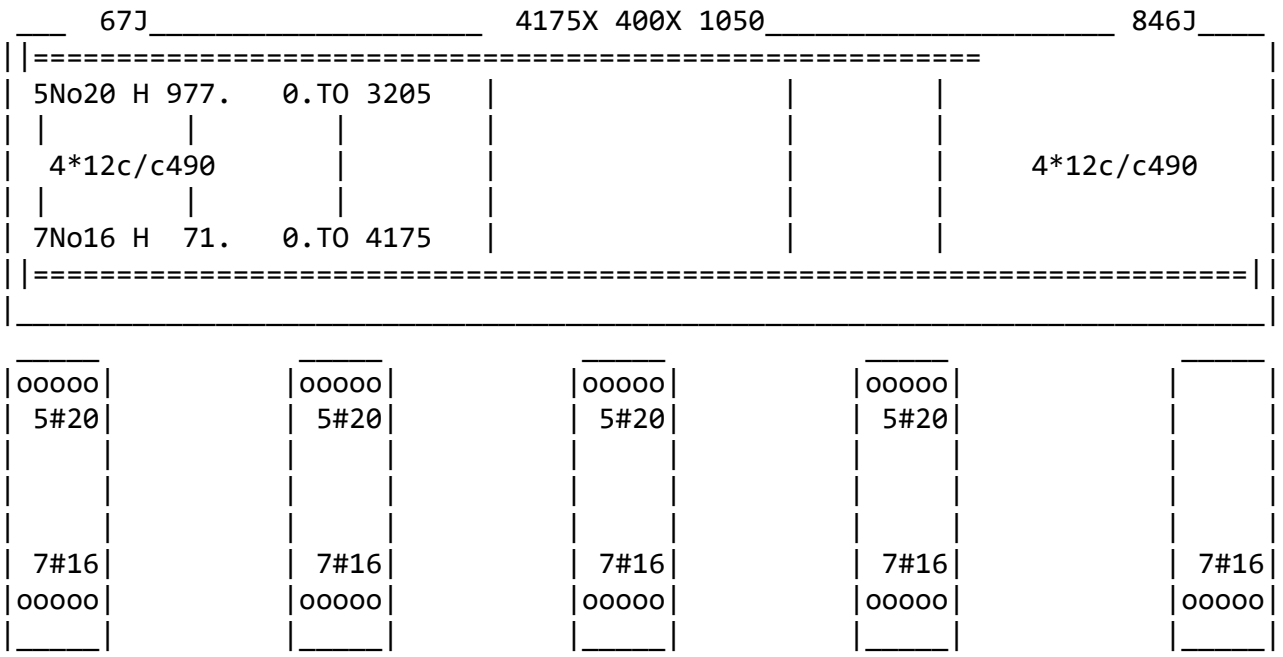
LEN - 4175. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	7 - 16MM	0.	4175.	YES	YES
2	977.	5 - 20MM	0.	3205.	YES	NO

B E A M N O. 2220 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 181.26 KNS Vc= 379.11 KNS Vs= 0.00 KNS
Tu= 10.59 KN-MET Tc= 22.1 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 1113. MM

AT END SUPPORT - Vu= 147.06 KNS Vc= 309.93 KNS Vs= 0.00 KNS
Tu= 10.26 KN-MET Tc= 18.2 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 1113. MM



ACI 318-11 BEAM NO. 2221 DESIGN RESULTS

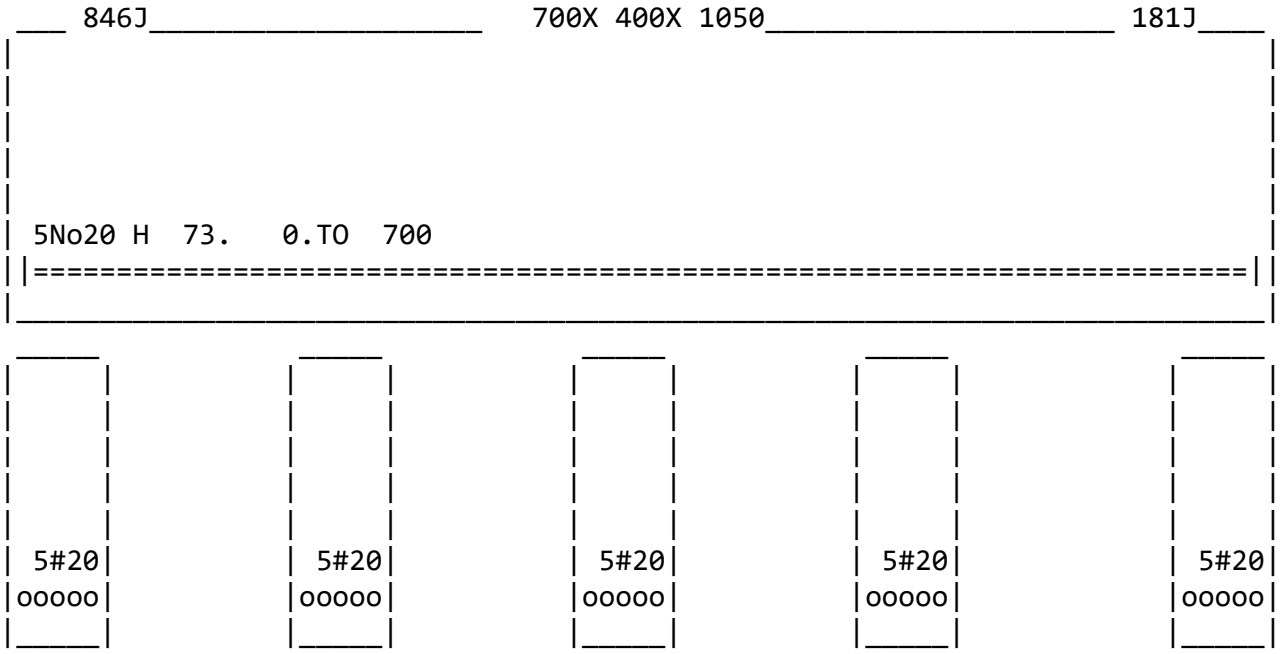
LEN - 700. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	5 - 20MM	0.	700.	YES	YES

B E A M N O. 2221 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2221 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2221 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2222 DESIGN RESULTS

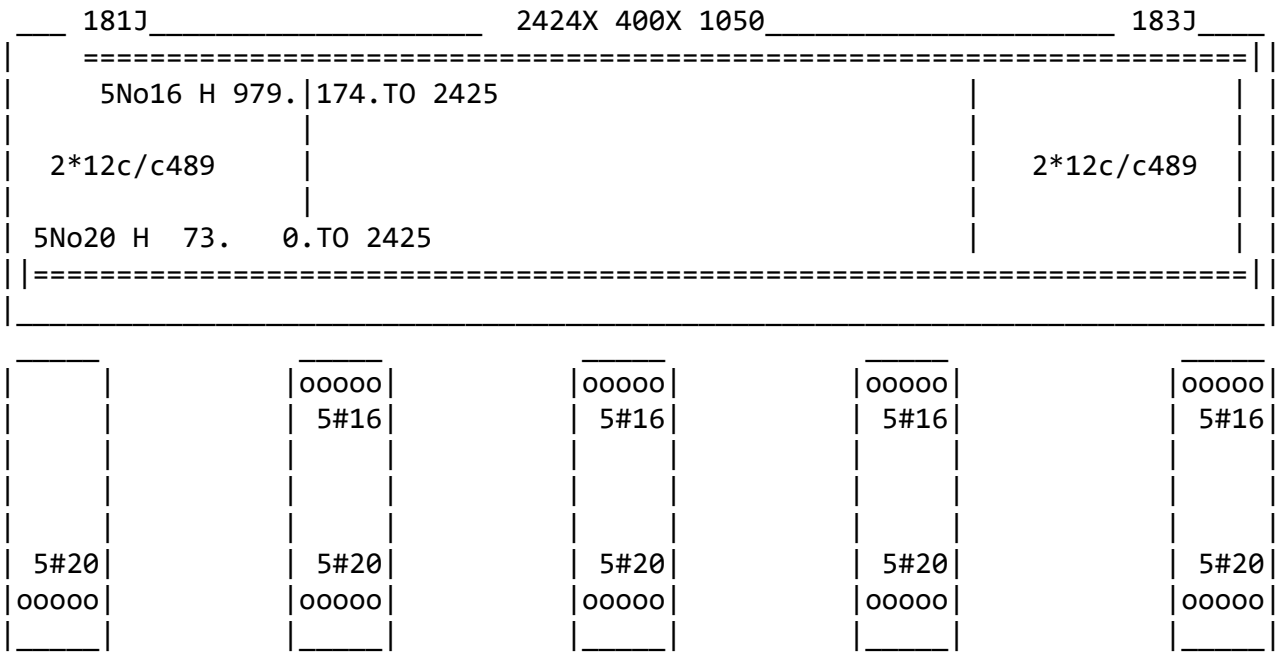
LEN - 2425. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	5 - 20MM	0.	2425.	YES	YES
2	979.	5 - 16MM	174.	2425.	NO	YES

B E A M N O. 2222 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 139.73 KNS Vc= 363.80 KNS Vs= 0.00 KNS
Tu= 7.57 KN-MET Tc= 21.3 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 238. MM

AT END SUPPORT - Vu= 151.09 KNS Vc= 363.80 KNS Vs= 0.00 KNS
Tu= 7.57 KN-MET Tc= 21.3 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 489. MM C/C FOR 238. MM



ACI 318-11 BEAM NO. 2223 DESIGN RESULTS

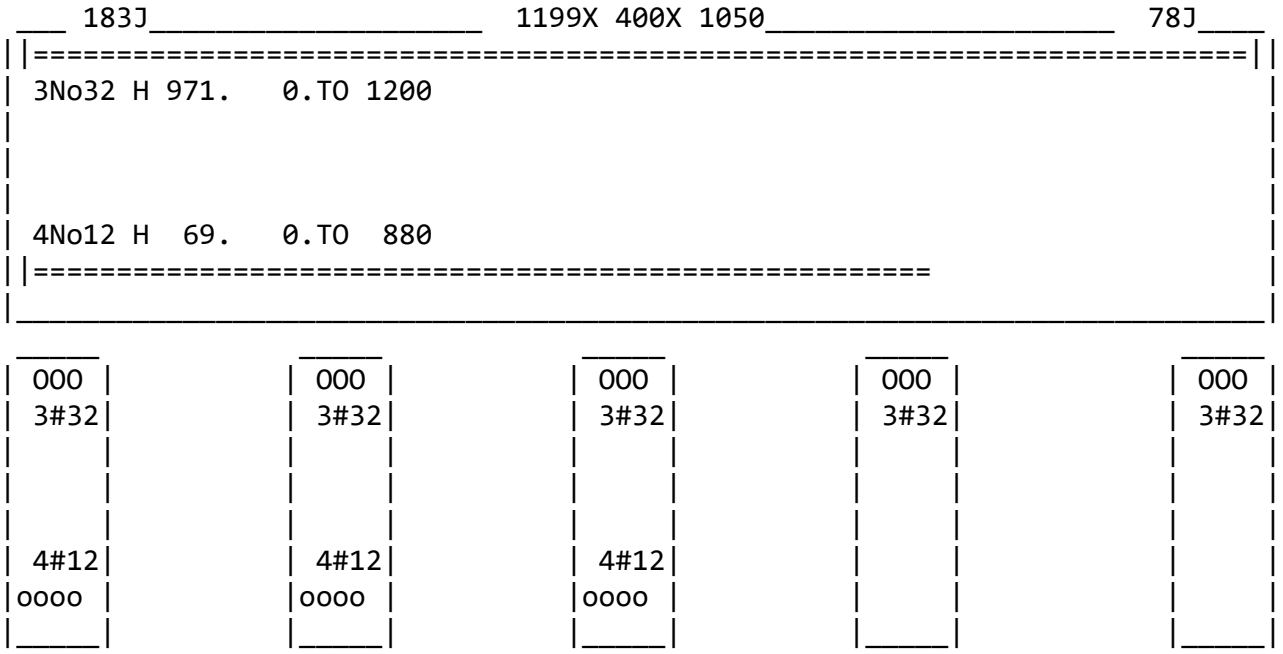
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	880.	YES	NO
2	971.	3 - 32MM	0.	1200.	YES	YES

B E A M N O. 2223 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2223 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2223 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2224 DESIGN RESULTS

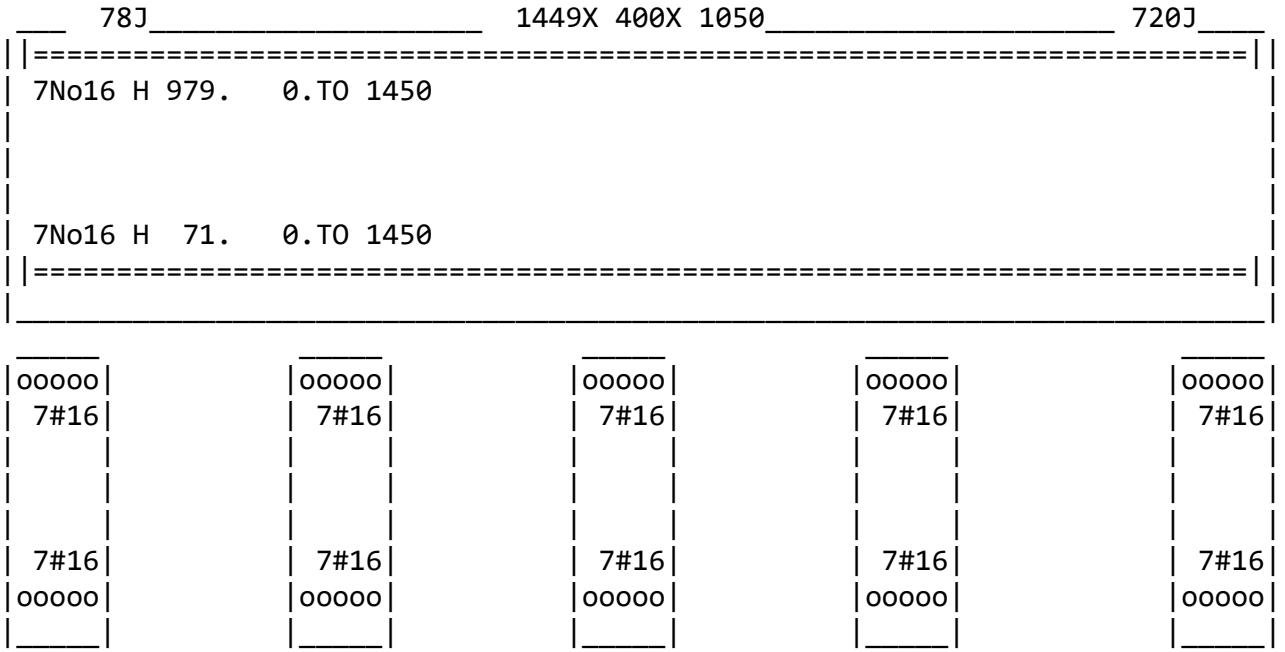
LEN - 1450. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	7 - 16MM	0.	1450.	YES	YES
2	979.	7 - 16MM	0.	1450.	YES	YES

B E A M N O. 2224 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2224 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2224 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2225 DESIGN RESULTS

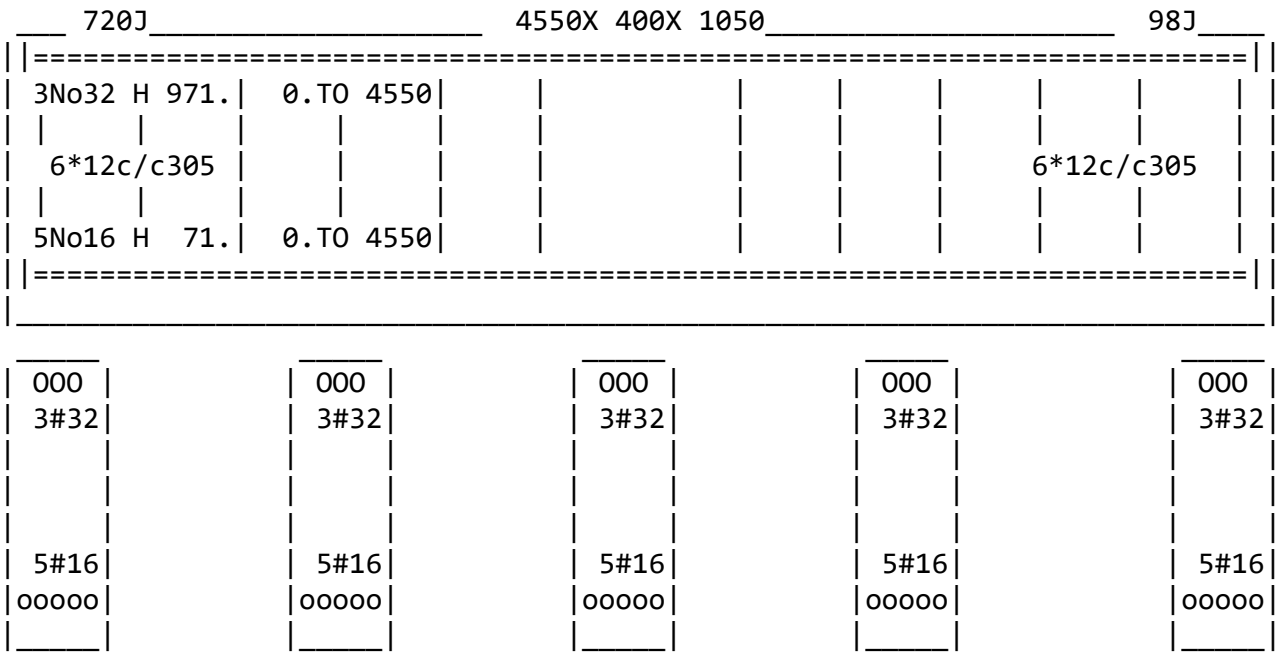
LEN - 4550. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	5 - 16MM	0.	4550.	YES	YES
2	971.	3 - 32MM	0.	4550.	YES	YES

B E A M N O. 2225 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 53.28 KNS Vc= 369.84 KNS Vs= 0.00 KNS
Tu= 52.88 KN-MET Tc= 21.6 KN-MET Ts= 70.5 KN-MET LOAD 410
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 8.96 SQ.CM.

AT END SUPPORT - Vu= 84.16 KNS Vc= 369.84 KNS Vs= 0.00 KNS
Tu= 52.88 KN-MET Tc= 21.6 KN-MET Ts= 70.5 KN-MET LOAD 410
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 8.96 SQ.CM.



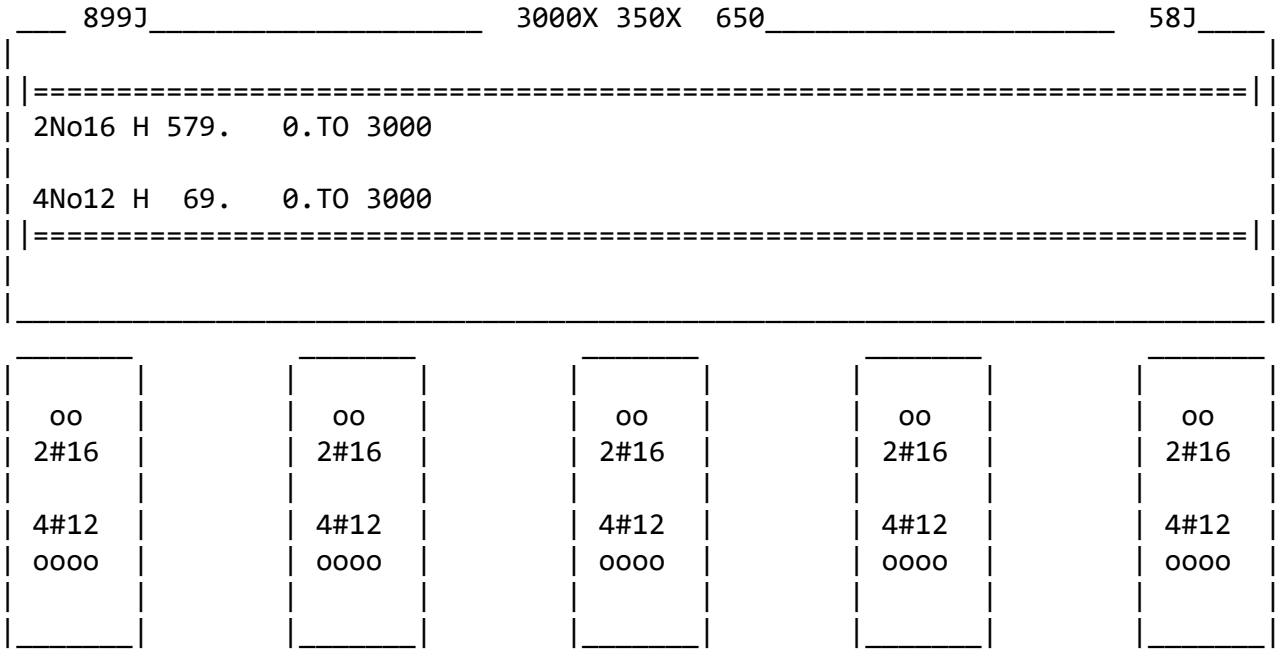
ACI 318-11 BEAM NO. 2226 DESIGN RESULTS

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	3000.	YES	YES
2	579.	2 - 16MM	0.	3000.	YES	YES

B E A M N O. 2226 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.28 KNS Vc= 198.60 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.59 KNS Vc= 198.60 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2227 DESIGN RESULTS

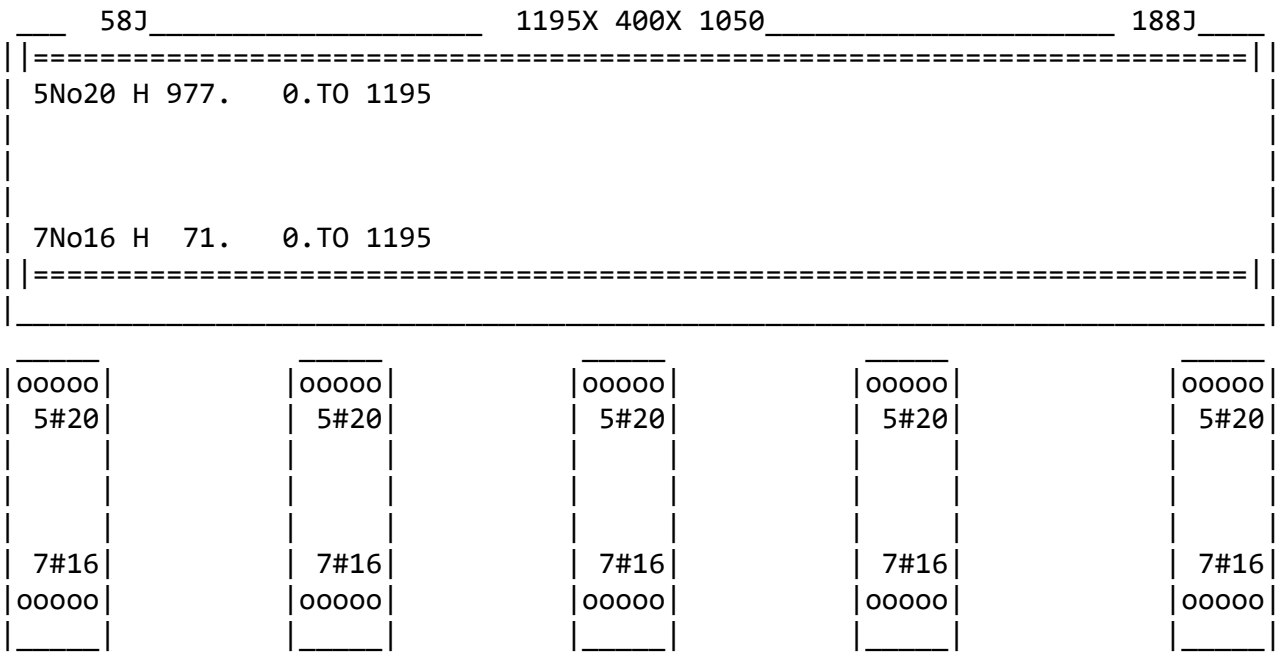
LEN - 1195. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	7 - 16MM	0.	1195.	YES	YES
2	977.	5 - 20MM	0.	1195.	YES	YES

B E A M N O. 2227 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2227 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2227 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2228 DESIGN RESULTS

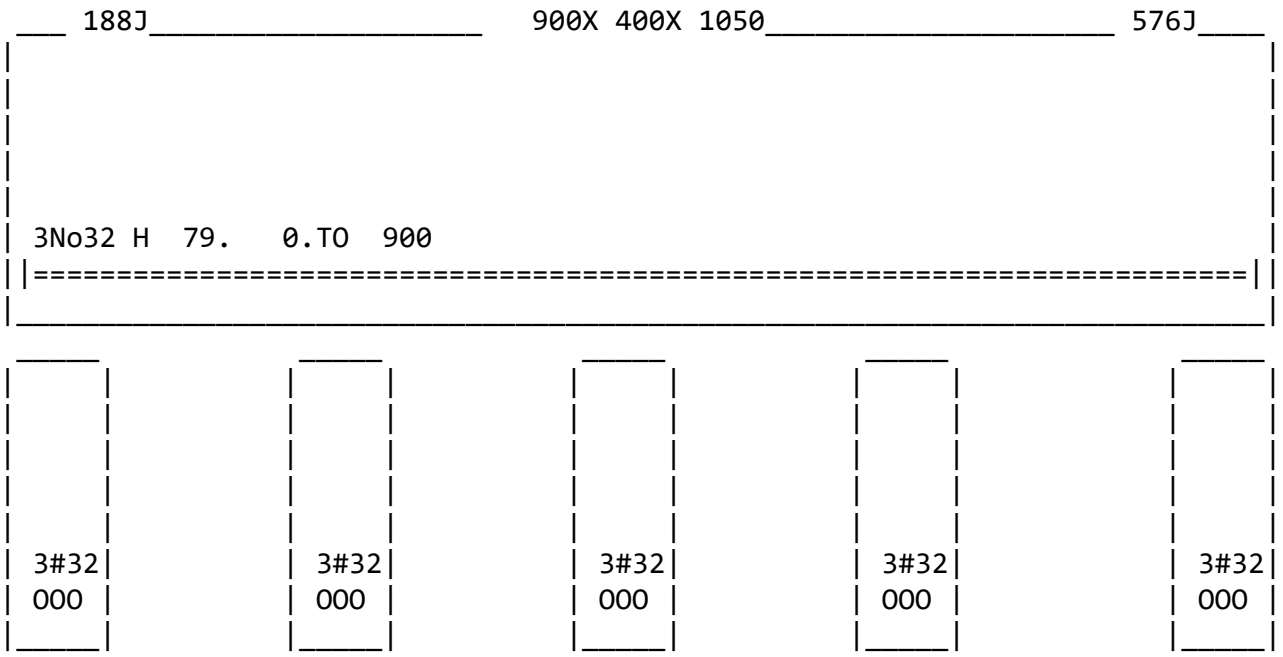
LEN - 900. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	900.	YES	YES

B E A M N O. 2228 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2228 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2228 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



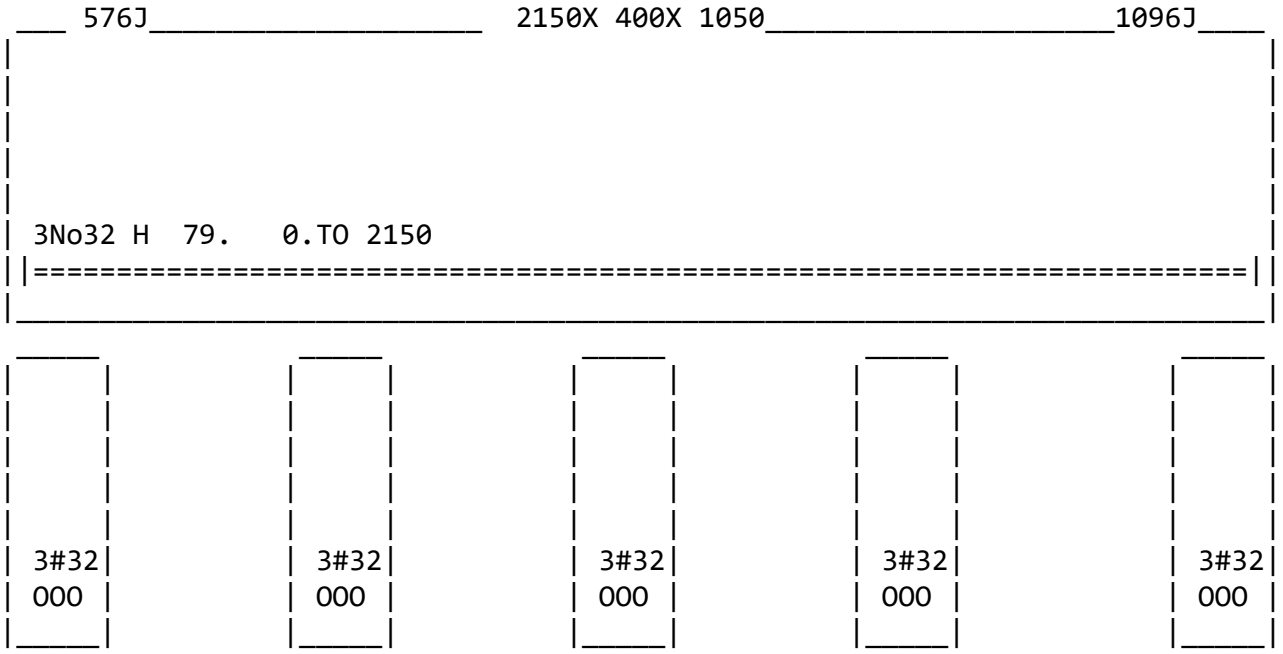
ACI 318-11 BEAM NO. 2229 DESIGN RESULTS

LEN - 2150. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	2150.	YES	YES

B E A M N O. 2229 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.51 KNS Vc= 363.78 KNS Vs= 0.00 KNS
Tu= 1.95 KN-MET Tc= 22.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.24 KNS Vc= 364.23 KNS Vs= 0.00 KNS
Tu= 1.95 KN-MET Tc= 22.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2230 DESIGN RESULTS

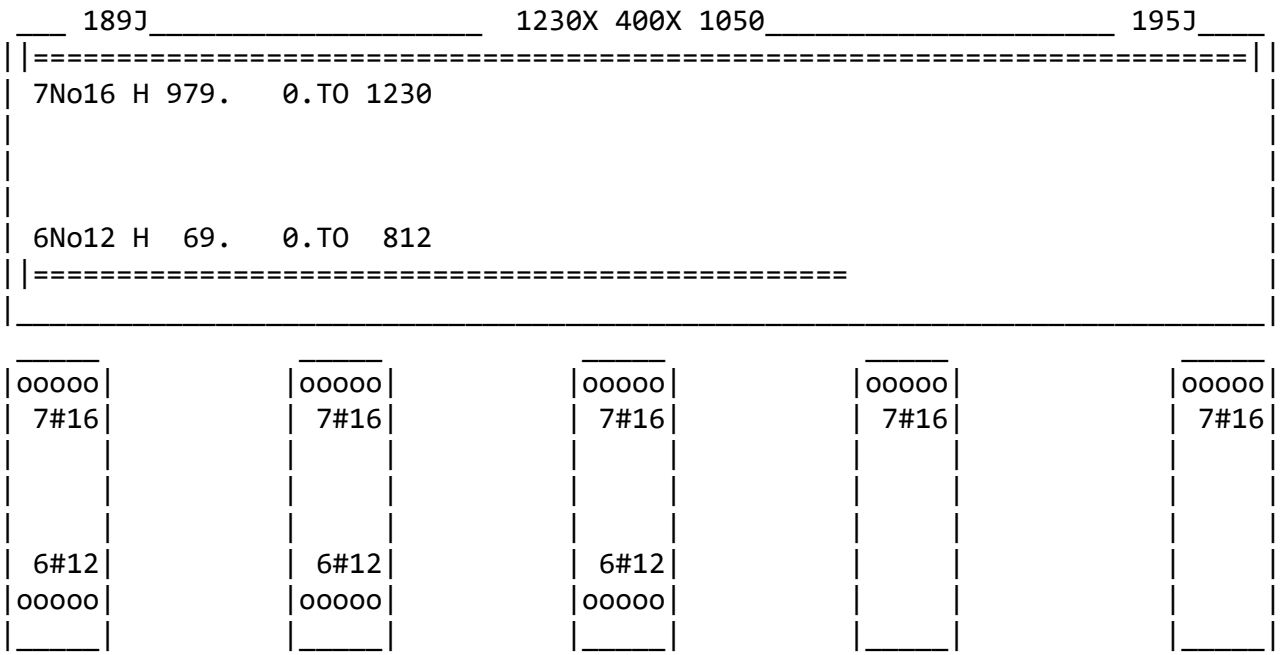
LEN - 1230. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	6 - 12MM	0.	812.	YES	NO
2	979.	7 - 16MM	0.	1230.	YES	YES

B E A M N O. 2230 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2230 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2230 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2231 DESIGN RESULTS

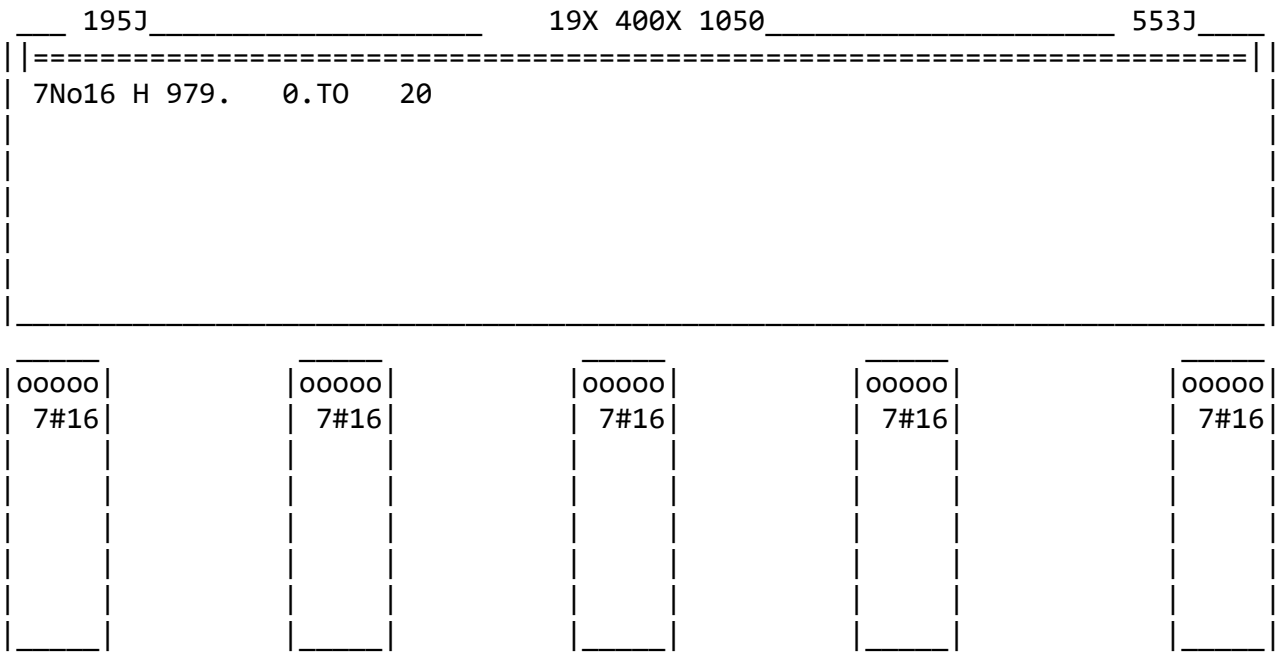
LEN - 20. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	979.	7 - 16MM	0.	20.	YES	YES

B E A M N O. 2231 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2231 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2231 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2232 DESIGN RESULTS

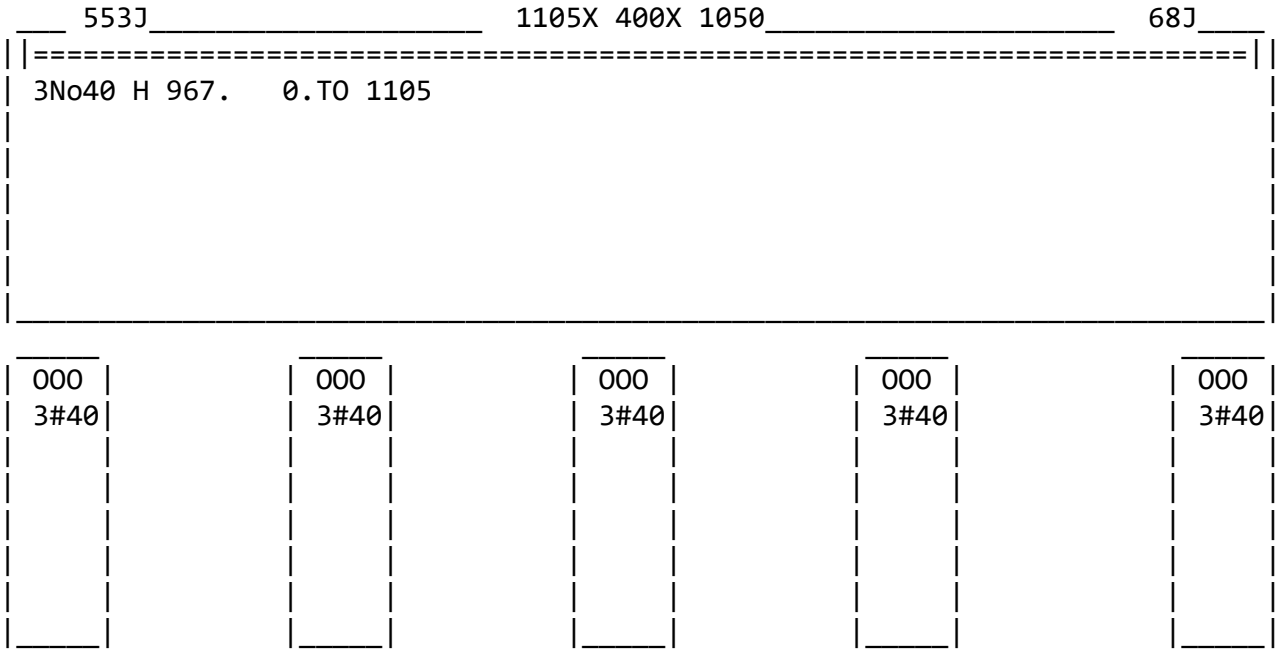
LEN - 1105. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	967.	3 - 40MM	0.	1105.	YES	YES

B E A M N O. 2232 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2232 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2232 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2233 DESIGN RESULTS

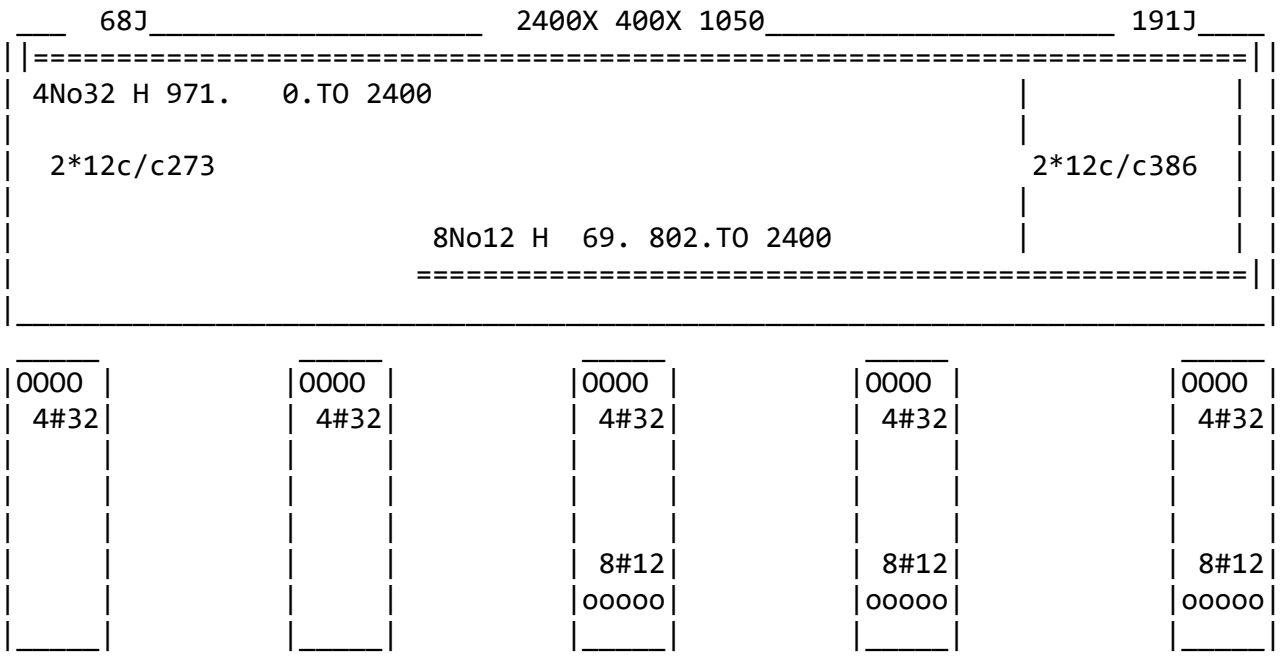
LEN - 2400. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	8 - 12MM	802.	2400.	NO	YES
2	971.	4 - 32MM	0.	2400.	YES	YES

B E A M N O. 2233 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 504.10 KNS Vc= 330.14 KNS Vs= 342.00 KNS
Tu= 4.66 KN-MET Tc= 19.4 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 273. MM C/C FOR 225. MM

AT END SUPPORT - Vu= 429.02 KNS Vc= 330.14 KNS Vs= 241.88 KNS
Tu= 4.66 KN-MET Tc= 19.4 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 386. MM C/C FOR 225. MM



ACI 318-11 BEAM NO. 2234 DESIGN RESULTS

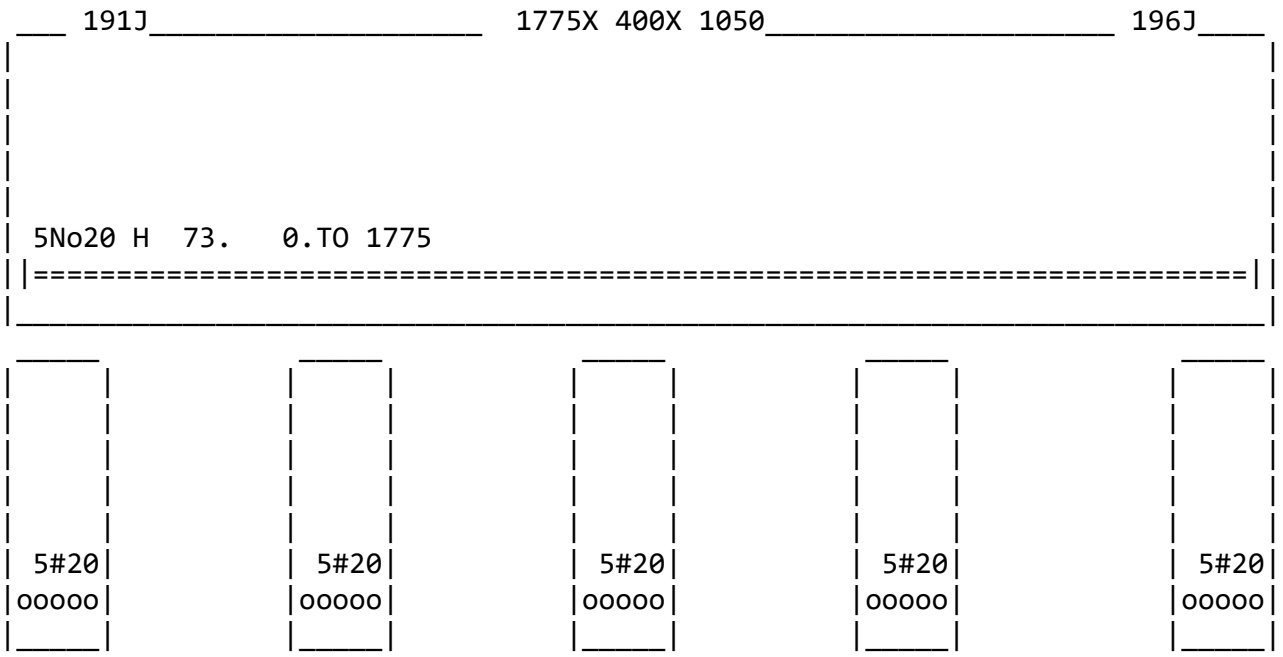
LEN - 1775. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	5 - 20MM	0.	1775.	YES	YES

B E A M N O. 2234 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2234 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2234 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2235 DESIGN RESULTS

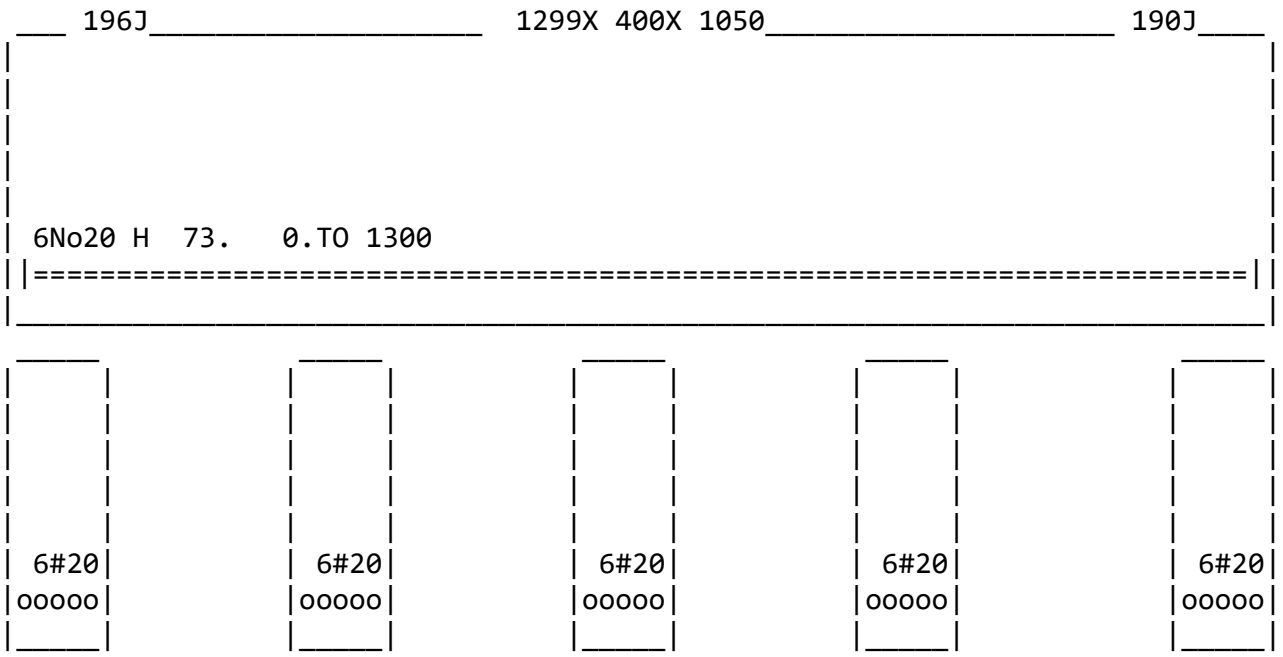
LEN - 1300. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	6 - 20MM	0.	1300.	YES	YES

B E A M N O. 2235 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2235 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2235 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2236 DESIGN RESULTS

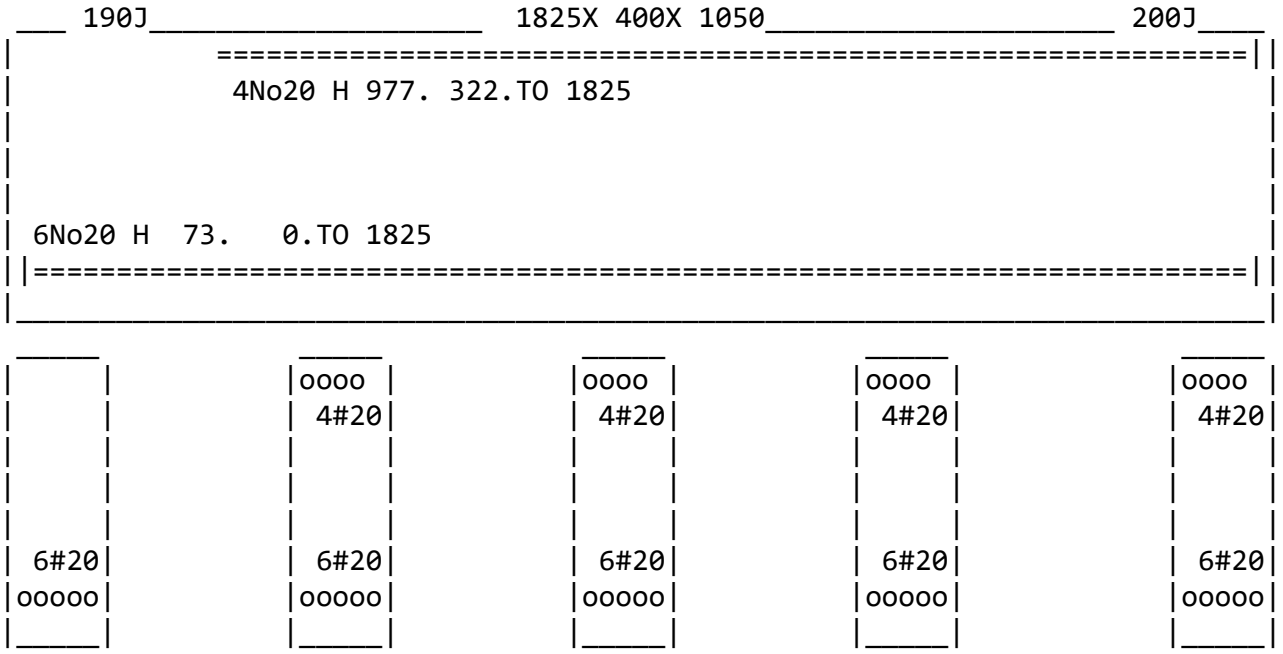
LEN - 1825. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	6 - 20MM	0.	1825.	YES	YES
2	977.	4 - 20MM	322.	1825.	NO	YES

B E A M N O. 2236 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2236 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2236 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2237 DESIGN RESULTS

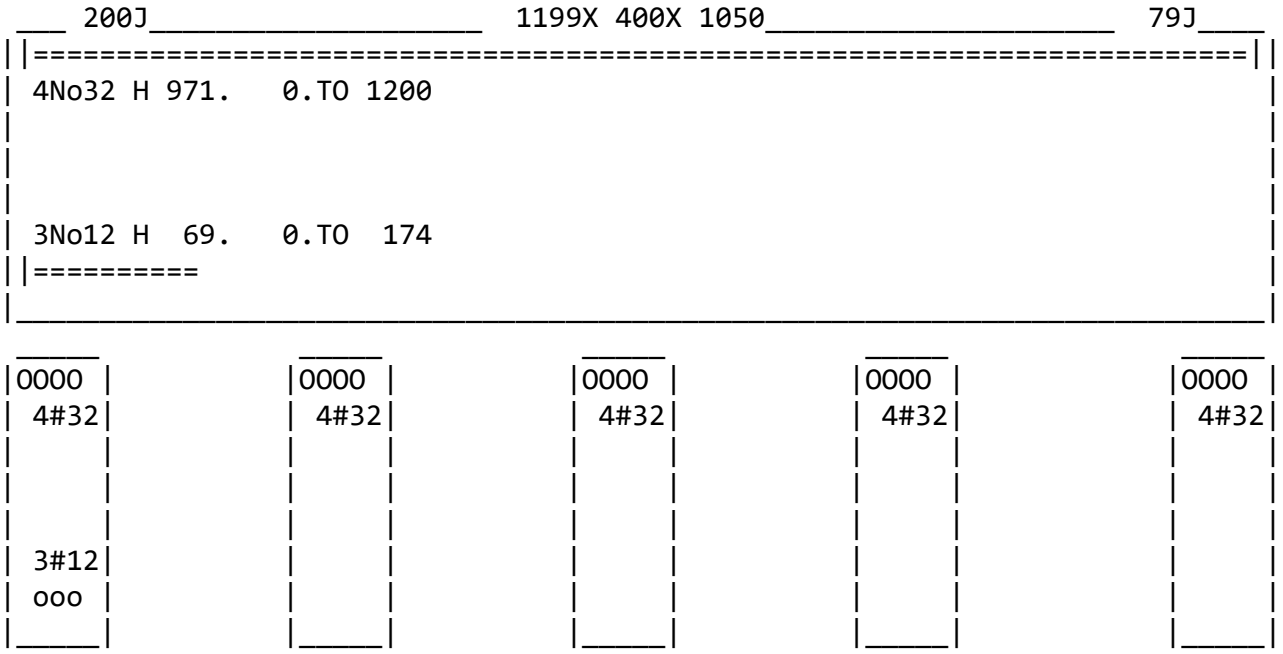
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	174.	YES	NO
2	971.	4 - 32MM	0.	1200.	YES	YES

B E A M N O. 2237 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2237 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2237 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2238 DESIGN RESULTS

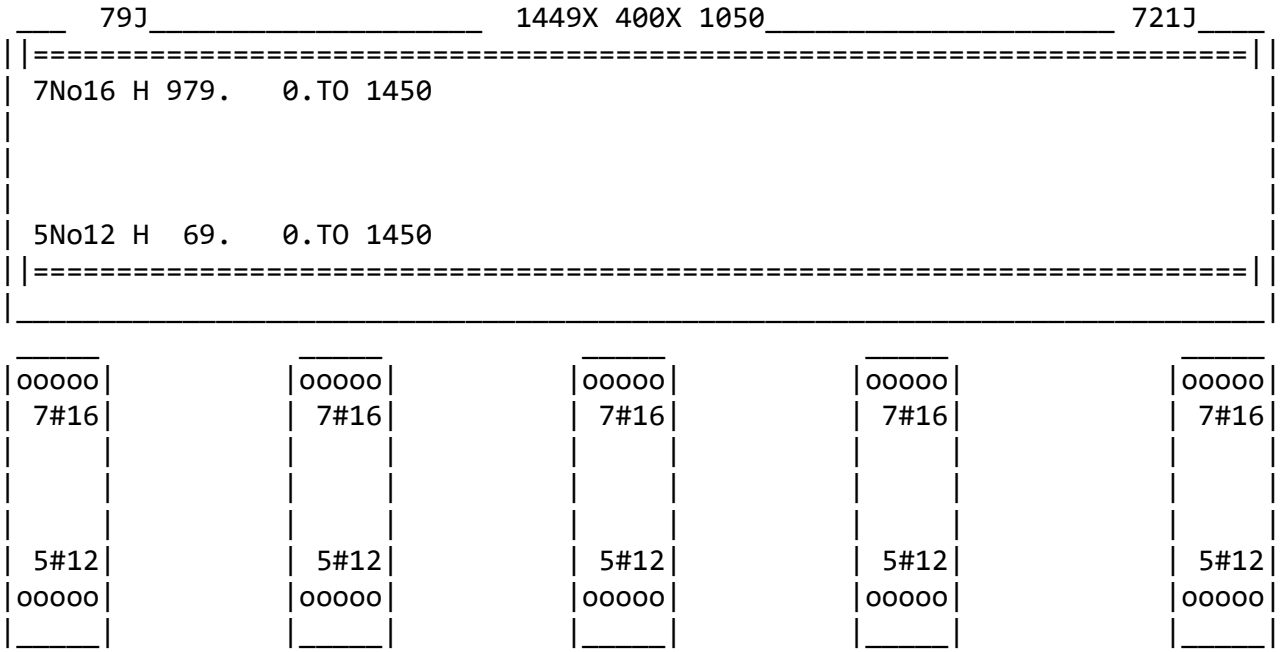
LEN - 1450. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1450.	YES	YES
2	979.	7 - 16MM	0.	1450.	YES	YES

B E A M N O. 2238 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2238 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2238 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2239 DESIGN RESULTS

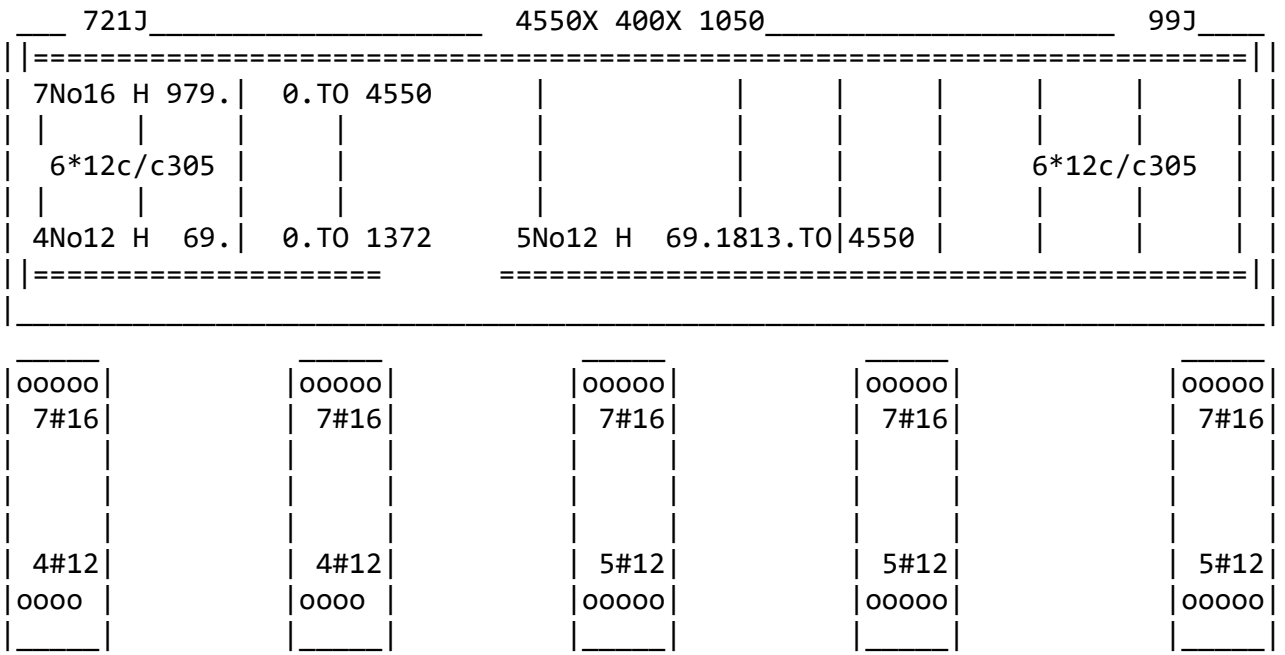
LEN - 4550. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	1372.	YES	NO
2	69.	5 - 12MM	1813.	4550.	NO	YES
3	979.	7 - 16MM	0.	4550.	YES	YES

B E A M N O. 2239 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.53 KNS Vc= 380.87 KNS Vs= 0.00 KNS
Tu= 45.64 KN-MET Tc= 22.2 KN-MET Ts= 60.8 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 7.74 SQ.CM.

AT END SUPPORT - Vu= 17.35 KNS Vc= 380.87 KNS Vs= 0.00 KNS
Tu= 45.64 KN-MET Tc= 22.2 KN-MET Ts= 60.8 KN-MET LOAD 413
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 305. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 7.74 SQ.CM.



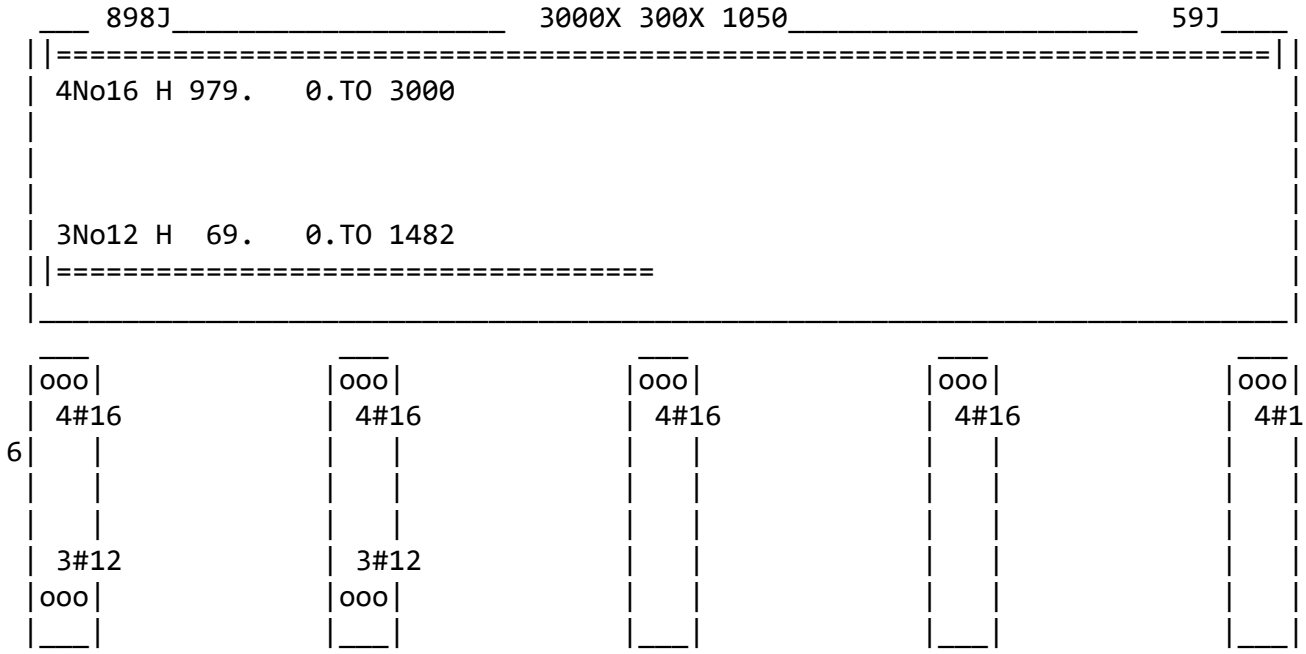
ACI 318-11 BEAM NO. 2240 DESIGN RESULTS

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1482.	YES	NO
2	979.	4 - 16MM	0.	3000.	YES	YES

B E A M N O. 2240 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 49.62 KNS Vc= 280.68 KNS Vs= 0.00 KNS
Tu= 1.11 KN-MET Tc= 13.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 60.53 KNS Vc= 280.68 KNS Vs= 0.00 KNS
Tu= 1.11 KN-MET Tc= 13.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2241 DESIGN RESULTS

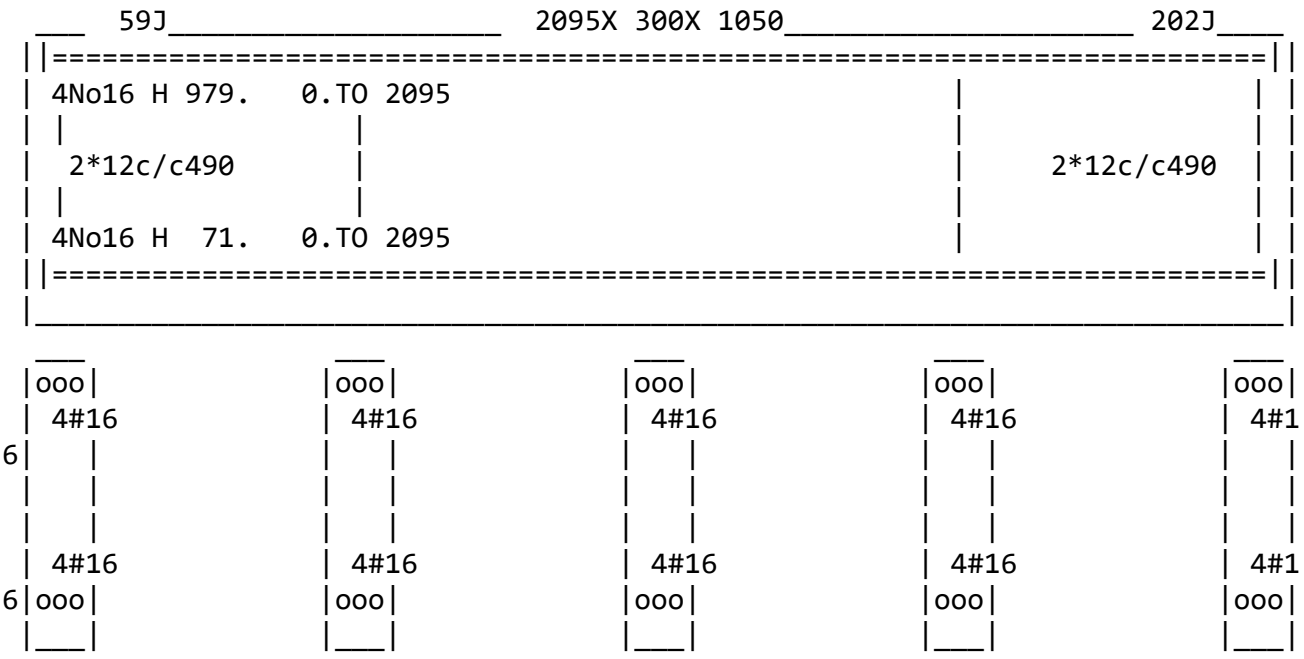
LEN - 2095. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	2095.	YES	YES
2	979.	4 - 16MM	0.	2095.	YES	YES

B E A M N O. 2241 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 113.13 KNS Vc= 274.46 KNS Vs= 0.00 KNS
Tu= 1.61 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 73. MM

AT END SUPPORT - Vu= 111.62 KNS Vc= 274.46 KNS Vs= 0.00 KNS
Tu= 1.61 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 73. MM



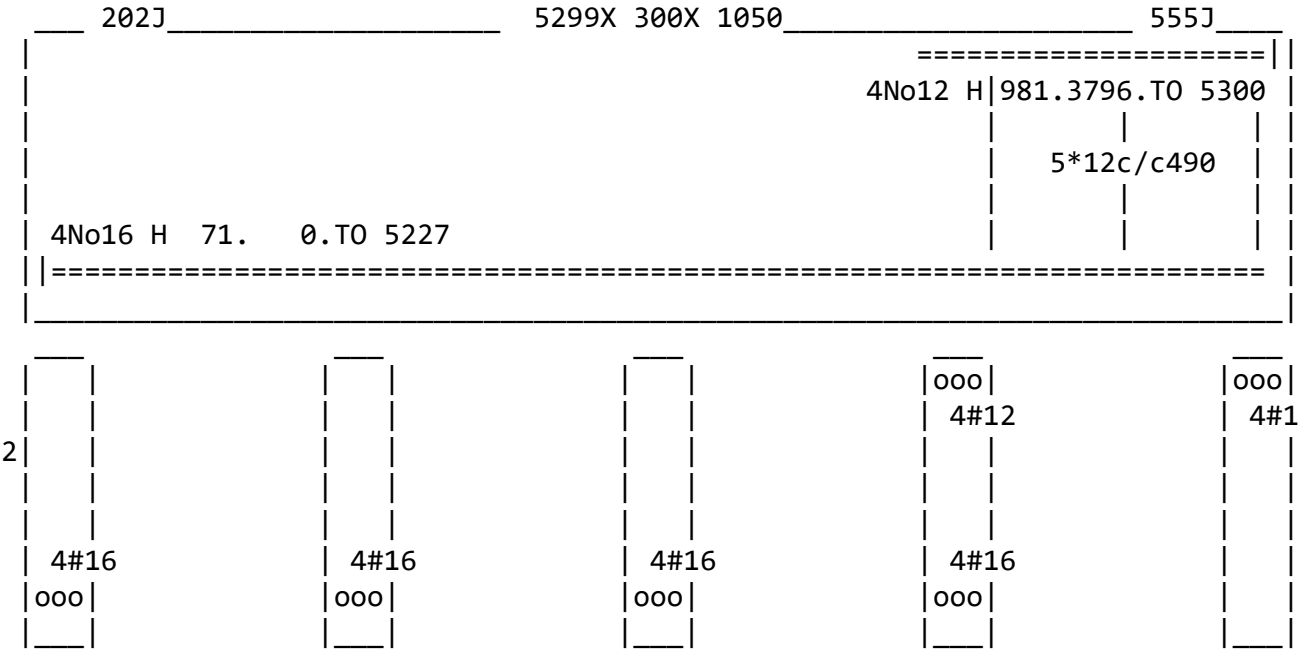
ACI 318-11 BEAM NO. 2242 DESIGN RESULTS

LEN - 5300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	5227.	YES	NO
2	981.	4 - 12MM	3796.	5300.	NO	YES

B E A M N O. 2242 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.04 KNS Vc= 274.49 KNS Vs= 0.00 KNS
Tu= 1.61 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 106.11 KNS Vc= 274.49 KNS Vs= 0.00 KNS
Tu= 1.61 KN-MET Tc= 12.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 490. MM C/C FOR 1675. MM



ACI 318-11 BEAM NO. 2243 DESIGN RESULTS

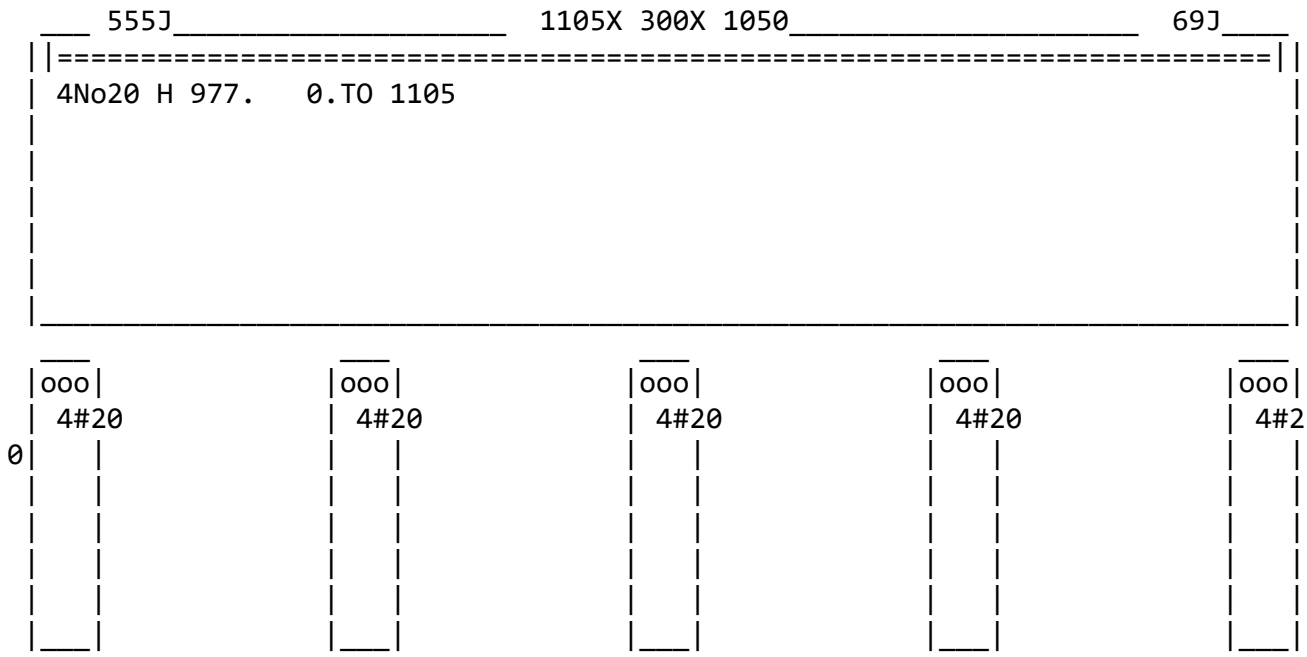
LEN - 1105. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	977.	4 - 20MM	0.	1105.	YES	YES

B E A M N O. 2243 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2243 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2243 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2244 DESIGN RESULTS

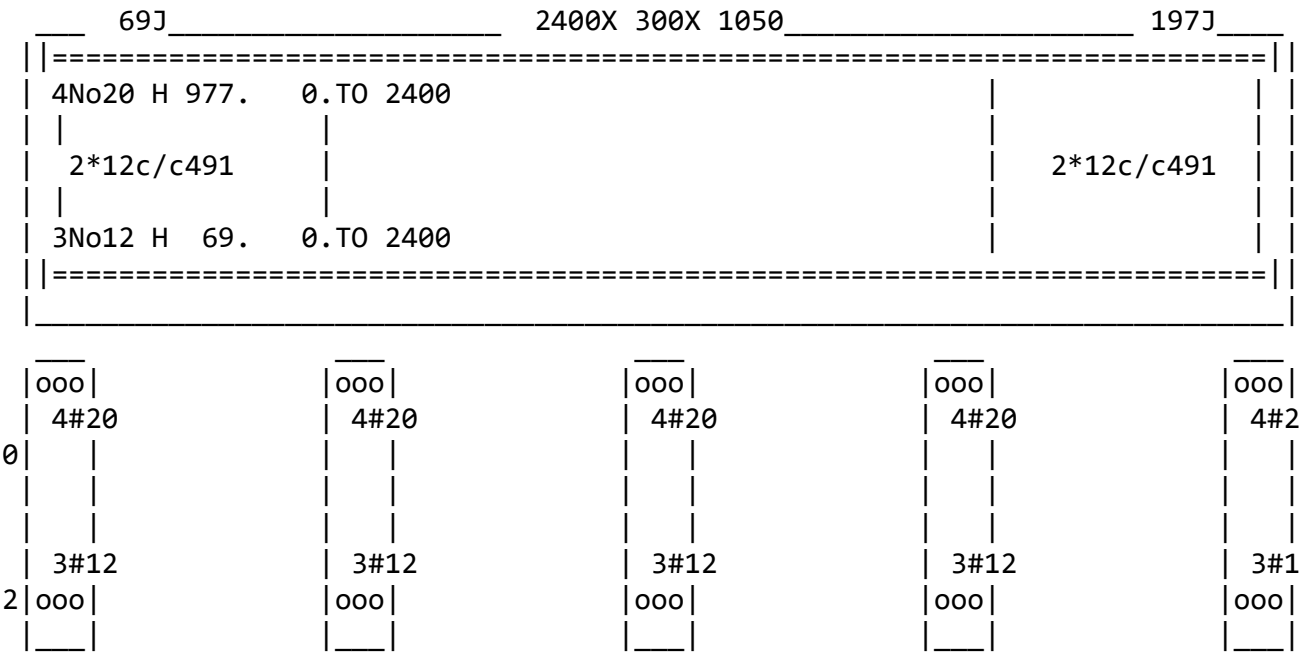
LEN - 2400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	2400.	YES	YES
2	977.	4 - 20MM	0.	2400.	YES	YES

B E A M N O. 2244 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 138.53 KNS Vc= 263.01 KNS Vs= 0.00 KNS
Tu= 2.35 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 491. MM C/C FOR 225. MM

AT END SUPPORT - Vu= 128.01 KNS Vc= 263.01 KNS Vs= 0.00 KNS
Tu= 2.35 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 491. MM C/C FOR 225. MM



ACI 318-11 BEAM NO. 2245 DESIGN RESULTS

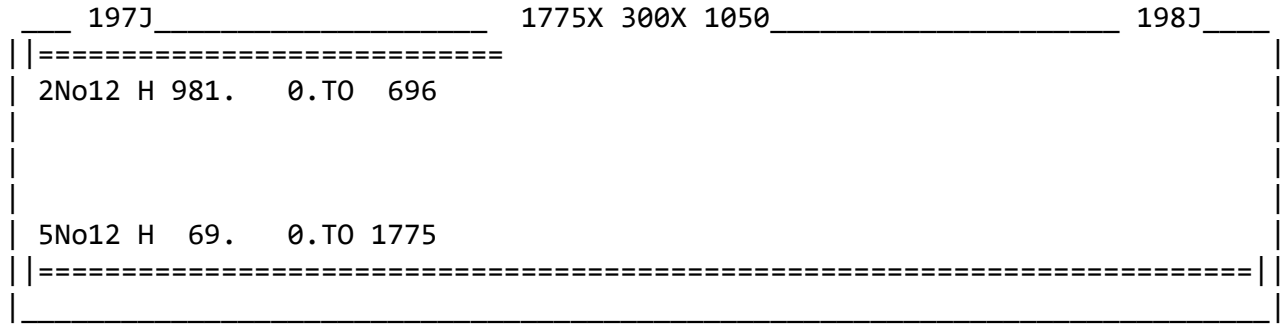
LEN - 1775. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1775.	YES	YES
2	981.	2 - 12MM	0.	696.	YES	NO

B E A M N O. 2245 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2245 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2245 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



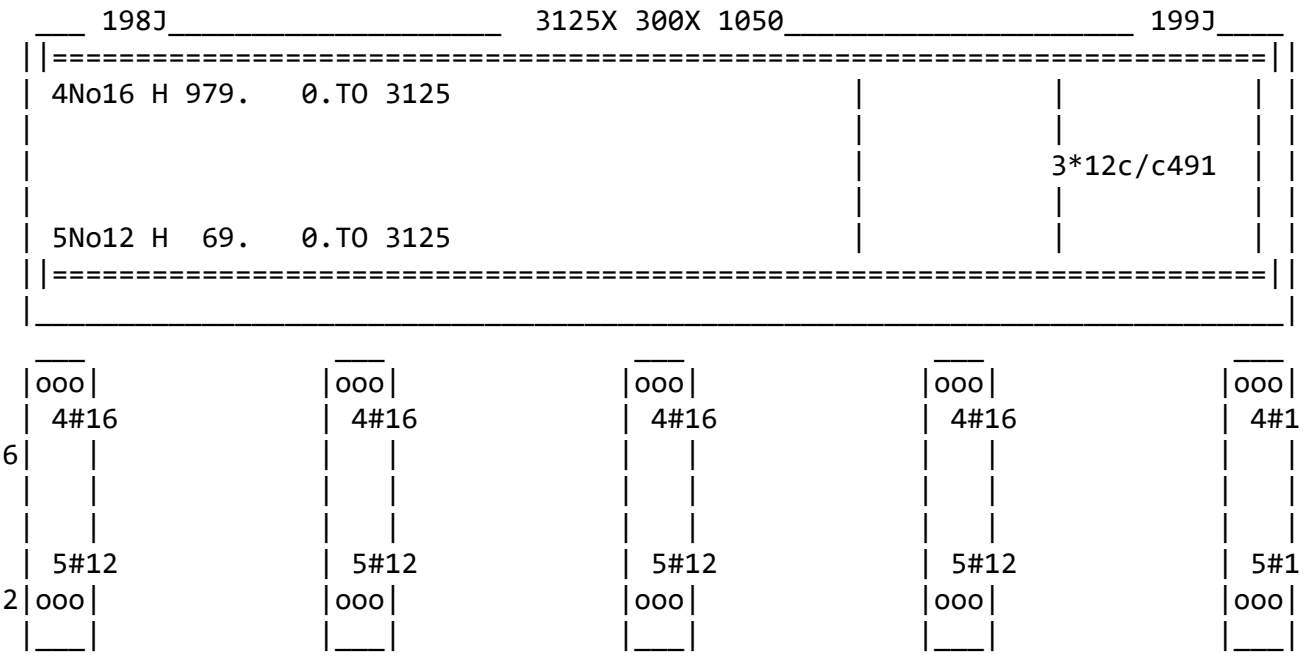
ACI 318-11 BEAM NO. 2246 DESIGN RESULTS

LEN - 3125. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3125.	YES	YES
2	979.	4 - 16MM	0.	3125.	YES	YES

B E A M N O. 2246 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 39.86 KNS Vc= 261.94 KNS Vs= 0.00 KNS
Tu= 0.80 KN-MET Tc= 12.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 90.49 KNS Vc= 236.03 KNS Vs= 0.00 KNS
Tu= 0.29 KN-MET Tc= 11.1 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 491. MM C/C FOR 588. MM



ACI 318-11 BEAM NO. 2247 DESIGN RESULTS

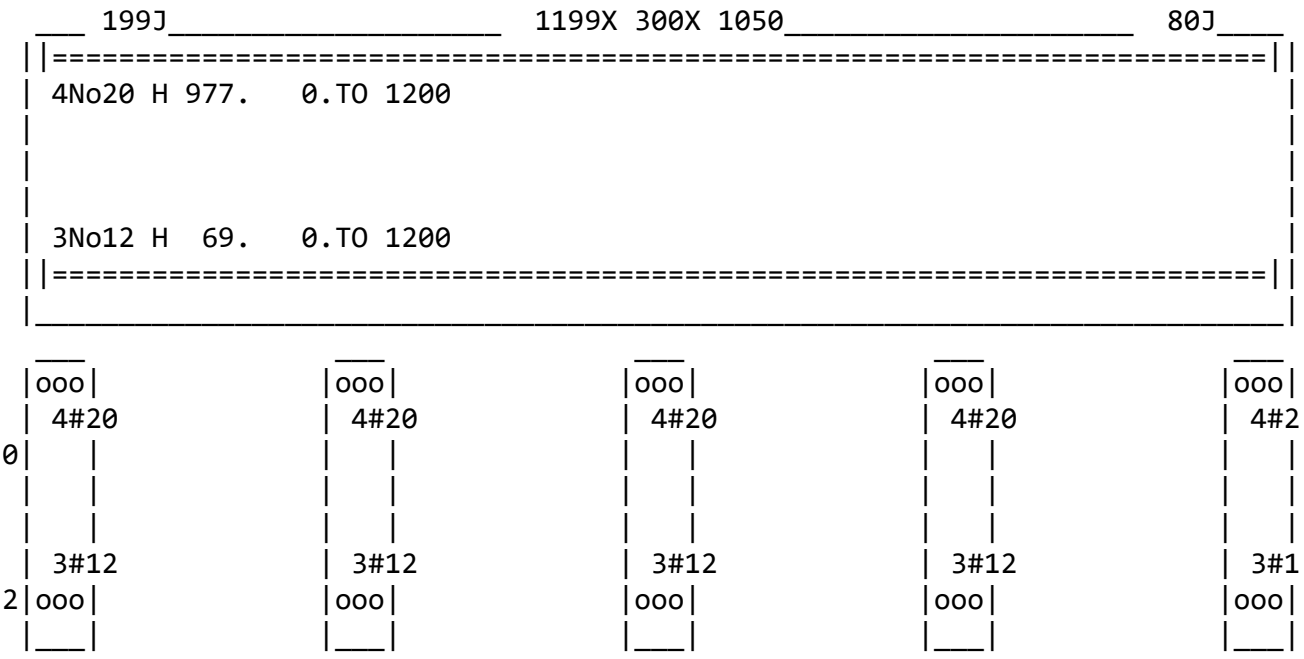
LEN - 1200. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1200.	YES	YES
2	977.	4 - 20MM	0.	1200.	YES	YES

B E A M N O. 2247 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2247 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2247 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2248 DESIGN RESULTS

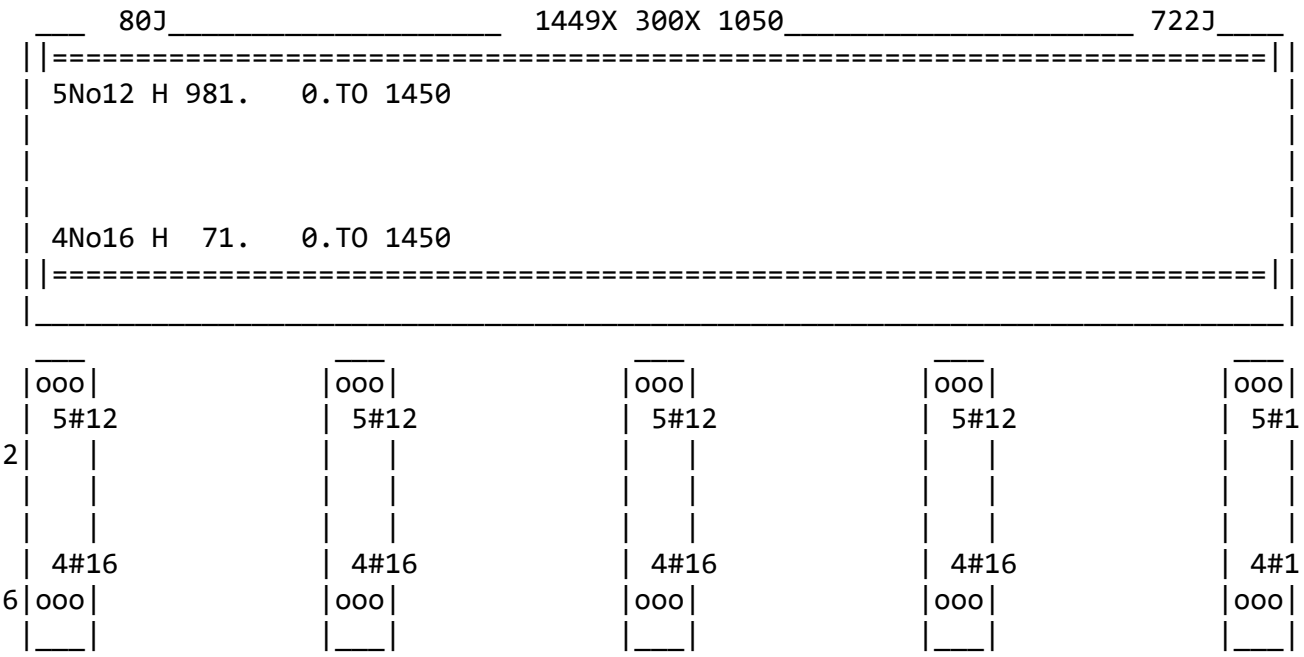
LEN - 1450. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	1450.	YES	YES
2	981.	5 - 12MM	0.	1450.	YES	YES

B E A M N O. 2248 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2248 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2248 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2249 DESIGN RESULTS

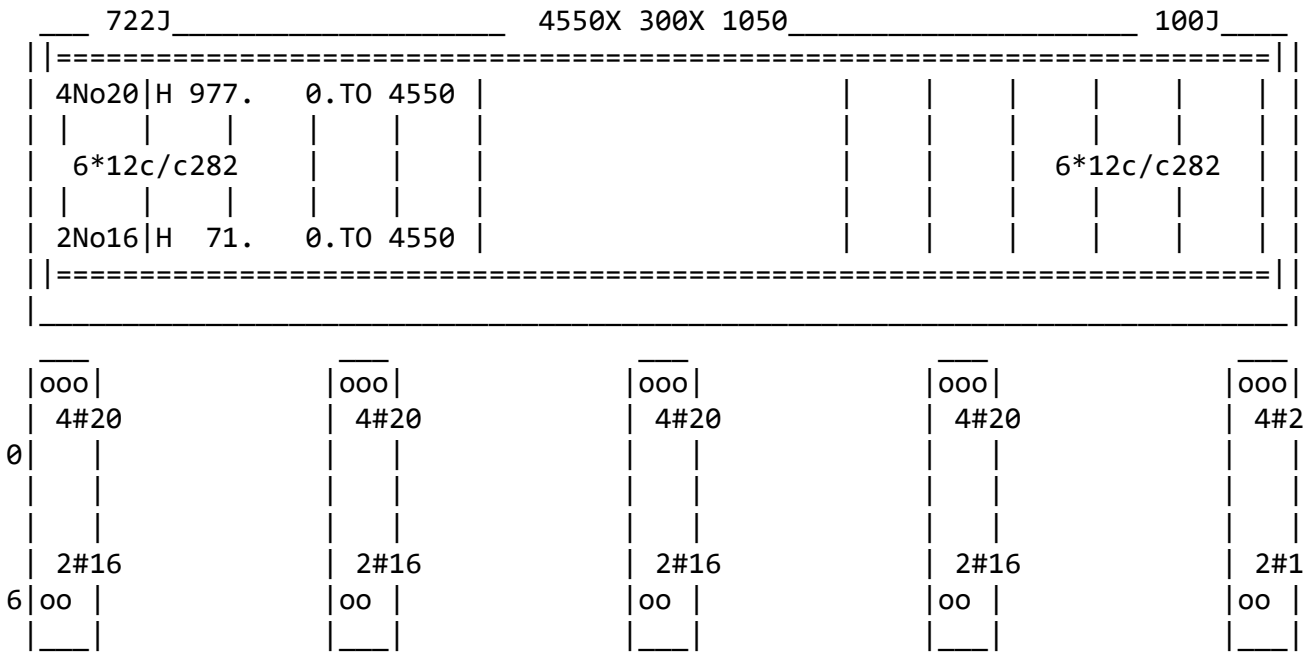
LEN - 4550. MM FY - 420. FC - 35. MPA, SIZE - 300. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	4550.	YES	YES
2	977.	4 - 20MM	0.	4550.	YES	YES

B E A M N O. 2249 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 31.68 KNS Vc= 274.59 KNS Vs= 0.00 KNS
Tu= 17.38 KN-MET Tc= 12.9 KN-MET Ts= 23.2 KN-MET LOAD 409
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 282. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.14 SQ.CM.

AT END SUPPORT - Vu= 54.84 KNS Vc= 274.59 KNS Vs= 0.00 KNS
Tu= 17.38 KN-MET Tc= 12.9 KN-MET Ts= 23.2 KN-MET LOAD 409
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 282. MM C/C FOR 1300. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.14 SQ.CM.



ACI 318-11 BEAM NO. 2250 DESIGN RESULTS

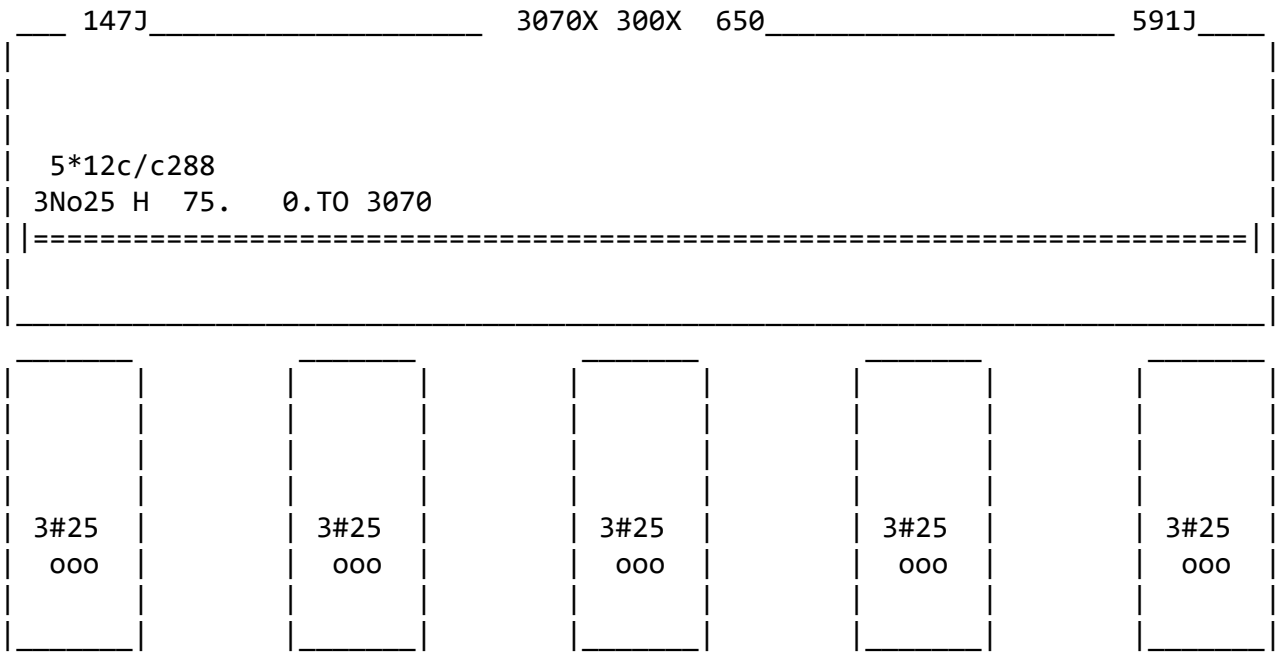
LEN - 3070. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	3070.	YES	YES

B E A M N O. 2250 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 85.65 KNS Vc= 184.93 KNS Vs= 0.00 KNS
Tu= 0.53 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 288. MM C/C FOR 960. MM

AT END SUPPORT - Vu= 29.09 KNS Vc= 163.59 KNS Vs= 0.00 KNS
Tu= 0.53 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2251 DESIGN RESULTS

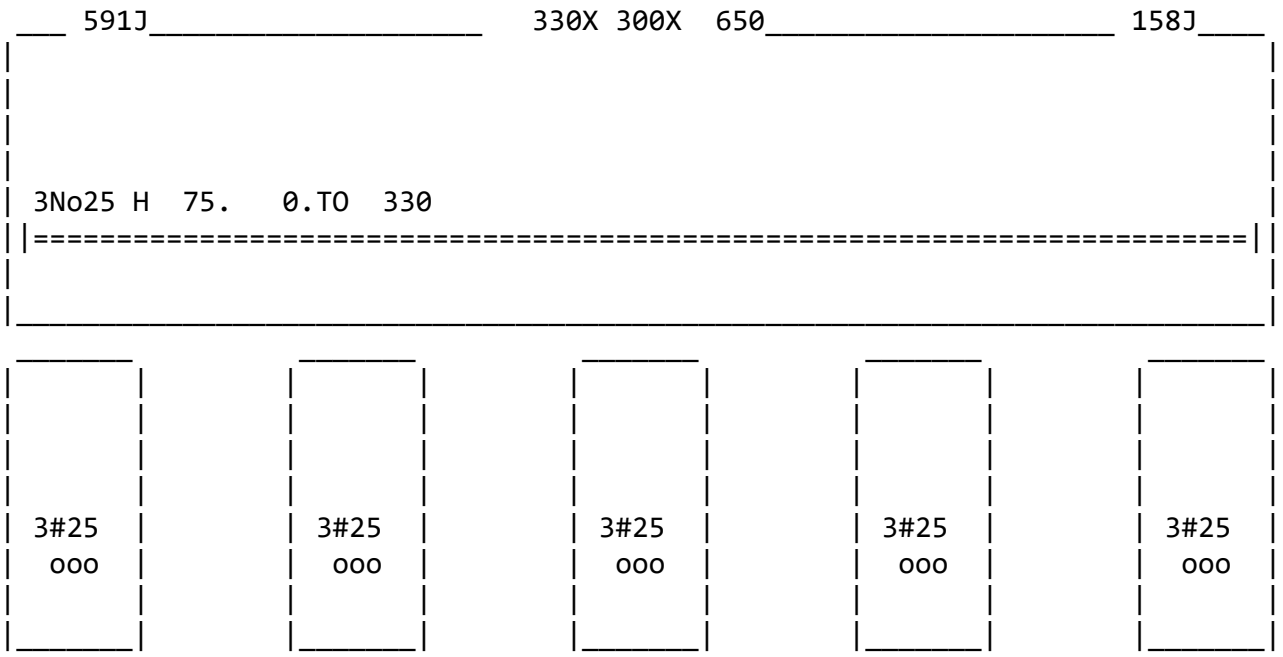
LEN - 330. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	330.	YES	YES

B E A M N O. 2251 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2251 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2251 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2252 DESIGN RESULTS

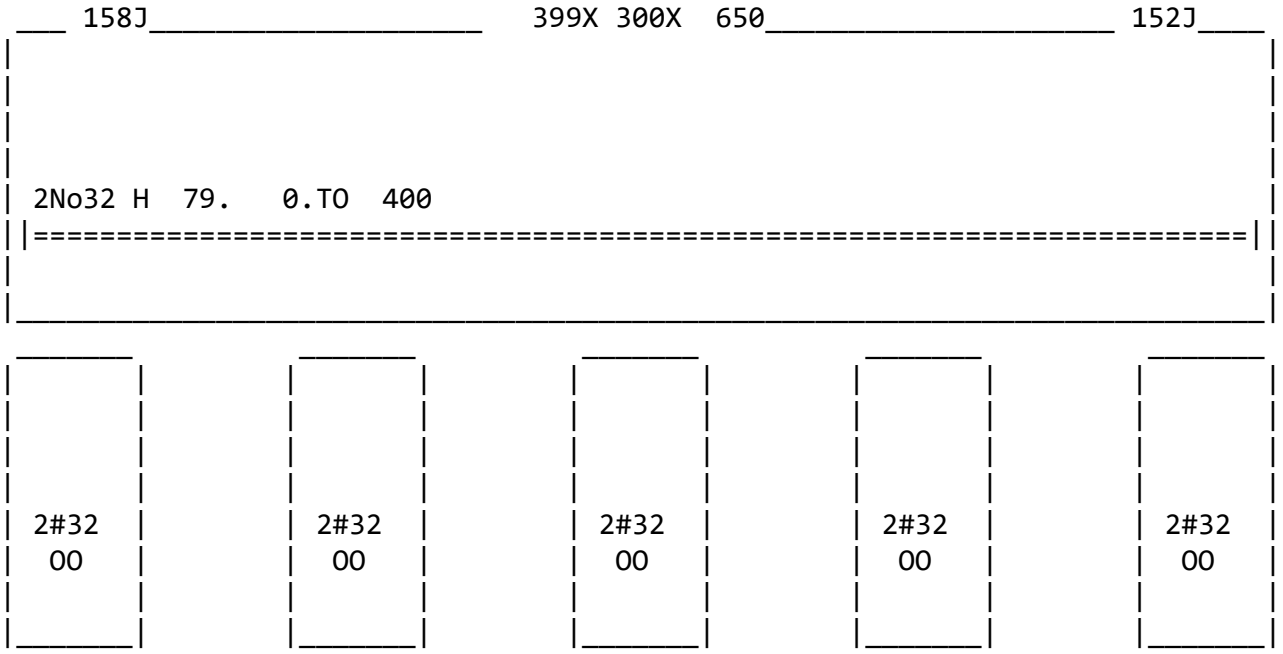
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	2 - 32MM	0.	400.	YES	YES

B E A M N O. 2252 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2252 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2252 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2253 DESIGN RESULTS

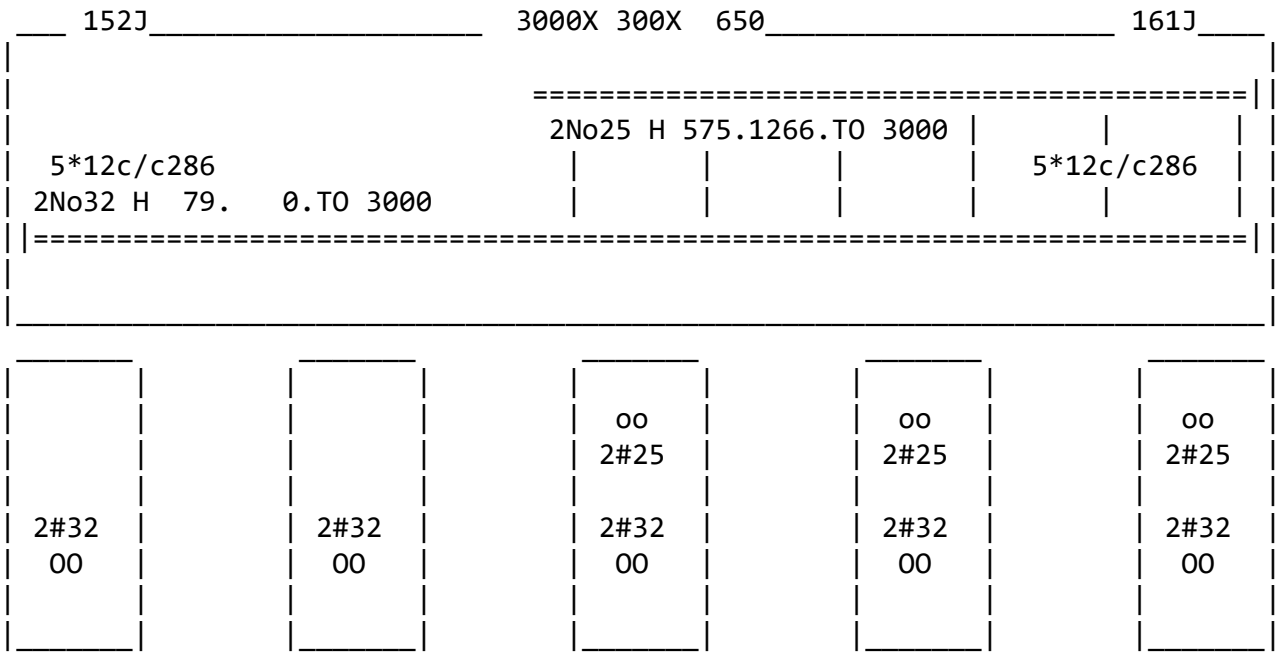
LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	2 - 32MM	0.	3000.	YES	YES
2	575.	2 - 25MM	1266.	3000.	NO	YES

B E A M N O. 2253 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 74.72 KNS Vc= 168.37 KNS Vs= 0.00 KNS
Tu= 0.53 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 286. MM C/C FOR 925. MM

AT END SUPPORT - Vu= 228.02 KNS Vc= 168.07 KNS Vs= 135.96 KNS
Tu= 0.18 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 406
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 286. MM C/C FOR 925. MM



ACI 318-11 BEAM NO. 2254 DESIGN RESULTS

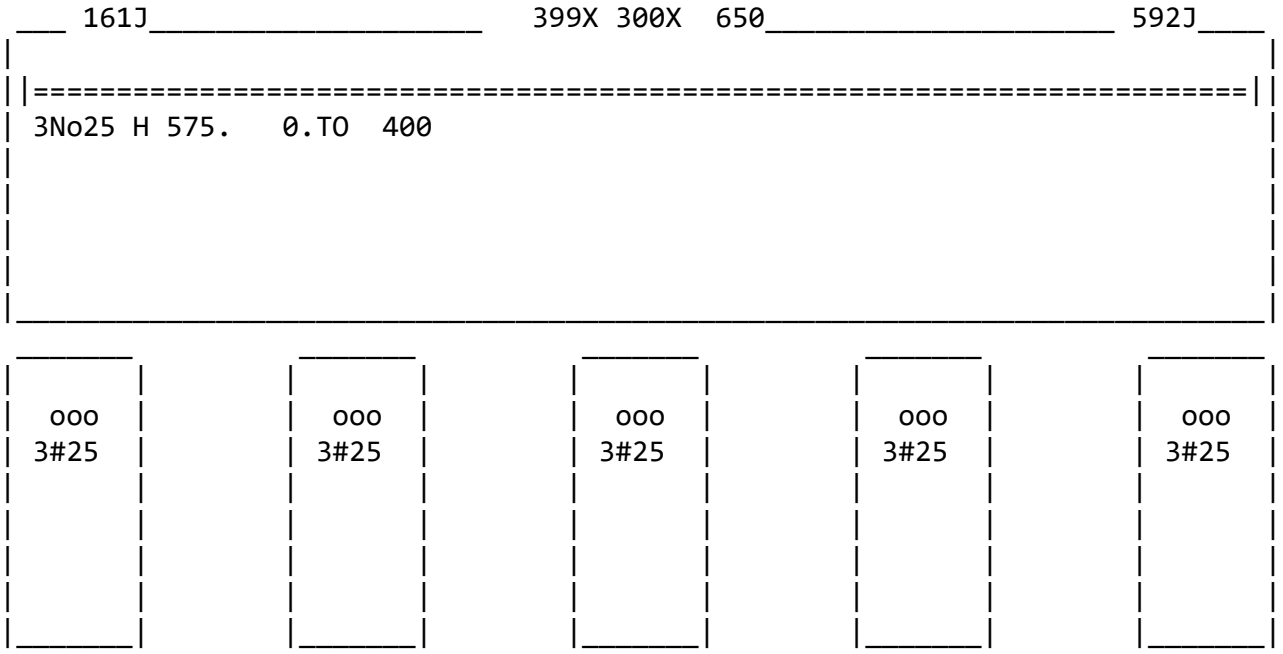
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	575.	3 - 25MM	0.	400.	YES	YES

B E A M N O. 2254 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2254 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2254 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2255 DESIGN RESULTS

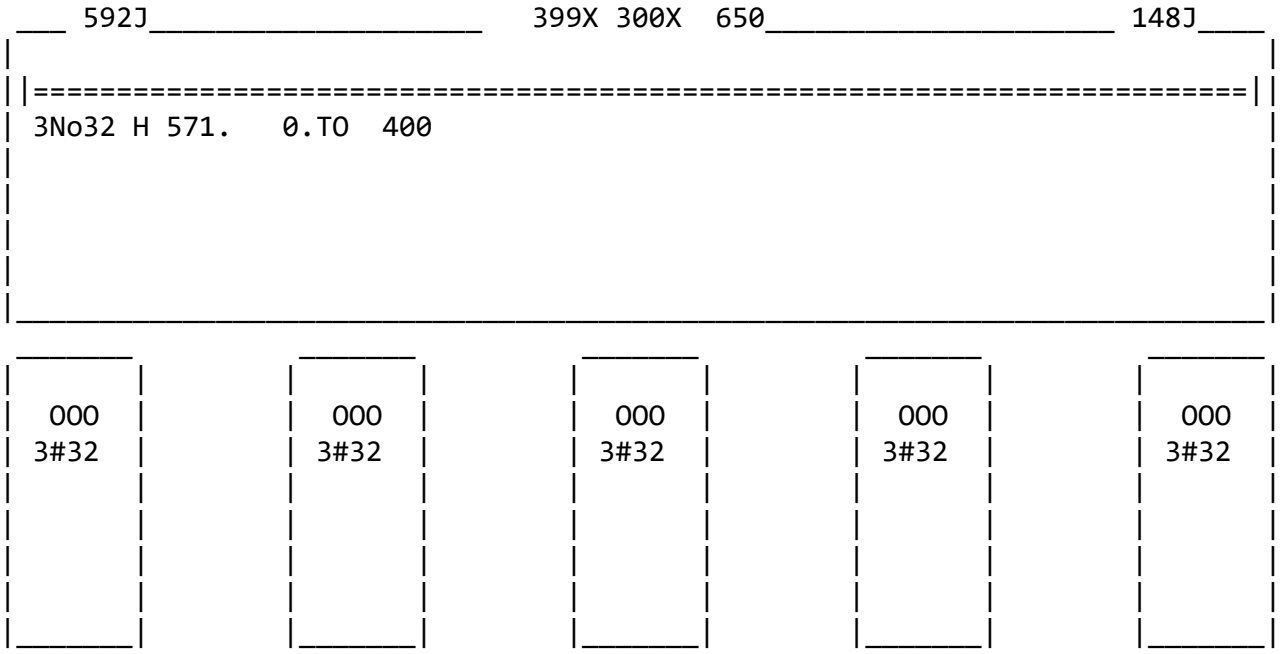
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	571.	3 - 32MM	0.	400.	YES	YES

B E A M N O. 2255 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2255 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2255 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2256 DESIGN RESULTS

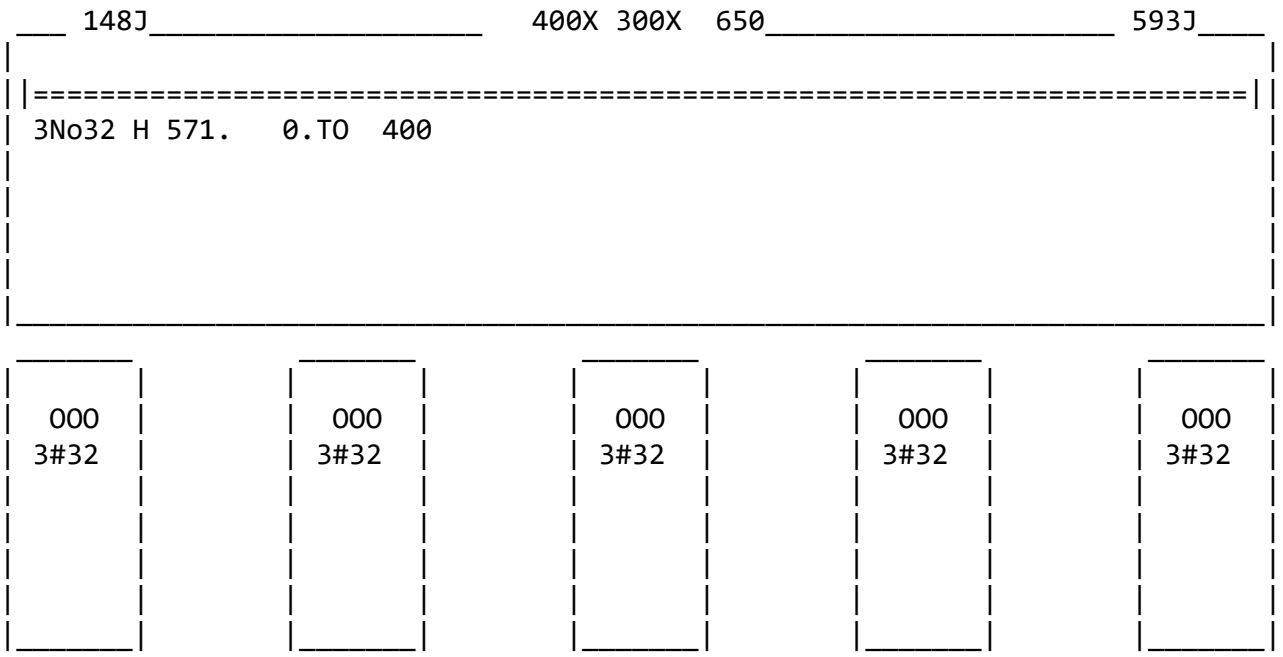
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	571.	3 - 32MM	0.	400.	YES	YES

B E A M N O. 2256 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2256 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2256 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2257 DESIGN RESULTS

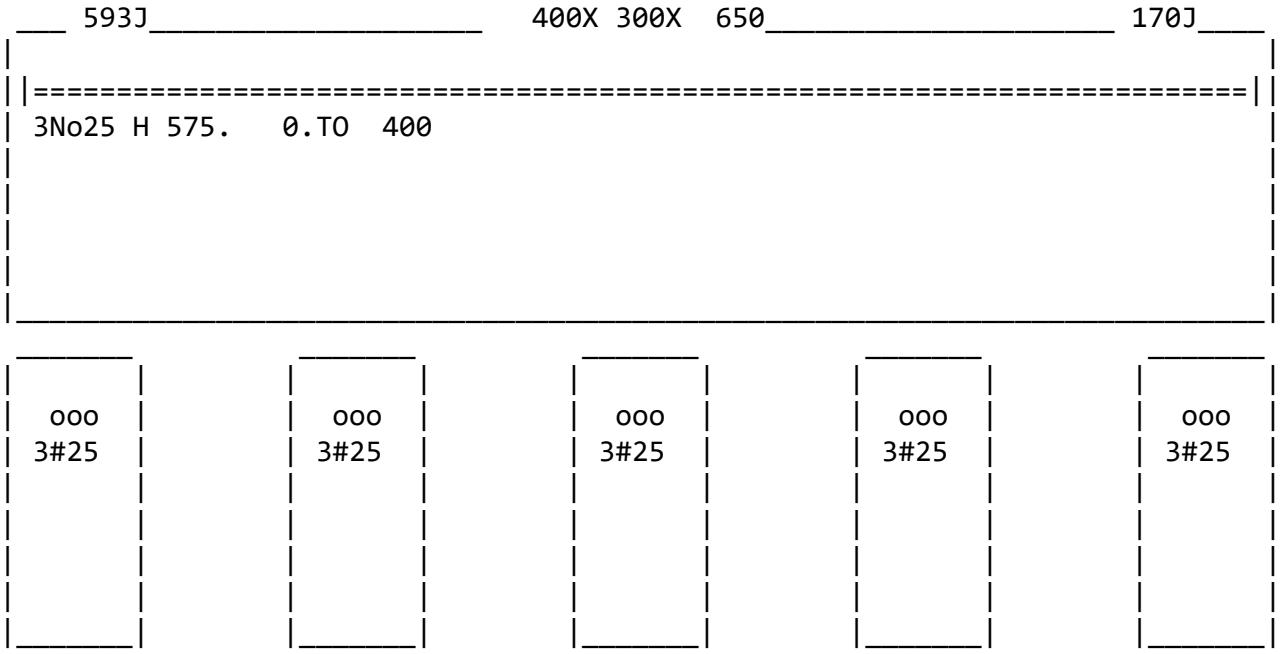
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	575.	3 - 25MM	0.	400.	YES	YES

B E A M N O. 2257 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2257 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2257 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2258 DESIGN RESULTS

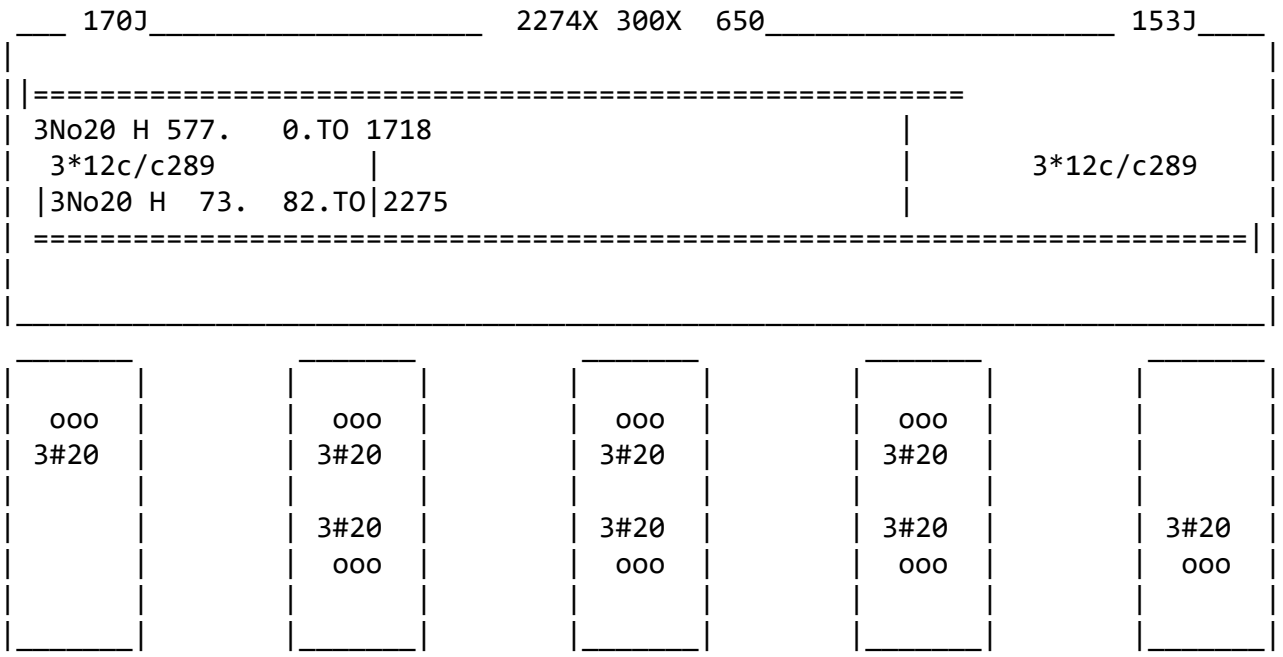
LEN - 2275. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	82.	2275.	NO	YES
2	577.	3 - 20MM	0.	1718.	YES	NO

B E A M N O. 2258 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 203.05 KNS Vc= 196.13 KNS Vs= 74.60 KNS
Tu= 0.25 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 289. MM C/C FOR 563. MM

AT END SUPPORT - Vu= 83.88 KNS Vc= 170.31 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 289. MM C/C FOR 563. MM



ACI 318-11 BEAM NO. 2259 DESIGN RESULTS

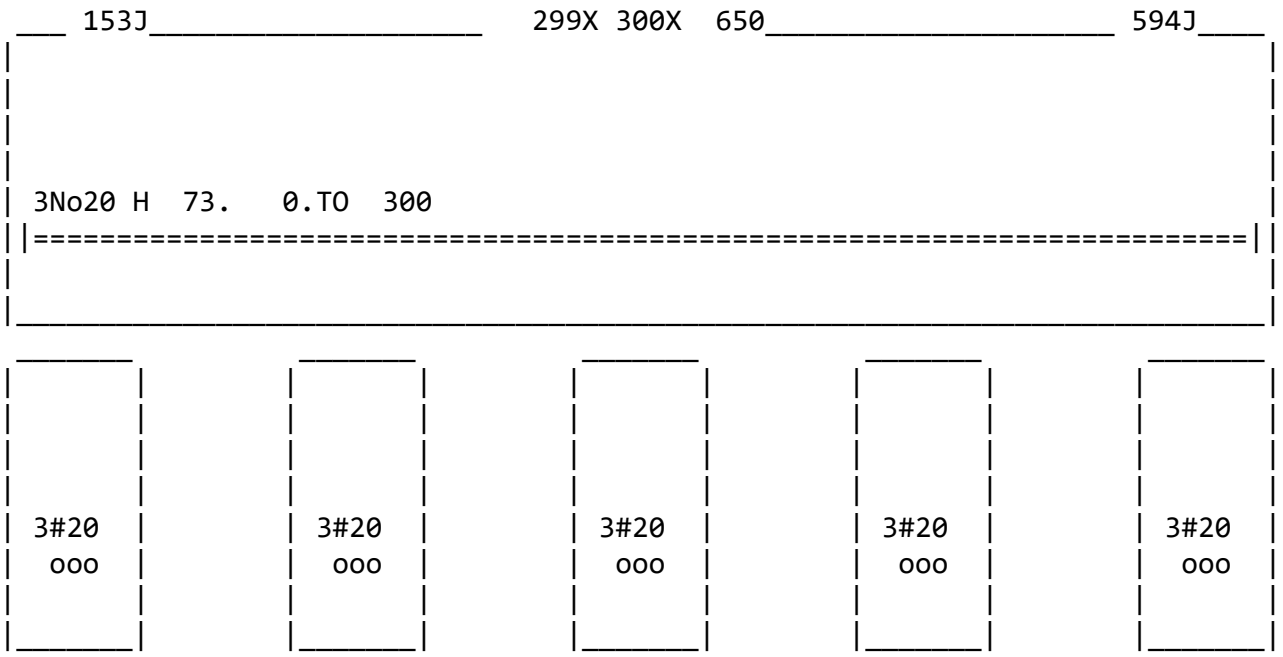
LEN - 300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	300.	YES	YES

B E A M N O. 2259 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2259 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2259 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2260 DESIGN RESULTS

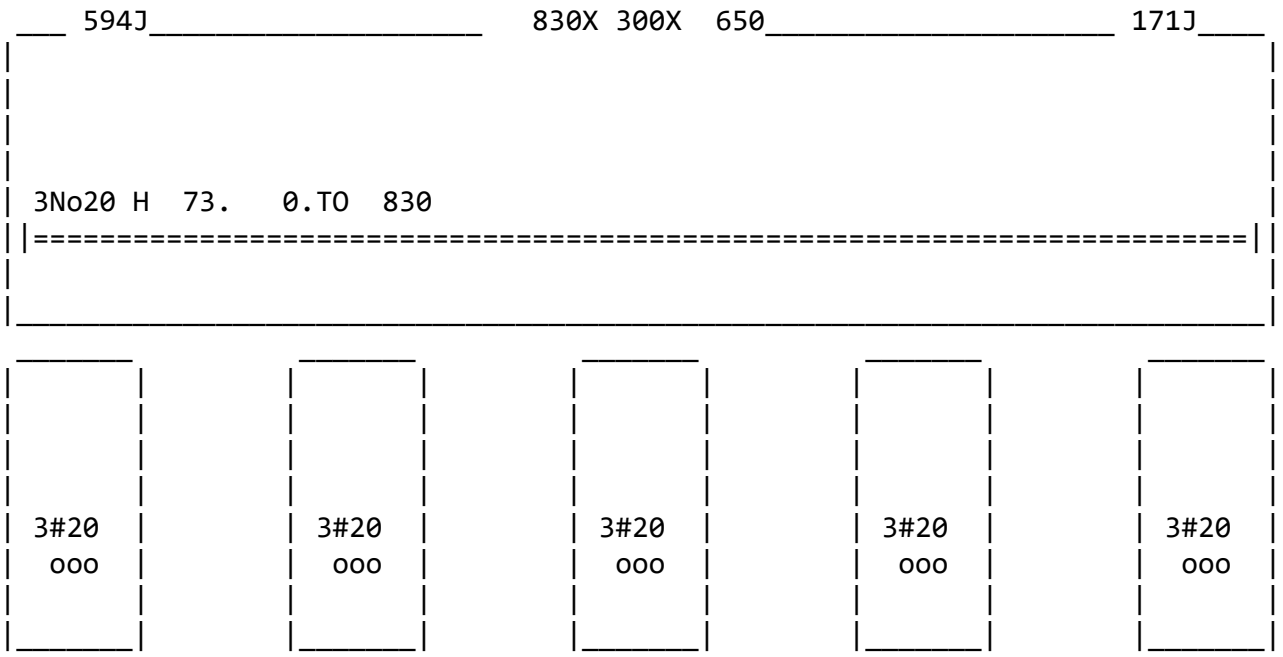
LEN - 830. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	830.	YES	YES

B E A M N O. 2260 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2260 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2260 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2261 DESIGN RESULTS

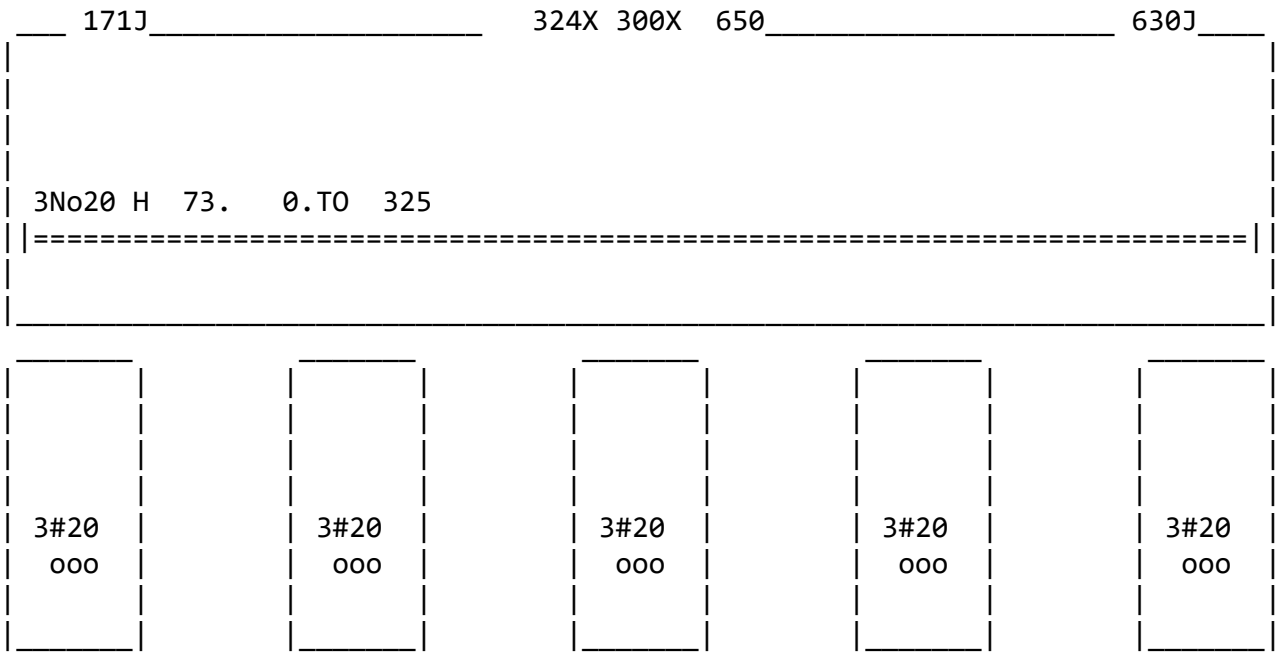
LEN - 325. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	325.	YES	YES

B E A M N O. 2261 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2261 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2261 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2262 DESIGN RESULTS

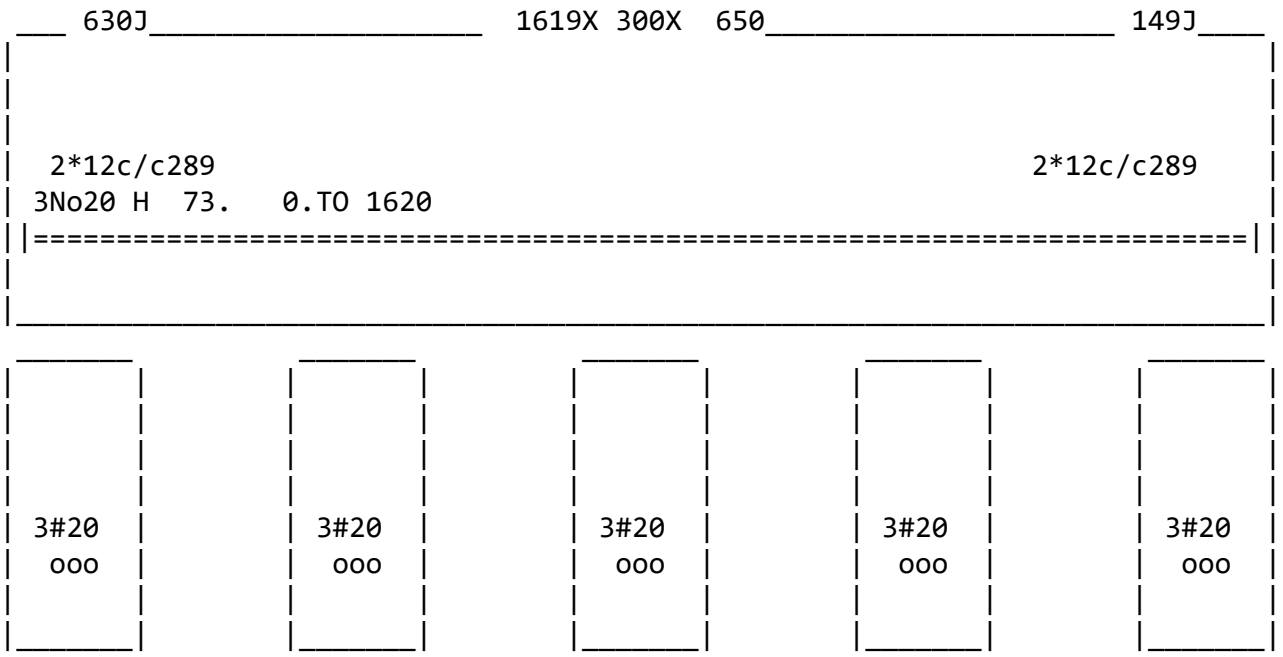
LEN - 1620. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	1620.	YES	YES

B E A M N O. 2262 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 72.57 KNS Vc= 169.04 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 289. MM C/C FOR 235. MM

AT END SUPPORT - Vu= 80.42 KNS Vc= 176.10 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 289. MM C/C FOR 235. MM



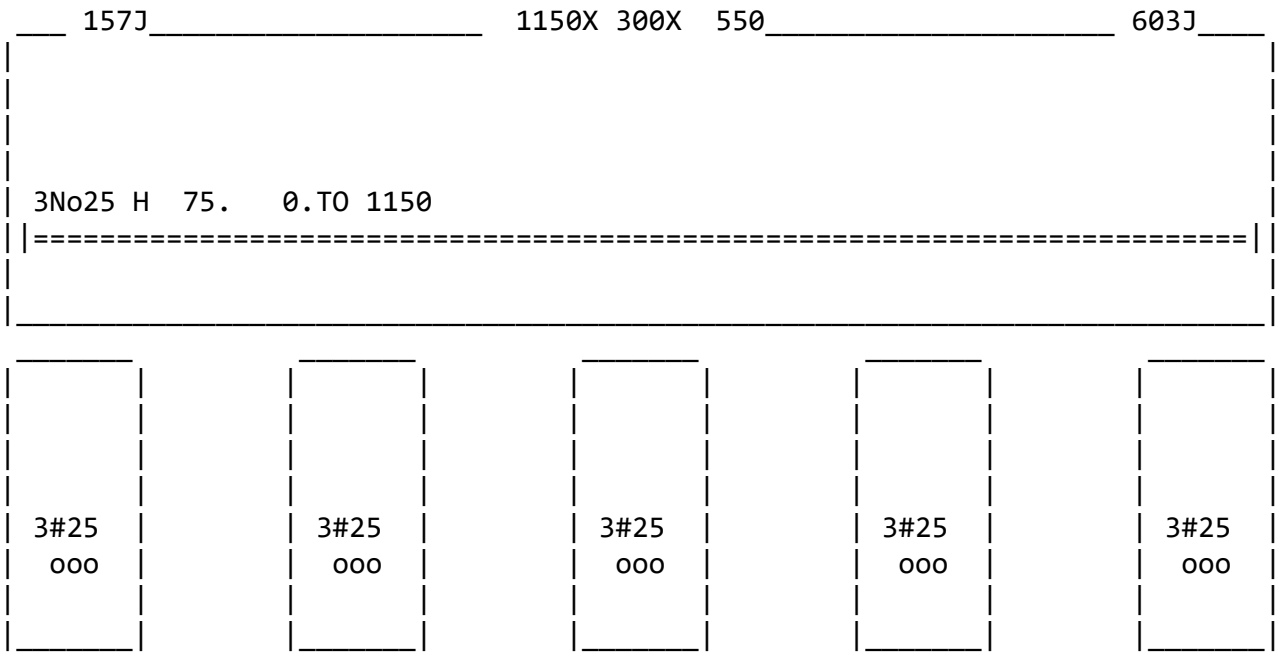
ACI 318-11 BEAM NO. 2263 DESIGN RESULTS

LEN - 1150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1150.	YES	YES

B E A M N O. 2263 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 200.88 KNS Tu= 37.7 KN-MET
Vc= 158.8 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 405 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
AT END SUPPORT - Vu= 195.60 KNS Tu= 37.7 KN-MET
Vc= 150.3 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 405 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.



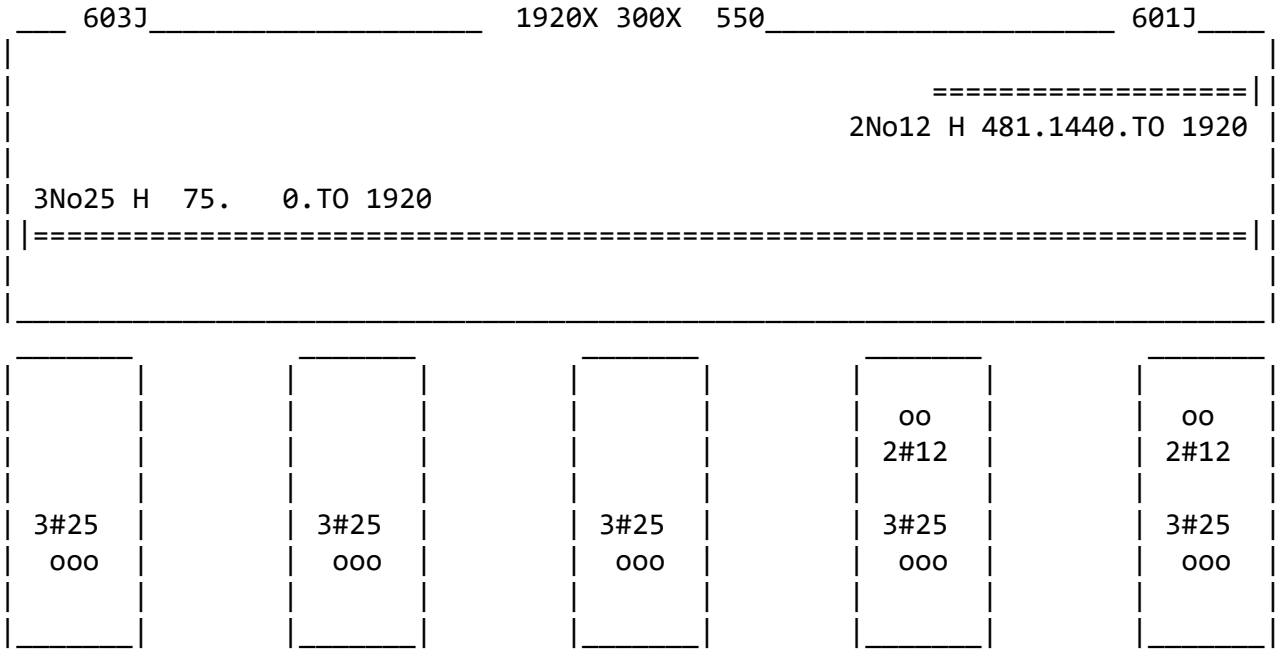
ACI 318-11 BEAM NO. 2264 DESIGN RESULTS

LEN - 1920. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1920.	YES	YES
2	481.	2 - 12MM	1440.	1920.	NO	YES

B E A M N O. 2264 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 102.00 KNS Tu= 37.7 KN-MET
Vc= 139.4 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 405 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
AT END SUPPORT - Vu= 131.80 KNS Tu= 37.7 KN-MET
Vc= 154.9 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 405 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.



ACI 318-11 BEAM NO. 2265 DESIGN RESULTS

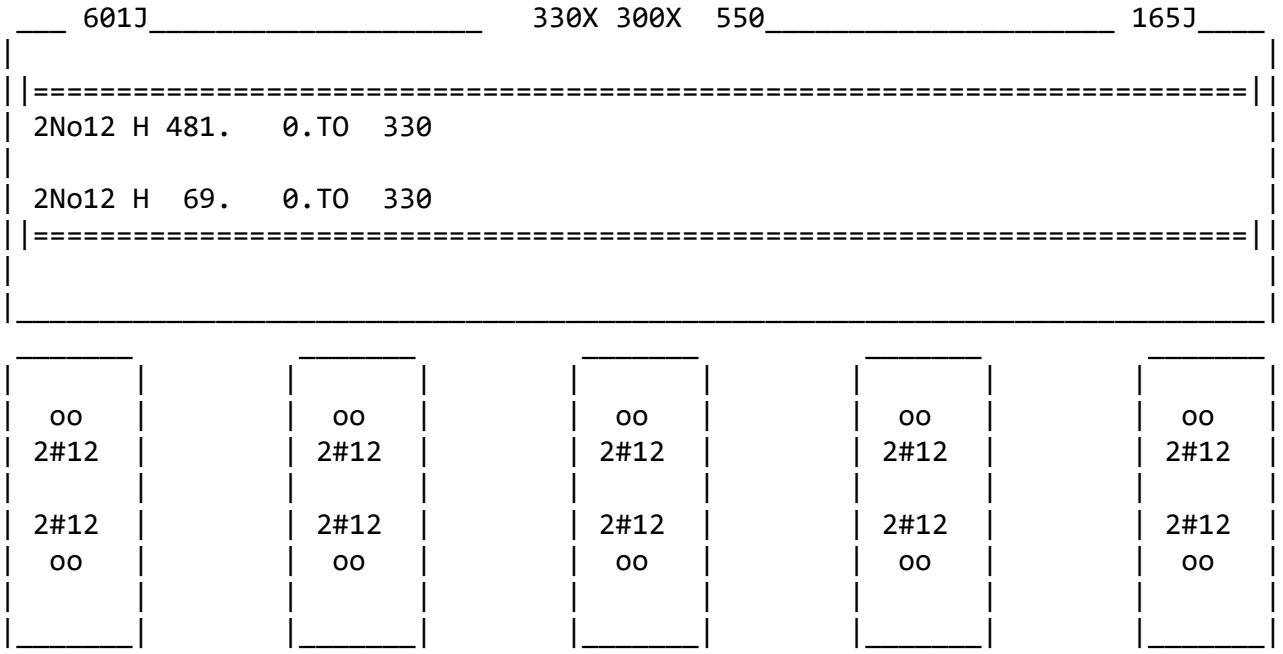
LEN - 330. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	330.	YES	YES
2	481.	2 - 12MM	0.	330.	YES	YES

B E A M N O. 2265 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2265 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2265 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2266 DESIGN RESULTS

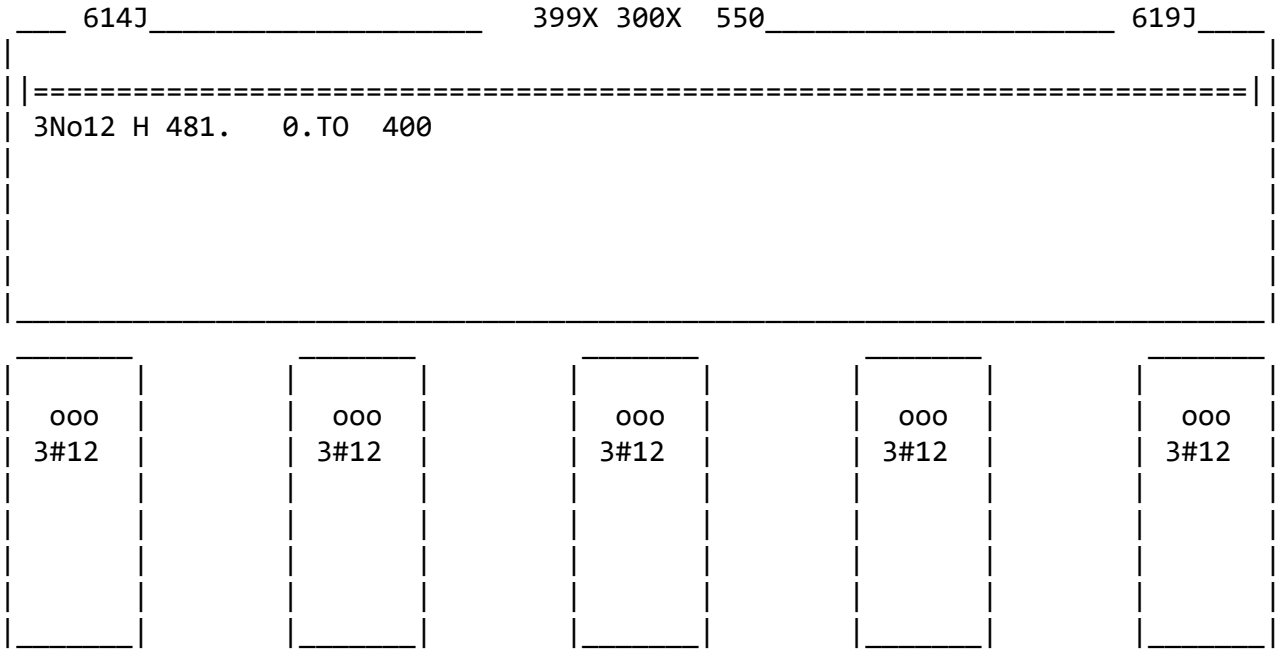
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	3 - 12MM	0.	400.	YES	YES

B E A M N O. 2266 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2266 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2266 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2267 DESIGN RESULTS

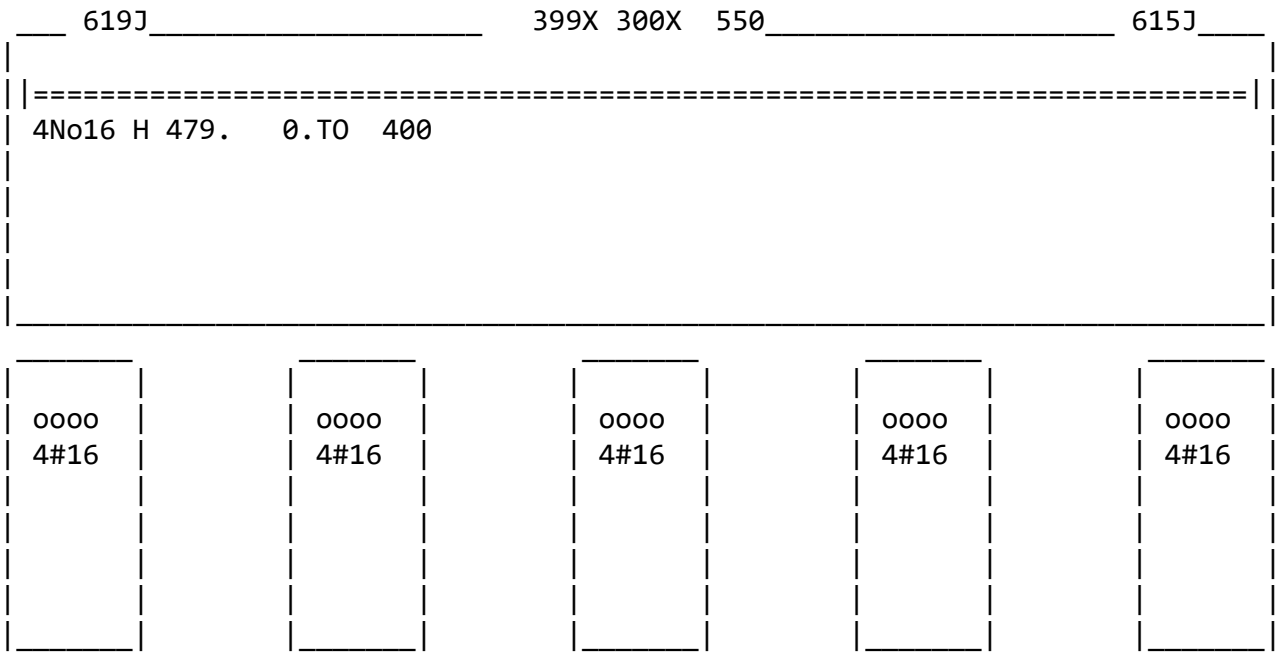
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	479.	4 - 16MM	0.	400.	YES	YES

B E A M N O. 2267 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2267 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2267 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2268 DESIGN RESULTS

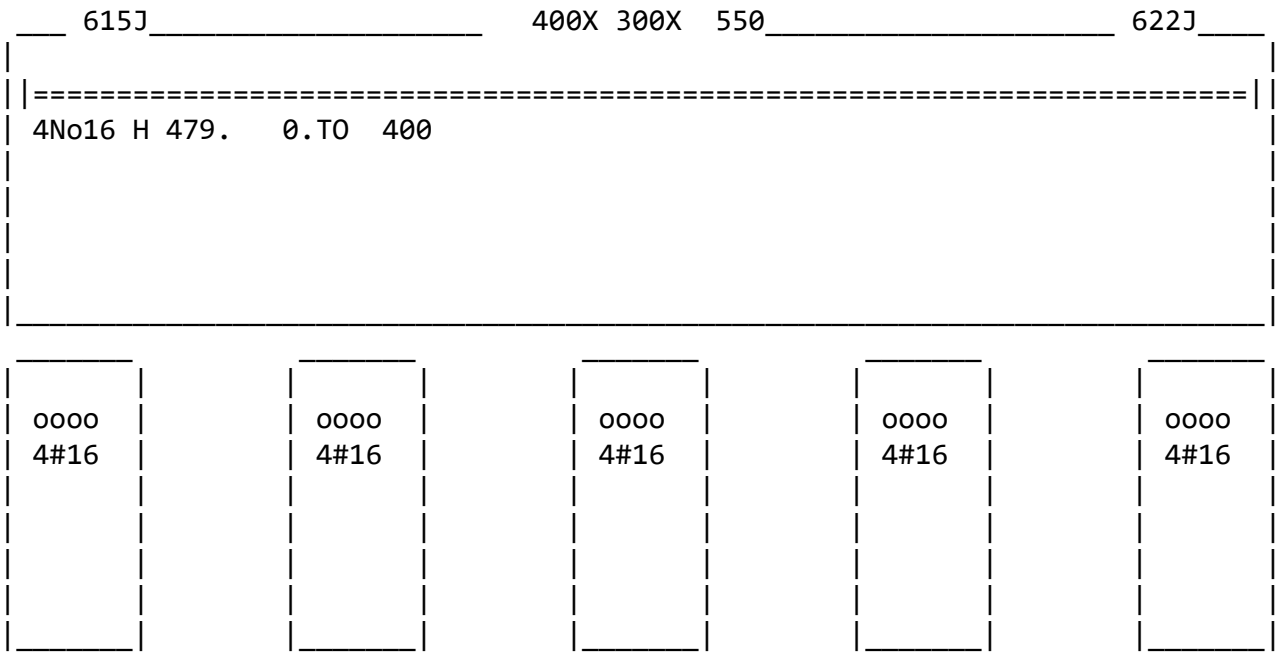
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	479.	4 - 16MM	0.	400.	YES	YES

B E A M N O. 2268 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2268 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2268 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2269 DESIGN RESULTS

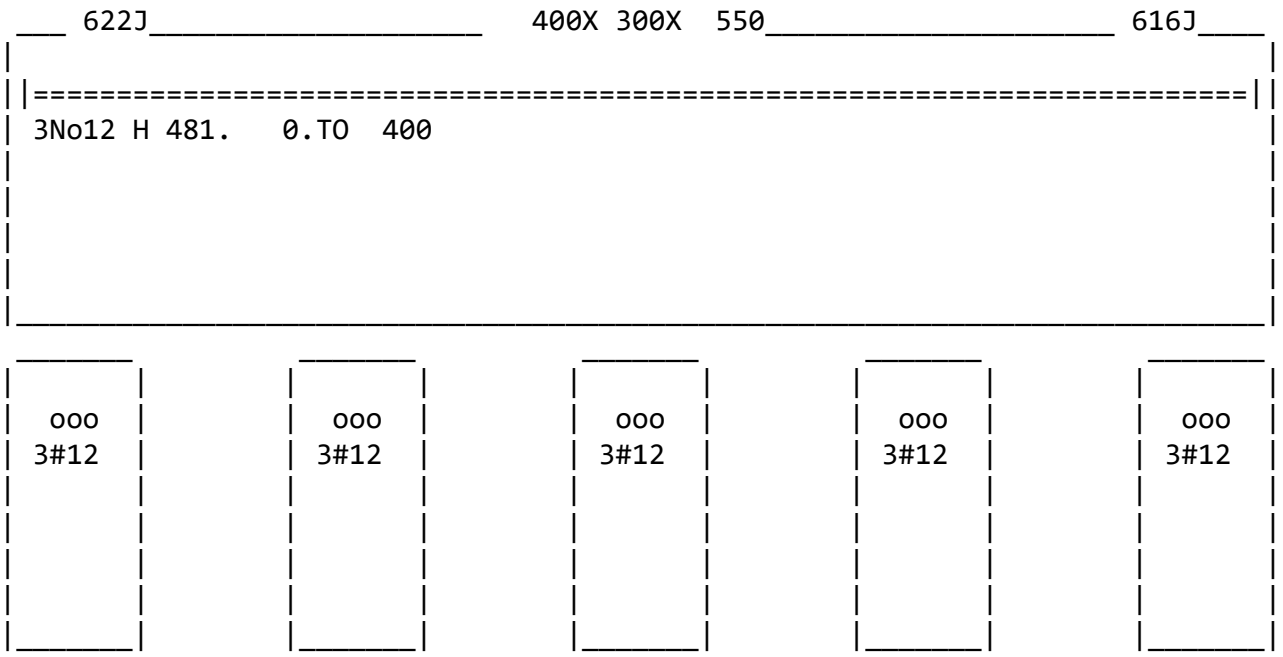
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	3 - 12MM	0.	400.	YES	YES

B E A M N O. 2269 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2269 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2269 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2270 DESIGN RESULTS

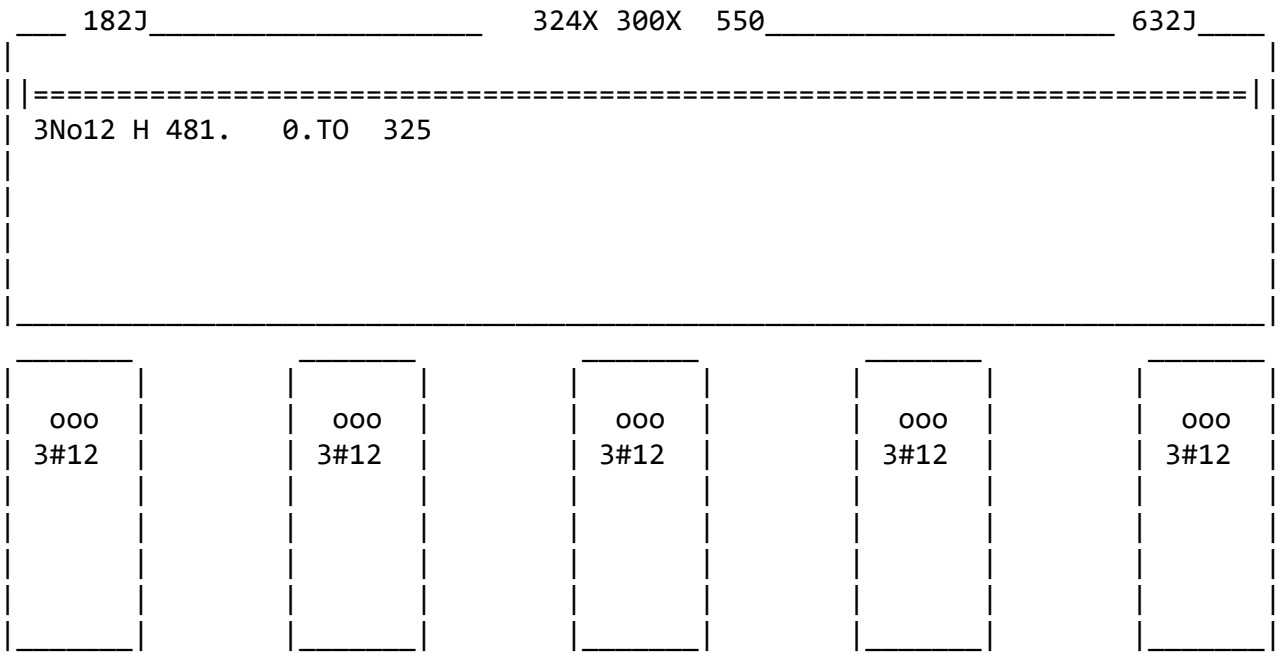
LEN - 325. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	3 - 12MM	0.	325.	YES	YES

B E A M N O. 2270 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2270 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2270 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2271 DESIGN RESULTS

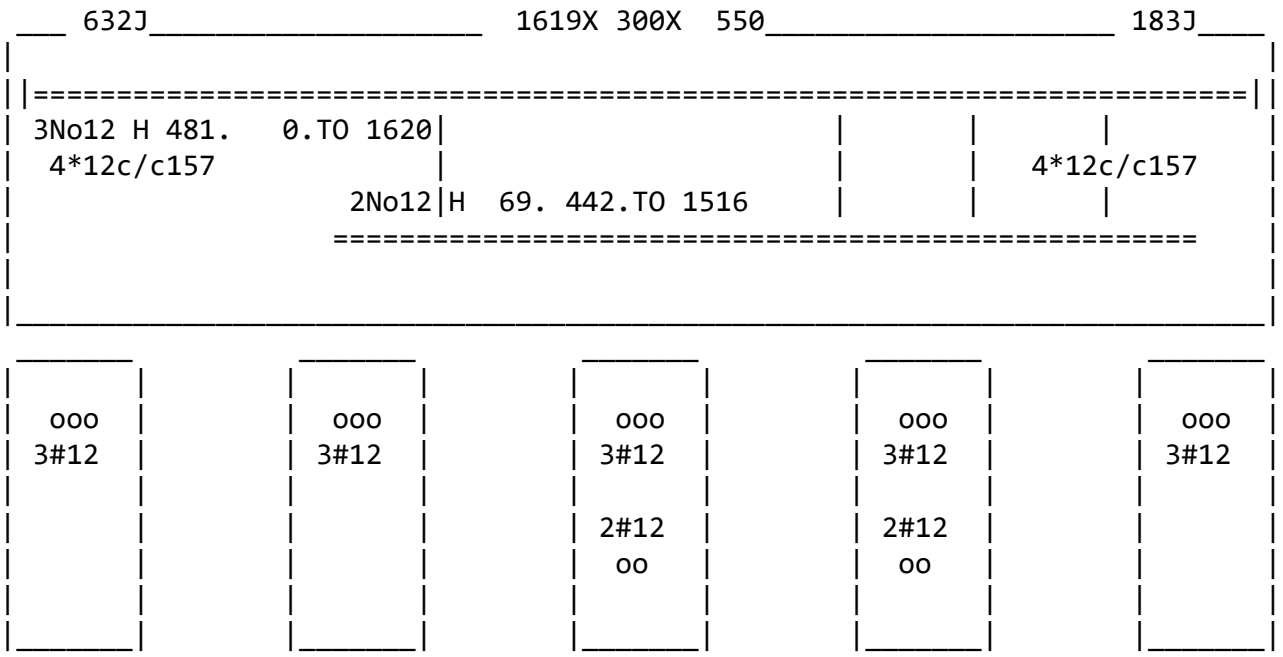
LEN - 1620. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	442.	1516.	NO	NO
2	481.	3 - 12MM	0.	1620.	YES	YES

B E A M N O. 2271 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.09 KNS Vc= 133.92 KNS Vs= 0.00 KNS
Tu= 19.69 KN-MET Tc= 5.6 KN-MET Ts= 26.3 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 335. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.59 SQ.CM.

AT END SUPPORT - Vu= 0.20 KNS Vc= 133.92 KNS Vs= 0.00 KNS
Tu= 19.69 KN-MET Tc= 5.6 KN-MET Ts= 26.3 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 335. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.59 SQ.CM.



ACI 318-11 BEAM NO. 2272 DESIGN RESULTS

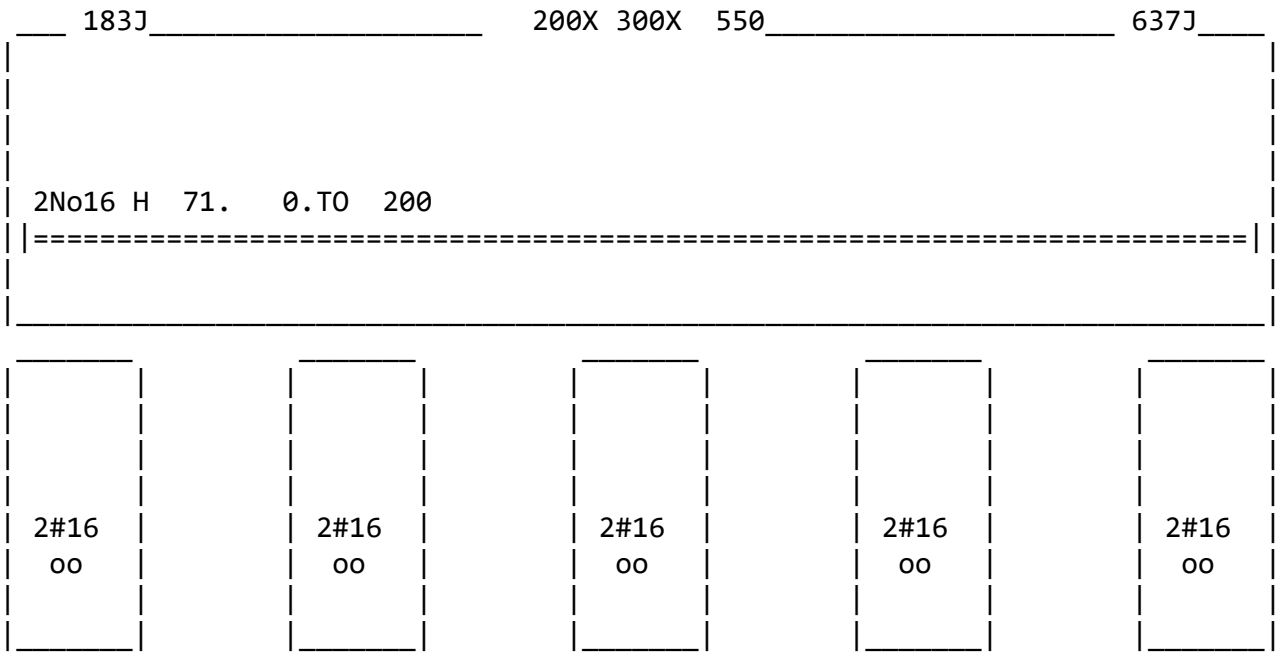
LEN - 200. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	200.	YES	YES

B E A M N O. 2272 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2272 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2272 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2273 DESIGN RESULTS

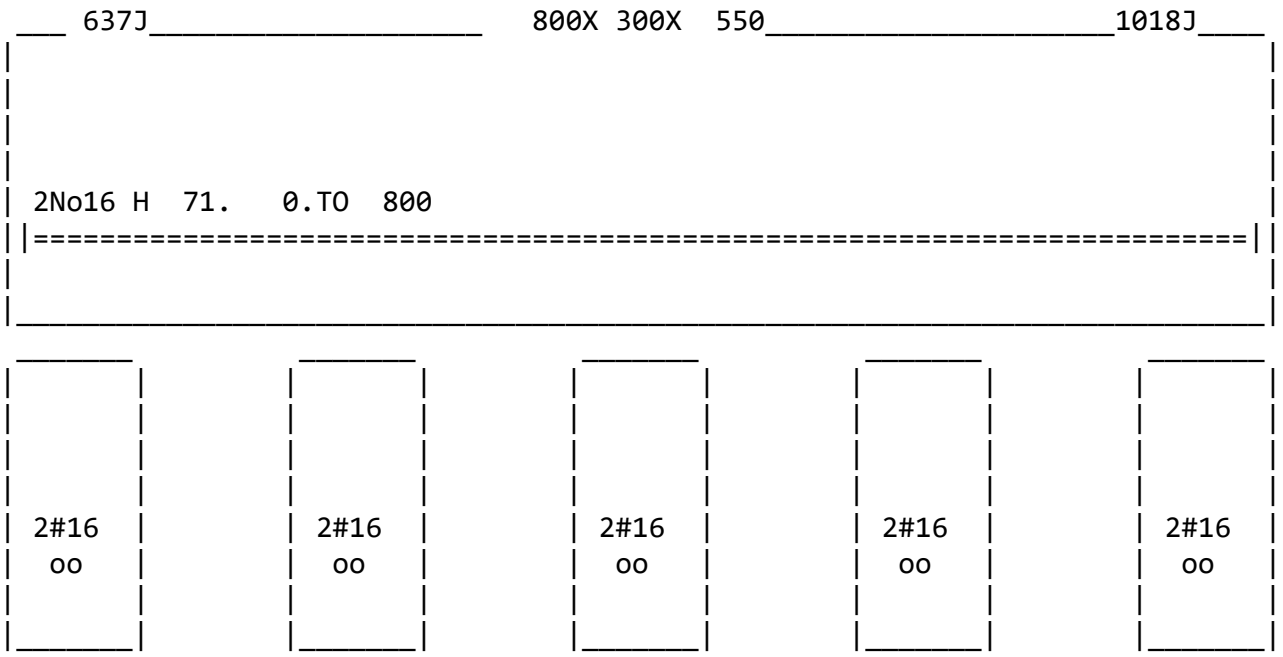
LEN - 800. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	800.	YES	YES

B E A M N O. 2273 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2273 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2273 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2274 DESIGN RESULTS

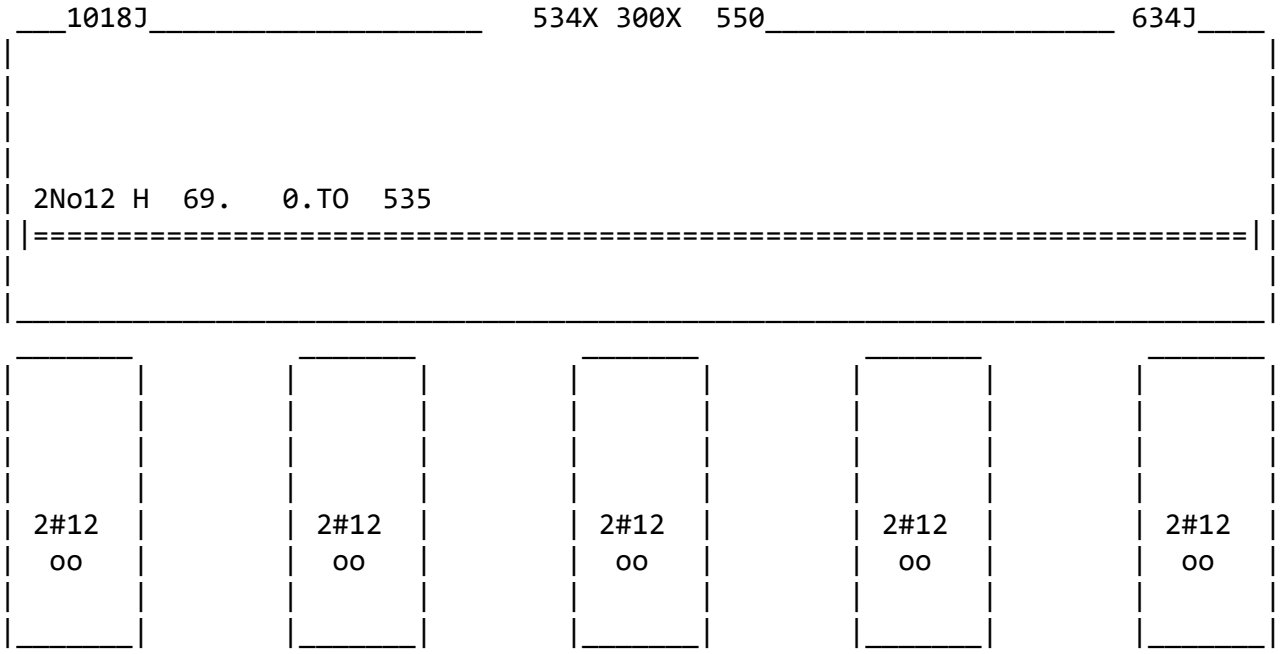
LEN - 535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	535.	YES	YES

B E A M N O. 2274 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2274 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2274 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2275 DESIGN RESULTS

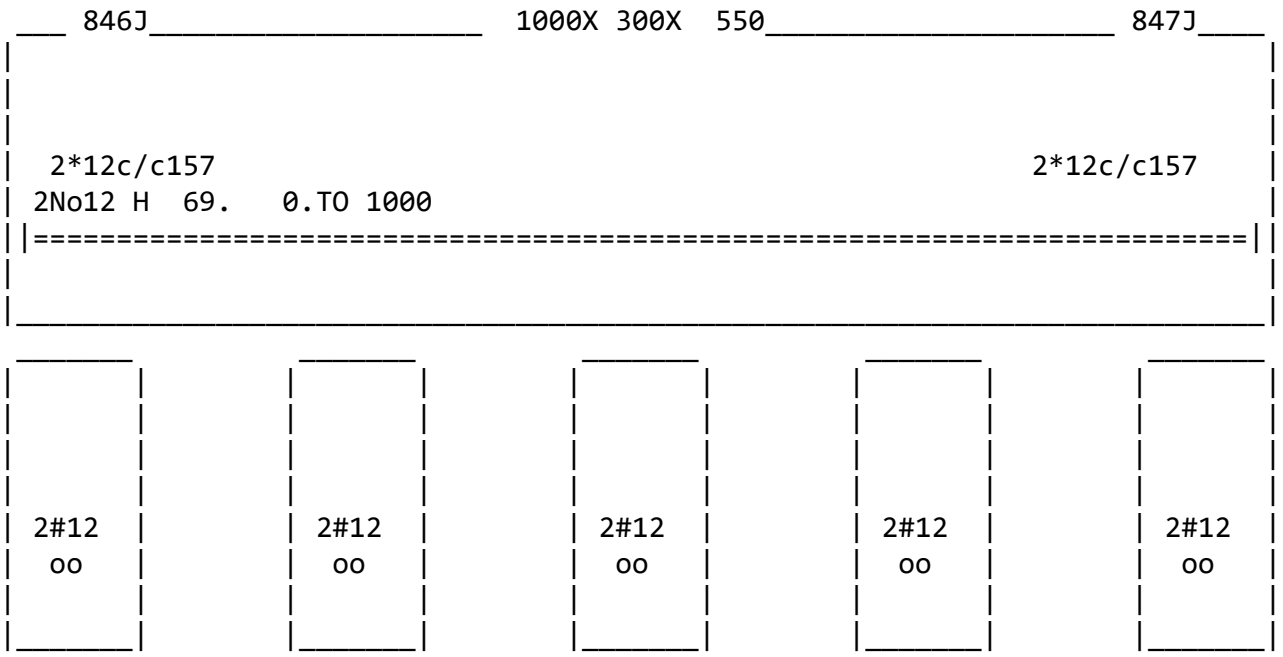
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1000.	YES	YES

B E A M N O. 2275 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.83 KNS Vc= 141.68 KNS Vs= 0.00 KNS
Tu= 13.22 KN-MET Tc= 5.9 KN-MET Ts= 17.6 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.75 SQ.CM.

AT END SUPPORT - Vu= 4.90 KNS Vc= 141.68 KNS Vs= 0.00 KNS
Tu= 13.22 KN-MET Tc= 5.9 KN-MET Ts= 17.6 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.75 SQ.CM.



ACI 318-11 BEAM NO. 2276 DESIGN RESULTS

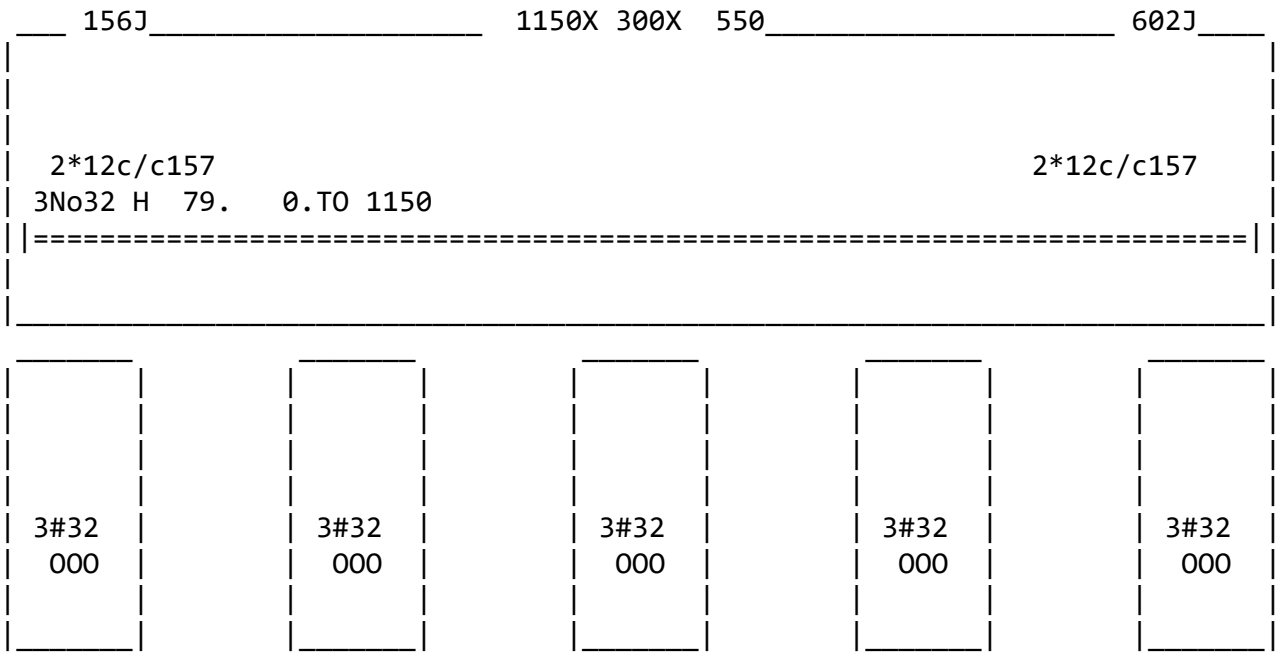
LEN - 1150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	1150.	YES	YES

B E A M N O. 2276 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 260.06 KNS Vc= 162.99 KNS Vs= 183.75 KNS
Tu= 7.98 KN-MET Tc= 5.9 KN-MET Ts= 10.6 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 100. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.27 SQ.CM.

AT END SUPPORT - Vu= 254.21 KNS Vc= 153.47 KNS Vs= 185.49 KNS
Tu= 7.98 KN-MET Tc= 5.9 KN-MET Ts= 10.6 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 100. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.27 SQ.CM.



ACI 318-11 BEAM NO. 2277 DESIGN RESULTS

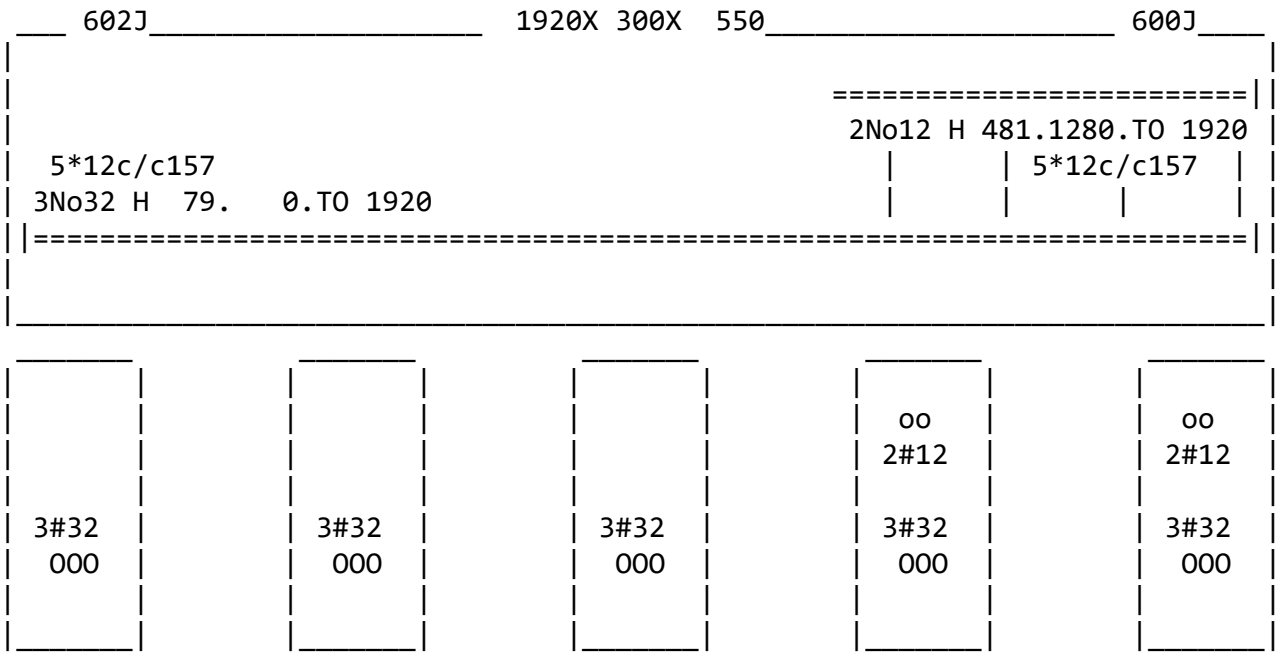
LEN - 1920. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	1920.	YES	YES
2	481.	2 - 12MM	1280.	1920.	NO	YES

B E A M N O. 2277 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 147.39 KNS Vc= 126.92 KNS Vs= 69.61 KNS
Tu= 7.98 KN-MET Tc= 5.4 KN-MET Ts= 10.6 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 485. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.27 SQ.CM.

AT END SUPPORT - Vu= 186.30 KNS Vc= 126.92 KNS Vs= 121.48 KNS
Tu= 7.98 KN-MET Tc= 5.4 KN-MET Ts= 10.6 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 485. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.27 SQ.CM.



ACI 318-11 BEAM NO. 2278 DESIGN RESULTS

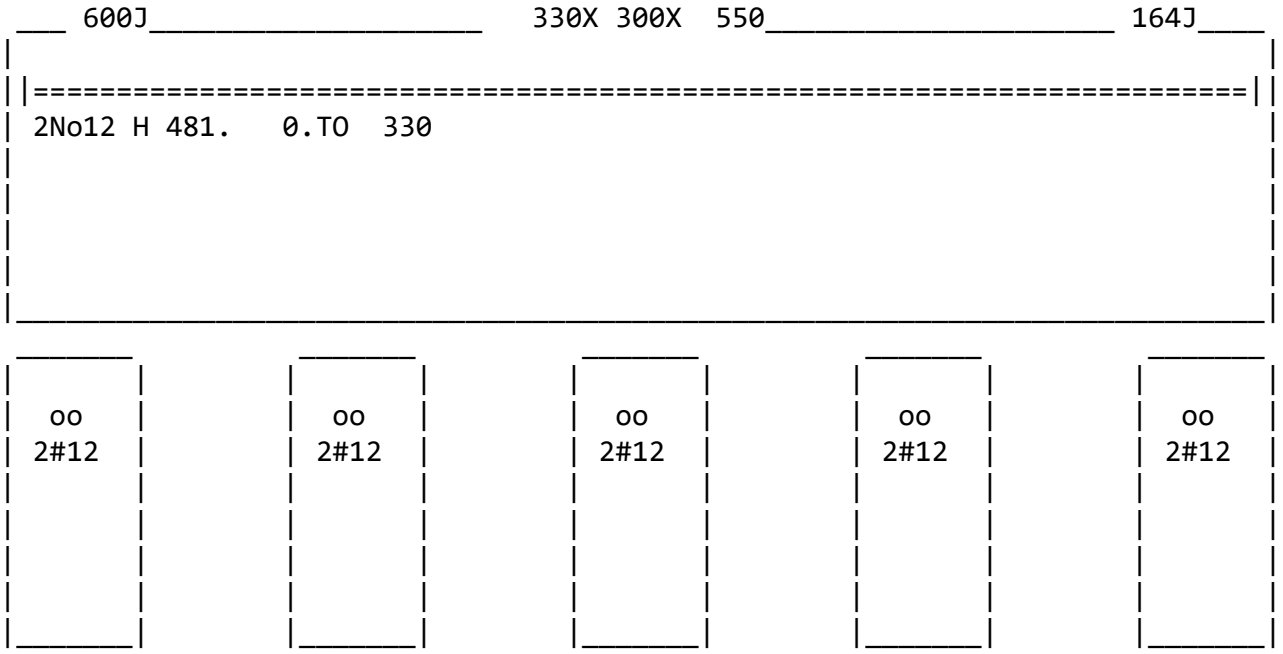
LEN - 330. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	2 - 12MM	0.	330.	YES	YES

B E A M N O. 2278 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2278 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2278 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2279 DESIGN RESULTS

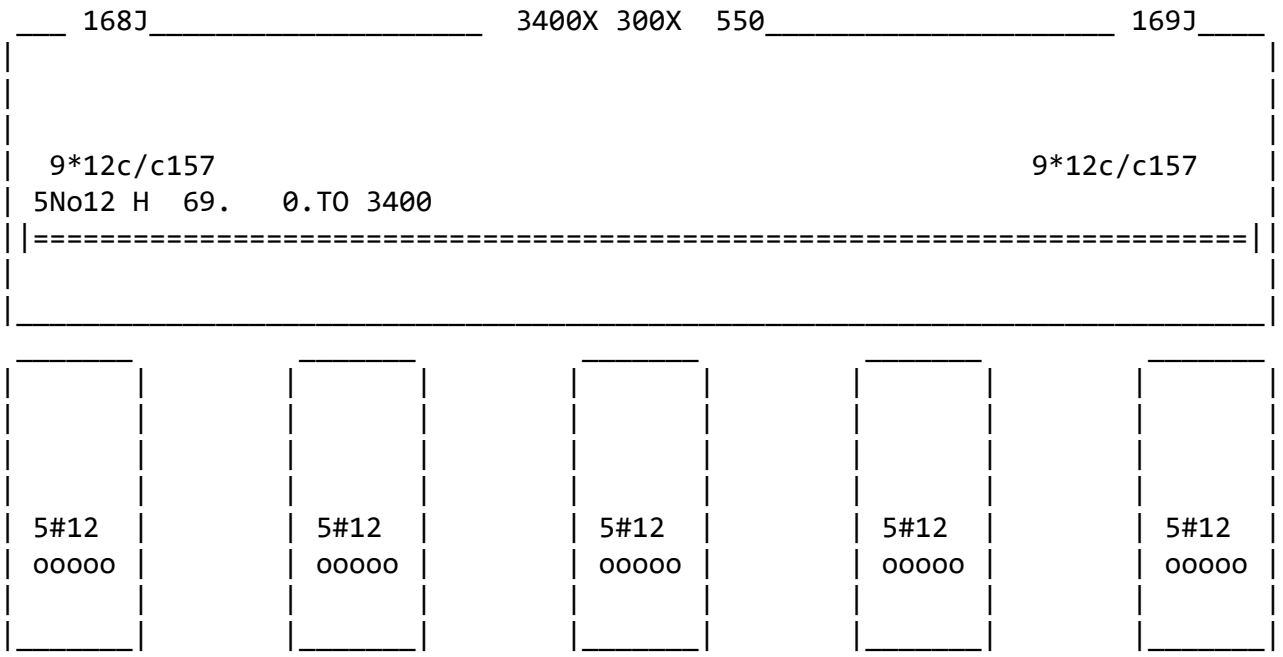
LEN - 3400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3400.	YES	YES

B E A M N O. 2279 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 101.73 KNS Vc= 106.42 KNS Vs= 29.22 KNS
Tu= 11.35 KN-MET Tc= 4.4 KN-MET Ts= 15.1 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1225. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.22 SQ.CM.

AT END SUPPORT - Vu= 89.38 KNS Vc= 106.42 KNS Vs= 12.75 KNS
Tu= 11.35 KN-MET Tc= 4.4 KN-MET Ts= 15.1 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1225. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.22 SQ.CM.



ACI 318-11 BEAM NO. 2280 DESIGN RESULTS

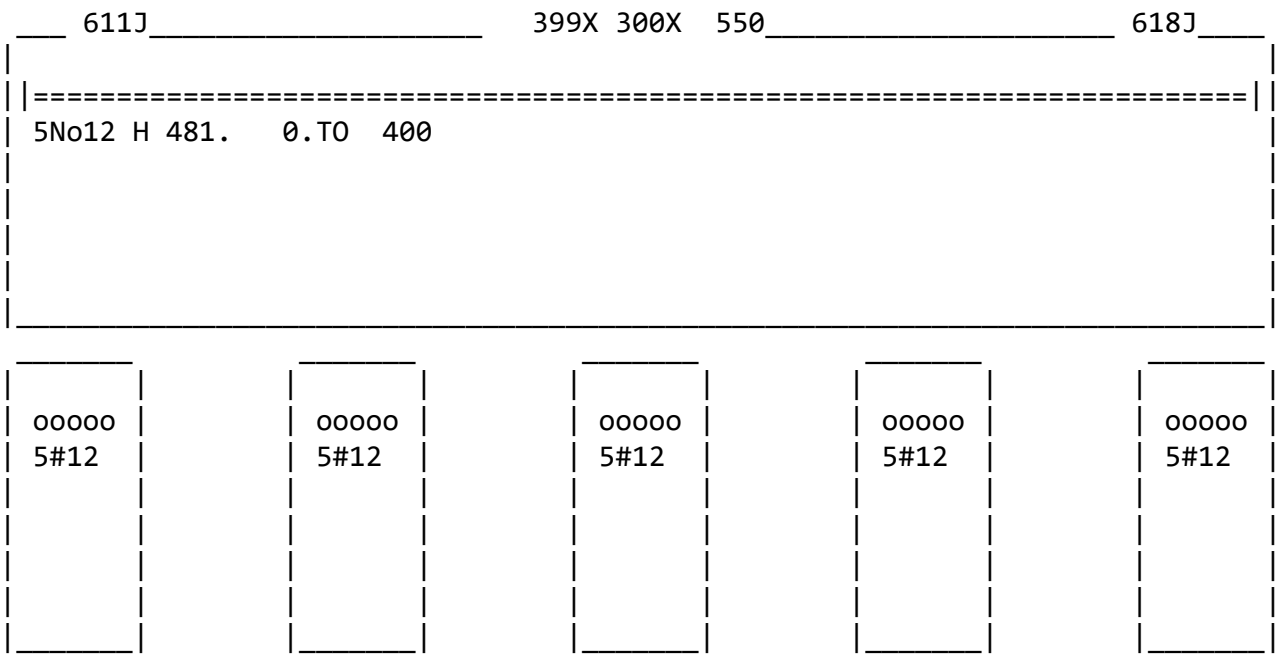
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	5 - 12MM	0.	400.	YES	YES

B E A M N O. 2280 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2280 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2280 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2281 DESIGN RESULTS

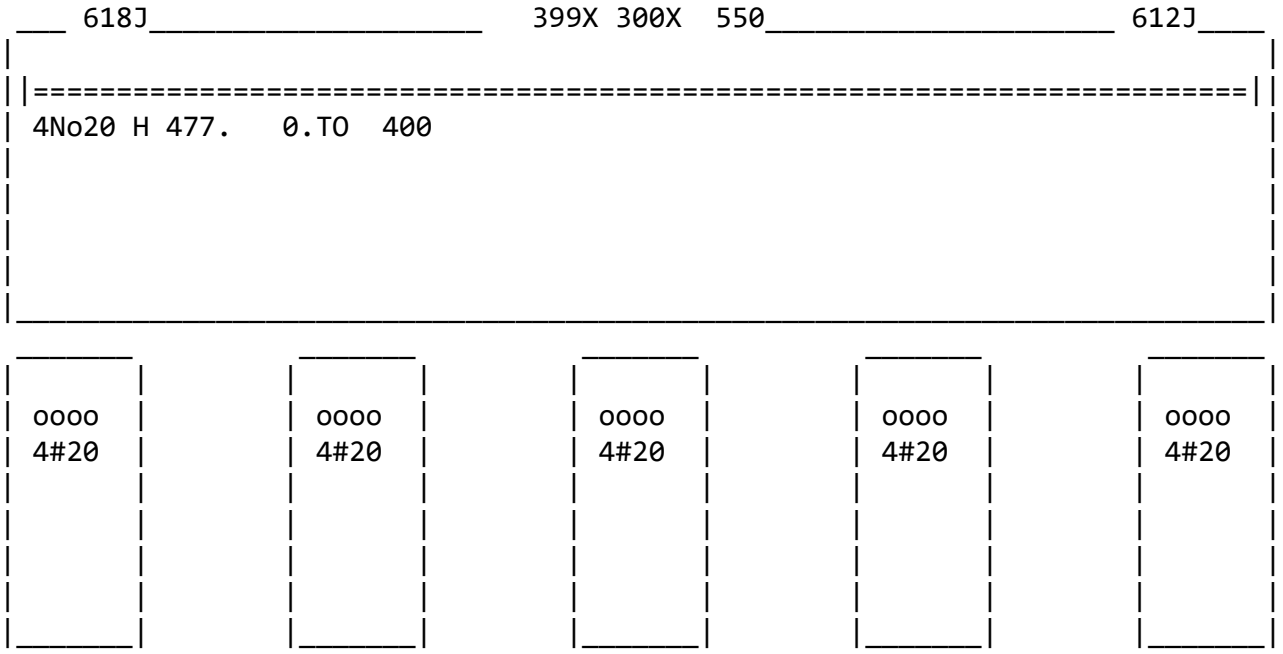
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	477.	4 - 20MM	0.	400.	YES	YES

B E A M N O. 2281 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2281 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2281 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2282 DESIGN RESULTS

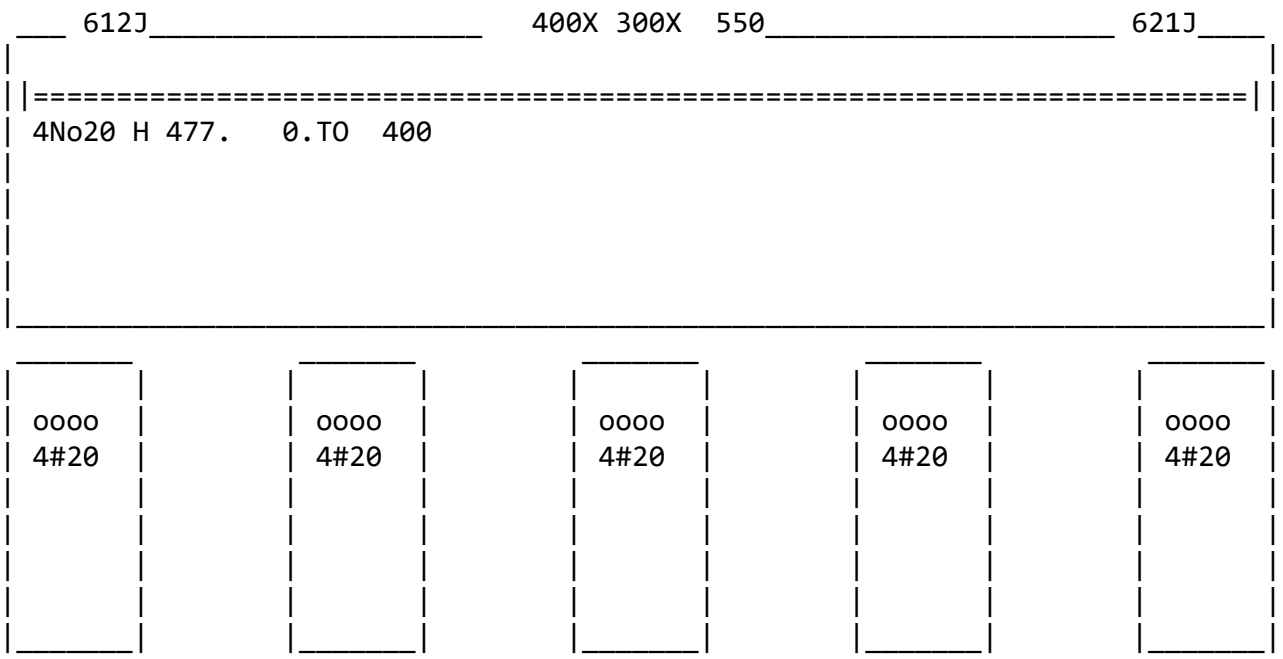
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	477.	4 - 20MM	0.	400.	YES	YES

B E A M N O. 2282 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2282 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2282 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2283 DESIGN RESULTS

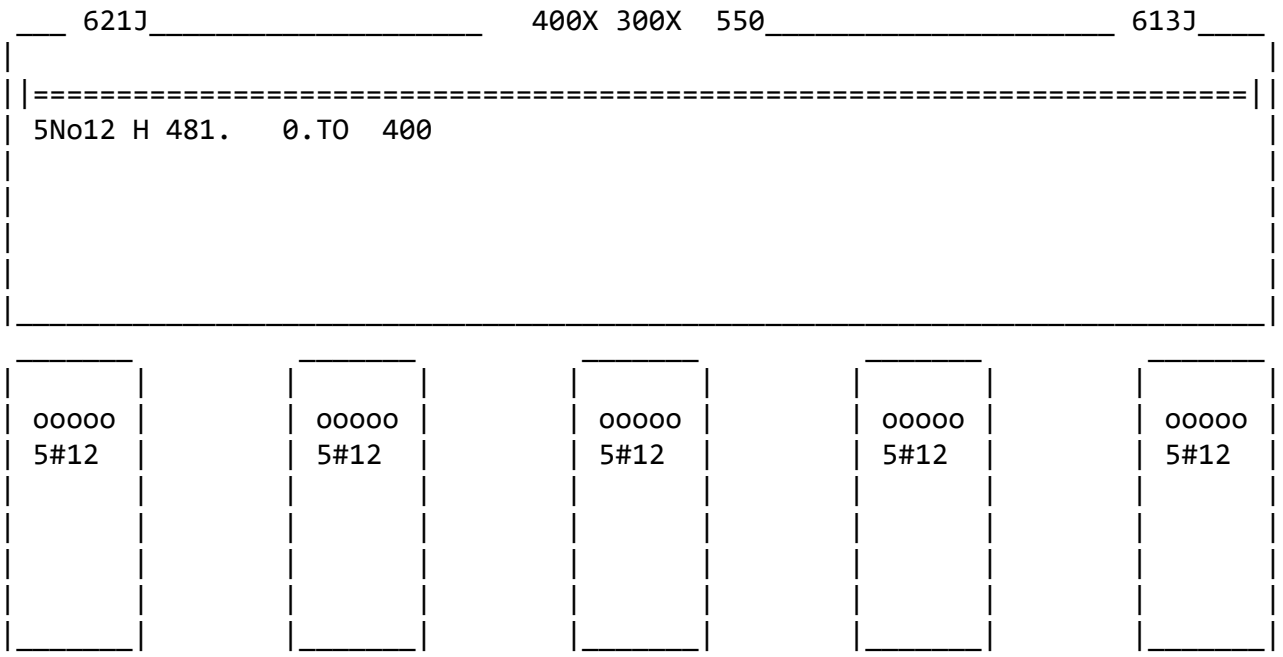
LEN - 400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	5 - 12MM	0.	400.	YES	YES

B E A M N O. 2283 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2283 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2283 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2284 DESIGN RESULTS

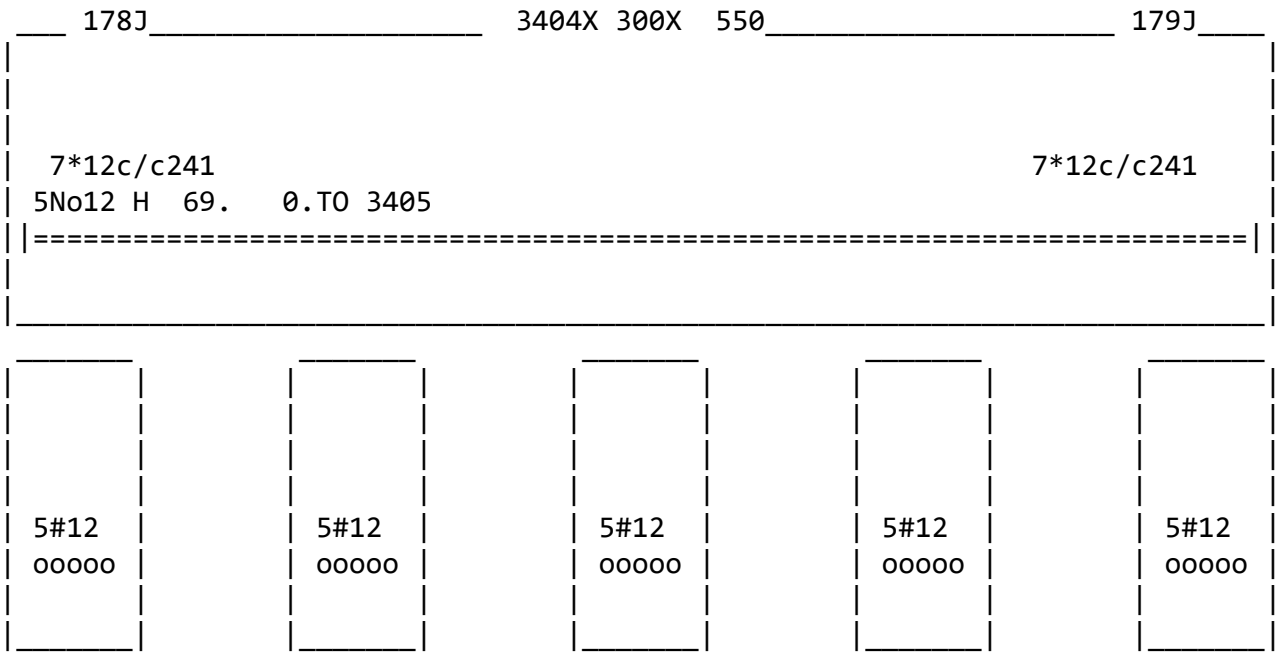
LEN - 3405. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3405.	YES	YES

B E A M N O. 2284 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 90.70 KNS Vc= 114.16 KNS Vs= 6.77 KNS
Tu= 0.57 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 1228. MM

AT END SUPPORT - Vu= 78.16 KNS Vc= 114.16 KNS Vs= 0.00 KNS
Tu= 0.57 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 1228. MM



ACI 318-11 BEAM NO. 2285 DESIGN RESULTS

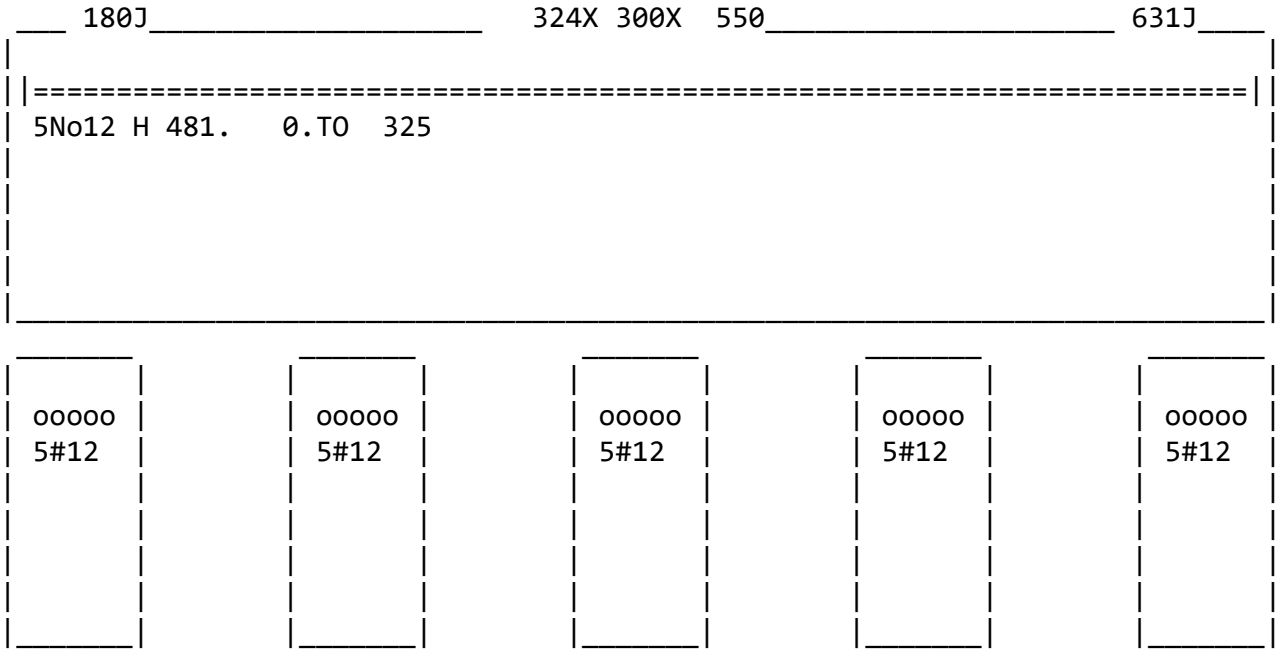
LEN - 325. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	5 - 12MM	0.	325.	YES	YES

B E A M N O. 2285 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2285 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2285 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2286 DESIGN RESULTS

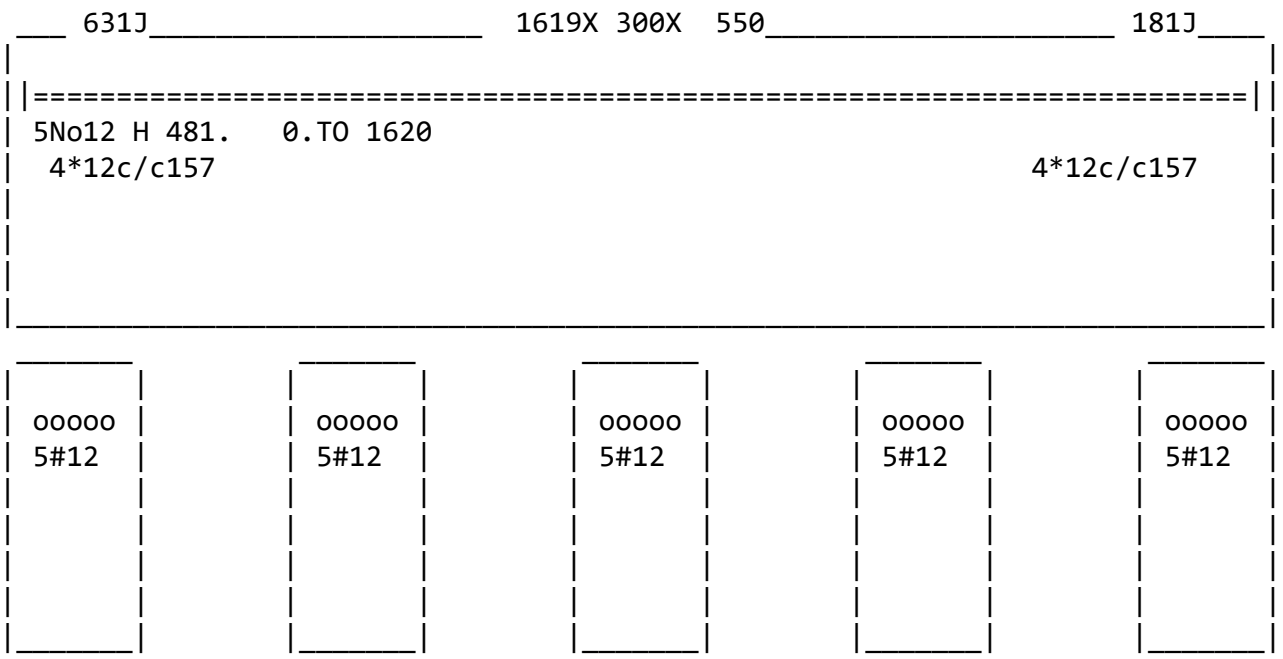
LEN - 1620. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	481.	5 - 12MM	0.	1620.	YES	YES

B E A M N O. 2286 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 36.06 KNS Vc= 132.16 KNS Vs= 0.00 KNS
Tu= 8.56 KN-MET Tc= 5.5 KN-MET Ts= 11.4 KN-MET LOAD 412
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 335. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.

AT END SUPPORT - Vu= 25.11 KNS Vc= 132.16 KNS Vs= 0.00 KNS
Tu= 8.56 KN-MET Tc= 5.5 KN-MET Ts= 11.4 KN-MET LOAD 412
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 335. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.43 SQ.CM.



ACI 318-11 BEAM NO. 2287 DESIGN RESULTS

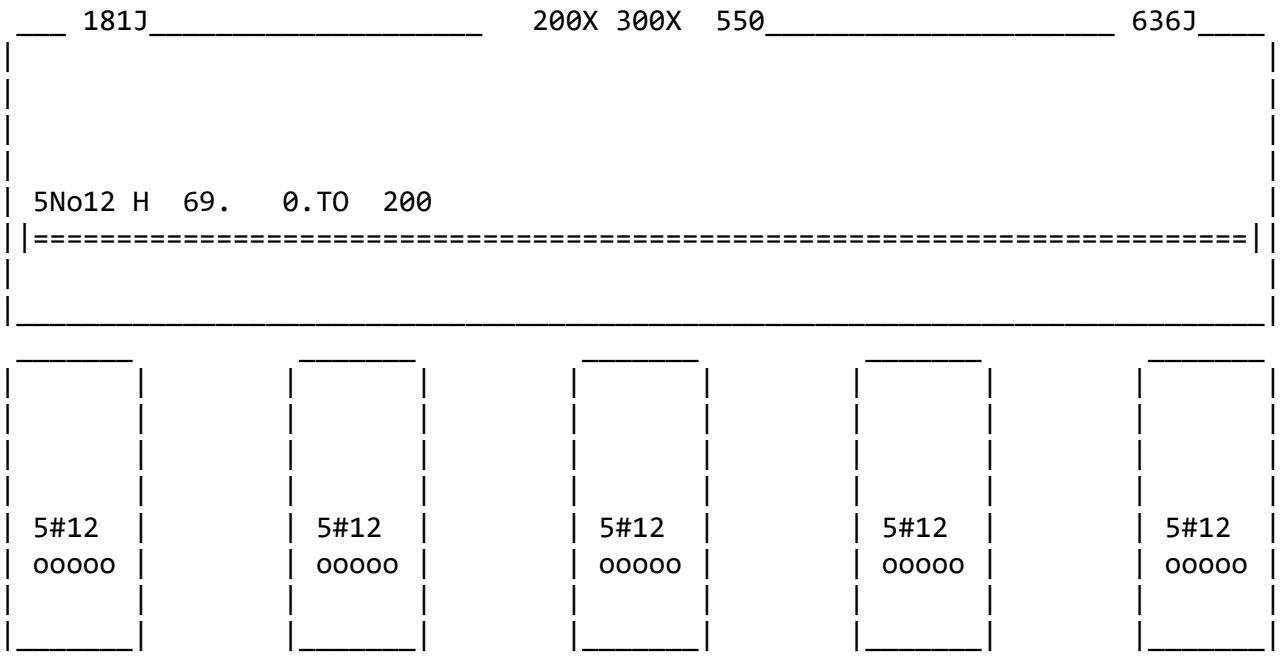
LEN - 200. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	200.	YES	YES

B E A M N O. 2287 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2287 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2287 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2288 DESIGN RESULTS

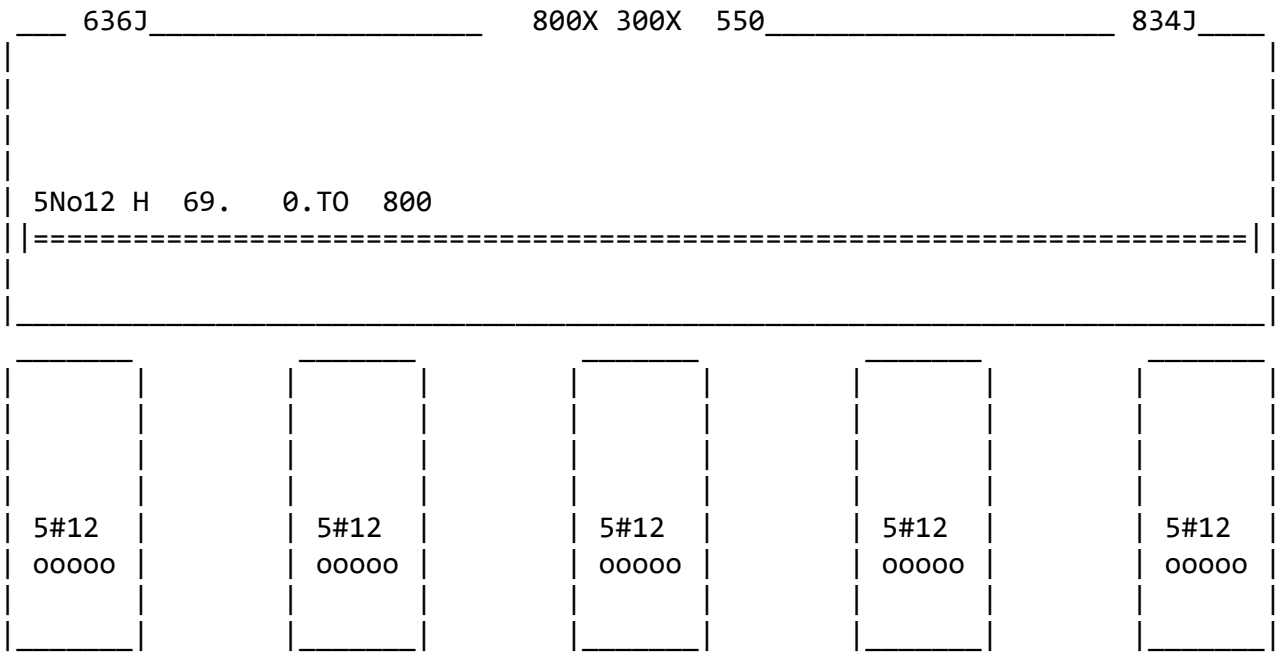
LEN - 800. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	800.	YES	YES

B E A M N O. 2288 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2288 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2288 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2289 DESIGN RESULTS

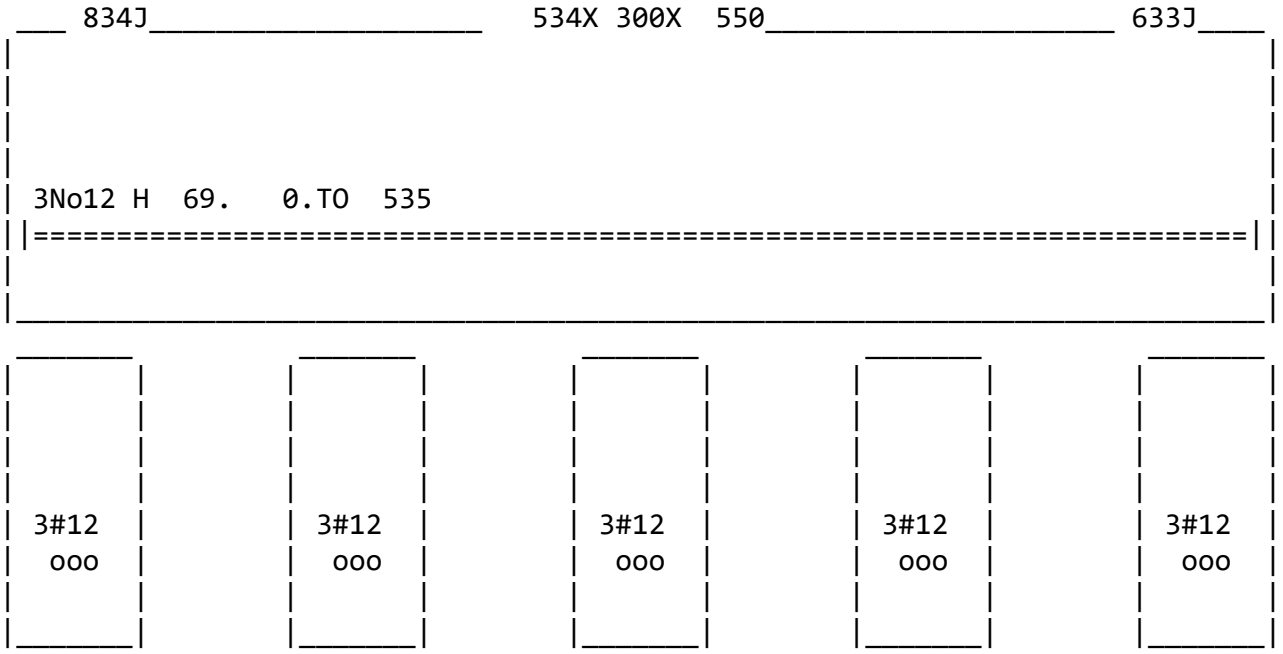
LEN - 535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	535.	YES	YES

B E A M N O. 2289 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2289 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2289 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2290 DESIGN RESULTS

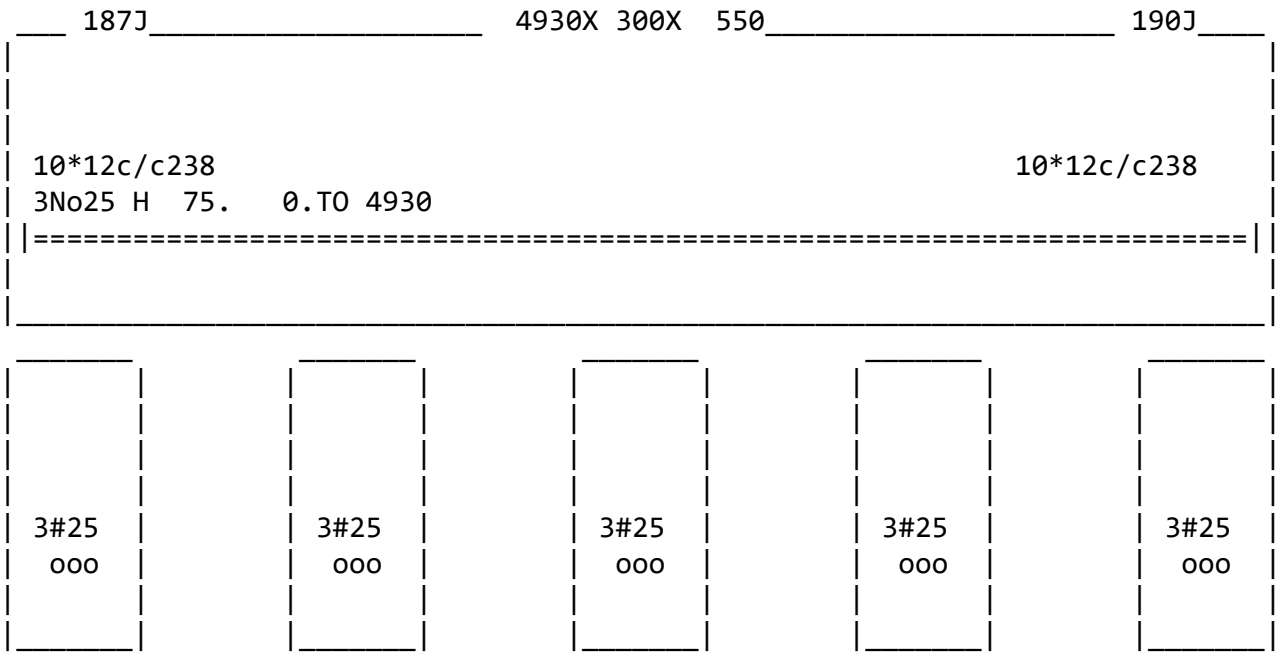
LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	4930.	YES	YES

B E A M N O. 2290 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.44 KNS Vc= 155.84 KNS Vs= 0.00 KNS
Tu= 0.71 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 1990. MM

AT END SUPPORT - Vu= 67.39 KNS Vc= 155.84 KNS Vs= 0.00 KNS
Tu= 0.71 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 1990. MM



ACI 318-11 BEAM NO. 2291 DESIGN RESULTS

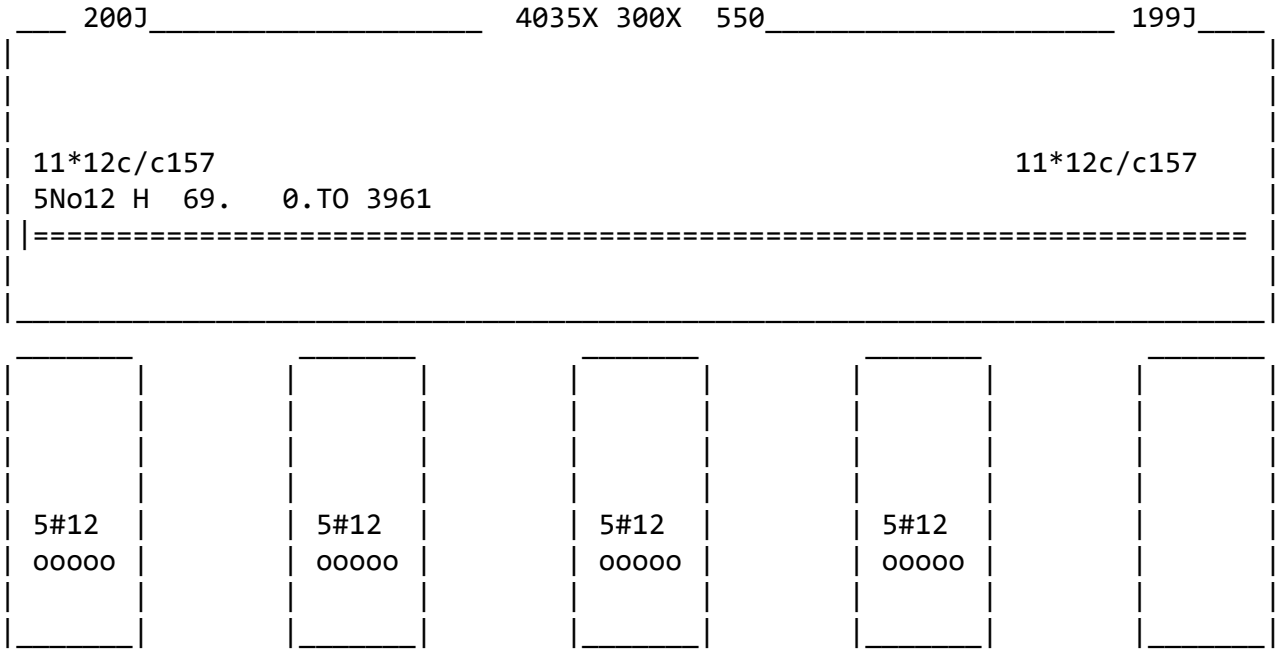
LEN - 4035. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	3961.	YES	NO

B E A M N O. 2291 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 47.63 KNS Vc= 143.16 KNS Vs= 0.00 KNS
Tu= 10.01 KN-MET Tc= 5.9 KN-MET Ts= 13.3 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1543. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.84 SQ.CM.

AT END SUPPORT - Vu= 47.63 KNS Vc= 143.16 KNS Vs= 0.00 KNS
Tu= 10.01 KN-MET Tc= 5.9 KN-MET Ts= 13.3 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1543. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.84 SQ.CM.



ACI 318-11 BEAM NO. 2292 DESIGN RESULTS

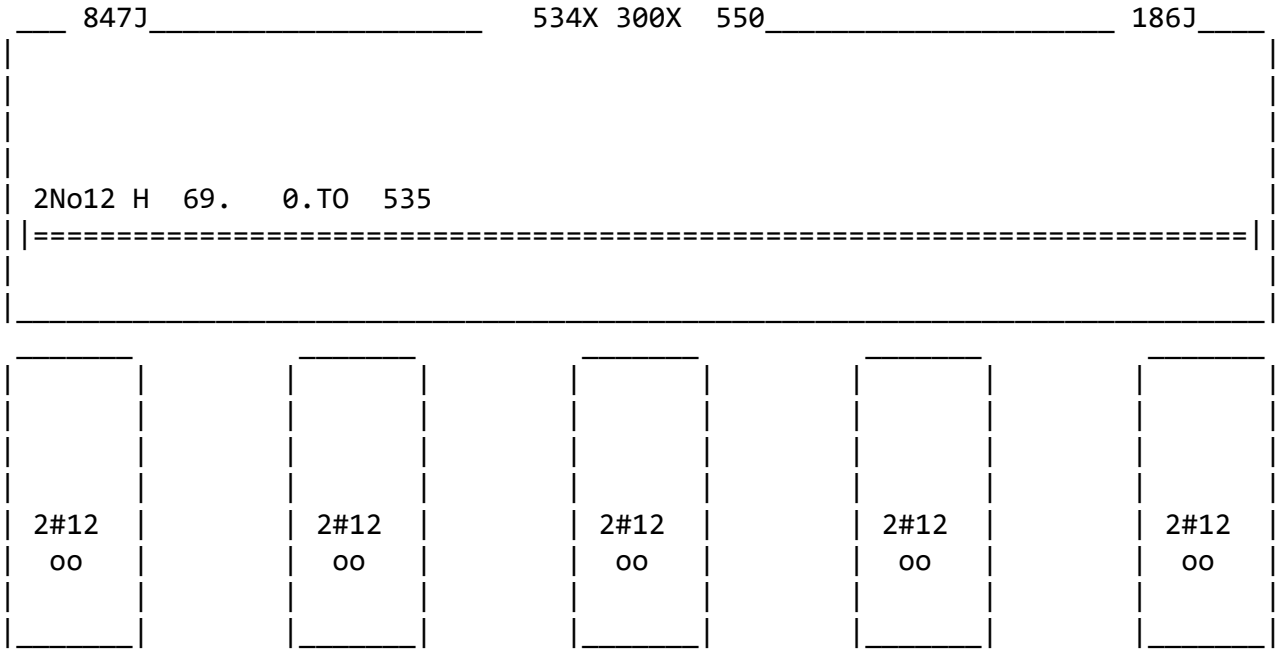
LEN - 535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	535.	YES	YES

B E A M N O. 2292 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2292 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2292 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2293 DESIGN RESULTS

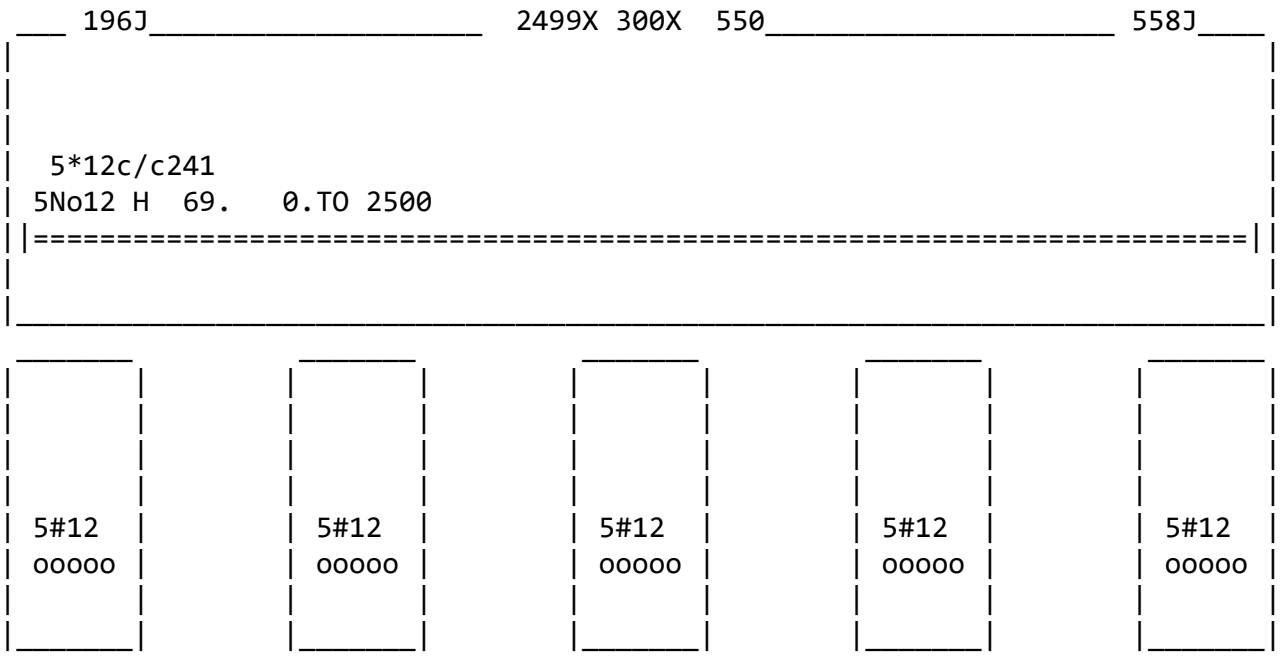
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	2500.	YES	YES

B E A M N O. 2293 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 58.43 KNS Vc= 143.24 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 775. MM

AT END SUPPORT - Vu= 0.79 KNS Vc= 135.05 KNS Vs= 0.00 KNS
Tu= 0.51 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



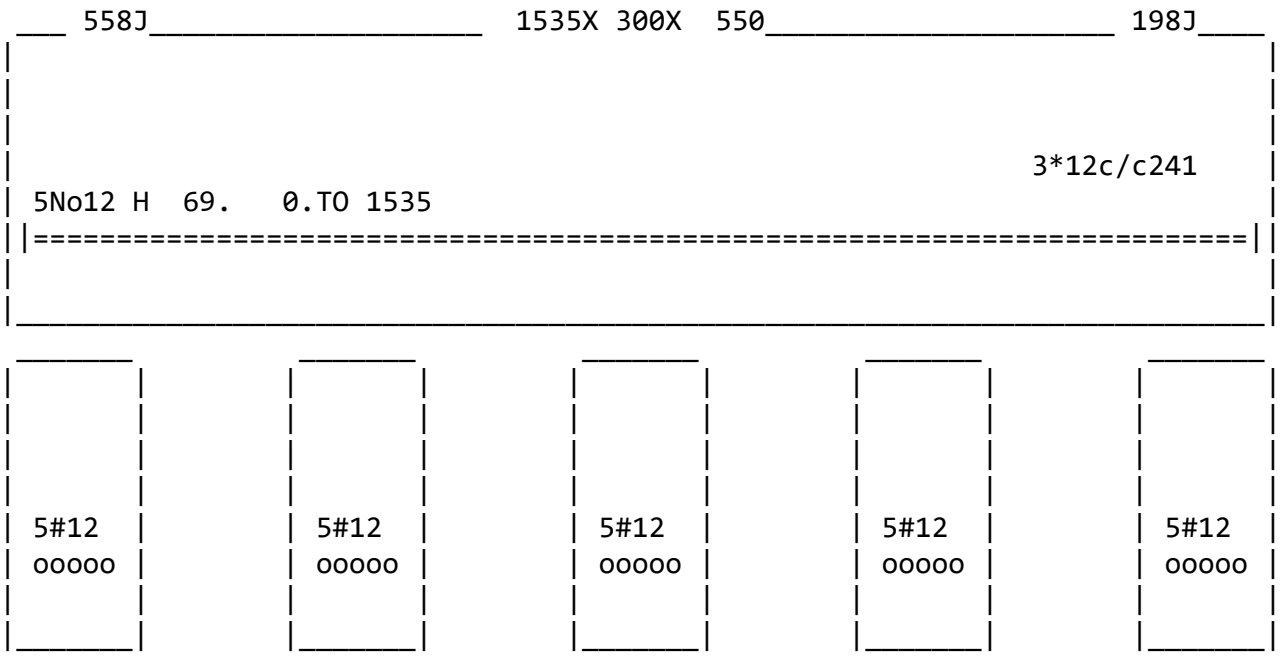
ACI 318-11 BEAM NO. 2294 DESIGN RESULTS

LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1535.	YES	YES

B E A M N O. 2294 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 21.48 KNS Vc= 137.97 KNS Vs= 0.00 KNS
Tu= 0.51 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 57.77 KNS Vc= 143.29 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 293. MM



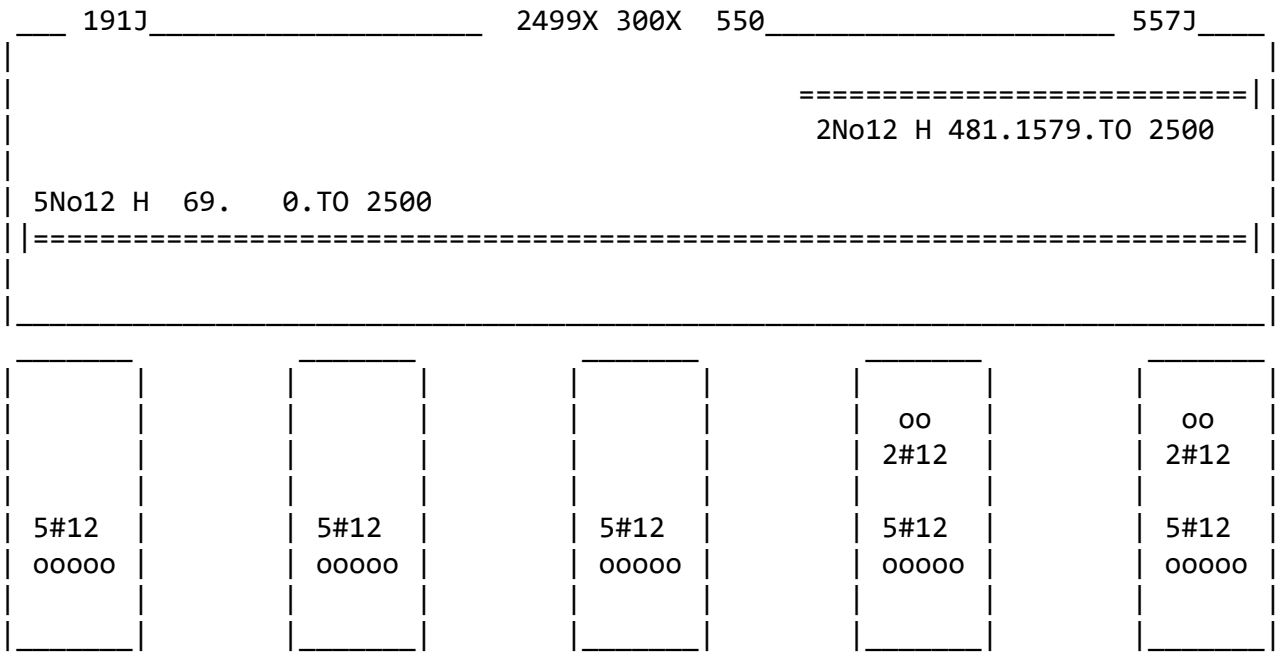
ACI 318-11 BEAM NO. 2295 DESIGN RESULTS

LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	2500.	YES	YES
2	481.	2 - 12MM	1579.	2500.	NO	YES

B E A M N O. 2295 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.56 KNS Vc= 142.05 KNS Vs= 0.00 KNS
Tu= 3.53 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 4.87 KNS Vc= 142.05 KNS Vs= 0.00 KNS
Tu= 3.53 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



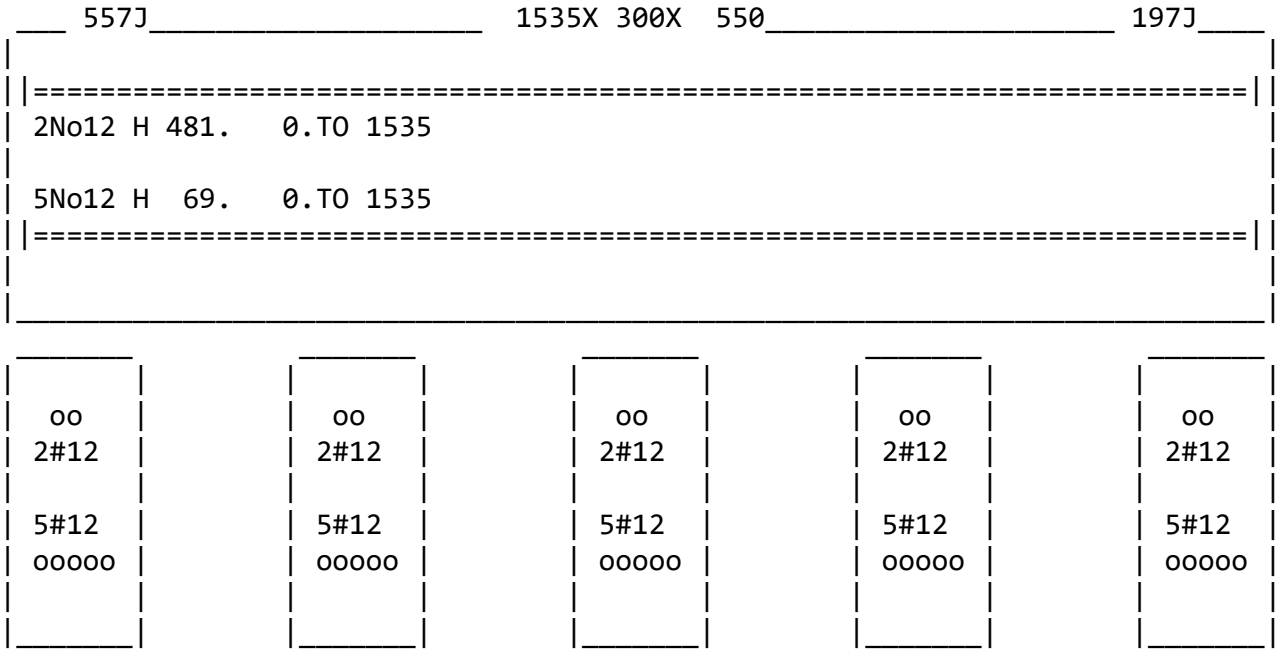
ACI 318-11 BEAM NO. 2296 DESIGN RESULTS

LEN - 1535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1535.	YES	YES
2	481.	2 - 12MM	0.	1535.	YES	YES

B E A M N O. 2296 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.42 KNS Vc= 142.04 KNS Vs= 0.00 KNS
Tu= 3.53 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 19.02 KNS Vc= 142.04 KNS Vs= 0.00 KNS
Tu= 3.53 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2297 DESIGN RESULTS

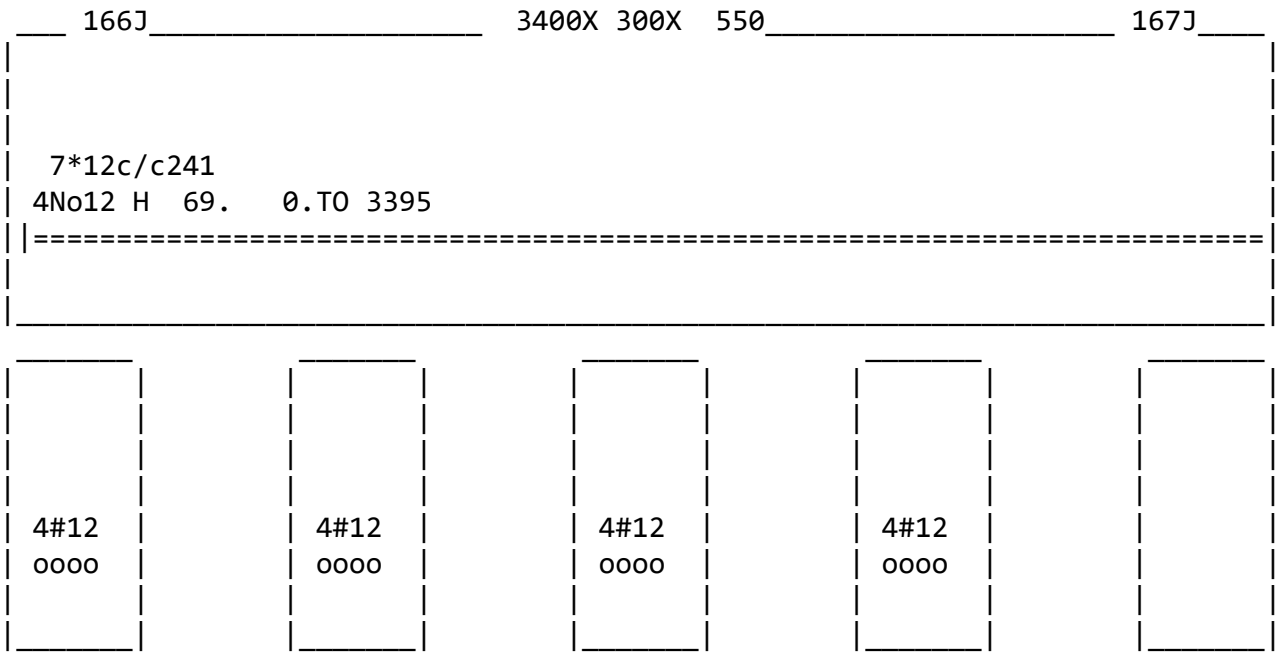
LEN - 3400. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	3395.	YES	NO

B E A M N O. 2297 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 58.10 KNS Vc= 148.55 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 1225. MM

AT END SUPPORT - Vu= 21.77 KNS Vc= 148.18 KNS Vs= 0.00 KNS
Tu= 1.24 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2298 DESIGN RESULTS

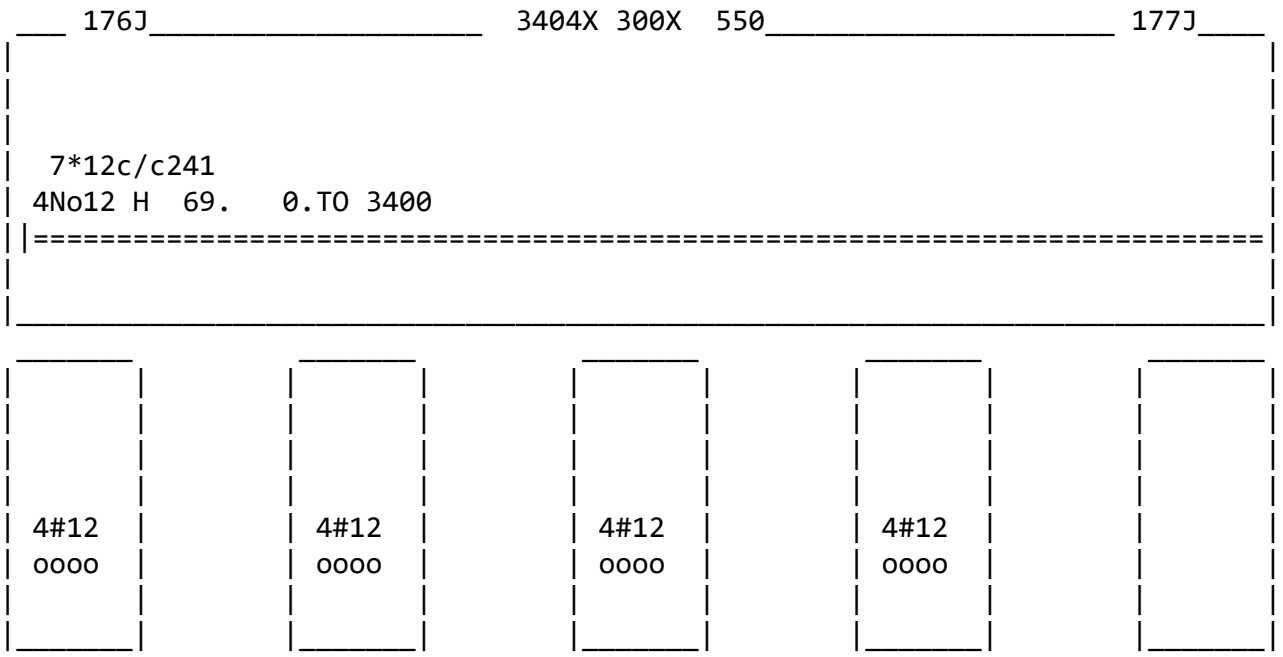
LEN - 3405. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	0.	3400.	YES	NO

B E A M N O. 2298 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 54.79 KNS Vc= 143.50 KNS Vs= 0.00 KNS
Tu= 0.35 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 241. MM C/C FOR 1228. MM

AT END SUPPORT - Vu= 21.77 KNS Vc= 142.94 KNS Vs= 0.00 KNS
Tu= 0.36 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2299 DESIGN RESULTS

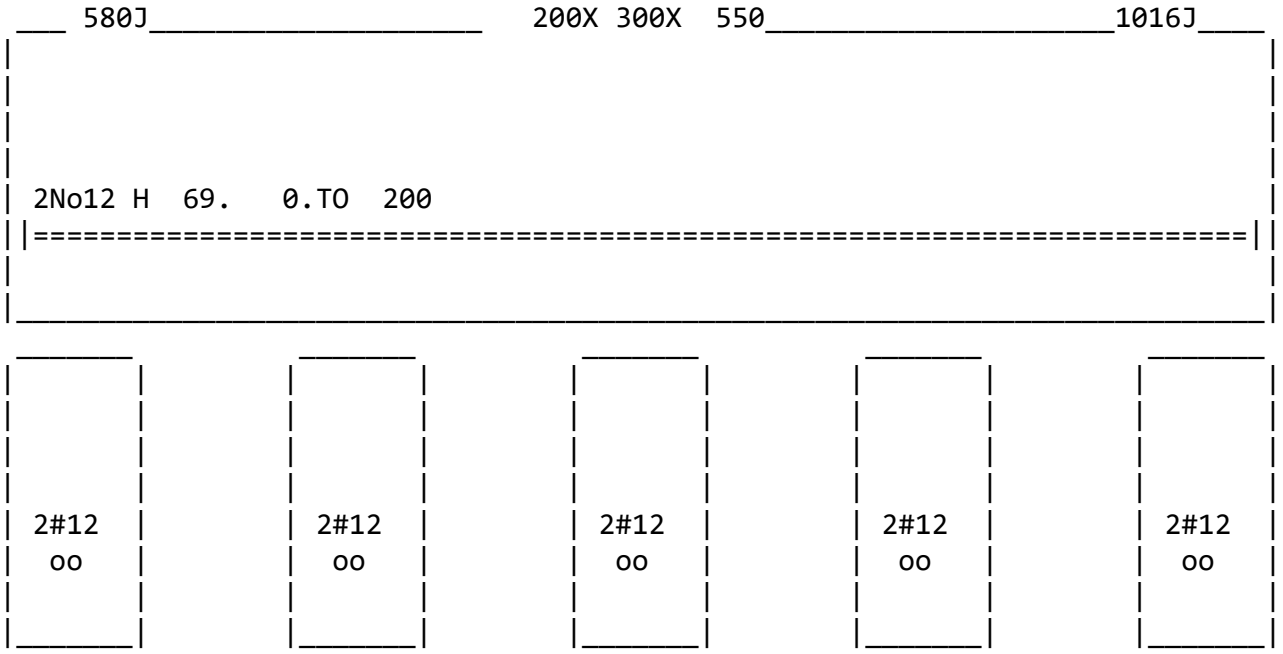
LEN - 200. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	200.	YES	YES

B E A M N O. 2299 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2299 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2299 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2300 DESIGN RESULTS

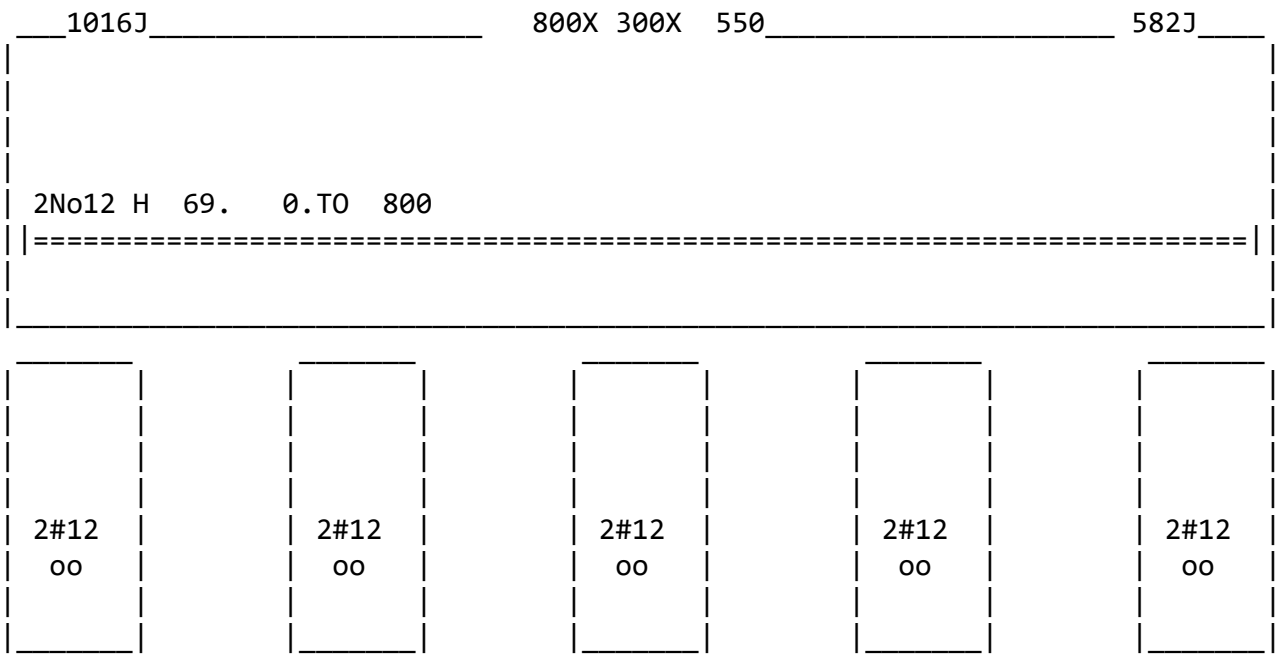
LEN - 800. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	800.	YES	YES

B E A M N O. 2300 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2300 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2300 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2301 DESIGN RESULTS

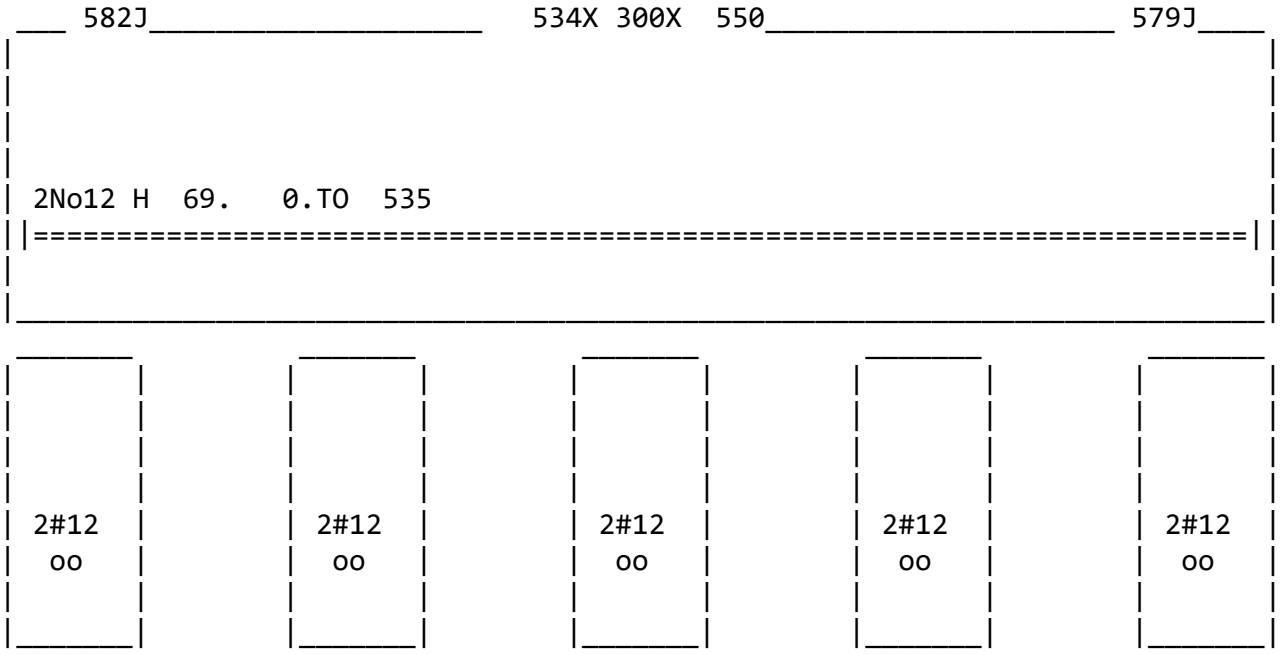
LEN - 535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	535.	YES	YES

B E A M N O. 2301 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2301 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2301 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



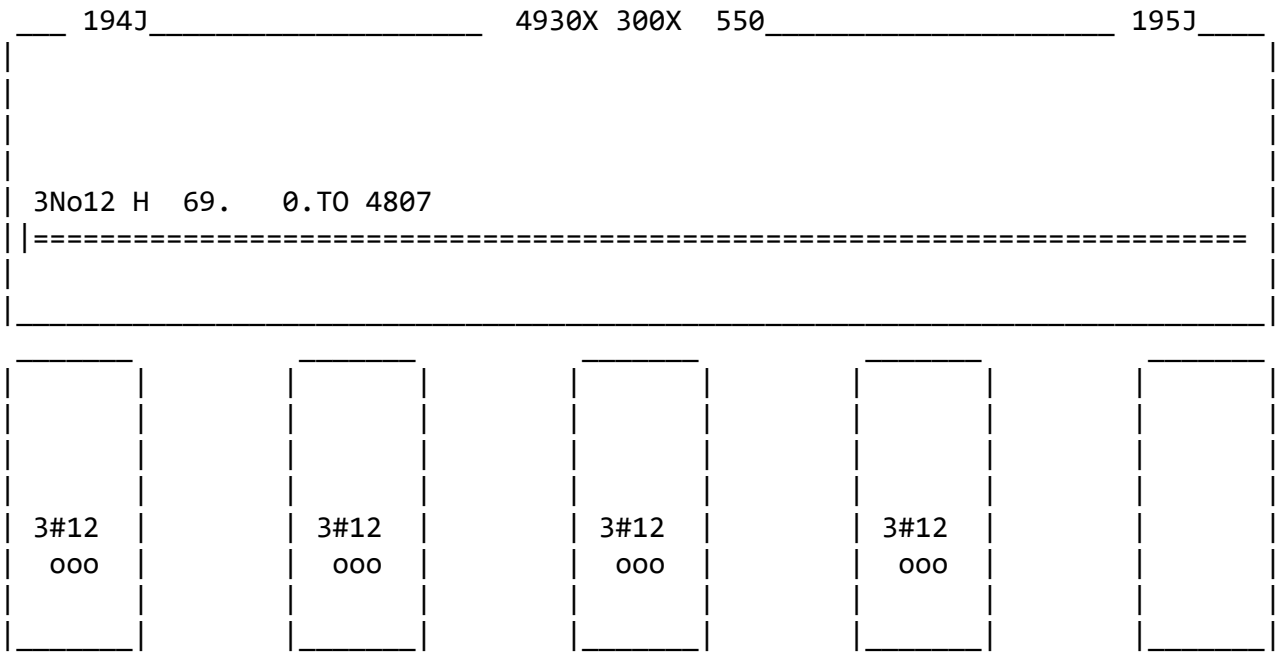
ACI 318-11 BEAM NO. 2302 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	4807.	YES	NO

B E A M N O. 2302 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.98 KNS Vc= 141.67 KNS Vs= 0.00 KNS
Tu= 2.00 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.98 KNS Vc= 141.67 KNS Vs= 0.00 KNS
Tu= 2.00 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2303 DESIGN RESULTS

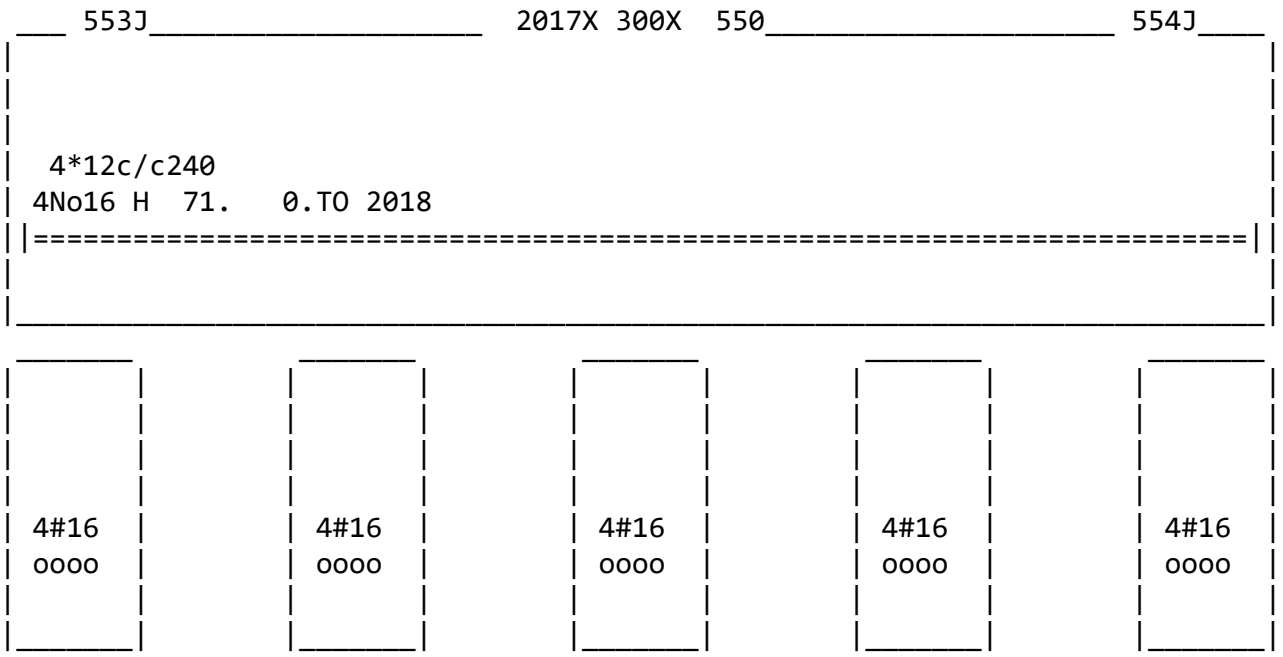
LEN - 2018. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	2018.	YES	YES

B E A M N O. 2303 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 66.31 KNS Vc= 145.98 KNS Vs= 0.00 KNS
Tu= 1.24 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 534. MM

AT END SUPPORT - Vu= 23.18 KNS Vc= 136.93 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2304 DESIGN RESULTS

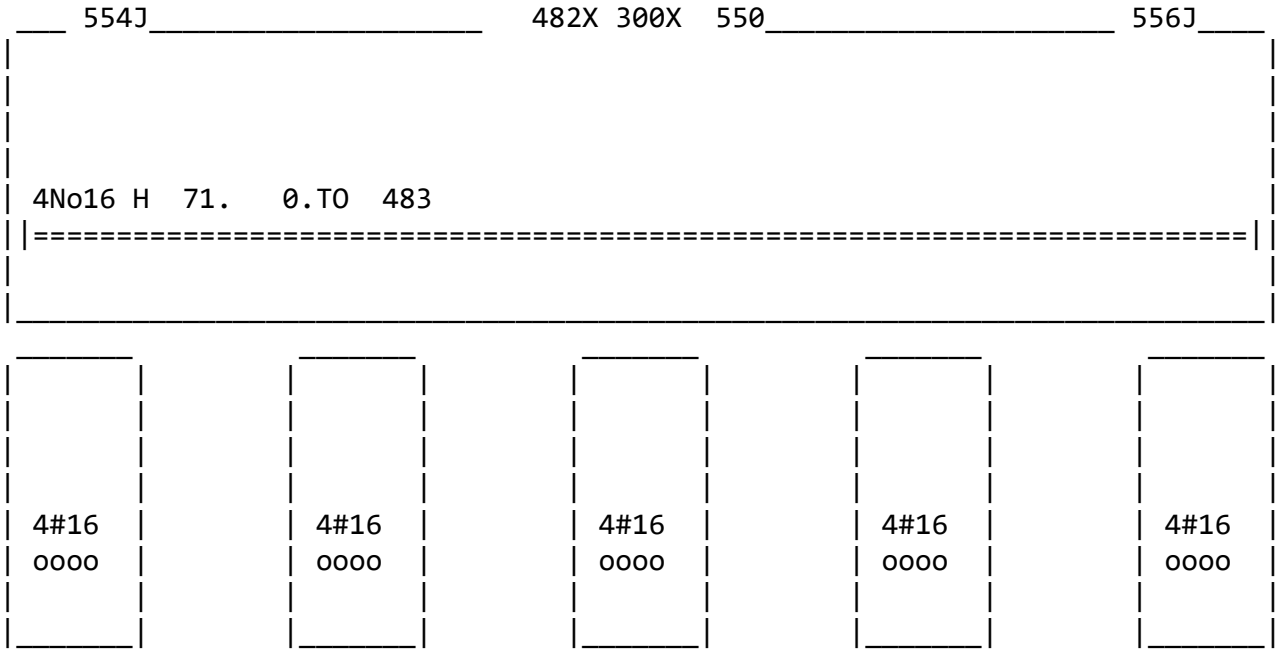
LEN - 483. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	483.	YES	YES

B E A M N O. 2304 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2304 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2304 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2305 DESIGN RESULTS

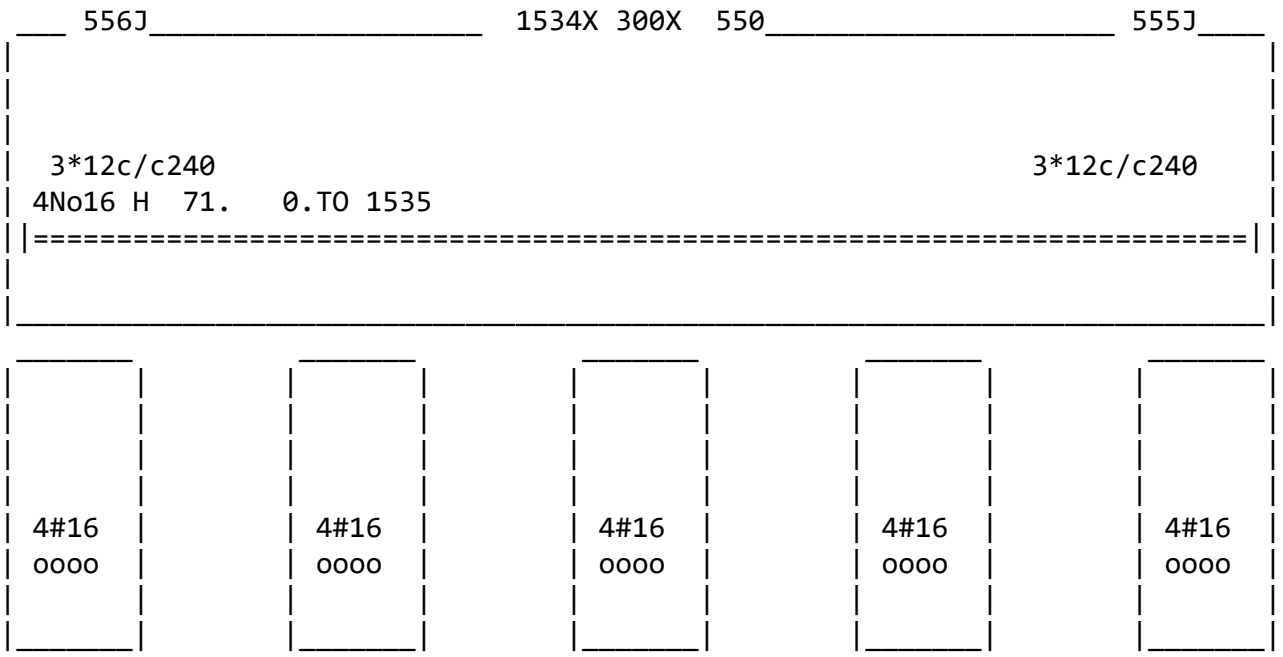
LEN - 1534. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	1535.	YES	YES

B E A M N O. 2305 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 53.36 KNS Vc= 139.73 KNS Vs= 0.00 KNS
Tu= 4.72 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 293. MM

AT END SUPPORT - Vu= 60.69 KNS Vc= 147.69 KNS Vs= 0.00 KNS
Tu= 4.72 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 240. MM C/C FOR 293. MM



ACI 318-11 BEAM NO. 2306 DESIGN RESULTS

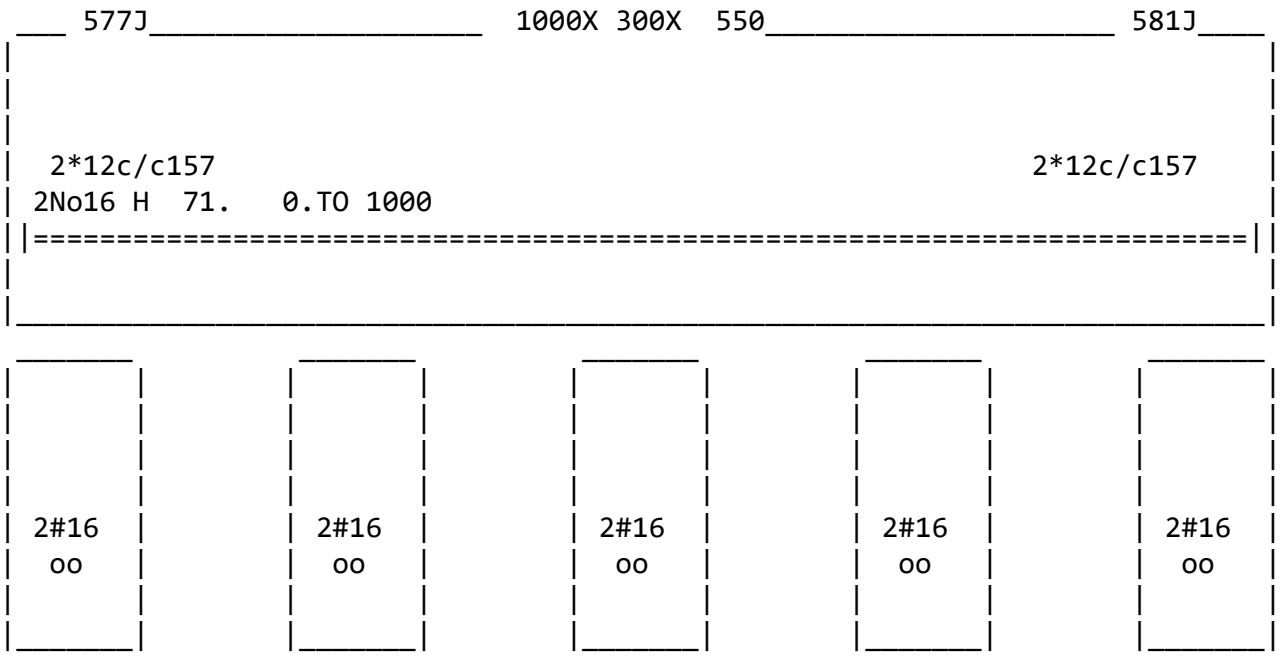
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	1000.	YES	YES

B E A M N O. 2306 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 40.34 KNS Vc= 140.44 KNS Vs= 0.00 KNS
Tu= 10.42 KN-MET Tc= 6.0 KN-MET Ts= 13.9 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.96 SQ.CM.

AT END SUPPORT - Vu= 39.74 KNS Vc= 139.77 KNS Vs= 0.00 KNS
Tu= 10.42 KN-MET Tc= 6.0 KN-MET Ts= 13.9 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.96 SQ.CM.



ACI 318-11 BEAM NO. 2307 DESIGN RESULTS

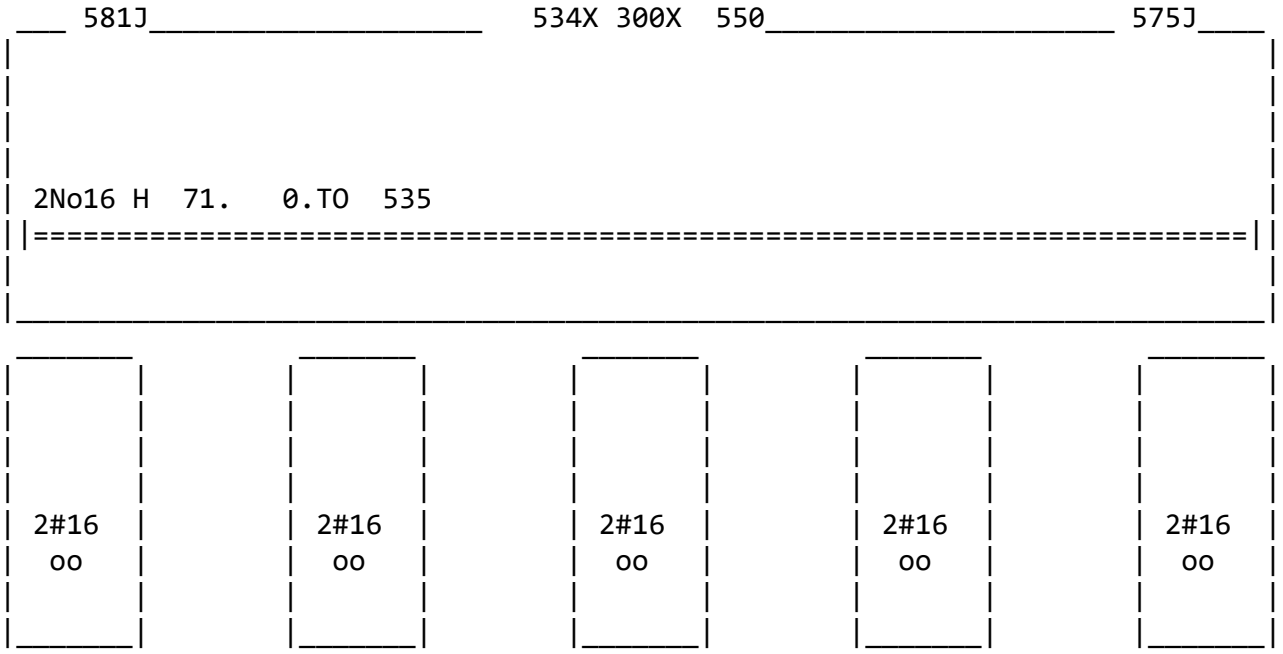
LEN - 535. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	0.	535.	YES	YES

B E A M N O. 2307 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2307 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2307 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



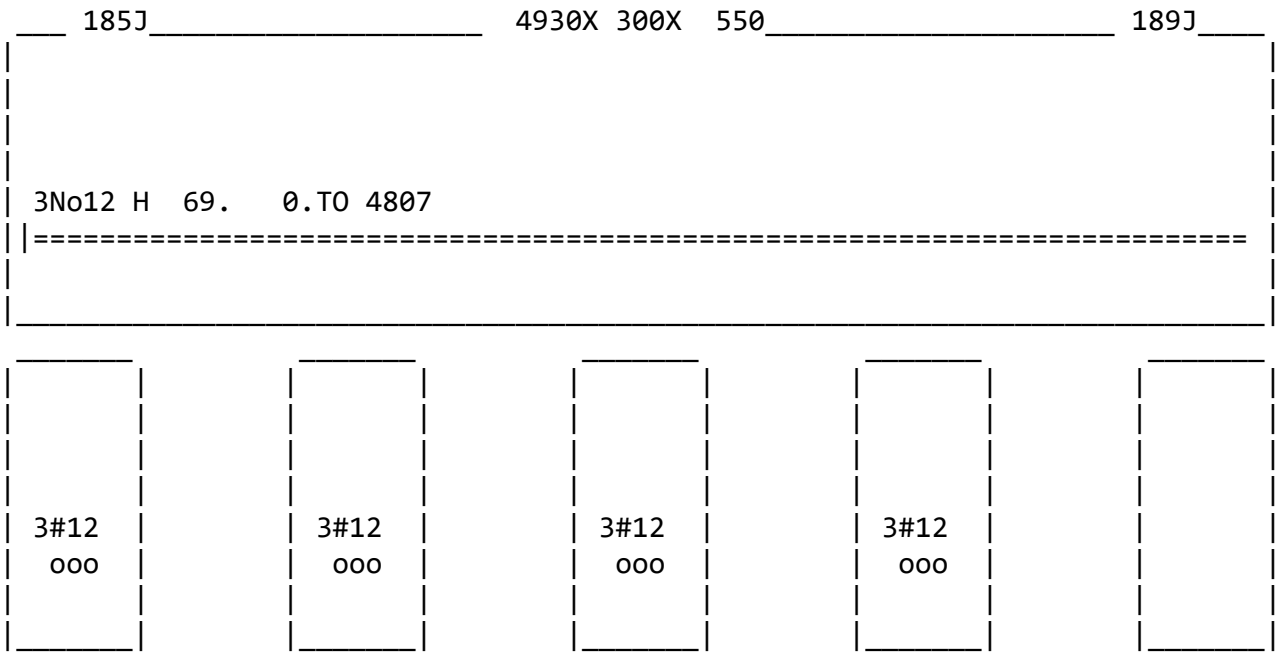
ACI 318-11 BEAM NO. 2308 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	4807.	YES	NO

B E A M N O. 2308 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.98 KNS Vc= 142.05 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.98 KNS Vc= 142.05 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2309 DESIGN RESULTS

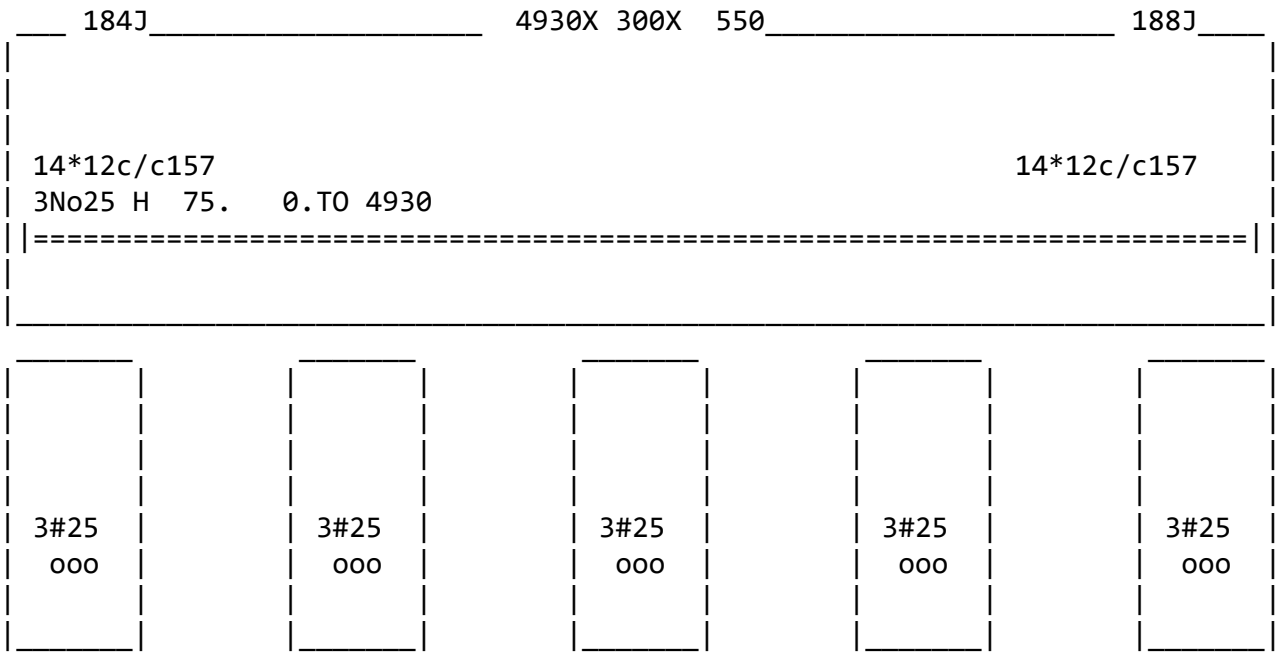
LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	4930.	YES	YES

B E A M N O. 2309 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 59.44 KNS Vc= 139.31 KNS Vs= 0.00 KNS
Tu= 5.93 KN-MET Tc= 5.9 KN-MET Ts= 7.9 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1990. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.68 SQ.CM.

AT END SUPPORT - Vu= 59.36 KNS Vc= 139.31 KNS Vs= 0.00 KNS
Tu= 5.93 KN-MET Tc= 5.9 KN-MET Ts= 7.9 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1990. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.68 SQ.CM.



ACI 318-11 BEAM NO. 2310 DESIGN RESULTS

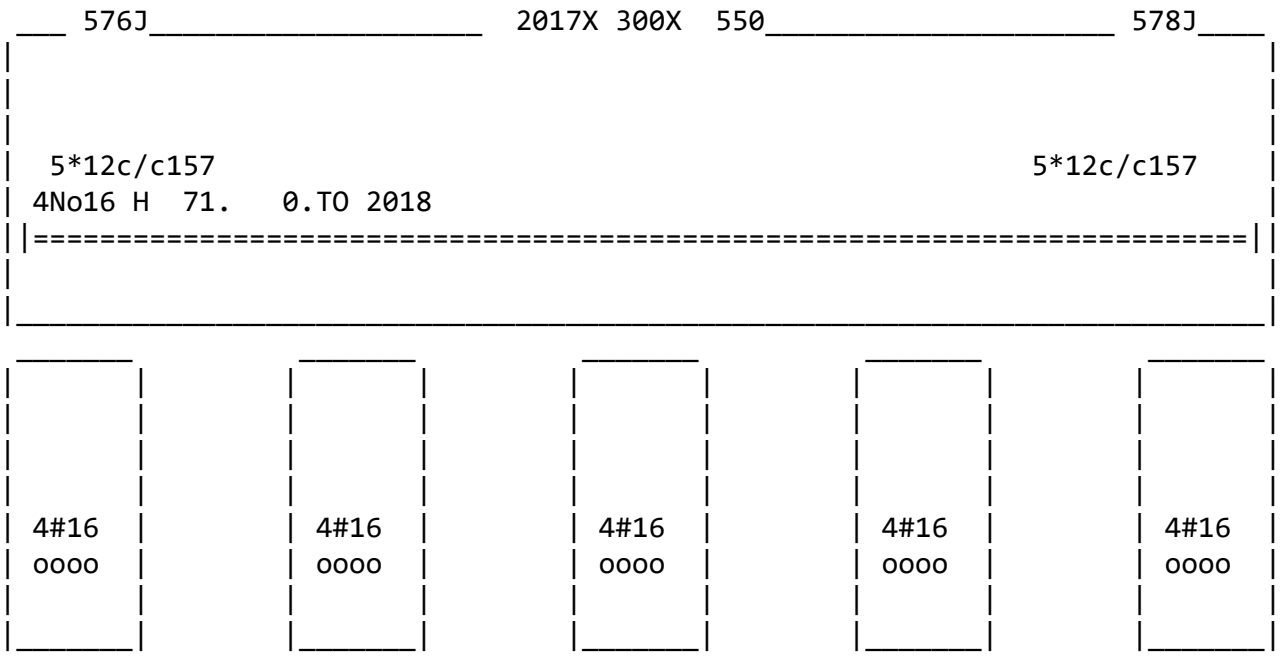
LEN - 2018. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	2018.	YES	YES

B E A M N O. 2310 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 73.17 KNS Vc= 140.94 KNS Vs= 0.00 KNS
Tu= 8.58 KN-MET Tc= 5.9 KN-MET Ts= 11.4 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 534. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.44 SQ.CM.

AT END SUPPORT - Vu= 51.43 KNS Vc= 140.94 KNS Vs= 0.00 KNS
Tu= 8.58 KN-MET Tc= 5.9 KN-MET Ts= 11.4 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 534. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.44 SQ.CM.



ACI 318-11 BEAM NO. 2311 DESIGN RESULTS

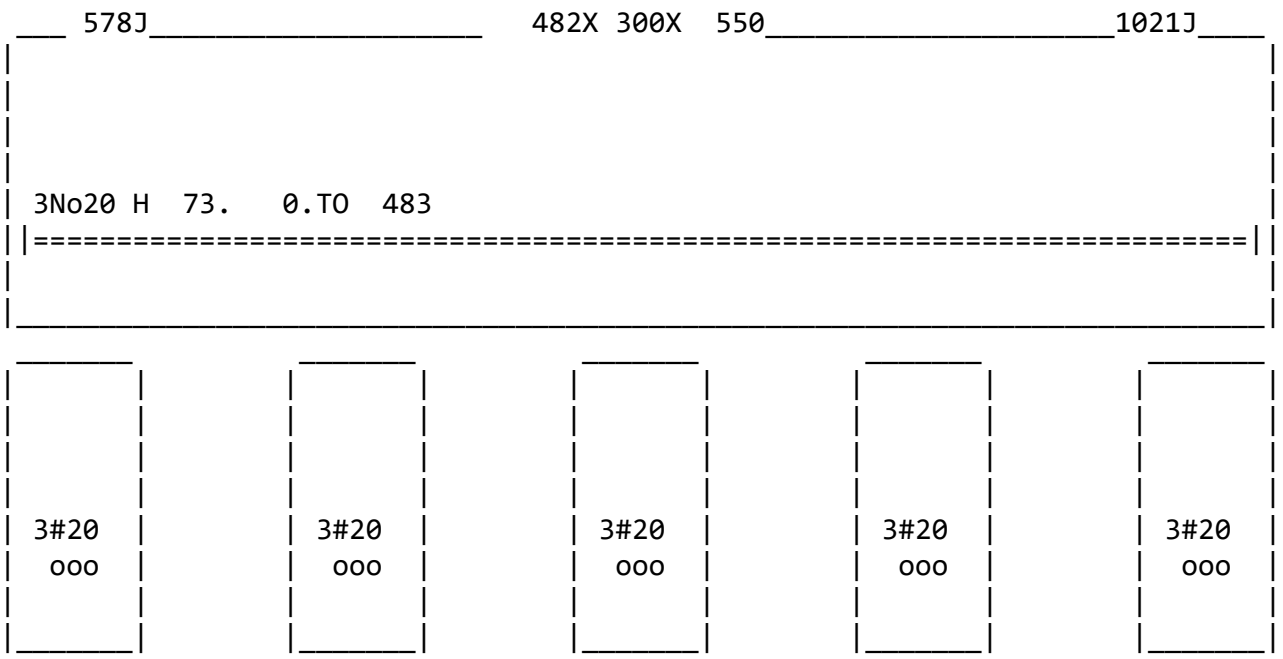
LEN - 483. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	483.	YES	YES

B E A M N O. 2311 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2311 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2311 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2312 DESIGN RESULTS

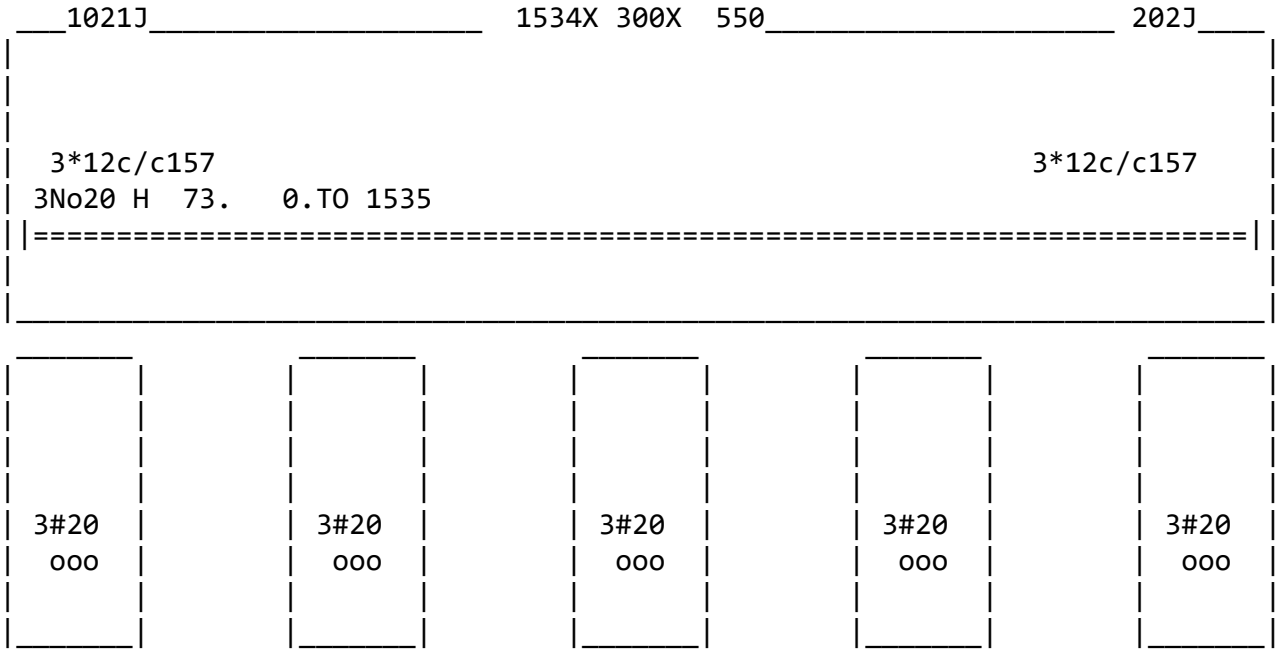
LEN - 1534. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	1535.	YES	YES

B E A M N O. 2312 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 87.47 KNS Vc= 140.34 KNS Vs= 0.00 KNS
Tu= 8.58 KN-MET Tc= 5.9 KN-MET Ts= 11.4 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 293. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.44 SQ.CM.

AT END SUPPORT - Vu= 96.11 KNS Vc= 140.34 KNS Vs= 0.00 KNS
Tu= 8.58 KN-MET Tc= 5.9 KN-MET Ts= 11.4 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 293. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.44 SQ.CM.



ACI 318-11 BEAM NO. 2328 DESIGN RESULTS

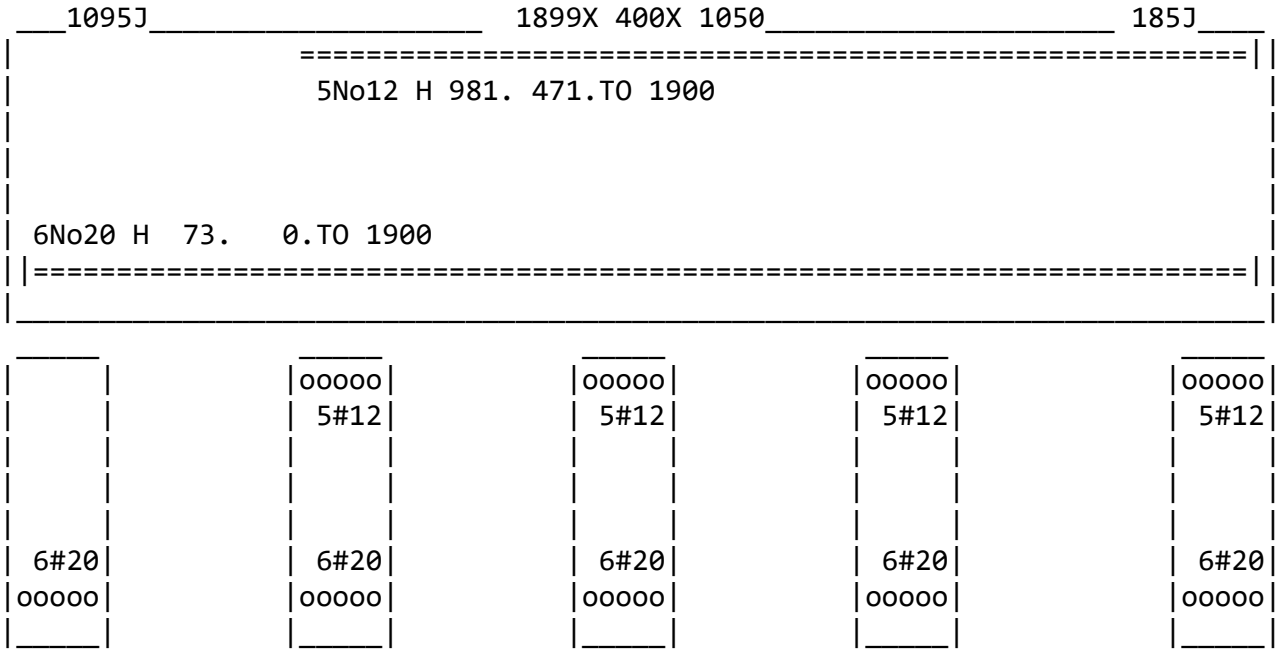
LEN - 1900. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	6 - 20MM	0.	1900.	YES	YES
2	981.	5 - 12MM	471.	1900.	NO	YES

B E A M N O. 2328 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2328 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2328 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2329 DESIGN RESULTS

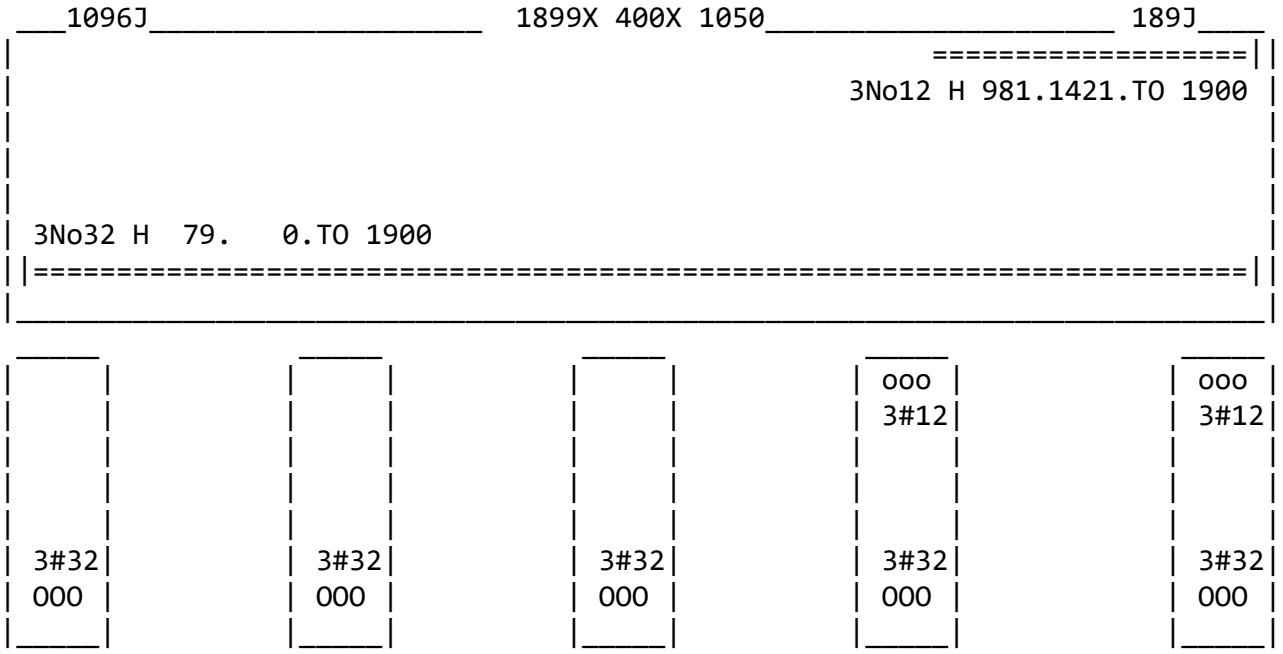
LEN - 1900. MM FY - 420. FC - 35. MPA, SIZE - 400. X 1050. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	1900.	YES	YES
2	981.	3 - 12MM	1421.	1900.	NO	YES

B E A M N O. 2329 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2329 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2329 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



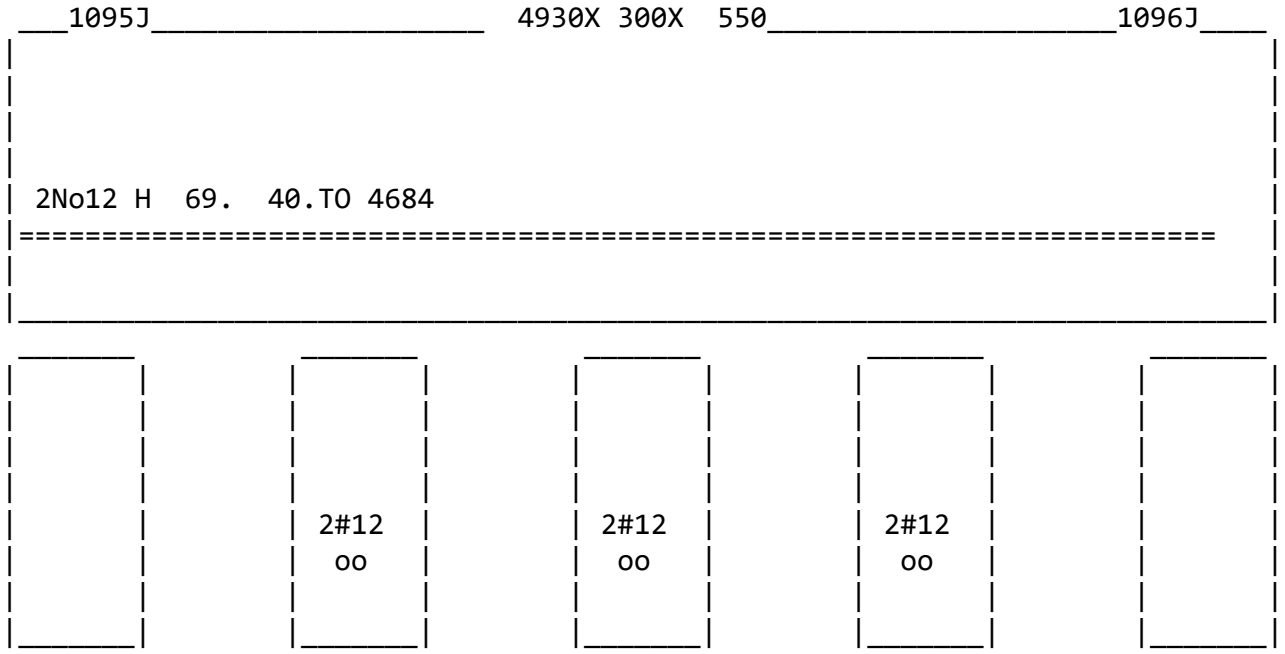
ACI 318-11 BEAM NO. 2330 DESIGN RESULTS

LEN - 4930. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	40.	4684.	NO	NO

B E A M N O. 2330 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.83 KNS Vc= 141.90 KNS Vs= 0.00 KNS
Tu= 1.08 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.83 KNS Vc= 141.90 KNS Vs= 0.00 KNS
Tu= 1.08 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

2227. DESIGN BEAM 2001 TO 2024 2035 TO 2056

ACI 318-11 BEAM NO. 2001 DESIGN RESULTS

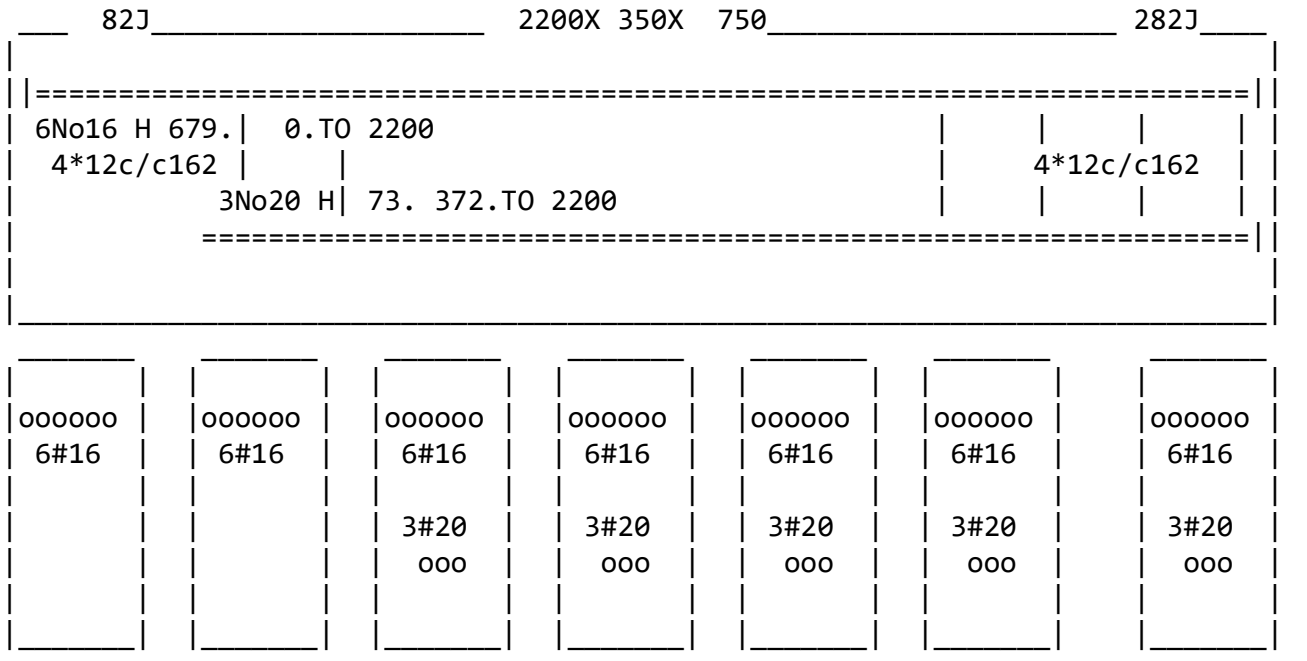
LEN - 2200. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	372.	2200.	NO	YES
2	679.	6 - 16MM	0.	2200.	YES	YES

B E A M N O. 2001 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 127.12 KNS Vc= 236.85 KNS Vs= 0.00 KNS
Tu= 56.75 KN-MET Tc= 12.6 KN-MET Ts= 75.7 KN-MET LOAD 412
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 162. MM C/C FOR 425. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 12.23 SQ.CM.

AT END SUPPORT - Vu= 110.45 KNS Vc= 430.74 KNS Vs= 0.00 KNS
Tu= 56.75 KN-MET Tc= 12.6 KN-MET Ts= 75.7 KN-MET LOAD 412
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 162. MM C/C FOR 425. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 12.23 SQ.CM.



ACI 318-11 BEAM NO. 2002 DESIGN RESULTS

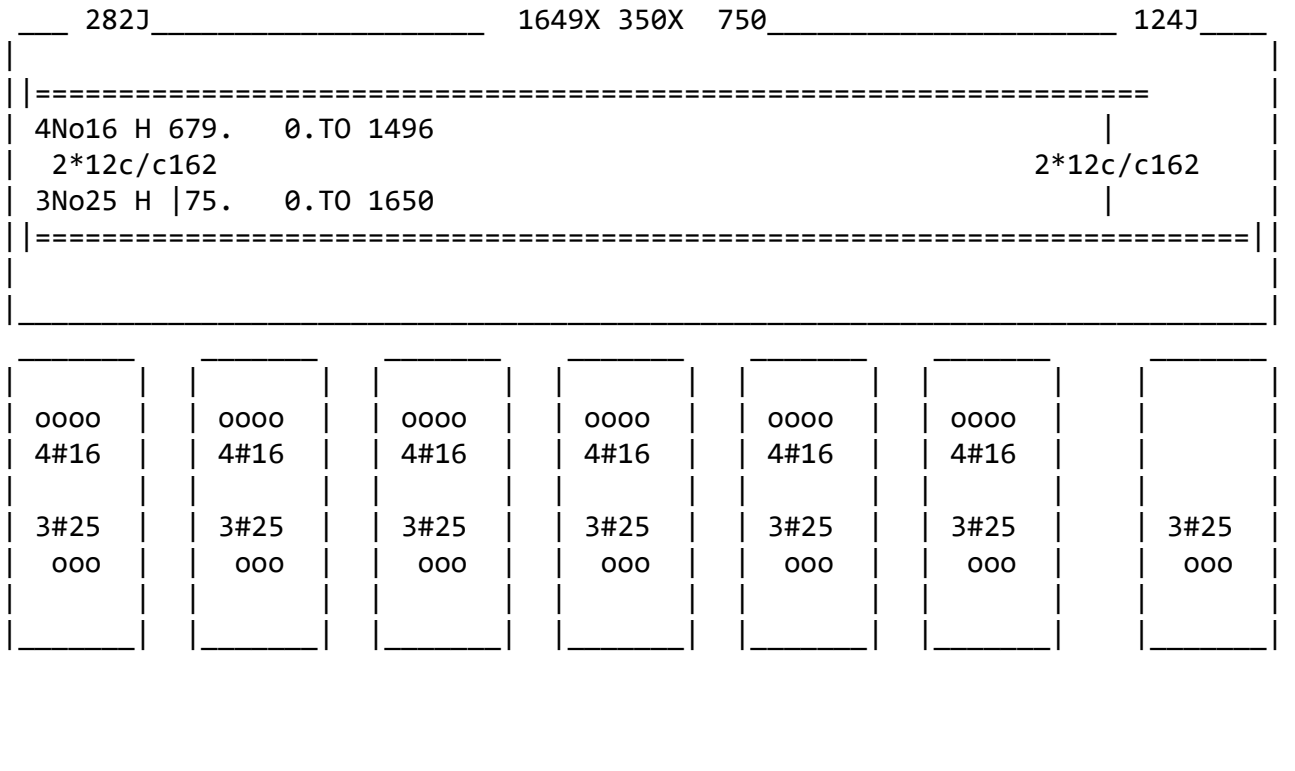
LEN - 1650. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	1650.	YES	YES
2	679.	4 - 16MM	0.	1496.	YES	NO

B E A M N O. 2002 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 92.86 KNS Vc= 236.12 KNS Vs= 0.00 KNS
Tu= 56.75 KN-MET Tc= 12.6 KN-MET Ts= 75.7 KN-MET LOAD 412
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 162. MM C/C FOR 150. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 12.23 SQ.CM.

AT END SUPPORT - Vu= 86.93 KNS Vc= 231.98 KNS Vs= 0.00 KNS
Tu= 56.75 KN-MET Tc= 12.6 KN-MET Ts= 75.7 KN-MET LOAD 412
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 162. MM C/C FOR 150. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 12.23 SQ.CM.



ACI 318-11 BEAM NO. 2003 DESIGN RESULTS

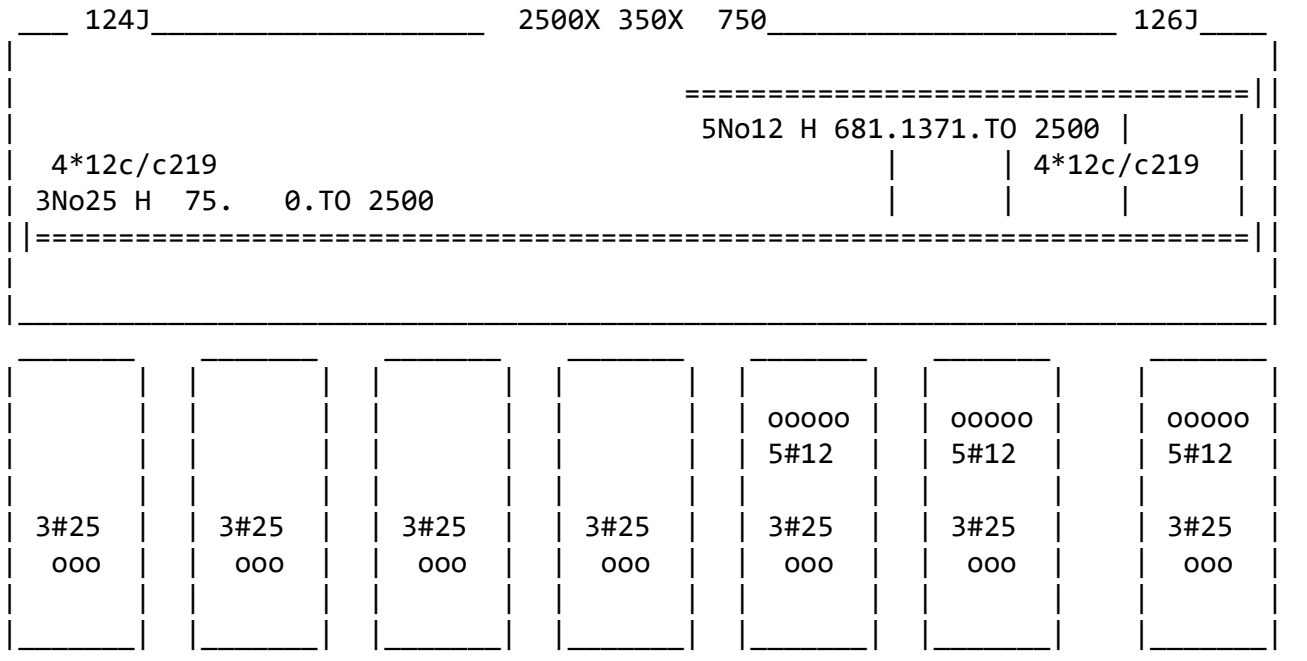
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	2500.	YES	YES
2	681.	5 - 12MM	1371.	2500.	NO	YES

B E A M N O. 2003 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 108.32 KNS Vc= 231.83 KNS Vs= 0.00 KNS
Tu= 31.58 KN-MET Tc= 14.1 KN-MET Ts= 42.1 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 575. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.80 SQ.CM.

AT END SUPPORT - Vu= 142.66 KNS Vc= 459.85 KNS Vs= 0.00 KNS
Tu= 31.58 KN-MET Tc= 14.1 KN-MET Ts= 42.1 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 575. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.80 SQ.CM.



ACI 318-11 BEAM NO. 2004 DESIGN RESULTS

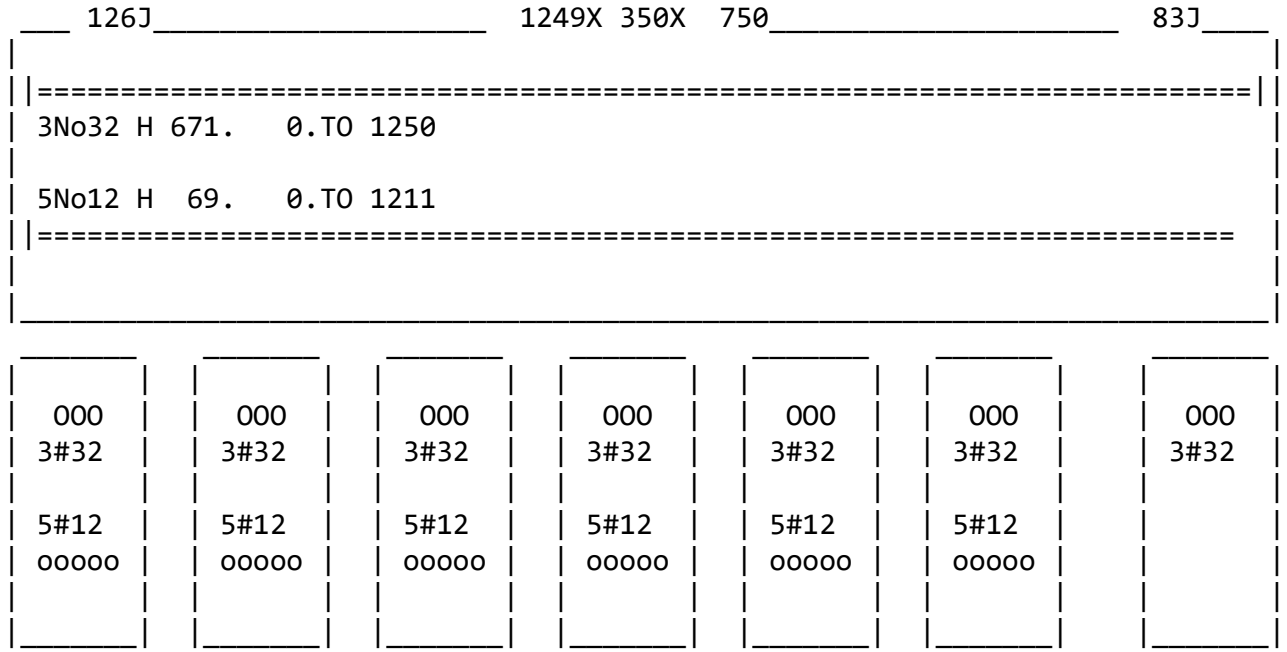
LEN - 1250. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1211.	YES	NO
2	671.	3 - 32MM	0.	1250.	YES	YES

B E A M N O. 2004 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2004 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2004 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2005 DESIGN RESULTS

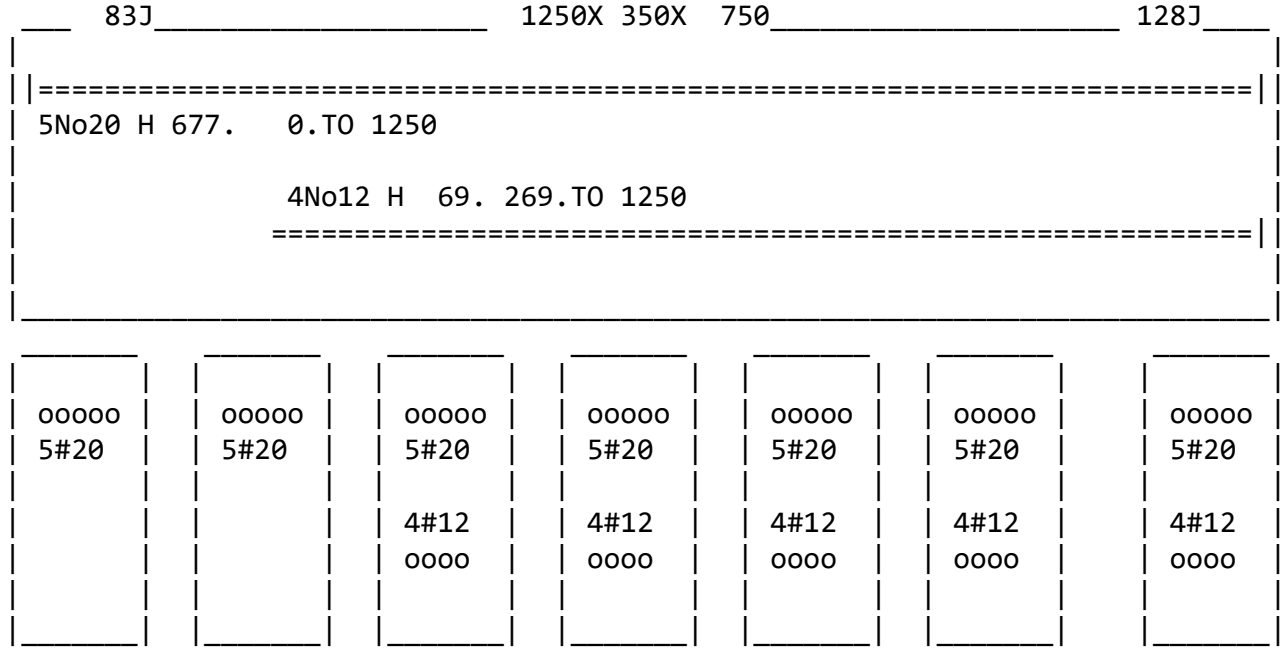
LEN - 1250. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	4 - 12MM	269.	1250.	NO	YES
2	677.	5 - 20MM	0.	1250.	YES	YES

B E A M N O. 2005 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2005 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2005 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2006 DESIGN RESULTS

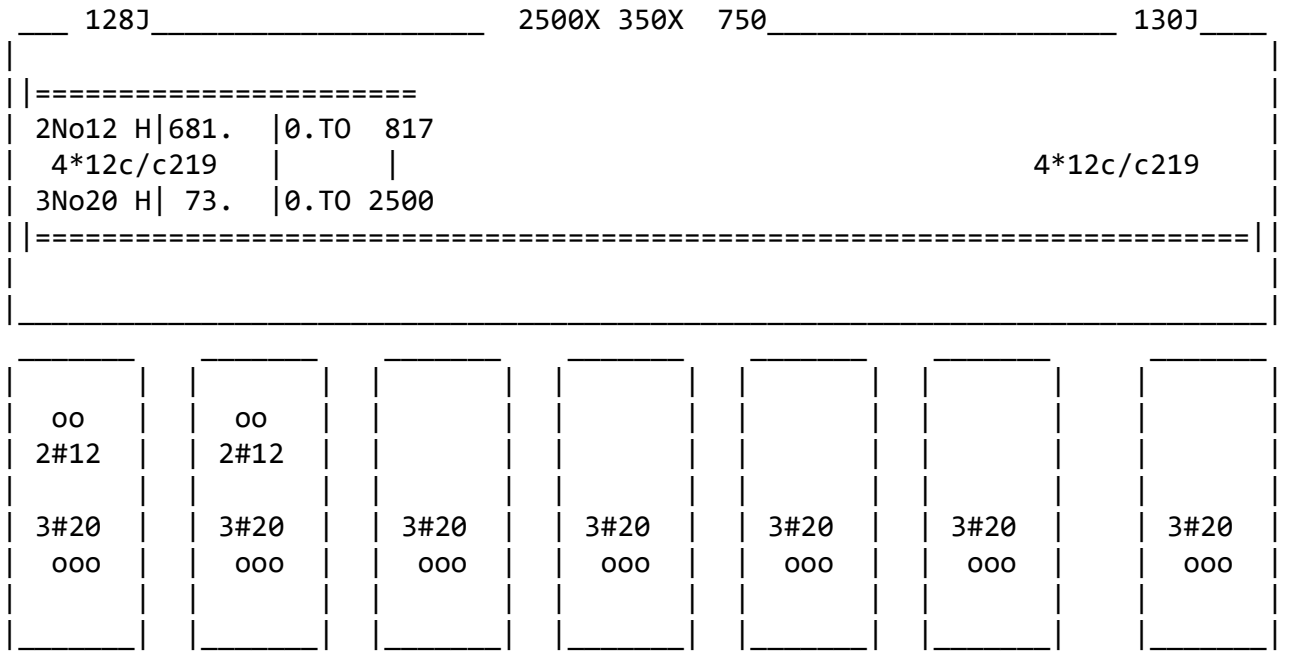
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	2500.	YES	YES
2	681.	2 - 12MM	0.	817.	YES	NO

B E A M N O. 2006 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.09 KNS Vc= 245.80 KNS Vs= 0.00 KNS
Tu= 21.13 KN-MET Tc= 13.2 KN-MET Ts= 28.2 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 575. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.55 SQ.CM.

AT END SUPPORT - Vu= 42.75 KNS Vc= 225.65 KNS Vs= 0.00 KNS
Tu= 21.13 KN-MET Tc= 13.2 KN-MET Ts= 28.2 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 575. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.55 SQ.CM.



ACI 318-11 BEAM NO. 2007 DESIGN RESULTS

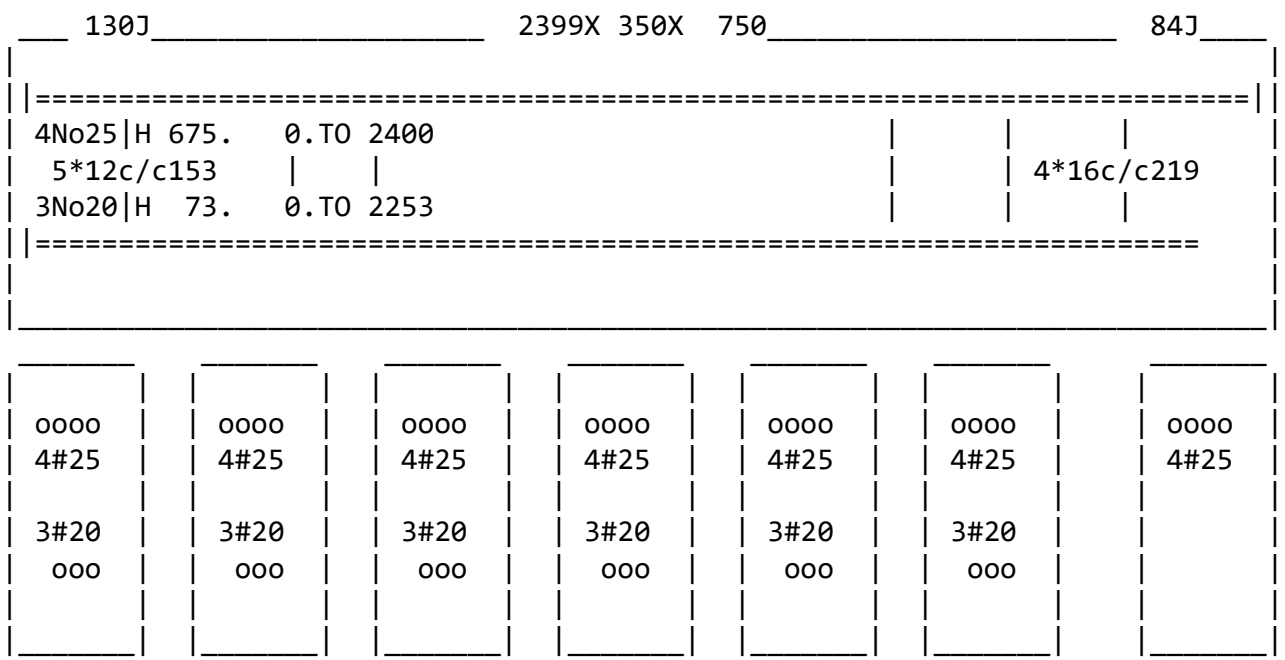
LEN - 2400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	2253.	YES	NO
2	675.	4 - 25MM	0.	2400.	YES	YES

B E A M N O. 2007 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 180.01 KNS Vc= 213.34 KNS Vs= 26.67 KNS
Tu= 56.37 KN-MET Tc= 10.6 KN-MET Ts= 75.2 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 153. MM C/C FOR 525. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 12.15 SQ.CM.

AT END SUPPORT - Vu= 211.11 KNS Vc= 213.34 KNS Vs= 68.13 KNS
Tu= 56.37 KN-MET Tc= 10.6 KN-MET Ts= 75.2 KN-MET LOAD 405
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 16 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 525. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 12.15 SQ.CM.



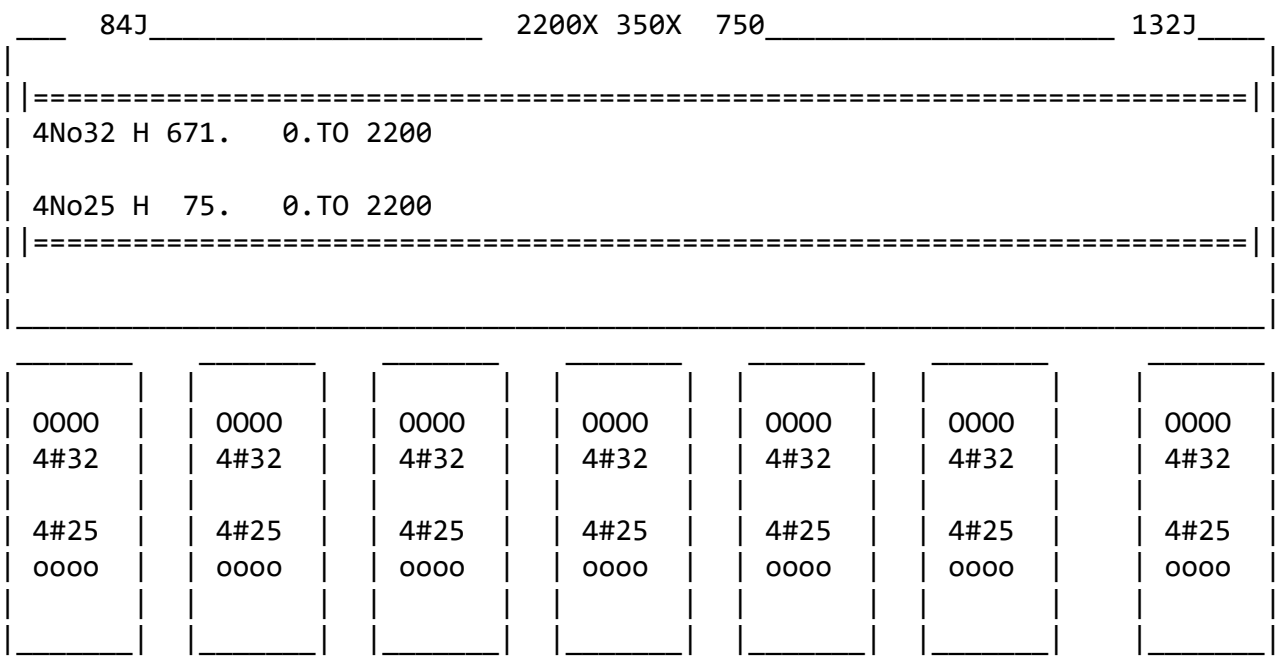
ACI 318-11 BEAM NO. 2008 DESIGN RESULTS

LEN - 2200. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	2200.	YES	YES
2	671.	4 - 32MM	0.	2200.	YES	YES

B E A M N O. 2008 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 279.70 KNS Tu= 81.3 KN-MET
Vc= 219.4 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 405 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
AT END SUPPORT - Vu= 254.83 KNS Tu= 81.3 KN-MET
Vc= 219.4 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD 405 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.



ACI 318-11 BEAM NO. 2009 DESIGN RESULTS

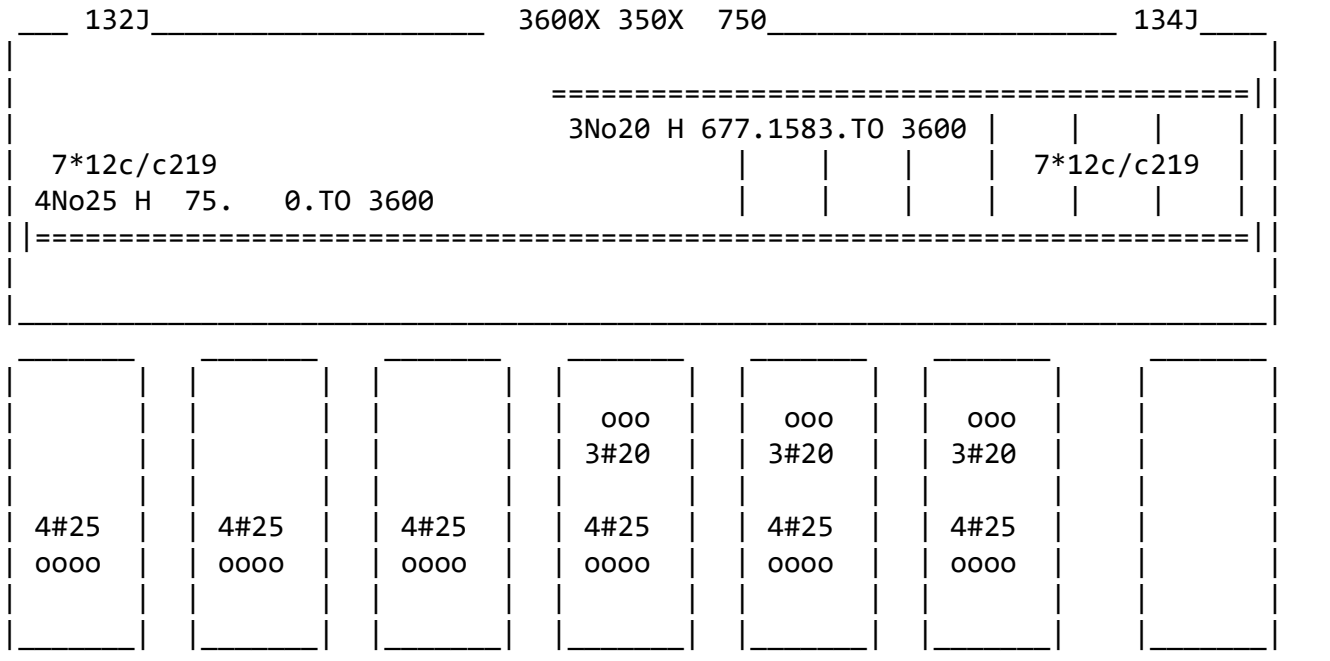
LEN - 3600. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	4 - 25MM	0.	3600.	YES	YES
2	677.	3 - 20MM	1583.	3600.	NO	YES

B E A M N O. 2009 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 66.76 KNS Vc= 228.81 KNS Vs= 0.00 KNS
Tu= 28.34 KN-MET Tc= 13.5 KN-MET Ts= 37.8 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 1125. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.11 SQ.CM.

AT END SUPPORT - Vu= 141.21 KNS Vc= 447.36 KNS Vs= 0.00 KNS
Tu= 28.34 KN-MET Tc= 13.5 KN-MET Ts= 37.8 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 1125. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.11 SQ.CM.



ACI 318-11 BEAM NO. 2010 DESIGN RESULTS

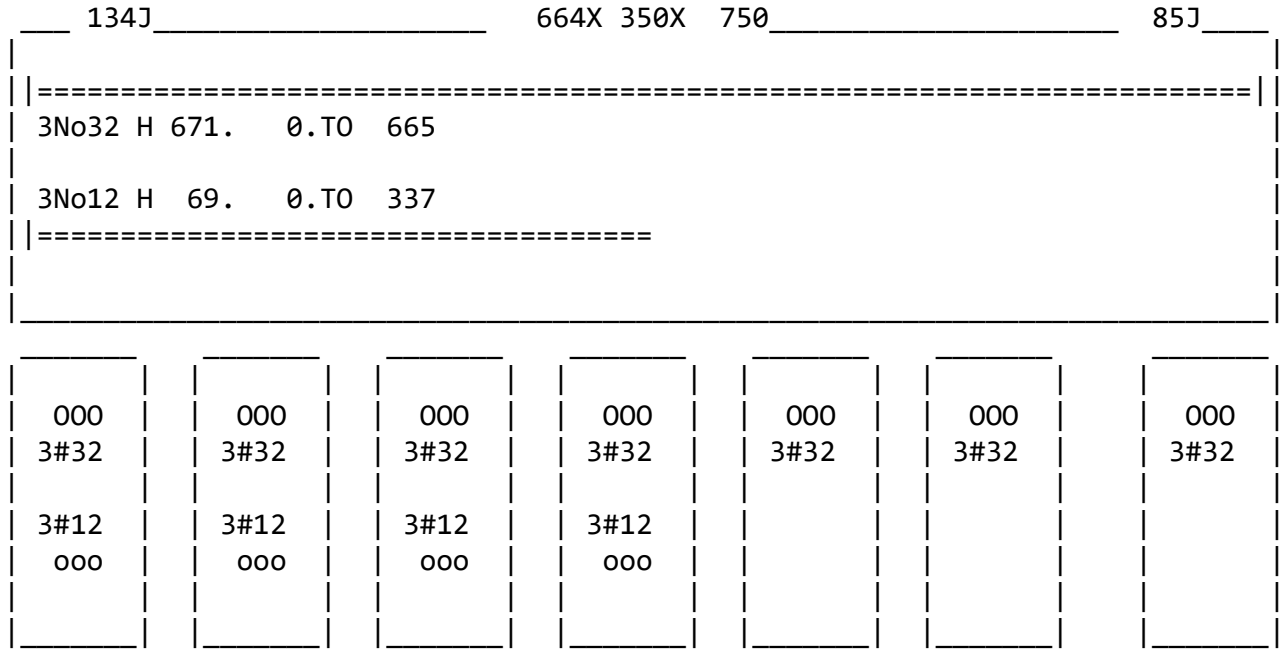
LEN - 665. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	337.	YES	NO
2	671.	3 - 32MM	0.	665.	YES	YES

B E A M N O. 2010 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2010 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2010 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2011 DESIGN RESULTS

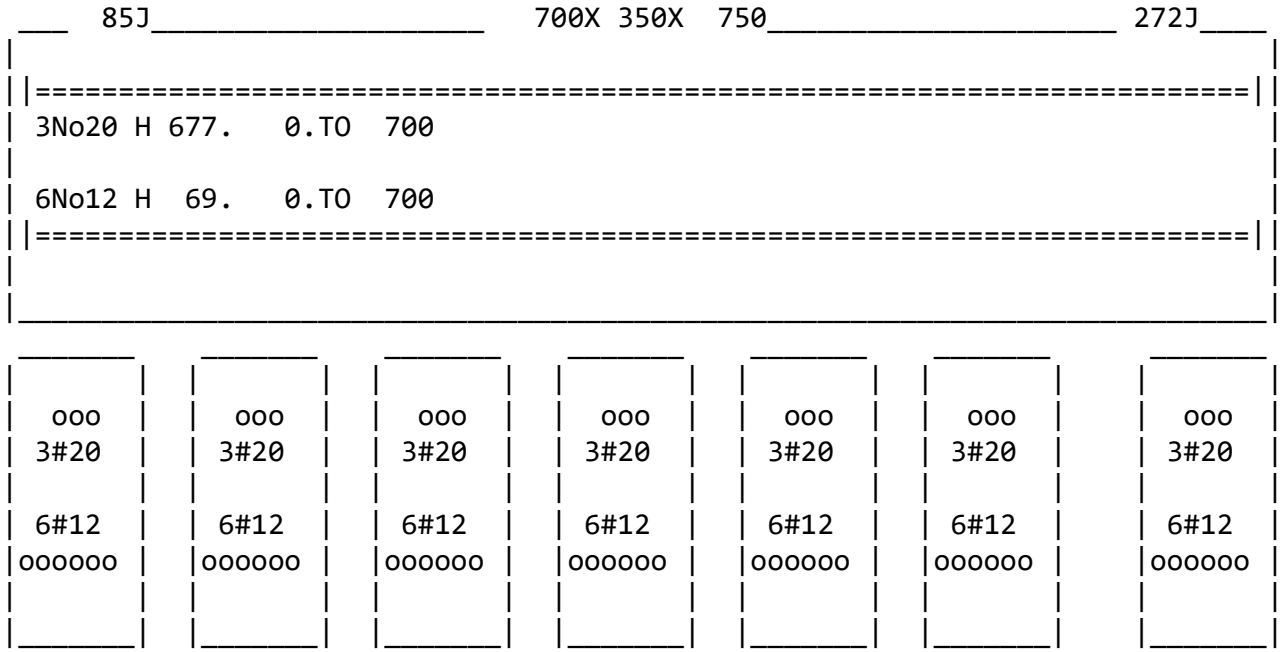
LEN - 700. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	6 - 12MM	0.	700.	YES	YES
2	677.	3 - 20MM	0.	700.	YES	YES

B E A M N O. 2011 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2011 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2011 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2012 DESIGN RESULTS

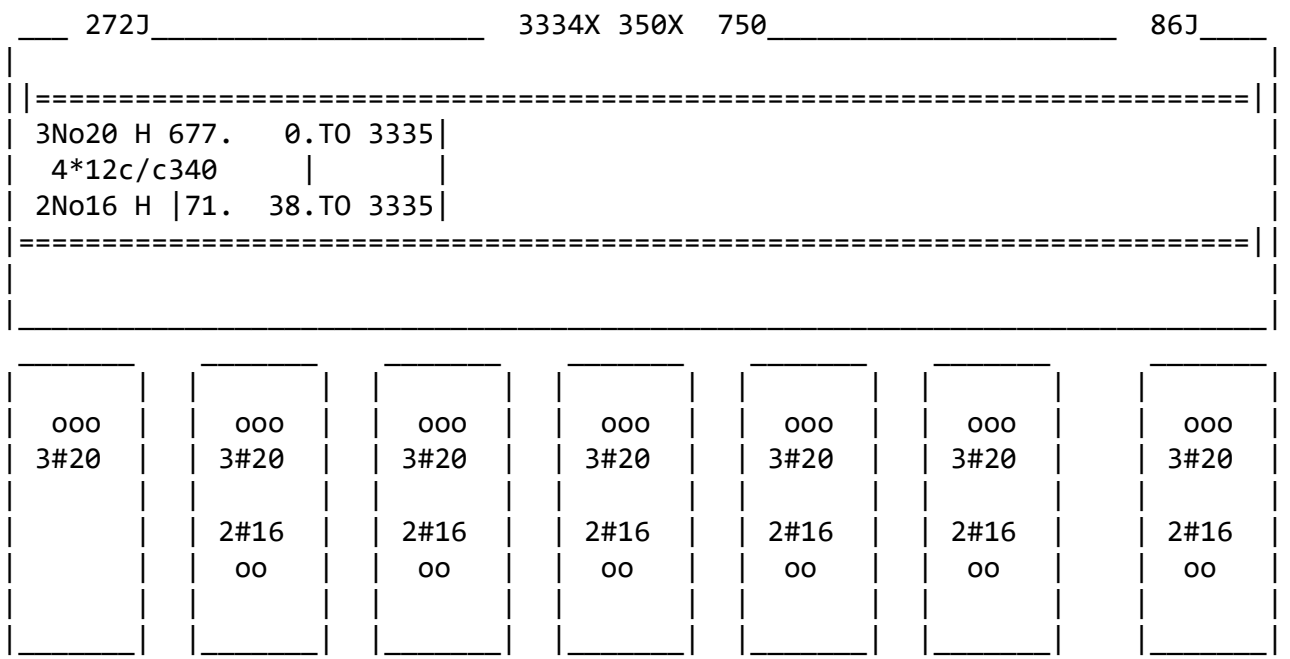
LEN - 3335. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	2 - 16MM	38.	3335.	NO	YES
2	677.	3 - 20MM	0.	3335.	YES	YES

B E A M N O. 2012 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 74.27 KNS Vc= 189.04 KNS Vs= 0.00 KNS
Tu= 1.43 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 340. MM C/C FOR 993. MM

AT END SUPPORT - Vu= 11.12 KNS Vc= 224.35 KNS Vs= 0.00 KNS
Tu= 5.22 KN-MET Tc= 11.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2013 DESIGN RESULTS

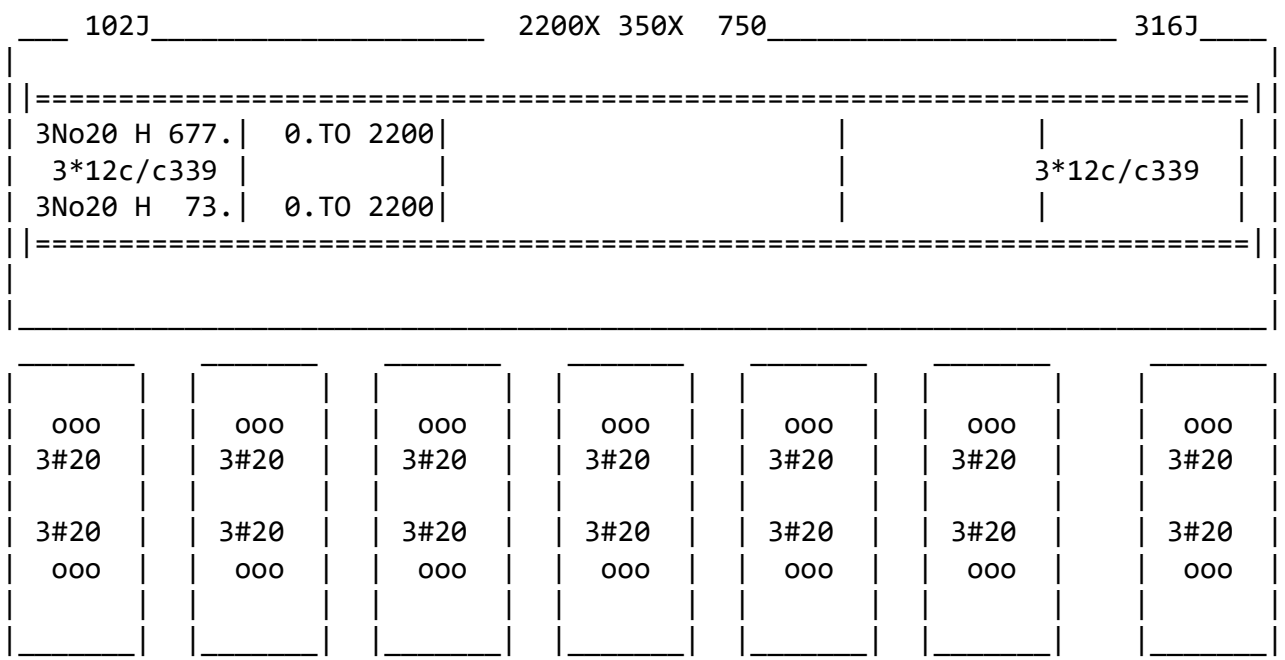
LEN - 2200. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	3 - 20MM	0.	2200.	YES	YES
2	677.	3 - 20MM	0.	2200.	YES	YES

B E A M N O. 2013 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 92.46 KNS Vc= 205.68 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 10.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 339. MM C/C FOR 425. MM

AT END SUPPORT - Vu= 107.74 KNS Vc= 253.89 KNS Vs= 0.00 KNS
Tu= 2.03 KN-MET Tc= 13.0 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 339. MM C/C FOR 425. MM



ACI 318-11 BEAM NO. 2014 DESIGN RESULTS

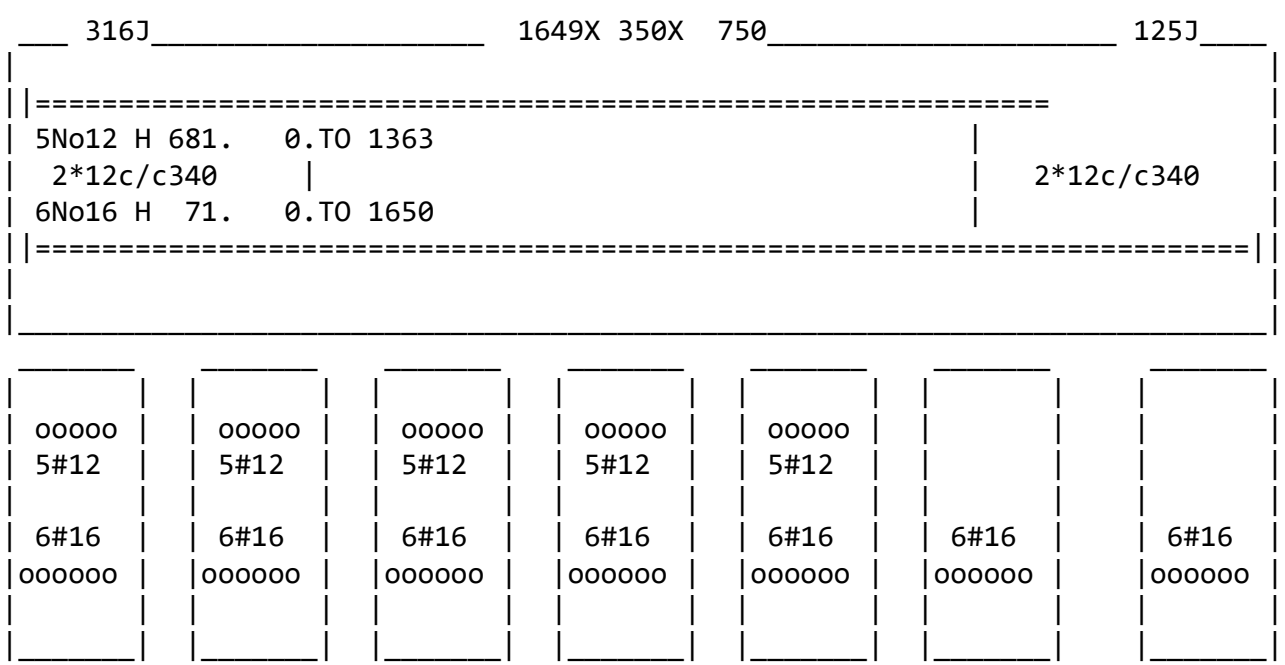
LEN - 1650. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	1650.	YES	YES
2	681.	5 - 12MM	0.	1363.	YES	NO

B E A M N O. 2014 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 116.92 KNS Vc= 206.30 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 10.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 340. MM C/C FOR 150. MM

AT END SUPPORT - Vu= 107.98 KNS Vc= 206.30 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 10.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 340. MM C/C FOR 150. MM



ACI 318-11 BEAM NO. 2015 DESIGN RESULTS

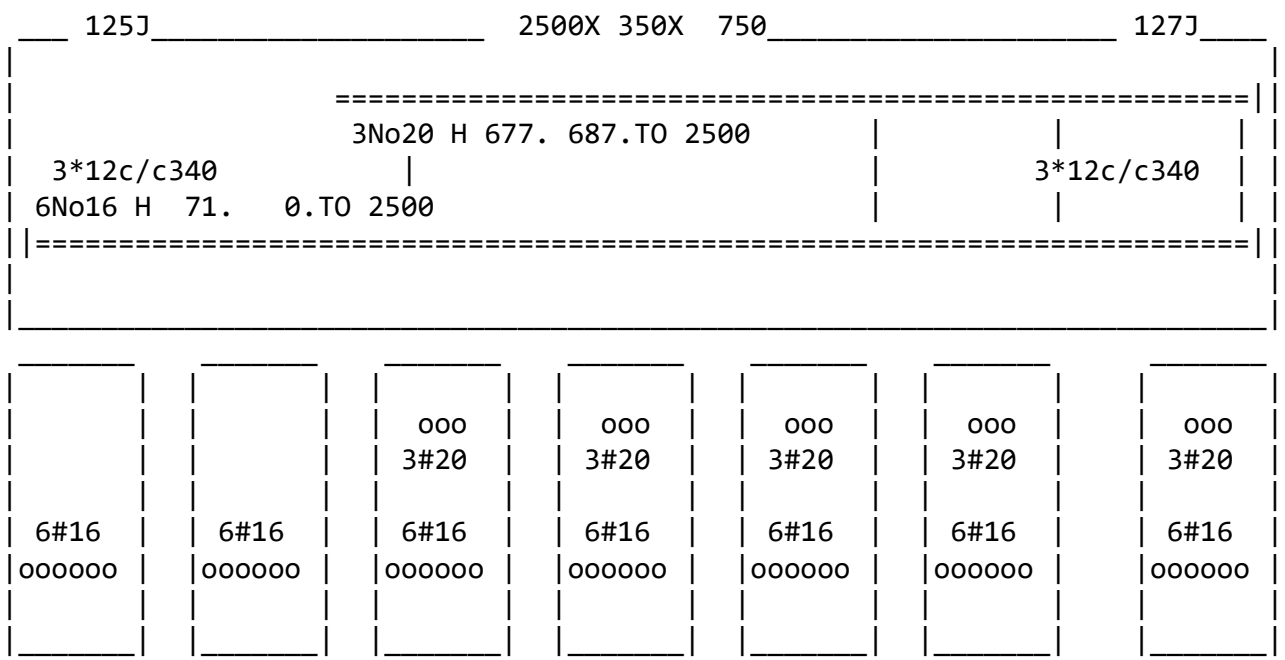
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	2500.	YES	YES
2	677.	3 - 20MM	687.	2500.	NO	YES

B E A M N O. 2015 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 85.09 KNS Vc= 206.43 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 10.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 340. MM C/C FOR 575. MM

AT END SUPPORT - Vu= 116.69 KNS Vc= 206.43 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 10.3 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 340. MM C/C FOR 575. MM



ACI 318-11 BEAM NO. 2016 DESIGN RESULTS

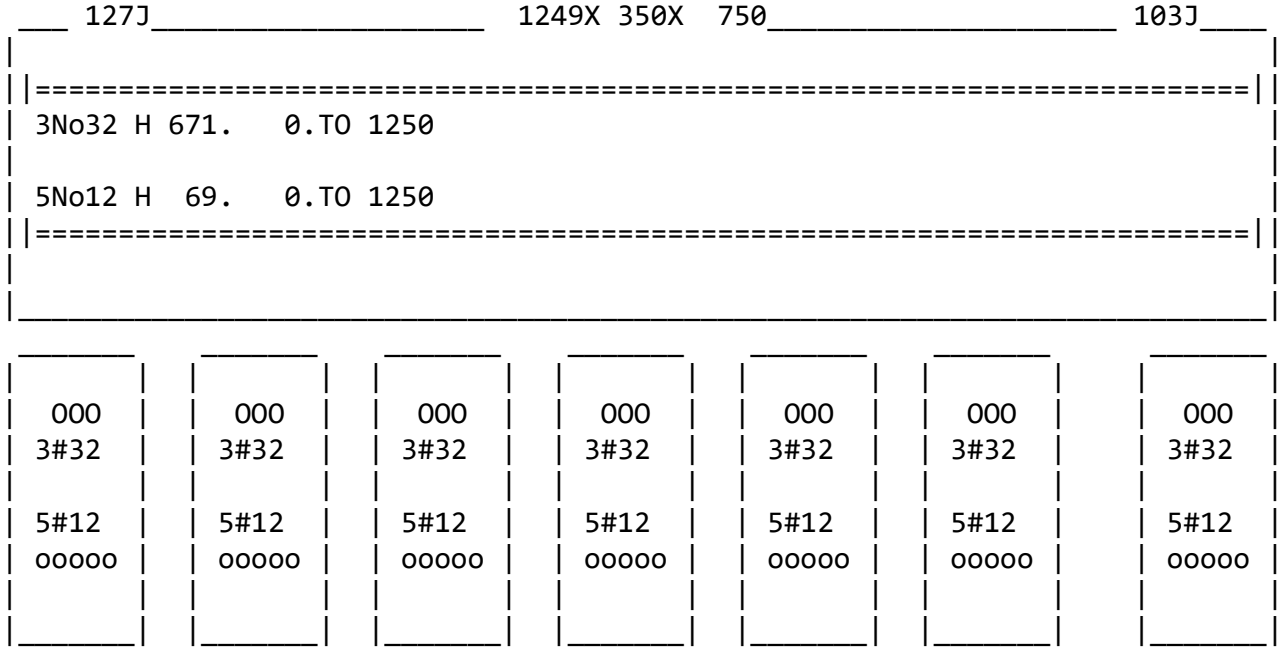
LEN - 1250. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1250.	YES	YES
2	671.	3 - 32MM	0.	1250.	YES	YES

B E A M N O. 2016 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2016 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2016 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2017 DESIGN RESULTS

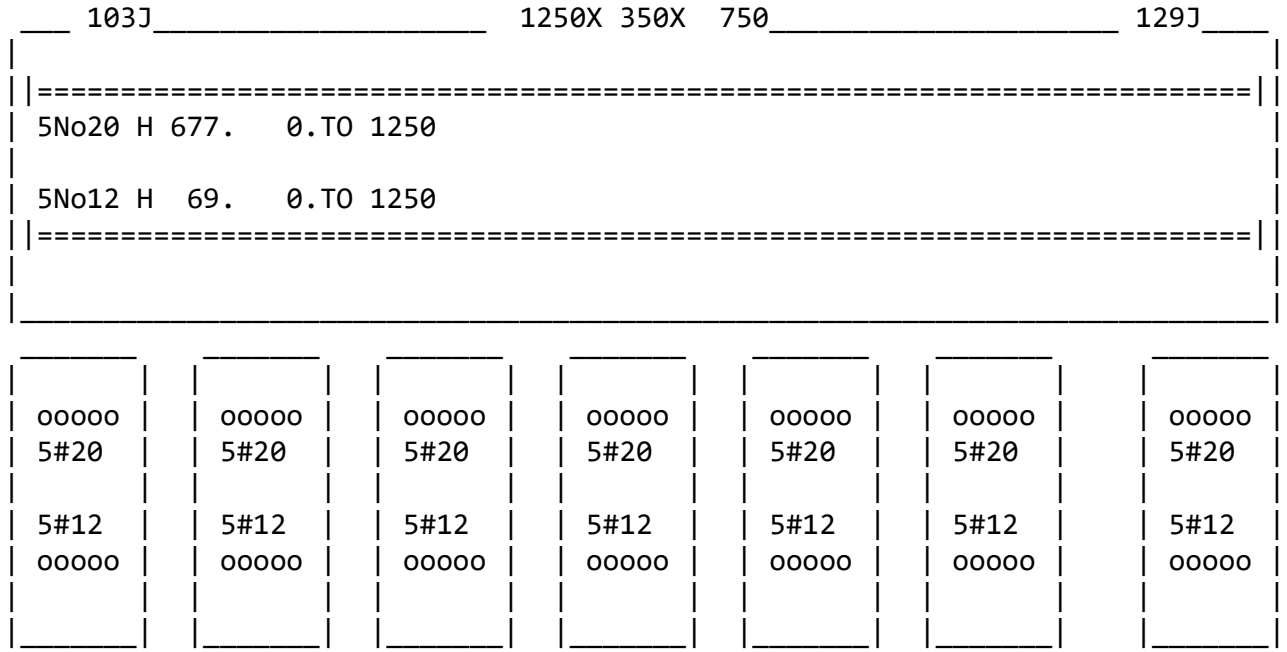
LEN - 1250. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	5 - 12MM	0.	1250.	YES	YES
2	677.	5 - 20MM	0.	1250.	YES	YES

B E A M N O. 2017 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2017 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2017 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2018 DESIGN RESULTS

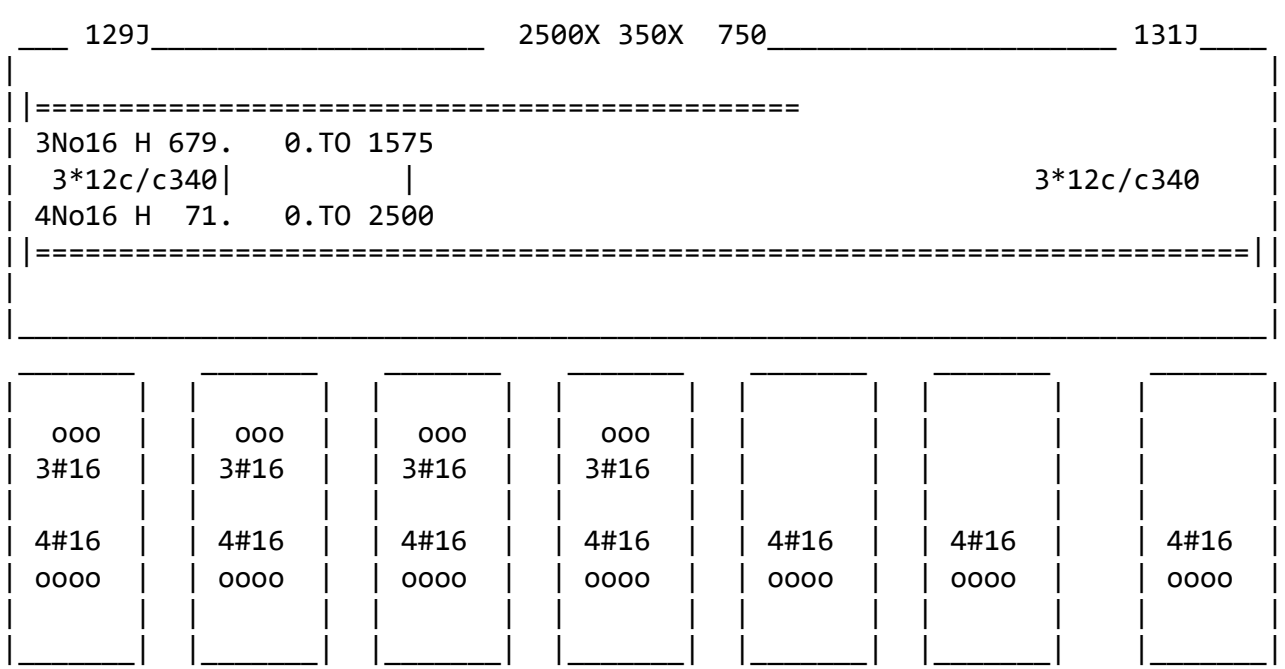
LEN - 2500. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	0.	2500.	YES	YES
2	679.	3 - 16MM	0.	1575.	YES	NO

B E A M N O. 2018 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 102.28 KNS Vc= 212.83 KNS Vs= 0.00 KNS
Tu= 6.37 KN-MET Tc= 10.6 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 340. MM C/C FOR 575. MM

AT END SUPPORT - Vu= 83.31 KNS Vc= 212.83 KNS Vs= 0.00 KNS
Tu= 6.37 KN-MET Tc= 10.6 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 340. MM C/C FOR 575. MM



ACI 318-11 BEAM NO. 2019 DESIGN RESULTS

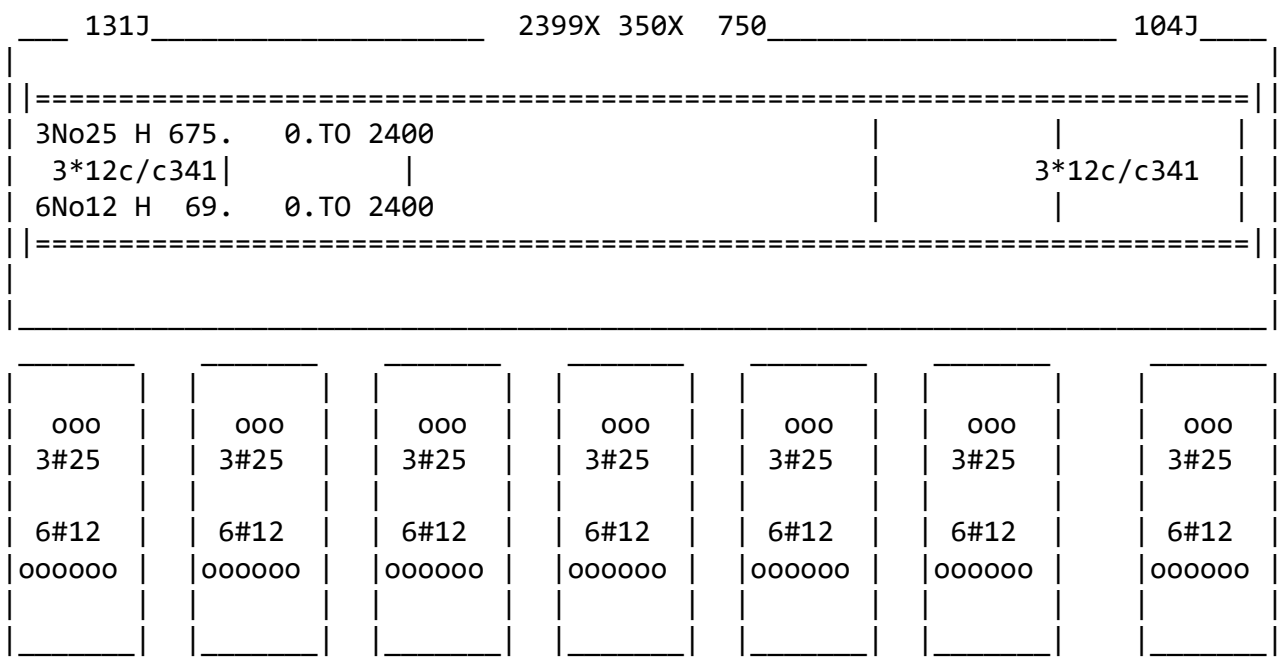
LEN - 2400. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	6 - 12MM	0.	2400.	YES	YES
2	675.	3 - 25MM	0.	2400.	YES	YES

B E A M N O. 2019 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 99.72 KNS Vc= 230.36 KNS Vs= 0.00 KNS
Tu= 7.40 KN-MET Tc= 11.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 341. MM C/C FOR 525. MM

AT END SUPPORT - Vu= 115.32 KNS Vc= 230.36 KNS Vs= 0.00 KNS
Tu= 7.40 KN-MET Tc= 11.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 341. MM C/C FOR 525. MM



ACI 318-11 BEAM NO. 2020 DESIGN RESULTS

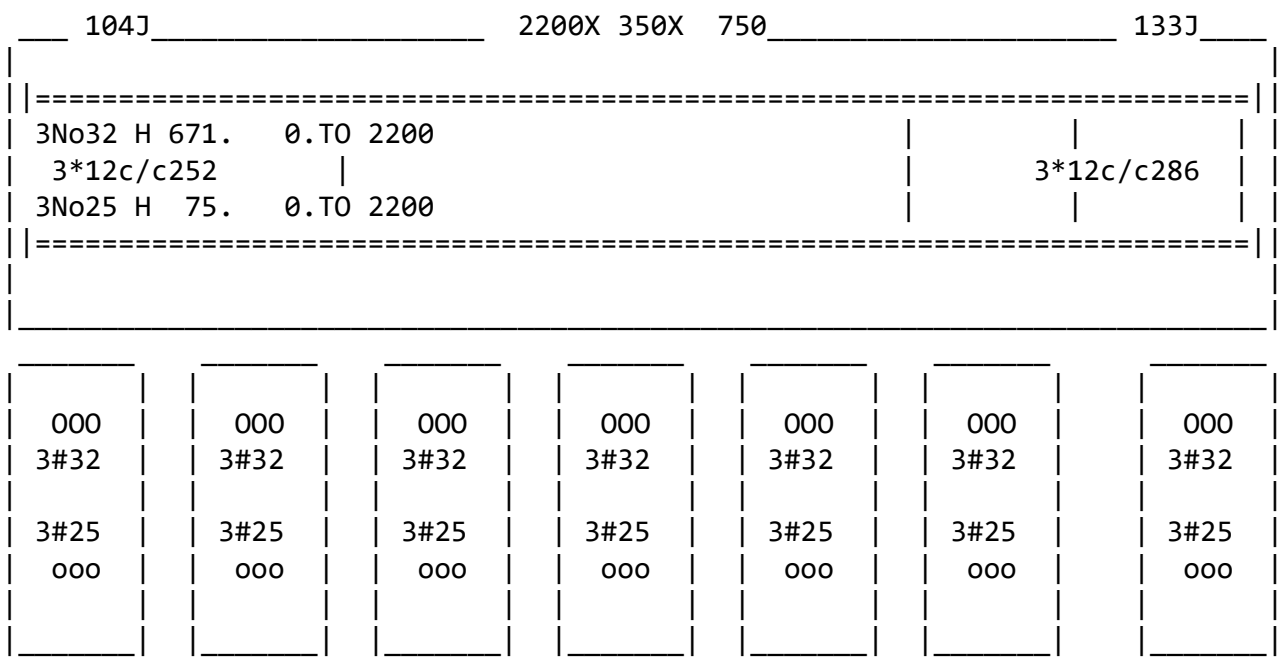
LEN - 2200. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	2200.	YES	YES
2	671.	3 - 32MM	0.	2200.	YES	YES

B E A M N O. 2020 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 355.54 KNS Vc= 219.17 KNS Vs= 254.88 KNS
Tu= 9.14 KN-MET Tc= 11.0 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 252. MM C/C FOR 425. MM

AT END SUPPORT - Vu= 332.70 KNS Vc= 219.17 KNS Vs= 224.42 KNS
Tu= 9.14 KN-MET Tc= 11.0 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 286. MM C/C FOR 425. MM



ACI 318-11 BEAM NO. 2021 DESIGN RESULTS

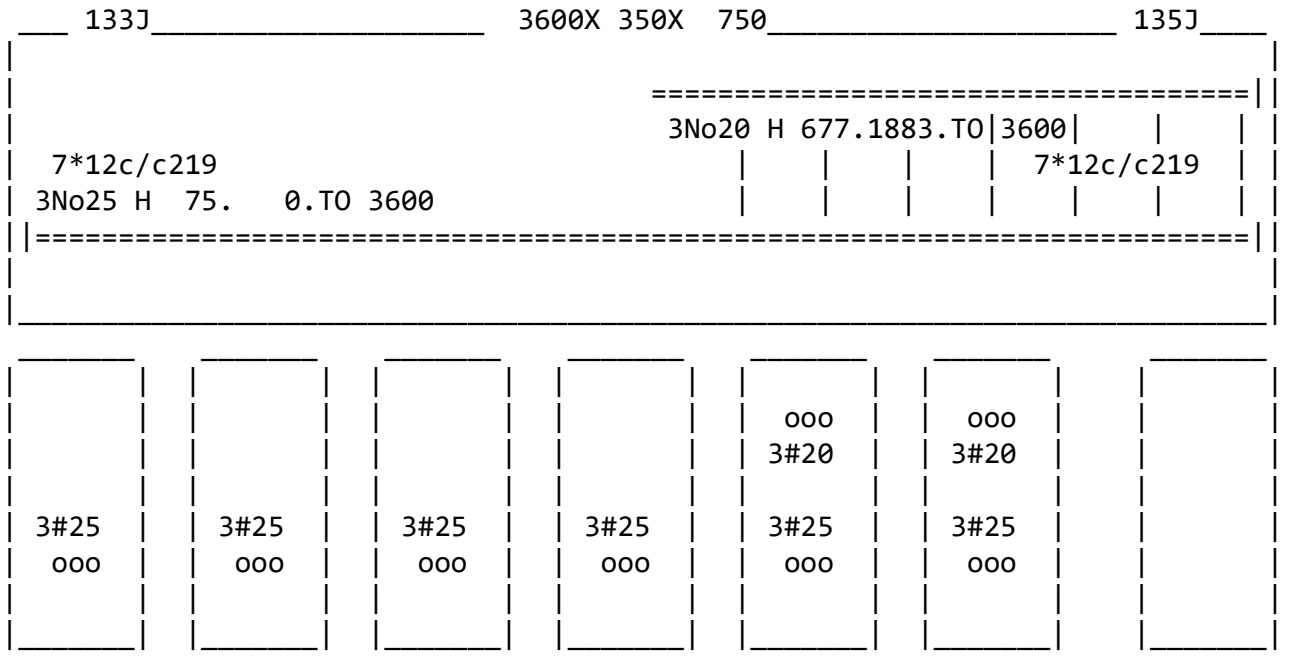
LEN - 3600. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	3600.	YES	YES
2	677.	3 - 20MM	1883.	3600.	NO	YES

B E A M N O. 2021 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.04 KNS Vc= 225.98 KNS Vs= 0.00 KNS
Tu= 13.91 KN-MET Tc= 11.9 KN-MET Ts= 18.5 KN-MET LOAD 407
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 1125. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.00 SQ.CM.

AT END SUPPORT - Vu= 107.97 KNS Vc= 398.05 KNS Vs= 0.00 KNS
Tu= 13.91 KN-MET Tc= 11.9 KN-MET Ts= 18.5 KN-MET LOAD 407
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 1125. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.00 SQ.CM.



ACI 318-11 BEAM NO. 2022 DESIGN RESULTS

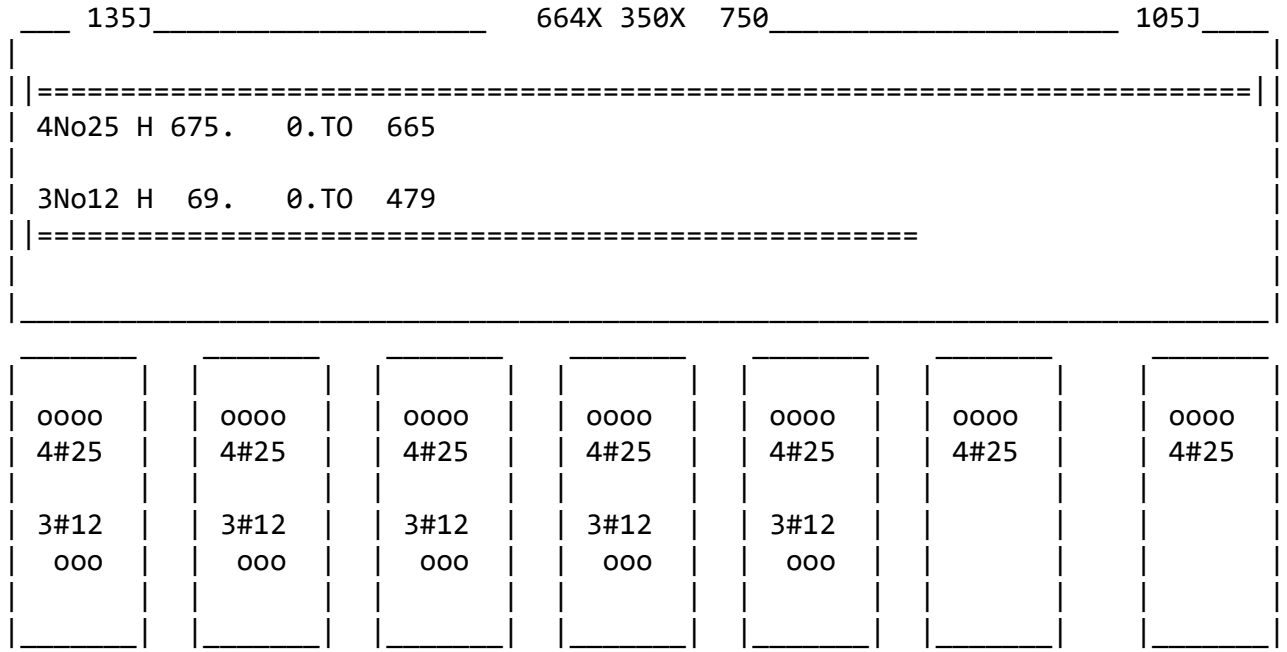
LEN - 665. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	479.	YES	NO
2	675.	4 - 25MM	0.	665.	YES	YES

B E A M N O. 2022 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2022 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2022 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2023 DESIGN RESULTS

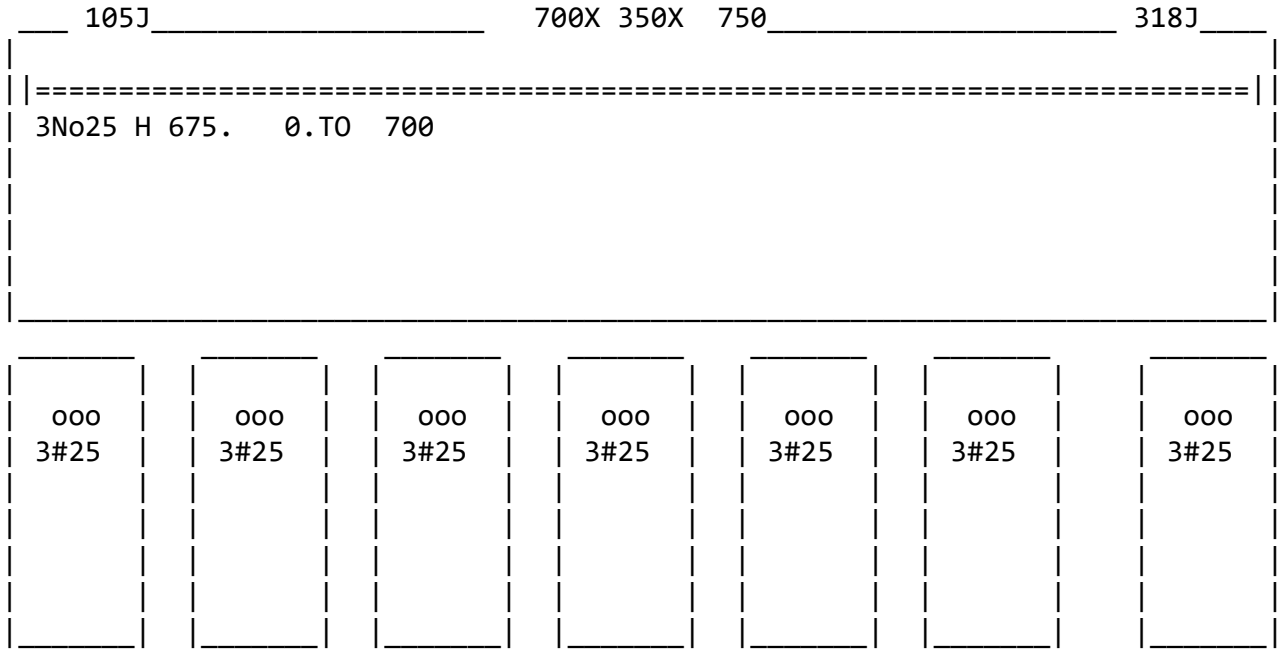
LEN - 700. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	675.	3 - 25MM	0.	700.	YES	YES

B E A M N O. 2023 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2023 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2023 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2024 DESIGN RESULTS

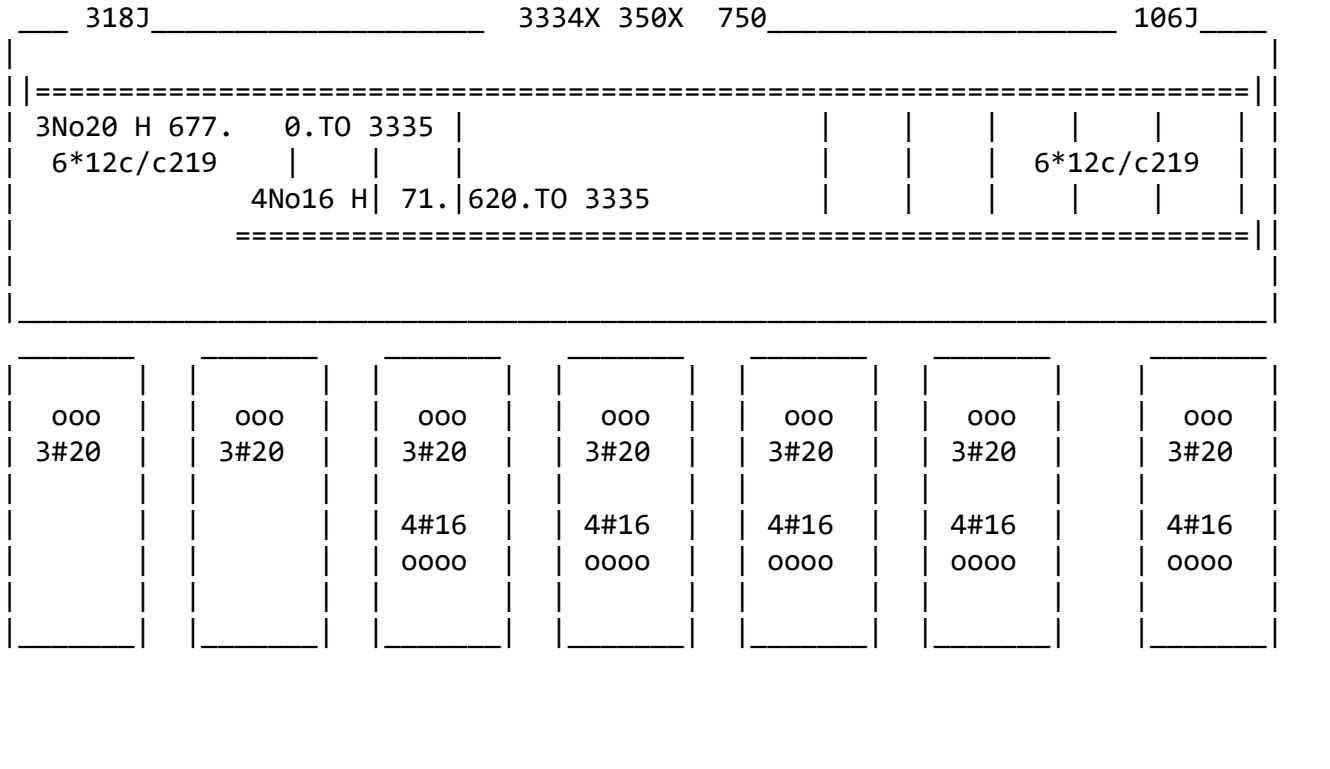
LEN - 3335. MM FY - 420. FC - 35. MPA, SIZE - 350. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	4 - 16MM	620.	3335.	NO	YES
2	677.	3 - 20MM	0.	3335.	YES	YES

B E A M N O. 2024 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 71.12 KNS Vc= 229.32 KNS Vs= 0.00 KNS
Tu= 17.20 KN-MET Tc= 11.6 KN-MET Ts= 22.9 KN-MET LOAD 407
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 993. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.71 SQ.CM.

AT END SUPPORT - Vu= 25.78 KNS Vc= 238.77 KNS Vs= 0.00 KNS
Tu= 17.20 KN-MET Tc= 11.6 KN-MET Ts= 22.9 KN-MET LOAD 407
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 219. MM C/C FOR 993. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.71 SQ.CM.



ACI 318-11 BEAM NO. 2035 DESIGN RESULTS

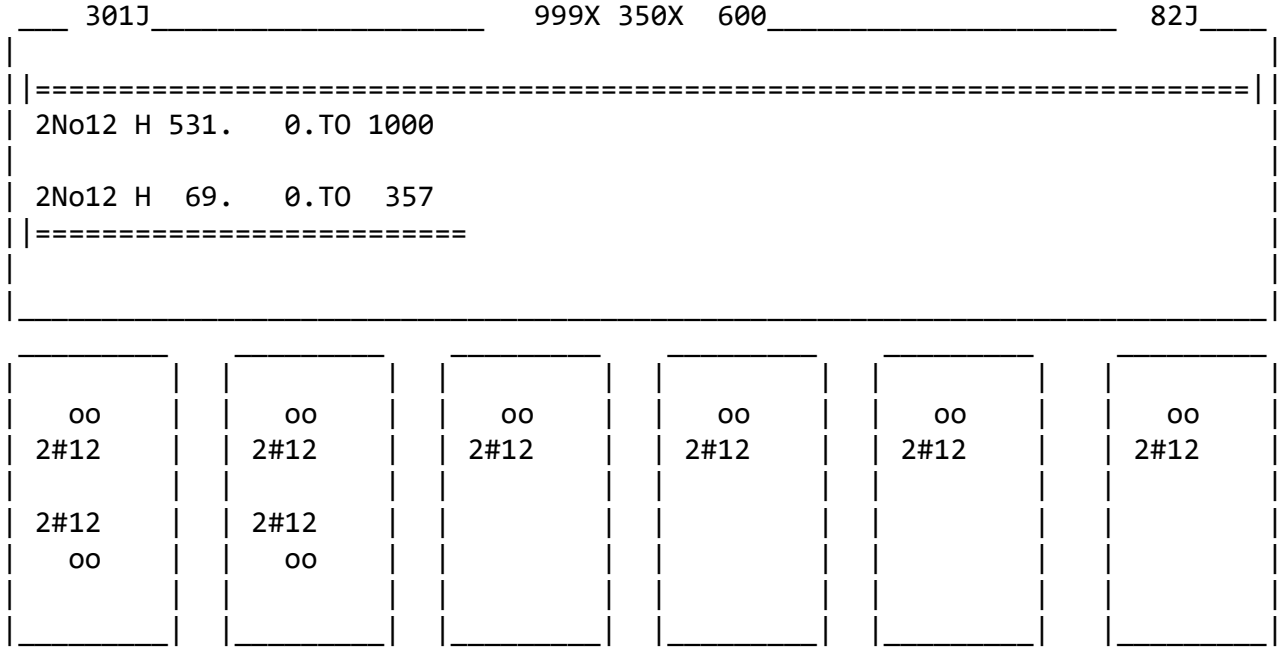
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	357.	YES	NO
2	531.	2 - 12MM	0.	1000.	YES	YES

B E A M N O. 2035 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2035 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2035 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2036 DESIGN RESULTS

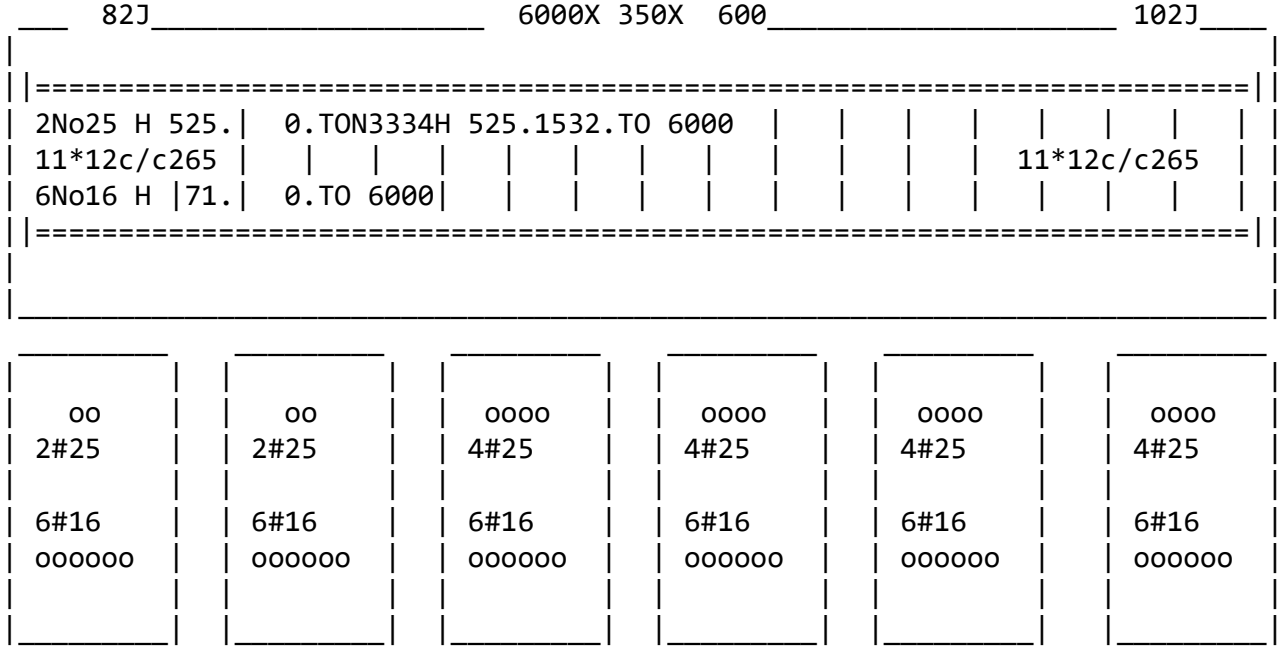
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	6000.	YES	YES
2	525.	2 - 25MM	0.	3334.	YES	NO
3	525.	4 - 25MM	1532.	6000.	NO	YES

B E A M N O. 2036 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 81.69 KNS Vc= 182.63 KNS Vs= 0.00 KNS
Tu= 1.32 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 408
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 2475. MM

AT END SUPPORT - Vu= 67.94 KNS Vc= 163.33 KNS Vs= 0.00 KNS
Tu= 2.04 KN-MET Tc= 7.7 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 2037 DESIGN RESULTS

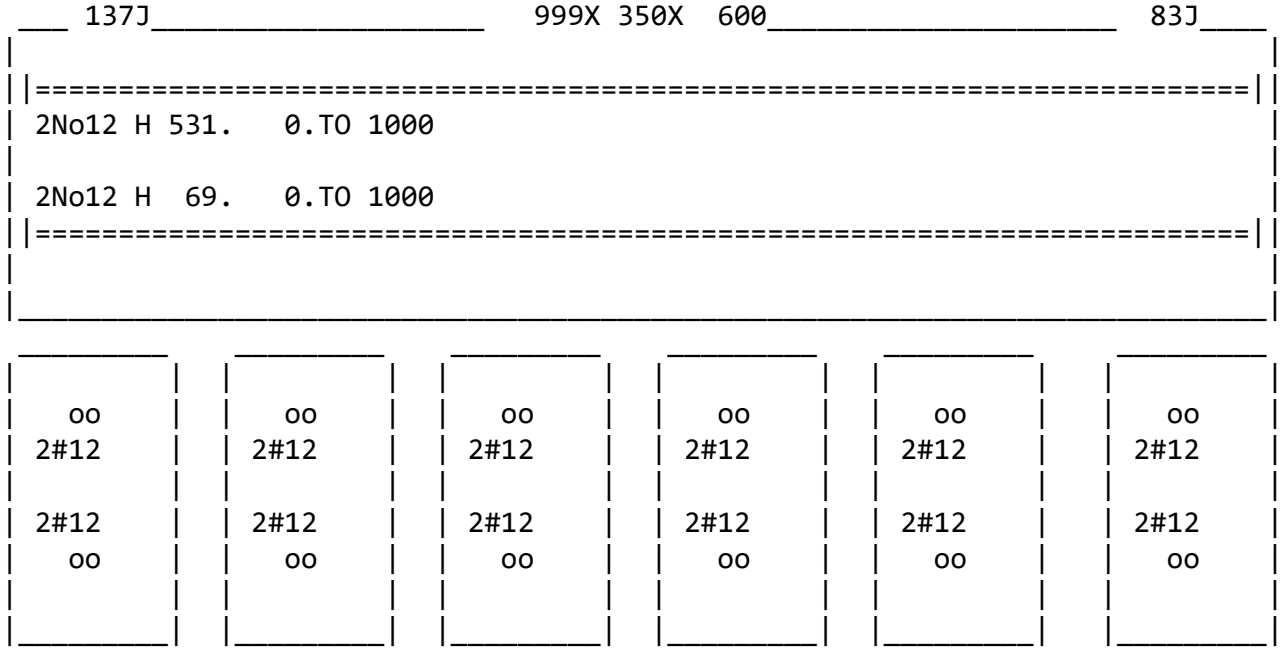
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1000.	YES	YES
2	531.	2 - 12MM	0.	1000.	YES	YES

B E A M N O. 2037 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2037 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2037 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



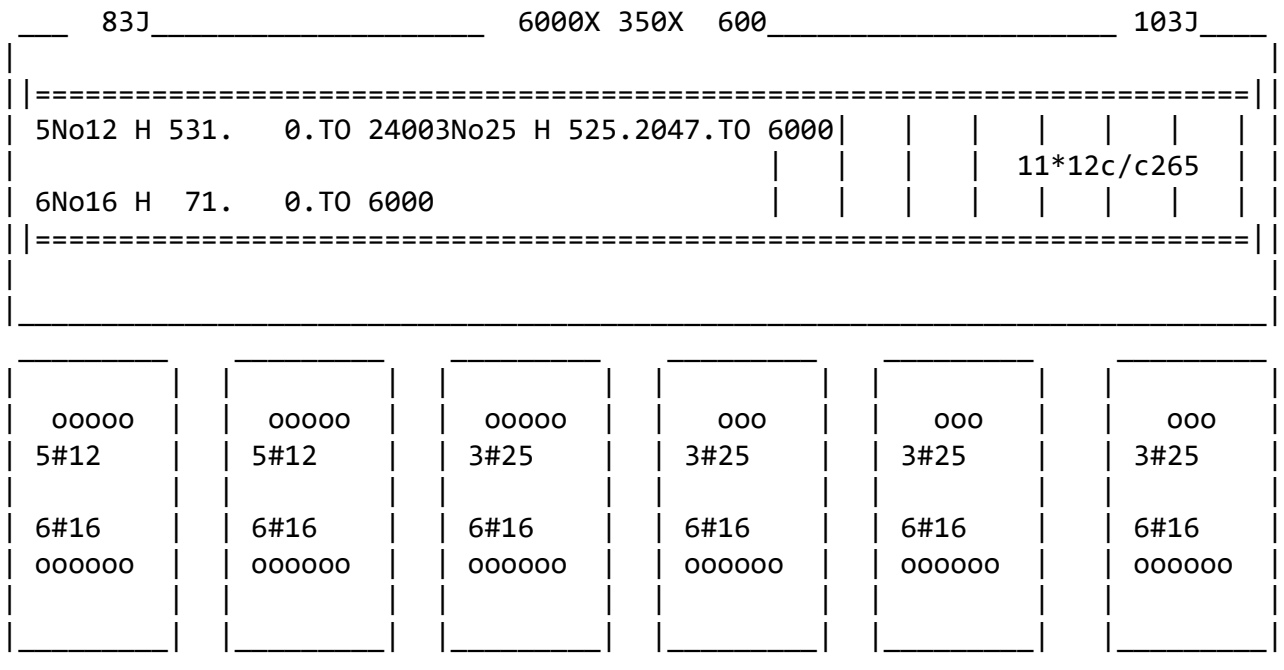
ACI 318-11 BEAM NO. 2038 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	6000.	YES	YES
2	531.	5 - 12MM	0.	2400.	YES	NO
3	525.	3 - 25MM	2047.	6000.	NO	YES

B E A M N O. 2038 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.48 KNS Vc= 173.20 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 8.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 70.36 KNS Vc= 180.24 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 8.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 2039 DESIGN RESULTS

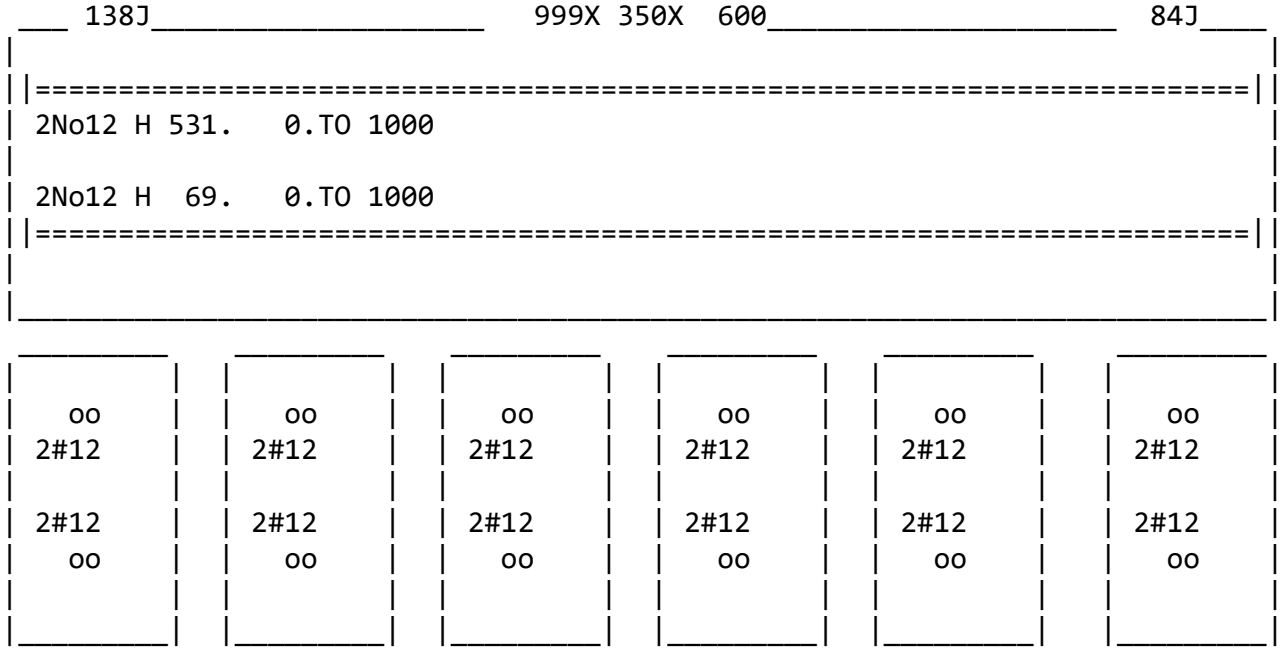
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1000.	YES	YES
2	531.	2 - 12MM	0.	1000.	YES	YES

B E A M N O. 2039 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2039 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2039 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



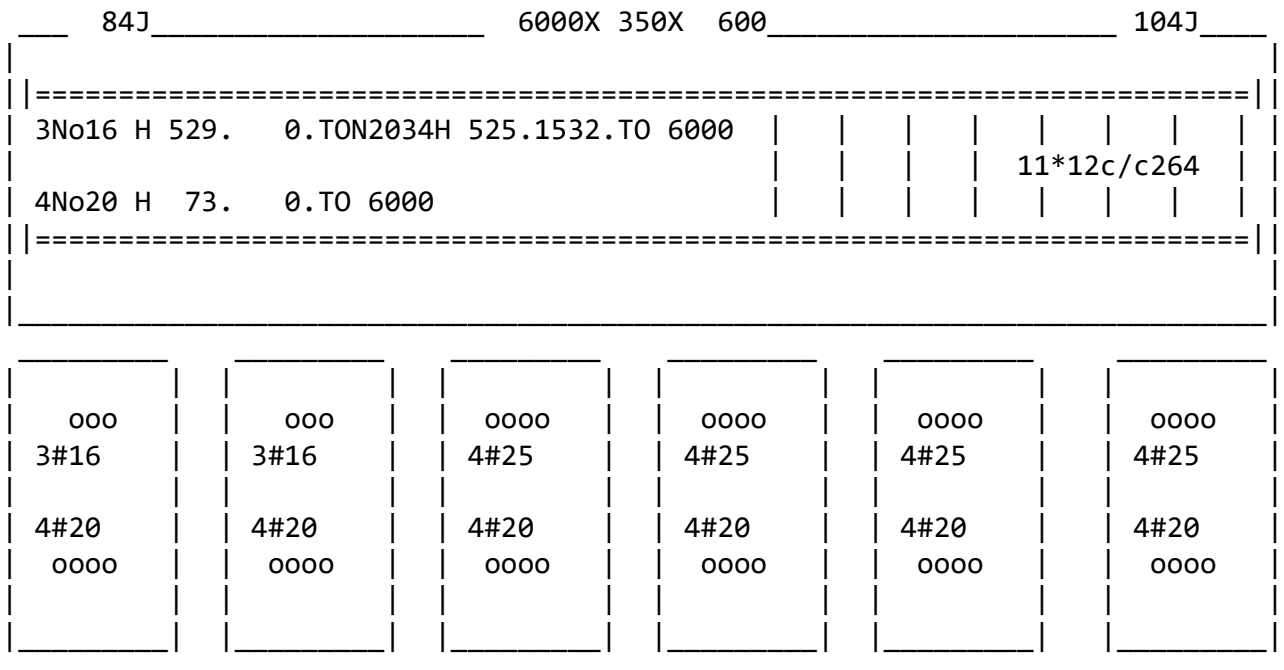
ACI 318-11 BEAM NO. 2040 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	73.	4 - 20MM	0.	6000.	YES	YES
2	529.	3 - 16MM	0.	2034.	YES	NO
3	525.	4 - 25MM	1532.	6000.	NO	YES

B E A M N O. 2040 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 3.18 KNS Vc= 172.82 KNS Vs= 0.00 KNS
Tu= 0.66 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 99.63 KNS Vc= 180.52 KNS Vs= 0.00 KNS
Tu= 0.66 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 264. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 2041 DESIGN RESULTS

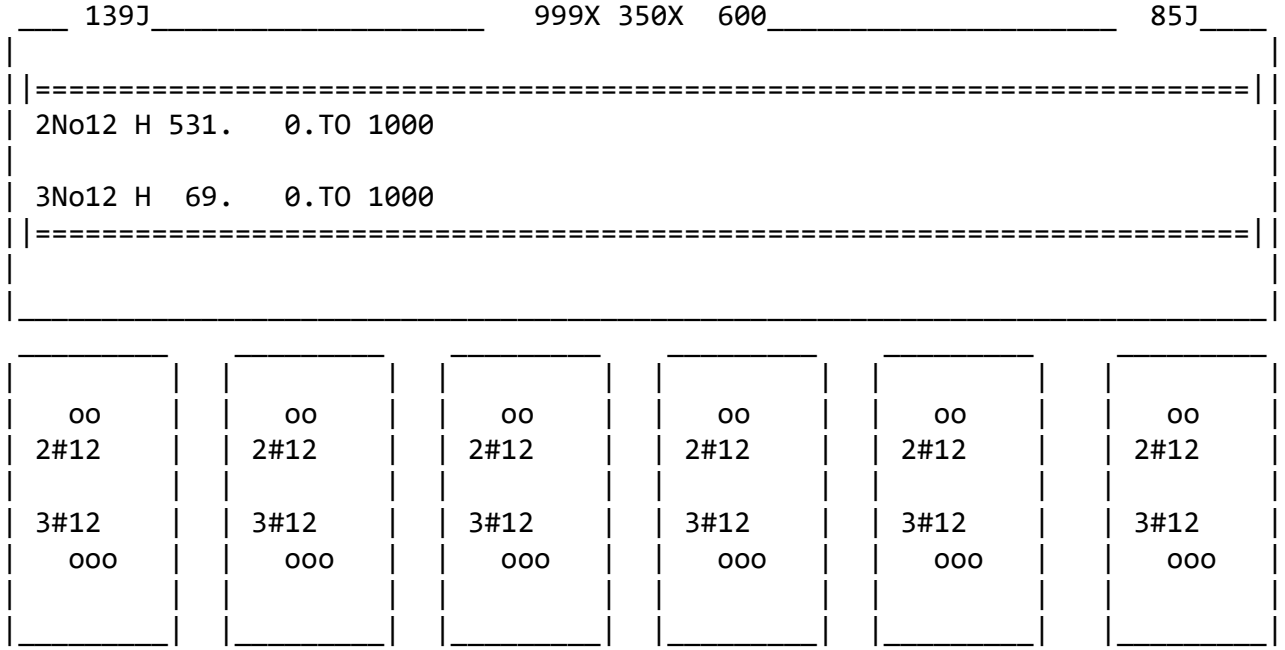
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1000.	YES	YES
2	531.	2 - 12MM	0.	1000.	YES	YES

B E A M N O. 2041 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2041 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2041 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



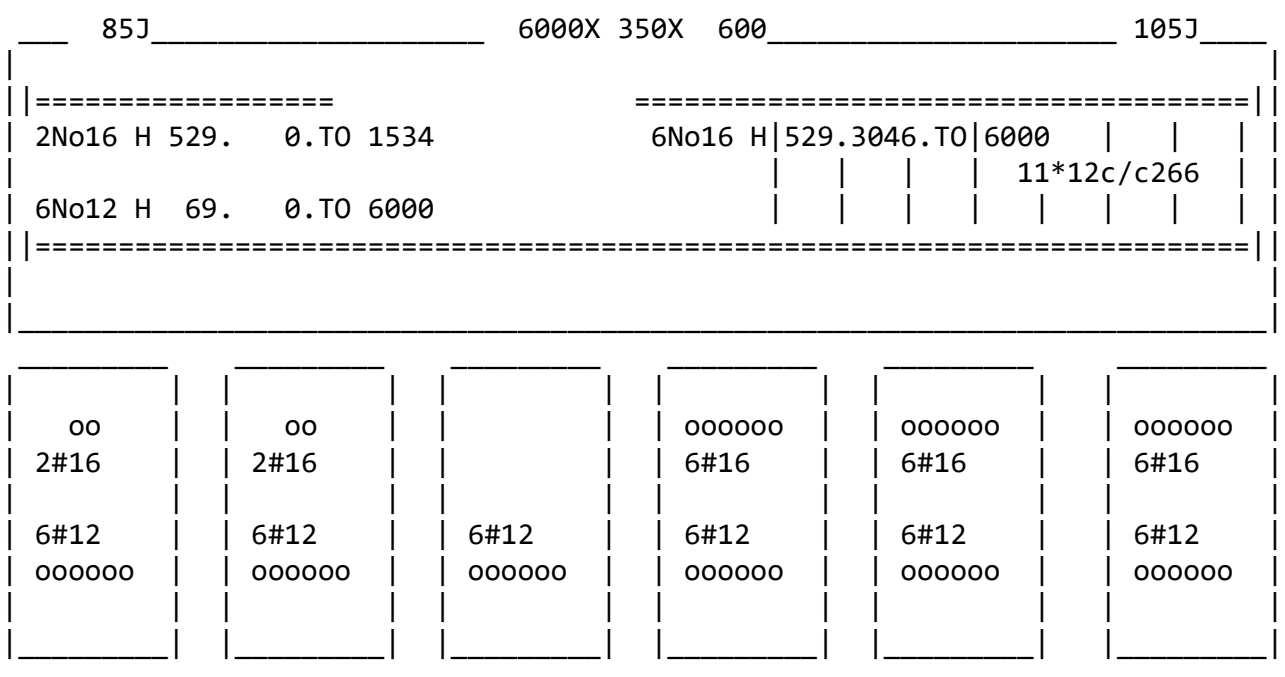
ACI 318-11 BEAM NO. 2042 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	6 - 12MM	0.	6000.	YES	YES
2	529.	2 - 16MM	0.	1534.	YES	NO
3	529.	6 - 16MM	3046.	6000.	NO	YES

B E A M N O. 2042 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.61 KNS Vc= 175.18 KNS Vs= 0.00 KNS
Tu= 1.30 KN-MET Tc= 8.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 74.89 KNS Vc= 179.28 KNS Vs= 0.00 KNS
Tu= 1.30 KN-MET Tc= 8.6 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 2043 DESIGN RESULTS

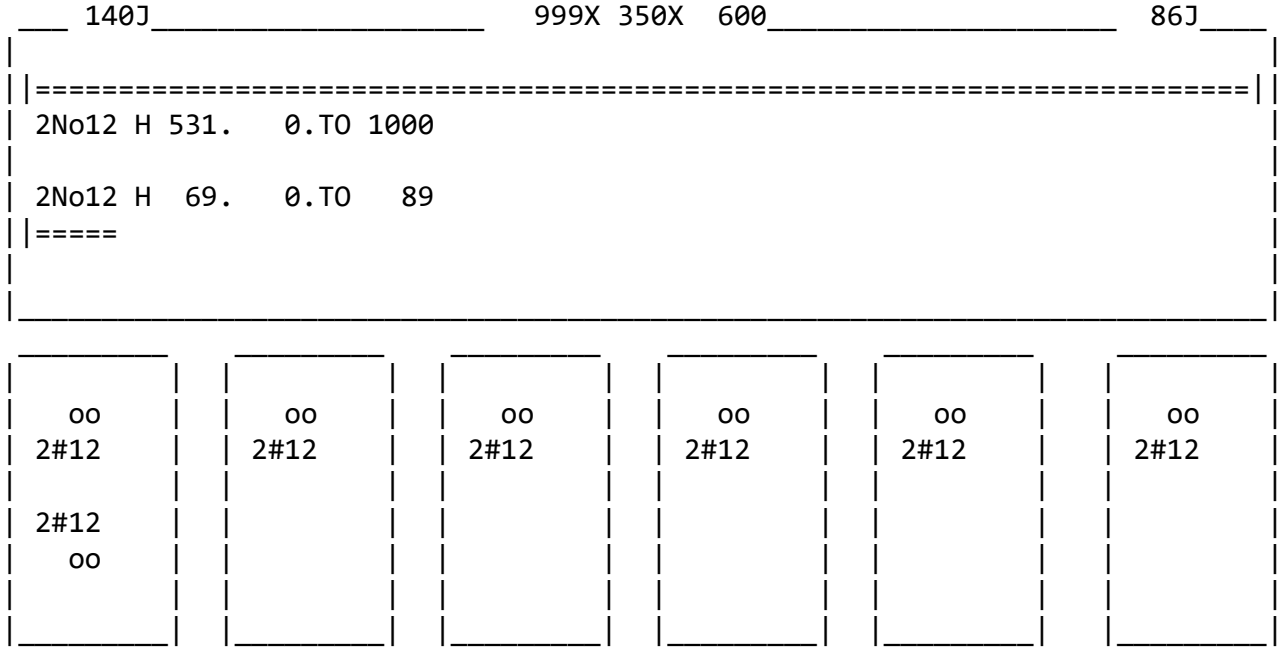
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	89.	YES	NO
2	531.	2 - 12MM	0.	1000.	YES	YES

B E A M N O. 2043 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2043 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2043 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



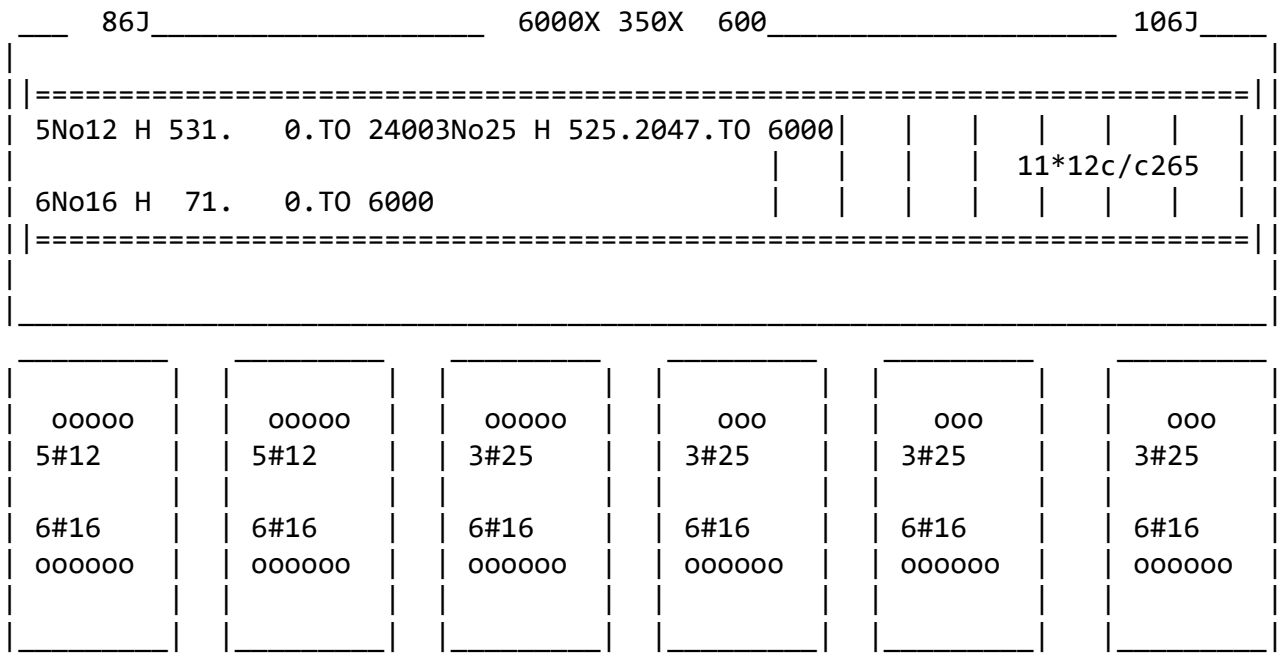
ACI 318-11 BEAM NO. 2044 DESIGN RESULTS

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	71.	6 - 16MM	0.	6000.	YES	YES
2	531.	5 - 12MM	0.	2400.	YES	NO
3	525.	3 - 25MM	2047.	6000.	NO	YES

B E A M N O. 2044 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 4.54 KNS Vc= 177.17 KNS Vs= 0.00 KNS
Tu= 1.63 KN-MET Tc= 8.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 84.36 KNS Vc= 177.17 KNS Vs= 0.00 KNS
Tu= 1.63 KN-MET Tc= 8.3 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 265. MM C/C FOR 2475. MM



ACI 318-11 BEAM NO. 2045 DESIGN RESULTS

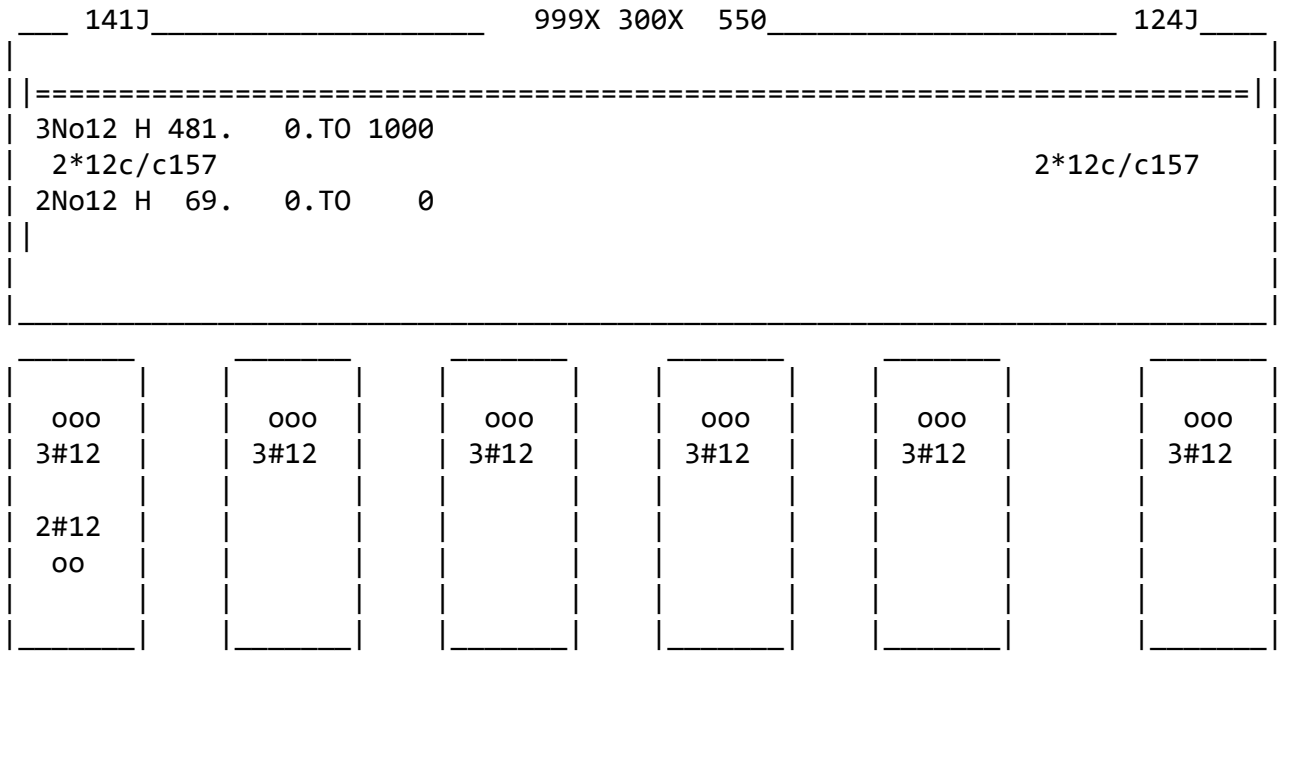
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	0.	YES	NO
2	481.	3 - 12MM	0.	1000.	YES	YES

B E A M N O. 2045 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 35.12 KNS Vc= 141.19 KNS Vs= 0.00 KNS
Tu= 10.94 KN-MET Tc= 5.9 KN-MET Ts= 14.6 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.11 SQ.CM.

AT END SUPPORT - Vu= 35.91 KNS Vc= 141.19 KNS Vs= 0.00 KNS
Tu= 10.94 KN-MET Tc= 5.9 KN-MET Ts= 14.6 KN-MET LOAD 403
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.11 SQ.CM.



ACI 318-11 BEAM NO. 2046 DESIGN RESULTS

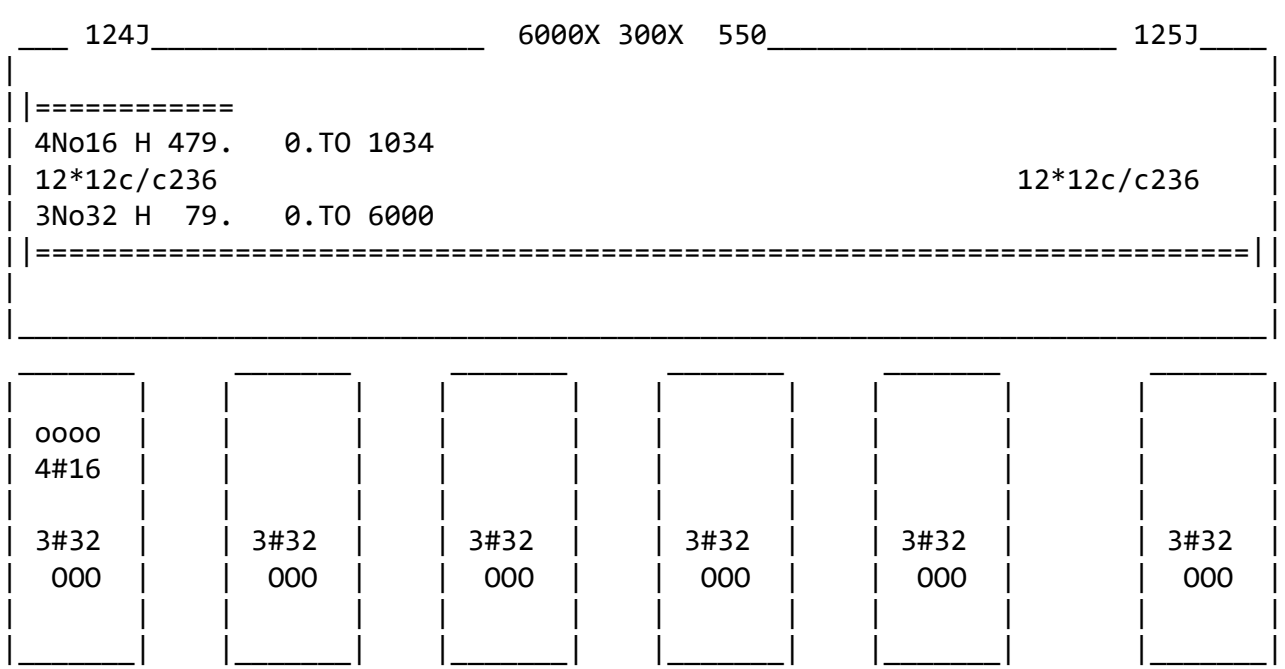
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	6000.	YES	YES
2	479.	4 - 16MM	0.	1034.	YES	NO

B E A M N O. 2046 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 179.36 KNS Vc= 136.89 KNS Vs= 102.26 KNS
Tu= 1.76 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 236. MM C/C FOR 2525. MM

AT END SUPPORT - Vu= 152.15 KNS Vc= 136.89 KNS Vs= 65.98 KNS
Tu= 1.76 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 236. MM C/C FOR 2525. MM



ACI 318-11 BEAM NO. 2047 DESIGN RESULTS

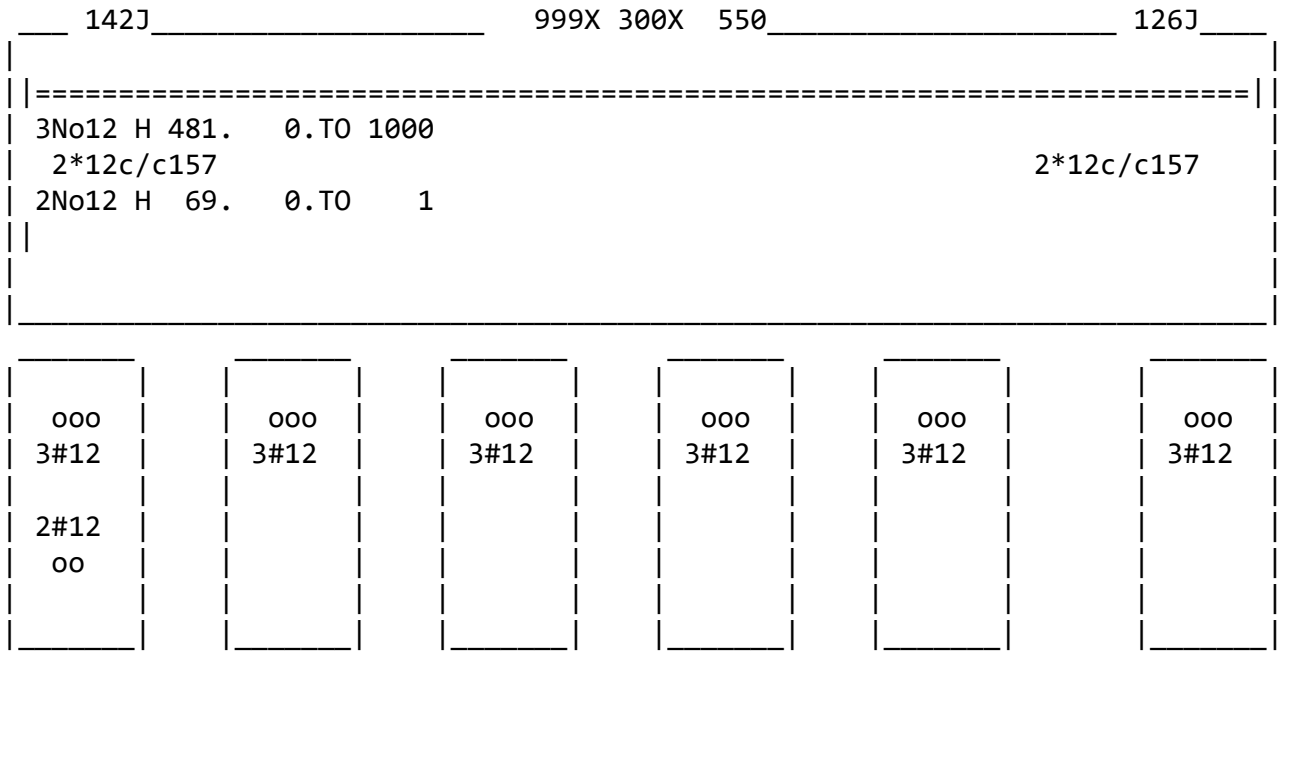
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1.	YES	NO
2	481.	3 - 12MM	0.	1000.	YES	YES

B E A M N O. 2047 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.36 KNS Vc= 142.06 KNS Vs= 0.00 KNS
Tu= 10.34 KN-MET Tc= 5.9 KN-MET Ts= 13.8 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.94 SQ.CM.

AT END SUPPORT - Vu= 35.15 KNS Vc= 142.06 KNS Vs= 0.00 KNS
Tu= 10.34 KN-MET Tc= 5.9 KN-MET Ts= 13.8 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.94 SQ.CM.



ACI 318-11 BEAM NO. 2048 DESIGN RESULTS

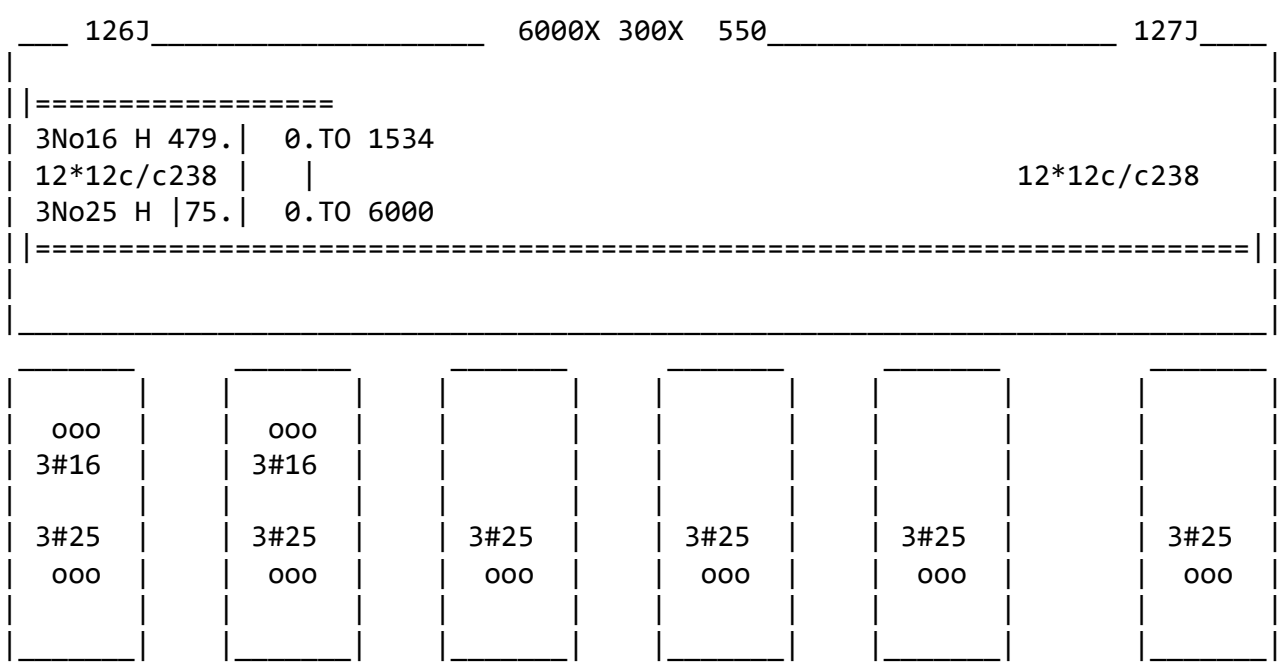
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	6000.	YES	YES
2	479.	3 - 16MM	0.	1534.	YES	NO

B E A M N O. 2048 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 143.52 KNS Vc= 137.29 KNS Vs= 54.06 KNS
Tu= 0.02 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 2525. MM

AT END SUPPORT - Vu= 90.51 KNS Vc= 139.13 KNS Vs= 0.00 KNS
Tu= 0.73 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 2525. MM



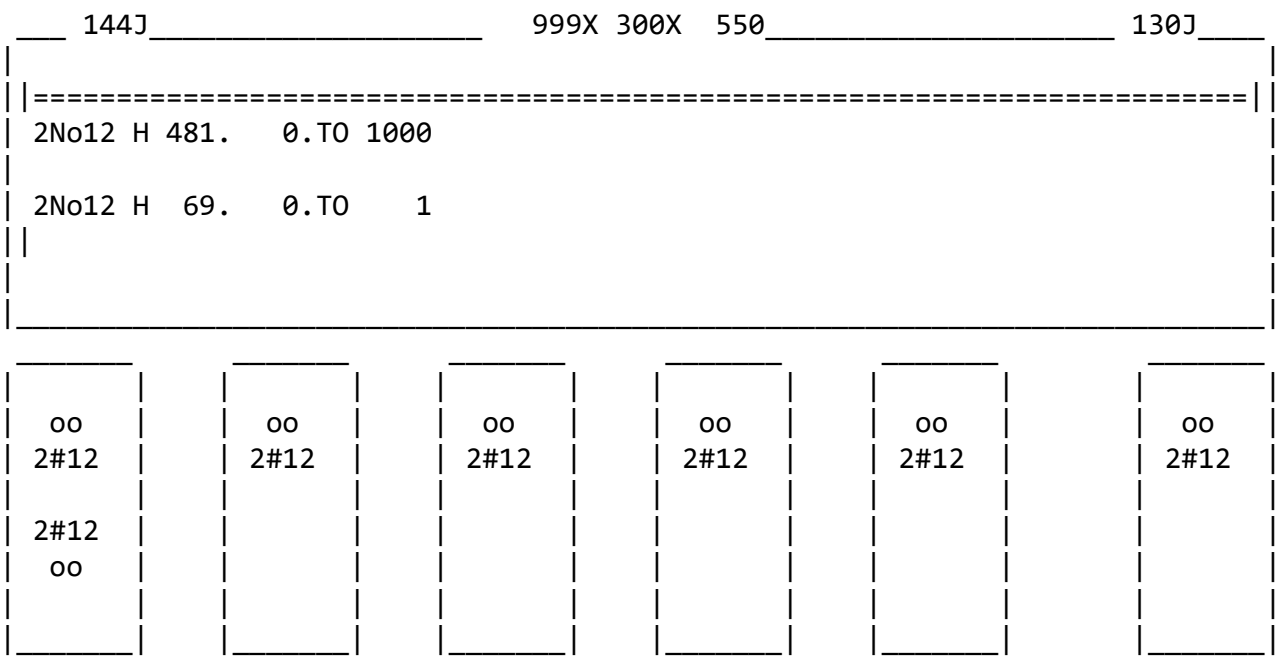
ACI 318-11 BEAM NO. 2049 DESIGN RESULTS

LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1.	YES	NO
2	481.	2 - 12MM	0.	1000.	YES	YES

B E A M N O. 2049 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.97 KNS Vc= 142.03 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.46 KNS Vc= 142.03 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 2050 DESIGN RESULTS

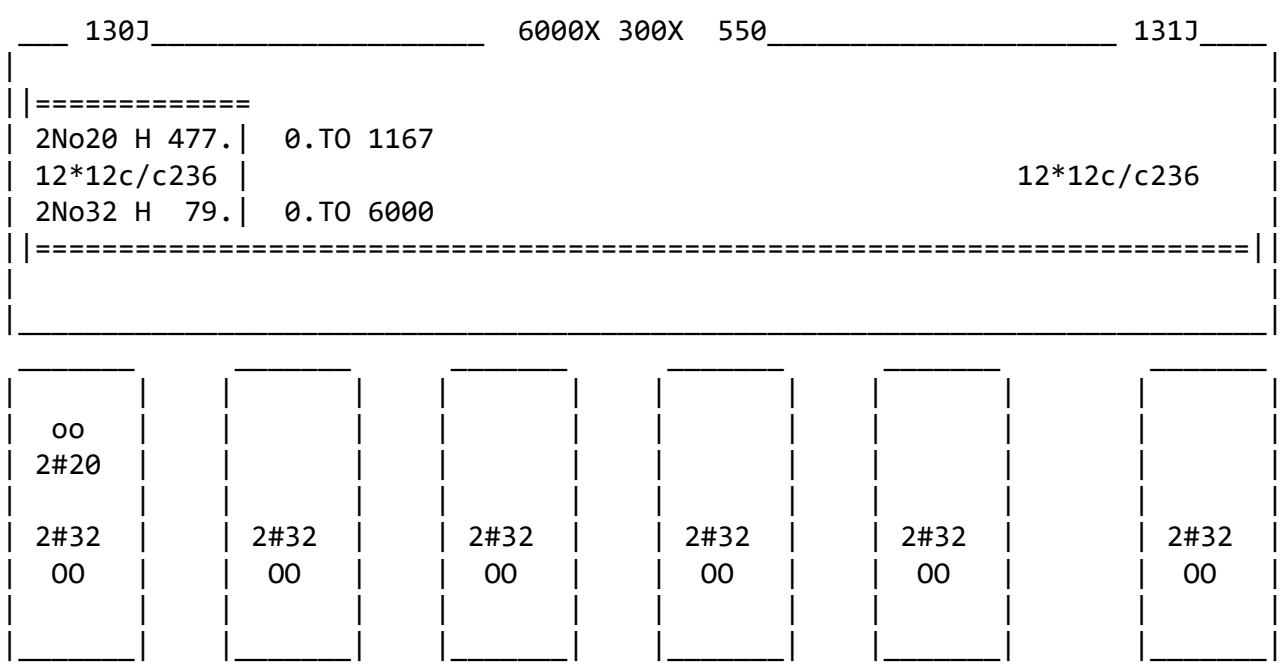
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	2 - 32MM	0.	6000.	YES	YES
2	477.	2 - 20MM	0.	1167.	YES	NO

B E A M N O. 2050 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 95.94 KNS Vc= 245.20 KNS Vs= 0.00 KNS
Tu= 2.28 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 236. MM C/C FOR 2525. MM

AT END SUPPORT - Vu= 98.44 KNS Vc= 158.07 KNS Vs= 0.00 KNS
Tu= 1.55 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 236. MM C/C FOR 2525. MM



ACI 318-11 BEAM NO. 2051 DESIGN RESULTS

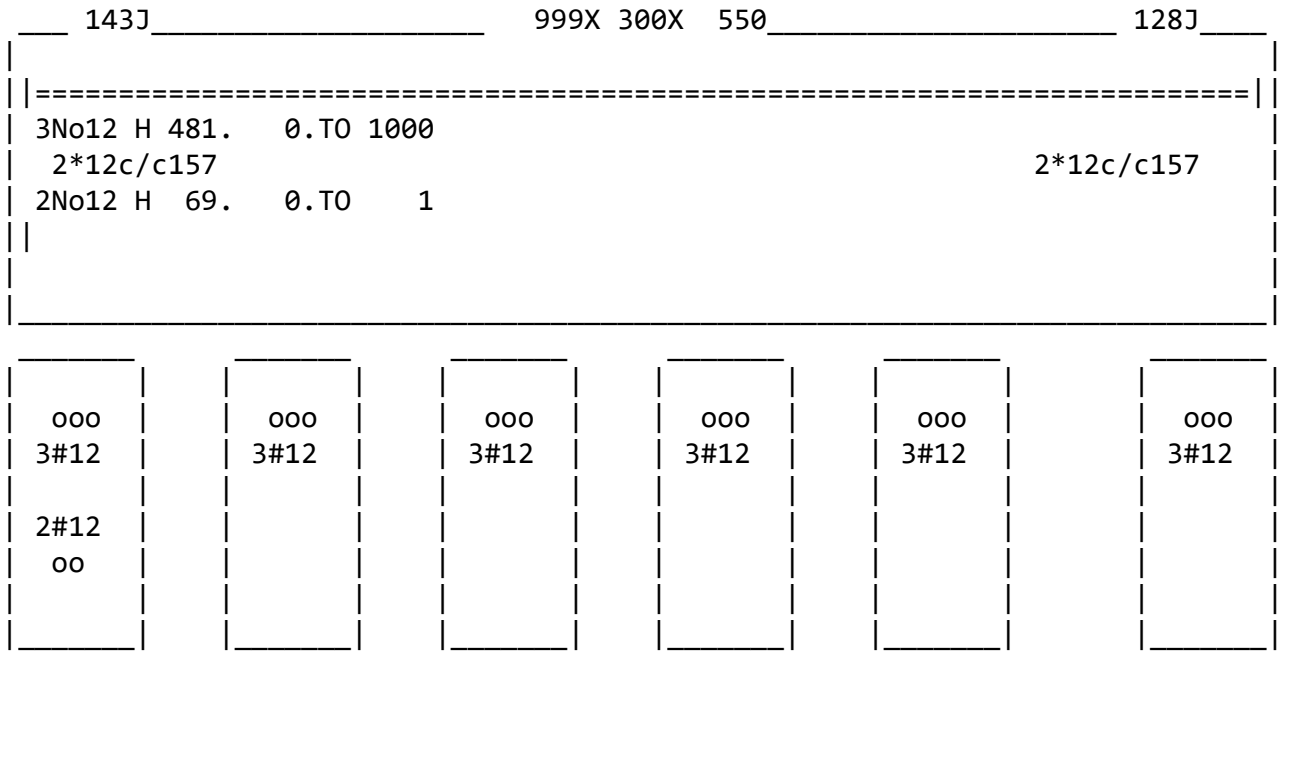
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1.	YES	NO
2	481.	3 - 12MM	0.	1000.	YES	YES

B E A M N O. 2051 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.27 KNS Vc= 141.42 KNS Vs= 0.00 KNS
Tu= 8.63 KN-MET Tc= 5.9 KN-MET Ts= 11.5 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.45 SQ.CM.

AT END SUPPORT - Vu= 35.06 KNS Vc= 141.42 KNS Vs= 0.00 KNS
Tu= 8.63 KN-MET Tc= 5.9 KN-MET Ts= 11.5 KN-MET LOAD 406
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 25. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.45 SQ.CM.



ACI 318-11 BEAM NO. 2052 DESIGN RESULTS

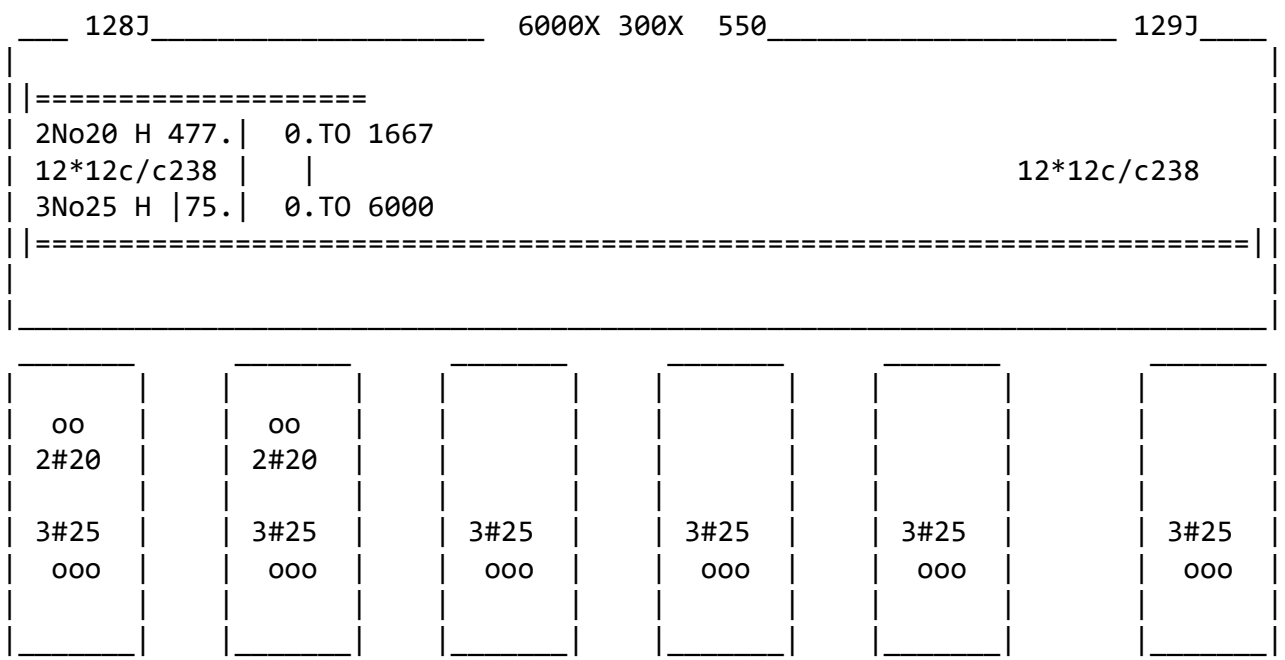
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 550. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	75.	3 - 25MM	0.	6000.	YES	YES
2	477.	2 - 20MM	0.	1667.	YES	NO

B E A M N O. 2052 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 144.00 KNS Vc= 135.98 KNS Vs= 56.03 KNS
Tu= 0.29 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 2525. MM

AT END SUPPORT - Vu= 90.36 KNS Vc= 136.94 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 402
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 238. MM C/C FOR 2525. MM



ACI 318-11 BEAM NO. 2053 DESIGN RESULTS

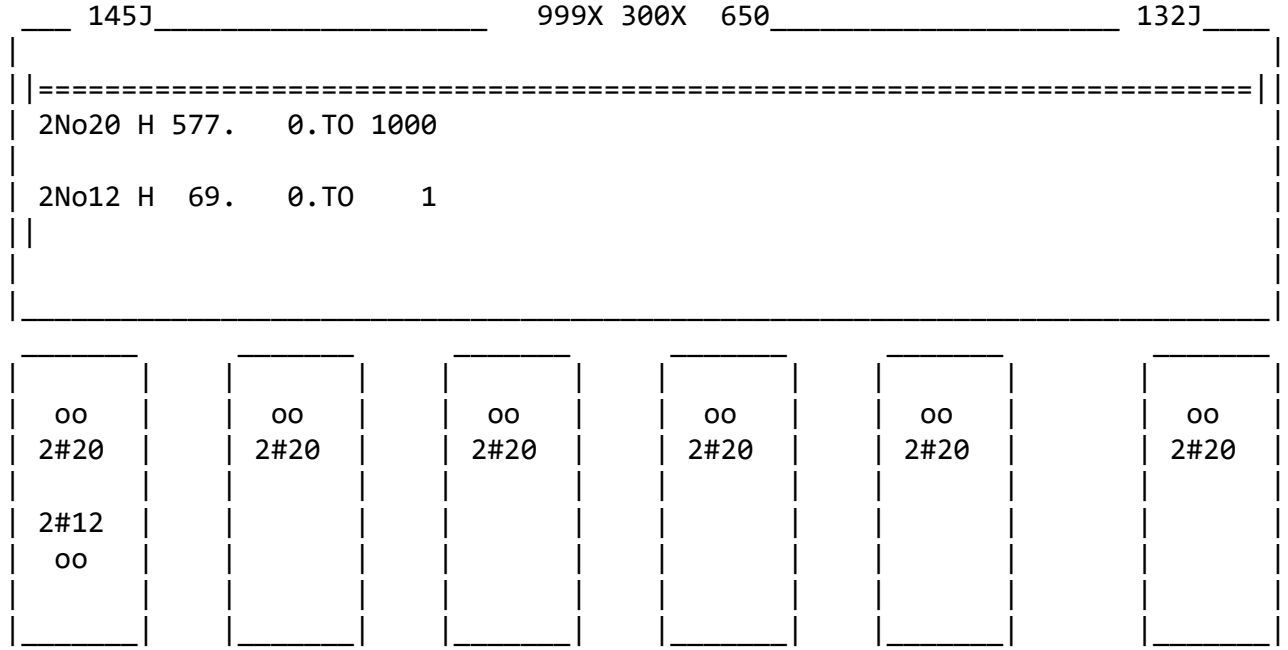
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	0.	1.	YES	NO
2	577.	2 - 20MM	0.	1000.	YES	YES

B E A M N O. 2053 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2053 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2053 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2054 DESIGN RESULTS

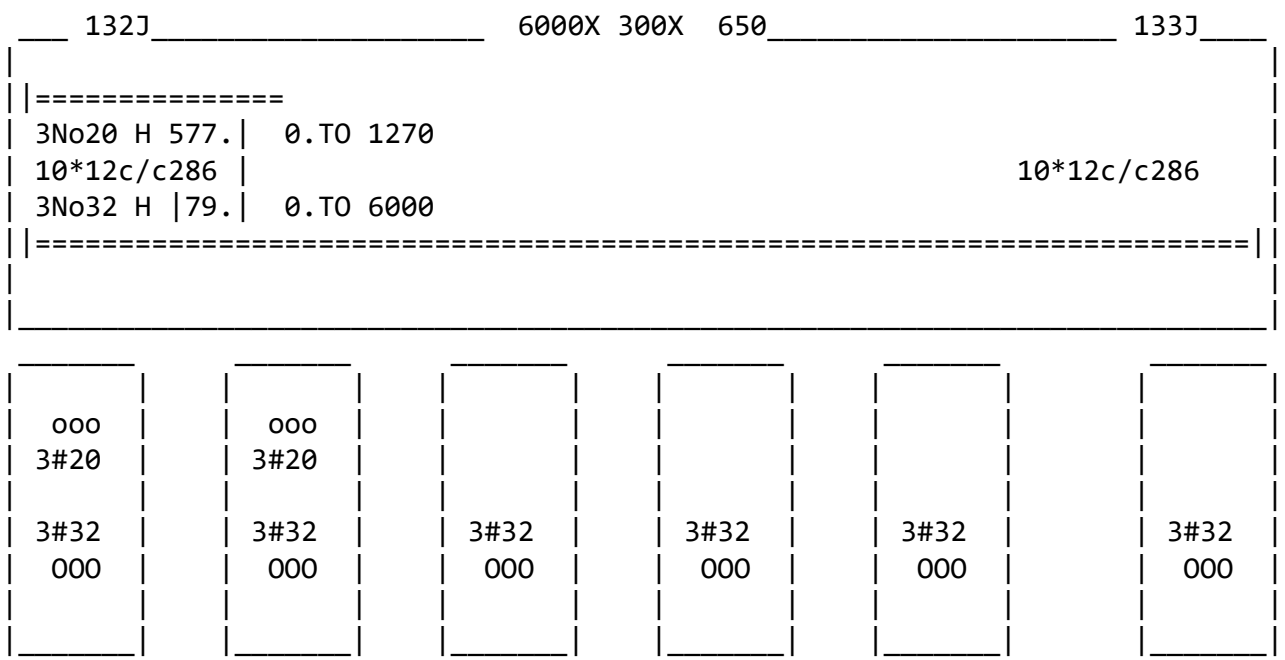
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	6000.	YES	YES
2	577.	3 - 20MM	0.	1270.	YES	NO

B E A M N O. 2054 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 258.27 KNS Vc= 168.30 KNS Vs= 176.06 KNS
Tu= 1.61 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 286. MM C/C FOR 2425. MM

AT END SUPPORT - Vu= 112.79 KNS Vc= 198.66 KNS Vs= 0.00 KNS
Tu= 1.67 KN-MET Tc= 7.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 286. MM C/C FOR 2425. MM



ACI 318-11 BEAM NO. 2055 DESIGN RESULTS

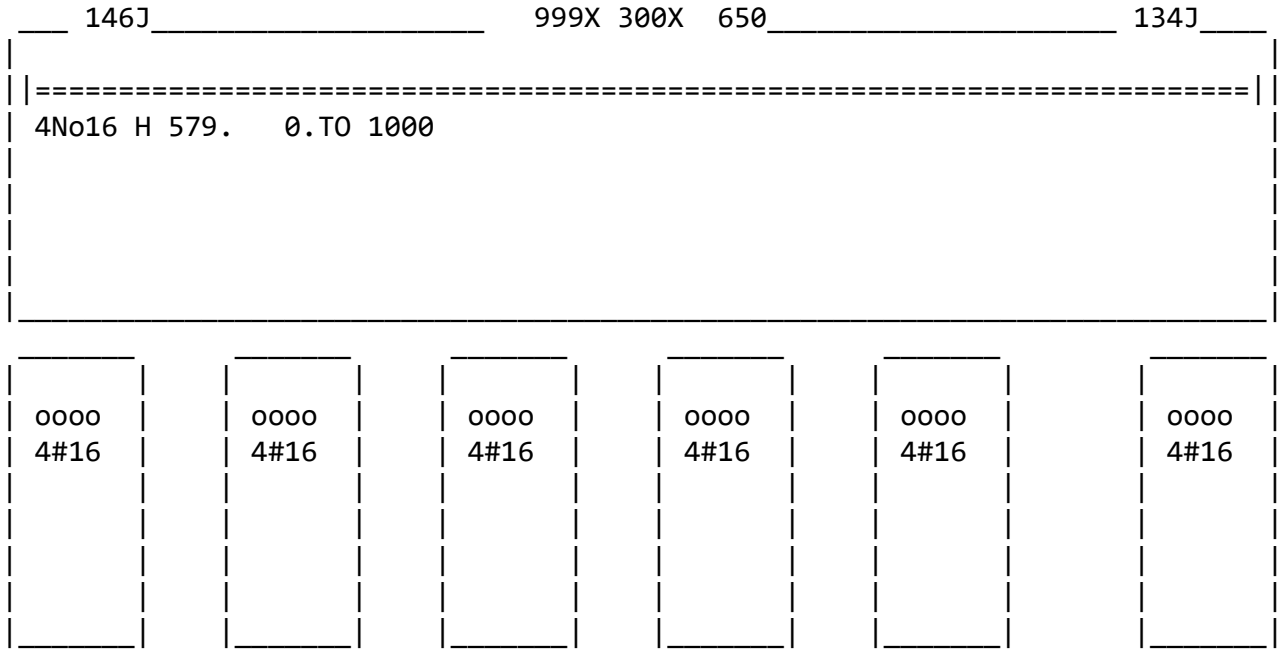
LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	579.	4 - 16MM	0.	1000.	YES	YES

B E A M N O. 2055 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 2055 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 2055 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 2056 DESIGN RESULTS

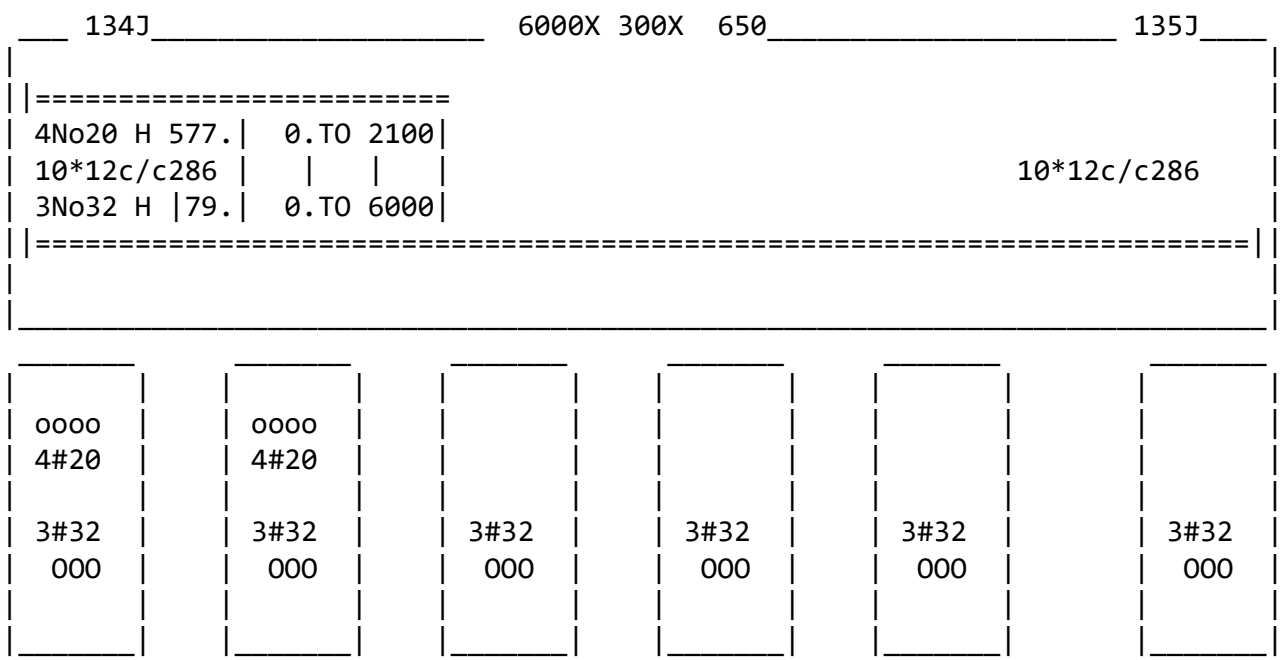
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 650. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	79.	3 - 32MM	0.	6000.	YES	YES
2	577.	4 - 20MM	0.	2100.	YES	NO

B E A M N O. 2056 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 238.21 KNS Vc= 158.69 KNS Vs= 158.93 KNS
Tu= 0.08 KN-MET Tc= 7.0 KN-MET Ts= 0.0 KN-MET LOAD 403
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 286. MM C/C FOR 2425. MM

AT END SUPPORT - Vu= 98.32 KNS Vc= 161.41 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 7.1 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 286. MM C/C FOR 2425. MM



**********END OF BEAM DESIGN**********

2228. DESIGN COLUMN 1301 TO 1316

ACI 318-11 COLUMN NO. 1301 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1302 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
AREA OF STEEL REQUIRED = 2335.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	407	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1303 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
AREA OF STEEL REQUIRED = 2144.1 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

12 - 16 MM 1.149 407 STA 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1304 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1305 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1306 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1307 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1308 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
AREA OF STEEL REQUIRED = 2114.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	402	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1309 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
AREA OF STEEL REQUIRED = 2114.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	408	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1310 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
AREA OF STEEL REQUIRED = 2232.3 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	407	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1311 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
AREA OF STEEL REQUIRED = 2570.4 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 32 MM	1.532	407	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1312 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
AREA OF STEEL REQUIRED = 2158.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	407	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1313 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1314 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1315 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1316 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 600.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.149	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

2229. DESIGN COLUMN 1331 TO 1387 2313 2316 2319

ACI 318-11 COLUMN NO. 1331 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1332 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1333 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1334 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 6135.3 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 20 MM	1.232	412	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1335 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1336 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1337 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1338 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 10240.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 25 MM	2.310	411	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1339 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1340 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1341 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1342 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1343 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 6313.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 32 MM	1.262	423	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1344 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 8241.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 20 MM	1.725	411	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1345 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1346 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1347 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 5135.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	419	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1348 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 5135.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	419	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1349 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 6528.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 20 MM	1.478	406	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)

TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1350 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 5135.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	407	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1351 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3194.4 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	410	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1352 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

28 - 12 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1353 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1354 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3617.9 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.196	407	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1355 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 4613.1 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.595	407	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1356 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1357 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1358 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1359 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1360 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1361 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 5635.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 25 MM	1.155	407	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1362 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1363 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

28 - 16 MM 1.104 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1364 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1365 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1366 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1367 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1368 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1369 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1370 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1371 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1372 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1373 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 5135.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	427	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1374 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

28 - 16 MM 1.104 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1375 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1376 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1377 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 5135.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	410	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1378 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1379 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 8777.1 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 20 MM	1.725	407	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1380 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1381 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 5242.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	403	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1382 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1383 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1384 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1385 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1386 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1387 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 2313 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 2316 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 2319 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 850.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5100.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.104	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

2230. DESIGN COLUMN 1388 TO 1446 2320

ACI 318-11 COLUMN NO. 1388 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1389 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1390 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1391 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1392 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1393 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1394 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3935.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.329	412	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1395 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1396 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1397 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1398 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 4697.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

24 - 16 MM 1.595 406 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1399 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1400 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3046.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	429	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1401 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 12 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1402 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 12 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1403 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 8001.1 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 20 MM 2.908 406 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1404 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1405 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1406 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1407 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1408 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1409 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 12 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1410 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 12 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1411 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 12 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1412 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1413 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1414 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1415 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1416 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1417 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3427.3 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

32 - 12 MM 1.196 407 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1418 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 5269.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.861	407	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1419 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1420 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 4570.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.595	407	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1421 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1422 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1423 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1424 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3046.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	407	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1425 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1426 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1427 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1428 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1429 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1430 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3406.1 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.196	407	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1431 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1432 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1433 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1434 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1435 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1436 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1437 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1438 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3109.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	406	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1439 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1440 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1441 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1442 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1443 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1444 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3046.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

28 - 12 MM 1.047 415 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1445 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 1446 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 3808.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.298	409	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 2320 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

2231. DESIGN COLUMN 1951 TO 1964

ACI 318-11 COLUMN NO. 1951 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

ACI 318-11 COLUMN NO. 1952 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

ACI 318-11 COLUMN NO. 1953 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.077 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

ACI 318-11 COLUMN NO. 1954 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.077 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

ACI 318-11 COLUMN NO. 1955 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.077 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

ACI 318-11 COLUMN NO. 1956 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
AREA OF STEEL REQUIRED = 3622.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	425	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

ACI 318-11 COLUMN NO. 1957 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
AREA OF STEEL REQUIRED = 4382.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.379	425	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1958 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
AREA OF STEEL REQUIRED = 5264.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.609	425	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 1959 DESIGN RESULTS